

Feynman's Preface



These are the lectures in physics that I gave last year and the year before to the freshman and sophomore classes at Caltech. The lectures are, of course, not verbatim—they have been edited, sometimes extensively and sometimes less so. The lectures form only part of the complete course. The whole group of 180 students gathered in a big lecture room twice a week to hear these lectures and then they broke up into small groups of 15 to 20 students in recitation sections under the guidance of a teaching assistant. In addition, there was a laboratory session once a week.

The special problem we tried to get at with these lectures was to maintain the interest of the very enthusiastic and rather smart students coming out of the high schools and into Caltech. They have heard a lot about how interesting and exciting physics is—the theory of relativity, quantum mechanics, and other modern ideas. By the end of two years of our previous course, many would be very discouraged because there were really very few grand, new, modern ideas presented to them. They were made to study inclined planes, electrostatics, and so forth, and after two years it was quite stultifying. The problem was whether or not we could make a course which would save the more advanced and excited student by maintaining his enthusiasm.

The lectures here are not in any way meant to be a survey course, but are very serious. I thought to address them to the most intelligent in the class and to make sure, if possible, that even the most intelligent student was unable to completely encompass everything that was in the lectures—by putting in suggestions of applications of the ideas and concepts in various directions outside the main line of attack. For this reason, though, I tried very hard to make all the statements as accurate as possible, to point out in every case where the equations and ideas fitted into the body of physics, and how—when they learned more—things would be modified. I also felt that for such students it is important to indicate what it is that they should—if they are sufficiently clever—be able to understand by deduction from what has been said before, and what is being put in as something new. When new ideas came in, I would try either to deduce them if they were deducible, or to explain that it was a new idea which hadn't any basis in terms of things they had already learned and which was not supposed to be provable—but was just added in.

At the start of these lectures, I assumed that the students knew something when they came out of high school—such things as geometrical optics, simple chemistry ideas, and so on. I also didn't see that there was any reason to make the lectures

in a definite order, in the sense that I would not be allowed to mention something until I was ready to discuss it in detail. There was a great deal of mention of things **to** come, without complete discussions. These more complete discussions would come later when the preparation became more advanced. Examples are the discussions of inductance, and of energy levels, which are at first brought in in a very qualitative way and are later developed more completely.

At the same time that I was aiming at the more active student, I also wanted to take care of the fellow for whom the extra fireworks and side applications are merely disquieting and who cannot be expected to learn most of the material in the lecture at all. For such students I wanted there to be at least a central core or backbone of material which he *could* get. Even if he didn't understand everything in a lecture, I hoped he wouldn't get nervous. I didn't expect him to understand everything, but only the central and most direct features. It takes, of course, a certain intelligence on his part to see which are the central theorems and central ideas, and which are the more advanced side issues and applications which he may understand only in later years.

In giving these lectures there was one serious difficulty: in the way the course was given, there wasn't any feedback from the students to the lecturer to indicate how well the lectures were going over. This is indeed a very serious difficulty, and I don't know how good the lectures really are. The whole thing was essentially an experiment. And if I did it again I wouldn't do it the same way—I hope I *don't* have to do it again! I think, though, that things worked out—so far as the physics is concerned—quite satisfactorily in the first year.

In the second year I was not so satisfied. In the first part of the course, dealing with electricity and magnetism, I couldn't think of any really unique or different way of doing it—of any way that would be particularly more exciting than the usual way of presenting it. So I don't think I did very much in the lectures on electricity and magnetism. At the end of the second year I had originally intended to go on, after the electricity and magnetism, by giving some more lectures on the properties of materials, but mainly to take up things like fundamental modes, solutions of the diffusion equation, vibrating systems, orthogonal functions,... developing the first stages of what are usually called "the mathematical methods of physics." In retrospect, I think that if I were doing it again I would go back to that original idea. But since it was not planned that I would be giving these lectures again, it was suggested that it might be a good idea to try to give an introduction to the quantum mechanics—what you will find in Volume III.

It is perfectly clear that students who will major in physics can wait until their third year for quantum mechanics. On the other hand, the argument was made that many of the students in our course study physics as a background for their primary interest in other fields. And the usual way of dealing with quantum mechanics makes that subject almost unavailable for the great majority of students because they have to take so long to learn it. Yet, in its real applications—especially in its more complex applications, such as in electrical engineering and chemistry—the full machinery of the differential equation approach is not actually used. So I tried to describe the principles of quantum mechanics in a way which wouldn't require that one first know the mathematics of partial differential equations. Even for a physicist I think that is an interesting thing to try to do—to present quantum mechanics in this reverse fashion—for several reasons which may be apparent in the lectures themselves. However, I think that the experiment in the quantum mechanics part was not completely successful—in large part because I really did not have enough time at the end (I should, for instance, have had three or four more lectures in order to deal more completely with such matters as energy bands and the spatial dependence of amplitudes). Also, I had never presented the subject this way before, so the lack of feedback was particularly serious. I now believe the quantum mechanics should be given at a later time. Maybe I'll have a chance to do it again someday. Then I'll do it right.

The reason there are no lectures on how to solve problems is because there **were** recitation sections. Although I did put in three lectures in the first year on how to solve problems, they are not included here. Also there was a lecture on inertial

guidance which certainly belongs after the lecture on rotating systems, but which was, unfortunately, omitted. The fifth and sixth lectures are actually due to Matthew Sands, as I was out of town.

The question, of course, is how well this experiment has succeeded. My own point of view—which, however, does not seem to be shared by most of the people who worked with the students—is pessimistic. I don't think I did very well by the students. When I look at the way the majority of the students handled the problems on the examinations, I think that the system is a failure. Of course, my friends point out to me that there were one or two dozen students who—very surprisingly—understood almost everything in all of the lectures, and who were quite active in working with the material and worrying about the many points in an excited and interested way. These people have now, I believe, a first-rate background in physics—and they are, after all, the ones I was trying to get at. But then, "The power of instruction is seldom of much efficacy except in those happy dispositions where it is almost superfluous." (Gibbon)

Still, I didn't want to leave any student completely behind, as perhaps I did. I think one way we could help the students more would be by putting more hard work into developing a set of problems which would elucidate some of the ideas in the lectures. Problems give a good opportunity to fill out the material of the lectures and make more realistic, more complete, and more settled in the mind the ideas that have been exposed.

I think, however, that there isn't any solution to this problem of education other than to realize that the best teaching can be done only when there is a direct individual relationship between a student and a good teacher—a situation in which the student discusses the ideas, thinks about the things, and talks about the things. It's impossible to learn very much by simply sitting in a lecture, or even by simply doing problems that are assigned. But in our modern times we have so many students to teach that we have to try to find some substitute for the ideal. Perhaps my lectures can make some contribution. Perhaps in some small place where there are individual teachers and students, they may get some inspiration or some ideas from the lectures. Perhaps they will have fun thinking them through—or going on to develop some of the ideas further.

RICHARD P. FEYNMAN

June, 1963

Foreword

This book is based upon a course of lectures in introductory physics given by Prof. R. P. Feynman at the California Institute of Technology during the academic year 1961-62; it covers the first year of the two-year introductory course taken by all Caltech freshmen and sophomores, and was followed in 1962-63 by a similar series covering the second year. The lectures constitute a major part of a fundamental revision of the introductory course, carried out over a four-year period.

The need for a basic revision arose both from the rapid development of physics in recent decades and from the fact that entering freshmen have shown a steady increase in mathematical ability as a result of improvements in high school mathematics course content. We hoped to take advantage of this improved mathematical background, and also to introduce enough modern subject matter to make the course challenging, interesting, and more representative of present-day physics.

In order to generate a variety of ideas on what material to include and how to present it, a substantial number of the physics faculty were encouraged to offer their ideas in the form of topical outlines for a revised course. Several of these were presented and were thoroughly and critically discussed. It was agreed almost at once that a basic revision of the course could not be accomplished either by merely adopting a different textbook, or even by writing one *ab initio*, but that the new course should be centered about a set of lectures, to be presented at the rate of two or three per week; the appropriate text material would then be produced as a secondary operation as the course developed, and suitable laboratory experiments would also be arranged to fit the lecture material. Accordingly, a rough outline of the course was established, but this was recognized as being incomplete, tentative, and subject to considerable modification by whoever was to bear the responsibility for actually preparing the lectures.

Concerning the mechanism by which the course would finally be brought to life, several plans were considered. These plans were mostly rather similar, involving a cooperative effort by N staff members who would share the total burden symmetrically and equally: each man would take responsibility for $1/N$ of the material, deliver the lectures, and write text material for his part. However, the unavailability of sufficient staff, and the difficulty of maintaining a uniform point of view because of differences in personality and philosophy of individual participants, made such plans seem unworkable.

The realization that we actually possessed the means to create not just a new and different physics course, but possibly a unique one, came as a happy inspiration to Professor Sands. He suggested that Professor R. P. Feynman prepare and deliver the lectures, and that these be tape-recorded. When transcribed and edited, they would then become the textbook for the new course. This is essentially the plan that was adopted.

It was expected that the necessary editing would be minor, mainly consisting of supplying figures, and checking punctuation and grammar; it was to be done by one or two graduate students on a part-time basis. Unfortunately, this expectation was short-lived. It was, in fact, a major editorial operation to transform the verbatim transcript into readable form, even without the reorganization or revision of the subject matter that was sometimes required. Furthermore, it was not a job for a technical editor or for a graduate student, but one that required the close attention of a professional physicist for from ten to twenty hours per lecture!

The difficulty of the editorial task, together with the need to place the material in the hands of the students as soon as possible, set a strict limit upon the amount of "polishing" of the material that could be accomplished, and thus we were forced to aim toward a preliminary but technically correct product that could be used immediately, rather than one that might be considered final or finished. Because of an urgent need for more copies for our students, and a heartening interest on the part of instructors and students at several other institutions, we decided to publish the material in its preliminary form rather than wait for a further major revision which might never occur. We have no illusions as to the completeness, smoothness, or logical organization of the material; in fact, we plan several minor modifications in the course in the immediate future, and we hope that it will not become static in form or content.

In addition to the lectures, which constitute a centrally important part of the course, it was necessary also to provide suitable exercises to develop the students' experience and ability, and suitable experiments to provide first-hand contact with the lecture material in the laboratory. Neither of these aspects is in as advanced a state as the lecture material, but considerable progress has been made. Some exercises were made up as the lectures progressed, and these were expanded and amplified for use in the following year. However, because we are not yet satisfied that the exercises provide sufficient variety and depth of application of the lecture material to make the student fully aware of the tremendous power being placed at his disposal, the exercises are published separately in a less permanent form in order to encourage frequent revision.

A number of new experiments for the new course have been devised by Professor H. V. Neher. Among these are several which utilize the extremely low friction exhibited by a gas bearing: a novel linear air trough, with which quantitative measurements of one-dimensional motion, impacts, and harmonic motion can be made, and an air-supported, air-driven Maxwell top, with which accelerated rotational motion and gyroscopic precession and nutation can be studied. The development of new laboratory experiments is expected to continue **for a** considerable period of time.

The revision program was under the direction of Professors R. B. Leighton, H. V. Neher, and M. Sands. Officially participating in the program were Professors R. P. Feynman, G. Neugebauer, R. M. Sutton, H. P. Stabler,* F. Strong, and R. Vogt, from the division of Physics, Mathematics and Astronomy, and Professors T. Caughey, M. Plesset, and C. H. Wilts from the division of Engineering Science. The valuable assistance of all those contributing to the revision program is gratefully acknowledged. We are particularly indebted to the Ford Foundation, without whose financial assistance this program could not have been carried out.

ROBERT B. LEIGHTON

July, 1963

* 1961-62, while on leave from Williams College, Williamstown, Mass.

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Atoms in Motion

1-1 Introduction

This two-year course in physics is presented from the point of view that you, the reader, are going to be a physicist. This is not necessarily the case of course, but that is what every professor in every subject assumes! If you are going to be a physicist, you will have a lot to study: two hundred years of the most rapidly developing field of knowledge that there is. So much knowledge, in fact, that you might think that you cannot learn all of it in four years, and truly you cannot; you will have to go to graduate school too!

Surprisingly enough, in spite of the tremendous amount of work that has been done for all this time it is possible to condense the enormous mass of results to a large extent—that is, to find *laws* which summarize all our knowledge. Even so, the laws are so hard to grasp that it is unfair to you to start exploring this tremendous subject without some kind of map or outline of the relationship of one part of the subject of science to another. Following these preliminary remarks, the first three chapters will therefore outline the relation of physics to the rest of the sciences, the relations of the sciences to each other, and the meaning of science, to help us develop a "feel" for the subject.

You might ask why we cannot teach physics by just giving the basic laws on page one and then showing how they work in all possible circumstances, as we do in Euclidean geometry, where we state the axioms and then make all sorts of deductions. (So, not satisfied to learn physics in four years, you want to learn it in four minutes?) We cannot do it in this way for two reasons. First, we do not yet *know* all the basic laws: there is an expanding frontier of ignorance. Second, the correct statement of the laws of physics involves some very unfamiliar ideas which require advanced mathematics for their description. Therefore, one needs a considerable amount of preparatory training even to learn what the *words* mean. No, it is not possible to do it that way. We can only do it piece by piece.

Each piece, or part, of the whole of nature is always merely an *approximation* to the complete truth, or the complete truth so far as we know it. In fact, everything we know is only some kind of approximation, because *we know that we do not know all the laws* as yet. Therefore, things must be learned only to be unlearned again or, more likely, to be corrected.

The principle of science, the definition, almost, is the following: *The test of all knowledge is experiment.* Experiment is the *sole judge* of scientific "truth." But what is the source of knowledge? Where do the laws that are to be tested come from? Experiment, itself, helps to produce these laws, in the sense that it gives us hints. But also needed is *imagination* to create from these hints the great generalizations—to guess at the wonderful, simple, but very strange patterns beneath them all, and then to experiment to check again whether we have made the right guess. This imagining process is so difficult that there is a division of labor in physics: there are *theoretical* physicists who imagine, deduce, and guess at new laws, but do not experiment; and then there are *experimental* physicists who experiment, imagine, deduce, and guess.

We said that the laws of nature are approximate: that we first find the "wrong" ones, and then we find the "right" ones. Now, how can an experiment be "wrong"? First, in a trivial way: if something is wrong with the apparatus that you did not notice. But these things are easily fixed, and checked back and forth. So without snatching at such minor things, how *can* the results of an experiment be wrong? Only by being inaccurate. For example, the mass of an object never seems to

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change; a spinning top has the same weight as a still one. So a "law" was invented: mass is constant, independent of speed. That "law" is now found to be incorrect. Mass is found to increase with velocity, but appreciable increases require velocities near that of light. A *true* law is: if an object moves with a speed of less than one hundred miles a second the mass is constant to within one part in a million. In some such approximate form this is a correct law. So in practice one might think that the new law makes no significant difference. Well, yes and no. For ordinary speeds we can certainly forget it and use the simple constant-mass law as a good approximation. But for high speeds we are wrong, and the higher the speed, the more wrong we are.

Finally, and most interesting, *philosophically we are completely wrong* with the approximate law. Our entire picture of the world has to be altered even though the mass changes only by a little bit. This is a very peculiar thing about the philosophy, or the ideas, behind the laws. Even a very small effect sometimes requires profound changes in our ideas.

Now, what should we teach first? Should we teach the *correct* but unfamiliar law with its strange and difficult conceptual ideas, for example the theory of relativity, four-dimensional space-time, and so on? Or should we first teach the simple "constant-mass" law, which is only approximate, but does not involve such difficult ideas? The first is more exciting, more wonderful, and more fun, but the second is easier to get at first, and is a first step to a real understanding of the second idea. This point arises again and again in teaching physics. At different times we shall have to resolve it in different ways, but at each stage it is worth learning what is now known, how accurate it is, how it fits into everything else, and how it may be changed when we learn more.

Let us now proceed with our outline, or general map, of our understanding of science today (in particular, physics, but also of other sciences on the periphery), so that when we later concentrate on some particular point we will have some idea of the background, why that particular point is interesting, and how it fits into the big structure. So, what is our over-all picture of the world?

1-2 Matter is made of atoms

If, in some cataclysm, all of scientific knowledge were to be destroyed, and only one sentence passed on to the next generations of creatures, what statement would contain the most information in the fewest words? I believe it is the *atomic hypothesis* (or the *atomic fact*, or whatever you wish to call it) that *all things are made of atoms—little particles that move around in perpetual motion, attracting each other when they are a little distance apart, but repelling upon being squeezed into one another*. In that one sentence, you will see, there is an *enormous* amount of information about the world, if just a little imagination and thinking are applied.

To illustrate the power of the atomic idea, suppose that we have a drop of water a quarter of an inch on the side. If we look at it very closely we see nothing but water—smooth, continuous water. Even if we magnify it with the best optical microscope available—roughly two thousand times—then the water drop will be roughly forty feet across, about as big as a large room, and if we looked rather closely, we would *still* see relatively smooth water—but here and there small football-shaped things swimming back and forth. Very interesting. These are paramecia. You may stop at this point and get so curious about the paramecia with their wiggling cilia and twisting bodies that you go no further, except perhaps to magnify the paramecia still more and see inside. This, of course, is a subject for biology, but for the present we pass on and look still more closely at the water material itself, magnifying it two thousand times again. Now the drop of water extends about fifteen miles across, and if we look very closely at it we see a kind of teeming, something which no longer has a smooth appearance—it looks something like a crowd at a football game as seen from a very great distance. In order to see what this teeming is about, we will magnify it another two hundred and fifty times and we will see something similar to what is shown in Fig. 1-1. This is a picture of water magnified a billion times, but idealized in several ways.

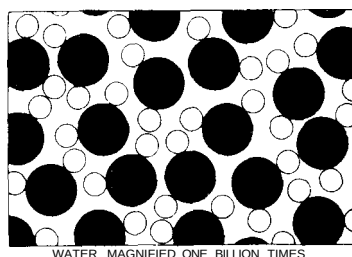


Figure 1-1

In the first place, the particles are drawn in a simple manner with sharp edges, which is inaccurate. Secondly, for simplicity, they are sketched almost schematically in a two-dimensional arrangement, but of course they are moving around in three dimensions. Notice that there are two kinds of "blobs" or circles to represent the atoms of oxygen (black) and hydrogen (white), and that each oxygen has two hydrogens tied to it. (Each little group of an oxygen with its two hydrogens is called a molecule.) The picture is idealized further in that the real particles in nature are continually jiggling and bouncing, turning and twisting around one another. You will have to imagine this as a dynamic rather than a static picture. Another thing that cannot be illustrated in a drawing is the fact that the particles are "stuck together"—that they attract each other, this one pulled by that one, etc. The whole group is "glued together," so to speak. On the other hand, the particles do not squeeze through each other. If you try to squeeze two of them too close together, they repel.

The atoms are 1 or 2×10^{-8} cm in radius. Now 10^{-8} cm is called an *angstrom* (just as another name), so we say they are 1 or 2 angstroms ($\text{\AA}$) in radius. Another way to remember their size is this: if an apple is magnified to the size of the earth, then the atoms in the apple are approximately the size of the original apple.

Now imagine this great drop of water with all of these jiggling particles stuck together and tagging along with each other. The water keeps its volume; it does not fall apart, because of the attraction of the molecules for each other. If the drop is on a slope, where it can move from one place to another, the water will flow, but it does not just disappear—things do not just fly apart—because of the molecular attraction. Now the jiggling motion is what we represent as *heat*: when we increase the temperature, we increase the motion. If we heat the water, the jiggling increases and the volume between the atoms increases, and if the heating continues there comes a time when the pull between the molecules is not enough to hold them together and they *do* fly apart and become separated from one another. Of course, this is how we manufacture steam out of water—by increasing the temperature; the particles fly apart because of the increased motion.

In Fig. 1-2 we have a picture of steam. This picture of steam fails in one respect: at ordinary atmospheric pressure there might be only a few molecules in a whole room, and there certainly would not be as many as three in this figure. Most squares this size would contain none—but we accidentally have two and a half or three in the picture (just so it would not be completely blank). Now in the case of steam we see the characteristic molecules more clearly than in the case of water. For simplicity, the molecules are drawn so that there is a 120° angle between them. In actual fact the angle is $105^\circ 3'$, and the distance between the center of a hydrogen and the center of the oxygen is $0.957 \text{ \AA}$, so we know this molecule very well.

Let us see what some of the properties of steam vapor or any other gas are. The molecules, being separated from one another, will bounce against the walls. Imagine a room with a number of tennis balls (a hundred or so) bouncing around in perpetual motion. When they bombard the wall, this pushes the wall away. (Of course we would have to push the wall back.) This means that the gas exerts a jittery force which our coarse senses (not being ourselves magnified a billion times) feels only as an *average push*. In order to confine a gas we must apply a pressure. Figure 1-3 shows a standard vessel for holding gases (used in all textbooks), a cylinder with a piston in it. Now, it makes no difference what the shapes of water molecules are, so for simplicity we shall draw them as tennis balls or little dots. These things are in perpetual motion in all directions. So many of them are hitting the top piston all the time that to keep it from being patiently knocked out of the tank by this continuous banging, we shall have to hold the piston down by a certain force, which we call the *pressure* (really, the pressure times the area is the force). Clearly, the force is proportional to the area, for if we increase the area but keep the number of molecules per cubic centimeter the same, we increase the number of collisions with the piston in the same proportion as the area was increased.

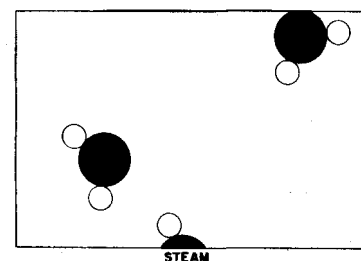


Figure 1-2

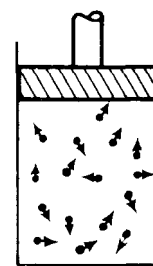


Figure 1-3

Now let us put twice as many molecules in this tank, so as to double the density, and let them have the same speed, i.e., the same temperature. Then, to a close approximation, the number of collisions will be doubled, and since each will be just as "energetic" as before, the pressure is proportional to the density. If we consider the true nature of the forces between the atoms, we would expect a slight decrease in pressure because of the attraction between the atoms, and a slight increase because of the finite volume they occupy. Nevertheless, to an excellent approximation, if the density is low enough that there are not many atoms, *the pressure is proportional to the density.*

We can also see something else: If we increase the temperature without changing the density of the gas, i.e., if we increase the speed of the atoms, what is going to happen to the pressure? Well, the atoms hit harder because they are moving faster, and in addition they hit more often, so the pressure increases. You see how simple the ideas of atomic theory are.

Let us consider another situation. Suppose that the piston moves inward, so that the atoms are slowly compressed into a smaller space. What happens when an atom hits the moving piston? Evidently it picks up speed from the collision. You can try it by bouncing a ping-pong ball from a forward-moving paddle, for example, and you will find that it comes off with more speed than that with which it struck. (Special example: if an atom happens to be standing still and the piston hits it, it will certainly move.) So the atoms are "hotter" when they come away from the piston than they were before they struck it. Therefore all the atoms which are in the vessel will have picked up speed. This means that *when we compress a gas slowly, the temperature of the gas increases.* So, under slow compression, a gas will increase in temperature, and under slow expansion it will decrease in temperature.

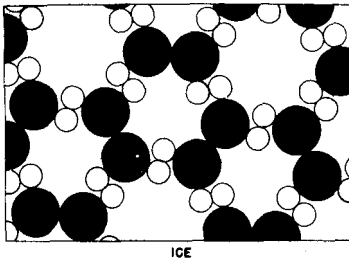


Figure 1-4

We now return to our drop of water and look in another direction. Suppose that we decrease the temperature of our drop of water. Suppose that the jiggling of the molecules of the atoms in the water is steadily decreasing. We know that there are forces of attraction between the atoms, so that after a while they will not be able to jiggle so well. What will happen at very low temperatures is indicated in Fig. 1-4: the molecules lock into a new pattern which is ice. This particular schematic diagram of ice is wrong because it is in two dimensions, but it is right qualitatively. The interesting point is that the material has a *definite place for every atom*, and you can easily appreciate that if somehow or other we were to hold all the atoms at one end of the drop in a certain arrangement, each atom in a certain place, then because of the structure of interconnections, which is rigid, the other end miles away (at our magnified scale) will have a definite location. So if we hold a needle of ice at one end, the other end resists our pushing it aside, unlike the case of water, in which the structure is broken down because of the increased jiggling so that the atoms all move around in different ways. The difference between solids and liquids is, then, that in a solid the atoms are arranged in some kind of an array, called a *crystalline array*, and they do not have a random position at long distances; the position of the atoms on one side of the crystal is determined by that of other atoms millions of atoms away on the other side of the crystal. Figure 1-4 is an invented arrangement for ice, and although it contains many of the correct features of ice, it is not the true arrangement. One of the correct features is that there is a part of the symmetry that is hexagonal. You can see that if we turn the picture around an axis by 120° , the picture returns to itself. So there is a *symmetry* in the ice which accounts for the six-sided appearance of snowflakes. Another thing we can see from Fig. 1-4 is why ice shrinks when it melts. The particular crystal pattern of ice shown here has many "holes" in it, as does the true ice structure. When the organization breaks down, these holes can be occupied by molecules. Most simple substances, with the exception of water and type metal, *expand* upon melting, because the atoms are closely packed in the solid crystal and upon melting need more room to jiggle around, but an open structure collapses, as in the case of water.

Now although ice has a "rigid" crystalline form, its temperature can change—ice has heat. If we wish, we can change the amount of heat. What is the heat in

the case of ice? The atoms are not standing still. They are jiggling and vibrating. So even though there is a definite order to the crystal—a definite structure—all of the atoms are vibrating "in place." As we increase the temperature, they vibrate with greater and greater amplitude, until they shake themselves out of place. We call this *melting*. As we decrease the temperature, the vibration decreases and decreases until, at absolute zero, there is a minimum amount of vibration that the atoms can have, but *not zero*. This minimum amount of motion that atoms can have is not enough to melt a substance, with one exception: helium. Helium merely decreases the atomic motions as much as it can, but even at absolute zero there is still enough motion to keep it from freezing. Helium, even at absolute zero, does not freeze, unless the pressure is made so great as to make the atoms squash together. If we increase the pressure, we *can* make it solidify.

1-3 Atomic processes

So much for the description of solids, liquids, and gases from the atomic point of view. However, the atomic hypothesis also describes *processes*, and so we shall now look at a number of processes from an atomic standpoint. The first process that we shall look at is associated with the surface of the water. What happens at the surface of the water? We shall now make the picture more complicated—and more realistic—by imagining that the surface is in air. Figure 1-5 shows the surface of water in air. We see the water molecules as before, forming a body of liquid water, but now we also see the surface of the water. Above the surface we find a number of things: First of all there are water molecules, as in steam. This is *water vapor*, which is always found above liquid water. (There is an equilibrium between the steam vapor and the water which will be described later.) In addition we find some other molecules—here two oxygen atoms stuck together by themselves, forming an *oxygen molecule*, there two nitrogen atoms also stuck together to make a nitrogen molecule. Air consists almost entirely of nitrogen, oxygen, some water vapor, and lesser amounts of carbon dioxide, argon, and other things. So above the water surface is the air, a gas, containing some water vapor. Now what is happening in this picture? The molecules in the water are always jiggling around. From time to time, one on the surface happens to be hit a little harder than usual, and gets knocked away. It is hard to see that happening in the picture because it is a *still* picture. But we can imagine that one molecule near the surface has just been hit and is flying out, or perhaps another one has been hit and is flying out. Thus, molecule by molecule, the water disappears—it evaporates. But if we *close* the vessel above, after a while we shall find a large number of molecules of water amongst the air molecules. From time to time, one of these vapor molecules comes flying down to the water and gets stuck again. So we see that what looks like a dead, uninteresting thing—a glass of water with a cover, that has been sitting there for perhaps twenty years—really contains a dynamic and interesting phenomenon which is going on all the time. To our eyes, our crude eyes, nothing is changing, but if we could see it a billion times magnified, we would see that from its own point of view it is always changing: molecules are leaving the surface, molecules are coming back.

Why do we see *no change*? Because just as many molecules are leaving as are coming back! In the long run "nothing happens." If we then take the top of the vessel off and blow the moist air away, replacing it with dry air, then the number of molecules leaving is just the same as it was before, because this depends on the jiggling of the water, but the number coming back is greatly reduced because there are so many fewer water molecules above the water. Therefore there are more going out than coming in, and the water evaporates. Hence, if you wish to evaporate water turn on the fan!

Here is something else: Which molecules leave? When a molecule leaves it is due to an accidental, extra accumulation of a little bit more than ordinary energy, which it needs if it is to break away from the attractions of its neighbors. Therefore, since those that leave have more energy than the average, the ones that are left have *less* average motion than they had before. So the liquid gradually

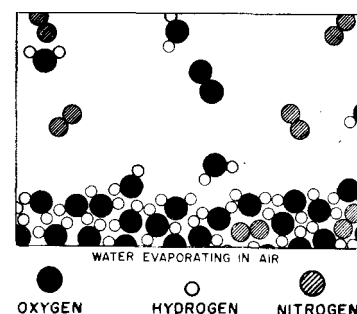


Figure 1-5

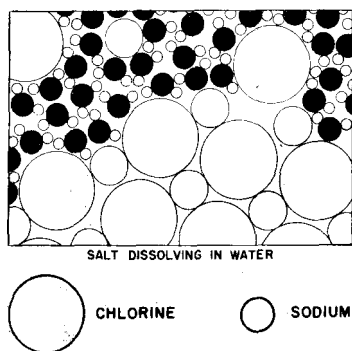


Figure 1-6

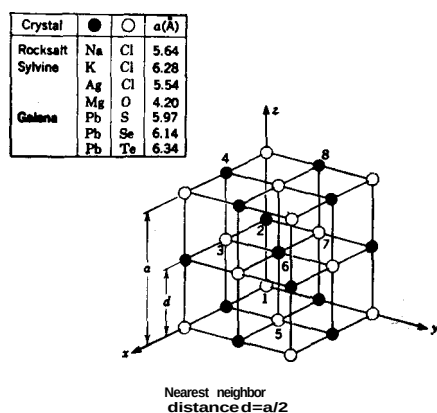


Figure 1-7

cools if it evaporates. Of course, when a molecule of vapor comes from the air to the water below there is a sudden great attraction as the molecule approaches the surface. This speeds up the incoming molecule and results in generation of heat. So when they leave they take away heat; when they come back they generate heat. Of course when there is no net evaporation the result is nothing—the water is not changing temperature. If we blow on the water so as to maintain a continuous preponderance in the number evaporating, then the water is cooled. Hence, blow on soup to cool it!

Of course you should realize that the processes just described are more complicated than we have indicated. Not only does the water go into the air, but also, from time to time, one of the oxygen or nitrogen molecules will come in and "get lost" in the mass of water molecules, and work its way into the water. Thus the air dissolves in the water; oxygen and nitrogen molecules will work their way into the water and the water will contain air. If we suddenly take the air away from the vessel, then the air molecules will leave more rapidly than they come in, and in doing so will make bubbles. This is very bad for divers, as you may know.

Now we go on to another process. In Fig. 1-6 we see, from an atomic point of view, a solid dissolving in water. If we put a crystal of salt in the water, what will happen? Salt is a solid, a crystal, an organized arrangement of "salt atoms." Figure 1-7 is an illustration of the three-dimensional structure of common salt, sodium chloride. Strictly speaking, the crystal is not made of atoms, but of what we call *ions*. An ion is an atom which either has a few extra electrons or has lost a few electrons. In a salt crystal we find chlorine ions (chlorine atoms with an extra electron) and sodium ions (sodium atoms with one electron missing). The ions all stick together by electrical attraction in the solid salt, but when we put them in the water we find, because of the attractions of the negative oxygen and positive hydrogen for the ions, that some of the ions jiggle loose. In Fig. 1-6 we see a chlorine ion getting loose, and other atoms floating in the water in the form of ions. This picture was made with some care. Notice, for example, that the hydrogen ends of the water molecules are more likely to be near the chlorine ion, while near the sodium ion we are more likely to find the oxygen end, because the sodium is positive and the oxygen end of the water is negative, and they attract electrically. Can we tell from this picture whether the salt is *dissolving in* water or *crystallizing out* of water? Of course we *cannot* tell, because while some of the atoms are leaving the crystal other atoms are rejoining it. The process is a *dynamic* one, just as in the case of evaporation, and it depends on whether there is more or less salt in the water than the amount needed for equilibrium. By equilibrium we mean that situation in which the rate at which atoms are leaving just matches the rate at which they are coming back. If there is almost no salt in the water, more atoms leave than return, and the salt dissolves. If, on the other hand, there are too many "salt atoms," more return than leave, and the salt is crystallizing.

In passing, we mention that the concept of a *molecule* of a substance is only approximate and exists only for a certain class of substances. It is clear in the case of water that the three atoms are actually stuck together. It is not so clear in the case of sodium chloride in the solid. There is just an arrangement of sodium and chlorine ions in a cubic pattern. There is no natural way to group them as "molecules of salt."

Returning to our discussion of solution and precipitation, if we increase the temperature of the salt solution, then the rate at which atoms are taken away is increased, and so is the rate at which atoms are brought back. It turns out to be very difficult, in general, to predict which way it is going to go, whether more or less of the solid will dissolve. Most substances dissolve more, but some substances dissolve less, as the temperature increases.

1-4 Chemical reactions

In all of the processes which have been described so far, the atoms and the ions have not changed partners, but of course there are circumstances in which the atoms do change combinations, forming new molecules. This is illustrated in

Fig. 1-8. A process in which the rearrangement of the atomic partners occurs is what we call a *chemical reaction*. The other processes so far described are called physical processes, but there is no sharp distinction between the two. (Nature does not care what we call it, she just keeps on doing it.) This figure is supposed to represent carbon burning in oxygen. In the case of oxygen, two oxygen atoms stick together very strongly. (Why do not *three* or *even four* stick together? That is one of the very peculiar characteristics of such atomic processes. Atoms are very special: they like certain particular partners, certain particular directions, and so on. It is the job of physics to analyze why each one wants what it wants. At any rate, two oxygen atoms form, saturated and happy, a molecule.)

The carbon atoms are supposed to be in a solid crystal (which could be graphite or diamond*). Now, for example, one of the oxygen molecules can come over to the carbon, and each atom can pick up a carbon atom and go flying off in a new combination—"carbon-oxygen"—which is a molecule of the gas called carbon monoxide. It is given the chemical name CO. It is very simple: the letters "CO" are practically a picture of that molecule. But carbon attracts oxygen much more than oxygen attracts oxygen or carbon attracts carbon. Therefore in this process the oxygen may arrive with only a little energy, but the oxygen and carbon will snap together with a tremendous vengeance and commotion, and everything near them will pick up the energy. A large amount of motion energy, kinetic energy, is thus generated. This of course is *burning*; we are getting *heat* from the combination of oxygen and carbon. The heat is ordinarily in the form of the molecular motion of the hot gas, but in certain circumstances it can be so enormous that it generates *light*. That is how one *gets flames*.

In addition, the carbon monoxide is not quite satisfied. It is possible for it to attach another oxygen, so that we might have a much more complicated reaction in which the oxygen is combining with the carbon, while at the same time there happens to be a collision with a carbon monoxide molecule. One oxygen atom could attach itself to the CO and ultimately form a molecule, composed of one carbon and two oxygens, which is designated CO_2 and called carbon dioxide. If we burn the carbon with very little oxygen in a very rapid reaction (for example, in an automobile engine, where the explosion is so fast that there is not time for it to make carbon dioxide) a considerable amount of carbon monoxide is formed. In many such rearrangements, a very large amount of energy is released, forming explosions, flames, etc., depending on the reactions. Chemists have studied these arrangements of the atoms, and found that every substance is some type of *arrangement of atoms*.

To illustrate this idea, let us consider another example. If we go into a field of small violets, we know what "that smell" is. It is some kind of *molecule*, or arrangement of atoms, that has worked its way into our noses. First of all, *how* did it work its way in? That is rather easy. If the smell is some kind of molecule in the air, jiggling around and being knocked every which way, it might have *accidentally* worked its way into the nose. Certainly it has no particular desire to get into our nose. It is merely one helpless part of a jostling crowd of molecules, and in its aimless wanderings this particular chunk of matter happens to find itself in the nose.

Now chemists can take special molecules like the odor of violets, and analyze them and tell us the *exact arrangement* of the atoms in space. We know that the carbon dioxide molecule is straight and symmetrical: $\text{O}-\text{C}-\text{O}$. (That can be determined easily, too, by physical methods.) However, even for the vastly more complicated arrangements of atoms that there are in chemistry, one can, by a long, remarkable process of detective work, find the arrangements of the atoms. Figure 1-9 is a picture of the air in the neighborhood of a violet; again we find nitrogen and oxygen in the air, and water vapor. (Why is there water vapor? Because the violet is *wet*. All plants transpire.) However, we also see a "monster" composed of carbon atoms, hydrogen atoms, and oxygen atoms, which have picked a certain particular pattern in which to be arranged. It is a much more complicated arrange-

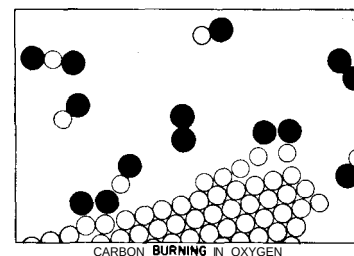


Figure 1-8

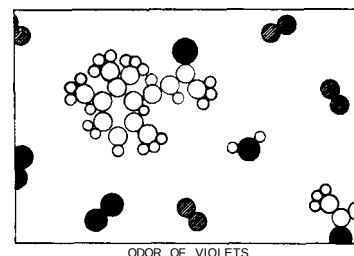


Figure 1-9

*One *can* burn a diamond in air.

ment than that of carbon dioxide; in fact, it is an enormously complicated arrangement. Unfortunately, we cannot picture all that is really known about it chemically, because the precise arrangement of all the atoms is actually known in three dimensions, while our picture is in only two dimensions. The six carbons which form a ring do not form a flat ring, but a kind of "puckered" ring. All of the angles and distances are known. So a chemical *formula* is merely a picture of such a molecule. When the chemist writes such a thing on the blackboard, he is trying to "draw," roughly speaking, in two dimensions. For example, we see a "ring" of six carbons, and a "chain" of carbons hanging on the end, with an oxygen second from the end, three hydrogens tied to that carbon, two carbons and three hydrogens sticking up here, etc.

How does the chemist find what the arrangement is? He mixes bottles full of stuff together, and if it turns red, it tells him that it consists of one hydrogen and two carbons tied on here; if it turns blue, on the other hand, that is not the way it is at all. This is one of the most fantastic pieces of detective work that has ever been done—organic chemistry. To discover the arrangement of the atoms in these enormously complicated arrays the chemist looks at what happens when he mixes two different substances together. The physicist could never quite believe that the chemist knew what he was talking about when he described the arrangement of the atoms. For about twenty years it has been possible, in some cases, to look at such molecules (not quite as complicated as this one, but some which contain parts of it) by a physical method, and it has been possible to locate every atom, not by looking at colors, but by *measuring where they are*. And lo and behold!, the chemists are almost always correct.

It turns out, in fact, that in the odor of violets there are three slightly different molecules, which differ only in the arrangement of the hydrogen atoms.

One problem of chemistry is to name a substance, so that we will know what it is. Find a name for this shape! Not only must the name tell the shape, but it must also tell that here is an oxygen atom, there a hydrogen—exactly what and where each atom is. So we can appreciate that the chemical names must be complex in order to be complete. You see that the name of this thing in the more complete form that will tell you the structure of it is 4-(2, 2, 3, 6 tetramethyl-5-cyclohexanyl)-3-buten-2-one, and that tells you that this is the arrangement. We can appreciate the difficulties that the chemists have, and also appreciate the reason for such long names. It is not that they wish to be obscure, but they have an extremely difficult problem in trying to describe the molecules in words!

How do we *know* that there are atoms? By one of the tricks mentioned earlier: we make the *hypothesis* that there are atoms, and one after the other results come out the way we predict, as they ought to if things *are* made of atoms. There is also somewhat more direct evidence, a good example of which is the following: The atoms are so small that you cannot see them with a light microscope—in fact, not even with an *electron* microscope. (With a light microscope you can only see things which are much bigger.) Now if the atoms are always in motion, say in water, and we put a big ball of something in the water, a ball much bigger than the atoms, the ball will jiggle around—much as in a push ball game, where a great big ball is pushed around by a lot of people. The people are pushing in various directions, and the ball moves around the field in an irregular fashion. So, in the same way, the "large ball" will move because of the inequalities of the collisions on one side to the other, from one moment to the next. Therefore, if we look at very tiny particles (colloids) in water through an excellent microscope, we see a perpetual jiggling of the particles, which is the result of the bombardment of the atoms. This is called the *Brownian motion*.

We can see further evidence for atoms in the structure of crystals. In many cases the structures deduced by x-ray analysis agree in their spatial "shapes" with the forms actually exhibited by crystals as they occur in nature. The angles between the various "faces" of a crystal agree, within seconds of arc, with angles deduced on the assumption that a crystal is made of many "layers" of atoms.

Everything is made of atoms. That is the key hypothesis. The most important hypothesis in all of biology, for example, is that *everything that animals do, atoms*

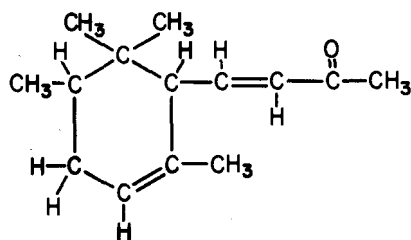


Fig. 1-10. The substance pictured is α -ionone.

do. In other words, *there is nothing that living things do that cannot be understood from the point of view that they are made of atoms acting according to the laws of physics*. This was not known from the beginning: it took some experimenting and theorizing to suggest this hypothesis, but now it is accepted, and it is the most useful theory for producing new ideas in the field of biology.

If a piece of steel or a piece of salt, consisting of atoms one next to the other, can have such interesting properties; if water—which is nothing but these little blobs, mile upon mile of the same thing over the earth—can form waves and foam, and make rushing noises and strange patterns as it runs over cement; if all of this, all the life of a stream of water, can be nothing but a pile of atoms, *how much more is possible?* If instead of arranging the atoms in some definite pattern, again and again repeated, on and on, or even forming little lumps of complexity like the odor of violets, we make an arrangement which is *always different* from place to place, with different kinds of atoms arranged in many ways, continually changing, not repeating, how much more marvelously is it possible that this thing might behave? Is it possible that that "thing" walking back and forth in front of you, talking to you, is a great glob of these atoms in a very complex arrangement, such that the sheer complexity of it staggers the imagination as to what it can do? When we say we are a pile of atoms, we do not mean we are *merely* a pile of atoms, because a pile of atoms which is not repeated from one to the other might well have the possibilities which you see before you in the mirror.

Basic Physics

2-1 Introduction

In this chapter, we shall examine the most fundamental ideas that we have about physics—the nature of things as we see them at the present time. We shall not discuss the history of how we know that all these ideas are true; you will learn these details in due time.

The things with which we concern ourselves in science appear in myriad forms, and with a multitude of attributes. For example, if we stand on the shore and look at the sea, we see the water, the waves breaking, the foam, the sloshing motion of the water, the sound, the air, the winds and the clouds, the sun and the blue sky, and light; there is sand and there are rocks of various hardness and permanence, color and texture. There are animals and seaweed, hunger and disease, and the observer on the beach; there may be even happiness and thought. Any other spot in nature has a similar variety of things and influences. It is always as complicated as that, no matter where it is. Curiosity demands that we ask questions, that we try to put things together and try to understand this multitude of aspects as perhaps resulting from the action of a relatively small number of elemental things and forces acting in an infinite variety of combinations.

For example: Is the sand other than the rocks? That is, is the sand perhaps nothing but a great number of very tiny stones? Is the moon a great rock? If we understood rocks, would we also understand the sand and the moon? Is the wind a sloshing of the air analogous to the sloshing motion of the water in the sea? What common features do different movements have? What is common to different kinds of sound? How many different colors are there? And so on. In this way we try gradually to analyze all things, to put together things which at first sight look different, with the hope that we may be able to *reduce* the number of *different* things and thereby understand them better.

A few hundred years ago, a method was devised to find partial answers to such questions. *Observation*, *reason*, and *experiment* make up what we call the *scientific method*. We shall have to limit ourselves to a bare description of **our** basic view of what is sometimes called *fundamental physics*, or fundamental ideas which have arisen from the application of the scientific method.

What do we mean by "understanding" something? We can imagine that this complicated array of moving things which constitutes "the world" is something like a great chess game being played by the gods, and we are observers of the game. We do not know what the rules of the game are; all we are allowed to do is to *watch* the playing. Of course, if we watch long enough, we may eventually catch on to a few of the rules. *The rules of the game* are what we mean by *fundamental physics*. Even if we knew every rule, however, we might not be able to understand why a particular move is made in the game, merely because it is too complicated and our minds are limited. If you play chess you must know that it is easy to learn all the rules, and yet it is often very hard to select the best move or to understand why a player moves as he does. So it is in nature, only much more so; but we may be able at least to find all the rules. Actually, we do not have all the rules now. (Every once in a while something like castling is going on that we still do not understand.) Aside from not knowing all of the rules, what we really can explain in terms of those rules is very limited, because almost all situations are so enormously complicated that we cannot follow the plays of the game using the rules, much less tell what is going to happen next. We must, therefore, limit ourselves to the more basic question of the rules of the game. If we know the rules, we consider that we "understand" the world.

2-1 Introduction

2-2 Physics before 1920

2-3 Quantum physics

2-4 Nuclei and particles

How can we tell whether the rules which we "guess" at are really right if we cannot analyze the game very well? There are, roughly speaking, three ways. First, there may be situations where nature has arranged, or we arrange nature, to be simple and to have so few parts that we can predict exactly what will happen, and thus we can check how our rules work. (In one corner of the board there may be only a few chess pieces at work, and that we can figure out exactly.)

A second good way to check rules is in terms of less specific rules derived from them. For example, the rule on the move of a bishop on a chessboard is that it moves only on the diagonal. One can deduce, no matter how many moves may be made, that a certain bishop will always be on a red square. So, without being able to follow the details, we can always check our idea about the bishop's motion by finding out whether it is always on a red square. Of course it will be, for a long time, until all of a sudden we find that it is on a *black* square (what happened of course, is that in the meantime it was captured, another pawn crossed for queening, and it turned into a bishop on a black square). That is the way it is in physics. For a long time we will have a rule that works excellently in an over-all way, even when we cannot follow the details, and then some time we may discover a *new rule*. From the point of view of basic physics, the most interesting phenomena are of course in the *new* places, the places where the rules do not work—not the places where they *do* work! That is the way in which we discover new rules.

The third way to tell whether our ideas are right is relatively crude but probably the most powerful of them all. That is, by rough *approximation*. While we may not be able to tell why Alekhine moves *this particular piece*, perhaps we can *roughly* understand that he is gathering his pieces around the king to protect it, more or less, since that is the sensible thing to do in the circumstances. In the same way, we can often understand nature, more or less, without being able to see what *every little piece* is doing, in terms of our understanding of the game.

At first the phenomena of nature were roughly divided into classes, like heat, electricity, mechanics, magnetism, properties of substances, chemical phenomena, light or optics, x-rays, nuclear physics, gravitation, meson phenomena, etc. However, the aim is to see *complete nature* as different aspects of *one set* of phenomena. That is the problem in basic theoretical physics, today—to *find the laws behind experiment*; to *amalgamate these classes*. Historically, we have always been able to amalgamate them, but as time goes on new things are found. We were amalgamating very well, when all of a sudden x-rays were found. Then we amalgamated some more, and mesons were found. Therefore, at any stage of the game, it always looks rather messy. A great deal is amalgamated, but there are always many wires or threads hanging out in all directions. That is the situation today, which we shall try to describe.

Some historic examples of amalgamation are the following. First, take *heat* and *mechanics*. When atoms are in motion, the more motion, the more heat the system contains, and so *heat and all temperature effects can be represented by the laws of mechanics*. Another tremendous amalgamation was the discovery of the relation between electricity, magnetism, and light, which were found to be different aspects of the same thing, which we call today the *electromagnetic field*. Another amalgamation is the unification of chemical phenomena, the various properties of various substances, and the behavior of atomic particles, which is in the *quantum mechanics of chemistry*.

The question is, of course, is it going to be possible to amalgamate *everything*, and merely discover that this world represents different aspects of *one thing*? Nobody knows. All we know is that as we go along, we find that we can amalgamate pieces, and then we find some pieces that do not fit, and we keep trying to put the jigsaw puzzle together. Whether there are a finite number of pieces, and whether there is even a border to the puzzle, is of course unknown. It will never be known until we finish the picture, if ever. What we wish to do here is to see to what extent this amalgamation process has gone on, and what the situation is at present, in understanding basic phenomena in terms of the smallest set of principles. To express it in a simple manner, *what are things made of and how few elements are there?*

2-2 Physics before 1920

It is a little difficult to begin at once with the present view, so we shall first see how things looked in about 1920 and then take a few things out of that picture. Before 1920, our world picture was something like this: The "stage" on which the universe goes is the three-dimensional *space* of geometry, as described by Euclid, and things change in a medium called *time*. The elements on the stage are *particles*, for example the atoms, which have some *properties*. First, the property of inertia: if a particle is moving it keeps on going in the same direction unless *forces* act upon it. The second element, then, is *forces*, which were then thought to be of two varieties: First, an enormously complicated, detailed kind of interaction force which held the various atoms in different combinations in a complicated way, which determined whether salt would dissolve faster or slower when we raise the temperature. The other force that was known was a long-range interaction—a smooth and quiet attraction—which varied inversely as the square of the distance, and was called *gravitation*. This law was known and was very simple. *Why* things remain in motion when they are moving, or *why* there is a law of gravitation was, of course, not known.

A description of nature is what we are concerned with here. From this point of view, then, a gas, and indeed *all* matter, is a myriad of moving particles. Thus many of the things we saw while standing at the seashore can immediately be connected. First the pressure: this comes from the collisions of the atoms with the walls or whatever; the drift of the atoms, if they are all moving in one direction on the average, is wind; the *random* internal motions are the *heat*. There are waves of excess density, where too many particles have collected, and so as they Tush off they push up piles of particles farther out, and so on. This wave of excess density is *sound*. It is a tremendous achievement to be able to understand so much. Some of these things were described in the previous chapter.

What *kinds* of particles are there? There were considered to be 92 at that time: 92 different kinds of atoms were ultimately discovered. They had different names associated with their chemical properties.

The next part of the problem was, *what are the short-range forces*? Why does carbon attract one oxygen or perhaps two oxygens, but not three oxygens? What is the machinery of interaction between atoms? Is it gravitation? The answer is no. Gravity is entirely too weak. But imagine a force analogous to gravity, varying inversely with the square of the distance, but enormously more powerful and having one difference. In gravity everything attracts everything else, but now imagine that there are *two kinds* of "things," and that this new force (which is the electrical force, of course) has the property that likes *repel* but unlikes *attract*. The "thing" that carries this strong interaction is called *charge*.

Then what do we have? Suppose that we have two unlikes that attract each other, a plus and a minus, and that they stick very close together. Suppose we have another charge some distance away. Would it feel any attraction? It would feel *practically none*, because if the first two are equal in size, the attraction for the one and the repulsion for the other balance out. Therefore there is very little force at any appreciable distance. On the other hand, if we get *very close* with the extra charge, *attraction* arises, because the repulsion of likes and attraction of unlikes will tend to bring unlikes closer together and push likes farther apart. Then the repulsion will be *less* than the attraction. This is the reason why the atoms, which are constituted out of plus and minus electric charges, feel very little force when they are separated by appreciable distance (aside from gravity). When they come close together, they can "see inside" each other and rearrange their charges, with the result that they have a very strong interaction. The ultimate basis of an interaction between the atoms is *electrical*. Since this force is so enormous, all the plusses and all minuses will normally come together in as intimate a combination as they can. All things, even ourselves, are made of fine-grained, enormously strongly interacting plus and minus parts, all neatly balanced out. Once in a while, by accident, we may rub off a few minuses or a few plusses (usually it is easier to rub off minuses), and in those circumstances we find the force of electricity *unbalanced*, and we can then see the effects of these electrical attractions.

To give an idea of how much stronger electricity is than gravitation, consider two grains of sand, a millimeter across, thirty meters apart. If the force between them were not balanced, if everything attracted everything else instead of likes repelling, so that there were no cancellation, how much force would there be? There would be a force of *three million tons* between the two! You see, there is very, very little excess or deficit of the number of negative or positive charges necessary to produce appreciable electrical effects. This is, of course, the reason why you cannot see the difference between an electrically charged or uncharged thing—so few particles are involved that they hardly make a difference in the weight or size of an object.

With this picture the atoms were easier to understand. They were thought to have a "nucleus" at the center, which is positively electrically charged and very massive, and the nucleus is surrounded by a certain number of "electrons" which are very light and negatively charged. Now we go a little ahead in our story to remark that in the nucleus itself there were found two kinds of particles, protons and neutrons, almost of the same weight and very heavy. The protons are electrically charged and the neutrons are neutral. If we have an atom with six protons inside its nucleus, and this is surrounded by six electrons (the negative particles in the ordinary world of matter are all electrons, and these are very light compared with the protons and neutrons which make nuclei), this would be atom number six in the chemical table, and it is called carbon. Atom number eight is called oxygen, etc., because the chemical properties depend upon the electrons on the *outside*, and in fact only upon *how many* electrons there are. So the *chemical* properties of a substance depend only on a number, the number of electrons. (The whole list of elements of the chemists really could have been called 1, 2, 3, 4, 5, etc. Instead of saying "carbon," we could say "element six," meaning six electrons, but of course, when the elements were first discovered, it was not known that they could be numbered that way, and secondly, it would make everything look rather complicated. It is better to have names and symbols for these things, rather than to call everything by number.)

More was discovered about the electrical force. The natural interpretation of electrical interaction is that two objects simply attract each other: plus against minus. However, this was discovered to be an inadequate idea to represent it. A more adequate representation of the situation is to say that the existence of the positive charge, in some sense, distorts, or creates a "condition" in space, so that when we put the negative charge in, it feels a force. This potentiality for producing a force is called an *electric field*. When we put an electron in an electric field, we say it is "pulled." We then have two rules: (a) charges make a field, and (b) charges in fields have forces on them and move. The reason for this will become clear when we discuss the following phenomena: If we were to charge a body, say a comb, electrically, and then place a charged piece of paper at a distance and move the comb back and forth, the paper will respond by always pointing to the comb. If we shake it faster, it will be discovered that the paper is a little behind, *there is a delay* in the action. (At the first stage, when we move the comb rather slowly, we find a complication which is *magnetism*. Magnetic influences have to do with *charges in relative motion*, so magnetic forces and electric forces can really be attributed to one field, as two different aspects of exactly the same thing. A changing electric field cannot exist without magnetism.) If we move the charged paper farther out, the delay is greater. Then an interesting thing is observed. Although the forces between two charged objects should go inversely as the *square* of the distance, it is found, when we shake a charge, that the influence extends *very much farther out* than we would guess at first sight. That is, the effect falls off more slowly than the inverse square.

Here is an analogy: If we are in a pool of water and there is a floating cork very close by, we can move it "directly" by pushing the water with another cork. If you looked only at the two *corks*, all you would see would be that one moved immediately in response to the motion of the other—there is some kind of "*interaction*" between them. Of course, what we really do is to disturb the *water*; the *water* then disturbs the other cork. We could make up a "law" that if you pushed

the water a little bit, an object close by in the water would move. If it were farther away, of course, the second cork would scarcely move, for we move the water *locally*. On the other hand, if we jiggle the cork a new phenomenon is involved, in which the motion of the water moves the water there, etc., and *waves* travel away, so that by jiggling, there is an influence *wry much farther out*, an oscillatory influence, that cannot be understood from the direct interaction. Therefore the idea of direct interaction must be replaced with the existence of the water, or—in the electrical case, with what we call the *electromagnetic field*.

The electromagnetic field can carry waves; some of these waves are *light*, others are used in *radio broadcasts*, but the general name is *electromagnetic waves*. These oscillatory waves can have various *frequencies*. The only thing that is really different from one wave to another is the *frequency of oscillation*. If we shake a charge back and forth more and more rapidly, and look at the effects, we get a whole series of different kinds of effects, which are all unified by specifying but one number, the number of oscillations per second. The usual "pickup" that we get from electric currents in the circuits in the walls of a building have a frequency of about one hundred cycles per second. If we increase the frequency to 500 or 1000 kilocycles (1 kilocycle = 1000 cycles) per second, we are "on the air," for this is the frequency range which is used for radio broadcasts. (Of course it has nothing to do with the *air*! We can have radio broadcasts without any air.) If we again increase the frequency, we come into the range that is used for FM and TV. Going still further, we use certain short waves, for example for *radar*. Still higher, and we do not need an instrument to "see" the stuff, we can see it with the human eye. In the range of frequency from 5×10^{14} to 5×10^{15} cycles per second our eyes would see the oscillation of the charged comb, if we could shake it that fast, as red, blue, or violet light, depending on the frequency. Frequencies below this range are called infrared, and above it, ultraviolet. The fact that we can see in a particular frequency range makes that part of the electromagnetic spectrum no more impressive than the other parts from a physicist's standpoint, but from a human standpoint, of course, it is more interesting. If we go up even higher in frequency, we get x-rays. X-rays are nothing but very high-frequency light. If we go still higher, we get gamma rays. These two terms, x-rays and gamma rays, are used almost synonymously. Usually electromagnetic rays coming from nuclei are called gamma rays, while those of high energy from atoms are called x-rays, but at the same frequency they are indistinguishable physically, no matter what their source. If we go to still higher frequencies, say to 10^{24} cycles per second, we find that we can make those waves artificially, for example with the synchrotron here at Caltech. We can find electromagnetic waves with stupendously high frequencies—with even a thousand times more rapid oscillation—in the waves found in *cosmic rays*. These waves cannot be controlled by us.

Table 2-1
The Electromagnetic Spectrum

Frequency in oscillations/sec	Name	Rough behavior
10^2	Electrical disturbance	Field
$5 \times 10^5 - 10^6$	Radio broadcast	} Waves
10^8	FM — TV	
10^{10}	Radar	
$5 \times 10^{14} - 10^{15}$	Light	
10^{18}	X-rays	} Particle
10^{21}	γ -rays, nuclear	
10^{24}	γ -rays, "artificial"	
10^{27}	γ -rays, in cosmic rays	

2-3 Quantum physics

Having described the idea of the electromagnetic field, and that this field can carry waves, we soon learn that these waves actually behave in a strange way which seems very unwavelike. At higher frequencies they behave much more like *particles*! It is *quantum mechanics*, discovered just after 1920, which explains this strange behavior. In the years before 1920, the picture of space as a three-dimensional space, and of time as a separate thing, was changed by Einstein, first into a combination which we call space-time, and then still further into a *curved* space-time to represent gravitation. So the "stage" is changed into space-time, and gravitation is presumably a modification of space-time. Then it was also found that the rules for the motions of particles were incorrect. The mechanical rules of "inertia" and "forces" are *wrong*—Newton's laws are *wrong*—in the world of atoms. Instead, it was discovered that things on a small scale behave *nothing like* things on a large scale. That is what makes physics difficult—and very interesting. It is hard because the way things behave on a small scale is so "unnatural"; we have no direct experience with it. Here things behave like nothing we know of, so that it is impossible to describe this behavior in any other than analytic ways. It is difficult, and takes a lot of imagination.

Quantum mechanics has many aspects. In the first place, the idea that a particle has a definite location and a definite speed is no longer allowed; that is wrong. To give an example of how wrong classical physics is, there is a rule in quantum mechanics that says that one cannot know both where something is and how fast it is moving. The uncertainty of the momentum and the uncertainty of the position are complementary, and the product of the two is constant. We can write the law like this: $\Delta x \Delta p \geq \hbar/2$, but we shall explain it in more detail later. This rule is the explanation of a very mysterious paradox: if the atoms are made out of plus and minus charges, why don't the minus charges simply sit on top of the plus charges (they attract each other) and get so close as to completely cancel them out? *Why are atoms so big?* Why is the nucleus at the center with the electrons around it? It was first thought that this was because the nucleus was so big; but no, the nucleus is *very small*. An atom has a diameter of about 10^{-8} cm. The nucleus has a diameter of about 10^{-13} cm. If we had an atom and wished to see the nucleus, we would have to magnify it until the whole atom was the size of a large room, and then the nucleus would be a bare speck which you could just about make out with the eye, but very nearly *all the weight* of the atom is in that infinitesimal *nucleus*. What keeps the electrons from simply falling in? This principle: If they were in the nucleus, we would know their position precisely, and the uncertainty principle would then require that they have a very *large* (but uncertain) momentum, i.e., a very large *kinetic energy*. With this energy they would break away from the nucleus. They make a compromise: they leave themselves a little room for this uncertainty and then jiggle with a certain amount of minimum motion in accordance with this rule. (Remember that when a crystal is cooled to absolute zero, we said that the atoms do not stop moving, they still jiggle. Why? If they stopped moving, we would know where they were and that they had zero motion, and that is against the uncertainty principle. We cannot know where they are and how fast they are moving, so they must be continually wiggling in there!)

Another most interesting change in the ideas and philosophy of science brought about by quantum mechanics is this: it is not possible to predict *exactly* what will happen in any circumstance. For example, it is possible to arrange an atom which is ready to emit light, and we can measure when it has emitted light by picking up a photon particle, which we shall describe shortly. We cannot, however, predict *when* it is going to emit the light or, with several atoms, *which one* is going to. You may say that this is because there are some internal "wheels" which we have not looked at closely enough. No, there *are* no internal wheels; nature, as we understand it today, behaves in such a way that *it is fundamentally impossible* to make a precise prediction of *exactly what will happen* in a given experiment. This is a horrible thing; in fact, philosophers have said before that one of the fundamental requisites of science is that whenever you set **up** the same

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conditions, the same thing must happen. This is simply *not true*, it is *not* a fundamental condition of science. The fact is that the same thing does not happen, that we can find only an average, statistically, **as to** what happens. Nevertheless, science has not completely collapsed. Philosophers, incidentally, say a great deal about what is *absolutely necessary* for science, and it is always, so far as one can see, rather naive, and probably wrong. For example, some philosopher or other said it is fundamental to the scientific effort that if an experiment is performed in, say, Stockholm, and then the same experiment is done in, say, Quito, the *same results* must occur. That is quite false. It is not necessary that *science* do that; it may be a *fact of experience*, but it is not necessary. For example, if one of the experiments is to look out at the sky and see the aurora borealis in Stockholm, you do not see it in Quito; that is a different phenomenon. "But," you say, "that is something that has to do with the outside; can you close yourself up in a box in Stockholm and pull down the shade and get any difference?" Surely. If we take a pendulum on a universal joint, and pull it out and let go, then the pendulum will swing almost in a plane, but not quite. Slowly the plane keeps changing in Stockholm, but not in Quito. The blinds are down, too. The fact that this happened does not bring on the destruction of science. What is the fundamental hypothesis of science, the fundamental philosophy? We stated it in the first chapter: *the sole test of the validity of any idea is experiment*. If it turns out that most experiments work out the same in Quito as they do in Stockholm, then those "most experiments" will be used to formulate some general law, and those experiments which do not come out the same we will say were a result of the environment near Stockholm. We will invent some way to summarize the results of the experiment, and we do not have to be told ahead of time what this way will look like. If we are told that the same experiment will always produce the same result, that is all very well, but if when we try it, it does *not*, then it does *not*. We just have to take what we see, and then formulate all the rest of our ideas in terms of our actual experience.

Returning again to quantum mechanics and fundamental physics, we cannot go into details of the quantum-mechanical principles at this time, of course, because these are rather difficult to understand. We shall assume that they are there, and go on to describe what some of the consequences are. One of the consequences is that things which we used to consider as waves also behave like particles, and particles behave like waves; in fact everything behaves the same way. There is no distinction between a wave and a particle. So quantum mechanics *unifies* the idea of the field and its waves, and the particles, all into one. Now it is true that when the frequency is low, the field aspect of the phenomenon is more evident, or more useful as an approximate description in terms of everyday experiences. But as the frequency increases, the particle aspects of the phenomenon become more evident with the equipment with which we usually make the measurements. In fact, although we mentioned many frequencies, no phenomenon directly involving a frequency has yet been detected above approximately 10^{12} cycles per second. We only *deduce* the higher frequencies from the energy of the particles, by a rule which assumes that the particle-wave idea of quantum mechanics is valid.

Thus we have a new view of electromagnetic interaction. We have a new kind of *particle* to add to the electron, the proton, and the neutron. That new particle is called a *photon*. The new view of the interaction of electrons and protons that is electromagnetic theory, but with everything quantum-mechanically correct, is called *quantum electrodynamics*. This fundamental theory of the interaction of light and matter, or electric field and charges, is our greatest success so far in physics. In this one theory we have the basic rules for all ordinary phenomena except for gravitation and nuclear processes. For example, out of quantum electrodynamics come all known electrical, mechanical, and chemical laws: the laws for the collision of billiard balls, the motions of wires in magnetic fields, the specific heat of carbon monoxide, the color of neon signs, the density of salt, and the reactions of hydrogen and oxygen to make water are all consequences of this one law. All these details can be worked out if the situation is simple enough for us to make an approximation, which is almost never, but often we can understand more

or less what is happening. At the present time no exceptions are found to the quantum-electrodynamic laws outside the nucleus, and there we do not know whether there is an exception because we simply do not know what is going on in the nucleus.

In principle, then, quantum electrodynamics is the theory of all chemistry, and of life, if life is ultimately reduced to chemistry and therefore just to physics because chemistry is already reduced (the part of physics which is involved in chemistry being already known). Furthermore, the same quantum electrodynamics, this great thing, predicts a lot of new things. In the first place, it tells the properties of very high-energy photons, gamma rays, etc. It predicted another very remarkable thing: besides the electron, there should be another particle of the same mass, but of opposite charge, called a *positron*, and these two, coming together, could annihilate each other with the emission of light or gamma rays. (After all, light and gamma rays are all the same, they are just different points on a frequency scale.) The generalization of this, that for each particle there is an antiparticle, turns out to be true. In the case of electrons, the antiparticle has another name—it is called a positron, but for most other particles, it is called anti-so-and-so, like antiproton or antineutron. In quantum electrodynamics, *two numbers* are put in and most of the other numbers in the world are supposed to come out. The two numbers that are put in are called the mass of the electron and the charge of the electron. Actually, that is not quite true, for we have a whole set of numbers for chemistry which tells how heavy the nuclei are. That leads us to the next part.

2-4 Nuclei and particles

What are the nuclei made of, and how are they held together? It is found that the nuclei are held together by enormous forces. When these are released, the energy released is tremendous compared with chemical energy, in the same ratio as the atomic bomb explosion is to a TNT explosion, because, of course, the atomic bomb has to do with changes inside the nucleus, while the explosion of TNT has to do with the changes of the electrons on the outside of the atoms. The question is, what are the forces which hold the protons and neutrons together in the nucleus? Just as the electrical interaction can be connected to a particle, a photon, Yukawa suggested that the forces between neutrons and protons also have a field of some kind, and that when this field jiggles it behaves like a particle. Thus there could be some other particles in the world besides protons and neutrons, and he was able to deduce the properties of these particles from the already known characteristics of nuclear forces. For example, he predicted they should have a mass of two or three hundred times that of an electron; and lo and behold, in cosmic rays there was discovered a particle of the right mass! But it later turned out to be the wrong particle. It was called a *m-meson*, or *muon*.

However, a little while later, in 1947 or 1948, another particle was found, the *p-meson*, or *pion*, which satisfied Yukawa's criterion. Besides the proton and the neutron, then, in order to get nuclear forces we must add the pion. Now, you say, "Oh great!, with this theory we make quantum nucleodynamics using the pions just like Yukawa wanted to do, and see if it works, and everything will be explained." Bad luck. It turns out that the calculations that are involved in this theory are so difficult that no one has ever been able to figure out what the consequences of the theory are, or to check it against experiment, and this has been going on now for almost twenty years!

So we are stuck with a theory, and we do not know whether it is right or wrong, but we do know that it is a *little* wrong, or at least incomplete. While we have been dawdling around theoretically, trying to calculate the consequences of this theory, the experimentalists have been discovering some things. For example, they had already discovered this *m-meson* or *muon*, and we do not yet know where it fits. Also, in cosmic rays, a large number of other "extra" particles were found. It turns out that today we have approximately thirty particles, and it is very difficult to understand the relationships of all these particles, and what nature

, wants them for, or what the connections are from one to another. We do not today understand these various particles as different aspects of the same thing, and the fact that we have so many unconnected particles is a representation of the fact that we have so much unconnected information without a good theory. After the great successes of quantum electrodynamics, there is a certain amount of knowledge of nuclear physics which is rough knowledge, sort of half experience and half theory, assuming a type of force between protons and neutrons and seeing what will happen, but not really understanding where the force comes from. Aside from that, we have made very little progress. We have collected an enormous number of chemical elements. In the chemical case, there suddenly appeared a relationship among these elements which was unexpected, and which is embodied in the periodic table of Mendeleev. For example, sodium and potassium are about the same in their chemical properties and are found in the same column in the Mendeleev chart. We have been seeking a Mendeleev-type chart for the new particles. One such chart of the new particles was made independently by Gell-Mann in the U.S.A. and Nishijima in Japan. The basis of their classification is a new number, like the electric charge, which can be assigned to each particle, called its "strangeness," S . This number is conserved, like the electric charge, in reactions which take place by nuclear forces.

In Table 2-2 are listed all the particles. We cannot discuss them much at this stage, but the table will at least show you how much we do not know. Underneath each particle its mass is given in a certain unit, called the Mev. One Mev is equal to 1.782×10^{-27} gram. The reason this unit was chosen is historical, and we shall not go into it now. More massive particles are put higher up on the chart; we see that a neutron and a proton have almost the same mass. In vertical columns we have put the particles with the same electrical charge, all neutral objects in one column, all positively charged ones to the right of this one, and all negatively charged objects to the left.

Particles are shown with a solid line and "resonances" with a dashed one. Several particles have been omitted from the table. These include the important zero-mass, zero-charge particles, the photon and the graviton, which do not fall into the baryon-meson-lepton classification scheme, and also some of the newer resonances (K^* , Δ , ρ , η). The antiparticles of the mesons are listed in the table, but the antiparticles of the leptons and baryons would have to be listed in another table which would look exactly like this one reflected on the zero-charge column. Although all of the particles except the electron, neutrino, photon, graviton, and proton are unstable, decay products have been shown only for the resonances. Strangeness assignments are not applicable for leptons, since they do not interact strongly with nuclei.

All particles which are together with the neutrons and protons are called *baryons*, and the following ones exist: There is a "lambda," with a mass of 1154 Mev, and three others, called sigmas, minus, neutral, and plus, with several masses almost the same. There are groups or multiplets with almost the same mass, within one or two percent. Each particle in a multiplet has the same strangeness. The first multiplet is the proton-neutron doublet, and then there is a singlet (the lambda) then the sigma triplet, and finally the xi doublet. Very recently, in 1961, even a few more particles were found. Or *are* they particles? They live so short a time, they disintegrate almost instantaneously, as soon as they are formed, that we do not know whether they should be considered as new particles, or some kind of "resonance" interaction of a certain definite energy between the A and T products into which they disintegrate.

In addition to the baryons the other particles which are involved in the nuclear interaction are called *mesons*. There are first the pions, which come in three varieties, positive, negative, and neutral; they form another multiplet. We have also found some new things called A'-mesons, and they occur as a doublet, K^+ and K^0 . Also, every particle has its antiparticle, unless a particle is its own antiparticle. For example, the π^- and the π^+ are antiparticles, but the π^0 is its own antiparticle. The K^- and K^+ are antiparticles, and the K^0 and $\bar{K}^0$. In addition, in 1961 we also found some more mesons or *maybe* mesons which disintegrate almost immediately.

Table 2-2. Elementary Particles

MASS in Gev.	-e	CHARGE 0	+e	GROUP STRANG
1.4	$\Upsilon^- \Lambda^0 \pi^-$	$\Upsilon^0 \Lambda^0 \pi^0$	$\Upsilon^+ \Lambda^0 \pi^+$ 1395	$S = 2$
1.3	Ξ^- 1319	Ξ^0 1311		$S = 2$
1.2	Σ^- 1196	Σ^0 1191	Σ^+ 1189	$S = 1$
1.1		Λ^0 1115		$S = 1$
1.0				
0.9		n 939	p 938	$S = 0$
0.8		$\omega^0 \rightarrow \pi^+ \pi^- \pi^0$		$S = 0$
0.7	$\rho^- \rightarrow \pi^+ \pi^-$	$\rho^0 \rightarrow \pi^+ \pi^-$	$\rho^+ \rightarrow \pi^+ \pi^0$	$S = 0$
0.6				
0.5	K^- 494	$K^0 \bar{K}^0$ 498	K^+ 494	$S = -1$ $+1$
0.4				
0.3				
0.2				
0.1	π^- 139.6	π^0 135.0	π^+ 139.6	$S = 0$
0	μ^- 105.6			LEPTONS
	e^- 0.51	ν^0 0		

A thing called w which goes into three pions has a mass 780 on this scale, and somewhat less certain is an object which disintegrates into two pions. These particles, called mesons and baryons, and the antiparticles of the mesons are on the same chart, but the antiparticles of the baryons must be put on another chart, "reflected" through the charge-zero column.

Just as Mendeleev's chart was very good, except for the fact that there were a number of rare earth elements which were hanging out loose from it, so we have a number of things hanging out loose from this chart—particles which do not interact strongly in nuclei, have nothing to do with a nuclear interaction, and do not have a strong interaction (I mean the powerful kind of interaction of nuclear energy). These are called leptons, and they are the following: there is the electron, which has a very small mass on this scale, only 0.510 Mev. Then there is that other, the μ -meson, the muon, which has a mass much higher, 206 times as heavy as an electron. So far as we can tell, by all experiments so far, the difference between the electron and the muon is nothing but the mass. Everything works exactly the same for the muon as for the electron, except that one is heavier than the other. Why is there another one heavier; what is the use for it? We do not know. In addition, there is a lepton which is neutral, called a neutrino, and this particle has zero mass. In fact, it is now known that there are two different kinds of neutrinos, one related to electrons and the other related to muons.

Finally, we have two other particles which do not interact strongly with the nuclear ones: one is a photon, and perhaps, if the field of gravity also has a quantum-mechanical analog (a quantum theory of gravitation has not yet been worked out), then there will be a particle, a graviton, which will have zero mass.

What is this "zero mass"? The masses given here are the masses of the particles *at rest*. The fact that a particle has zero mass means, in a way, that it cannot *be* at rest. A photon is never at rest, it is always moving at 186,000 miles a second. We will understand more what mass means when we understand the theory of relativity, which will come in due time.

Thus we are confronted with a large number of particles, which together seem to be the fundamental constituents of matter. Fortunately, these particles are not *all* different in their *interactions* with one another. In fact, there seem to be just *four kinds* of interaction between particles which, in the order of decreasing strength, are the nuclear force, electrical interactions, the beta-decay interaction, and gravity. The photon is coupled to all charged particles and the strength of the interaction is measured by some number, which is $1/137$. The detailed law of this coupling is known, that is quantum electrodynamics. • Gravity is coupled to all *energy*, but its coupling is extremely weak, much weaker than that of electricity. This law is also known. Then there are the so-called weak decays—beta decay, which causes the neutron to disintegrate into proton, electron, and neutrino, relatively slowly. This law is only partly known. The so-called strong interaction, the meson-baryon interaction, has a strength of 1 in this scale, and the law is completely unknown, although there are a number of known rules, such as that the number of baryons does not change in any reaction.

Table 2-3. Elementary Interactions

Coupling	Strength*	Law
Photon to charged particles	$\sim 10^{-2}$	Law known
Gravity to all energy	$\sim 10^{-40}$	Law known
Weak decays	$\sim 10^{-5}$	Law partly known
Mesons to baryons	~ 1	Law unknown (some rules known)

* The "strength" is a dimensionless measure of the coupling constant involved in each interaction ($\sim$ means "approximately").

This then, is the horrible condition of our physics today. To summarize it, I would say this: outside the nucleus, we seem to know all; inside it, quantum mechanics is valid—the principles of quantum mechanics have not been found to fail. The stage on which we put all of our knowledge, we would say, is relativistic space-time; perhaps gravity is involved in space-time. We do not know how the universe got started, and we have never made experiments which check our ideas of space and time accurately, below some tiny distance, so we only *know* that our ideas work above that distance. We should also add that the rules of the game are the quantum-mechanical principles, and those principles apply, so far as we can tell, to the new particles as well as to the old. The origin of the forces in nuclei leads us to new particles, but unfortunately they appear in great profusion and we lack a complete understanding of their interrelationship, although we already know that there are some very surprising relationships among them. We seem gradually to be groping toward an understanding of the world of sub-atomic particles, but we really do not know how far we have yet to go in this task.

The Relation of Physics to Other Sciences

3-1 Introduction

Physics is the most fundamental and all-inclusive of the sciences, and has had a profound effect on all scientific development. In fact, physics is the present-day equivalent of what used to be called *natural philosophy*, from which most of our modern sciences arose. Students of many fields find themselves studying physics because of the basic role it plays in all phenomena. In this chapter we shall try to explain what the fundamental problems in the other sciences are, but of course it is impossible in so small a space really to deal with the complex, subtle, beautiful matters in these other fields. Lack of space also prevents our discussing the relation of physics to engineering, industry, society, and war, or even the most remarkable relationship between mathematics and physics. (Mathematics is not a science from our point of view, in the sense that it is not a *natural* science. The test of its validity is not experiment.) We must, incidentally, make it clear from the beginning that if a thing is not a science, it is not necessarily bad. For example, love is not a science. So, if something is said not to be a science, it does not mean that there is something wrong with it; it just means that it is not a science.

3-2 Chemistry

The science which is perhaps the most deeply affected by physics is chemistry. Historically, the early days of chemistry dealt almost entirely with what we now call inorganic chemistry, the chemistry of substances which are not associated with living things. Considerable analysis was required to discover the existence of the many elements and their relationships—how they make the various relatively simple compounds found in rocks, earth, etc. This early chemistry was very important for physics. The interaction between the two sciences was very great because the theory of atoms was substantiated to a large extent by experiments in chemistry. The theory of chemistry, i.e., of the reactions themselves, was summarized to a large extent in the periodic chart of Mendeleev, which brings out many strange relationships among the various elements, and it was the collection of rules as to which substance is combined with which, and how, that constituted inorganic chemistry. All these rules were ultimately explained in principle by quantum mechanics, so that theoretical chemistry is in fact physics. On the other hand, it must be emphasized that this explanation is *in principle*. We have already discussed the difference between knowing the rules of the game of chess, and being able to play. So it is that we may know the rules, but we cannot play very well. It turns out to be very difficult to predict precisely what will happen in a given chemical reaction; nevertheless, the deepest part of theoretical chemistry must end up in quantum mechanics.

There is also a branch of physics and chemistry which was developed by both sciences together, and which is extremely important. This is the method of statistics applied in a situation in which there are mechanical laws, which is aptly called *statistical mechanics*. In any chemical situation a large number of atoms are involved, and we have seen that the atoms are all jiggling around in a very random and complicated way. If we could analyze each collision, and be able to follow in detail the motion of each molecule, we might hope to figure out what would happen, but the many numbers needed to keep track of all these molecules exceeds so enormously the capacity of any computer, and certainly the capacity of

3-1 Introduction

3-2 Chemistry

3-3 Biology

3-4 Astronomy

3-5 Geology

3-6 Psychology

3-7 How did it get that way?

the mind, that it was important to develop a method for dealing with such complicated situations. Statistical mechanics, then, is the science of the phenomena of heat, or thermodynamics. Inorganic chemistry is, as a science, now reduced essentially to what are called physical chemistry and quantum chemistry; physical chemistry to study the rates at which reactions occur and what is happening in detail (How do the molecules hit? Which pieces fly off first?, etc.), and quantum chemistry to help us understand what happens in terms of the physical laws.

The other branch of chemistry is *organic chemistry*, the chemistry of the substances which are associated with living things. For a time it was believed that the substances which are associated with living things were so marvelous that they could not be made by hand, from inorganic materials. This is not at all true—they are just the same as the substances made in inorganic chemistry, but more complicated arrangements of atoms are involved. Organic chemistry obviously has a very close relationship to the biology which supplies its substances, and to industry, and furthermore, much physical chemistry and quantum mechanics can be applied to organic as well as to inorganic compounds. However, the main problems of organic chemistry are not in these aspects, but rather in the analysis and synthesis of the substances which are formed in biological systems, in living things. This leads imperceptibly, in steps, toward biochemistry, and then into biology itself, or molecular biology.

3-3 Biology

Thus we come to the science of *biology*, which is the study of living things. In the early days of biology, the biologists had to deal with the purely descriptive problem of finding out *what* living things there were, and so they just had to count such things as the hairs of the limbs of fleas. After these matters were worked out with a great deal of interest, the biologists went into the *machinery* inside the living bodies, first from a gross standpoint, naturally, because it takes some effort to get into the finer details.

There was an interesting early relationship between physics and biology in which biology helped physics in the discovery of the *conservation of energy*, which was first demonstrated by Mayer in connection with the amount of heat taken in and given out by a living creature.

If we look at the processes of biology of living animals more closely, we see *many* physical phenomena: the circulation of blood, pumps, pressure, etc. There are nerves: we know what is happening when we step on a sharp stone, and that somehow or other the information goes from the leg up. It is interesting how that happens. In their study of nerves, the biologists have come to the conclusion that nerves are very fine tubes with a complex wall which is very thin; through this wall the cell pumps ions, so that there are positive ions on the outside and negative ions on the inside, like a capacitor. Now this membrane has an interesting property; if it "discharges" in one place, i.e., if some of the ions were able to move through one place, so that the electric voltage is reduced there, that electrical influence makes itself felt on the ions in the neighborhood, and it affects the membrane in such a way that it lets the ions through at neighboring points also. This in turn affects it farther along, etc., and so there is a wave of "penetrability" of the membrane which runs down the fiber when it is "excited" at one end by stepping on the sharp stone. This wave is somewhat analogous to a long sequence of vertical dominoes; if the end one is pushed over, that one pushes the next, etc. Of course this will transmit only one message unless the dominoes are set up again; and similarly in the nerve cell, there are processes which pump the ions slowly out again, to get the nerve ready for the next impulse. So it is that we know what we are doing (or at least where we are). Of course the electrical effects associated with this nerve impulse can be picked up with electrical instruments, and because there *are* electrical effects, obviously the physics of electrical effects has had a great deal of influence on understanding the phenomenon.

The opposite effect is that, from somewhere in the brain, a message is sent out along a nerve. What happens at the end of the nerve? There the nerve branches

out into fine little things, connected to a structure near a muscle, called an end-plate. For reasons which are not exactly understood, when the impulse reaches the end of the nerve, little packets of a chemical called acetylcholine are shot off (five or ten molecules at a time) and they affect the muscle fiber and make it contract—how simple! What makes a muscle contract? A muscle is a very large number of fibers close together, containing two different substances, myosin and actomyosin, but the machinery by which the chemical reaction induced by acetylcholine can modify the dimensions of the molecule is not yet known. Thus the fundamental processes in the muscle that make mechanical motions are not known.

Biology is such an enormously wide field that there are hosts of other problems that we cannot mention at all—problems on how vision works (what the light does in the eye), how hearing works, etc. (The way in which *thinking* works we shall discuss later under psychology.) Now, these things concerning biology which we have just discussed are, from a biological standpoint, really not fundamental, at the bottom of life, in the sense that even if we understood them we still would not understand life itself. To illustrate: the men who study nerves feel their work is very important, because after all you cannot have animals without nerves. But you *can* have *life* without nerves. Plants have neither nerves nor muscles, but they are working, they are alive, just the same. So for the fundamental problems of biology we must look deeper; when we do, we discover that all living things have a great many characteristics in common. The most common feature is that they are made of *cells*, within each of which is complex machinery for doing things chemically. In plant cells, for example, there is machinery for picking up light and generating sucrose, which is consumed in the dark to keep the plant alive. When the plant is eaten the sucrose itself generates in the animal a series of chemical reactions very closely related to photosynthesis (and its opposite effect in the dark) in plants.

In the cells of living systems there are many elaborate chemical reactions, in which one compound is changed into another and another. To give some impression of the enormous efforts that have gone into the study of biochemistry, the chart in Fig. 3-1 summarizes our knowledge to date on just one small part of the many series of reactions which occur in cells, perhaps a percent or so of it.

Here we see a whole series of molecules which change from one to another in a sequence or cycle of rather small steps. It is called the Krebs cycle, the respiratory cycle. Each of the chemicals and each of the steps is fairly simple, in terms of what change is made in the molecule, but—and this is a centrally important discovery in biochemistry—these changes are *relatively difficult to accomplish in a laboratory*. If we have one substance and another very similar substance, the one does not just turn into the other, because the two forms are usually separated by

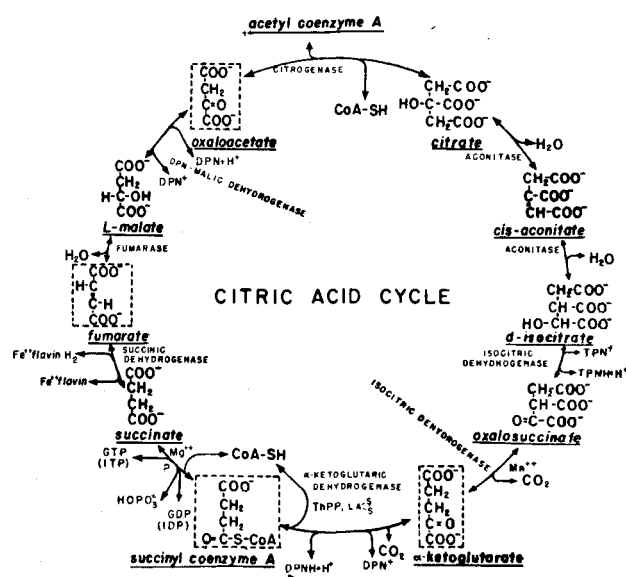


Fig. 3-1. The Krebs cycle.

an energy barrier or "hill." Consider this analogy: If we wanted to take an object from one place to another, at the same level but on the other side of a hill, we could push it over the top, but to do so requires the addition of some energy. Thus most chemical reactions do not occur, because there is what is called an *activation energy* in the way. In order to add an extra atom to our chemical requires that we get it *close* enough that some rearrangement can occur; then it will stick. But if we cannot give it enough energy to get it close enough, it will not go to completion, it will just go part way up the "hill" and back down again. However, if we could literally take the molecules in our hands and push and pull the atoms around in such a way as to open a hole to let the new atom in, and then let it snap back, we would have found another way, *around* the hill, which would not require extra energy, and the reaction would go easily. Now there actually *are*, in the cells, *very* large molecules, much larger than the ones whose changes we have been describing, which in some complicated way hold the smaller molecules just right, so that the reaction can occur easily. These very large and complicated things are called *enzymes*. (They were first called ferments, because they were originally discovered in the fermentation of sugar. In fact, some of the first reactions in the cycle were discovered there.) In the presence of an enzyme the reaction will go.

An enzyme is made of another substance called *protein*. Enzymes are very big and complicated, and each one is different, each being built to control a certain special reaction. The names of the enzymes are written in Fig. 3-1 at each reaction. (Sometimes the same enzyme may control two reactions.) We emphasize that the enzymes themselves are not involved in the reaction directly. They do not change; they merely let an atom go from one place to another. Having done so, the enzyme is ready to do it to the next molecule, like a machine in a factory. Of course, there must be a supply of certain atoms and a way of disposing of other atoms. Take hydrogen, for example: there are enzymes which have special units on them which carry the hydrogen for all chemical reactions. For example, there are three or four hydrogen-reducing enzymes which are used all over our cycle in different places. It is interesting that the machinery which liberates some hydrogen at one place will take that hydrogen and use it somewhere else.

The most important feature of the cycle of Fig. 3-1 is the transformation from GDP to GTP (guanadine-di-phosphate to guanadine-tri-phosphate) because the one substance has much more energy in it than the other. Just as there is a "box" in certain enzymes for carrying hydrogen atoms around, there are special *energy-carrying* "boxes" which involve the triphosphate group. So, GTP has more energy than GDP and if the cycle is going one way, we are producing molecules which have extra energy and which can go drive some other cycle which *requires* energy, for example the contraction of muscle. The muscle will not contract unless there is GTP. We can take muscle fiber, put it in water, and add GTP, and the fibers contract, changing GTP to GDP if the right enzymes are present. So the real system is in the GDP-GTP transformation; in the dark the GTP which has been stored up during the day is used to run the whole cycle around the other way. An enzyme you see, does not care in which direction the reaction goes, for if it did it would violate one of the laws of physics.

Physics is of great importance in biology and other sciences for still another reason, that has to do with *experimental techniques*. In fact, if it were not for the great development of experimental physics, these biochemistry charts would not be known today. The reason is that the most useful tool of all for analyzing this fantastically complex system is to *label* the atoms which are used in the reactions. Thus, if we could introduce into the cycle some carbon dioxide which has a "green mark" on it, and then measure after three seconds where the green mark is, and again measure after ten seconds, etc., we could trace out the course of the reactions. What are the "green marks"? They are different *isotopes*. We recall that the chemical properties of atoms are determined by the number of *electrons*, not by the mass of the nucleus. But there can be, for example in carbon, six neutrons or seven neutrons, together with the six protons which all carbon nuclei have. Chemically, the two atoms C₁₂ and C₁₃ are the same, but they differ in weight and they have different nuclear properties, and so they are distinguishable.

By using these isotopes of different weights, or even radioactive isotopes like C^{14} , which provide a more sensitive means for tracing very small quantities, it is possible to trace the reactions.

Now, we return to the description of enzymes and proteins. All proteins are not enzymes, but all enzymes are proteins. There are many proteins, such as the proteins in muscle, the structural proteins which are, for example, in cartilage and hair, skin, etc., that are not themselves enzymes. However, proteins are a very characteristic substance of life: first of all they make up all the enzymes, and second, they make up much of the rest of living material. Proteins have a very interesting and simple structure. They are a series, or chain, of different *amino acids*. There are twenty different amino acids, and they all can combine with each other to form chains in which the backbone is CO-NH, etc. Proteins are nothing but chains of various ones of these twenty amino acids. Each of the amino acids probably serves some special purpose. Some, for example, have a sulphur atom at a certain place; when two sulphur atoms are in the same protein, they form a bond, that is, they tie the chain together at two points and form a loop. Another has extra oxygen atoms which make it an acidic substance, another has a basic characteristic. Some of them have big groups hanging out to one side, so that they take up a lot of space. One of the amino acids, called proline, is not really an amino acid, but imino acid. There is a slight difference, with the result that when proline is in the chain, there is a kink in the chain. If we wished to manufacture a particular protein, we would give these instructions: put one of those sulphur hooks here; next, add something to take up space; then attach something to put a kink in the chain. In this way, we will get a complicated-looking chain, hooked together and having some complex structure; this is presumably just the manner in which all the various enzymes are made. One of the great triumphs in recent times (since 1960), was at last to discover the exact spatial atomic arrangement of certain proteins, which involve some fifty-six or sixty amino acids in a row. Over a thousand atoms (more nearly two thousand, if we count the hydrogen atoms) have been located in a complex pattern in two proteins. The first was hemoglobin. One of the sad aspects of this discovery is that we cannot see anything from the pattern; we do not understand why it works the way it does. Of course, that is the next problem to be attacked.

Another problem is how do the enzymes know what to be? A red-eyed fly makes a red-eyed fly baby, and so the information for the whole pattern of enzymes to make red pigment must be passed from one fly to the next. This is done by a substance in the nucleus of the cell, not a protein, called DNA (short for deoxyribose nucleic acid). This is the key substance which is passed from one cell to another (for instance, sperm cells consist mostly of DNA) and carries the information as to how to make the enzymes. DNA is the "blueprint." What does the blueprint look like and how does it work? First, the blueprint must be able to reproduce itself. Secondly, it must be able to instruct the protein. Concerning the reproduction, we might think that this proceeds like cell reproduction. Cells simply grow bigger and then divide in half. Must it be thus with DNA molecules, then, that they too grow bigger and divide in half? Every *atom* certainly does not grow bigger and divide in half! No, it is impossible to reproduce a molecule except by some more clever way.

The structure of the substance DNA was studied for a long time, first chemically to find the composition, and then with x-rays to find the pattern in space. The result was the following remarkable discovery: The DNA molecule is a pair of chains, twisted upon each other. The backbone of each of these chains, which are analogous to the chains of proteins but chemically quite different, is a series of sugar and phosphate groups, as shown in Fig. 3-2. Now we see how the chain can contain instructions, for if we could split this chain down the middle, we would have a series BAADC . . . and every living thing could have a different series. Thus perhaps, in some way, the specific *instructions* for the manufacture of proteins are contained in the specific *series* of the DNA.

Attached to each sugar along the line, and linking the two chains together, are certain pairs of cross-links. However, they are not all of the same kind; there are

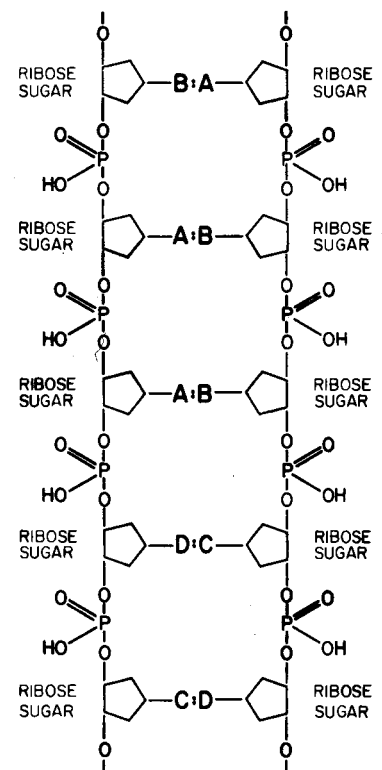


Fig. 3-2. Schematic diagram of DNA.

four kinds, called adenine, thymine, cytosine, and guanine, but let us call them *A*, *B*, *C*, and *D*. The interesting thing is that only certain pairs can sit opposite each other, for example *A* with *B* and *C* with *D*. These pairs are put on the two chains in such a way that they "fit together," and have a strong energy of interaction. However, *C* will not fit with *A*, and *B* will not fit with *C*; they will only fit in pairs, *A* against *B* and *C* against *D*. Therefore if one is *C*, the other must be *D*, etc. Whatever the letters may be in one chain, each one must have its specific complementary letter on the other chain.

What then about reproduction? Suppose we split this chain in two. How can we make another one just like it? If, in the substances of the cells, there is a manufacturing department which brings up phosphate, sugar, and *A*, *B*, *C*, *D* units not connected in a chain, the only ones which will attach to our split chain will be the correct ones, the complements of *BAADC* . . . , namely, *ABBCD* ... Thus what happens is that the chain splits down the middle during cell division, one half ultimately to go with one cell, the other half to end up in the other cell; when separated, a new complementary chain is made by each half-chain.

Next comes the question, precisely how does the order of the *A*, *B*, *C*, *D* units determine the arrangement of the amino acids in the protein? This is the central unsolved problem in biology today. The first clues, or pieces of information, however, are these: There are in the cell tiny particles called microsomes, and it is now known that that is the place where proteins are made. But the microsomes are not in the nucleus, where the DNA and its instructions are. Something seems to be the matter. However, it is also known that little molecule pieces come off the DNA—not as long as the big DNA molecule that carries all the information itself, but like a small section of it. This is called RNA, but that is not essential. It is a kind of copy of the DNA, a short copy. The RNA, which somehow carries a message as to what kind of protein to make goes over to the microsome; that is known. When it gets there, protein is synthesized at the microsome. That is also known. However, the details of how the amino acids come in and are arranged in accordance with a code that is on the RNA are, as yet, still unknown. We do not know how to read it. If we knew, for example, the "lineup" *A*, *B*, *C*, *C*, *A*, we could not tell you what protein is to be made.

Certainly no subject or field is making more progress on so many fronts at the present moment, than biology, and if we were to name the most powerful assumption of all, which leads one on and on in an attempt to understand life, it is that *all things are made of atoms*, and that everything that living things do can be understood in terms of the jiggings and wiggings of atoms.

3-4 Astronomy

In this rapid-fire explanation of the whole world, we must now turn to astronomy. Astronomy is older than physics. In fact, it got physics started by showing the beautiful simplicity of the motion of the stars and planets, the understanding of which was the *beginning* of physics. But the most remarkable discovery in all of astronomy is that *the stars are made of atoms of the same kind as those on the earth** How was this done? Atoms liberate light which has definite fre-

* How I'm rushing through this! How much each sentence in this brief story contains. "The stars are made of the same atoms as the earth." I usually pick one small topic like this to give a lecture on. Poets say science takes away from the beauty of the stars—mere globs of gas atoms. Nothing is "mere." I too can see the stars on a desert night, and feel them. But do I see less or more? The vastness of the heavens stretches my imagination—stuck on this carousel my little eye can catch one-million-year-old light. A vast pattern—of which I am a part—perhaps my stuff was belched from some forgotten star, as one is belching there. Or see them with the greater eye of Palomar, rushing all apart from some common starting point when they were perhaps all together. What is the pattern, or the meaning, or the *why*? It does not do harm to the mystery to know a little about it. For far more marvelous is the truth than any artists of the past imagined! Why do the poets of the present not speak of it? What men are poets who can speak of Jupiter if he were like a man, but if he is an immense spinning sphere of methane and ammonia must be silent?

quencies, something like the timbre of a musical instrument, which has definite pitches or frequencies of sound. When we are listening to several different tones we can tell them apart, but when we look with our eyes at a mixture of colors we cannot tell the parts from which it was made, because the eye is nowhere near as discerning as the ear in this connection. However, with a spectroscope we *can* analyze the frequencies of the light waves and in this way we can see the very tunes of the atoms that are in the different stars. As a matter of fact, two of the chemical elements were discovered on a star before they were discovered on the earth. Helium was discovered on the sun, whence its name, and technetium was discovered in certain cool stars. This, of course, permits us to make headway in understanding the stars, because they are made of the same kinds of atoms which are on the earth. Now we know a great deal about the atoms, especially concerning their behavior under conditions of high temperature but not very great density, so that we can analyze by statistical mechanics the behavior of the stellar substance. Even though we cannot reproduce the conditions on the earth, using the basic physical laws we often can tell precisely, or very closely, what will happen. So it is that physics aids astronomy. Strange as it may seem, we understand the distribution of matter in the interior of the sun far better than we understand the interior of the earth. What goes on *inside* a star is better understood than one might guess from the difficulty of having to look at a little dot of light through a telescope, because we can *calculate* what the atoms in the stars should do in most circumstances.

One of the most impressive discoveries was the origin of the energy of the stars, that makes them continue to burn. One of the men who discovered this was out with his girl friend the night after he realized that *nuclear reactions* must be going on in the stars in order to make them shine. She said "Look at how pretty the stars shine!" He said "Yes, and right now I am the only man in the world who knows *why* they shine." She merely laughed at him. She was not impressed with being out with the only man who, at that moment, knew why stars shine. Well, it is sad to be alone, but that is the way it is in this world.

It is the nuclear "burning" of hydrogen which supplies the energy of the sun; the hydrogen is converted into helium. Furthermore, ultimately, the manufacture of various chemical elements proceeds in the centers of the stars, from hydrogen. The stuff of which we are made, was "cooked" once, in a star, and spit out. How do we know? Because there is a clue. The proportion of the different isotopes—how much C^{12} , how much C^{13} , etc., is something which is never changed by *chemical* reactions, because the chemical reactions are so much the same for the two. The proportions are purely the result of *nuclear* reactions. By looking at the proportions of the isotopes in the cold, dead ember which we are, we can discover what the *furnace* was like in which the stuff of which we are made was formed. That furnace was like the stars, and so it is very likely that our elements were "made" in the stars and spit out in the explosions which we call novae and supernovae. Astronomy is so close to physics that we shall study many astronomical things as we go along.

3-5 Geology

We turn now to what are called *earth sciences*, or *geology*. First, meteorology and the weather. Of course the *instruments* of meteorology are physical instruments, and the development of experimental physics made these instruments possible, as was explained before. However, the theory of meteorology has never been satisfactorily worked out by the physicist. "Well," you say, "there is nothing but air, and we know the equations of the motions of air." Yes we do. "So if we know the condition of air today, why can't we figure out the condition of the air tomorrow?" First, we do not *really* know what the condition is today, because the air is swirling and twisting everywhere. It turns out to be very sensitive, and even unstable. If you have ever seen water run smoothly over a dam, and then turn into a large number of blobs and drops as it falls, you will understand what I mean by unstable. You know the condition of the water before it goes over the

spillway; it is perfectly smooth; but the moment it begins to fall, where do the drops begin? What determines how big the lumps are going to be and where they will be? That is not known, because the water is unstable. Even a smooth moving mass of air, in going over a mountain turns into complex whirlpools and eddies. In many fields we find this situation of *turbulent flow* that we cannot analyze today. Quickly we leave the subject of weather, and discuss geology!

The question basic to geology is, what makes the earth the way it is? The most obvious processes are in front of your very eyes, the erosion processes of the rivers, the winds, etc. It is easy enough to understand these, but for every bit of erosion there is an equal amount of something else going on. Mountains are no lower today, on the average, than they were in the past. There must be *mountain-forming* processes. You will find, if you study geology, that there *are* mountain-forming processes and vulcanism, which nobody understands but which is half of geology. The phenomenon of volcanoes is really not understood. What makes an earthquake is, ultimately, not understood. It is understood that if something is pushing something else, it snaps and will slide—that is all right. But what pushes, and why? The theory is that there are currents inside the earth—circulating currents, due to the difference in temperature inside and outside—which, in their motion, push the surface slightly. Thus if there are two opposite circulations next to each other, the matter will collect in the region where they meet and make belts of mountains which are in unhappy stressed conditions, and so produce volcanoes and earthquakes.

What about the inside of the earth? A great deal is known about the speed of earthquake waves through the earth and the density of distribution of the earth. However, physicists have been unable to get a good theory as to how dense a substance should be at the pressures that would be expected at the center of the earth. In other words, we cannot figure out the properties of matter very well in these circumstances. We do much less well with the earth than we do with the conditions of matter in the stars. The mathematics involved seems a little too difficult, so far, but perhaps it will not be too long before someone realizes that it is an important problem, and really work it out. The other aspect, of course, is that even if we did know the density, we cannot figure out the circulating currents. Nor can we really work out the properties of rocks at high pressure. We cannot tell how fast the rocks should "give"; that must all be worked out by experiment.

3-6 Psychology

Next, we consider the science of *psychology*. Incidentally, psychoanalysis is not a science: it is at best a medical process, and perhaps even more like witch-doctoring. It has a theory as to what causes disease—lots of different "spirits," etc. The witch doctor has a theory that a disease like malaria is caused by a spirit which comes into the air; it is not cured by shaking a snake over it, but quinine does help malaria. So, if you are sick, I would advise that you go to the witch doctor because he is the man in the tribe who knows the most about the disease; on the other hand, his knowledge is not science. Psychoanalysis has not been checked carefully by experiment, and there is no way to find a list of the number of cases in which it works, the number of cases in which it does not work, etc.

The other branches of psychology, which involve things like the physiology of sensation—what happens in the eye, and what happens in the brain—are, if you wish, less interesting. But some small but real progress has been made in studying them. One of the most interesting technical problems may or may not be called psychology. The central problem of the mind, if you will, or the nervous system, is this: when an animal learns something, it can do something different than it could before, and its brain cell must have changed too, if it is made out of atoms. *In what way is it different?* We do not know where to look, or what to look for, when something is memorized. We do not know what it means, or what change there is in the nervous system, when a fact is learned. This is a very important problem which has not been solved at all. Assuming, however, that there is some kind of memory thing, the brain is such an enormous mass of interconnect-

ing wires and nerves that it probably cannot be analyzed in a straightforward manner. There is an analog of this to computing machines and computing elements, in that they also have a lot of lines, and they have some kind of element, analogous, perhaps, to the synapse, or connection of one nerve to another. This is a very interesting subject which we have not the time to discuss further—the relationship between thinking and computing machines. It must be appreciated, of course, that this subject will tell us very little about the real complexities of ordinary human behavior. All human beings are so different. It will be a long time before we get there. We must start much further back. If we could even figure out how a *dog* works, we would have gone pretty far. Dogs are easier to understand, but nobody yet knows how dogs work.

3-7 How did it get that way?

In order for physics to be useful to other sciences in a *theoretical* way, other than in the invention of instruments, the science in question must supply to the physicist a description of the object in a physicist's language. They can say "why does a frog jump?," and the physicist cannot answer. If they tell him what a frog is, that there are so many molecules, there is a nerve here, etc., that is different. If they will tell us, more or less, what the earth or the stars are like, then we can figure it out. In order for physical theory to be of any use, we must know where the atoms are located. In order to understand the chemistry, we must know exactly what atoms are present, for otherwise we cannot analyze it. That is but one limitation, of course.

There is another *kind* of problem in the sister sciences which does not exist in physics; we might call it, for lack of a better term, the historical question. How did it get that way? If we understand all about biology, we will want **to** know how all the things which are on the earth got there. There is the theory of evolution, an important part of biology. In geology, we not only want to know how the mountains are forming, but how the entire earth was formed in the beginning, the origin of the solar system, etc. That, of course, leads us to want to know what kind of matter there was in the world. How did the stars evolve? What were the initial conditions? That is the problem of astronomical history. A great deal has been found out about the formation of stars, the formation of elements from which we were made, and even a little about the origin of the universe.

There is no historical question being studied in physics at the present time. We do not have a question, "Here are the laws of physics, how did they get that way?" We do not imagine, at the moment, that the laws of physics are somehow changing with time, that they were different in the past than they are at present. Of course they *may* be, and the moment we find they *are*, the historical question of physics will be wrapped up with the rest of the history of the universe, and then the physicist will be talking about the same problems as astronomers, geologists, and biologists.

Finally, there is a physical problem that is common to many fields, that is very old, and that has not been solved. It is not the problem of finding new fundamental particles, but something left over from a long time ago—over a hundred years. Nobody in physics has really been able to analyze it mathematically satisfactorily in spite of its importance to the sister sciences. It is the analysis of *circulating or turbulent fluids*. If we watch the evolution of a star, there comes a point where we can deduce that it is going to start convection, and thereafter we can no longer deduce what should happen. A few million years later the star explodes, but we cannot figure out the reason. We cannot analyze the weather. We do not know the patterns of motions that there should be inside the earth. The simplest form of the problem is to take a pipe that is very long and push water through it at high speed. We ask: to push a given amount of water through that pipe, how much pressure is needed? No one can analyze it from first principles and the properties of water. If the water flows very slowly, or if we use a thick goo like honey, then we can do it nicely. You will find that in your textbook.

What we really cannot do is deal with actual, wet water running through a pipe. That is the central problem which we ought to solve some day, and we have not.

A poet once said, "The whole universe is in a glass of wine." We will probably never know in what sense he meant that, for poets do not write to be understood. But it is true that if we look at a glass of wine closely enough we see the entire universe. There are the things of physics: the twisting liquid which evaporates depending on the wind and weather, the reflections in the glass, and our imagination adds the atoms. The glass is a distillation of the earth's rocks, and in its composition we see the secrets of the universe's age, and the evolution of stars. What strange array of chemicals are in the wine? How did they come to be? There are the ferments, the enzymes, the substrates, and the products. There in wine is found the great generalization: all life is fermentation. Nobody can discover the chemistry of wine without discovering, as did Louis Pasteur, the cause of much disease. How vivid is the claret, pressing its existence into the consciousness that watches it! If our small minds, for some convenience, divide this glass of wine, this universe, into parts—physics, biology, geology, astronomy, psychology, and so on—remember that nature does not know it! So let us put it all back together, not forgetting ultimately what it is for. Let it give us one more final pleasure: drink it and forget it all!

Conservation of Energy

4-1 What is energy?

In this chapter, we begin our more detailed study of the different aspects of physics, having finished our description of things in general. To illustrate the ideas and the kind of reasoning that might be used in theoretical physics, we shall now examine one of the most basic laws of physics, the conservation of energy.

There is a fact, or if you wish, a *law*, governing all natural phenomena that are known to date. There is no known exception to this law—it is exact so far as we know. The law is called the *conservation of energy*. It states that there is a certain quantity, which we call energy, that does not change in the manifold changes which nature undergoes. That is a most abstract idea, because it is a mathematical principle; it says that there is a numerical quantity which does not change when something happens. It is not a description of a mechanism, or anything concrete; it is just a strange fact that we can calculate some number and when we finish watching nature go through her tricks and calculate the number again, it is the same. (Something like the bishop on a red square, and after a number of moves—details unknown—it is still on some red square. It is a law of this nature.) Since it is an abstract idea, we shall illustrate the meaning of it by an analogy.

Imagine a child, perhaps "Dennis the Menace," who has blocks which are absolutely indestructible, and cannot be divided into pieces. Each is the same as the other. Let us suppose that he has 28 blocks. His mother puts him with his 28 blocks into a room at the beginning of the day. At the end of the day, being curious, she counts the blocks very carefully, and discovers a phenomenal law—no matter what he does with the blocks, there are always 28 remaining! This continues for a number of days, until one day there are only 27 blocks, but a little investigating shows that there is one under the rug—she must look everywhere to be sure that the number of blocks has not changed. One day, however, the number appears to change—there are only 26 blocks. Careful investigation indicates that the window was open, and upon looking outside, the other two blocks are found. Another day, careful count indicates that there are 30 blocks! This causes considerable consternation, until it is realized that Bruce came to visit, bringing his blocks with him, and he left a few at Dennis' house. After she has disposed of the extra blocks, she closes the window, does not let Bruce in, and then everything is going along all right, until one time she counts and finds only 25 blocks. However, there is a box in the room, a toy box, and the mother goes to open the toy box, but the boy says "No, do not open my toy box," and screams. Mother is not allowed to open the toy box. Being extremely curious, and somewhat ingenious, she invents a scheme! She knows that a block weighs three ounces, so she weighs the box at a time when she sees 28 blocks, and it weighs 16 ounces. The next time she wishes to check, she weighs the box again, subtracts sixteen ounces and divides by three. She discovers the following:

$$\left(\frac{\text{number of}}{\text{blocks seen}} \right) + \frac{(\text{weight of box}) - 16 \text{ ounces}}{3 \text{ ounces}} = \text{constant.} \quad (4.1)$$

There then appear to be some new deviations, but careful study indicates that the dirty water in the bathtub is changing its level. The child is throwing blocks into the water, and she cannot see them because it is so dirty, but she can find out how many blocks are in the water by adding another term to her formula. Since the original height of the water was 6 inches and each block raises the water a quarter

4-1 What is energy?

4-2 Gravitational potential energy

4-3 Kinetic energy

4-4 Other forms of energy

of an inch, this new formula would be:

$$\left(\frac{\text{number of}}{\text{blocks seen}} \right) + \frac{(\text{weight of box}) - 16 \text{ ounces}}{3 \text{ ounces}} + \frac{(\text{height of water}) - 6 \text{ inches}}{1/4 \text{ inch}} = \text{constant.} \quad (4.2)$$

In the gradual increase in the complexity of her world, she finds a whole series of terms representing ways of calculating how many blocks are in places where she is not allowed to look. As a result, she finds a complex formula, a quantity which *has to be computed*, which always stays the same in her situation.

What is the analogy of this to the conservation of energy? The most remarkable aspect that must be abstracted from this picture is that *there are no blocks*. Take away the first terms in (4.1) and (4.2) and we find ourselves calculating more or less abstract things. The analogy has the following points. First, when we are calculating the energy, sometimes some of it leaves the system and goes away, or sometimes some comes in. In order to verify the conservation of energy, we must be careful that we have not put any in or taken any out. Second, the energy has a large number of *different forms*, and there is a formula for each one. These are: gravitational energy, kinetic energy, heat energy, elastic energy, electrical energy, chemical energy, radiant energy, nuclear energy, mass energy. If we total up the formulas for each of these contributions, it will not change except for energy going in and out.

It is important to realize that in physics today, we have no knowledge of what energy *is*. We do not have a picture that energy comes in little blobs of a definite amount. It is not that way. However, there are formulas for calculating some numerical quantity, and when we add it all together it gives "28"—always the same number. It is an abstract thing in that it does not tell us the mechanism or the *reasons* for the various formulas.

4-2 Gravitational potential energy

Conservation of energy can be understood only if we have the formula for all of its forms. I wish to discuss the formula for gravitational energy near the surface of the Earth, and I wish to derive this formula in a way which has nothing to do with history but is simply a line of reasoning invented for this particular lecture to give you an illustration of the remarkable fact that a great deal about nature can be extracted from a few facts and close reasoning. It is an illustration of the kind of work theoretical physicists become involved in. It is patterned after a most excellent argument by Mr. Carnot on the efficiency of steam engines.*

Consider weight-lifting machines—machines which have the property that they lift one weight by lowering another. Let us also make a hypothesis: that *there is no such thing as perpetual motion* with these weight-lifting machines. (In fact, that there is no perpetual motion at all is a general statement of the law of conservation of energy.) We must be careful to define perpetual motion. First, let us do it for weight-lifting machines. If, when we have lifted and lowered a lot of weights and restored the machine to the original condition, we find that the net result is to have *lifted a weight*, then we have a perpetual motion machine because we can use that lifted weight to run something else. That is, *provided* the machine which lifted the weight is brought back to its exact *original condition*, and furthermore that it is completely *self-contained*—that it has not received the energy to lift that weight from some external source—like Bruce's blocks.

A very simple weight-lifting machine is shown in Fig. 4-1. This machine lifts weights three units "strong." We place three units on one balance pan, and one unit on the other. However, in order to get it actually to work, we must lift a little weight off the left pan. On the other hand, we could lift a one-unit weight



Fig. 4-1. Simple weight-lifting machine.

* Our point here is not so much the result, (4.3), which in fact you may already know, as the possibility of arriving at it by theoretical reasoning.

by lowering the three-unit weight, if we cheat a little by lifting a little weight off the other pan. Of course, we realize that with any *actual* lifting machine, we must add a little extra to get it to run. This we disregard, *temporarily*. Ideal machines, although they do not exist, do not require anything extra. A machine that we actually use can be, in a sense, *almost* reversible: that is, if it will lift the weight of three by lowering a weight of one, then it will also lift nearly the weight of one the same amount by lowering the weight of three.

We imagine that there are two classes of machines, those that are *not* reversible, which includes all real machines, and those that *are* reversible, which of course are actually not attainable no matter how careful we may be in our design of bearings, levers, etc. We suppose, however, that there is such a thing—a reversible machine—which lowers one unit of weight (a pound or any other unit) by one unit of distance, and at the same time lifts a three-unit weight. Call this reversible machine, Machine A. Suppose this particular reversible machine lifts the three-unit weight a distance X . Then suppose we have another machine, Machine B, which is not necessarily reversible, which also lowers a unit weight a unit distance, but which lifts three units a distance Y . We can now prove that Y is not higher than X ; that is, it is impossible to build a machine that will lift a weight *any higher* than it will be lifted by a reversible machine. Let us see why. Let us suppose that Y were higher than X . We take a one-unit weight and lower it one unit height with Machine B, and that lifts the three-unit weight up a distance Y . Then we could lower the weight from Y to X , *obtaining free power*, and use the reversible Machine A, running backwards, to lower the three-unit weight a distance X and lift the one-unit weight by one unit height. This will put the one-unit weight back where it was before, and leave both machines ready to be used again! We would therefore have perpetual motion if Y were higher than X , which we assumed was impossible. With those assumptions, we thus deduce that Y is not higher than X , so that of all machines that can be designed, the reversible machine is the best.

We can also see that all reversible machines must lift to *exactly the same height*. Suppose that B were really reversible also. The argument that Y is not higher than X is, of course, just as good as it was before, but we can also make our argument the other way around, using the machines in the opposite order, and prove that X is not higher than Y . This, then, is a very remarkable observation because it permits us to analyze the height to which different machines are going to lift something *without looking at the interior mechanism*. We know at once that if somebody makes an enormously elaborate series of levers that lift three units a certain distance by lowering one unit by one unit distance, and we compare it with a simple lever which does the same thing and is fundamentally reversible, his machine will lift it no higher, but perhaps less high. If his machine is reversible, we also know exactly *how high* it will lift. To summarize: every reversible machine, no matter how it operates, which drops one pound one foot and lifts a three-pound weight always lifts it the same distance, X . This is clearly a universal law of great utility. The next question is, of course, what is X ?

Suppose we have a reversible machine which is going to lift this distance X , three for one. We set up three balls in a rack which does not move, as shown in Fig. 4-2. One ball is held on a stage at a distance one foot above the ground. The machine can lift three balls, lowering one by a distance 1. Now, we have arranged that the platform which holds three balls has a floor and two shelves, exactly spaced at distance X , and further, that the rack which holds the balls is spaced at distance X , (a). First we roll the balls horizontally from the rack to the shelves, (b), and we suppose that this takes no energy because we do not change the height. The reversible machine then operates: it lowers the single ball to the floor, and it lifts the rack a distance X , (c). Now we have ingeniously arranged the rack so that these balls are again even with the platforms. Thus we unload the balls onto the rack, (d); having unloaded the balls, we can restore the machine to its original condition. Now we have three balls on the upper three shelves and one at the bottom. But the strange thing is that, in a certain way of speaking, we have not lifted two of them at all because, after all, there were balls on shelves 2 and 3

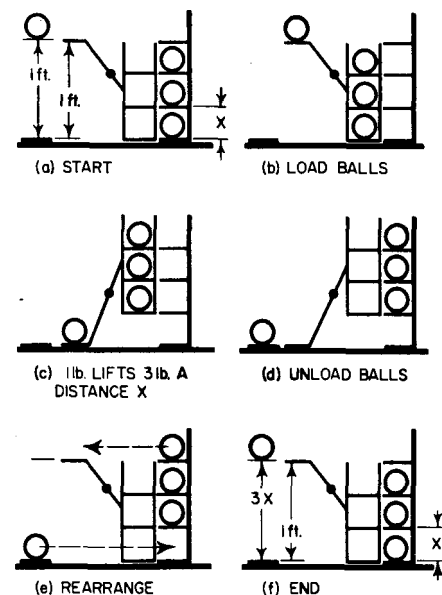


Fig. 4-2. A reversible machine.

before. The resulting effect has been to lift *one ball* a distance $3X$. Now, if $3X$ exceeds one foot, then we can *lower* the ball to return the machine to the initial condition, (f), and we can run the apparatus again. Therefore $3X$ *cannot* exceed one foot, for if $3X$ exceeds one foot we can make perpetual motion. Likewise, we can prove that *one foot cannot exceed $3X$* , by making the whole machine run the opposite way, since it is a reversible machine. Therefore $3X$ is neither *greater nor less than a foot*, and we discover then, by argument alone, the law that $X = \frac{1}{3}$ foot. The generalization is clear: one pound falls a certain distance in operating a reversible machine; then the machine can lift p pounds this distance divided by p . Another way of putting the result is that three pounds times the height lifted, which in our problem was X , is equal to one pound times the distance lowered, which is one foot in this case. If we take all the weights and multiply them by the heights at which they are now, above the floor, let the machine operate, and then multiply all the weights by all the heights again, *there will be no change*. (We have to generalize the example where we moved only one weight to the case where when we lower one we lift several different ones—but that is easy.)

We call the sum of the weights times the heights *gravitational potential energy*—the energy which an object has because of its relationship in space, relative to the earth. The formula for gravitational energy, then, so long as we are not too far from the earth (the force weakens as we go higher) is

$$\left(\begin{array}{c} \text{gravitational} \\ \text{potential energy} \\ \text{for one object} \end{array} \right) = (\text{weight}) \times (\text{height}). \quad (4.3)$$

It is a very beautiful line of reasoning. The only problem is that perhaps it is not true. (After all, nature does not *have* to go along with our reasoning.) For example, perhaps perpetual motion is, in fact, possible. Some of the assumptions may be wrong, or we may have made a mistake in reasoning, so it is always necessary to check. *It turns out experimentally*, in fact, to be true.

The general name of energy which has to do with location relative to something else is called *potential energy*. In this particular case, of course, we call it *gravitational potential energy*. If it is a question of electrical forces against which we are working, instead of gravitational forces, if we are "lifting" charges away from other charges with a lot of levers, then the energy content is called *electrical potential energy*. The general principle is that the change in the energy is the force times the distance that the force is pushed, and that this is a change in energy in general:

$$\left(\begin{array}{c} \text{change in} \\ \text{energy} \end{array} \right) = (\text{force}) \times \left(\begin{array}{c} \text{distance force} \\ \text{acts through} \end{array} \right). \quad (4.4)$$

We will return to many of these other kinds of energy as we continue the course.

The principle of the conservation of energy is very useful for deducing what will happen in a number of circumstances. In high school we learned a lot of laws about pulleys and levers used in different ways. We can now see that these "laws" are *all the same thing*, and that we did not have to memorize 75 rules to figure it out. A simple example is a smooth inclined plane which is, happily, a three-four-five triangle (Fig. 4-3). We hang a one-pound weight on the inclined plane with a pulley, and on the other side of the pulley, a weight W . We want to know how heavy W must be to balance the one pound on the plane. How can we figure that out? If we say it is just balanced, it is reversible and so can move up and down, and we can consider the following situation. In the initial circumstance, (a), the one pound weight is at the bottom and weight W is at the top. When W has slipped down in a reversible way, we have a one-pound weight at the top and the weight W the slant distance, (b), or five feet, from the plane in which it was before. We *lifted* the one-pound weight only *three* feet and we lowered W pounds by *five* feet. Therefore $W = \frac{3}{5}$ of a pound. Note that we deduced this from the *conservation of energy*, and not from force components. Cleverness, however, is relative. It can be deduced in a way which is even more brilliant, discovered by

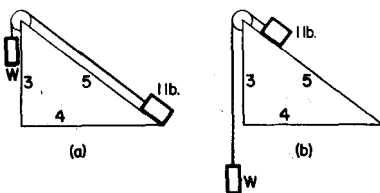


Fig. 4-3. Inclined plane.

Stevinus and inscribed on his tombstone. Figure 4-4 explains that it has to be $\frac{2}{3}$ of a pound, because the chain does not go around. It is evident that the lower part of the chain is balanced by itself, so that the pull of the five weights on one side must balance the pull of three weights on the other, or whatever the ratio of the legs. You see, by looking at this diagram, that W must be $\frac{2}{3}$ of a pound. (If you get an epitaph like that on your gravestone, you are doing fine.)

Let us now illustrate the energy principle with a more complicated problem, the screw jack shown in Fig. 4-5. A handle 20 inches long is used to turn the screw, which has 10 threads to the inch. We would like to know how much force would be needed at the handle to lift one ton (2000 pounds). If we want to lift the ton one inch, say, then we must turn the handle around ten times. When it goes around once it goes approximately 126 inches. The handle must thus travel 1260 inches, and if we used various pulleys, etc., we would be lifting our one ton with an unknown smaller weight W applied to the end of the handle. So we find out that W is about 1.6 pounds. This is a result of the conservation of energy.

Take now the somewhat more complicated example shown in Fig. 4-6. A rod or bar, 8 feet long, is supported at one end. In the middle of the bar is a weight of 60 pounds, and at a distance of two feet from the support there is a weight of 100 pounds. How hard do we have to lift the end of the bar in order to keep it balanced, disregarding the weight of the bar? Suppose we put a pulley at one end and hang a weight on the pulley. How big would the weight W have to be in order for it to balance? We imagine that the weight falls any arbitrary distance—to make it easy for ourselves suppose it goes down 4 inches—how high would the two load weights rise? The center rises 2 inches, and the point a quarter of the way from the fixed end lifts 1 inch. Therefore, the principle that the sum of the heights times the weights does not change tells us that the weight W times 4 inches down, plus 60 pounds times 2 inches up, plus 100 pounds times 1 inch has to add up to nothing:

$$-4W + (2)(60) + (1)(100) = 0, \quad W = 55 \text{ lb.} \quad (4.5)$$

Thus we must have a 55-pound weight to balance the bar. In this way we can work out the laws of "balance"—the statics of complicated bridge arrangements, and so on. This approach is called the *principle of virtual work*, because in order to apply this argument we had to *imagine* that the structure moves a little—even though it is not *really* moving or even *movable*. We use the very small imagined motion to apply the principle of conservation of energy.

4-3 Kinetic energy

To illustrate another type of energy we consider a pendulum (Fig. 4-7). If we pull the mass aside and release it, it swings back and forth. In its motion, it loses height in going from either end to the center. Where does the potential energy go? Gravitational energy disappears when it is down at the bottom; nevertheless, it will climb up again. The gravitational energy must have gone into another form. Evidently it is by virtue of its *motion* that it is able to climb up again, so we have the conversion of gravitational energy into some other form when it reaches the bottom.

We must get a formula for the energy of motion. Now, recalling our arguments about reversible machines, we can easily see that in the motion at the bottom must be a quantity of energy which permits it to rise a certain height, and which has nothing to do with the *machinery* by which it comes up or the *path* by which it comes up. So we have an equivalence formula something like the one we wrote for the child's blocks. We have another form to represent the energy. It is easy to say what it is. The kinetic energy at the bottom equals the weight times the height that it could go, corresponding to its velocity: $K.E. = WH$. What we need is the formula which tells us the height by some rule that has to do with the motion of objects. If we start something out with a certain velocity, say straight up, it will reach a certain height; we do not know what it is yet, but it depends on the velocity—there is a formula for that. Then to find the formula for kinetic energy

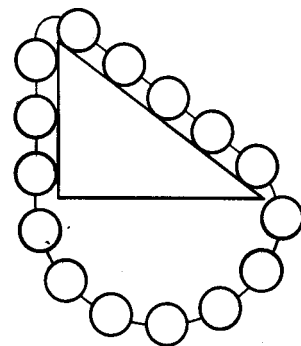


Fig. 4-4. The epitaph of Stevinus.

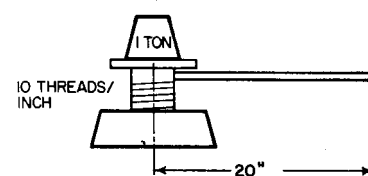


Fig. 4-5. A screw jack.

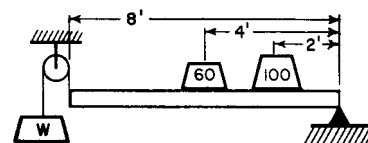


Fig. 4-6. Weighted rod supported on one end.

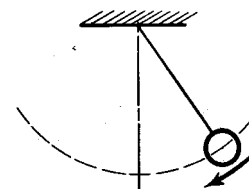


Fig. 4-7. Pendulum.

for an object moving with velocity V , we must calculate the height that it could reach, and multiply by the weight. We shall soon find that we can write it this way:

$$\text{K.E.} = WV^2/2g. \quad (4.6)$$

Of course, the fact that motion has energy has nothing to do with the fact that we are in a gravitational field. It makes no difference *where* the motion came from. This is a general formula for various velocities. Both (4.3) and (4.6) are approximate formulas, the first because it is incorrect when the heights are great, i.e., when the heights are so high that gravity is weakening; the second, because of the relativistic correction at high speeds. However, when we do finally get the exact formula for the energy, then the law of conservation of energy is correct.

4-4 Other forms of energy

We can continue in this way to illustrate the existence of energy in other forms. First, consider elastic energy. If we pull down on a spring, we must do some work, for when we have it down, we can lift weights with it. Therefore in its stretched condition it has a possibility of doing some work. If we were to evaluate the sums of weights times heights, it would not check out—we must add something else to account for the fact that the spring is under tension. Elastic energy is the formula for a spring when it is stretched. How much energy is it? If we let go, the elastic energy, as the spring passes through the equilibrium point, is converted to kinetic energy and it goes back and forth between compressing or stretching the spring and kinetic energy of motion. (There is also some gravitational energy going in and out, but we can do this experiment "sideways" if we like.) It keeps going until the losses—Aha! We have cheated all the way through by putting on little weights to move things or saying that the machines are reversible, or that they go on forever, but we can see that things do stop, eventually. Where is the energy when the spring has finished moving up and down? This brings in *another* form of energy: *heat energy*.

Inside a spring or a lever there are crystals which are made up of lots of atoms, and with great care and delicacy in the arrangement of the parts one can try to adjust things so that as something rolls on something else, none of the atoms do any jiggling at all. But one must be very careful. Ordinarily when things roll, there is bumping and jiggling because of the irregularities of the material, and the atoms start to wiggle inside. So we lose track of that energy; we find the atoms are wiggling inside in a random and confused manner after the motion slows down. There is still kinetic energy, all right, but it is not associated with visible motion. What a dream! How do we *know* there is still kinetic energy? It turns out that with thermometers you can find out that, in fact, the spring or the lever is *warmer*, and that there is really an increase of kinetic energy by a definite amount. We call this form of energy *heat energy*, but we know that it is not really a new form, it is just kinetic energy—internal motion. (One of the difficulties with all these experiments with matter that we do on a large scale is that we cannot really demonstrate the conservation of energy and we cannot really make our reversible machines, because every time we move a large clump of stuff, the atoms do not remain absolutely undisturbed, and so a certain amount of random motion goes into the atomic system. We cannot see it, but we can measure it with thermometers, etc.)

There are many other forms of energy, and of course we cannot describe them in any more detail just now. There is electrical energy, which has to do with pushing and pulling by electric charges. There is radiant energy, the energy of light, which we know is a form of electrical energy because light can be represented as wiggles in the electromagnetic field. There is chemical energy, the energy which is released in chemical reactions. Actually, elastic energy is, to a certain extent, like chemical energy, because chemical energy is the energy of the attraction of the atoms, one for the other, and so is elastic energy. Our modern understanding is the following: chemical energy has two parts, kinetic energy of the electrons inside the atoms, so part of it is kinetic, and electrical energy of interaction of the

electrons and the protons—the rest of it, therefore, is electrical. Next we come to nuclear energy, the energy which is involved with the arrangement of particles inside the nucleus, and we have formulas for that, but we do not have the fundamental laws. We know that it is not electrical, not gravitational, and not purely chemical, but we do not know what it is. It seems to be an additional form of energy. Finally, associated with the relativity theory, there is a modification of the laws of kinetic energy, or whatever you wish to call it, so that kinetic energy is combined with another thing called *mass energy*. An object has energy from its sheer *existence*. If I have a positron and an electron, standing still doing nothing—never mind gravity, never mind anything—and they come together and disappear, radiant energy will be liberated, in a definite amount, and the amount can be calculated. All we need know is the mass of the object. It does not depend on what it is—we make two things disappear, and we get a certain amount of energy. The formula was first found by Einstein; it is $E = mc^2$.

It is obvious from our discussion that the law of conservation of energy is enormously useful in making analyses, as we have illustrated in a few examples without knowing all the formulas. If we had all the formulas for all kinds of energy, we could analyze how many processes should work without having to go into the details. Therefore conservation laws are very interesting. The question naturally arises as to what other conservation laws there are in physics. There are two other conservation laws which are analogous to the conservation of energy. One is called the conservation of linear momentum. The other is called the conservation of angular momentum. We will find out more about these later. In the last analysis, we do not understand the conservation laws deeply. We do not understand the conservation of energy. We do not understand energy as a certain number of little blobs. You may have heard that photons come out in blobs and that the energy of a photon is Planck's constant times the frequency. That is true, but since the frequency of light can be anything, there is no law that says that energy has to be a certain definite amount. Unlike Dennis' blocks, there can be any amount of energy, at least as presently understood. So we do not understand this energy as counting something at the moment, but just as a mathematical quantity, which is an abstract and rather peculiar circumstance. In quantum mechanics it turns out that the conservation of energy is very closely related to another important property of the world, *things do not depend on the absolute time*. We can set up an experiment at a given moment and try it out, and then do the same experiment at a later moment, and it will behave in exactly the same way. Whether this is strictly true or not, we do not know. If we assume that it is true, and add the principles of quantum mechanics, then we can deduce the principle of the conservation of energy. It is a rather subtle and interesting thing, and it is not easy to explain. The other conservation laws are also linked together. The conservation of momentum is associated in quantum mechanics with the proposition that it makes no difference *where* you do the experiment, the results will always be the same. As independence in space has to do with the conservation of momentum, independence of time has to do with the conservation of energy, and finally, if we *turn* our apparatus, this too makes no difference, and so the invariance of the world to angular orientation is related to the conservation of *angular momentum*. Besides these, there are three other conservation laws, that are exact so far as we can tell today, which *are* much simpler to understand because they are in the nature of counting blocks.

The first of the three is the *conservation of charge*, and that merely means that you count how many positive, minus how many negative electrical charges you have, and the number is never changed. You may get rid of a positive with a negative, but you do not create any net excess of positives over negatives. Two other laws are analogous to this one—one is called the *conservation of baryons*. There are a number of strange particles, a neutron and a proton are examples, which are called baryons. In any reaction whatever in nature, if we count how many baryons are coming into a process, the number of baryons* which come out

*Counting antibaryons as —1 baryon.

will be exactly the same. There is another law, the *conservation of leptons*. We can say that the group of particles called leptons are: electron, mu meson, and neutrino. There is an antielectron which is a positron, that is, a -1 lepton. Counting the total number of leptons in a reaction reveals that the number in and out never changes, at least so far as we know at present.

These are the six conservation laws, three of them subtle, involving space and time, and three of them simple, in the sense of counting something.

With regard to the conservation of energy, we should note that *available* energy is another matter—there is a lot of jiggling around in the atoms of the water of the sea, because the sea has a certain temperature, but it is impossible to get them herded into a definite motion without taking energy from somewhere else. That is, although we know for a fact that energy is conserved, the energy available for human utility is not conserved so easily. The laws which govern how much energy is available are called the *laws of thermodynamics* and involve a concept called entropy for irreversible thermodynamic processes.

Finally, we remark on the question of where we can get our supplies of energy today. Our supplies of energy are from the sun, rain, coal, uranium, and hydrogen. The sun makes the rain, and the coal also, so that all these are from the sun. Although energy is conserved, nature does not seem to be interested in it; she liberates a lot of energy from the sun, but only one part in two billion falls on the earth. Nature has conservation of energy, but does not really care; she spends a lot of it in all directions. We have already obtained energy from uranium; we can also get energy from hydrogen, but at present only in an explosive and dangerous condition. If it can be controlled in thermonuclear reactions, it turns out that the energy that can be obtained from 10 quarts of water per second is equal to all of the electrical power generated in the United States. With 150 gallons of running water a minute, you have enough fuel to supply all the energy which is used in the United States today! Therefore it is up to the physicist to figure out how to liberate us from the need for having energy. It can be done.

Time and Distance

5-1 Motion

In this chapter we shall consider some aspects of the concepts of *time* and *distance*. It has been emphasized earlier that physics, as do all the sciences, depends on *observation*. One might also say that the development of the physical sciences to their present form has depended to a large extent on the emphasis which has been placed on the making of *quantitative* observations. Only with quantitative observations can one arrive at quantitative relationships, which are the heart of physics.

Many people would like to place the beginnings of physics with the work done 350 years ago by Galileo, and to call him the first physicist. Until that time, the study of motion had been a philosophical one based on arguments that could be thought up in one's head. Most of the arguments had been presented by Aristotle and other Greek philosophers, and were taken as "proven." Galileo was skeptical, and did an experiment on motion which was essentially this: **He** allowed a ball to roll down an inclined trough and observed the motion. He did not, however, just look; he measured *how far* the ball went in *how long a time*.

The way to measure a distance was well known long before Galileo, but there were no accurate ways of measuring time, particularly short times. Although he later devised more satisfactory clocks (though not like the ones we know), Galileo's first experiments on motion were done by using his pulse to count off equal intervals of time. Let us do the same.

We may count off beats of a pulse as the ball rolls down the track: "one . . . two . . . three . . . four . . . five . . . six . . . seven . . . eight..." We ask a friend to make a small mark at the location of the ball at each count; we can then measure the *distance* the ball travelled from the point of release in one, or two, or three, etc., equal intervals of time. Galileo expressed the result of *his* observations in this way: if the location of the ball is marked at 1, 2, 3, 4, . . . units of time from the instant of its release, those marks are distant from the starting point in proportion to the numbers 1, 4, 9, 16, . . . Today we would say the distance is proportional to the square of the time:

$$D \propto t^2.$$

The study of *motion*, which is basic to all of physics, treats with the questions: where? and when?

5-2 Time

Let us consider first what we mean by *time*. What *is* time? It would be nice if we could find a good definition of time. Webster defines "a time" as "a period," and the latter as "a time," which doesn't seem to be very useful. Perhaps we should say: "Time is what happens when nothing else happens." Which also doesn't get us very far. Maybe it is just as well if we face the fact that time is one of the things we probably cannot define (in the dictionary sense), and just say that it is what we already know it to be: it is how long we wait!

What really matters anyway is not how we *define* time, but how we measure it. One way of measuring time is to utilize something which happens over and over again in a regular fashion—something which is *periodic*. For example, a day. A day seems to happen over and over again. But when you begin to think

5-1 Motion

5-2 Time

5-3 Short times

5-4 Long times

5-5 Units and standards of time

5-6 Large distances

5-7 Short distances

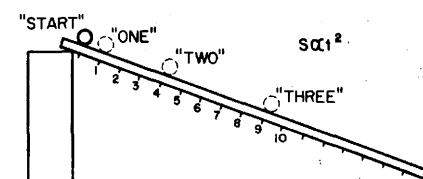


Fig. 5-1. A ball rolls down an inclined track.

about it, you might well ask: "Are days periodic; are they regular? Are all days the same length?" One certainly has the impression that days in summer are longer than days in winter. Of course, some of the days in winter seem to get awfully long if one is very bored. You have certainly heard someone say, "My, but this has been a long day!"

It does seem, however, that days are about the same length *on the average*. Is there any way we can test whether the days are the same length—either from one day to the next, or at least on the average? One way is to make a comparison with some other periodic phenomenon. Let us see how such a comparison might be made with an hour glass. With an hour glass, we can "create" a periodic occurrence if we have someone standing by it day and night to turn it over whenever the last grain of sand runs out.

We could then count the turnings of the glass from each morning to the next. We would find, this time, that the number of "hours" (i.e., turnings of the glass) was not the same each "day." We should distrust the sun, or the glass, or both. After some thought, it might occur to us to count the "hours" from noon to noon. (Noon is here defined *not* as 12:00 o'clock, but that instant when the sun is at its highest point.) We would find, this time, that the number of "hours" each day is the same.

We now have some confidence that both the "hour" and the "day" have a regular periodicity, i.e., mark off successive equal intervals of time, although we have not *proved* that either one is "really" periodic. Someone might question whether there might not be some omnipotent being who would slow down the flow of sand every night and speed it up during the day. Our experiment does not, of course, give us an answer to this sort of question. All we can say is that we find that a regularity of one kind fits together with a regularity of another kind. We can just say that we base our *definition* of time on the repetition of some apparently periodic event.

5-3 Short times

We should now notice that in the process of checking on the reproducibility of the day, we have received an important by-product. We have found a way of measuring, more accurately, *fractions* of a day. We have found a way of counting time in smaller pieces. Can we carry the process further, and learn to measure even smaller intervals of time?

Galileo decided that a given pendulum always swings back and forth in equal intervals of time so long as the size of the swing is kept small. A test comparing the number of swings of a pendulum in one "hour" shows that such is indeed the case. We can in this way mark fractions of an hour. If we use a mechanical device to count the swings—and to keep them going—we have the pendulum clock of our grandfathers.

Let us agree that if our pendulum oscillates 3600 times in one hour (and if there are 24 such hours in a day), we shall call each period of the pendulum one "second." We have then divided our original unit of time into approximately 10⁵ parts. We can apply the same principles to divide the second into smaller and smaller intervals. It is, you will realize, not practical to make mechanical pendulums which go arbitrarily fast, but we can now make *electrical* pendulums, called oscillators, which can provide a periodic occurrence with a very short period of swing. In these electronic oscillators it is an electrical current which swings to and fro, in a manner analogous to the swinging of the bob of the pendulum.

We can make a series of such electronic oscillators, each with a period 10 times shorter than the previous one. We may "calibrate" each oscillator against the next slower one by counting the number of swings it makes for one swing of the slower oscillator. When the period of oscillation of our clock is shorter than a fraction of a second, we cannot count the oscillations without the help of some device which extends our powers of observation. One such device is the electron-beam oscilloscope, which acts as a sort of microscope for short times. This device plots on a fluorescent screen a graph of electrical current (or voltage) versus time.

By connecting the oscilloscope to two of our oscillators in sequence, so that it plots a graph first of the current in one of our oscillators and then of the current in the other, we get two graphs like those shown in Fig. 5-2. We can readily determine the number of periods of the faster oscillator in one period of the slower oscillator.

With modern electronic techniques, oscillators have been built with periods as short as about 10^{-12} second, and they have been calibrated (by comparison methods such as we have described) in terms of our standard unit of time, the second. With the invention and perfection of the "laser," or light amplifier, in the past few years, it has become possible to make oscillators with even shorter periods than 10^{-12} second, but it has not yet been possible to calibrate them by the methods which have been described, although it will no doubt soon be possible.

Times shorter than 10^{-12} second have been measured, but by a different technique. In effect, a different *definition* of "time" has been used. One way has been to observe the *distance* between two happenings on a moving object. If, for example, the headlights of a moving automobile are turned on and then off, we can figure out *how long* the lights were on if we know *where* they were turned on and off and how fast the car was moving. The time is the distance over which the lights were on divided by the speed.

Within the past few years, just such a technique was used to measure the lifetime of the π^0 meson. By observing in a microscope the minute tracks left in a photographic emulsion in which π^0 -mesons had been created one saw that a π^0 -meson (known to be travelling at a certain speed nearly that of light) went a distance of about 10^{-7} meter, on the average, before disintegrating. It lived for only about 10^{-16} sec. It should be emphasized that we have here used a somewhat different definition of "time" than before. So long as there are no inconsistencies in our understanding, however, we feel fairly confident that our definitions are sufficiently equivalent.

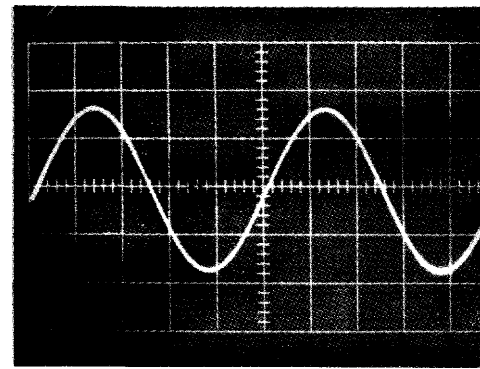
By extending our techniques—and if necessary our definitions—still further we can infer the time duration of still faster physical events. We can speak of the period of a nuclear vibration. We can speak of the lifetime of the newly discovered strange resonances (particles) mentioned in Chapter 2. Their complete life occupies a time span of only 10^{-24} second, approximately the time it would take light (which moves at the fastest known speed) to cross the nucleus of hydrogen (the smallest known object).

What about still smaller times? Does "time" exist on a still smaller scale? Does it make any sense to speak of smaller times if we cannot measure—or perhaps even think sensibly about—something which happens in a shorter time? Perhaps not. These are some of the open questions which you will be asking and perhaps answering in the next twenty or thirty years.

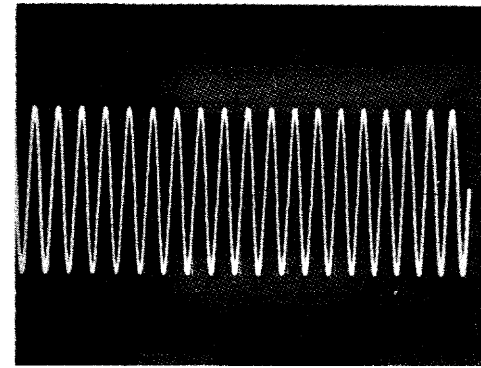
5-4 Long times

Let us now consider times longer than one day. Measurement of longer times is easy; we just count the days—so long as there is someone around to do the counting. First we find that there is another natural periodicity: the year, about 365 days. We have also discovered that nature has sometimes provided a counter for the years, in the form of tree rings or river-bottom sediments. In some cases we can use these natural time markers to determine the time which has passed since some early event.

When we cannot count the years for the measurement of long times, we must look for other ways to measure. One of the most successful is the use of radioactive material as a "clock." In this case we do not have a periodic occurrence, as for the day or the pendulum, but a new kind of "regularity." We find that the radioactivity of a particular sample of material decreases by the same *fraction* for successive equal increases in its age. If we plot a graph of the radioactivity observed as a function of time (say in days), we obtain a curve like that shown in Fig. 5-3. We observe that if the radioactivity decreases to one-half in T days (called the "half-life"), then it decreases to one-quarter in another T days, and so



(a)



(b)

Fig. 5-2. Two views of an oscilloscope screen. In (a) the oscilloscope is connected to one oscillator, in (b) it is connected to an oscillator with a period one-tenth as long.

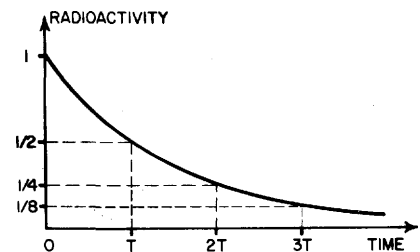


Fig. 5-3. The decrease with time of radioactivity. The activity decreases by one-half in each "half-life," T .

		TIMES	MEAN LIFE OF
YEARS	SECONDS		
		?? ? ? ? ? ? ?	
	10^{18}	Age of universe	
10^9		Age of earth	U^{238}
	10^{15}		
10^6		Earliest men	
	10^{12}	Age of pyramids	
10^3			Ra^{226}
		Age of U.S.	
	10^9	Life of a man	H^3
1			
	10^6		
		One day	
	10^3	Light goes from sun to earth	Neutron
	1	One heart beat	
	10^{-3}	Period of sound wave	
	10^{-6}	Period of radiowave	Muon $\pi^\pm$ -meson
	10^{-9}	Light travels one foot	
	10^{-12}	Period of molecular rotation	
	10^{-15}	Period of atomic vibration	π^0 -meson
	10^{-18}		
		Light crosses an atom	
	10^{-21}		
		Period of nuclear vibration	
	10^{-24}	Light crosses a nucleus	Strange particle
		?? ? ? ? ? ? ?	

on. In an arbitrary time interval t there are t/T "half-lives," and the fraction left after this time t is $(\frac{1}{2})^{t/T}$.

If we knew that a piece of material, say a piece of wood, had contained an amount A of radioactive material when it was formed, and we found out by a direct measurement that it now contains the amount B , we could compute the age of the object, t , by solving the equation

$$\left(\frac{1}{2}\right)^{t/T} = B/A.$$

There are, fortunately, cases in which we can know the amount of radioactivity that was in an object when it was formed. We know, for example, that the carbon dioxide in the air contains a certain small fraction of the radioactive carbon isotope C^{14} (replenished continuously by the action of cosmic rays). If we measure the *total* carbon content of an object, we know that a certain fraction of that amount was originally the radioactive C^{14} ; we know, therefore, the starting amount A to use in the formula above. Carbon-14 has a half-life of 5000 years. By careful measurements we can measure the amount left after 20 half-lives or so and can therefore "date" organic objects which grew as long as 100,000 years ago.

We would like to know, and we think we do know, the life of still older things. Much of our knowledge is based on the measurements of other radioactive isotopes which have different half-lives. If we make measurements with an isotope with a longer half-life, then we are able to measure longer times. Uranium, for example, has an isotope whose half-life is about 10^9 years, so that if some material was formed with uranium in it 10^9 years ago, only half the uranium would remain today. When the uranium disintegrates, it changes into lead. Consider a piece of rock which was formed a long time ago in some chemical process. Lead, being of a chemical nature different from uranium, would appear in one part of the rock and uranium would appear in another part of the rock. The uranium and lead

would be separate. If we look at that piece of rock today, where there should only be uranium we will now find a certain fraction of uranium and a certain fraction of lead. By comparing these fractions, we can tell what percent of the uranium disappeared and changed into lead. By this method, the age of certain rocks has been determined to be several billion years. An extension of this method, not using particular rocks but looking at the uranium and lead in the oceans and using averages over the earth, has been used to determine (within the past few years) that the age of the earth itself is approximately 5.5 billion years.

It is encouraging that the age of the earth is found to be the same as the age of the meteorites which land on the earth, as determined by the uranium method. It appears that the earth was formed out of rocks floating in space, and that the meteorites are, quite likely, some of that material left over. At some time more than five billion years ago, the universe started. It is now believed that at least our part of the universe had its beginning about ten or twelve billion years ago. We do not know what happened before then. In fact, we may well ask again: Does the question make any sense? Does an earlier time have any meaning?

5-5 Units and standards of time

We have implied that it is convenient if we start with some standard unit of time, say a day or a second, and refer all other times to some multiple or fraction of this unit. What shall we take as our basic standard of time? Shall we take the human pulse? If we compare pulses, we find that they seem to vary a lot. On comparing two clocks, one finds they do not vary so much. You might then say, well, let us take a clock. But whose clock? There is a story of a Swiss boy who wanted all of the clocks in his town to ring noon at the same time. So he went around trying to convince everyone of the value of this. Everyone thought it was a marvelous idea so long as all of the other clocks rang noon when his did! It is rather difficult to decide whose clock we should take as a standard. Fortunately, we all share one clock—the earth. For a long time the rotational period of the earth has been taken as the basic standard of time. As measurements have been made more and more precise, however, it has been found that the rotation of the earth is not exactly periodic, when measured in terms of the best clocks. These "best" clocks are those which we have reason to believe are accurate because they agree with each other. We now believe that, for various reasons, some days are longer than others, some days are shorter, and on the average the period of the earth becomes a little longer as the centuries pass.

Until very recently we had found nothing much better than the earth's period, so all clocks have been related to the length of the day, and the second has been defined as $1/86400$ of an average day. Recently we have been gaining experience with some natural oscillators which we now believe would provide a more constant time reference than the earth, and which are also based on a natural phenomenon available to everyone. These are the so-called "atomic clocks." Their basic internal period is that of an atomic vibration which is very insensitive to the temperature or any other external effects. These clocks keep time to an accuracy of one part in 10^9 or better. Within the past two years an improved atomic clock which operates on the vibration of the hydrogen atom has been designed and built by Professor Norman Ramsey at Harvard University. He believes that this clock might be 100 times more accurate still. Measurements now in progress will show whether this is true or not.

We may expect that since it has been possible to build clocks much more accurate than astronomical time, there will soon be an agreement among scientists to define the unit of time in terms of one of the atomic clock standards.

5-6 Large distances

Let us now turn to the question of *distance*. How far, or how big, are things? Everybody knows that the way you measure distance is to start with a stick and count. Or start with a thumb and count. You begin with a unit and count. How

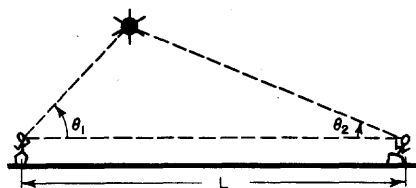


Fig. 5-4. The height of a Sputnik is determined by triangulation

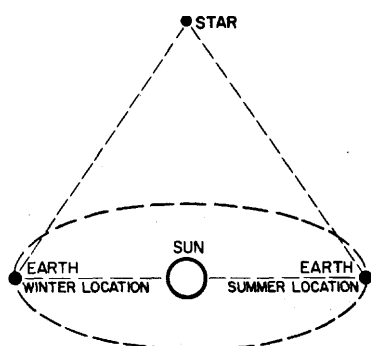


Fig. 5-5. The distance of nearby stars can be measured by triangulation, using the diameter of the earth's orbit as a baseline.

does one measure smaller things? How does one subdivide distance? In the same way that we subdivided time: we take a smaller unit and count the number of such units it takes to make up the longer unit. So we can measure smaller and smaller lengths.

But we do not always mean by distance what one gets by counting off with a meter stick. It would be difficult to measure the horizontal distance between two mountain tops using only a meter stick. We have found by experience that distance can be measured in another fashion: by triangulation. Although this means that we are really using a different definition of distance, when they can both be used they agree with each other. Space is more or less what Euclid thought it was, so the two types of definitions of distance agree. Since they do agree on the earth it gives us some confidence in using triangulation for still larger distances. For example, we were able to use triangulation to measure the height of the first Sputnik. We found that it was roughly 5×10^5 meters high. By more careful measurements the distance to the moon can be measured in the same way. Two telescopes at different places on the earth can give us the two angles we need. It has been found in this way that the moon is 4×10^8 meters away.

We cannot do the same with the sun, or at least no one has been able to yet. The accuracy with which one can focus on a given point on the sun and with which one can measure angles is not good enough to permit us to measure the distance to the sun. Then how can we measure the distance to the sun? We must invent an extension of the idea of triangulation. We measure the relative distances of all the planets by astronomical observations of where the planets appear to be, and we get a picture of the solar system with the proper *relative* distances of everything, but with no *absolute* distance. One absolute measurement is then required, which has been obtained in a number of ways. One of the ways, which was believed until recently to be the most accurate, was to measure the distance from the earth to Eros, one of the small planetoids which passes near the earth every now and then. By triangulation on this little object, one could get the one required scale measurement. Knowing the relative distances of the rest, we can then tell the distance, for example, from the earth to the sun, or from the earth to Pluto.

Within the past year there has been a big improvement in our knowledge of the scale of the solar system. At the Jet Propulsion Laboratory the distance from the earth to Venus was measured quite accurately by a direct radar observation. This, of course, is a still different type of inferred distance. We say we know the speed at which light travels (and therefore, at which radar waves travel), and we assume that it is the same speed everywhere between the earth and Venus. We send the radio wave out, and count the time until the reflected wave comes back. From the *time* we infer a *distance*, assuming we know the speed. We have really another definition of a measurement of distance.

How do we measure the distance to a star, which is much farther away? Fortunately, we can go back to our triangulation method, because the earth moving around the sun gives us a large baseline for measurements of objects outside the solar system. If we focus a telescope on a star in summer and in winter, we might hope to determine these two angles accurately enough to be able to measure the distance to a star.

What if the stars are too far away for us to use triangulation? Astronomers are always inventing new ways of measuring distance. They find, for example, that they can estimate the size and brightness of a star by its color. The color and brightness of many nearby stars—whose distances are known by triangulation—have been measured, and it is found that there is a smooth relationship between the color and the intrinsic brightness of stars (in most cases). If one now measures the color of a distant star, one may use the color-brightness relationship to determine the intrinsic brightness of the star. By measuring how bright the star *appears* to us at the earth (or perhaps we should say how *dim* it appears), we can compute how far away it is. (For a given intrinsic brightness, the apparent brightness decreases with the square of the distance.) A nice confirmation of the correctness of this method of measuring stellar distances is given by the results obtained for groups of stars known as globular clusters. A photograph of such a group is

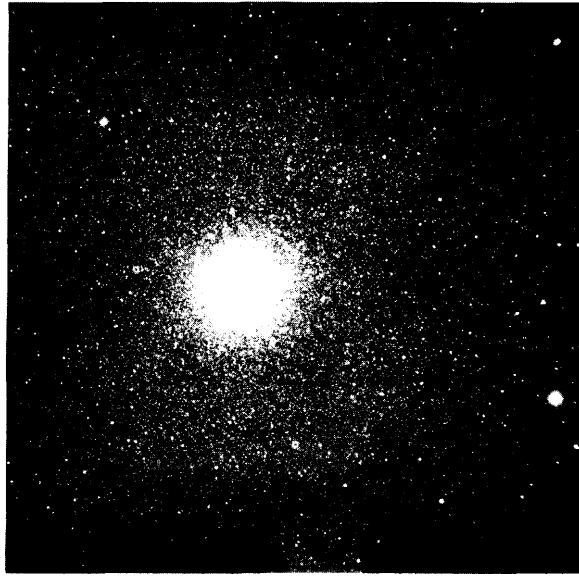


Fig. 5-6. A cluster of stars near the center of our galaxy. Their distance from the earth is 30,000 light-years, or about 3×10^{20} meters.

shown in Fig. 5-6. Just from looking at the photograph one is convinced that these stars are all together. The same result is obtained from distance measurements by the color-brightness method.

A study of many globular clusters gives another important bit of information. It is found that there is a high concentration of such clusters in a certain part of the sky and that most of them are about the same distance from us. Coupling this information with other evidence, we conclude that this concentration of clusters marks the center of our galaxy. We then know the distance to the center of the galaxy—about 10^{20} meters.

Knowing the size of our own galaxy, we have a key to the measurement of still larger distances—the distances to other galaxies. Figure 5-7 is a photograph of a galaxy, which has much the same shape as our own. Probably it is the same size, too. (Other evidence supports the idea that galaxies are all about the same size.) If it is the same size as ours, we can tell its distance. We measure the angle it subtends in the sky; we know its diameter, and we compute its distance—triangulation again!



Fig. 5-7. A spiral galaxy like our own. Presuming that its diameter is similar to that of our own galaxy, we may compute its distance from its apparent size. It is 30 million light-years (3×10^{23} meters) from the earth.

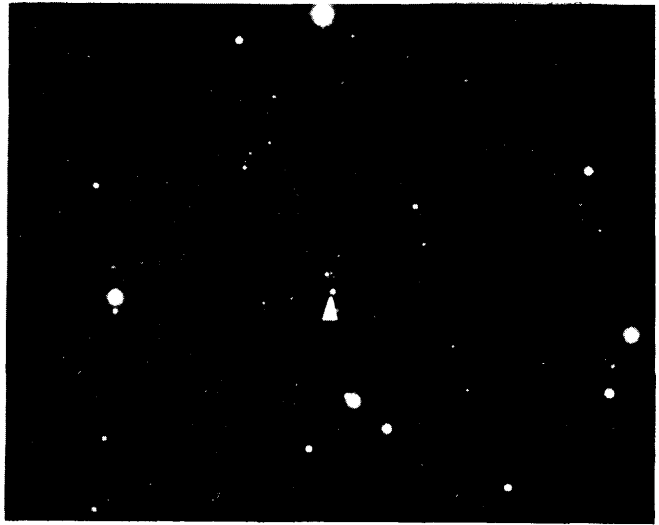


Fig. 5-8. The most distant object, 3C295 in BOOTES (indicated by the arrow), measured by the 200-inch telescope to date (1960).

Photographs of exceedingly distant galaxies have recently been obtained with the giant Palomar telescope. One is shown in Fig. 5-8. It is now believed that some of these galaxies are about halfway to the limit of the universe— 10^{26} meters away—the largest distance we can contemplate!

5-7 Short distances

Now let's think about smaller distances. Subdividing the meter is easy. Without much difficulty we can mark off one thousand equal spaces which add up to one meter. With somewhat more difficulty, but in a similar way (using a good microscope), we can mark off a thousand equal subdivisions of the millimeter to make a scale of microns (millionths of a meter). It is difficult to continue to smaller scales, because we cannot "see" objects smaller than the wavelength of visible light (about 5×10^{-7} meter).

We need not stop, however, at what we can see. With an electron microscope, we can continue the process by making photographs on a still smaller scale, say down to 10^{-8} meter (Fig. 5-9). By indirect measurements—by a kind of triangulation on a microscopic scale—we can continue to measure to smaller and smaller scales. First, from an observation of the way light of short wavelength (x-radiation) is reflected from a pattern of marks of known separation, we determine the wave-

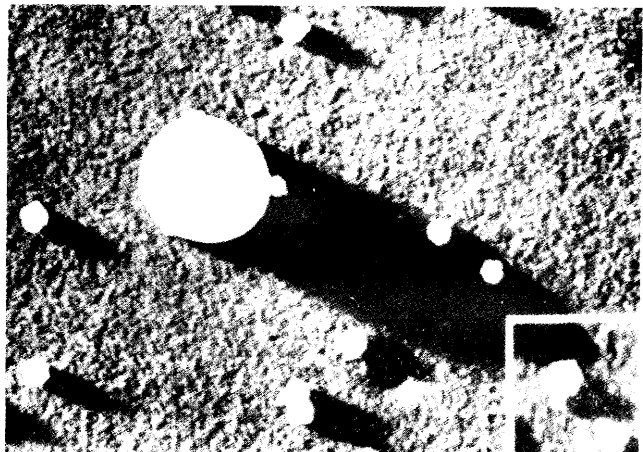


Fig. 5-9. Electron micrograph of some virus molecules. The "large" sphere is for calibration and is known to have a diameter of 2×10^{-7} meter (2000 Å).

DISTANCES		
LIGHT-YEARS	METERS	
	10^{27}	?? ? ? ? ? ? ?
10^9		Edge of universe
10^6	10^{24}	
	10^{21}	To nearest neighbor galaxy
10^3		To center of our galaxy
	10^{18}	
1		To nearest star
	10^{15}	
	10^{12}	Radius of orbit of Pluto
	10^9	To the sun
	10^6	To the moon
	10^3	Height of a Sputnik
	1	Height of a TV antenna tower
	10^{-3}	Height of a child
		A grain of salt
	10^{-6}	
	10^{-9}	A virus
	10^{-12}	Radius of an atom
	10^{-15}	Radius of a nucleus
		?? ? ? ? ? ? ?

length of the light vibrations. Then, from the pattern of the scattering of the same light from a crystal, we can determine the relative location of the atoms in the crystal, obtaining results which agree with the atomic spacings also determined by chemical means. We find in this way that atoms have a diameter of about 10^{-10} meter.

There is a large "gap" in physical sizes between the typical atomic dimension of about 10^{-10} meter and the nuclear dimensions 10^{-15} meter, 10^{-5} times smaller. For nuclear sizes, a different way of measuring size becomes convenient. We measure the *apparent area*, σ , called the effective *cross section*. If we wish the radius, we can obtain it from $\sigma = \pi r^2$, since nuclei are nearly spherical.

Measurement of a nuclear cross section can be made by passing a beam of high-energy particles through a thin slab of material and observing the number of particles which do not get through. These high-energy particles will plow right through the thin cloud of electrons and will be stopped or deflected only if they hit the concentrated weight of a nucleus. Suppose we have a piece of material 1 centimeter thick. There will be about 10^8 atomic layers. But the nuclei are so small that there is little chance that any nucleus will lie behind another. We might *imagine* that a highly magnified view of the situation—looking along the particle beam—would look like Fig. 5-10.

The chance that a very small particle will hit a nucleus on the trip through is just the total area covered by the profiles of the nuclei divided by the total area in the picture. Suppose that we know that in an area A of our slab of material there are N atoms (each with one nucleus, of course). Then the total area "covered" by the nuclei is $N\sigma/A$. Now let the number of particles of our beam which arrive at the slab be n_1 and the number which come out the other side be n_2 . The fraction which do *not* get through is $(n_1 - n_2)/n_1$, which should just equal the

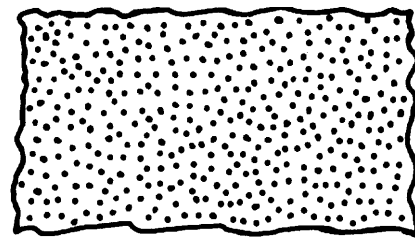


Fig. 5-10. Imagined view through a block of carbon 1 cm thick if only the nuclei were observed.

fraction of the area covered. We can obtain the radius of the nucleus from the equation*

$$\pi r^2 = \sigma = \frac{A}{N} \frac{n_1 - n_2}{n_1}.$$

From such an experiment we find that the radii of the nuclei are from about 1 to 6 times 10^{-15} meter. The length unit 10^{-15} meter is called *the fermi*, in honor of Enrico Fermi (1901-1958).

What do we find if we go to smaller distances? Can we measure smaller distances? Such questions are not yet answerable. It has been suggested that the still unsolved mystery of nuclear forces may be unravelled only by some modification of our idea of space, or measurement, at such small distances.

It might be thought that it would be a good idea to use some natural length as our unit of length—say the radius of the earth or some fraction of it. The meter was originally intended to be such a unit and was defined to be $(\pi/2) \times 10^{-7}$ times the earth's radius. It is neither convenient nor very accurate to determine the unit of length in this way. For a long time it has been agreed internationally that the meter would be defined as the distance between two scratches on a bar kept in a special laboratory in France. More recently, it has been realized that this definition is neither as precise as would be useful, nor as permanent or universal as one would like. It is currently being considered that a new definition be adopted, an agreed-upon (arbitrary) number of wavelengths of a chosen spectral line.

Measurements of distance and of time give results which depend on the observer. Two observers moving with respect to each other will not measure the same distances and times when measuring what appear to be the same things. Distances and time intervals have different magnitudes, depending on the coordinate system (or "frame of reference") used for making the measurements. We shall study this subject in more detail in a later chapter.

Perfectly precise measurements of distances or times are not permitted by the laws of nature. We have mentioned earlier that the errors in a measurement of the position of an object must be at least as large as

$$\Delta x = h/\Delta p,$$

where h is a small quantity called "Planck's constant" and Δp is the error in our knowledge of the momentum (mass times velocity) of the object whose position we are measuring. It was also mentioned that the uncertainty in position measurements is related to the wave nature of particles.

The relativity of space and time implies that time measurements have also a minimum error, given in fact by

$$\Delta t = h/\Delta E,$$

where ΔE is the error in our knowledge of the energy of the process whose time period we are measuring. If we wish to know *more* precisely *when* something happened we must know less about *what* happened, because our knowledge of the energy involved will be less. The time uncertainty is also related to the wave nature of matter.

* This equation is right only if the area covered by the nuclei is a small fraction of the total, i.e., if $(n_1 - n_2)/n_1$ is much less than 1. Otherwise we must make a correction for the fact that some nuclei will be partly obscured by the nuclei in front of them.

Probability

"The true logic of this world is in the calculus of probabilities."

—James Clerk Maxwell

6-1 Chance and likelihood

"Chance" is a word which is in common use in everyday living. The radio reports speaking of tomorrow's weather may say: "There is a sixty percent chance of rain." You might say: "There is a small chance that I shall live to be one hundred years old." Scientists also use the word chance. A seismologist may be interested in the question: "What is the chance that there will be an earthquake of a certain size in Southern California next year?" A physicist might ask the question: "What is the chance that a particular geiger counter will register twenty counts in the next ten seconds?" A politician or statesman might be interested in the question: "What is the chance that there will be a nuclear war within the next ten years?" You may be interested in the chance that you will learn something from this chapter.

By *chance*, we mean something like a guess. Why do we make guesses? We make guesses when we wish to make a judgment but have incomplete information or uncertain knowledge. We want to make a guess as to what things are, or what things are likely to happen. Often we wish to make a guess because we have to make a decision. For example: Shall I take my raincoat with me tomorrow? For what earth movement should I design a new building? Shall I build myself a fallout shelter? Shall I change my stand in international negotiations? Shall I go to class today?

Sometimes we make guesses because we wish, with our limited knowledge, to say as much as we *can* about some situation. Really, any generalization is in the nature of a guess. Any physical theory is a kind of guesswork. There are good guesses and there are bad guesses. The theory of probability is a system for making better guesses. The language of probability allows us to speak quantitatively about some situation which may be highly variable, but which does have some consistent average behavior.

Let us consider the flipping of a coin. If the toss—and the coin—are "honest," we have no way of knowing what to expect for the outcome of any particular toss. Yet we would feel that in a large number of tosses there should be about equal numbers of heads and tails. We say: "The probability that a toss will land heads is 0.5."

We speak of probability only for observations that we contemplate being made in the future. By the "*probability*" of a particular outcome of an observation we mean our estimate for the most likely fraction of a number of repeated observations that will yield that particular outcome. If we imagine repeating an observation—such as looking at a freshly tossed coin— N times, and if we call N_A our estimate of the most likely number of our observations that will give some specified result A , say the result "heads," then by $P(A)$, the probability of observing A , we mean

$$P(A) = N_A/N. \quad (6.1)$$

Our definition requires several comments. First of all, we may speak of a probability of something happening only if the occurrence is a possible outcome of some *repeatable* observation. It is not clear that it would make any sense to ask: "What is the probability that there is a ghost in that house?"

6-1 Chance and likelihood

6-2 Fluctuations

6-3 The random walk

6-4 A probability distribution

6-5 The uncertainty principle

You may object that no situation is *exactly* repeatable. That is right. Every different observation must at least be at a different time or place. All we can say is that the “repeated” observations should, for our intended purposes, *appear to be equivalent*. We should assume, at least, that each observation was made from an equivalently prepared situation, and especially with the same degree of ignorance at the start. (If we sneak a look at an opponent’s hand in a card game, our estimate of our chances of winning are different than if we do not!)

We should emphasize that N and N_A in Eq. (6.1) are *not* intended to represent numbers based on actual observations. N_A is our best *estimate* of what *would* occur in N *imagined* observations. Probability depends, therefore, on our knowledge and on our ability to make estimates. In effect, on our common sense! Fortunately, there is a certain amount of agreement in the common sense of many things, so that different people will make the same estimate. Probabilities need not, however, be “absolute” numbers. Since they depend on our ignorance, they may become different if our knowledge changes.

You may have noticed another rather “subjective” aspect of our definition of probability. We have referred to N_A as “our estimate of the most likely number . . .” We do not mean that we expect to observe *exactly* N_A , but that we expect a number *near* N_A , and that the number N_A is *more likely* than any other number in the vicinity. If we toss a coin, say, 30 times, we should expect that the number of heads would not be very likely to be exactly 15, but rather only some number near to 15, say 12, 13, 14, 15, 16, or 17. However, if we *must* choose, we would decide that 15 heads is *more likely* than any other number. We would write $P(\text{heads}) = 0.5$.

Why did we choose 15 as more likely than any other number? We must have argued with ourselves in the following manner: If the most likely number of heads is N_H in a total number of tosses N , then the most likely number of tails N_T is $(N - N_H)$. (We are assuming that every toss gives *either* heads *or* tails, and no “other” result!) But if the coin is “honest,” there is no preference for heads or tails. Until we have some reason to think the coin (or toss) is dishonest, we must give equal likelihoods for heads and tails. So we must set $N_T = N_H$. It follows that $N_T = N_H = N/2$, or $P(H) = p(T) = 0.5$.

We can generalize our reasoning to *any* situation in which there are m different but “equivalent” (that is, equally likely) possible results of an observation. If an observation can yield m different results, and we have reason to believe that any one of them is as likely as any other, then the probability of a *particular* outcome A is $P(A) = 1/m$.

If there are seven different-colored balls in an opaque box and we pick one out “at random” (that is, without looking), the probability of getting a ball of a particular color is $\frac{1}{7}$. The probability that a “blind draw” from a shuffled deck of 52 cards will show the ten of hearts is $\frac{1}{52}$. The probability of throwing a double-one with dice is $\frac{1}{36}$.

In Chapter 5 we described the size of a nucleus in terms of its apparent area, or “cross section.” When we did so we were really talking about probabilities. When we shoot a high-energy particle at a thin slab of material, there is some chance that it will pass right through and some chance that it will hit a nucleus. (Since the nucleus is so small that we cannot *see* it, we cannot aim right at a nucleus. We must “shoot blind.”) If there are n atoms in our slab and the nucleus of each atom has a cross-sectional area σ , then the total area “shadowed” by the nuclei is $n\sigma$. In a large number N of random shots, we expect that the number of hits N_C of *some* nucleus will be in the ratio to N as the shadowed area is to the total area of the slab:

$$N_C/N = n\sigma/A. \quad (6.2)$$

We may say, therefore, that the *probability* that any one projectile particle will suffer a collision in passing through the slab is

$$P_C = \frac{n}{A} \sigma, \quad (6.3)$$

where n/A is the number of atoms per unit area in our slab.

6-2 Fluctuations

We would like now to use our ideas about probability to consider in some greater detail the question: "How many heads do I really *expect* to get if I toss a coin N times?" Before answering the question, however, let us look at what does happen in such an "experiment." Figure 6-1 shows the results obtained in the first three "runs" of such an experiment in which $N = 30$. The sequences of "heads" and "tails" are shown just as they were obtained. The first game gave 11 heads; the second also 11; the third 16. In three trials we did not once get 15 heads. Should we begin to suspect the coin? Or were we wrong in thinking that the most likely number of "heads" in such a game is 15? Ninety-seven more runs were made to obtain a total of 100 experiments of 30 tosses each. The results of the experiment are given in Table 6-1.*

```

H   x x x x       x x x x x x x x   11
T   x x x x x x x x x x x x x x   19
H   x   x   x x x x x x x x x   11
T   x x x x x x x x x x x x x x   19
H   x x x x x x x x x x x x x x   16
T   x x x x x x x x x x x x x x   14

```

Fig. 6-1. Observed sequences of heads and tails in three games of 30 tosses each.

Table 6-1

Number of heads in successive trials of 30 tosses of a coin.

11	16	17	15	17	16	19	18	15	13
11	17	17	12	20	23	11	16	17	14
16	12	15	10	18	17	13	15	14	15
16	12	11	22	12	20	12	15	16	12
16	10	15	13	14	16	15	16	13	18
14	14	13	16	15	19	21	14	12	15
16	11	16	14	17	14	11	16	17	16
19	15	14	12	18	15	14	21	11	16
17	17	12	13	14	17	9	13	16	13

90 trials

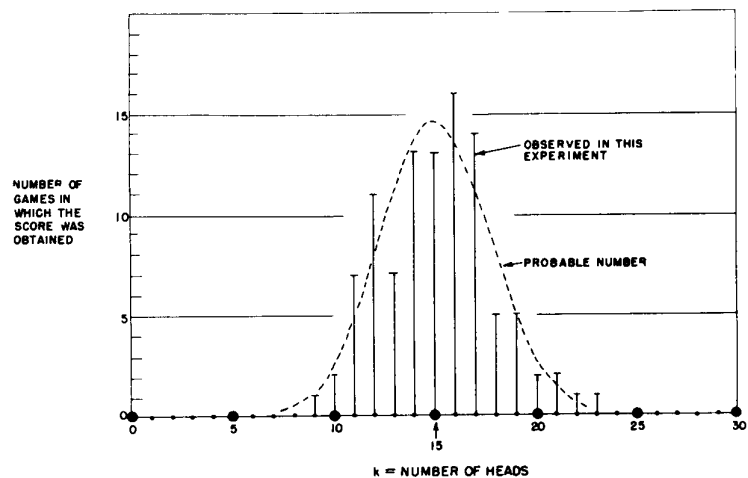


Fig. 6-2. Summary of the results of 100 games of 30 tosses each. The vertical bars show the number of games in which a score of k heads was obtained. The dashed curve shows the expected numbers of games with the score k obtained by a probability computation.

Looking at the numbers in Table 6-1, we see that most of the results are "near" 15, in that they are between 12 and 18. We can get a better feeling for the details of these results if we plot a graph of the *distribution* of the results. We count the number of games in which a score of k was obtained, and plot this number for each k . Such a graph is shown in Fig. 6-2. A score of 15 heads was obtained in 13 games. A score of 14 heads was also obtained 13 times. Scores of 16 and 17 were each obtained *more* than 13 times. Are we to conclude that there is some bias toward heads? Was our "best estimate" not good enough? Should

* After the first three games, the experiment was actually done by shaking 30 pennies violently in a box and then counting the number of heads that showed.

We may ask the question: "What is the probability that a game of 30 tosses will yield 15 heads—or 16, or any other number?" We have said that in a game of one toss, the probability of obtaining *one* head is 0.5, and the probability of obtaining no head is 0.5. In a game of two tosses there are *four* possible outcomes: *HH, HT, TH, TT*. Since each of these sequences is equally likely, we conclude that (a) the probability of a score of two heads is $\frac{1}{4}$, (b) the probability of a score of one head is $\frac{2}{4}$, (c) the probability of a zero score is $\frac{1}{4}$. There are *two* ways of obtaining one head, but only one of obtaining either zero or two heads.

The argument we have been making can be summarized by a diagram like that in Fig. 6-3. It is clear how the diagram should be continued for games with a larger number of tosses. Figure 6-4 shows such a diagram for a game of 6 tosses. The number of "ways" to any point on the diagram is just the number of different "paths" (sequences of heads and tails) which can be taken from the starting point. The vertical position gives us the total number of heads thrown. The set of numbers which appears in such a diagram is known as *Pascal's triangle*. The numbers are also known as the *binomial coefficients*, because they also appear in the expansion of $(a + b)^n$. If we call n the number of tosses and k the number of heads thrown, then the numbers in the diagram are usually designated by the symbol $\binom{n}{k}$. We may remark in passing that the binomial coefficients can also be computed from

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}, \quad (6.4)$$

where $n!$, called “ n -factorial,” represents the product $(n)(n - 1)(n - 2) \dots (3)(2)(1)$.

We are now ready to compute the probability $P(k, n)$ of throwing k heads in n tosses, using our definition Eq. (6.1). The total number of possible sequences is 2^n (since there are 2 outcomes for each toss), and the number of ways of obtaining k heads is $\binom{n}{k}$, all equally likely, so we have

$$P(k, n) = \frac{\binom{n}{k}}{2^n}. \quad (6.5)$$

Since $P(k, n)$ is the fraction of games which we expect to yield k heads, then in 100 games we should expect to find k heads $100 \cdot P(k, n)$ times. The dashed curve in Fig. 6-2 passes through the points computed from $100 \cdot P(k, 30)$. We see that we *expect* to obtain a score of 15 heads in 14 or 15 games, whereas this score was observed in 13 games. We *expect* a score of 16 in 13 or 14 games, but we obtained that score in 16 games. Such fluctuations are "part of the game."

The method we have just used can be applied to the most general situation in which there are only two possible outcomes of a single observation. Let us designate the two outcomes by W (for "win") and L (for "lose"). In the general case, the probability of W or L in a single event need not be equal. Let p be the probability of obtaining the result W . Then q , the probability of L , is necessarily

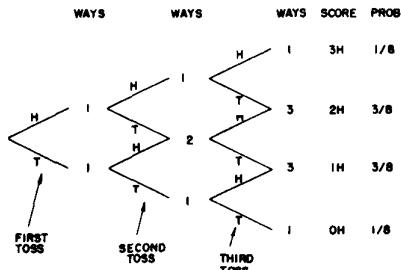


Fig. 6-3. A diagram for showing the number of ways a score of 0, 1, 2, or 3 heads can be obtained in a game of 3 tosses.

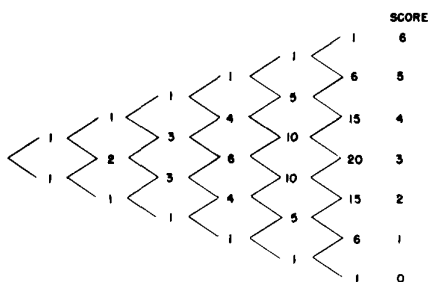


Fig. 6-4. A diagram like that of Fig. 6-3, for a game of 6 tosses.

$(1 - p)$. In a set of n trials, the probability $P(k, n)$ that W will be obtained k times is

$$P(k, n) = \binom{n}{k} p^k q^{n-k}. \quad (6.6)$$

This probability function is called the *Bernoulli* or, also, the *binomial* probability.

6-3 The random walk

There is another interesting problem in which the idea of probability is required. It is the problem of the “random walk.” In its simplest version, we imagine a “game” in which a “player” starts at the point $x = 0$ and at each “move” is required to take a step *either* forward (toward $+x$) *or* backward (toward $-x$). The choice is to be made *randomly*, determined, for example, by the toss of a coin. How shall we describe the resulting motion? In its general form the problem is related to the motion of atoms (or other particles) in a gas—called Brownian motion—and also to the combination of errors in measurements. You will see that the random-walk problem is closely related to the coin-tossing problem we have already discussed.

First, let us look at a few examples of a random walk. We may characterize the walker’s progress by the net distance D_N traveled in N steps. We show in the graph of Fig. 6-5 three examples of the path of a random walker. (We have used for the random sequence of choices the results of the coin tosses shown in Fig. 6-1.)

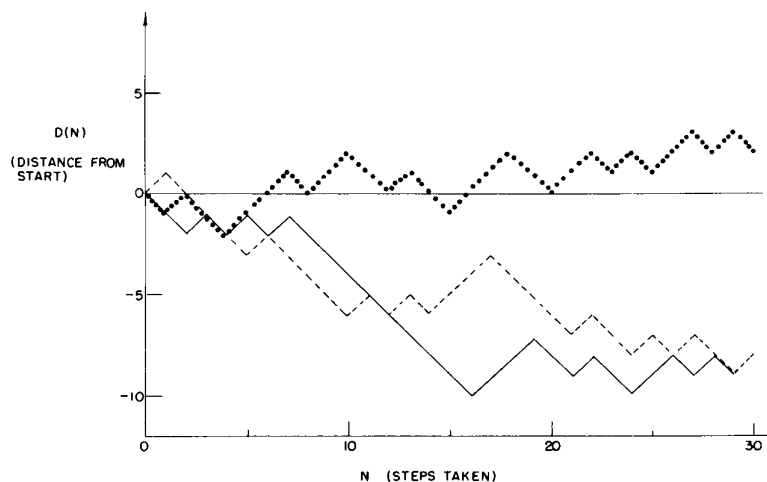


Fig. 6-5. The progress made in a random walk. The horizontal coordinate N is the total number of steps taken; the vertical coordinate $D(N)$ is the net distance moved from the starting position.

What can we say about such a motion? We might first ask: “How far does he get on the average?” We must *expect* that his average progress will be zero, since he is equally likely to go either forward or backward. But we have the feeling that as N increases, he is more likely to have strayed farther from the starting point. We might, therefore, ask what is his average distance travelled in *absolute value*, that is, what is the average of $|D|$. It is, however, more convenient to deal with another measure of “progress,” the square of the distance: D^2 is positive for either positive or negative motion, and is therefore a reasonable *measure* of such random wandering.

We can show that the expected value of D_N^2 is just N , the number of steps taken. By “expected value” we mean the probable value (our best guess), which we can think of as the *expected* average behavior in *many repeated* sequences. We represent such an expected value by $\langle D_N^2 \rangle$, and may refer to it also as the “mean square distance.” After one step, D^2 is always $+1$, so we have certainly $\langle D_1^2 \rangle = 1$. (All distances will be measured in terms of a unit of one step. We shall not continue to write the units of distance.)

The expected value of D_N^2 for $N > 1$ can be obtained from D_{N-1} . If, after $(N - 1)$ steps, we have D_{N-1} , then after N steps we have $D_N = D_{N-1} + 1$ or $D_N = D_{N-1} - 1$. For the squares,

$$D_N^2 = \begin{cases} D_{N-1}^2 + 2D_{N-1} + 1, \\ \text{or} \\ D_{N-1}^2 - 2D_{N-1} + 1. \end{cases} \quad (6.7)$$

In a number of independent sequences, we expect to obtain each value one-half of the time, so our average expectation is just the average of the two possible values. The expected value of D_N^2 is then $D_{N-1}^2 + 1$. In general, we should expect for D_{N-1}^2 its “expected value” $\langle D_{N-1}^2 \rangle$ (by definition!). So

$$\langle D_N^2 \rangle = \langle D_{N-1}^2 \rangle + 1. \quad (6.8)$$

We have already shown that $\langle D_1^2 \rangle = 1$; it follows then that

$$D_N^2 = N, \quad (6.9)$$

a particularly simple result!

If we wish a number like a distance, rather than a distance squared, to represent the “progress made away from the origin” in a random walk, we can use the “root-mean-square distance” D_{rms} :

$$D_{\text{rms}} = \sqrt{\langle D^2 \rangle} = \sqrt{N}. \quad (6.10)$$

We have pointed out that the random walk is closely similar in its mathematics to the coin-tossing game we considered at the beginning of the chapter. If we imagine the direction of each step to be in correspondence with the appearance of heads or tails in a coin toss, then D is just $N_H - N_T$, the difference in the number of heads and tails. Since $N_H + N_T = N$, the total number of steps (and tosses), we have $D = 2N_H - N$. We have derived earlier an expression for the expected distribution of N_H (also called k) and obtained the result of Eq. (6.5). Since N is just a constant, we have the corresponding distribution for D . (Since for every head more than $N/2$ there is a tail “missing,” we have the factor of 2 between N_H and D .) The graphs of Fig. 6-2 represent the distribution of distances we might get in 30 random steps (where $k = 15$ is to be read $D = 0$; $k = 16$, $D = 2$; etc.).

The variation of N_H from its expected value $N/2$ is

$$N_H - \frac{N}{2} = \frac{D}{2}. \quad (6.11)$$

The rms deviation is

$$\left(N_H - \frac{N}{2} \right)_{\text{rms}} = \frac{1}{2} \sqrt{N}. \quad (6.12)$$

According to our result for D_{rms} , we expect that the “typical” distance in 30 steps ought to be $\sqrt{30} = 5.5$, or a typical k should be about $5.5/2 = 2.8$ units from 15. We see that the “width” of the curve in Fig. 6-2, measured from the center, is just about 3 units, in agreement with this result.

We are now in a position to consider a question we have avoided until now. How shall we tell whether a coin is “honest” or “loaded”? We can give now at least a partial answer. For an honest coin, we expect the fraction of the times heads appears to be 0.5, that is,

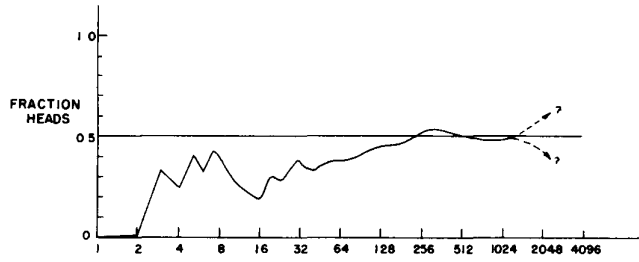
$$\frac{\langle N_H \rangle}{N} = 0.5. \quad (6.13)$$

We also expect an actual N_H to deviate from $N/2$ by about $\sqrt{N}/2$, or the fraction to deviate by

$$\frac{1}{N} \frac{\sqrt{N}}{2} = \frac{1}{2\sqrt{N}}.$$

The larger N is, the closer we expect the fraction N_H/N to be to one-half.

Fig. 6-6. The fraction of the tosses that gave heads in a particular sequence of N tosses of a penny.



In Fig. 6-6 we have plotted the fraction N_H/N for the coin tosses reported earlier in this chapter. We see the tendency for the fraction of heads to approach 0.5 for large N . Unfortunately, for any given run or combination of runs there is no *guarantee* that the observed deviation will be even *near* the *expected* deviation. There is always the finite chance that a large fluctuation—a long string of heads or tails—will give an arbitrarily large deviation. All we can say is that *if* the deviation is near the expected $1/2\sqrt{N}$ (say within a factor of 2 or 3), we have no reason to suspect the honesty of the coin. If it is much larger, we may be suspicious, but cannot prove, that the coin is loaded (or that the tosser is clever!).

We have also not considered how we should treat the case of a “coin” or some similar “chancy” object (say a stone that always lands in either of two positions) that we have good reason to believe *should* have a different probability for heads and tails. We have defined $P(H) = \langle N_H \rangle / N$. How shall we know what to *expect* for N_H ? In some cases, the best we can do is to observe the number of heads obtained in large numbers of tosses. For want of anything better, we must set $\langle N_H \rangle = N_H(\text{observed})$. (How could we expect anything else?) We must understand, however, that in such a case a different experiment, or a different observer, might conclude that $P(H)$ was different. We would *expect*, however, that the various answers should agree within the deviation $1/2\sqrt{N}$ [if $P(H)$ is near one-half]. An experimental physicist usually says that an “experimentally determined” probability has an “error,” and writes

$$P(H) = \frac{N_H}{N} \pm \frac{1}{2\sqrt{N}}. \quad (6.14)$$

There is an implication in such an expression that there *is* a “true” or “correct” probability which *could* be computed if we knew enough, and that the observation may be in “error” due to a fluctuation. There is, however, no way to make such thinking logically consistent. It is probably better to realize that the probability concept is in a sense subjective, that it is always based on uncertain knowledge, and that its quantitative evaluation is subject to change as we obtain more information.

6-4 A probability distribution

Let us return now to the random walk and consider a modification of it. Suppose that in addition to a random choice of the *direction* (+ or −) of each step, the *length* of each step also varied in some unpredictable way, the only condition being that *on the average* the step length was one unit. This case is more representative of something like the thermal motion of a molecule in a gas. If we call the length of a step S , then S may have any value at all, but most often will be “near” 1. To be specific, we shall let $\langle S^2 \rangle = 1$ or, equivalently, $S_{\text{rms}} = 1$. Our derivation for $\langle D^2 \rangle$ would proceed as before except that Eq. (6.8) would be changed now to read

$$\langle D_N^2 \rangle = \langle D_{N-1}^2 \rangle + \langle S^2 \rangle = \langle D_{N-1}^2 \rangle + 1. \quad (6.15)$$

We have, as before, that

$$\langle D_N^2 \rangle = N. \quad (6.16)$$

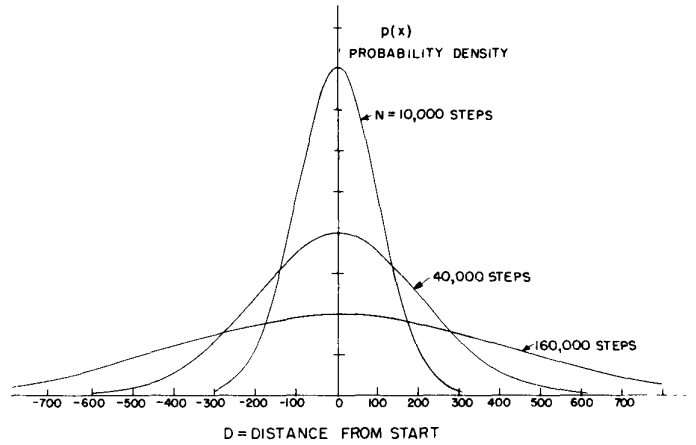


Fig. 6-7. The probability density for ending up at the distance D from the starting place in a random walk of N steps. (D is measured in units of the rms step length.)

What would we expect now for the distribution of distances D ? What is, for example, the probability that $D = 0$ after 30 steps? The answer is zero! The probability is zero that D will be *any particular* value, since there is no chance at all that the sum of the backward steps (of varying lengths) would exactly equal the sum of forward steps. We cannot plot a graph like that of Fig. 6-2.

We can, however, obtain a representation similar to that of Fig. 6-2, if we ask, not what is the probability of obtaining D exactly equal to 0, 1, or 2, but instead what is the probability of obtaining D *near* 0, 1, or 2. Let us define $P(x, \Delta x)$ as the probability that D will lie in the interval Δx located at x (say from x to $x + \Delta x$). We expect that for small Δx the chance of D landing in the interval is proportional to Δx , the width of the interval. So we can write

$$P(x, \Delta x) = p(x) \Delta x. \quad (6.17)$$

The function $p(x)$ is called the *probability density*.

The form of $p(x)$ will depend on N , the number of steps taken, and also on the distribution of individual step lengths. We cannot demonstrate the proofs here, but for large N , $p(x)$ is the *same* for all reasonable distributions in individual step lengths, and depends only on N . We plot $p(x)$ for three values of N in Fig. 6-7. You will notice that the “half-widths” (typical spread from $x = 0$) of these curves is $\sqrt{N}$, as we have shown it should be.

You may notice also that the value of $p(x)$ near zero is inversely proportional to $\sqrt{N}$. This comes about because the curves are all of a similar shape and their areas under the curves must all be equal. Since $p(x) \Delta x$ is the probability of finding D in Δx when Δx is small, we can determine the chance of finding D *somewhere* inside an arbitrary interval from x_1 to x_2 , by cutting the interval in a number of small increments Δx and evaluating the sum of the terms $p(x) \Delta x$ for each increment. The probability that D lands somewhere between x_1 and x_2 , which we may write $P(x_1 < D < x_2)$, is equal to the shaded area in Fig. 6-8. The smaller we take the increments Δx , the more correct is our result. We can write, therefore,

$$P(x_1 < D < x_2) = \sum p(x) \Delta x = \int_{x_1}^{x_2} p(x) dx. \quad (6.18)$$

The area under the whole curve is the probability that D lands somewhere (that is, has *some* value between $x = -\infty$ and $x = +\infty$). That probability is surely 1. We must have that

$$\int_{-\infty}^{+\infty} p(x) dx = 1. \quad (6.19)$$

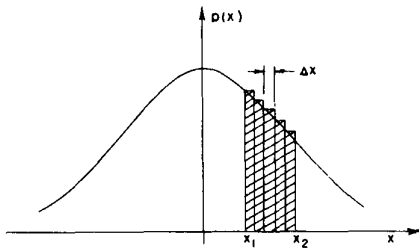


Fig. 6-8. The probability that the distance D traveled in a random walk is between x_1 and x_2 is the area under the curve of $p(x)$ from x_1 to x_2 .

Since the curves in Fig. 6-7 get wider in proportion to $\sqrt{N}$, their heights must be proportional to $1/\sqrt{N}$ to maintain the total area equal to 1.

The probability density function we have been describing is one that is encountered most commonly. It is known as the *normal* or *gaussian* probability density. It has the mathematical form

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-x^2/2\sigma^2}, \quad (6.20)$$

where σ is called the *standard deviation* and is given, in our case, by $\sigma = \sqrt{N}$ or, if the rms step size is different from 1, by $\sigma = \sqrt{N}S_{\text{rms}}$.

We remarked earlier that the motion of a molecule, or of any particle, in a gas is like a random walk. Suppose we open a bottle of an organic compound and let some of its vapor escape into the air. If there are air currents, so that the air is circulating, the currents will also carry the vapor with them. But even in *perfectly still air*, the vapor will gradually spread out—will diffuse—until it has penetrated throughout the room. We might detect it by its color or odor. The individual molecules of the organic vapor spread out in still air because of the molecular motions caused by collisions with other molecules. If we know the average “step” size, and the number of steps taken per second, we can find the probability that one, or several, molecules will be found at some distance from their starting point after any particular passage of time. As time passes, more steps are taken and the gas spreads out as in the successive curves of Fig. 6-7. In a later chapter, we shall find out how the step sizes and step frequencies are related to the temperature and pressure of a gas.

Earlier, we said that the pressure of a gas is due to the molecules bouncing against the walls of the container. When we come later to make a more quantitative description, we will wish to know how fast the molecules are going when they bounce, since the impact they make will depend on that speed. We cannot, however, speak of *the* speed of the molecules. It is necessary to use a probability description. A molecule may have any speed, but some speeds are more likely than others. We describe what is going on by saying that the probability that any particular molecule will have a speed between v and $v + \Delta v$ is $p(v) \Delta v$, where $p(v)$, a probability density, is a given function of the speed v . We shall see later how Maxwell, using common sense and the ideas of probability, was able to find a mathematical expression for $p(v)$. The form* of the function $p(v)$ is shown in Fig. 6-9. Velocities may have any value, but are most likely to be near the most probable or expected value $\langle v \rangle$.

We often think of the curve of Fig. 6-9 in a somewhat different way. If we consider the molecules in a typical container (with a volume of, say, one liter), then there are a very large number N of molecules present ($N \approx 10^{22}$). Since $p(v) \Delta v$ is the probability that *one* molecule will have its velocity in Δv , by our definition of probability we mean that the *expected* number $\langle \Delta N \rangle$ to be found with a velocity in the interval Δv is given by

$$\langle \Delta N \rangle = Np(v) \Delta v. \quad (6.21)$$

We call $Np(v)$ the “distribution in velocity.” The area under the curve between two velocities v_1 and v_2 , for example the shaded area in Fig. 6-9, represents [for the curve $Np(v)$] the expected number of molecules with velocities between v_1 and v_2 . Since with a gas we are usually dealing with large numbers of molecules, we expect the deviations from the expected numbers to be small (like $1/\sqrt{N}$), so we often neglect to say the “expected” number, and say instead: “The number of molecules with velocities between v_1 and v_2 is the area under the curve.” We should remember, however, that such statements are always about *probable* numbers.

* Maxwell's expression is $p(v) = Cv^2e^{-av^2}$, where a is a constant related to the temperature and C is chosen so that the total probability is one.

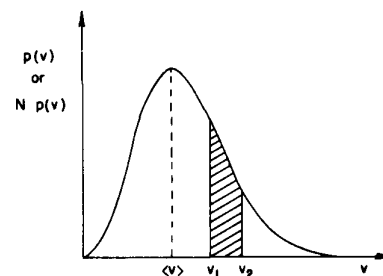


Fig. 6-9. The distribution of velocities of the molecules in a gas.

6-5 The uncertainty principle

The ideas of probability are certainly useful in describing the behavior of the 10^{22} or so molecules in a sample of a gas, for it is clearly impractical even to attempt to write down the position or velocity of each molecule. When probability was first applied to such problems, it was considered to be a *convenience*—a way of dealing with very complex situations. We now believe that the ideas of probability are *essential* to a description of atomic happenings. According to quantum mechanics, the mathematical theory of particles, there is always some uncertainty in the *specification* of positions and velocities. We can, at best, say that there is a certain probability that any particle will have a position near some coordinate x .

We can give a probability density $p_1(x)$, such that $p_1(x) \Delta x$ is the probability that the particle will be found between x and $x + \Delta x$. If the particle is reasonably well localized, say near x_1 , the function $p_1(x)$ might be given by the graph of Fig. 6-10(a). Similarly, we must specify the velocity of the particle by means of a probability density $p_2(v)$, with $p_2(v) \Delta v$ the probability that the velocity will be found between v and $v + \Delta v$.

It is one of the fundamental results of quantum mechanics that the two functions $p_1(x)$ and $p_2(v)$ cannot be chosen independently and, in particular, cannot both be made arbitrarily narrow. If we call the typical “width” of the $p_1(x)$ curve $[\Delta x]$, and that of the $p_2(v)$ curve $[\Delta v]$ (as shown in the figure), nature demands that the *product* of the two widths be at least as big as the number h/m , where m is the mass of the particle and h is a fundamental physical constant called *Planck’s constant*. We may write this basic relationship as

$$[\Delta x] \cdot [\Delta v] \geq h/m. \quad (6.22)$$

This equation is a statement of the *Heisenberg uncertainty principle* that we mentioned earlier.

Since the right-hand side of Eq. (6.22) is a constant, this equation says that if we try to “pin down” a particle by forcing it to be at a particular place, it ends up by having a high speed. Or if we try to force it to go very slowly, or at a precise velocity, it “spreads out” so that we do not know very well just where it is. Particles behave in a funny way!

The uncertainty principle describes an inherent fuzziness that must exist in any attempt to describe nature. Our most precise description of nature *must* be in terms of *probabilities*. There are some people who do not like this way of describing nature. They feel somehow that if they could only tell what is *really* going on with a particle, they could know its speed and position simultaneously. In the early days of the development of quantum mechanics, Einstein was quite worried about this problem. He used to shake his head and say, “But, surely God does not throw dice in determining how electrons should go!” He worried about that problem for a long time and he probably never really reconciled himself to the fact that this is the best description of nature that one can give. There are still one or two physicists who are working on the problem who have an intuitive conviction that it is possible somehow to describe the world in a different way and that all of this uncertainty about the way things are can be removed. No one has yet been successful.

The necessary uncertainty in our specification of the position of a particle becomes most important when we wish to describe the structure of atoms. In the hydrogen atom, which has a nucleus of one proton with one electron outside of the nucleus, the uncertainty in the position of the electron is as large as the atom itself! We cannot, therefore, properly speak of the electron moving in some “orbit” around the proton. The most we can say is that there is a certain *chance* $p(r) \Delta V$, of observing the electron in an element of volume ΔV at the distance r from the proton. The probability density $p(r)$ is given by quantum mechanics. For an undisturbed hydrogen atom $p(r) = Ae^{-r^2/a^2}$, which is a bell-shaped function like that in Fig. 6-8. The number a is the “typical” radius, where the function is decreasing rapidly. Since there is a small probability of finding the electron at distances

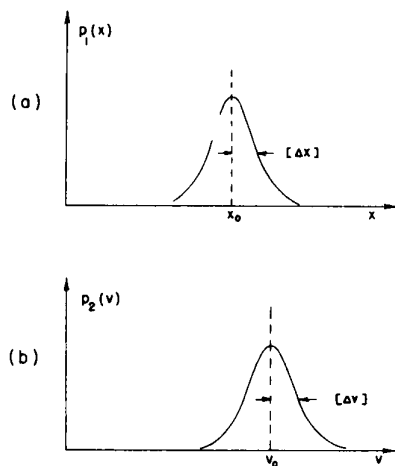


Fig. 6-10. Probability densities for observation of the position and velocity of a particle.

Fig. 6-11. A way of visualizing a hydrogen atom. The density (whiteness) of the cloud represents the probability density for observing the electron.



from the nucleus much greater than a , we may think of a as “the radius of the atom,” about 10^{-10} meter.

We can form an image of the hydrogen atom by imagining a “cloud” whose density is proportional to the probability density for observing the electron. A sample of such a cloud is shown in Fig. 6-11. Thus our best “picture” of a hydrogen atom is a nucleus surrounded by an “electron cloud” (although we *really* mean a “probability cloud”). The electron is there somewhere, but nature permits us to know only the *chance* of finding it at any particular place.

In its efforts to learn as much as possible about nature, modern physics has found that certain things can never be “known” with certainty. Much of our knowledge must always remain uncertain. The *most* we can know is in terms of probabilities.

The Theory of Gravitation

7-1 Planetary motions

In this chapter we shall discuss one of the most far-reaching generalizations of the human mind. While we are admiring the human mind, we should take some time off to stand in awe of a *nature* that could follow with such completeness and generality such an elegantly simple principle as the law of gravitation. What is this law of gravitation? It is that every object in the universe attracts every other object with a force which for any two bodies is proportional to the mass of each and varies inversely as the square of the distance between them. This statement can be expressed mathematically by the equation

$$F = G \frac{mm'}{r^2}.$$

If to this we add the fact that an object responds to a force by accelerating in the direction of the force by an amount that is inversely proportional to the mass of the object, we shall have said everything required, for a sufficiently talented mathematician could then deduce all the consequences of these two principles. However, since you are not assumed to be sufficiently talented yet, we shall discuss the consequences in more detail, and not just leave you with only these two bare principles. We shall briefly relate the story of the discovery of the law of gravitation and discuss some of its consequences, its effects on history, the mysteries that such a law entails, and some refinements of the law made by Einstein; we shall also discuss the relationships of the law to the other laws of physics. All this cannot be done in one chapter, but these subjects will be treated in due time in subsequent chapters.

The story begins with the ancients observing the motions of planets among the stars, and finally deducing that they went around the sun, a fact that was rediscovered later by Copernicus. Exactly *how* the planets went around the sun, with exactly *what motion*, took a little more work to discover. In the beginning of the fifteenth century there were great debates as to whether they really went around the sun or not. Tycho Brahe had an idea that was different from anything proposed by the ancients: his idea was that these debates about the nature of the motions of the planets would best be resolved if the actual positions of the planets in the sky were measured sufficiently accurately. If measurement showed exactly how the planets moved, then perhaps it would be possible to establish one or another viewpoint. This was a tremendous idea—that to find something out, it is better to perform some careful experiments than to carry on deep philosophical arguments. Pursuing this idea, Tycho Brahe studied the positions of the planets for many years in his observatory on the island of Hven, near Copenhagen. He made voluminous tables, which were then studied by the mathematician Kepler, after Tycho's death. Kepler discovered from the data some very beautiful and remarkable, but simple, laws regarding planetary motion.

7-2 Kepler's laws

First of all, Kepler found that each planet goes around the sun in a curve called an *ellipse*, with the sun at a focus of the ellipse. An ellipse is not just an oval, but is a very specific and precise curve that can be obtained by using two tacks, one at each focus, a loop of string, and a pencil; more mathematically, it

7-1 Planetary motions

7-2 Kepler's laws

7-3 Development of dynamics

7-4 Newton's law of gravitation

7-5 Universal gravitation

7-6 Cavendish's experiment

7-7 What is gravity?

7-8 Gravity and relativity

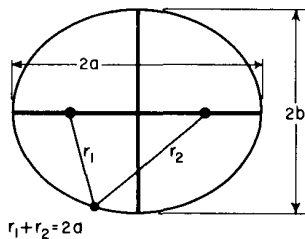


Fig. 7-1. An ellipse.

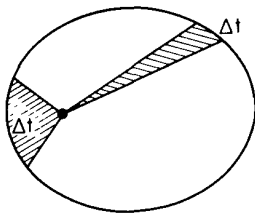


Fig. 7-2. Kepler's law of areas.

is the locus of all points the sum of whose distances from two fixed points (the foci) is a constant. Or, if you will, it is a foreshortened circle (Fig. 7-1).

Kepler's second observation was that the planets do not go around the sun at a uniform speed, but move faster when they are nearer the sun and more slowly when they are farther from the sun, in precisely this way: Suppose a planet is observed at any two successive times, let us say a week apart, and that the radius vector* is drawn to the planet for each observed position. The orbital arc traversed by the planet during the week, and the two radius vectors, bound a certain plane area, the shaded area shown in Fig. 7-2. If two similar observations are made a week apart, at a part of the orbit farther from the sun (where the planet moves more slowly), the similarly bounded area is exactly the same as in the first case. So, in accordance with the second law, the orbital speed of each planet is such that the radius "sweeps out" equal areas in equal times.

Finally, a third law was discovered by Kepler much later; this law is of a different category from the other two, because it deals not with only a single planet, but relates one planet to another. This law says that when the orbital period and orbit size of any two planets are compared, the periods are proportional to the $3/2$ power of the orbit size. In this statement the period is the time interval it takes a planet to go completely around its orbit, and the size is measured by the length of the greatest diameter of the elliptical orbit, technically known as the major axis. More simply, if the planets went in circles, as they nearly do, the time required to go around the circle would be proportional to the $3/2$ power of the diameter (or radius). Thus Kepler's three laws are:

- I. Each planet moves around the sun in an ellipse, with the sun at one focus.
- II. The radius vector from the sun to the planet sweeps out equal areas in equal intervals of time.
- III. The squares of the periods of any two planets are proportional to the cubes of the semimajor axes of their respective orbits: $T \sim a^{3/2}$.

7-3 Development of dynamics

While Kepler was discovering these laws, Galileo was studying the laws of motion. The problem was, what makes the planets go around? (In those days, one of the theories proposed was that the planets went around because behind them were invisible angels, beating their wings and driving the planets forward. You will see that this theory is now modified! It turns out that in order to keep the planets going around, the invisible angels must fly in a different direction and they have no wings. Otherwise, it is a somewhat similar theory!) Galileo discovered a very remarkable fact about motion, which was essential for understanding these laws. That is the principle of *inertia*—if something is moving, with nothing touching it and completely undisturbed, it will go on forever, coasting at a uniform speed in a straight line. (*Why* does it keep on coasting? We do not know, but that is the way it is.)

Newton modified this idea, saying that the only way to change the motion of a body is to use *force*. If the body speeds up, a force has been applied *in the direction of motion*. On the other hand, if its motion is changed to a new *direction*, a force has been applied *sideways*. Newton thus added the idea that a force is needed to change the speed *or the direction* of motion of a body. For example, if a stone is attached to a string and is whirling around in a circle, it takes a force to keep it in the circle. We have to *pull* on the string. In fact, the law is that the acceleration produced by the force is inversely proportional to the mass, or the force is proportional to the mass times the acceleration. The more massive a thing is, the stronger the force required to produce a given acceleration. (The mass can be measured by putting other stones on the end of the same string and making them go around the same circle at the same speed. In this way it is found that more or less force is required, the more massive object requiring more force.)

* A radius vector is a line drawn from the sun to any point in a planet's orbit.

The brilliant idea resulting from these considerations is that no *tangential* force is needed to keep a planet in its orbit (the angels do not have to fly tangentially) because the planet would coast in that direction anyway. If there were nothing at all to disturb it, the planet would go off in a *straight line*. But the actual motion deviates from the line on which the body would have gone if there were no force, the deviation being essentially *at right angles* to the motion, not in the direction of the motion. In other words, because of the principle of inertia, the force needed to control the motion of a planet around the sun is not a force *around* the sun but *toward* the sun. (If there is a force toward the sun, the sun might be the angel, of course!)

7-4 Newton's law of gravitation

From his better understanding of the theory of motion, Newton appreciated that *the sun* could be the seat or organization of forces that govern the motion of the planets. Newton proved to himself (and perhaps we shall be able to prove it soon) that the very fact that equal areas are swept out in equal times is a precise sign post of the proposition that all deviations are precisely *radial*—that the law of areas is a direct consequence of the idea that all of the forces are directed exactly *toward the sun*.

Next, by analyzing Kepler's third law it is possible to show that the farther away the planet, the weaker the forces. If two planets at different distances from the sun are compared, the analysis shows that the forces are inversely proportional to the squares of the respective distances. With the combination of the two laws, Newton concluded that there must be a force, inversely as the square of the distance, directed in a line between the two objects.

Being a man of considerable feeling for generalities, Newton supposed, of course, that this relationship applied more generally than just to the sun holding the planets. It was already known, for example, that the planet Jupiter had moons going around it as the moon of the earth goes around the earth, and Newton felt certain that each planet held its moons with a force. He already knew of the force holding *us* on the earth, so he proposed that this was a *universal* force—that *everything pulls everything else*.

The next problem was whether the pull of the earth on its people was the "same" as its pull on the moon, i.e., inversely as the square of the distance. If an object on the surface of the earth falls 16 feet in the first second after it is released from rest, how far does the moon fall in the same time? We might say that the moon does not fall at all. But if there were no force on the moon, it would go off in a straight line, whereas it goes in a circle instead, so it really *falls in* from where it would have been if there were no force at all. We can calculate from the radius of the moon's orbit (which is about 240,000 miles) and how long it takes to go around the earth (approximately 29 days), how far the moon moves in its orbit in 1 second, and can then calculate how far it falls in one second.* This distance turns out to be roughly 1/20 of an inch in a second. That fits very well with the inverse square law, because the earth's radius is 4000 miles, and if something which is 4000 miles from the center of the earth falls 16 feet in a second, something 240,000 miles, or 60 times as far away, should fall only 1/3600 of 16 feet, which also is roughly 1/20 of an inch. Wishing to put this theory of gravitation to a test by similar calculations, Newton made his calculations very carefully and found a discrepancy so large that he regarded the theory as contradicted by facts, and did not publish his results. Six years later a new measurement of the size of the earth showed that the astronomers had been using an incorrect distance to the moon. When Newton heard of this, he made the calculation again, with the corrected figures, and obtained beautiful agreement.

This idea that the moon "falls" is somewhat confusing, because, as you see, it does not come any *closer*. The idea is sufficiently interesting to merit further

* That is, how far the circle of the moon's orbit falls below the straight line tangent to it at the point where the moon was one second before.

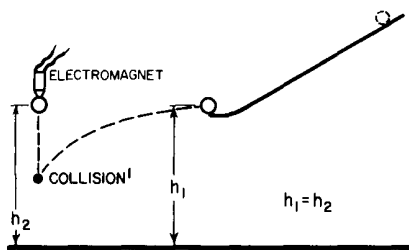


Fig. 7-3. Apparatus for showing the independence of vertical and horizontal motions.

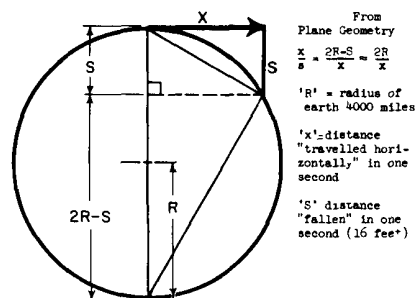


Fig. 7-4. Acceleration toward the center of a circular path. From plane geometry, $x/s = (2R - S)/x \approx 2R/x$, where R is the radius of the earth, 4000 miles; x is the distance "travelled horizontally" in one second; and S is the distance "fallen" in one second (16 feet).

explanation: the moon falls in the sense that *it falls away from the straight line that it would pursue if there were no forces*. Let us take an example on the surface of the earth. An object released near the earth's surface will fall 16 feet in the first second. An object shot out *horizontally* will also fall 16 feet; even though it is moving horizontally, it still falls the same 16 feet in the same time. Figure 7-3 shows an apparatus which demonstrates this. On the horizontal track is a ball which is going to be driven forward a little distance away. At the same height is a ball which is going to fall vertically, and there is an electrical switch arranged so that at the moment the first ball leaves the track, the second ball is released. That they come to the same depth at the same time is witnessed by the fact that they collide in midair. An object like a bullet, shot horizontally, might go a long way in one second—perhaps 2000 feet—but it will still fall 16 feet if it is aimed horizontally. What happens if we shoot a bullet faster and faster? Do not forget that the earth's surface is curved. If we shoot it fast enough, then when it falls 16 feet it may be at just the same height above the ground as it was before. How can that be? It still falls, but the earth curves away, so it falls "around" the earth. The question is, how far does it have to go in one second so that the earth is 16 feet below the horizon? In Fig. 7-4 we see the earth with its 4000-mile radius, and the tangential, straightline path that the bullet would take if there were no force. Now, if we use one of those wonderful theorems in geometry, which says that our tangent is the mean proportional between the two parts of the diameter cut by an equal chord, we see that the horizontal distance travelled is the mean proportional between the 16 feet fallen and the 8000-mile diameter of the earth. The square root of $(16/5280) \times 8000$ comes out very close to 5 miles. Thus we see that if the bullet moves at 5 miles a second, it then will continue to fall toward the earth at the same rate of 16 feet each second, but will never get any closer because the earth keeps curving away from it. Thus it was that Mr. Gagarin maintained himself in space while going 25,000 miles around the earth at approximately 5 miles per second. (He took a little longer because he was a little higher.)

Any great discovery of a new law is useful only if we can take more out than we put in. Now, Newton *used* the second and third of Kepler's laws to deduce his law of gravitation. What did he *predict*? First, his analysis of the moon's motion was a prediction because it connected the falling of objects on the earth's surface with that of the moon. Second, the question is, *is the orbit an ellipse?* We shall see in a later chapter how it is possible to calculate the motion exactly, and indeed one can prove that it should be an ellipse,* so no extra fact is needed to explain Kepler's *first* law. Thus Newton made his first powerful prediction.

The law of gravitation explains many phenomena not previously understood. For example, the pull of the moon on the earth causes the tides, hitherto mysterious. The moon pulls the water up under it and makes the tides—people had thought of that before, but they were not as clever as Newton, and so they thought there ought to be only one tide during the day. The reasoning was that the moon pulls the water up under it, making a high tide and a low tide, and since the earth spins underneath, that makes the tide at one station go up and down every 24 hours. Actually the tide goes up and down in 12 hours. Another school of thought claimed that the high tide should be on the other side of the earth because, so they argued, the moon pulls the earth away from the water! Both of these theories are wrong. It actually works like this: the pull of the moon for the earth and for the water is "balanced" at the center. But the water which is closer to the moon is pulled *more* than the average and the water which is farther away from it is pulled *less* than the average. Furthermore, the water can flow while the more rigid earth cannot. The true picture is a combination of these two things.

What do we mean by "balanced"? What balances? If the moon pulls the whole earth toward it, why doesn't the earth fall right "up" to the moon? Because the earth does the same trick as the moon, it goes in a circle around a point which is inside the earth but not at its center. The moon does not just go around the

* The proof is not given in this course.

earth, the earth and the moon both go around a central position, each falling toward this common position, as shown in Fig. 7-5. This motion around the common center is what balances the fall of each. So the earth is not going in a straight line either; it travels in a circle. The water on the far side is "unbalanced" because the moon's attraction there is weaker than it is at the center of the earth, where it just balances the "centrifugal force." The result of this imbalance is that the water rises up, away from the center of the earth. On the near side, the attraction from the moon is stronger, and the imbalance is in the opposite direction in space, but again *away* from the center of the earth. The net result is that we get *two* tidal bulges.

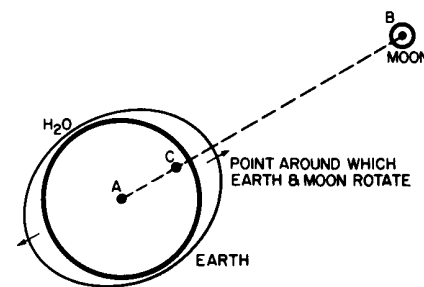


Fig. 7-5. The earth-moon system, with tides.

7-5 Universal gravitation

What else can we understand when we understand gravity? Everyone knows the earth is round. Why is the earth round? That is easy; it is due to gravitation. The earth can be understood to be round merely because everything attracts everything else and so it has attracted itself together as far as it can! If we go even further, the earth is not *exactly* a sphere because it is rotating, and this brings in centrifugal effects which tend to oppose gravity near the equator. It turns out that the earth should be elliptical, and we even get the right shape for the ellipse. We can thus deduce that the sun, the moon, and the earth should be (nearly) spheres, just from the law of gravitation.

What else can you do with the law of gravitation? If we look at the moons of Jupiter we can understand everything about the way they move around that planet. Incidentally, there was once a certain difficulty with the moons of Jupiter that is worth remarking on. These satellites were studied very carefully by Roemer, who noticed that the moons sometimes seemed to be ahead of schedule, and sometimes behind. (One can find their schedules by waiting a very long time and finding out how long it takes on the average for the moons to go around.) Now they were *ahead* when Jupiter was particularly *close* to the earth and they were *behind* when Jupiter was *farther* from the earth. This would have been a very difficult thing to explain according to the law of gravitation—it would have been, in fact, the death of this wonderful theory if there were no other explanation. If a law does not work even in *one place* where it ought to, it is just wrong. But the reason for this discrepancy was very simple and beautiful: it takes a little while to *see* the moons of Jupiter because of the time it takes light to travel from Jupiter to the earth. When Jupiter is closer to the earth the time is a little less, and when it is farther from the earth, the time is more. This is why moons appear to be, on the average, a little ahead or a little behind, depending on whether they are closer to or farther from the earth. This phenomenon showed that light does not travel instantaneously, and furnished the first estimate of the speed of light. This was done in 1656.

If all of the planets push and pull on each other, the force which controls, let us say, Jupiter in going around the sun is not just the force from the sun; there is also a pull from, say, Saturn. This force is not really strong, since the sun is much more massive than Saturn, but there is *some* pull, so the orbit of Jupiter should not be a perfect ellipse, and it is not; it is slightly off, and "wobbles" around the correct elliptical orbit. Such a motion is a little more complicated. Attempts were made to analyze the motions of Jupiter, Saturn, and Uranus on the basis of the law of gravitation. The effects of each of these planets on each other were calculated to see whether or not the tiny deviations and irregularities in these motions could be completely understood from this one law. Lo and behold, for Jupiter and Saturn, all was well, but Uranus was "weird." It behaved in a very peculiar manner. It was not travelling in an exact ellipse, but that was understandable, because of the attractions of Jupiter and Saturn. But even if allowance were made for these attractions, Uranus *still* was not going right, so the laws of gravitation were in danger of being overturned, a possibility that could not be ruled out. Two men, Adams and Leverrier, in England and France, independently,

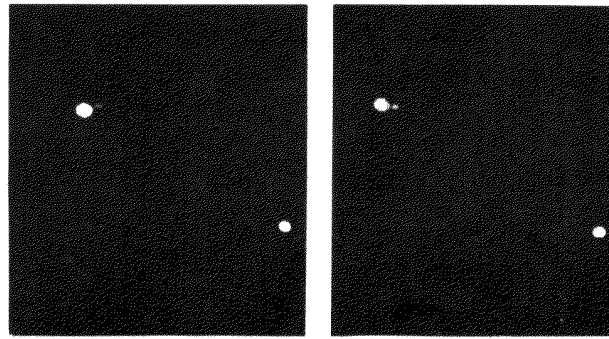


Fig. 7-6. A double-star system.

arrived at another possibility: perhaps there is *another* planet, dark and invisible, which men had not seen. This planet, *N*, could pull on Uranus. They calculated where such a planet would have to be in order to cause the observed perturbations. They sent messages to the respective observatories, saying, "Gentlemen, point your telescope to such and such a place, and you will see a new planet." It often depends on with whom you are working as to whether they pay any attention to you or not. They did pay attention to Leverrier; they looked, and there planet *N* was! The other observatory then also looked very quickly in the next few days and saw it too.

This discovery shows that Newton's laws are absolutely right in the solar system; but do they extend beyond the relatively small distances of the nearest planets? The first test lies in the question, do *stars* attract *each other* as well as planets? We have definite evidence that they do in the *double stars*. Figure 7-6 shows a double star—two stars very close together (there is also a third star in the picture so that we will know that the photograph was not turned). The stars are also shown as they appeared several years later. We see that, relative to the "fixed" star, the axis of the pair has rotated, i.e., the two stars are going around each other. Do they rotate according to Newton's laws? Careful measurements of the relative positions of one such double star system are shown in Fig. 7-7. There we see a beautiful ellipse, the measures starting in 1862 and going all the way around to 1904 (by now it must have gone around once more). Everything coincides with Newton's laws, except that the star Sirius A is *not at the focus*. Why should that be? Because the plane of the ellipse is not in the "plane of the sky." We are not looking at right angles to the orbit plane, and when an ellipse is viewed at a tilt, it remains an ellipse but the focus is no longer at the same place. Thus we can analyze double stars, moving about each other, according to the requirements of the gravitational law.

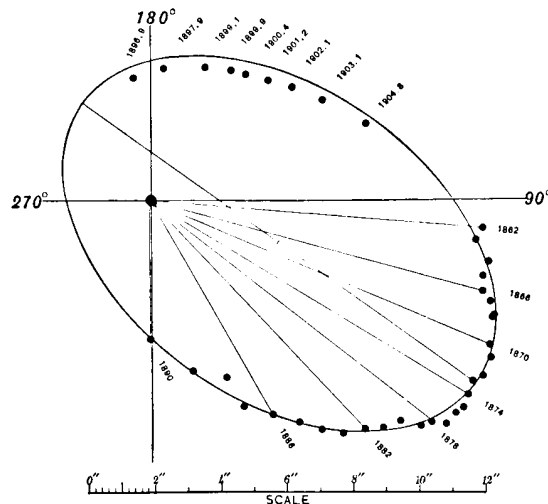


Fig. 7-7. Orbit of Sirius B with respect to Sirius A.

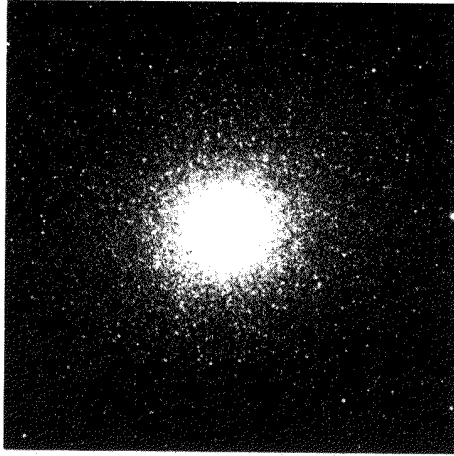


Fig. 7-8. A globular star cluster.

That the law of gravitation is true at even bigger distances is indicated in Fig. 7-8. If one cannot see gravitation acting here, he has no soul. This figure shows one of the most beautiful things in the sky—a globular star cluster. All of the dots are stars. Although they look as if they are packed solid toward the center, that is due to the fallibility of our instruments. Actually, the distances between even the centermost stars are very great and they very rarely collide. There are more stars in the interior than farther out, and as we move outward there are fewer and fewer. It is obvious that there is an attraction among these stars. It is clear that gravitation exists at these enormous dimensions, perhaps 100,000 times the size of the solar system. Let us now go further, and look at an *entire galaxy*, shown in Fig. 7-9. The shape of this galaxy indicates an obvious tendency for its matter to agglomerate. Of course we cannot prove that the law here is precisely inverse square, only that there is still an attraction, at this enormous dimension, that holds the whole thing together. One may say, “Well, that is all very clever but why is it not just a ball?” Because it is *spinning* and has *angular momentum* which it cannot give up as it contracts; it must contract mostly in a plane. (Incidentally, if you are looking for a good problem, the exact details of how the arms are formed and what determines the shapes of these galaxies has not been worked out.) It is, however, clear that the shape of the galaxy is due to gravitation even though the complexities of its structure have not yet allowed



Fig. 7-9. A galaxy.

us to analyze it completely. In a galaxy we have a scale of perhaps 50,000 to 100,000 light years. The earth's distance from the sun is $8\frac{1}{3}$ light *minutes*, so you can see how large these dimensions are.

Gravity appears to exist at even bigger dimensions, as indicated by Fig. 7-10, which shows many "little" things clustered together. This is a *cluster of galaxies*, just like a star cluster. Thus galaxies attract each other at such distances that they too are agglomerated into clusters. Perhaps gravitation exists even over distances of *tens of millions* of light years; so far as we now know, gravity seems to go out forever inversely as the square of the distance.

Not only can we understand the nebulae, but from the law of gravitation we can even get some ideas about the origin of the stars. If we have a big cloud of dust and gas, as indicated in Fig. 7-11, the gravitational attractions of the pieces of dust for one another might make them form little lumps. Barely visible in the figure are "little" black spots which may be the beginning of the accumulations of dust and gases which, due to their gravitation, begin to form stars. Whether we have ever seen a star form or not is still debatable. Figure 7-12 shows the one piece of evidence which suggests that we have. At the left is a picture of a region of gas with some stars in it taken in 1947, and at the right is another picture, taken only 7 years later, which shows two new bright spots. Has gas accumulated, has gravity acted hard enough and collected it into a ball big enough that the stellar nuclear reaction starts in the interior and turns it into a star? Perhaps, and perhaps not. It is unreasonable that in only seven years we should be so lucky as to see a star change itself into visible form; it is much less probable that we should see *two*!



Fig. 7-10. A cluster of galaxies.

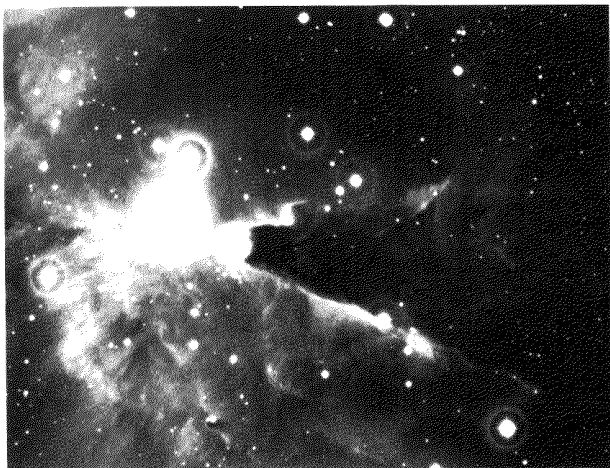


Fig. 7-11. An interstellar dust cloud.

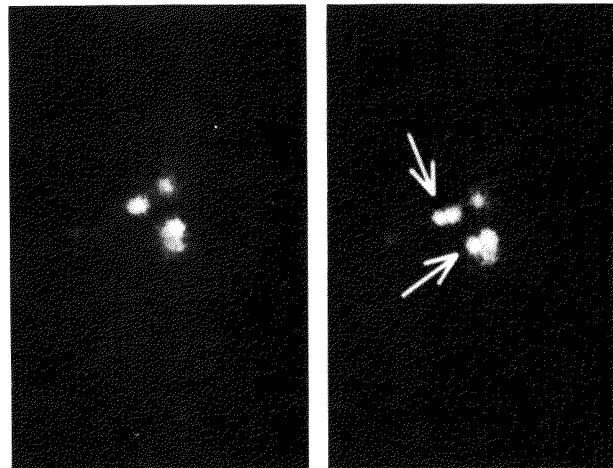


Fig. 7-12. The formation of new stars?

7-6 Cavendish's experiment

Gravitation, therefore, extends over enormous distances. But if there is a force between *any* pair of objects, we ought to be able to measure the force between our own objects. Instead of having to watch the stars go around each other, why can we not take a ball of lead and a marble and watch the marble go toward the ball of lead? The difficulty of this experiment when done in such a simple manner is the very weakness or delicacy of the force. It must be done with extreme care, which means covering the apparatus to keep the air out, making sure it is not electrically charged, and so on; then the force can be measured. It was first measured by Cavendish with an apparatus which is schematically indicated in Fig. 7-13. This first demonstrated the direct force between two large, fixed balls of lead and two smaller balls of lead on the ends of an arm supported by a very fine fiber, called a torsion fiber. By measuring how much the fiber gets twisted, one can measure the strength of the force, verify that it is inversely proportional to the square of the distance, and determine how strong it is. Thus, one may accurately determine the coefficient G in the formula

$$F = G \frac{mm'}{r^2}.$$

All the masses and distances are known. You say, "We knew it already for the earth." Yes, but we did not know the *mass* of the earth. By knowing G from this experiment and by knowing how strongly the earth attracts, we can indirectly learn how great is the mass of the earth! This experiment has been called "weighing the earth." Cavendish claimed he was weighing the earth, but what he was measuring was the coefficient G of the gravity law. This is the only way in which the mass of the earth can be determined. G turns out to be

$$6.670 \times 10^{-11} \text{ newton} \cdot \text{m}^2/\text{kg}^2.$$

It is hard to exaggerate the importance of the effect on the history of science produced by this great success of the theory of gravitation. Compare the confusion, the lack of confidence, the incomplete knowledge that prevailed in the earlier ages, when there were endless debates and paradoxes, with the clarity and simplicity of this law—this fact that all the moons and planets and stars have such a *simple rule* to govern them, and further that man could *understand* it and deduce how the planets should move! This is the reason for the success of the sciences in following years, for it gave hope that the other phenomena of the world might also have such beautifully simple laws.

7-7 What is gravity?

But is this such a simple law? What about the machinery of it? All we have done is to describe *how* the earth moves around the sun, but we have not said *what makes it go*. Newton made no hypotheses about this; he was satisfied to find *what* it did without getting into the machinery of it. *No one has since given any machinery*. It is characteristic of the physical laws that they have this abstract character. The law of conservation of energy is a theorem concerning quantities that have to be calculated and added together, with no mention of the machinery, and likewise the great laws of mechanics are quantitative mathematical laws for which no machinery is available. Why can we use mathematics to describe nature without a mechanism behind it? No one knows. We have to keep going because we find out more that way.

Many mechanisms for gravitation have been suggested. It is interesting to consider one of these, which many people have thought of from time to time. At first, one is quite excited and happy when he "discovers" it, but he soon finds that it is not correct. It was first discovered about 1750. Suppose there were many particles moving in space at a very high speed in all directions and being only slightly absorbed in going through matter. When they *are* absorbed, they give an impulse to the earth. However, since there are as many going one way as another, the

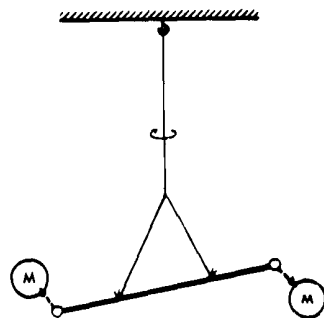


Fig. 7-13. A simplified diagram of the apparatus used by Cavendish to verify the law of universal gravitation for small objects and to measure the gravitational constant G .

turns out that if we consider the structure of the sun—the balance between the weight of its material and the rate at which radiant energy is generated inside it—we can deduce that if the gravity were 10 percent stronger, the sun would be much more than 10 percent brighter—by the *sixth power* of the gravity constant! If we calculate what happens to the orbit of the earth when the gravity is changing, we find that the earth was then *closer in*. Altogether, the earth would be about 100 degrees centigrade hotter, and all of the water would not have been in the sea, but vapor in the air, so life would not have started in the sea. So we do *not* now believe that the gravity constant is changing with the age of the universe. But such arguments as the one we have just given are not very convincing, and the subject is not completely closed.

It is a fact that the force of gravitation is proportional to the *mass*, the quantity which is fundamentally a measure of *inertia*—of how hard it is to hold something which is going around in a circle. Therefore two objects, one heavy and one light, going around a larger object in the same circle at the same speed because of gravity, will stay together because to go in a circle *requires* a force which is stronger for a bigger mass. That is, the gravity is stronger for a given mass in *just the right proportion* so that the two objects will go around together. If one object were inside the other it would *stay* inside; it is a perfect balance. Therefore, Gagarin or Titov would find things “weightless” inside a space ship; if they happened to let go of a piece of chalk, for example, it would go around the earth in exactly the same way as the whole space ship, and so it would appear to remain suspended before them in space. It is very interesting that this force is *exactly* proportional to the mass with great precision, because if it were not exactly proportional there would be some effect by which inertia and weight would differ. The absence of such an effect has been checked with great accuracy by an experiment done first by Eotvos in 1909 and more recently by Dicke. For all substances tried, the masses and weights are exactly proportional within 1 part in 1,000,000,000, or less. This is a remarkable experiment.

7-8 Gravity and relativity

Another topic deserving discussion is Einstein’s modification of Newton’s law of gravitation. In spite of all the excitement it created, Newton’s law of gravitation is not correct! It was modified by Einstein to take into account the theory of relativity. According to Newton, the gravitational effect is instantaneous, that is, if we were to move a mass, we would at once feel a new force because of the new position of that mass; by such means we could send signals at infinite speed. Einstein advanced arguments which suggest that we *cannot send signals faster than the speed of light*, so the law of gravitation must be wrong. By correcting it to take the delays into account, we have a new law, called Einstein’s law of gravitation. One feature of this new law which is quite easy to understand is this: In the Einstein relativity theory, anything which has *energy* has mass—mass in the sense that it is attracted gravitationally. Even light, which has an energy, has a “mass.” When a light beam, which has energy in it, comes past the sun there is an attraction on it by the sun. Thus the light does not go straight, but is deflected. During the eclipse of the sun, for example, the stars which are around the sun should appear displaced from where they would be if the sun were not there, and this has been observed.

Finally, let us compare gravitation with other theories. In recent years we have discovered that all mass is made of tiny particles and that there are several kinds of interactions, such as nuclear forces, etc. None of these nuclear or electrical forces has yet been found to explain gravitation. The quantum-mechanical aspects of nature have not yet been carried over to gravitation. When the scale is so small that we need the quantum effects, the gravitational effects are so weak that the need for a quantum theory of gravitation has not yet developed. On the other hand, for consistency in our physical theories it would be important to see whether Newton’s law modified to Einstein’s law can be further modified to be consistent with the uncertainty principle. This last modification has not yet been completed.

Motion

8-1 Description of motion

In order to find the laws governing the various changes that take place in bodies as time goes on, we must be able to *describe* the changes and have some way to record them. The simplest change to observe in a body is the apparent change in its position with time, which we call motion. Let us consider some solid object with a permanent mark, which we shall call a point, that we can observe. We shall discuss the motion of the little marker, which might be the radiator cap of an automobile or the center of a falling ball, and shall try to describe the fact that it moves and how it moves.

These examples may sound trivial, but many subtleties enter into the description of change. Some changes are more difficult to describe than the motion of a point on a solid object, for example the speed of drift of a cloud that is drifting very slowly, but rapidly forming or evaporating, or the change of a woman's mind. We do not know a simple way to analyze a change of mind, but since the cloud can be represented or described by many molecules, perhaps we can describe the motion of the cloud in principle by describing the motion of all its individual molecules. Likewise, perhaps even the changes in the mind may have a parallel in changes of the atoms inside the brain, but we have no such knowledge yet.

At any rate, that is why we begin with the motion of points; perhaps we should think of them as atoms, but it is probably better to be more rough in the beginning and simply to think of some kind of small objects—small, that is, compared with the distance moved. For instance, in describing the motion of a car that is going a hundred miles, we do not have to distinguish between the front and the back of the car. To be sure, there are slight differences, but for rough purposes we say “the car,” and likewise it does not matter that our points are not absolute points; for our present purposes it is not necessary to be extremely precise. Also, while we take a first look at this subject we are going to forget about the three dimensions of the world. We shall just concentrate on moving in one direction, as in a car on one road. We shall return to three dimensions after we see how to describe motion in one dimension. Now, you may say, “This is all some kind of trivia,” and indeed it is. How can we describe such a one-dimensional motion—let us say, of a car? Nothing could be simpler. Among many possible ways, one would be the following. To determine the position of the car at different times, we measure its distance from the starting point and record all the observations. In Table 8-1, s represents the distance of the car, in feet, from the starting point, and t represents the time in minutes. The first line in the table represents zero distance and zero time—the car has not started yet. After one minute it has started and has gone 1200 feet. Then in two minutes, it goes farther—notice that it picked up more distance in the second minute—it has accelerated; but something happened between 3 and 4 and even more so at 5—it stopped at a light perhaps? Then it speeds up again and goes 13,000 feet by the end of 6 minutes, 18,000 feet at the end of 7 minutes, and 23,500 feet in 8 minutes; at 9 minutes it has advanced to only 24,000 feet, because in the last minute it was stopped by a cop.

That is one way to describe the motion. Another way is by means of a graph. If we plot the time horizontally and the distance vertically, we obtain a curve something like that shown in Fig. 8-1. As the time increases, the distance increases, at first very slowly and then more rapidly, and very slowly again for a little while at 4 minutes; then it increases again for a few minutes and finally, at 9 minutes, appears to have stopped increasing. These observations can be made from the

8-1 Description of motion

8-2 Speed

8-3 Speed as a derivative

8-4 Distance as an integral

8-5 Acceleration

Table 8-1

t (min)	s (ft)
0	0
1	1200
2	4000
3	9000
4	9500
5	9600
6	13000
7	18000
8	23500
9	24000

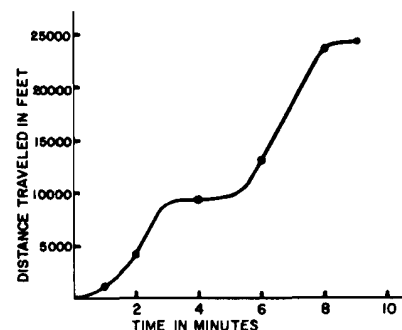


Fig. 8-1. Graph of distance versus time for the car.

Table 8-2

t (sec)	s (ft)
0	0
1	16
2	64
3	144
4	256
5	400
6	576

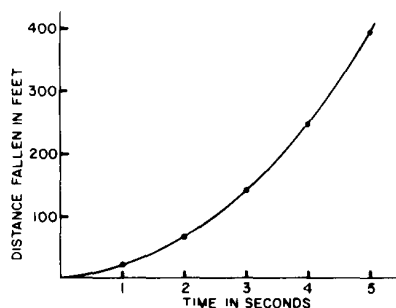


Fig. 8-2. Graph of distance versus time for a falling body.

graph, without a table. Obviously, for a complete description one would have to know where the car is at the half-minute marks, too, but we suppose that the graph means something, that the car has some position at all the intermediate times.

The motion of a car is complicated. For another example we take something that moves in a simpler manner, following more simple laws: a falling ball. Table 8-2 gives the time in seconds and the distance in feet for a falling body. At zero seconds the ball starts out at zero feet, and at the end of 1 second it has fallen 16 feet. At the end of 2 seconds, it has fallen 64 feet, at the end of 3 seconds, 144 feet, and so on; if the tabulated numbers are plotted, we get the nice parabolic curve shown in Fig. 8-2. The formula for this curve can be written as

$$s = 16t^2. \quad (8.1)$$

This formula enables us to calculate the distances at any time. You might say there ought to be a formula for the first graph too. Actually, one may write such a formula abstractly, as

$$s = f(t), \quad (8.2)$$

meaning that s is some quantity depending on t or, in mathematical phraseology, s is a function of t . Since we do not know what the function is, there is no way we can write it in definite algebraic form.

We have now seen two examples of motion, adequately described with very simple ideas, no subtleties. However, there *are* subtleties—several of them. In the first place, what do we mean by *time* and *space*? It turns out that these deep philosophical questions have to be analyzed very carefully in physics, and this is not so easy to do. The theory of relativity shows that our ideas of space and time are not as simple as one might think at first sight. However, for our present purposes, for the accuracy that we need at first, we need not be very careful about defining things precisely. Perhaps you say, “That’s a terrible thing—I learned that in science we have to define *everything* precisely.” We cannot define *anything* precisely! If we attempt to, we get into that paralysis of thought that comes to philosophers, who sit opposite each other, one saying to the other, “You don’t know what you are talking about!” The second one says, “What do you mean by *know*? What do you mean by *talking*? What do you mean by *you*?,” and so on. In order to be able to talk constructively, we just have to agree that we are talking about roughly the same thing. You know as much about time as we need for the present, but remember that there are some subtleties that have to be discussed; we shall discuss them later.

Another subtlety involved, and already mentioned, is that it should be possible to imagine that the moving point we are observing is always located somewhere. (Of course when we are looking at it, there it is, but maybe when we look away it isn’t there.) It turns out that in the motion of atoms, that idea also is false—we cannot find a marker on an atom and watch it move. That subtlety we shall have to get around in quantum mechanics. But we are first going to learn what the problems are before introducing the complications, and *then* we shall be in a better position to make corrections, in the light of the more recent knowledge of the subject. We shall, therefore, take a simple point of view about time and space. We know what these concepts are in a rough way, and those who have driven a car know what speed means.

8-2 Speed

Even though we know roughly what “speed” means, there are still some rather deep subtleties; consider that the learned Greeks were never able to adequately describe problems involving velocity. The subtlety comes when we try to comprehend exactly what is meant by “speed.” The Greeks got very confused about this, and a new branch of mathematics had to be discovered beyond the geometry and algebra of the Greeks, Arabs, and Babylonians. As an illustration of the difficulty, try to solve this problem by sheer algebra: A balloon is being inflated so

that the volume of the balloon is increasing at the rate of 100 cm^3 per second; at what speed is the radius increasing when the volume is 1000 cm^3 ? The Greeks were somewhat confused by such problems, being helped, of course, by some very confusing Greeks. To show that there were difficulties in reasoning about speed at the time, Zeno produced a large number of paradoxes, of which we shall mention one to illustrate his point that there are obvious difficulties in thinking about motion. "Listen," he says, "to the following argument: Achilles runs 10 times as fast as a tortoise, nevertheless he can never catch the tortoise. For, suppose that they start in a race where the tortoise is 100 meters ahead of Achilles; then when Achilles has run the 100 meters to the place where the tortoise was, the tortoise has proceeded 10 meters, having run one-tenth as fast. Now, Achilles has to run another 10 meters to catch up with the tortoise, but on arriving at the end of that run, he finds that the tortoise is still 1 meter ahead of him; running another meter, he finds the tortoise 10 centimeters ahead, and so on, *ad infinitum*. Therefore, at any moment the tortoise is always ahead of Achilles and Achilles can never catch up with the tortoise." What is wrong with that? It is that a finite amount of time can be divided into an infinite number of pieces, just as a length of line can be divided into an infinite number of pieces by dividing repeatedly by two. And so, although there are an infinite number of steps (in the argument) to the point at which Achilles reaches the tortoise, it doesn't mean that there is an infinite amount of *time*. We can see from this example that there are indeed some subtleties in reasoning about speed.

In order to get to the subtleties in a clearer fashion, we remind you of a joke which you surely must have heard. At the point where the lady in the car is caught by a cop, the cop comes up to her and says, "Lady, you were going 60 miles an hour!" She says, "That's impossible, sir, I was travelling for only seven minutes. It is ridiculous—how can I go 60 miles an hour when I wasn't going an hour?" How would you answer her if you were the cop? Of course, if you were really the cop, then no subtleties are involved; it is very simple: you say, "Tell that to the judge!" But let us suppose that we do not have that escape and we make a more honest, intellectual attack on the problem, and try to explain to this lady what we mean by the idea that she was going 60 miles an hour. Just what *do* we mean? We say, "What we mean, lady, is this: if you kept on going the same way as you are going now, in the next hour you would go 60 miles." She could say, "Well, my foot was off the accelerator and the car was slowing down, so if I kept on going that way it would not go 60 miles." Or consider the falling ball and suppose we want to know its speed at the time three seconds if the ball kept on going the way it is going. What does that mean—kept on *accelerating*, going faster? No—kept on going with the same *velocity*. But that is what we are trying to define! For if the ball keeps on going the way it is going, it will just keep on going the way it is going. Thus we need to define the velocity better. What has to be kept the same? The lady can also argue this way: "If I kept on going the way I'm going for one more hour, I would run into that wall at the end of the street!" It is not so easy to say what we mean.

Many physicists think that measurement is the only definition of anything. Obviously, then, we should use the instrument that measures the speed—the speedometer—and say, "Look, lady, your speedometer reads 60." So she says, "My speedometer is broken and didn't read at all." Does that mean the car is standing still? We believe that there is something to measure before we build the speedometer. Only then can we say, for example, "The speedometer isn't working right," or "the speedometer is broken." That would be a meaningless sentence if the velocity had no meaning independent of the speedometer. So we have in our minds, obviously, an idea that is independent of the speedometer, and the speedometer is meant only to measure this idea. So let us see if we can get a better definition of the idea. We say, "Yes, of course, before you went an hour, you would hit that wall, but if you went one second, you would go 88 feet; lady, you were going 88 feet per second, and if you kept on going, the next second it would be 88 feet, and the wall down there is farther away than that." She says, "Yes, but there's no law against going 88 feet per second! There is only a law

against going 60 miles an hour.” “But,” we reply, “it’s the same thing.” If it is the same thing, it should not be necessary to go into this circumlocution about 88 feet per second. In fact, the falling ball could not keep going the same way even one second because it would be changing speed, and we shall have to define speed somehow.

Now we seem to be getting on the right track; it goes something like this: If the lady kept on going for another $1/1000$ of an hour, she would go $1/1000$ of 60 miles. In other words, she does not have to keep on going for the whole hour; the point is that *for a moment* she is going at that speed. Now what that means is that if she went just a little bit more in time, the extra distance she goes would be the same as that of a car that goes at a *steady* speed of 60 miles an hour. Perhaps the idea of the 88 feet per second is right; we see how far she went in the last second, divide by 88 feet, and if it comes out 1 the speed was 60 miles an hour. In other words, we can find the speed in this way: We ask, how far do we go in a very short time? We divide that distance by the time, and that gives the speed. But the time should be made as short as possible, the shorter the better, because some change could take place during that time. If we take the time of a falling body as an hour, the idea is ridiculous. If we take it as a second, the result is pretty good for a car, because there is not much change in speed, but not for a falling body; so in order to get the speed more and more accurately, we should take a smaller and smaller time interval. What we should do is take a millionth of a second, and divide that distance by a millionth of a second. The result gives the distance per second, which is what we mean by the velocity, so we can define it that way. That is a successful answer for the lady, or rather, that is the definition that we are going to use.

The foregoing definition involves a new idea, an idea that was not available to the Greeks in a general form. That idea was to take an *infinitesimal distance* and the corresponding *infinitesimal time*, form the ratio, and watch what happens to that ratio as the time that we use gets smaller and smaller and smaller. In other words, take a limit of the distance travelled divided by the time required, as the time taken gets smaller and smaller, *ad infinitum*. This idea was invented by Newton and by Leibnitz, independently, and is the beginning of a new branch of mathematics, called the *differential calculus*. Calculus was invented in order to describe motion, and its first application was to the problem of defining what is meant by going “60 miles an hour.”

Let us try to define velocity a little better. Suppose that in a short time, ϵ , the car or other body goes a short distance x ; then the velocity, v , is defined as

$$v = x/\epsilon,$$

an approximation that becomes better and better as the ϵ is taken smaller and smaller. If a mathematical expression is desired, we can say that the velocity equals the limit as the ϵ is made to go smaller and smaller in the expression x/ϵ , or

$$v = \lim_{\epsilon \rightarrow 0} \frac{x}{\epsilon}. \quad (8.3)$$

We cannot do the same thing with the lady in the car, because the table is incomplete. We know only where she was at intervals of one minute; we can get a rough idea that she was going 5000 ft/min during the 7th minute, but we do not know, at exactly the moment 7 minutes, whether she had been speeding up and the speed was 4900 ft/min at the beginning of the 6th minute, and is now 5100 ft/min, or something else, because we do not have the exact details in between. So only if the table were completed with an infinite number of entries could we really calculate the velocity from such a table. On the other hand, when we have a complete mathematical formula, as in the case of a falling body (Eq. 8.1), then it is possible to calculate the velocity, because we can calculate the position at any time whatsoever.

Let us take as an example the problem of determining the velocity of the falling ball at the particular time 5 seconds. One way to do this is to see from

Table 8-2 what it did in the 5th second; it went $400 - 256 = 144$ ft, so it is going 144 ft/sec; however, that is wrong, because the speed is changing; *on the average* it is 144 ft/sec during this interval, but the ball is speeding up and is really going faster than 144 ft/sec. We want to find out *exactly how fast*. The technique involved in this process is the following: We know where the ball was at 5 sec. At 5.1 sec, the distance that it has gone all together is $16(5.1)^2 = 416.16$ ft (see Eq. 8.1). At 5 sec it had already fallen 400 ft; in the last tenth of a second it fell $416.16 - 400 = 16.16$ ft. Since 16.16 ft in 0.1 sec is the same as 161.6 ft/sec, that is the speed more or less, but it is not exactly correct. Is that the speed at 5, or at 5.1, or halfway between at 5.05 sec, or when *is* that the speed? Never mind—the problem was to find the speed *at 5 seconds*, and we do not have exactly that; we have to do a better job. So, we take one-thousandth of a second more than 5 sec, or 5.001 sec, and calculate the total fall as

$$s = 16(5.001)^2 = 16(25.010001) = 400.160016 \text{ ft.}$$

In the last 0.001 sec the ball fell 0.160016 ft, and if we divide this number by 0.001 sec we obtain the speed as 160.016 ft/sec. That is closer, very close, but it is *still not exact*. It should now be evident what we must do to find the speed exactly. To perform the mathematics we state the problem a little more abstractly: to find the velocity at a special time, t_0 , which in the original problem was 5 sec. Now the distance at t_0 , which we call s_0 , is $16t_0^2$, or 400 ft in this case. In order to find the velocity, we ask, “At the time $t_0 +$ (a little bit), or $t_0 + \epsilon$, where is the body?” The new position is $16(t_0 + \epsilon)^2 = 16t_0^2 + 32t_0\epsilon + 16\epsilon^2$. So it is farther along than it was before, because before it was only $16t_0^2$. This distance we shall call $s_0 +$ (a little bit more), or $s_0 + x$ (if x is the extra bit). Now if we subtract the distance at t_0 from the distance at $t_0 + \epsilon$, we get x , the extra distance gone, as $x = 32t_0 \cdot \epsilon + 16\epsilon^2$. Our first approximation to the velocity is

$$v = \frac{x}{\epsilon} = 32t_0 + 16\epsilon. \quad (8.4)$$

The true velocity is the value of this ratio, x/ϵ , when ϵ becomes vanishingly small. In other words, after forming the ratio, we take the limit as ϵ gets smaller and smaller, that is, approaches 0. The equation reduces to,

$$v \text{ (at time } t_0) = 32t_0.$$

In our problem, $t_0 = 5$ sec, so the solution is $v = 32 \times 5 = 160$ ft/sec. A few lines above, where we took ϵ as 0.1 and 0.01 sec successively, the value we got for v was a little more than this, but now we see that the actual velocity is precisely 160 ft/sec.

8-3 Speed as a derivative

The procedure we have just carried out is performed so often in mathematics that for convenience special notations have been assigned to our quantities ϵ and x . In this notation, the ϵ used above becomes Δt and x becomes Δs . This Δt means “an extra bit of t ,” and carries an implication that it can be made smaller. The prefix Δ is not a multiplier, any more than $\sin \theta$ means $s \cdot i \cdot n \cdot \theta$ —it simply defines a time increment, and reminds us of its special character. Δs has an analogous meaning for the distance s . Since Δ is not a factor, it cannot be cancelled in the ratio $\Delta s/\Delta t$ to give s/t , any more than the ratio $\sin \theta/\sin 2\theta$ can be reduced to $1/2$ by cancellation. In this notation, velocity is equal to the limit of $\Delta s/\Delta t$ when Δt gets smaller, or

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t}. \quad (8.5)$$

This is really the same as our previous expression (8.3) with ϵ and x , but it has the advantage of showing that something is changing, and it keeps track of what is changing.

Incidentally, to a good approximation we have another law, which says that the change in distance of a moving point is the velocity times the time interval, or $\Delta s = v \Delta t$. This statement is true only if the velocity is not changing during that time interval, and this condition is true only in the limit as Δt goes to 0. Physicists like to write it $ds = v dt$, because by dt they mean Δt in circumstances in which it is very small; with this understanding, the expression is valid to a close approximation. If Δt is too long, the velocity might change during the interval, and the approximation would become less accurate. For a time dt , approaching zero, $ds = v dt$ precisely. In this notation we can write (8.5) as

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}.$$

The quantity ds/dt which we found above is called the “derivative of s with respect to t ” (this language helps to keep track of what was changed), and the complicated process of finding it is called finding a derivative, or differentiating. The ds ’s and dt ’s which appear separately are called *differentials*. To familiarize you with the words, we say we found the derivative of the function $16t^2$, or the derivative (with respect to t) of $16t^2$ is $32t$. When we get used to the words, the ideas are more easily understood. For practice, let us find the derivative of a more complicated function. We shall consider the formula $s = At^3 + Bt + C$, which might describe the motion of a point. The letters A , B , and C represent constant numbers, as in the familiar general form of a quadratic equation. Starting from the formula for the motion, we wish to find the velocity at any time. To find the velocity in the more elegant manner, we change t to $t + \Delta t$ and note that s is then changed to $s + \text{some } \Delta s$; then we find the Δs in terms of Δt . That is to say,

$$\begin{aligned} s + \Delta s &= A(t + \Delta t)^3 + B(t + \Delta t) + C \\ &= At^3 + Bt + C + 3At^2\Delta t + B\Delta t + 3At(\Delta t)^2 + A(\Delta t)^3, \end{aligned}$$

but since

$$s = At^3 + Bt + C,$$

we find that

$$\Delta s = 3At^2\Delta t + B\Delta t + 3At(\Delta t)^2 + A(\Delta t)^3.$$

But we do not want Δs —we want Δs divided by Δt . We divide the preceding equation by Δt , getting

$$\frac{\Delta s}{\Delta t} = 3At^2 + B + 3At(\Delta t) + A(\Delta t)^2.$$

Table 8-3. A Short Table of Derivatives

s, u, v, w are arbitrary functions of t ; a, b, c , and n are arbitrary constants

Function	Derivative
$s = t^n$	$\frac{ds}{dt} = nt^{n-1}$
$s = cu$	$\frac{ds}{dt} = c \frac{du}{dt}$
$s = u + v + w + \dots$	$\frac{ds}{dt} = \frac{du}{dt} + \frac{dv}{dt} + \frac{dw}{dt} + \dots$
$s = c$	$\frac{ds}{dt} = 0$
$s = u^a v^b w^c \dots$	$\frac{ds}{dt} = s \left(\frac{a}{u} \frac{du}{dt} + \frac{b}{v} \frac{dv}{dt} + \frac{c}{w} \frac{dw}{dt} + \dots \right)$

As Δt goes toward 0 the limit of $\Delta s/\Delta t$ is ds/dt and is equal to

$$\frac{ds}{dt} = 3At^2 + B.$$

This is the fundamental process of calculus, differentiating functions. The process is even more simple than it appears. Observe that when these expansions contain any term with a square or a cube or any higher power of Δt , such terms may be dropped at once, since they will go to 0 when the limit is taken. After a little practice the process gets easier because one knows what to leave out. There are many rules or formulas for differentiating various types of functions. These can be memorized, or can be found in tables. A short list is found in Table 8-3.

8-4 Distance as an integral

Now we have to discuss the inverse problem. Suppose that instead of a table of distances, we have a table of speeds at different times, starting from zero. For the falling ball, such speeds and times are shown in Table 8-4. A similar table could be constructed for the velocity of the car, by recording the speedometer reading every minute or half-minute. If we know how fast the car is going at any time, can we determine how far it goes? This problem is just the inverse of the one solved above; we are given the velocity and asked to find the distance. How can we find the distance if we know the speed? If the speed of the car is not constant, and the lady goes sixty miles an hour for a moment, then slows down, speeds up, and so on, how can we determine how far she has gone? That is easy. We use the same idea, and express the distance in terms of infinitesimals. Let us say, "In the first second her speed was such and such, and from the formula $\Delta s = v \Delta t$ we can calculate how far the car went the first second at that speed." Now in the next second her speed is nearly the same, but slightly different; we can calculate how far she went in the next second by taking the new speed times the time. We proceed similarly for each second, to the end of the run. We now have a number of little distances, and the total distance will be the sum of all these little pieces. That is, the distance will be the sum of the velocities times the times, or $s = \sum v \Delta t$, where the Greek letter $\sum$ (sigma) is used to denote addition. To be more precise, it is the sum of the velocity at a certain time, let us say the i -th time, multiplied by Δt .

$$s = \sum_i v(t_i) \Delta t. \quad (8.6)$$

The rule for the times is that $t_{i+1} = t_i + \Delta t$. However, the distance we obtain by this method will not be correct, because the velocity changes during the time interval Δt . If we take the times short enough, the sum is precise, so we take them smaller and smaller until we obtain the desired accuracy. The true s is

$$s = \lim_{\Delta t \rightarrow 0} \sum_i v(t_i) \Delta t. \quad (8.7)$$

The mathematicians have invented a symbol for this limit, analogous to the symbol for the differential. The Δ turns into a d to remind us that the time is as small as it can be; the velocity is then called v at the time t , and the addition is written as a sum with a great "s," $\int$ (from the Latin *summa*), which has become distorted and is now unfortunately just called an integral sign. Thus we write

$$s = \int v(t) dt. \quad (8.8)$$

This process of adding all these terms together is called integration, and it is the opposite process to differentiation. The derivative of this integral is v , so one operator (d) undoes the other ($\int$). One can get formulas for integrals by taking the formulas for derivatives and running them backwards, because they are related to each other inversely. Thus one can work out his own table of integrals by differentiating all sorts of functions. For every formula with a differential, we get an integral formula if we turn it around.

Table 8-4
Velocity of a Falling Ball

t (sec)	v (ft/sec)
0	0
1	32
2	64
3	96
4	128

Every function can be differentiated analytically, i.e., the process can be carried out algebraically, and leads to a definite function. But it is not possible in a simple manner to write an analytical value for any integral at will. You can calculate it, for instance, by doing the above sum, and then doing it again with a finer interval Δt and again with a finer interval until you have it nearly right. In general, given some particular function, it is not possible to find, analytically, what the integral is. One may always try to find a function which, when differentiated, gives some desired function; but one may not find it, and it may not exist, in the sense of being expressible in terms of functions that have already been given names.

8-5 Acceleration

The next step in developing the equations of motion is to introduce another idea which goes beyond the concept of velocity to that of *change* of velocity, and we now ask, "How does the velocity *change*?" In previous chapters we have discussed cases in which forces produce changes in velocity. You may have heard with great excitement about some car that can get from rest to 60 miles an hour in ten seconds flat. From such a performance we can see how fast the speed changes, but only on the average. What we shall now discuss is the next level of complexity, which is how fast the velocity is changing. In other words, by how many feet per second does the velocity change in a second, that is, how many feet per second, per second? We previously derived the formula for the velocity of a falling body as $v = 32t$, which is charted in Table 8-4, and now we want to find out how much the velocity changes per second; this quantity is called the acceleration.

Acceleration is defined as the time rate of change of velocity. From the preceding discussion we know enough already to write the acceleration as the derivative dv/dt , in the same way that the velocity is the derivative of the distance. If we now differentiate the formula $v = 32t$ we obtain, for a falling body,

$$a = \frac{dv}{dt} = 32. \quad (8.9)$$

[To differentiate the term $32t$ we can utilize the result obtained in a previous problem, where we found that the derivative of Bt is simply B (a constant). So by letting $B = 32$, we have at once that the derivative of $32t$ is 32.] This means that the velocity of a falling body is changing by 32 feet per second, per second always. We also see from Table 8-4 that the velocity increases by 32 ft/sec in each second. This is a very simple case, for accelerations are usually not constant. The reason the acceleration is constant here is that the force on the falling body is constant, and Newton's law says that the acceleration is proportional to the force.

As a further example, let us find the acceleration in the problem we have already solved for the velocity. Starting with

$$s = At^3 + Bt + C$$

we obtained, for $v = ds/dt$,

$$v = 3At^2 + B.$$

Since acceleration is the derivative of the velocity with respect to the time, we need to differentiate the last expression above. Recall the rule that the derivative of the two terms on the right equals the sum of the derivatives of the individual terms. To differentiate the first of these terms, instead of going through the fundamental process again we note that we have already differentiated a quadratic term when we differentiated $16t^2$, and the effect was to double the numerical coefficient and change the t^2 to t ; let us assume that the same thing will happen this time, and you can check the result yourself. The derivative of $3At^2$ will then be $6At$. Next we differentiate B , a constant term; but by a rule stated previously, the derivative of B is zero; hence this term contributes nothing to the acceleration. The final result, therefore, is $a = dv/dt = 6At$.

For reference, we state two very useful formulas, which can be obtained by integration. If a body starts from rest and moves with a constant acceleration, g , its velocity v at any time t is given by

$$v = gt.$$

The distance it covers in the same time is

$$s = \frac{1}{2}gt^2.$$

Various mathematical notations are used in writing derivatives. Since velocity is ds/dt and acceleration is the time derivative of the velocity, we can also write

$$a = \frac{d}{dt} \left(\frac{ds}{dt} \right) = \frac{d^2s}{dt^2}, \quad (8.10)$$

which are common ways of writing a second derivative.

We have another law that the velocity is equal to the integral of the acceleration. This is just the opposite of $a = dv/dt$; we have already seen that distance is the integral of the velocity, so distance can be found by twice integrating the acceleration.

In the foregoing discussion the motion was in only one dimension, and space permits only a brief discussion of motion in three dimensions. Consider a particle P which moves in three dimensions in any manner whatsoever. At the beginning of this chapter, we opened our discussion of the one-dimensional case of a moving car by observing the distance of the car from its starting point at various times. We then discussed velocity in terms of changes of these distances with time, and acceleration in terms of changes in velocity. We can treat three-dimensional motion analogously. It will be simpler to illustrate the motion on a two-dimensional diagram, and then extend the ideas to three dimensions. We establish a pair of axes at right angles to each other, and determine the position of the particle at any moment by measuring how far it is from each of the two axes. Thus each position is given in terms of an x -distance and a y -distance, and the motion can be described by constructing a table in which both these distances are given as functions of time. (Extension of this process to three dimensions requires only another axis, at right angles to the first two, and measuring a third distance, the z -distance. The distances are now measured from coordinate *planes* instead of lines.) Having constructed a table with x - and y -distances, how can we determine the velocity? We first find the components of velocity in each direction. The horizontal part of the velocity, or x -component, is the derivative of the x -distance with respect to the time, or

$$v_x = dx/dt. \quad (8.11)$$

Similarly, the vertical part of the velocity, or y -component, is

$$v_y = dy/dt. \quad (8.12)$$

In the third dimension,

$$v_z = dz/dt. \quad (8.13)$$

Now, given the components of velocity, how can we find the velocity along the actual path of motion? In the two-dimensional case, consider two successive positions of the particle, separated by a short distance Δs and a short time interval $t_2 - t_1 = \Delta t$. In the time Δt the particle moves horizontally a distance $\Delta x \sim v_x \Delta t$, and vertically a distance $\Delta y \sim v_y \Delta t$. (The symbol “ $\sim$ ” is read “is approximately.”) The actual distance moved is approximately

$$\Delta s \sim \sqrt{(\Delta x)^2 + (\Delta y)^2}, \quad (8.14)$$

as shown in Fig. 8-3. The approximate velocity during this interval can be obtained by dividing by Δt and by letting Δt go to 0, as at the beginning of the chapter.

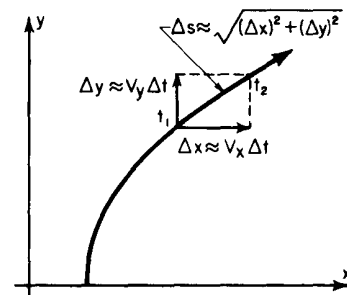


Fig. 8-3. Description of the motion of a body in two dimensions and the computation of its velocity.

We then get the velocity as

$$v = \frac{ds}{dt} = \sqrt{(dx/dt)^2 + (dy/dt)^2} = \sqrt{v_x^2 + v_y^2} \quad (8.15)$$

For three dimensions the result is

$$v = \sqrt{v_x^2 + v_y^2 + v_z^2}. \quad (8.16)$$

In the same way as we defined velocities, we can define accelerations: we have an x -component of acceleration a_x , which is the derivative of v_x , the x -component of the velocity (that is, $a_x = d^2x/dt^2$, the second derivative of x with respect to t), and so on.

Let us consider one nice example of compound motion in a plane. We shall take a motion in which a ball moves horizontally with a constant velocity u , and at the same time goes vertically downward with a constant acceleration $-g$; what is the motion? We can say $dx/dt = v_x = u$. Since the velocity v_x is constant,

$$x = ut, \quad (8.17)$$

and since the downward acceleration $-g$ is constant, the distance y the object falls can be written as

$$y = -\frac{1}{2}gt^2. \quad (8.18)$$

What is the curve of its path, i.e., what is the relation between y and x ? We can eliminate t from Eq. (8.18), since $t = x/u$. When we make this substitution we find that

$$y = -\frac{g}{2u^2} x^2. \quad (8.19)$$

This relation between y and x may be considered as the equation of the path of the moving ball. When this equation is plotted we obtain a curve that is called a parabola; any freely falling body that is shot out in any direction will travel in a parabola, as shown in Fig. 8-4.

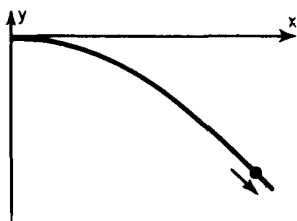


Fig. 8-4. The parabola described by a falling body with an initial horizontal velocity.

Newton's Laws of Dynamics

9-1 Momentum and force

The discovery of the laws of dynamics, or the laws of motion, was a dramatic moment in the history of science. Before Newton's time, the motions of things like the planets were a mystery, but after Newton there was complete understanding. Even the slight deviations from Kepler's laws, due to the perturbations of the planets, were computable. The motions of pendulums, oscillators with springs and weights in them, and so on, could all be analyzed completely after Newton's laws were enunciated. So it is with this chapter: before this chapter we could not calculate how a mass on a spring would move; much less could we calculate the perturbations on the planet Uranus due to Jupiter and Saturn. After this chapter we *will* be able to compute not only the motion of the oscillating mass, but also the perturbations on the planet Uranus produced by Jupiter and Saturn!

Galileo made a great advance in the understanding of motion when he discovered the *principle of inertia*: if an object is left alone, is not disturbed, it continues to move with a constant velocity in a straight line if it was originally moving, or it continues to stand still if it was just standing still. Of course this never appears to be the case in nature, for if we slide a block across a table it stops, but that is because it is *not* left to itself—it is rubbing against the table. It required a certain imagination to find the right rule, and that imagination was supplied by Galileo.

Of course, the next thing which is needed is a rule for finding how an object *changes* its speed if something *is* affecting it. That is the contribution of Newton. Newton wrote down three laws: The First Law was a mere restatement of the Galilean principle of inertia just described. The Second Law gave a specific way of determining how the velocity changes under different influences called *forces*. The Third Law describes the forces to some extent, and we shall discuss that at another time. Here we shall discuss only the Second Law, which asserts that the motion of an object is changed by forces in this way: *the time-rate-of-change of a quantity called momentum is proportional to the force*. We shall state this mathematically shortly, but let us first explain the idea.

Momentum is not the same as *velocity*. A lot of words are used in physics, and they all have precise meanings in physics, although they may not have such precise meanings in everyday language. Momentum is an example, and we must define it precisely. If we exert a certain push with our arms on an object that is light, it moves easily; if we push just as hard on another object that is much heavier in the usual sense, then it moves much less rapidly. Actually, we must change the words from "light" and "heavy" to *less massive* and *more massive*, because there is a difference to be understood between the *weight* of an object and its *inertia*. (How hard it is to get it going is one thing, and how much it weighs is something else.) Weight and inertia are *proportional*, and on the earth's surface are often taken to be numerically equal, which causes a certain confusion to the student. On Mars, weights would be different but the amount of force needed to overcome inertia would be the same.

We use the term *mass* as a quantitative measure of inertia, and we may measure mass, for example, by swinging an object in a circle at a certain speed and measuring how much force we need to keep it in the circle. In this way we find a certain quantity of mass for every object. Now the *momentum* of an object is a product of two parts: its *mass* and its *velocity*. Thus Newton's Second Law may

9-1 Momentum and force

9-2 Speed and velocity

9-3 Components of velocity, acceleration, and force

9-4 What is the force?

9-5 Meaning of the dynamical equations

9-6 Numerical solution of the equations

9-7 Planetary motions

be written mathematically this way:

$$F = \frac{d}{dt} (mv). \quad (9.1)$$

Now there are several points to be considered. In writing down any law such as this, we use many intuitive ideas, implications, and assumptions which are at first combined approximately into our "law." Later we may have to come back and study in greater detail exactly what each term means, but if we try to do this too soon we shall get confused. Thus at the beginning we take several things for granted. First, that the mass of an object is *constant*; it isn't really, but we shall start out with the Newtonian approximation that mass is constant, the same all the time, and that, further, when we put two objects together, their masses *add*. These ideas were of course implied by Newton when he wrote his equation, for otherwise it is meaningless. For example, suppose the mass varied inversely as the velocity; then the momentum would *never change* in any circumstance, so the law means nothing unless you know how the mass changes with velocity. At first we say, *it does not change*.

Then there are some implications concerning force. As a rough approximation we think of force as a kind of push or pull that we make with our muscles, but we can define it more accurately now that we have this law of motion. The most important thing to realize is that this relationship involves not only changes in the *magnitude* of the momentum or of the velocity but also in their *direction*. If the mass is constant, then Eq. (9.1) can also be written as

$$F = m \frac{dv}{dt} = ma. \quad (9.2)$$

The acceleration a is the rate of change of the velocity, and Newton's Second Law says more than that the effect of a given force varies inversely as the mass; it says also that the *direction* of the change in the velocity and the *direction* of the force are the same. Thus we must understand that a change in a velocity, or an acceleration, has a wider meaning than in common language: The velocity of a moving object can change by its speeding up, slowing down (when it slows down, we say it accelerates with a negative acceleration), or changing its direction of motion. An acceleration at right angles to the velocity was discussed in Chapter 7. There we saw that an object moving in a circle of radius R with a certain speed v along the circle falls away from a straightline path by a distance equal to $\frac{1}{2}(v^2/R)t^2$ if t is very small. Thus the formula for acceleration at right angles to the motion is

$$a = v^2/R, \quad (9.3)$$

and a force at right angles to the velocity will cause an object to move in a curved path whose radius of curvature can be found by dividing the force by the mass to get the acceleration, and then using (9.3).

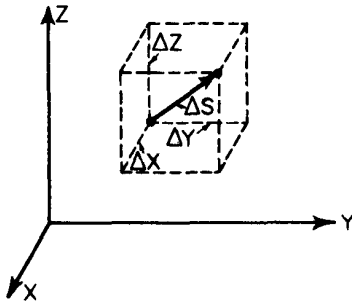


Fig. 9-1. A small displacement of an object.

9-2 Speed and velocity

In order to make our language more precise, we shall make one further definition in our use of the words *speed* and *velocity*. Ordinarily we think of speed and velocity as being the same, and in ordinary language they are the same. But in physics we have taken advantage of the fact that there *are* two words and have chosen to use them to distinguish two ideas. We carefully distinguish velocity, which has both magnitude and direction, from speed, which we choose to mean the magnitude of the velocity, but which does not include the direction. We can formulate this more precisely by describing how the x -, y -, and z -coordinates of an object change with time. Suppose, for example, that at a certain instant an object is moving as shown in Fig. 9-1. In a given small interval of time Δt it will move a certain distance Δx in the x -direction, Δy in the y -direction, and Δz in the z -direction. The total effect of these three coordinate changes is a displacement Δs along the diagonal of a parallelepiped whose sides are Δx , Δy , and Δz . In terms

of the velocity, the displacement Δx is the x -component of the velocity times Δt , and similarly for Δy and Δz :

$$\Delta x = v_x \Delta t, \quad \Delta y = v_y \Delta t, \quad \Delta z = v_z \Delta t. \quad (9.4)$$

9-3 Components of velocity, acceleration, and force

In Eq. (9.4) we have resolved the velocity into components by telling how fast the object is moving in the x -direction, the y -direction, and the z -direction. The velocity is completely specified, both as to magnitude and direction, if we give the numerical values of its three rectangular components:

$$v_x = dx/dt, \quad v_y = dy/dt, \quad v_z = dz/dt. \quad (9.5)$$

On the other hand, the speed of the object is

$$ds/dt = |v| = \sqrt{v_x^2 + v_y^2 + v_z^2}. \quad (9.6)$$

Next, suppose that, because of the action of a force, the velocity changes to some other direction and a different magnitude, as shown in Fig. 9-2. We can analyze this apparently complex situation rather simply if we evaluate the changes in the x -, y -, and z -components of velocity. The change in the component of the velocity in the x -direction in a time Δt is $\Delta v_x = a_x \Delta t$, where a_x is what we call the x -component of the acceleration. Similarly, we see that $\Delta v_y = a_y \Delta t$ and $\Delta v_z = a_z \Delta t$. In these terms, we see that Newton's Second Law, in saying that the force is in the same direction as the acceleration, is really three laws, in the sense that the component of the force in the x -, y -, or z -direction is equal to the mass times the rate of change of the corresponding component of velocity:

$$\begin{aligned} F_x &= m(dv_x/dt) = m(d^2x/dt^2) = ma_x, \\ F_y &= m(dv_y/dt) = m(d^2y/dt^2) = ma_y, \\ F_z &= m(dv_z/dt) = m(d^2z/dt^2) = ma_z. \end{aligned} \quad (9.7)$$

Just as the velocity and acceleration have been resolved into components by projecting a line segment representing the quantity and its direction onto three coordinate axes, so, in the same way, a force in a given direction is represented by certain components in the x -, y -, and z -directions:

$$\begin{aligned} F_x &= F \cos(x, F), \\ F_y &= F \cos(y, F), \\ F_z &= F \cos(z, F), \end{aligned} \quad (9.8)$$

where F is the magnitude of the force and (x, F) represents the angle between the x -axis and the direction of F , etc.

Newton's Second Law is given in complete form in Eq. (9.7). If we know the forces on an object and resolve them into x -, y -, and z -components, then we can find the motion of the object from these equations. Let us consider a simple example. Suppose there are no forces in the y - and z -directions, the only force being in the x -direction, say vertically. Equation (9.7) tells us that there would be changes in the velocity in the vertical direction, but no changes in the horizontal direction. This was demonstrated with a special apparatus in Chapter 7 (see Fig. 7-3). A falling body moves horizontally without any change in horizontal motion, while it moves vertically the same way as it would move if the horizontal motion were zero. In other words, motions in the x -, y -, and z -directions are independent if the *forces* are not connected.

9-4 What is the force?

In order to use Newton's laws, we have to have some formula for the force; these laws say *pay attention to the forces*. If an object is accelerating, some agency is at work; find it. Our program for the future of dynamics must be to *find the*

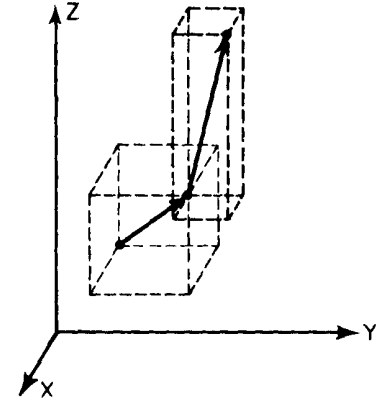


Fig. 9-2. A change in velocity in which both the magnitude and direction change.

laws for the force. Newton himself went on to give some examples. In the case of gravity he gave a specific formula for the force. In the case of other forces he gave some part of the information in his Third Law, which we will study in the next chapter, having to do with the equality of action and reaction.

Extending our previous example, what are the forces on objects near the earth's surface? Near the earth's surface, the force in the vertical direction due to gravity is proportional to the mass of the object and is nearly independent of height for heights small compared with the earth's radius R : $F = GmM/R^2 = mg$, where $g = GM/R^2$ is called the *acceleration of gravity*. Thus the law of gravity tells us that weight is proportional to mass; the force is in the vertical direction and is the mass times g . Again we find that the motion in the horizontal direction is at constant velocity. The interesting motion is in the vertical direction, and Newton's Second Law tells us

$$mg = m(d^2x/dt^2). \quad (9.9)$$

Cancelling the m 's, we find that the acceleration in the x -direction is constant and equal to g . This is of course the well known law of free fall under gravity, which leads to the equations

$$\begin{aligned} v_x &= v_0 + gt, \\ x &= x_0 + v_0t + \frac{1}{2}gt^2. \end{aligned} \quad (9.10)$$

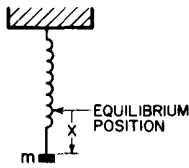


Fig. 9-3. A mass on a spring.

As another example, let us suppose that we have been able to build a gadget (Fig. 9-3) which applies a force proportional to the distance and directed oppositely—a spring. If we forget about gravity, which is of course balanced out by the initial stretch of the spring, and talk only about *excess* forces, we see that if we pull the mass down, the spring pulls up, while if we push it up the spring pulls down. This machine has been designed carefully so that the force is greater, the more we pull it up, in exact proportion to the displacement from the balanced condition, and the force upward is similarly proportional to how far we pull down. If we watch the dynamics of this machine, we see a rather beautiful motion—up, down, up, down, . . . The question is, will Newton's equations correctly describe this motion? Let us see whether we can exactly calculate how it moves with this periodic oscillation, by applying Newton's law (9.7). In the present instance, the equation is

$$-kx = m(dv_x/dt). \quad (9.11)$$

Here we have a situation where the velocity in the x -direction changes at a rate proportional to x . Nothing will be gained by retaining numerous constants, so we shall imagine either that the scale of time has changed or that there is an accident in the units, so that we happen to have $k/m = 1$. Thus we shall try to solve the equation

$$dv_x/dt = -x. \quad (9.12)$$

To proceed, we must know what v_x is, but of course we know that the velocity is the rate of change of the position.

9-5 Meaning of the dynamical equations

Now let us try to analyze just what Eq. (9.12) means. Suppose that at a given time t the object has a certain velocity v_x and position x . What is the velocity and what is the position at a slightly later time $t + \epsilon$? If we can answer this question our problem is solved, for then we can start with the given condition and compute how it changes for the first instant, the next instant, the next instant, and so on, and in this way we gradually evolve the motion. To be specific, let us suppose that at the time $t = 0$ we are given that $x = 1$ and $v_x = 0$. Why does the object move at all? Because there is a *force* on it when it is at any position except $x = 0$. If $x > 0$, that force is upward. Therefore the velocity which is zero starts to change, because of the law of motion. Once it starts to build up some velocity the object starts to move up, and so on. Now at any time t , if ϵ is very small,

we may express the position at time $t + \epsilon$ in terms of the position at time t and the velocity at time t to a very good approximation as

$$x(t + \epsilon) = x(t) + \epsilon v_x(t). \quad (9.13)$$

The smaller the ϵ , the more accurate this expression is, but it is still usefully accurate even if ϵ is not vanishingly small. Now what about the velocity? In order to get the velocity later, the velocity at the time $t + \epsilon$, we need to know how the velocity changes, the *acceleration*. And how are we going to find the acceleration? That is where the law of dynamics comes in. The law of dynamics tells us what the acceleration is. It says the acceleration is $-x$.

$$v_x(t + \epsilon) = v_x(t) + \epsilon a_x(t) \quad (9.14)$$

$$= v_x(t) - \epsilon x(t). \quad (9.15)$$

Equation (9.14) is merely kinematics; it says that a velocity changes because of the presence of acceleration. But Eq. (9.15) is *dynamics*, because it relates the acceleration to the force; it says that at this particular time for this particular problem, you can replace the acceleration by $-x(t)$. Therefore, if we know both the x and v at a given time, we know the acceleration, which tells us the new velocity, and we know the new position—this is how the machinery works. The velocity changes a little bit because of the force, and the position changes a little bit because of the velocity.

9-6 Numerical solution of the equations

Now let us really solve the problem. Suppose that we take $\epsilon = 0.100$ sec. After we do all the work if we find that this is not small enough we may have to go back and do it again with $\epsilon = 0.010$ sec. Starting with our initial value $x(0) = 1.00$, what is $x(0.1)$? It is the old position $x(0)$ plus the velocity (which is zero) times 0.10 sec. Thus $x(0.1)$ is still 1.00 because it has not yet started to move. But the new velocity at 0.10 sec will be the old velocity $v(0) = 0$ plus ϵ times the acceleration. The acceleration is $-x(0) = -1.00$. Thus

$$v(0.1) = 0.00 - 0.10 \times 1.00 = -0.10.$$

Now at 0.20 sec

$$\begin{aligned} x(0.2) &= x(0.1) + \epsilon v(0.1) \\ &= 1.00 - 0.10 \times 0.10 = 0.99 \end{aligned}$$

and

$$\begin{aligned} v(0.2) &= v(0.1) + \epsilon a(0.1) \\ &= -0.10 - 0.10 \times 1.00 = -0.20. \end{aligned}$$

And so, on and on and on, we can calculate the rest of the motion, and that is just what we shall do. However, for practical purposes there are some little tricks by which we can increase the accuracy. If we continued this calculation as we have started it, we would find the motion only rather crudely because $\epsilon = 0.100$ sec is rather crude, and we would have to go to a very small interval, say $\epsilon = 0.01$. Then to go through a reasonable total time interval would take a lot of cycles of computation. So we shall organize the work in a way that will increase the precision of our calculations, using the same coarse interval $\epsilon = 0.10$ sec. This can be done if we make a subtle improvement in the technique of the analysis.

Notice that the new position is the old position plus the time interval ϵ times the velocity. But the velocity *when*? The velocity at the beginning of the time interval is one velocity and the velocity at the end of the time interval is another velocity. Our improvement is to use the velocity *halfway between*. If we know the speed now, but the speed is changing, then we are not going to get the right answer by going at the same speed as now. We should use some speed between the “now” speed and the “then” speed at the end of the interval. The same considerations also apply to the velocity: to compute the velocity changes, we

Table 9-1

Solution of $dv_x/dt = -x$

Interval: $\epsilon = 0.10$ sec

t	x	v_x	a_x
0.0	1.000	0.000	-1.000
		-0.050	
0.1	0.995		-0.995
		-0.150	
0.2	0.980		-0.980
		-0.248	
0.3	0.955		-0.955
		-0.343	
0.4	0.921		-0.921
		-0.435	
0.5	0.877		-0.877
		-0.523	
0.6	0.825		-0.825
		-0.605	
0.7	0.764		-0.764
		-0.682	
0.8	0.696		-0.696
		-0.751	
0.9	0.621		-0.621
		-0.814	
1.0	0.540		-0.540
		-0.868	
1.1	0.453		-0.453
		-0.913	
1.2	0.362		-0.362
		-0.949	
1.3	0.267		-0.267
		-0.976	
1.4	0.169		-0.169
		-0.993	
1.5	0.070		-0.070
		-1.000	
1.6	-0.030		+0.030

should use the acceleration midway between the two times at which the velocity is to be found. Thus the equations that we shall actually use will be something like this: the position later is equal to the position before plus ϵ times the velocity *at the time in the middle of the interval*. Similarly, the velocity at this halfway point is the velocity at a time ϵ before (which is in the middle of the previous interval) plus ϵ times the acceleration at the time t . That is, we use the equations

$$\begin{aligned}x(t + \epsilon) &= x(t) + \epsilon v(t + \epsilon/2), \\v(t + \epsilon/2) &= v(t - \epsilon/2) + \epsilon a(t), \\a(t) &= -x(t).\end{aligned}\tag{9.16}$$

There remains only one slight problem: what is $v(\epsilon/2)$? At the start, we are given $v(0)$, not $v(-\epsilon/2)$. To get our calculation started, we shall use a special equation, namely, $v(\epsilon/2) = v(0) + (\epsilon/2)a(0)$.

Now we are ready to carry through our calculation. For convenience, we may arrange the work in the form of a table, with columns for the time, the position, the velocity, and the acceleration, and the in-between lines for the velocity, as shown in Table 9-1. Such a table is, of course, just a convenient way of representing the numerical values obtained from the set of equations (9.16), and in fact the equations themselves need never be written. We just fill in the various spaces in the table one by one. This table now gives us a very good idea of the motion: it starts from rest, first picks up a little upward (negative) velocity and it loses some of its distance. The acceleration is then a little bit less but it is still gaining speed. But as it goes on it gains speed more and more slowly, until as it passes $x = 0$ at about $t = 1.50$ sec we can confidently predict that it will keep going, but now it will be on the other side; the position x will become negative, the acceleration therefore positive. Thus the speed decreases. It is interesting to compare these numbers with the function $x = \cos t$, which is done in Fig. 9-4. The agreement is within the three significant figure accuracy of our calculation! We shall see later that $x = \cos t$ is the exact mathematical solution of our equation of motion, but it is an impressive illustration of the power of numerical analysis that such an easy calculation should give such precise results.

9-7 Planetary motions

The above analysis is very nice for the motion of an oscillating spring, but can we analyze the motion of a planet around the sun? Let us see whether we can arrive at an approximation to an ellipse for the orbit. We shall suppose that the sun is infinitely heavy, in the sense that we shall not include its motion. Suppose a planet starts at a certain place and is moving with a certain velocity; it goes around the sun in some curve, and we shall try to analyze, by Newton's laws of motion and his law of gravitation, what the curve is. How? At a given moment it is at some position in space. If the radial distance from the sun to this position is called r , then we know that there is a force directed inward which, according to the law of gravity, is equal to a constant times the product of the sun's mass and the planet's mass divided by the square of the distance. To analyze this further we must find out what acceleration will be produced by this force. We shall need the *components* of the acceleration along two directions, which we call x and y . Thus if we specify the position of the planet at a given moment by giving x and y (we shall suppose that z is always zero because there is no force in the z -direction and, if there is no initial velocity v_z , there will be nothing to make z other than zero), the force is directed along the line joining the planet to the sun, as shown in Fig. 9-5.

From this figure we see that the horizontal component of the force is related to the complete force in the same manner as the horizontal distance x is to the complete hypotenuse r , because the two triangles are similar. Also, if x is positive, F_x is negative. That is, $F_x/|F| = -x/r$, or $F_x = -|F|x/r = -GMmx/r^3$. Now we use the dynamical law to find that this force component is equal to the mass of

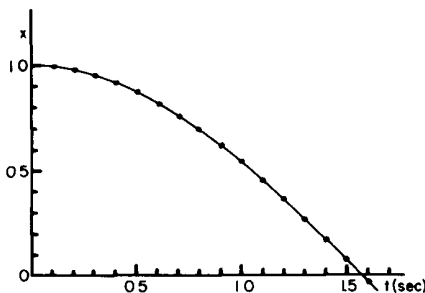


Fig. 9-4. Graph of the motion of a mass on a spring.

the planet times the rate of change of its velocity in the x -direction. Thus we find the following laws:

$$\begin{aligned} m(dv_x/dt) &= -GMmx/r^3, \\ m(dv_y/dt) &= -GMmy/r^3, \\ r &= \sqrt{x^2 + y^2}. \end{aligned} \quad (9.17)$$

This, then, is the set of equations we must solve. Again, in order to simplify the numerical work, we shall suppose that the unit of time, or the mass of the sun, has been so adjusted (or luck is with us) that $GM \equiv 1$. For our specific example we shall suppose that the initial position of the planet is at $x = 0.500$ and $y = 0.000$, and that the velocity is all in the y -direction at the start, and is of magnitude 1.6300. Now how do we make the calculation? We again make a table with columns for the time, the x -position, the x -velocity v_x , and the x -acceleration a_x ; then, separated by a double line, three columns for position, velocity, and acceleration in the y -direction. In order to get the accelerations we are going to need Eq. (9.17); it tells us that the acceleration in the x -direction is $-x/r^3$, and the acceleration in the y -direction is $-y/r^3$, and that r is the square root of $x^2 + y^2$. Thus, given x and y , we must do a little calculating on the side, taking the square root of the sum of the squares to find r and then, to get ready to calculate the two accelerations, it is useful also to evaluate $1/r^3$. This work can be done rather easily by using a table of squares, cubes, and reciprocals: then we need only multiply x by $1/r^3$, which we do on a slide rule.

Our calculation thus proceeds by the following steps, using time intervals $\epsilon = 0.100$: Initial values at $t = 0$:

$$\begin{aligned} x(0) &= 0.500 & y(0) &= 0.000 \\ v_x(0) &= 0.000 & v_y(0) &= +1.630 \end{aligned}$$

From these we find:

$$\begin{aligned} r(0) &= 0.500 & 1/r^3(0) &= 8.000 \\ a_x &= -4.000 & a_y &= 0.000 \end{aligned}$$

Thus we may calculate the velocities $v_x(0.05)$ and $v_y(0.05)$:

$$\begin{aligned} v_x(0.05) &= 0.000 - 4.000 \times 0.050 = -0.200; \\ v_y(0.05) &= 1.630 + 0.000 \times 0.100 = 1.630. \end{aligned}$$

Now our main calculations begin:

$$\begin{aligned} x(0.1) &= 0.500 - 0.20 \times 0.1 &= 0.480 \\ y(0.1) &= 0.0 + 1.63 \times 0.1 &= 0.163 \\ r &= \sqrt{0.480^2 + 0.163^2} &= 0.507 \\ 1/r^3 &= 7.67 \\ a_x(0.1) &= 0.480 \times 7.67 &= -3.68 \\ a_y(0.1) &= -0.163 \times 7.67 &= -1.256 \\ v_x(0.15) &= -0.200 - 3.68 \times 0.1 &= -0.568 \\ v_y(0.15) &= 1.630 - 1.26 \times 0.1 &= 1.505 \\ x(0.2) &= 0.480 - 0.568 \times 0.1 &= 0.423 \\ y(0.2) &= 0.163 + 1.50 \times 0.1 &= 0.313 \\ &&&\text{etc.} \end{aligned}$$

In this way we obtain the values given in Table 9-2, and in 20 steps or so we have chased the planet halfway around the sun! In Fig. 9-6 are plotted the x - and y -coordinates given in Table 9-2. The dots represent the positions at the succession of times a tenth of a unit apart; we see that at the start the planet moves rapidly

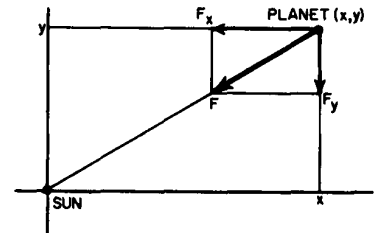


Fig. 9-5. The force of gravity on a planet.

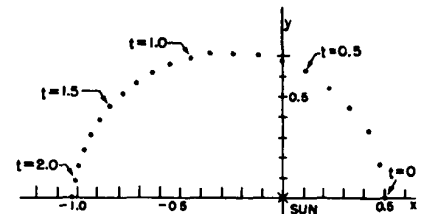


Fig. 9-6. The calculated motion of a planet around the sun.

Table 9-2

Solution of $dv_x/dt = -x/r^3$, $dv_y/dt = -y/r^3$, $r = \sqrt{x^2 + y^2}$.

Interval: $\epsilon = 0.100$

Orbit $v_y = 1.63$ $v_x = 0$ $x = 0.5$ $y = 0$ at $t = 0$

t	x	v_x	a_x	y	v_y	a_y	r	$1/r^3$
0.0	0.500		-4.00	0.000		0.00	0.500	8.000
		-0.200			1.630			
0.1	0.480		-3.68	0.163		-1.25	0.507	7.675
		-0.568			1.505			
0.2	0.423		-2.91	0.313		-2.15	0.526	6.873
		-0.859			1.290			
0.3	0.337		-1.96	0.442		-2.57	0.556	5.824
		-1.055			1.033			
0.4	0.232		-1.11	0.545		-2.62	0.592	4.81
		-1.166			0.771			
0.5	0.115		-0.453	0.622		-2.45	0.633	3.942
		-1.211			0.526			
0.6	-0.006		+0.020	0.675		-2.20	0.675	3.252
		-1.209			0.306			
0.7	-0.127		+0.344	0.706		-1.91	0.717	2.712
		-1.175			0.115			
0.8	-0.245		+0.562	0.718		-1.64	0.758	2.296
		-1.119			-0.049			
0.9	-0.357		+0.705	0.713		-1.41	0.797	1.975
		-1.048			-0.190			
1.0	-0.462		+0.796	0.694		-1.20	0.834	1.723
		-0.968			-0.310			
1.1	-0.559		+0.858	0.663		-1.02	0.867	1.535
		-0.882			-0.412			
1.2	-0.647		+0.90	0.622		-0.86	0.897	1.385
		-0.792			-0.499			
1.3	-0.726		+0.92	0.572		-0.72	0.924	1.267
		-0.700			-0.570			
1.4	-0.796		+0.93	0.515		-0.60	0.948	1.173
		-0.607			-0.630			
1.5	-0.857		+0.94	0.452		-0.50	0.969	1.099
		-0.513			-0.680			
1.6	-0.908		+0.95	0.384		-0.40	0.986	1.043
		-0.418			-0.720			
1.7	-0.950		+0.95	0.312		-0.31	1.000	1.000
		-0.323			-0.751			
1.8	-0.982		+0.95	0.237		-0.23	1.010	0.970
		-0.228			-0.773			
1.9	-1.005		+0.95	0.160		-0.15	1.018	0.948
		-0.113			-0.778			
2.0	-1.018		+0.96	0.081		-0.08	1.021	0.939
		-0.037			-0.796			
2.1	-1.022		+0.95	0.001		0.00	1.022	0.936
		+0.058			-0.796			
2.2	-1.016		+0.96	-0.079		+0.07	1.019	0.945
					-0.789			
2.3								

Crossed x-axis at 2.101 sec, $\therefore$ period = 4.20 sec.

$v_x = 0$ at 2.086 sec.

Cross x at 1.022, $\therefore$ semimajor axis = $\frac{1.022 + 0.500}{2} = 0.761$.

$v_y = 0.796$.

Predicted time $\pi(0.761)^{3/2} = \pi(0.663) = 2.082$.

and at the end it moves slowly, and so the shape of the curve is determined. Thus we see that we *really do* know how to calculate the motion of planets!

Now let us see how we can calculate the motion of Neptune, Jupiter, Uranus, or any other planet. If we have a great many planets, and let the sun move too, can we do the same thing? Of course we can. We calculate the force on a particular planet, let us say planet number i , which has a position x_i, y_i, z_i ($i = 1$ may represent the sun, $i = 2$ Mercury, $i = 3$ Venus, and so on). We must know the positions of all the planets. The force acting on one is due to all the other bodies which are located, let us say, at positions x_j, y_j, z_j . Therefore the equations are

$$\begin{aligned} m_i \frac{dv_{ix}}{dt} &= \sum_{j=1}^N - \frac{Gm_i m_j (x_i - x_j)}{r_{ij}^3}, \\ m_i \frac{dv_{iy}}{dt} &= \sum_{j=1}^N - \frac{Gm_i m_j (y_i - y_j)}{r_{ij}^3}, \\ m_i \frac{dv_{iz}}{dt} &= \sum_{j=1}^N - \frac{Gm_i m_j (z_i - z_j)}{r_{ij}^3}. \end{aligned} \quad (9.18)$$

Further, we define r_{ij} as the distance between the two planets i and j ; this is equal to

$$r_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2}. \quad (9.19)$$

Also, $\sum$ means a sum over all values of j —all other bodies—except, of course, for $j = i$. Thus all we have to do is to make more columns, *lots* more columns. We need nine columns for the motions of Jupiter, nine for the motions of Saturn, and so on. Then when we have all initial positions and velocities we can calculate all the accelerations from Eq. (9.18) by first calculating all the distances, using Eq. (9.19). How long will it take to do it? If you do it at home, it will take a very long time! But in modern times we have machines which do arithmetic very rapidly; a very good computing machine may take 1 microsecond, that is, a millionth of a second, to do an addition. To do a multiplication takes longer, say 10 microseconds. It may be that in one cycle of calculation, depending on the problem, we may have 30 multiplications, or something like that, so one cycle will take 300 microseconds. That means that we can do 3000 cycles of computation per second. In order to get an accuracy, of, say, one part in a billion, we would need 4×10^5 cycles to correspond to one revolution of a planet around the sun. That corresponds to a computation time of 130 seconds or about two minutes. Thus it takes only two minutes to follow Jupiter around the sun, with all the perturbations of all the planets correct to one part in a billion, by this method! (It turns out that the error varies about as the square of the interval ϵ . If we make the interval a thousand times smaller, it is a million times more accurate. So, let us make the interval 10,000 times smaller.)

So, as we said, we began this chapter not knowing how to calculate even the motion of a mass on a spring. Now, armed with the tremendous power of Newton's laws, we can not only calculate such simple motions but also, given only a machine to handle the arithmetic, even the tremendously complex motions of the planets, to as high a degree of precision as we wish!

Conservation of Momentum

10-1 Newton's Third Law

On the basis of Newton's second law of motion, which gives the relation between the acceleration of any body and the force acting on it, any problem in mechanics can be solved in principle. For example, to determine the motion of a few particles, one can use the numerical method developed in the preceding chapter. But there are good reasons to make a further study of Newton's laws. First, there are quite simple cases of motion which can be analyzed not only by numerical methods, but also by direct mathematical analysis. For example, although we know that the acceleration of a falling body is 32 ft/sec^2 , and from this fact could calculate the motion by numerical methods, it is much easier and more satisfactory to analyze the motion and find the general solution, $s = s_0 + v_0 t + 16t^2$. In the same way, although we can work out the positions of a harmonic oscillator by numerical methods, it is also possible to show analytically that the general solution is a simple cosine function of t , and so it is unnecessary to go to all that arithmetical trouble when there is a simple and more accurate way to get the result. In the same manner, although the motion of one body around the sun, determined by gravitation, can be calculated point by point by the numerical methods of Chapter 9, which show the general shape of the orbit, it is nice also to get the exact shape, which analysis reveals as a perfect ellipse.

Unfortunately, there are really very few problems which can be solved exactly by analysis. In the case of the harmonic oscillator, for example, if the spring force is not proportional to the displacement, but is something more complicated, one must fall back on the numerical method. Or if there are two bodies going around the sun, so that the total number of bodies is three, then analysis cannot produce a simple formula for the motion, and in practice the problem must be done numerically. That is the famous three-body problem, which so long challenged human powers of analysis; it is very interesting how long it took people to appreciate the fact that perhaps the powers of mathematical analysis were limited and it might be necessary to use the numerical methods. Today an enormous number of problems that cannot be done analytically are solved by numerical methods, and the old three-body problem, which was supposed to be so difficult, is solved as a matter of routine in exactly the same manner that was described in the preceding chapter, namely, by doing enough arithmetic. However, there are also situations where both methods fail: the simple problems we can do by analysis, and the moderately difficult problems by numerical, arithmetical methods, but the very complicated problems we cannot do by either method. A complicated problem is, for example, the collision of two automobiles, or even the motion of the molecules of a gas. There are countless particles in a cubic millimeter of gas, and it would be ridiculous to try to make calculations with so many variables (about 10^{17} —a hundred million billion). Anything like the motion of the molecules or atoms of a gas or a block of iron, or the motion of the stars in a globular cluster, instead of just two or three planets going around the sun—such problems we cannot do directly, so we have to seek other means.

In the situations in which we cannot follow details, we need to know some general properties, that is, general theorems or principles which are consequences of Newton's laws. One of these is the principle of conservation of energy, which was discussed in Chapter 4. Another is the principle of conservation of momentum, the subject of this chapter. Another reason for studying mechanics further is that there are certain patterns of motion that are repeated in many different circum-

10-1 Newton's Third Law

10-2 Conservation of momentum

10-3 Momentum is conserved!

10-4 Momentum and energy

10-5 Relativistic momentum

stances, so it is good to study these patterns in one particular circumstance. For example, we shall study collisions; different kinds of collisions have much in common. In the flow of fluids, it does not make much difference what the fluid is, the laws of the flow are similar. Other problems that we shall study are vibrations and oscillations and, in particular, the peculiar phenomena of mechanical waves—sound, vibrations of rods, and so on.

In our discussion of Newton's laws it was explained that these laws are a kind of program that says "Pay attention to the forces," and that Newton told us only two things about the nature of forces. In the case of gravitation, he gave us the complete law of the force. In the case of the very complicated forces between atoms, he was not aware of the right laws for the forces; however, he discovered one rule, one general property of forces, which is expressed in his Third Law, and that is the total knowledge that Newton had about the nature of forces—the law of gravitation and this principle, but no other details.

This principle is that *action equals reaction*.

What is meant is something of this kind: Suppose we have two small bodies, say particles, and suppose that the first one exerts a force on the second one, pushing it with a certain force. Then, simultaneously, according to Newton's Third Law, the second particle will push on the first with an equal force, in the opposite direction; furthermore, these forces effectively act in the same line. This is the hypothesis, or law, that Newton proposed, and it seems to be quite accurate, though not exact (we shall discuss the errors later). For the moment we shall take it to be true that action equals reaction. Of course, if there is a third particle, not on the same line as the other two, the law does *not* mean that the total force on the first one is equal to the total force on the second, since the third particle, for instance, exerts its own push on each of the other two. The result is that the total effect on the first two is in some other direction, and the forces on the first two particles are, in general, neither equal nor opposite. However, the forces on each particle can be resolved into parts, there being one contribution or part due to each other interacting particle. Then each *pair* of particles has corresponding components of mutual interaction that are equal in magnitude and opposite in direction.

10-2 Conservation of momentum

Now what are the interesting consequences of the above relationship? Suppose, for simplicity, that we have just two interacting particles, possibly of different mass, and numbered 1 and 2. The forces between them are equal and opposite; what are the consequences? According to Newton's Second Law, force is the time rate of change of the momentum, so we conclude that the rate of change of momentum p_1 of particle 1 is equal to minus the rate of change of momentum p_2 of particle 2, or

$$dp_1/dt = -dp_2/dt. \quad (10.1)$$

Now if the *rate of change* is always equal and opposite, it follows that the *total change* in the momentum of particle 1 is equal and opposite to the *total change* in the momentum of particle 2; this means that if we *add* the momentum of particle 1 to the momentum of particle 2, the rate of change of the sum of these, due to the mutual forces (called internal forces) between particles, is zero; that is

$$d(p_1 + p_2)/dt = 0. \quad (10.2)$$

There is assumed to be no other force in the problem. If the rate of change of this sum is always zero, that is just another way of saying that the quantity $(p_1 + p_2)$ does not change. (This quantity is also written $m_1v_1 + m_2v_2$, and is called the *total momentum* of the two particles.) We have now obtained the result that the total momentum of the two particles does not change because of any mutual interactions between them. This statement expresses the law of conservation of

momentum in that particular example. We conclude that if there is any kind of force, no matter how complicated, between two particles, and we measure or calculate $m_1v_1 + m_2v_2$, that is, the sum of the two momenta, both before and after the forces act, the results should be equal, i.e., the total momentum is a constant.

If we extend the argument to three or more interacting particles in more complicated circumstances, it is evident that so far as internal forces are concerned, the total momentum of all the particles stays constant, since an increase in momentum of one, due to another, is exactly compensated by the decrease of the second, due to the first. That is, all the internal forces will balance out, and therefore cannot change the total momentum of the particles. Then if there are no forces from the outside (external forces), there are no forces that can change the total momentum; hence the total momentum is a constant.

It is worth describing what happens if there are forces that do *not* come from the mutual actions of the particles in question: suppose we isolate the interacting particles. If there are only mutual forces, then, as before, the total momentum of the particles does not change, no matter how complicated the forces. On the other hand, suppose there are also forces coming from the particles outside the isolated group. Any force exerted by outside bodies on inside bodies, we call an *external* force. We shall later demonstrate that the sum of all external forces equals the rate of change of the total momentum of all the particles inside, a very useful theorem.

The conservation of the total momentum of a number of interacting particles can be expressed as

$$m_1v_1 + m_2v_2 + m_3v_3 + \cdots = \text{a constant}, \quad (10.3)$$

if there are no net external forces. Here the masses and corresponding velocities of the particles are numbered 1, 2, 3, 4, . . . The general statement of Newton's Second Law for each particle,

$$f = \frac{d}{dt}(mv), \quad (10.4)$$

is true specifically for the *components* of force and momentum in any given direction; thus the x -component of the force on a particle is equal to the x -component of the rate of change of momentum of that particle, or

$$f_x = \frac{d}{dt}(mv_x), \quad (10.5)$$

and similarly for the y - and z -directions. Therefore Eq. (10.3) is really three equations, one for each direction.

In addition to the law of conservation of momentum, there is another interesting consequence of Newton's Second Law, to be proved later, but merely stated now. This principle is that the laws of physics will look the same whether we are standing still or moving with a uniform speed in a straight line. For example, a child bouncing a ball in an airplane finds that the ball bounces the same as though he were bouncing it on the ground. Even though the airplane is moving with a very high velocity, unless it changes its velocity, the laws look the same to the child as they do when the airplane is standing still. This is the so-called *relativity principle*. As we use it here we shall call it "Galilean relativity" to distinguish it from the more careful analysis made by Einstein, which we shall study later.

We have just derived the law of conservation of momentum from Newton's laws, and we could go on from here to find the special laws that describe impacts and collisions. But for the sake of variety, and also as an illustration of a kind of reasoning that can be used in physics in other circumstances where, for example, one might not know Newton's laws and might take a different approach, we shall discuss the laws of impacts and collisions from a completely different point of view. We shall base our discussion on the principle of Galilean relativity, stated above, and shall end up with the law of conservation of momentum.

We shall start by assuming that nature would look the same if we run along at a certain speed and watch it as it would if we were standing still. Before dis-

cussing collisions in which two bodies collide and stick together, or come together and bounce apart, we shall first consider two bodies that are held together by a spring or something else, and are then suddenly released and pushed by the spring or perhaps by a little explosion. Further, we shall consider motion in only one direction. First, let us suppose that the two objects are exactly the same, are nice symmetrical objects, and then we have a little explosion between them. After the explosion, one of the bodies will be moving, let us say toward the right, with a velocity v . Then it appears reasonable that the other body is moving toward the left with a velocity v , because if the objects are alike there is no reason for right or left to be preferred and so the bodies would do something that is symmetrical. This is an illustration of a kind of thinking that is very useful in many problems but would not be brought out if we just started with the formulas.

The first result from our experiment is that equal objects will have equal speed, but now suppose that we have two objects made of different materials, say copper and aluminum, and we make the two *masses* equal. We shall now suppose that if we do the experiment with two masses that are equal, even though the objects are not identical, the velocities will be equal. Someone might object: "But you know, you could do it backwards, you did not have to *suppose* that. You could *define* equal masses to mean two masses that acquire equal velocities in this experiment." We follow that suggestion and make a little explosion between the copper and a very large piece of aluminum, so heavy that the copper flies out and the aluminum hardly budes. That is too much aluminum, so we reduce the amount until there is just a very tiny piece, then when we make the explosion the aluminum goes flying away, and the *copper* hardly budes. That is not enough aluminum. Evidently there is some right amount in between; so we keep adjusting the amount until the velocities come out equal. Very well then—let us turn it around, and say that when the velocities are equal, the masses are equal. This appears to be just a definition, and it seems remarkable that we can transform physical laws into mere definitions. Nevertheless, there *are* some physical laws involved, and if we accept this definition of equal masses, we immediately find one of the laws, as follows.

Suppose we know from the foregoing experiment that two pieces of matter, A and B (of copper and aluminum), have equal masses, and we compare a third body, say a piece of gold, with the copper in the same manner as above, making sure that its mass is equal to the mass of the copper. If we now make the experiment between the aluminum and the gold, there is nothing in logic that says *these* masses must be equal; however, the *experiment* shows that they actually are. So now, by experiment, we have found a new law. A statement of this law might be: If two masses are each equal to a third mass (as determined by equal velocities in this experiment), then they are equal to each other. (This statement does *not* follow at all from a similar statement used as a postulate regarding *mathematical* quantities.) From this example we can see how quickly we start to infer things if we are careless. It is *not* just a definition to say the masses are equal when the velocities are equal, because to say the masses are equal is to imply the mathematical laws of equality, which in turn makes a prediction about an experiment.

As a second example, suppose that A and B are found to be equal by doing the experiment with one strength of explosion, which gives a certain velocity; if we then use a stronger explosion, will it be true or not true that the velocities now obtained are equal? Again, in logic there is nothing that can decide this question, but experiment shows that it *is* true. So, here is another law, which might be stated: If two bodies have equal masses, as measured by equal velocities at one velocity, they will have equal masses when measured at another velocity. From these examples we see that what appeared to be only a definition really involved some laws of physics.

In the development that follows we shall assume it is true that equal masses have equal and opposite velocities when an explosion occurs between them. We shall make another assumption in the inverse case: If two identical objects, moving in opposite directions with equal velocities, collide and stick together by some kind of glue, then which way will they be moving after the collision? This is again a

symmetrical situation, with no preference between right and left, so we assume that they stand still. We shall also suppose that any two objects of equal mass, even if the objects are made of different materials, which collide and stick together, when moving with the same velocity in opposite directions will come to rest after the collision.

10-3 Momentum is conserved!

We can verify the above assumptions experimentally: first, that if two stationary objects of equal mass are separated by an explosion they will move apart with the same speed, and second, if two objects of equal mass, coming together with the same speed, collide and stick together they will stop. This we can do by means of a marvelous invention called an air trough,* which gets rid of friction, the thing which continually bothered Galileo (Fig. 10-1). He could not do experiments by sliding things because they do not slide freely, but, by adding a magic touch, we can today get rid of friction. Our objects will slide without difficulty, on and on at a constant velocity, as advertised by Galileo. This is done by supporting the objects on air. Because air has very low friction, an object glides along with practically constant velocity when there is no applied force. First, we use two glide blocks which have been made carefully to have the same weight, or mass (their weight was measured really, but we know that this weight is proportional to the mass), and we place a small explosive cap in a closed cylinder between the two blocks (Fig. 10-2). We shall start the blocks from rest at the center point of the track and force them apart by exploding the cap with an electric spark. What should happen? If the speeds are equal when they fly apart, they should arrive at the ends of the trough at the same time. On reaching the ends they will both bounce back with practically opposite velocity, and will come together and stop at the center where they started. It is a good test; when it is actually done the result is just as we have described (Fig. 10-3).

Now the next thing we would like to figure out is what happens in a less simple situation. Suppose we have two equal masses, one moving with velocity v and the other standing still, and they collide and stick; what is going to happen? There is a mass $2m$ altogether when we are finished, drifting with an unknown velocity. What velocity? That is the problem. To find the answer, we make the assumption that if we ride along in a car, physics will look the same as if we are standing still. We start with the knowledge that two equal masses, moving in opposite directions with equal speeds v , will stop dead when they collide. Now suppose that while this happens, we are riding by in an automobile, at a velocity $-v$. Then what does it look like? Since we are riding along with one of the two masses which are coming together, that one appears to us to have zero velocity. The other mass, however, going the other way with velocity v , will appear to be coming toward us at a velocity $2v$ (Fig. 10-4). Finally, the combined masses after collision will seem to be passing by with velocity v . We therefore conclude that an object with velocity $2v$, hitting an equal one at rest, will end up with velocity v , or what is mathematically exactly the same, an object with velocity v hitting and sticking to one at rest will produce an object moving with velocity $v/2$. Note that if we multiply the mass and the velocity beforehand and add them together, $mv + 0$, we get the same answer as when we multiply the mass and the velocity of everything afterwards, $2m$ times $v/2$. So that tells us what happens when a mass of velocity v hits one standing still.

In exactly the same manner we can deduce what happens when equal objects having *any* two velocities hit each other.

Suppose we have two equal bodies with velocities v_1 and v_2 , respectively, which collide and stick together. What is their velocity v after the collision? Again we ride by in an automobile, say at velocity v_2 , so that one body appears to be at rest. The other then appears to have a velocity $v_1 - v_2$, and we have the same case that we had before. When it is all finished they will be moving at $\frac{1}{2}(v_1 - v_2)$ with respect to the car. What then is the actual speed on the ground?

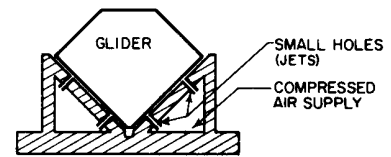


Fig. 10-1. End view of linear air trough.

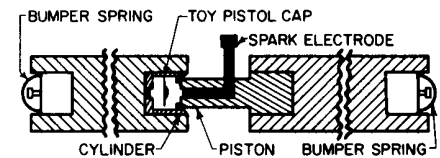


Fig. 10-2. Sectional view of gliders with explosive interaction cylinder attachment.

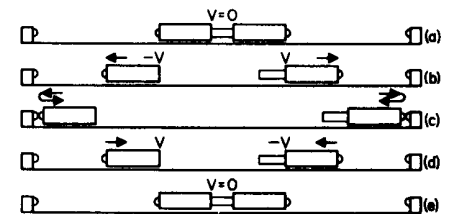


Fig. 10-3. Schematic view of action-reaction experiment with equal masses.

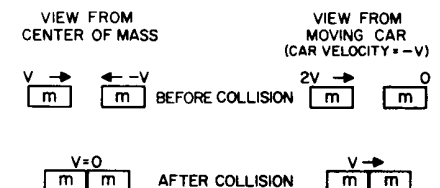


Fig. 10-4. Two views of an inelastic collision between equal masses.

* H. V. Neher and R. B. Leighton, *Amer. Jour. of Phys.* 31, 255 (1963).

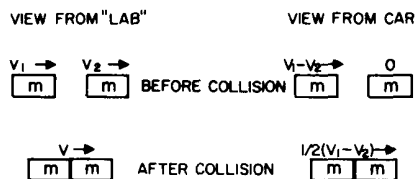


Fig. 10-5. Two views of another inelastic collision between equal masses.

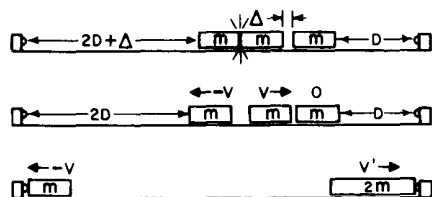


Fig. 10-6. An experiment to verify that a mass m with velocity v striking a mass m with zero velocity gives $2m$ with velocity $v/2$.

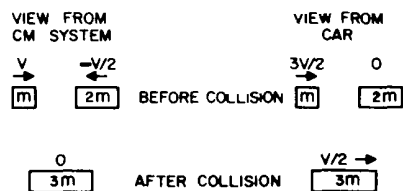


Fig. 10-7. Two views of an inelastic collision between m and $2m$.

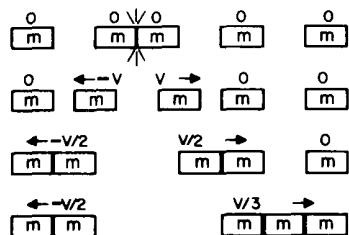


Fig. 10-8. Action and reaction between $2m$ and $3m$.

It is $v = \frac{1}{2}(v_1 - v_2) + v_2$ or $\frac{1}{2}(v_1 + v_2)$ (Fig. 10-5). Again we note that

$$mv_1 + mv_2 = 2m(v_1 + v_2)/2. \quad (10.6)$$

Thus, using this principle, we can analyze any kind of collision in which two bodies of equal mass hit each other and stick. In fact, although we have worked only in one dimension, we can find out a great deal about much more complicated collisions by imagining that we are riding by in a car in some oblique direction. The principle is the same, but the details get somewhat complicated.

In order to test experimentally whether an object moving with velocity v , colliding with an equal one at rest, forms an object moving with velocity $v/2$, we may perform the following experiment with our air-trough apparatus. We place in the trough three equally massive objects, two of which are initially joined together with our explosive cylinder device, the third being very near to but slightly separated from these and provided with a sticky bumper so that it will stick to another object which hits it. Now, a moment after the explosion, we have two objects of mass m moving with equal and opposite velocities v . A moment after that, one of these collides with the third object and makes an object of mass $2m$ moving, so we believe, with velocity $v/2$. How do we test whether it is really $v/2$? By arranging the initial positions of the masses on the trough so that the distances to the ends are not equal, but are in the ratio 2:1. Thus our first mass, which continues to move with velocity v , should cover twice as much distance in a given time as the two which are stuck together (allowing for the small distance travelled by the second object before it collided with the third). The mass m and the mass $2m$ should reach the ends at the same time, and when we try it, we find that they do (Fig. 10-6).

The next problem that we want to work out is what happens if we have two different masses. Let us take a mass m and a mass $2m$ and apply our explosive interaction. What will happen then? If, as a result of the explosion, m moves with velocity v , with what velocity does $2m$ move? The experiment we have just done may be repeated with zero separation between the second and third masses, and when we try it we get the same result, namely, the reacting masses m and $2m$ attain velocities $-v$ and $v/2$. Thus the direct reaction between m and $2m$ gives the same result as the symmetrical reaction between m and m , followed by a collision between m and a third mass m in which they stick together. Furthermore, we find that the masses m and $2m$ returning from the ends of the trough, with their velocities (nearly) exactly reversed, stop dead if they stick together.

Now the next question we may ask is this. What will happen if a mass m with velocity v , say, hits and sticks to another mass $2m$ at rest? This is very easy to answer using our principle of Galilean relativity, for we simply watch the collision which we have just described from a car moving with velocity $-v/2$ (Fig. 10-7). From the car, the velocities are

$$v'_1 = v - v(\text{car}) = v + v/2 = 3v/2$$

and

$$v'_2 = -v/2 - v(\text{car}) = -v/2 + v/2 = 0.$$

After the collision, the mass $3m$ appears to us to be moving with velocity $v/2$. Thus we have the answer, i.e., the ratio of velocities before and after collision is 3 to 1: if an object of mass m collides with a stationary object of mass $2m$, then the whole thing moves off, stuck together, with a velocity $1/3$ as much. The general rule again is that the sum of the products of the masses and the velocities stays the same: $mv + 0$ equals $3m$ times $v/3$, so we are gradually building up the theorem of the conservation of momentum, piece by piece.

Now we have one against two. Using the same arguments, we can predict the result of one against three, two against three, etc. The case of two against three, starting from rest, is shown in Fig. 10-8.

In every case we find that the mass of the first object times its velocity, plus the mass of the second object times its velocity, is equal to the total mass of the final object times its velocity. These are all examples, then, of the conservation of

momentum. Starting from simple, symmetrical cases, we have demonstrated the law for more complex cases. We could, in fact, do it for any rational mass ratio, and since every ratio is exceedingly close to a rational ratio, we can handle every ratio as precisely as we wish.

10-4 Momentum and energy

All the foregoing examples are simple cases where the bodies collide and stick together, or were initially stuck together and later separated by an explosion. However, there are situations in which the bodies do *not* cohere, as, for example, two bodies of equal mass which collide with equal speeds and then rebound. For a brief moment they are in contact and both are compressed. At the instant of maximum compression they both have zero velocity and energy is stored in the elastic bodies, as in a compressed spring. This energy is derived from the kinetic energy the bodies had before the collision, which becomes zero at the instant their velocity is zero. The loss of kinetic energy is only momentary, however. The compressed condition is analogous to the cap that releases energy in an explosion. The bodies are immediately decompressed in a kind of explosion, and fly apart again; but we already know that case—the bodies fly apart with equal speeds. However, this speed of rebound is less, in general, than the initial speed, because not all the energy is available for the explosion, depending on the material. If the material is putty no kinetic energy is recovered, but if it is something more rigid, some kinetic energy is usually regained. In the collision the rest of the kinetic energy is transformed into heat and vibrational energy—the bodies are hot and vibrating. The vibrational energy also is soon transformed into heat. It is possible to make the colliding bodies from highly elastic materials, such as steel, with carefully designed spring bumpers, so that the collision generates very little heat and vibration. In these circumstances the velocities of rebound are practically equal to the initial velocities; such a collision is called *elastic*.

That the velocities *before* and *after* an elastic collision are equal is not a matter of conservation of momentum, but a matter of conservation of *kinetic energy*. That the speeds of the bodies rebounding after a symmetrical collision are equal to *each other*, however, is a matter of conservation of momentum.

We might similarly analyze collisions between bodies of different masses, different initial velocities, and various degrees of elasticity, and determine the final velocities and the loss of kinetic energy, but we shall not go into the details of these processes.

Elastic collisions are especially interesting for systems that have no internal “gears, wheels, or parts.” Then when there is a collision there is nowhere for the energy to be impounded, because the objects that move apart are in the same condition as when they collided. Therefore, between very elementary objects, the collisions are always elastic or very nearly elastic. For instance, the collisions between atoms or molecules in a gas are said to be perfectly elastic. Although this is an excellent approximation, even such collisions are not *perfectly* elastic; otherwise one could not understand how energy in the form of light or heat radiation could come out of a gas. Once in a while, in a gas collision, a low-energy infrared ray is emitted, but this occurrence is very rare and the energy emitted is very small. So, for most purposes, collisions of molecules in gases are considered to be perfectly elastic.

As an interesting example, let us consider an *elastic* collision between two objects of *equal mass*. If they come together with the same speed, they would come apart at that same speed, by symmetry. But now look at this in another circumstance, in which one of them is moving with velocity v and the other one is at rest. What happens? We have been through this before. We watch the symmetrical collision from a car moving along with one of the objects, and we find that if a stationary body is struck elastically by another body of exactly the same mass, the moving body stops, and the one that was standing still now moves away with the same speed that the other one had; the bodies simply exchange velocities. This behavior can easily be demonstrated with a suitable impact apparatus. More

generally, if both bodies are moving, with different velocities, they simply exchange velocity at impact.

Another example of an almost elastic interaction is magnetism. If we arrange a pair of U-shaped magnets in our glide blocks, so that they repel each other, when one drifts quietly up to the other, it pushes it away and stands perfectly still, and now the other goes along, frictionlessly.

The principle of conservation of momentum is very useful, because it enables us to solve many problems without knowing the details. We did not know the details of the gas motions in the cap explosion, yet we could predict the velocities with which the bodies came apart, for example. Another interesting example is rocket propulsion. A rocket of large mass, M , ejects a small piece, of mass m , with a terrific velocity V relative to the rocket. After this the rocket, if it were originally standing still, will be moving with a small velocity, v . Using the principle of conservation of momentum, we can calculate this velocity to be

$$v = \frac{m}{M} \cdot V.$$

So long as material is being ejected, the rocket continues to pick up speed. Rocket propulsion is essentially the same as the recoil of a gun: there is no need for any air to push against.

10-5 Relativistic momentum

In modern times the law of conservation of momentum has undergone certain modifications. However, the law is still true today, the modifications being mainly in the definitions of things. In the theory of relativity it turns out that we do have conservation of momentum; the particles have mass and the momentum is still given by mv , the mass times the velocity, *but the mass changes with the velocity*, hence the momentum also changes. The mass varies with velocity according to the law

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}, \quad (10.7)$$

where m_0 is the mass of the body at rest and c is the speed of light. It is easy to see from the formula that there is negligible difference between m and m_0 unless v is very large, and that for ordinary velocities the expression for momentum reduces to the old formula.

The components of momentum for a single particle are written as

$$p_x = \frac{m_0 v_x}{\sqrt{1 - v^2/c^2}}, \quad p_y = \frac{m_0 v_y}{\sqrt{1 - v^2/c^2}}, \quad p_z = \frac{m_0 v_z}{\sqrt{1 - v^2/c^2}}, \quad (10.8)$$

where $v^2 = v_x^2 + v_y^2 + v_z^2$. If the x -components are summed over all the interacting particles, both before and after a collision, the sums are equal; that is, momentum is conserved in the x -direction. The same holds true in any direction.

In Chapter 4 we saw that the law of conservation of energy is not valid unless we recognize that energy appears in different forms, electrical energy, mechanical energy, radiant energy, heat energy, and so on. In some of these cases, heat energy for example, the energy might be said to be "hidden." This example might suggest the question, "Are there also hidden forms of momentum—perhaps heat momentum?" The answer is that it is very hard to hide momentum for the following reasons.

The random motions of the atoms of a body furnish a measure of heat energy, if the *squares* of the velocities are summed. This sum will be a positive result, having no directional character. The heat is there, whether or not the body moves as a whole, and conservation of energy in the form of heat is not very obvious. On the other hand, if one sums the *velocities*, which have direction, and finds a result that is not zero, that means that there is a drift of the entire body in some particular direction, and such a gross momentum is readily observed. Thus there is no random internal lost momentum, because the body has net momentum only

when it moves as a whole. Therefore momentum, as a mechanical quantity, is difficult to hide. Nevertheless, momentum *can* be hidden—in the electromagnetic field, for example. This case is another effect of relativity.

One of the propositions of Newton was that interactions at a distance are instantaneous. It turns out that such is not the case; in situations involving electrical forces, for instance, if an electrical charge at one location is suddenly moved, the effects on another charge, at another place, do not appear instantaneously—there is a little delay. In those circumstances, even if the forces are equal the momentum will not check out; there will be a short time during which there will be trouble, because for a while the first charge will feel a certain reaction force, say, and will pick up some momentum, but the second charge has felt nothing and has not yet changed its momentum. It takes time for the influence to cross the intervening distance, which it does at 186,000 miles a second. In that tiny time the momentum of the particles is not conserved. Of course after the second charge has felt the effect of the first one and all is quieted down, the momentum equation will check out all right, but during that small interval momentum is not conserved. We represent this by saying that during this interval there is another kind of momentum besides that of the particle, mv , and that is momentum in the electromagnetic field. If we add the field momentum to the momentum of the particles, then momentum is conserved at any moment all the time. The fact that the electromagnetic field can possess momentum and energy makes that field very real, and so, for better understanding, the original idea that there are just the forces between particles has to be modified to the idea that a particle makes a field, and a field acts on another particle, and the field itself has such familiar properties as energy content and momentum, just as particles can have. To take another example: an electromagnetic field has waves, which we call light; it turns out that light also carries momentum with it, so when light impinges on an object it carries in a certain amount of momentum per second; this is equivalent to a force, because if the illuminated object is picking up a certain amount of momentum per second, its momentum is changing and the situation is exactly the same as if there were a force on it. Light can exert pressure by bombarding an object; this pressure is very small, but with sufficiently delicate apparatus it is measurable.

Now in quantum mechanics it turns out that momentum is a different thing—it is no longer mv . It is hard to define exactly what is meant by the velocity of a particle, but momentum still exists. In quantum mechanics the difference is that when the particles are represented as particles, the momentum is still mv , but when the particles are represented as waves, the momentum is measured by the number of waves per centimeter: the greater this number of waves, the greater the momentum. In spite of the differences, the law of conservation of momentum holds also in quantum mechanics. Even though the law $f = ma$ is false, and all the derivations of Newton were wrong for the conservation of momentum, in quantum mechanics, nevertheless, in the end, that particular law maintains itself!

Vectors

11-1 Symmetry in physics

In this chapter we introduce a subject that is technically known in physics as *symmetry in physical law*. The word “symmetry” is used here with a special meaning, and therefore needs to be defined. When is a thing symmetrical—how can we define it? When we have a picture that is symmetrical, one side is somehow the same as the other side. Professor Hermann Weyl has given this definition of symmetry: a thing is symmetrical if one can subject it to a certain operation and it appears exactly the same after the operation. For instance, if we look at a vase that is left-and-right symmetrical, then turn it 180° around the vertical axis, it looks the same. We shall adopt the definition of symmetry in Weyl’s more general form, and in that form we shall discuss symmetry of physical laws.

Suppose we build a complex machine in a certain place, with a lot of complicated interactions, and balls bouncing around with forces between them, and so on. Now suppose we build exactly the same kind of equipment at some other place, matching part by part, with the same dimensions and the same orientation, everything the same only displaced laterally by some distance. Then, if we start the two machines in the same initial circumstances, in exact correspondence, we ask: will one machine behave exactly the same as the other? Will it follow all the motions in exact parallelism? Of course the answer may well be *no*, because if we choose the wrong place for our machine it might be inside a wall and interferences from the wall would make the machine not work.

All of our ideas in physics require a certain amount of common sense in their application; they are not purely mathematical or abstract ideas. We have to understand what we mean when we say that the phenomena are the same when we move the apparatus to a new position. We mean that we move everything that we believe is relevant; if the phenomenon is not the same, we suggest that something relevant has not been moved, and we proceed to look for it. If we never find it, then we claim that the laws of physics do not have this symmetry. On the other hand, we may find it—we expect to find it—if the laws of physics do have this symmetry; looking around, we may discover, for instance, that the wall is pushing on the apparatus. The basic question is, if we define things well enough, if all the essential forces are included inside the apparatus, if all the relevant parts are moved from one place to another, will the laws be the same? Will the machinery work the same way?

It is clear that what we want to do is to move all the equipment and *essential* influences, but not *everything* in the world—planets, stars, and all—for if we do that, we have the same phenomenon again for the trivial reason that we are right back where we started. No, we cannot move *everything*. But it turns out in practice that with a certain amount of intelligence about what to move, the machinery will work. In other words, if we do not go inside a wall, if we know the origin of the outside forces, and arrange that those are moved too, then the machinery *will* work the same in one location as in another.

11-2 Translations

We shall limit our analysis to just mechanics, for which we now have sufficient knowledge. In previous chapters we have seen that the laws of mechanics can be summarized by a set of three equations for each particle:

$$m(d^2x/dt^2) = F_x, \quad m(d^2y/dt^2) = F_y, \quad m(d^2z/dt^2) = F_z. \quad (11.1)$$

11-1 Symmetry in physics

11-2 Translations

11-3 Rotations

11-4 Vectors

11-5 Vector algebra

11-6 Newton’s laws in vector notation

11-7 Scalar product of vectors

Now this means that there exists a way to *measure* x , y , and z on three perpendicular axes, and the forces along those directions, such that these laws are true. These must be measured from some origin, but *where do we put the origin?* All that Newton would tell us at first is that there *is* some place that we can measure from, perhaps the center of the universe, such that these laws are correct. But we can show immediately that we can never find the center, because if we use some other origin it would make no difference. In other words, suppose that there are two people—Joe, who has an origin in one place, and Moe, who has a parallel system whose origin is somewhere else (Fig. 11-1). Now when Joe measures the location of the point in space, he finds it at x , y , and z (we shall usually leave z out because it is too confusing to draw in a picture). Moe, on the other hand, when measuring the same point, will obtain a different x (in order to distinguish it, we will call it x'), and in principle a different y , although in our example they are numerically equal. So we have

$$x' = x - a, \quad y' = y, \quad z' = z. \quad (11.2)$$

Now in order to complete our analysis we must know what Moe would obtain for the forces. The force is supposed to act along some line, and by the force in the x -direction we mean the part of the total which is in the x -direction, which is the magnitude of the force times this cosine of its angle with the x -axis. Now we see that Moe would use exactly the same projection as Joe would use, so we have a set of equations

$$F_{x'} = F_x, \quad F_{y'} = F_y, \quad F_{z'} = F_z. \quad (11.3)$$

These would be the relationships between quantities as seen by Joe and Moe.

The question is, if Joe knows Newton's laws, and if Moe tries to write down Newton's laws, will they also be correct for him? Does it make any difference from which origin we measure the points? In other words, assuming that equations (11.1) are true, and the Eqs. (11.2) and (11.3) give the relationship of the measurements, is it or is it not true that

$$\begin{aligned} (a) \quad m(d^2x'/dt^2) &= F_{x'}, \\ (b) \quad m(d^2y'/dt^2) &= F_{y'}, \\ (c) \quad m(d^2z'/dt^2) &= F_{z'}? \end{aligned} \quad (11.4)$$

In order to test these equations we shall differentiate the formula for x' twice. First of all

$$\frac{dx'}{dt} = \frac{d}{dt}(x - a) = \frac{dx}{dt} - \frac{da}{dt}.$$

Now we shall assume that Moe's origin is fixed (not moving) relative to Joe's; therefore a is a constant and $da/dt = 0$, so we find that

$$dx'/dt = dx/dt$$

and therefore

$$d^2x'/dt^2 = d^2x/dt^2;$$

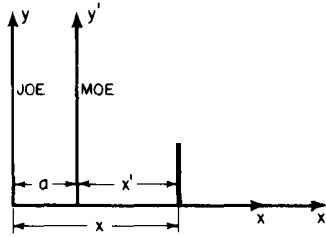
therefore we know that Eq. (11.4a) becomes

$$m(d^2x/dt^2) = F_{x'}.$$

(We also suppose that the masses measured by Joe and Moe are equal.) Thus the acceleration times the mass is the same as the other fellow's. We have also found the formula for $F_{x'}$, for, substituting from Eq. (11.1), we find that

$$F_{x'} = F_x.$$

Therefore the laws as seen by Moe appear the same; he can write Newton's laws too, with different coordinates, and they will still be right. That means that



there is no unique way to define the origin of the world, because the laws will appear the same, from whatever position they are observed.

This is also true: if there is a piece of equipment in one place with a certain kind of machinery in it, the same equipment in another place will behave in the same way. Why? Because one machine, when analyzed by Moe, has exactly the same equations as the other one, analyzed by Joe. Since the *equations* are the same, the *phenomena* appear the same. So the proof that an apparatus in a new position behaves the same as it did in the old position is the same as the proof that the equations when displaced in space reproduce themselves. Therefore we say that *the laws of physics are symmetrical for translational displacements*, symmetrical in the sense that the laws do not change when we make a translation of our coordinates. Of course it is quite obvious intuitively that this is true, but it is interesting and entertaining to discuss the mathematics of it.

11-3 Rotations

The above is the first of a series of ever more complicated propositions concerning the symmetry of a physical law. The next proposition is that it should make no difference in which *direction* we choose the axes. In other words, if we build a piece of equipment in some place and watch it operate, and nearby we build the same kind of apparatus but put it up on an angle, will it operate in the same way? Obviously it will not if it is a Grandfather clock, for example! If a pendulum clock stands upright, it works fine, but if it is tilted the pendulum falls against the side of the case and nothing happens. The theorem is then false in the case of the pendulum clock, unless we include the earth, which is pulling on the pendulum. Therefore we can make a prediction about pendulum clocks if we believe in the symmetry of physical law for rotation: something else is involved in the operation of a pendulum clock besides the machinery of the clock, something outside it that we should look for. We may also predict that pendulum clocks will not work the same way when located in different places relative to this mysterious source of asymmetry, perhaps the earth. Indeed, we know that a pendulum clock up in an artificial satellite, for example, would not tick either, because there is no effective force, and on Mars it would go at a different rate. Pendulum clocks *do* involve something more than just the machinery inside, they involve something on the outside. Once we recognize this factor, we see that we must turn the earth along with the apparatus. Of course we do not have to worry about that, it is easy to do; one simply waits a moment or two and the earth turns; then the pendulum clock ticks again in the new position the same as it did before. While we are rotating in space our angles are always changing, absolutely; this change does not seem to bother us very much, for in the new position we seem to be in the same condition as in the old. This has a certain tendency to confuse one, because it is true that in the new turned position the laws are the same as in the unturned position, but it is *not* true that *as we turn* a thing it follows the same laws as it does when we are not turning it. If we perform sufficiently delicate experiments, we can tell that the earth *is rotating*, but not that it *had rotated*. In other words, we cannot locate its angular position, but we can tell that it is changing.

Now we may discuss the effects of angular orientation upon physical laws. Let us find out whether the same game with Joe and Moe works again. This time, to avoid needless complication, we shall suppose that Joe and Moe use the same origin (we have already shown that the axes can be moved by translation to another place). Assume that Moe's axes have rotated relative to Joe's by an angle θ . The two coordinate systems are shown in Fig. 11-2, which is restricted to two dimensions. Consider any point P having coordinates (x, y) in Joe's system and (x', y') in Moe's system. We shall begin, as in the previous case, by expressing the coordinates x' and y' in terms of x, y , and θ . To do so, we first drop perpendiculars from P to all four axes and draw AB perpendicular to PQ . Inspection of the figure shows that x' can be written as the sum of two lengths along the x' -axis, and y' as the difference of two lengths along AB . All these lengths are expressed

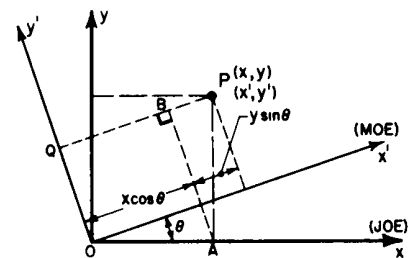


Fig. 11-2. Two coordinate systems having different angular orientations.

in terms of x , y , and θ in equations (11.5), to which we have added an equation for the third dimension.

$$\begin{aligned}x' &= x \cos \theta + y \sin \theta, \\y' &= y \cos \theta - x \sin \theta, \\z' &= z.\end{aligned}\tag{11.5}$$

The next step is to analyze the relationship of forces as seen by the two observers, following the same general method as before. Let us assume that a force F , which has already been analyzed as having components F_x and F_y (as seen by Joe), is acting on a particle of mass m , located at point P in Fig. 11-2. For simplicity, let us move both sets of axes so that the origin is at P , as shown in Fig. 11-3. Moe sees the components of F along his axes as $F_{x'}$ and $F_{y'}$. F_x has components along both the x' - and y' -axes, and F_y likewise has components along both these axes. To express $F_{x'}$ in terms of F_x and F_y , we sum these components along the x' -axis, and in a like manner we can express $F_{y'}$ in terms of F_x and F_y . The results are

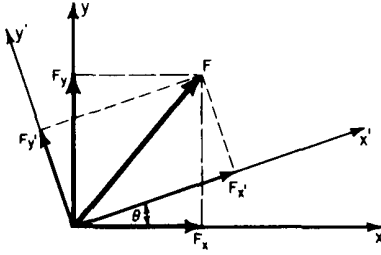


Fig. 11-3. Components of a force in the two systems.

$$\begin{aligned}F_{x'} &= F_x \cos \theta + F_y \sin \theta, \\F_{y'} &= F_y \cos \theta - F_x \sin \theta, \\F_{z'} &= F_z.\end{aligned}\tag{11.6}$$

It is interesting to note an accident of sorts, which is of extreme importance: the formulas (11.5) and (11.6), for coordinates of P and components of F , respectively, are of identical form.

As before, Newton's laws are assumed to be true in Joe's system, and are expressed by equations (11.1). The question, again, is whether Moe can apply Newton's laws—will the results be correct for his system of rotated axes? In other words, if we assume that Eqs. (11.5) and (11.6) give the relationship of the measurements, is it true or not true that

$$\begin{aligned}m(d^2x'/dt^2) &= F_{x'}, \\m(d^2y'/dt^2) &= F_{y'}, \\m(d^2z'/dt^2) &= F_{z'}?\end{aligned}\tag{11.7}$$

To test these equations, we calculate the left and right sides independently, and compare the results. To calculate the left sides, we multiply equations (11.5) by m , and differentiate twice with respect to time, assuming the angle θ to be constant. This gives

$$\begin{aligned}m(d^2x'/dt^2) &= m(d^2x/dt^2) \cos \theta + m(d^2y/dt^2) \sin \theta, \\m(d^2y'/dt^2) &= m(d^2y/dt^2) \cos \theta - m(d^2x/dt^2) \sin \theta, \\m(d^2z'/dt^2) &= m(d^2z/dt^2).\end{aligned}\tag{11.8}$$

We calculate the right sides of equations (11.7) by substituting equations (11.1) into equations (11.6). This gives

$$\begin{aligned}F_{x'} &= m(d^2x/dt^2) \cos \theta + m(d^2y/dt^2) \sin \theta, \\F_{y'} &= m(d^2y/dt^2) \cos \theta - m(d^2x/dt^2) \sin \theta, \\F_{z'} &= m(d^2z/dt^2).\end{aligned}\tag{11.9}$$

Behold! The right sides of Eqs. (11.8) and (11.9) are identical, so we conclude that if Newton's laws are correct on one set of axes, they are also valid on any other set of axes. This result, which has now been established for both translation and rotation of axes, has certain consequences: first, no one can claim his particular axes are unique, but of course they can be more *convenient* for certain particular problems. For example, it is handy to have gravity along one axis, but this is not physically necessary. Second, it means that any piece of equipment which is completely self-contained, with all the force-generating equipment completely inside the apparatus, would work the same when turned at an angle.

11-4 Vectors

Not only Newton's laws, but also the other laws of physics, so far as we know today, have the two properties which we call invariance (or symmetry) under translation of axes and rotation of axes. These properties are so important that a mathematical technique has been developed to take advantage of them in writing and using physical laws.

The foregoing analysis involved considerable tedious mathematical work. To reduce the details to a minimum in the analysis of such questions, a very powerful mathematical machinery has been devised. This system, called *vector analysis*, supplies the title of this chapter; strictly speaking, however, this is a chapter on the symmetry of physical laws. By the methods of the preceding analysis we were able to do everything required for obtaining the results that we sought, but in practice we should like to do things more easily and rapidly, so we employ the vector technique.

We began by noting some characteristics of two kinds of quantities that are important in physics. (Actually there are more than two, but let us start out with two.) One of them, like the number of potatoes in a sack, we call an ordinary quantity, or an undirected quantity, or a *scalar*. Temperature is an example of such a quantity. Other quantities that are important in physics do have direction, for instance velocity: we have to keep track of which way a body is going, not just its speed. Momentum and force also have direction, as does displacement: when someone steps from one place to another in space, we can keep track of how far he went, but if we wish also to know *where* he went, we have to specify a direction.

All quantities that have a direction, like a step in space, are called *vectors*.

A vector is three numbers. In order to represent a step in space, say from the origin to some particular point P whose location is (x, y, z) , we really need three numbers, but we are going to invent a single mathematical symbol, $\mathbf{r}$, which is unlike any other mathematical symbols we have so far used.* It is *not* a single number, it represents *three* numbers: x , y , and z . It means three numbers, but not really only *those* three numbers, because if we were to use a different coordinate system, the three numbers would be changed to x' , y' , and z' . However, we want to keep our mathematics simple and so we are going to use the *same mark* to represent the three numbers (x, y, z) and the three numbers (x', y', z') . That is, we use the same mark to represent the first set of three numbers for one coordinate system, but the second set of three numbers if we are using the other coordinate system. This has the advantage that when we change the coordinate system, we do not have to change the letters of our equations. If we write an equation in terms of x, y, z , and then use another system, we have to change to x', y', z' , but we shall just write $\mathbf{r}$, with the convention that it represents (x, y, z) if we use one set of axes, or (x', y', z') if we use another set of axes, and so on. The three numbers which describe the quantity in a given coordinate system are called the *components* of the vector in the direction of the coordinate axes of that system. That is, we use the same symbol for the three letters that correspond to the *same object, as seen from different axes*. The very fact that we can say "the same object" implies a physical intuition about the reality of a step in space, that is independent of the components in terms of which we measure it. So the symbol $\mathbf{r}$ will represent the same thing no matter how we turn the axes.

Now suppose there is another directed physical quantity, any other quantity, which also has three numbers associated with it, like force, and these three numbers change to three other numbers by a certain mathematical rule, if we change the axes. It must be the same rule that changes (x, y, z) into (x', y', z') . In other words, any physical quantity associated with three numbers which transform as do the components of a step in space is a vector. An equation like

$$\mathbf{F} = \mathbf{r}$$

would thus be true in *any* coordinate system if it were true in one. This equation,

* In type, vectors are represented by boldface; in handwritten form an arrow is used: $\vec{r}$.

of course, stands for the three equations

$$F_x = x, \quad F_y = y, \quad F_z = z,$$

or, alternatively, for

$$F_{x'} = x', \quad F_{y'} = y', \quad F_{z'} = z'.$$

The fact that a physical relationship can be expressed as a vector equation assures us the relationship is unchanged by a mere rotation of the coordinate system. That is the reason why vectors are so useful in physics.

Now let us examine some of the properties of vectors. As examples of vectors we may mention velocity, momentum, force, and acceleration. For many purposes it is convenient to represent a vector quantity by an arrow that indicates the direction in which it is acting. Why can we represent force, say, by an arrow? Because it has the same mathematical transformation properties as a "step in space." We thus represent it in a diagram as if it were a step, using a scale such that one unit of force, or one newton, corresponds to a certain convenient length. Once we have done this, all forces can be represented as lengths, because an equation like

$$\mathbf{F} = k\mathbf{r},$$

where k is some constant, is a perfectly legitimate equation. Thus we can always represent forces by lines, which is very convenient, because once we have drawn the line we no longer need the axes. Of course, we can quickly calculate the three components as they change upon turning the axes, because that is just a geometric problem.

11-5 Vector algebra

Now we must describe the laws, or rules, for combining vectors in various ways. The first such combination is the *addition* of two vectors: suppose that $\mathbf{a}$ is a vector which in some particular coordinate system has the three components (a_x, a_y, a_z) , and that $\mathbf{b}$ is another vector which has the three components (b_x, b_y, b_z) . Now let us invent three new numbers $(a_x + b_x, a_y + b_y, a_z + b_z)$. Do these form a vector? "Well," we might say, "they are three numbers, and every three numbers form a vector." No, *not* every three numbers form a vector! In order for it to be a vector, not only must there be three numbers, but these must be associated with a coordinate system in such a way that if we turn the coordinate system, the three numbers "revolve" on each other, get "mixed up" in each other, by the precise laws we have already described. So the question is, if we now rotate the coordinate system so that (a_x, a_y, a_z) become $(a_{x'}, a_{y'}, a_{z'})$ and (b_x, b_y, b_z) become $(b_{x'}, b_{y'}, b_{z'})$, what do $(a_x + b_x, a_y + b_y, a_z + b_z)$ become? Do they become $(a_{x'} + b_{x'}, a_{y'} + b_{y'}, a_{z'} + b_{z'})$ or not? The answer is, of course, yes, because the prototype transformations of Eq. (11.5) constitute what we call a *linear* transformation. If we apply those transformations to a_x and b_x to get $a_{x'} + b_{x'}$, we find that the transformed $a_x + b_x$ is indeed the same as $a_{x'} + b_{x'}$. When $\mathbf{a}$ and $\mathbf{b}$ are "added together" in this sense, they will form a vector which we may call $\mathbf{c}$. We would write this as

$$\mathbf{c} = \mathbf{a} + \mathbf{b}.$$

Now $\mathbf{c}$ has the interesting property

$$\mathbf{c} = \mathbf{b} + \mathbf{a},$$

as we can immediately see from its components. Thus also,

$$\mathbf{a} + (\mathbf{b} + \mathbf{c}) = (\mathbf{a} + \mathbf{b}) + \mathbf{c}.$$

We can add vectors in any order.

What is the geometric significance of $\mathbf{a} + \mathbf{b}$? Suppose that $\mathbf{a}$ and $\mathbf{b}$ were represented by lines on a piece of paper, what would $\mathbf{c}$ look like? This is shown in

Fig. 11-4. We see that we can add the components of $\mathbf{b}$ to those of $\mathbf{a}$ most conveniently if we place the rectangle representing the components of $\mathbf{b}$ next to that representing the components of $\mathbf{a}$ in the manner indicated. Since $\mathbf{b}$ just “fits” into its rectangle, as does $\mathbf{a}$ into its rectangle, this is the same as putting the “tail” of $\mathbf{b}$ on the “head” of $\mathbf{a}$, the arrow from the “tail” of $\mathbf{a}$ to the “head” of $\mathbf{b}$ being the vector $\mathbf{c}$. Of course, if we added $\mathbf{a}$ to $\mathbf{b}$ the other way around, we would put the “tail” of $\mathbf{a}$ on the “head” of $\mathbf{b}$, and by the geometrical properties of parallelograms we would get the same result for $\mathbf{c}$. Note that vectors can be added in this way without reference to any coordinate axes.

Suppose we multiply a vector by a number α , what does this mean? We define it to mean a new vector whose components are αa_x , αa_y , and αa_z . We leave it as a problem for the student to prove that it is a vector.

Now let us consider vector subtraction. We may define subtraction in the same way as addition, but instead of adding, we subtract the components. Or we might define subtraction by defining a negative vector, $-\mathbf{b} = -1\mathbf{b}$, and then we would add the components. It comes to the same thing. The result is shown in Fig. 11-5. This figure shows $\mathbf{d} = \mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b})$; we also note that the difference $\mathbf{a} - \mathbf{b}$ can be found very easily from $\mathbf{a}$ and $\mathbf{b}$ by using the equivalent relation $\mathbf{a} = \mathbf{b} + \mathbf{d}$. Thus the difference is even easier to find than the sum: we just draw the vector from $\mathbf{b}$ to $\mathbf{a}$, to get $\mathbf{a} - \mathbf{b}$!

Next we discuss velocity. Why is velocity a vector? If position is given by the three coordinates (x, y, z) , what is the velocity? The velocity is given by dx/dt , dy/dt , and dz/dt . Is that a vector, or not? We can find out by differentiating the expressions in Eq. (11.5) to find out whether dx'/dt transforms in the right way. We see that the components dx/dt and dy/dt do transform according to the same law as x and y , and therefore the time derivative is a vector. So the velocity is a vector. We can write the velocity in an interesting way as

$$\mathbf{v} = d\mathbf{r}/dt.$$

What the velocity is, and why it is a vector, can also be understood more pictorially: How far does a particle move in a short time Δt ? Answer: $\Delta \mathbf{r}$, so if a particle is “here” at one instant and “there” at another instant, then the vector difference of the positions $\Delta \mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1$, which is in the direction of motion shown in Fig. 11-6, divided by the time interval $\Delta t = t_2 - t_1$, is the “average velocity” vector.

In other words, by vector velocity we mean the limit, as Δt goes to 0, of the difference between the radius vectors at the time $t + \Delta t$ and the time t , divided by Δt :

$$\mathbf{v} = \lim_{\Delta t \rightarrow 0} (\Delta \mathbf{r} / \Delta t) = d\mathbf{r} / dt. \quad (11.10)$$

Thus velocity is a vector because it is the difference of two vectors. It is also the right definition of velocity because its components are dx/dt , dy/dt , and dz/dt . In fact, we see from this argument that if we differentiate *any* vector with respect to time we produce a new vector. So we have several ways of producing new vectors: (1) multiply by a constant, (2) differentiate with respect to time, (3) add or subtract two vectors.

11-6 Newton's laws in vector notation

In order to write Newton's laws in vector form, we have to go just one step further, and define the acceleration vector. This is the time derivative of the velocity vector, and it is easy to demonstrate that its components are the second derivatives of x , y , and z with respect to t :

$$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \left(\frac{d}{dt} \right) \left(\frac{d\mathbf{r}}{dt} \right) = \frac{d^2\mathbf{r}}{dt^2}, \quad (11.11)$$

$$a_x = \frac{dv_x}{dt} = \frac{d^2x}{dt^2}, \quad a_y = \frac{dv_y}{dt} = \frac{d^2y}{dt^2}, \quad a_z = \frac{dv_z}{dt} = \frac{d^2z}{dt^2}. \quad (11.12)$$

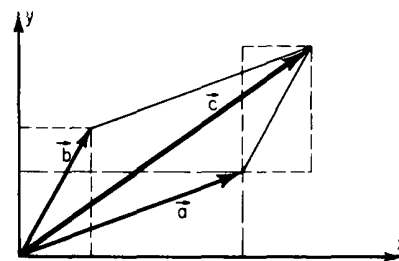


Fig. 11-4. The addition of vectors.

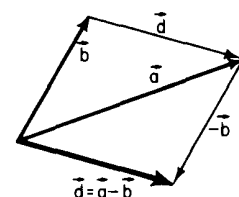


Fig. 11-5. The subtraction of vectors.

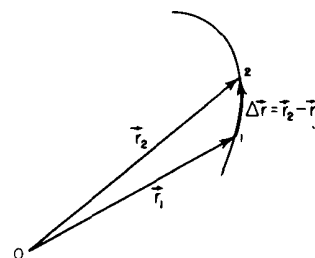


Fig. 11-6. The displacement of a particle in a short time interval $\Delta t = t_2 - t_1$.

With this definition, then, Newton's laws can be written in this way:

$$m\mathbf{a} = \mathbf{F} \quad (11.13)$$

or

$$m(d^2\mathbf{r}/dt^2) = \mathbf{F}. \quad (11.14)$$

Now the problem of proving the invariance of Newton's laws under rotation of coordinates is this: prove that $\mathbf{a}$ is a vector; this we have just done. Prove that $\mathbf{F}$ is a vector; we *suppose* it is. So if force is a vector, then, since we know acceleration is a vector, Eq. (11.13) will look the same in any coordinate system. Writing it in a form which does not explicitly contain x 's, y 's, and z 's has the advantage that from now on we need not write *three* laws every time we write Newton's equations or other laws of physics. We write what looks like *one* law, but really, of course, it is the three laws for any particular set of axes, because any vector equation involves the statement that *each of the components is equal*.

The fact that the acceleration is the rate of change of the vector velocity helps us to calculate the acceleration in some rather complicated circumstances. Suppose, for instance, that a particle is moving on some complicated curve (Fig. 11-7) and that, at a given instant t , it had a certain velocity $\mathbf{v}_1$, but that when we go to another instant t_2 a little later, it has a different velocity $\mathbf{v}_2$. What is the acceleration? Answer: Acceleration is the difference in the velocity divided by the small time interval, so we need the difference of the two velocities. How do we get the difference of the velocities? To subtract two vectors, we put the vector across the ends of $\mathbf{v}_2$ and $\mathbf{v}_1$; that is, we draw Δ as the difference of the two vectors, right? *No!* That only works when the *tails* of the vectors are in the same place! It has no meaning if we move the vector somewhere else and then draw a line across, so watch out! We have to draw a new diagram to subtract the vectors. In Fig. 11-8, $\mathbf{v}_1$ and $\mathbf{v}_2$ are both drawn parallel and equal to their counterparts in Fig. 11-7, and now we can discuss the acceleration. Of course the acceleration is simply $\Delta\mathbf{v}/\Delta t$. It is interesting to note that we can compose the velocity difference out of two parts; we can think of acceleration as having *two components*, $\Delta\mathbf{v}_{||}$ in the direction tangent to the path and $\Delta\mathbf{v}_{\perp}$ at right angles to the path, as indicated in Fig. 11-8. The acceleration tangent to the path is, of course, just the change in the *length* of the vector, i.e., the change in the *speed* v :

$$a_{||} = dv/dt. \quad (11.15)$$

The other component of acceleration, at right angles to the curve, is easy to calculate, using Figs. 11-7 and 11-8. In the short time Δt let the change in angle between $\mathbf{v}_1$ and $\mathbf{v}_2$ be the small angle $\Delta\theta$. If the magnitude of the velocity is called v , then of course

$$\Delta v_{\perp} = v \Delta\theta$$

and the acceleration a will be

$$a_{\perp} = v (\Delta\theta/\Delta t).$$

Now we need to know $\Delta\theta/\Delta t$, which can be found this way: If, at the given moment, the curve is approximated as a circle of a certain radius R , then in a time Δt the distance s is, of course, $v \Delta t$, where v is the speed.

$$\Delta\theta = v (\Delta t/R), \quad \text{or} \quad \Delta\theta/\Delta t = v/R.$$

Therefore, we find

$$a = v^2/R, \quad (11.16)$$

as we have seen before.

11-7 Scalar product of vectors

Now let us examine a little further the properties of vectors. It is easy to see that the *length* of a step in space would be the same in any coordinate system. That is, if a particular step $\mathbf{r}$ is represented by x , y , z , in one coordinate system,

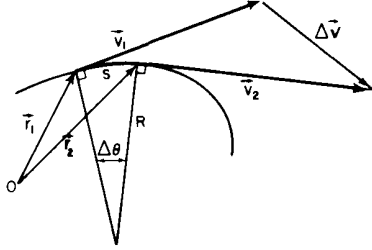


Fig. 11-7. A curved trajectory.

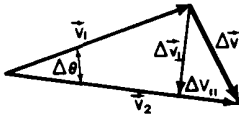


Fig. 11-8. Diagram for calculating the acceleration.

and by x' , y' , z' in another coordinate system, surely the distance $r = |\mathbf{r}|$ would be the same in both. Now

$$r = \sqrt{x^2 + y^2 + z^2}$$

and also

$$r' = \sqrt{x'^2 + y'^2 + z'^2}.$$

So what we wish to verify is that these two quantities are equal. It is much more convenient not to bother to take the square root, so let us talk about the square of the distance; that is, let us find out whether

$$x^2 + y^2 + z^2 = x'^2 + y'^2 + z'^2. \quad (11.17)$$

It had better be—and if we substitute Eq. (11.5) we do indeed find that it is. So we see that there are other kinds of equations which are true for any two coordinate systems.

Something new is involved. We can produce a new quantity, a function of x , y , and z , called a *scalar function*, a quantity which has no direction but which is the same in both systems. Out of a vector we can make a scalar. We have to find a general rule for that. It is clear what the rule is for the case just considered: add the squares of the components. Let us now define a new thing, which we call $\mathbf{a} \cdot \mathbf{a}$. This is not a vector, but a scalar; it is a number that is the same in all coordinate systems, and it is defined to be the sum of the squares of the three components of the vector:

$$\mathbf{a} \cdot \mathbf{a} = a_x^2 + a_y^2 + a_z^2. \quad (11.18)$$

Now you say, “But with what axes?” It does not depend on the axes, the answer is the same in *every* set of axes. So we have a new *kind* of quantity, a new *invariant* or *scalar* produced by one vector “squared.” If we now define the following quantity for any two vectors $\mathbf{a}$ and $\mathbf{b}$:

$$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z, \quad (11.19)$$

we find that this quantity, calculated in the primed and unprimed systems, also stays the same. To prove it we note that it is true of $\mathbf{a} \cdot \mathbf{a}$, $\mathbf{b} \cdot \mathbf{b}$, and $\mathbf{c} \cdot \mathbf{c}$, where $\mathbf{c} = \mathbf{a} + \mathbf{b}$. Therefore the sum of the squares $(a_x + b_x)^2 + (a_y + b_y)^2 + (a_z + b_z)^2$ will be invariant:

$$(a_x + b_x)^2 + (a_y + b_y)^2 + (a_z + b_z)^2 = (a_{x'} + b_{x'})^2 + (a_{y'} + b_{y'})^2 + (a_{z'} + b_{z'})^2. \quad (11.20)$$

If both sides of this equation are expanded, there will be cross products of just the type appearing in Eq. (11.19), as well as the sums of squares of the components of $\mathbf{a}$ and $\mathbf{b}$. The invariance of terms of the form of Eq. (11.18) then leaves the cross product terms (11.19) invariant also.

The quantity $\mathbf{a} \cdot \mathbf{b}$ is called the *scalar product* of two vectors, $\mathbf{a}$ and $\mathbf{b}$, and it has many interesting and useful properties. For instance, it is easily proved that

$$\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}. \quad (11.21)$$

Also, there is a simple geometrical way to calculate $\mathbf{a} \cdot \mathbf{b}$, without having to calculate the components of $\mathbf{a}$ and $\mathbf{b}$: $\mathbf{a} \cdot \mathbf{b}$ is the product of the length of $\mathbf{a}$ and the length of $\mathbf{b}$ times the cosine of the angle between them. Why? Suppose that we choose a special coordinate system in which the x -axis lies along $\mathbf{a}$; in those circumstances, the only component of $\mathbf{a}$ that will be there is a_x , which is of course the whole length of $\mathbf{a}$. Thus Eq. (11.19) reduces to $\mathbf{a} \cdot \mathbf{b} = a_x b_x$ for this case, and this is the length of $\mathbf{a}$ times the component of $\mathbf{b}$ in the direction of $\mathbf{a}$, that is, $b \cos \theta$:

$$\mathbf{a} \cdot \mathbf{b} = ab \cos \theta.$$

Therefore, in that special coordinate system, we have proved that $\mathbf{a} \cdot \mathbf{b}$ is the

length of $\mathbf{a}$ times the length of $\mathbf{b}$ times $\cos \theta$. But *if it is true in one coordinate system, it is true in all*, because $\mathbf{a} \cdot \mathbf{b}$ is independent of the coordinate system; that is our argument.

What good is the dot product? Are there any cases in physics where we need it? Yes, we need it all the time. For instance, in Chapter 4 the kinetic energy was called $\frac{1}{2}mv^2$, but if the object is moving in space it should be the velocity squared in the x -direction, the y -direction, and the z -direction, and so the formula for kinetic energy according to vector analysis is

$$\text{K.E.} = \frac{1}{2}m(\mathbf{v} \cdot \mathbf{v}) = \frac{1}{2}m(v_x^2 + v_y^2 + v_z^2). \quad (11.22)$$

Energy does not have direction. Momentum has direction; it is a vector, and it is the mass times the velocity vector.

Another example of a dot product is the work done by a force when something is pushed from one place to the other. We have not yet defined work, but it is equivalent to the energy change, the weights lifted, when a force $\mathbf{F}$ acts through a distance $\mathbf{s}$:

$$\text{Work} = \mathbf{F} \cdot \mathbf{s} \quad (11.23)$$

It is sometimes very convenient to talk about the component of a vector in a certain direction (say the vertical direction because that is the direction of gravity). For such purposes, it is useful to invent what we call a *unit vector* in the direction that we want to study. By a unit vector we mean one whose dot product with itself is equal to unity. Let us call this unit vector $\mathbf{i}$; then $\mathbf{i} \cdot \mathbf{i} = 1$. Then, if we want the component of some vector in the direction of $\mathbf{i}$, we see that the dot product $\mathbf{a} \cdot \mathbf{i}$ will be $a \cos \theta$, i.e., the component of $\mathbf{a}$ in the direction of $\mathbf{i}$. This is a nice way to get the component; in fact, it permits us to get *all* the components and to write a rather amusing formula. Suppose that in a given system of coordinates, x , y , and z , we invent three vectors: $\mathbf{i}$, a unit vector in the direction x ; $\mathbf{j}$, a unit vector in the direction y ; and $\mathbf{k}$, a unit vector in the direction z . Note first that $\mathbf{i} \cdot \mathbf{i} = 1$. What is $\mathbf{i} \cdot \mathbf{j}$? When two vectors are at right angles, their dot product is zero. Thus

$$\begin{aligned} \mathbf{i} \cdot \mathbf{i} &= 1 \\ \mathbf{i} \cdot \mathbf{j} &= 0 & \mathbf{j} \cdot \mathbf{j} &= 1 \\ \mathbf{i} \cdot \mathbf{k} &= 0 & \mathbf{j} \cdot \mathbf{k} &= 0 & \mathbf{k} \cdot \mathbf{k} &= 1 \end{aligned} \quad (11.24)$$

Now with these definitions, any vector whatsoever can be written this way:

$$\mathbf{a} = a_x \mathbf{i} + a_y \mathbf{j} + a_z \mathbf{k}. \quad (11.25)$$

By this means we can go from the components of a vector to the vector itself.

This discussion of vectors is by no means complete. However, rather than try to go more deeply into the subject now, we shall first learn to use in physical situations some of the ideas so far discussed. Then, when we have properly mastered this basic material, we shall find it easier to penetrate more deeply into the subject without getting too confused. We shall later find that it is useful to define another kind of product of two vectors, called the vector product, and written as $\mathbf{a} \times \mathbf{b}$. However, we shall undertake a discussion of such matters in a later chapter.

Characteristics of Force

12-1 What is a force?

Although it is interesting and worth while to study the physical laws simply because they help us to understand and to use nature, one ought to stop every once in a while and think, "What do they really mean?" The meaning of any statement is a subject that has interested and troubled philosophers from time immemorial, and the meaning of physical laws is even more interesting, because it is generally believed that these laws represent some kind of real knowledge. The meaning of knowledge is a deep problem in philosophy, and it is always important to ask, "What does it mean?"

Let us ask, "What is the meaning of the physical laws of Newton, which we write as $F = ma$? What is the meaning of force, mass, and acceleration?" Well, we can intuitively sense the meaning of mass, and we can *define* acceleration if we know the meaning of position and time. We shall not discuss those meanings, but shall concentrate on the new concept of *force*. The answer is equally simple: "If a body is accelerating, then there is a force on it." That is what Newton's laws say, so the most precise and beautiful definition of force imaginable might simply be to say that force is the mass of an object times the acceleration. Suppose we have a law which says that the conservation of momentum is valid if the sum of all the external forces is zero; then the question arises, "What does it *mean*, that the sum of all the external forces is zero?" A pleasant way to define that statement would be: "When the total momentum is a constant, then the sum of the external forces is zero." There must be something wrong with that, because it is just not saying anything new. If we have discovered a fundamental law, which asserts that the force is equal to the mass times the acceleration, and then *define* the force to be the mass times the acceleration, we have found out nothing. We could also define force to mean that a moving object with no force acting on it continues to move with constant velocity in a straight line. If we then observe an object *not* moving in a straight line with a constant velocity, we might say that there is a force on it. Now such things certainly cannot be the content of physics, because they are definitions going in a circle. The Newtonian statement above, however, seems to be a most precise definition of force, and one that appeals to the mathematician; nevertheless, it is completely useless, because no prediction whatsoever can be made from a definition. One might sit in an armchair all day long and define words at will, but to find out what happens when two balls push against each other, or when a weight is hung on a spring, is another matter altogether, because the way the bodies *behave* is something completely outside any choice of definitions.

For example, if we were to choose to say that an object left to itself keeps its position and does not move, then when we see something drifting, we could say that must be due to a "gorce"—a gorce is the rate of change of position. Now we have a wonderful new law, everything stands still except when a gorce is acting. You see, that would be analogous to the above definition of force, and it would contain no information. The real content of Newton's laws is this: that the force is supposed to have some *independent properties*, in addition to the law $F = ma$; but the *specific* independent properties that the force has were not completely described by Newton or by anybody else, and therefore the physical law $F = ma$ is an incomplete law. It implies that if we study the mass times the acceleration and call the product the force, i.e., if we study the characteristics of force as a program

12-1 What is a force?

12-2 Friction

12-3 Molecular forces

12-4 Fundamental forces. Fields

12-5 Pseudo forces

12-6 Nuclear forces

of interest, then we shall find that forces have some simplicity; the law is a good program for analyzing nature, it is a suggestion that the forces will be simple.

Now the first example of such forces was the complete law of gravitation, which was given by Newton, and in stating the law he answered the question, "What is the force?" If there were nothing but gravitation, then the combination of this law and the force law (second law of motion) would be a complete theory, but there is much more than gravitation, and we want to use Newton's laws in many different situations. Therefore in order to proceed we have to tell something about the properties of force.

For example, in dealing with force the tacit assumption is always made that the force is equal to zero unless some physical body is present, that if we find a force that is not equal to zero we also find something in the neighborhood that is a source of the force. This assumption is entirely different from the case of the "gorce" that we introduced above. One of the most important characteristics of force is that it has a material origin, and this is *not* just a definition.

Newton also gave one rule about the force: that the forces between interacting bodies are equal and opposite—action equals reaction; that rule, it turns out, is not exactly true. In fact, the law $F = ma$ is not exactly true; if it were a definition we should have to say that it is *always* exactly true; but it is not.

The student may object, "I do not like this imprecision, I should like to have everything defined exactly; in fact, it says in some books that any science is an exact subject, in which *everything* is defined." If you insist upon a precise definition of force, you will never get it! First, because Newton's Second Law is not exact, and second, because in order to understand physical laws you must understand that they are all some kind of approximation.

Any simple idea is approximate; as an illustration, consider an object, . . . what *is* an object? Philosophers are always saying, "Well, just take a chair for example." The moment they say that, you know that they do not know what they are talking about any more. What *is* a chair? Well, a chair is a certain thing over there . . . certain?, how certain? The atoms are evaporating from it from time to time—not many atoms, but a few—dirt falls on it and gets dissolved in the paint; so to define a chair precisely, to say exactly which atoms are chair, and which atoms are air, or which atoms are dirt, or which atoms are paint that belongs to the chair is impossible. So the mass of a chair can be defined only approximately. In the same way, to define the mass of a single object is impossible, because there are not any single, left-alone objects in the world—every object is a mixture of a lot of things, so we can deal with it only as a series of approximations and idealizations.

The trick is the idealizations. To an excellent approximation of perhaps one part in 10^{10} , the number of atoms in the chair does not change in a minute, and if we are not too precise we may idealize the chair as a definite thing; in the same way we shall learn about the characteristics of force, in an ideal fashion, if we are not too precise. One may be dissatisfied with the approximate view of nature that physics tries to obtain (the attempt is always to increase the accuracy of the approximation), and may prefer a mathematical definition; but mathematical definitions can never work in the real world. A mathematical definition will be good for mathematics, in which all the logic can be followed out completely, but the physical world is complex, as we have indicated in a number of examples, such as those of the ocean waves and a glass of wine. When we try to isolate pieces of it, to talk about one mass, the wine and the glass, how can we know which is which, when one dissolves in the other? The forces on a single thing already involve approximation, and if we have a system of discourse about the real world, then that system, at least for the present day, must involve approximations of some kind.

This system is quite unlike the case of mathematics, in which everything can be defined, and then we do not *know* what we are talking about. In fact, the glory of mathematics is that *we do not have to say what we are talking about*. The glory is that the laws, the arguments, and the logic are independent of what "it" is. If we have any other set of objects that obey the same system of axioms as Euclid's

geometry, then if we make new definitions and follow them out with correct logic, all the consequences will be correct, and it makes no difference what the subject was. In nature, however, when we draw a line or establish a line by using a light beam and a theodolite, as we do in surveying, are we measuring a line in the sense of Euclid? No, we are making an approximation; the cross hair has some width, but a geometrical line has no width, and so, whether Euclidean geometry can be used for surveying or not is a physical question, not a mathematical question. However, from an experimental standpoint, not a mathematical standpoint, we need to know whether the laws of Euclid apply to the kind of geometry that we use in measuring land; so we make a hypothesis that it does, and it works pretty well; but it is not precise, because our surveying lines are not really geometrical lines. Whether or not those lines of Euclid, which are really abstract, apply to the lines of experience is a question for experience; it is not a question that can be answered by sheer reason.

In the same way, we cannot just call $F = ma$ a definition, deduce everything purely mathematically, and make mechanics a mathematical theory, when mechanics is a description of nature. By establishing suitable postulates it is always possible to make a system of mathematics, just as Euclid did, but we cannot make a mathematics of the world, because sooner or later we have to find out whether the axioms are valid for the objects of nature. Thus we immediately get involved with these complicated and "dirty" objects of nature, but with approximations ever increasing in accuracy.

12-2 Friction

The foregoing considerations show that a true understanding of Newton's laws requires a discussion of forces, and it is the purpose of this chapter to introduce such a discussion, as a kind of completion of Newton's laws. We have already studied the definitions of acceleration and related ideas, but now we have to study the properties of force, and this chapter, unlike the previous chapters, will not be very precise, because forces are quite complicated.

To begin with a particular force, let us consider the drag on an airplane flying through the air. What is the law for that force? (Surely there is a law for every force, we *must* have a law!) One can hardly think that the law for that force will be simple. Try to imagine what makes a drag on an airplane flying through the air—the air rushing over the wings, the swirling in the back, the changes going on around the fuselage, and many other complications, and you see that there is not going to be a simple law. On the other hand, it is a remarkable fact that the drag force on an airplane is approximately a constant times the square of the velocity, or $F \sim cv^2$.

Now what is the status of such a law, is it analogous to $F = ma$? Not at all, because in the first place this law is an empirical thing that is obtained roughly by tests in a wind tunnel. You say, "Well $F = ma$ might be empirical too." That is not the reason that there is a difference. The difference is not that it is empirical, but that, as we understand nature, this law is the result of an enormous complexity of events and is not, fundamentally, a simple thing. If we continue to study it more and more, measuring more and more accurately, the law will continue to become *more* complicated, not *less*. In other words, as we study this law of the drag on an airplane more and more closely, we find out that it is "falser" and "falser," and the more deeply we study it, and the more accurately we measure, the more complicated the truth becomes; so in that sense we consider it not to result from a simple, fundamental process, which agrees with our original surmise. For example, if the velocity is extremely low, so low that an ordinary airplane is not flying, as when the airplane is dragged slowly through the air, then the law changes, and the drag friction depends more nearly linearly on the velocity. To take another example, the frictional drag on a ball or a bubble or anything that is moving slowly through a viscous liquid like honey, is proportional to the velocity, but for motion so fast that the fluid swirls around (honey does not but water and air do) then the drag becomes more nearly proportional to the square of the velocity ($F = cv^2$), and

if the velocity continues to increase, then even this law begins to fail. People who say, "Well the coefficient changes slightly," are dodging the issue. Second, there are other great complications: can this force on the airplane be divided or analyzed as a force on the wings, a force on the front, and so on? Indeed, this can be done, if we are concerned about the torques here and there, but then we have to get special laws for the force on the wings, and so on. It is an amazing fact that the force on a wing depends upon the other wing: in other words, if we take the airplane apart and put just one wing in the air, then the force is not the same as if the rest of the plane were there. The reason, of course, is that some of the wind that hits the front goes around to the wings and changes the force on the wings. It seems a miracle that there is such a simple, rough, empirical law that can be used in the design of airplanes, but this law is not in the same class as the *basic* laws of physics, and further study of it will only make it more and more complicated. A study of how the coefficient c depends on the shape of the front of the airplane is, to put it mildly, frustrating. There just is no simple law for determining the coefficient in terms of the shape of the airplane. In contrast, the law of gravitation is simple, and further study only indicates its greater simplicity.

We have just discussed two cases of friction, resulting from fast movement in air and slow movement in honey. There is another kind of friction, called dry friction or sliding friction, which occurs when one solid body slides on another. In this case a force is needed to maintain motion. This is called a frictional force, and its origin, also, is a very complicated matter. Both surfaces of contact are irregular, on an atomic level. There are many points of contact where the atoms seem to cling together, and then, as the sliding body is pulled along, the atoms snap apart and vibration ensues; something like that has to happen. Formerly the mechanism of this friction was thought to be very simple, that the surfaces were merely full of irregularities and the friction originated in lifting the slider over the bumps; but this cannot be, for there is no loss of energy in that process, whereas power is in fact consumed. The mechanism of power loss is that as the slider snaps over the bumps, the bumps deform and then generate waves and atomic motions and, after a while, heat, in the two bodies. Now it is very remarkable that again, empirically, this friction can be described approximately by a simple law. This law is that the force needed to overcome friction and to drag one object over another depends upon the normal force (i.e., perpendicular to the surface) between the two surfaces that are in contact. Actually, to a fairly good approximation, the frictional force is proportional to this normal force, and has a more or less constant coefficient; that is,

$$F = \mu N, \quad (12.1)$$

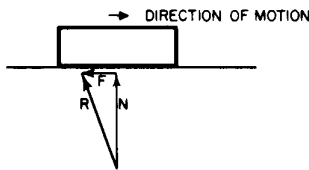


Fig. 12-1. The relation between frictional force and the normal force for sliding contact.

where μ is called the *coefficient of friction* (Fig. 12-1). Although this coefficient is not exactly constant, the formula is a good empirical rule for judging approximately the amount of force that will be needed in certain practical or engineering circumstances. If the normal force or the speed of motion gets too big, the law fails because of the excessive heat generated. It is important to realize that each of these empirical laws has its limitations, beyond which it does not really work.

That the formula $F = \mu N$ is approximately correct can be demonstrated by a simple experiment. We set up a plane, inclined at a small angle θ , and place a block of weight W on the plane. We then tilt the plane at a steeper angle, until the block just begins to slide from its own weight. The component of the weight downward along the plane is $W \sin \theta$, and this must equal the frictional force F when the block is sliding uniformly. The component of the weight normal to the plane is $W \cos \theta$, and this is the normal force N . With these values, the formula becomes $W \sin \theta = \mu W \cos \theta$, from which we get $\mu = \sin \theta / \cos \theta = \tan \theta$. If this law were exactly true, an object would start to slide at some definite inclination. If the same block is loaded by putting extra weight on it, then, although W is increased, all the forces in the formula are increased in the same proportion, and W cancels out. If μ stays constant, the loaded block will slide again at the same slope. When the angle θ is determined by trial with the original weight, it is found

that with the greater weight the block will slide at about the same angle. This will be true even when one weight is many times as great as the other, and so we conclude that the coefficient of friction is independent of the weight.

In performing this experiment it is noticeable that when the plane is tilted at about the correct angle θ , the block does not slide steadily but in a halting fashion. At one place it may stop, at another it may move with acceleration. This behavior indicates that the coefficient of friction is only roughly a constant, and varies from place to place along the plane. The same erratic behavior is observed whether the block is loaded or not. Such variations are caused by different degrees of smoothness or hardness of the plane, and perhaps dirt, oxides, or other foreign matter. The tables that list purported values of μ for "steel on steel," "copper on copper," and the like, are all false, because they ignore the factors mentioned above, which really determine μ . The friction is never due to "copper on copper," etc., but to the impurities clinging to the copper.

In experiments of the type described above, the friction is nearly independent of the velocity. Many people believe that the friction to be overcome to get something started (static friction) exceeds the force required to keep it sliding (sliding friction), but with dry metals it is very hard to show any difference. The opinion probably arises from experiences where small bits of oil or lubricant are present, or where blocks, for example, are supported by springs or other flexible supports so that they appear to bind.

It is quite difficult to do accurate quantitative experiments in friction, and the laws of friction are still not analyzed very well, in spite of the enormous engineering value of an accurate analysis. Although the law $F = \mu N$ is fairly accurate once the surfaces are standardized, the reason for this form of the law is not really understood. To show that the coefficient μ is nearly independent of velocity requires some delicate experimentation, because the apparent friction is much reduced if the lower surface vibrates very fast. When the experiment is done at very high speed, care must be taken that the objects do not vibrate relative to one another, since apparent decreases of the friction at high speed are often due to vibrations. At any rate, this friction law is another of those semiempirical laws that are not thoroughly understood, and in view of all the work that has been done it is surprising that more understanding of this phenomenon has not come about. At the present time, in fact, it is impossible even to estimate the coefficient of friction between two substances.

It was pointed out above that attempts to measure μ by sliding pure substances such as copper on copper will lead to spurious results, because the surfaces in contact are not pure copper, but are mixtures of oxides and other impurities. If we try to get absolutely pure copper, if we clean and polish the surfaces, outgas the materials in a vacuum, and take every conceivable precaution, we still do not get μ . For if we tilt the apparatus even to a vertical position, the slider will not fall off—the two pieces of copper stick together! The coefficient μ , which is ordinarily less than unity for reasonably hard surfaces, becomes several times unity! The reason for this unexpected behavior is that when the atoms in contact are all of the same kind, there is no way for the atoms to "know" that they are in different pieces of copper. When there are other atoms, in the oxides and greases and more complicated thin surface layers of contaminants in between, the atoms "know" when they are not on the same part. When we consider that it is forces between atoms that hold the copper together as a solid, it should become clear that it is impossible to get the right coefficient of friction for pure metals.

The same phenomenon can be observed in a simple home-made experiment with a flat glass plate and a glass tumbler. If the tumbler is placed on the plate and pulled along with a loop of string, it slides fairly well and one can feel the coefficient of friction; it is a little irregular, but it is a coefficient. If we now wet the glass plate and the bottom of the tumbler and pull again, we find that it binds, and if we look closely we shall find scratches, because the water is able to lift the grease and the other contaminants off the surface, and then we really have a glass-to-glass contact; this contact is so good that it holds tight and resists separation so much that the glass is torn apart; that is, it makes scratches.

12-3 Molecular forces

We shall next discuss the characteristics of molecular forces. These are forces between the atoms, and are the ultimate origin of friction. Molecular forces have never been satisfactorily explained on a basis of classical physics; it takes quantum mechanics to understand them fully. Empirically, however, the force between atoms is illustrated schematically in Fig. 12-2, where the force F between two atoms is plotted as a function of the distance r between them. There are different cases: in the water molecule, for example, the negative charges sit more on the oxygen, and the mean positions of the negative charges and of the positive charges are not at the same point; consequently, another molecule nearby feels a relatively large force, which is called a dipole-dipole force. However, for many systems the charges are very much better balanced, in particular for oxygen gas, which is perfectly symmetrical. In this case, although the minus charges and the plus charges are dispersed over the molecule, the distribution is such that the center of the minus charges and the center of the plus charges coincide. A molecule where the centers do not coincide is called a polar molecule, and charge times the separation between centers is called the dipole moment. A nonpolar molecule is one in which the centers of the charges coincide. For all nonpolar molecules, in which all the electrical forces are neutralized, it nevertheless turns out that the force at very large distances is an attraction and varies inversely as the seventh power of the distance, or $F = k/r^7$, where k is a constant that depends on the molecules. Why this is we shall learn only when we learn quantum mechanics. When there are dipoles the forces are greater. When atoms or molecules get too close they repel with a very large repulsion; that is what keeps us from falling through the floor!

These molecular forces can be demonstrated in a fairly direct way: one of these is the friction experiment with a sliding glass tumbler; another is to take two very carefully ground and lapped surfaces which are very accurately flat, so that the surfaces can be brought very close together. An example of such surfaces is the Johansson blocks that are used in machine shops as standards for making accurate length measurements. If one such block is slid over another very carefully and the upper one is lifted, the other one will adhere and also be lifted by the molecular forces, exemplifying the direct attraction between the atoms on one block for the atoms on the other block.

Nevertheless these molecular forces of attraction are still not fundamental in the sense that gravitation is fundamental; they are due to the vastly complex interactions of all the electrons and nuclei in one molecule with all the electrons and nuclei in another. Any simple-looking formula we get represents a summation of complications, so we still have not got the fundamental phenomena.

Since the molecular forces attract at large distances and repel at short distances, as shown in Fig. 12-2, we can make up solids in which all the atoms are held together by their attractions and held apart by the repulsion that sets in when they are too close together. At a certain distance d (where the graph in Fig. 12-2 crosses the axis) the forces are zero, which means that they are all balanced, so that the molecules stay that distance apart from one another. If the molecules are pushed closer together than the distance d they all show a repulsion, represented by the portion of the graph above the r -axis. To push the molecules only slightly closer together requires a great force, because the molecular repulsion rapidly becomes very great at distances less than d . If the molecules are pulled slightly apart there is a slight attraction, which increases as the separation increases. If they are pulled sufficiently hard, they will separate permanently—the bond is broken.

If the molecules are pushed only a *very small* distance closer, or pulled only a *very small* distance farther than d , the corresponding distance along the curve of Fig. 12-2 is also very small, and can then be approximated by a straight line. Therefore, in many circumstances, if the displacement is not too great *the force is proportional to the displacement*. This principle is known as Hooke's law, or the law of elasticity, which says that the force in a body which tries to restore the body

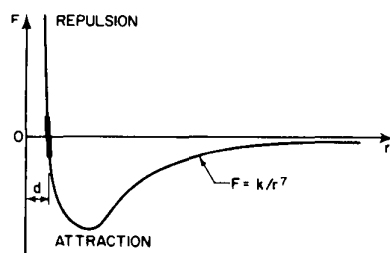


Fig. 12-2. The force between two atoms as a function of their distance of separation.

to its original condition when it is distorted is proportional to the distortion. This law, of course, holds true only if the distortion is relatively small; when it gets too large the body will be torn apart or crushed, depending on the kind of distortion. The amount of force for which Hooke's law is valid depends upon the material; for instance, for dough or putty the force is very small, but for steel it is relatively large. Hooke's law can be nicely demonstrated with a long coil spring, made of steel and suspended vertically. A suitable weight hung on the lower end of the spring produces a tiny twist throughout the length of the wire, which results in a small vertical deflection in each turn and adds up to a large displacement if there are many turns. If the total elongation produced, say, by a 100-gram weight, is measured, it is found that additional weights of 100 grams will each produce an additional elongation that is very nearly equal to the stretch that was measured for the first 100 grams. This constant ratio of force to displacement begins to change when the spring is overloaded, i.e., Hooke's law no longer holds.

12-4 Fundamental forces. Fields

We shall now discuss the only remaining forces that are fundamental. We call them fundamental in the sense that their laws are fundamentally simple. We shall first discuss electrical force. Objects carry electrical charges which consist simply of electrons or protons. If any two bodies are electrically charged, there is an electrical force between them, and if the magnitudes of the charges are q_1 and q_2 , respectively, the force varies inversely as the square of the distance between the charges, or $F = (\text{const}) q_1 q_2 / r^2$. For unlike charges, this law is like the law of gravitation, but for *like* charges the force is repulsive and the sign (direction) is reversed. The charges q_1 and q_2 can be intrinsically either positive or negative, and in any specific application of the formula the direction of the force will come out right if the q 's are given the proper plus or minus sign; the force is directed along the line between the two charges. The constant in the formula depends, of course, upon what units are used for the force, the charge, and the distance. In current practice the charge is measured in coulombs, the distance in meters, and the force in newtons. Then, in order to get the force to come out properly in newtons, the constant (which for historical reasons is written $1/4\pi\epsilon_0$) takes the numerical value

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ coul}^2/\text{newton} \cdot \text{m}^2$$

or

$$1/4\pi\epsilon_0 = 8.99 \times 10^9 \text{ n} \cdot \text{m}^2/\text{coul}^2.$$

Thus the force law for static charges is

$$\mathbf{F} = q_1 q_2 \mathbf{r} / 4\pi\epsilon_0 r^3. \quad (12.2)$$

In nature, the most important charge of all is the charge on a single electron, which is 1.60×10^{-19} coulomb. In working with electrical forces between fundamental particles rather than with large charges, many people prefer the combination $(q_{e1})^2 / 4\pi\epsilon_0$, in which q_{e1} is defined as the charge on an electron. This combination occurs frequently, and to simplify calculations it has been defined by the symbol e^2 ; its numerical value in the mks system of units turns out to be $(1.52 \times 10^{-14})^2$. The advantage of using the constant in this form is that the force between two electrons in newtons can then be written simply as e^2/r^2 , with r in meters, without all the individual constants. Electrical forces are much more complicated than this simple formula indicates, since the formula gives the force between two objects only when the objects are standing still. We shall consider the more general case shortly.

In the analysis of forces of the more fundamental kinds (not such forces as friction, but the electrical force or the gravitational force), an interesting and very important concept has been developed. Since at first sight the forces are very much more complicated than is indicated by the inverse-square laws and these laws hold true only when the interacting bodies are standing still, an improved

method is needed to deal with the very complex forces that ensue when the bodies start to move in a complicated way. Experience has shown that an approach known as the concept of a "field" is of great utility for the analysis of forces of this type. To illustrate the idea for, say, electrical force, suppose we have two electrical charges, q_1 and q_2 , located at points P and R respectively. Then the force between the charges is given by

$$\mathbf{F} = q_1 q_2 \mathbf{r} / r^3. \quad (12.3)$$

To analyze this force by means of the field concept, we say that the charge q_1 at P produces a "condition" at R , such that when the charge q_2 is placed at R it "feels" the force. This is one way, strange perhaps, of describing it; we say that the force $\mathbf{F}$ on q_2 at R can be written in two parts. It is q_2 multiplied by a quantity $\mathbf{E}$ that would be there whether q_2 were there or not (provided we keep all the other charges in their right places). $\mathbf{E}$ is the "condition" produced by q_1 , we say, and $\mathbf{F}$ is the response of q_2 to $\mathbf{E}$. $\mathbf{E}$ is called an *electric field*, and it is a vector. The formula for the electric field $\mathbf{E}$ that is produced at R by a charge q_1 at P is the charge q_1 times the constant $1/4\pi\epsilon_0$ divided by r^2 (r is the distance from P to R), and it is acting in the direction of the radius vector (the radius vector $\mathbf{r}$ divided by its own length). The expression for $\mathbf{E}$ is thus

$$\mathbf{E} = q_1 \mathbf{r} / 4\pi\epsilon_0 r^3. \quad (12.4)$$

We then write

$$\mathbf{F} = q_2 \mathbf{E}, \quad (12.5)$$

which expresses the force, the field, and the charge in the field. What is the point of all this? The point is to divide the analysis into two parts. One part says that something *produces* a field. The other part says that something is *acted on* by the field. By allowing us to look at the two parts independently, this separation of the analysis simplifies the calculation of a problem in many situations. If many charges are present, we first work out the total electric field produced at R by all the charges, and then, knowing the charge that is placed at R , we find the force on it.

In the case of gravitation, we can do exactly the same thing. In this case, where the force $\mathbf{F} = -Gm_1 m_2 \mathbf{r} / r^3$, we can make an analogous analysis, as follows: the force on a body in a gravitational field is the mass of that body times the field $\mathbf{C}$. The force on m_2 is the mass m_2 times the field $\mathbf{C}$ produced by m_1 ; that is, $\mathbf{F} = m_2 \mathbf{C}$. Then the field $\mathbf{C}$ produced by a body of mass m_1 is $\mathbf{C} = -Gm_1 \mathbf{r} / r^3$ and it is directed radially, as in the electrical case.

In spite of how it might at first seem, this separation of one part from another is not a triviality. It would be trivial, just another way of writing the same thing, if the laws of force were simple, but the laws of force are so complicated that it turns out that the fields have a reality that is almost independent of the objects which create them. One can do something like shake a charge and produce an effect, a field, at a distance; if one then stops moving the charge, the field keeps track of all the past, because the interaction between two particles is not instantaneous. It is desirable to have some way to remember what happened previously. If the force upon some charge depends upon where another charge was yesterday, which it does, then we need machinery to keep track of what went on yesterday, and that is the character of a field. So when the forces get more complicated, the field becomes more and more real, and this technique becomes less and less of an artificial separation.

In analyzing forces by the use of fields, we need two kinds of laws pertaining to fields. The first is the response to a field, and that gives the equations of motion. For example, the law of response of a mass to a gravitational field is that the force is equal to the mass times the gravitational field; or, if there is also a charge on the body, the response of the charge to the electric field equals the charge times the electric field. The second part of the analysis of nature in these situations is to formulate the laws which determine the strength of the field and how it is produced. These laws are sometimes called the *field equations*. We shall learn more about them in due time, but shall write down a few things about them now.

First, the most remarkable fact of all, which is true exactly and which can be easily understood, is that the total electric field produced by a number of sources is the vector sum of the electric fields produced by the first source, the second source, and so on. In other words, if we have numerous charges making a field, and if all by itself one of them would make the field E_1 , another would make the field E_2 , and so on, then we merely add the vectors to get the total field. This principle can be expressed as

$$\mathbf{E} = \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_3 + \cdots \quad (12.6)$$

or, in view of the definition given above,

$$\mathbf{E} = \sum_i \frac{q_i \mathbf{r}_i}{4\pi\epsilon_0 r_i^3} \quad (12.7)$$

Can the same methods be applied to gravitation? The force between two masses m_1 and m_2 was expressed by Newton as $\mathbf{F} = Gm_1m_2\mathbf{r}/r^3$. But according to the field concept, we may say that m_1 creates a field $\mathbf{C}$ in all the surrounding space, such that the force on m_2 is given by

$$\mathbf{F} = m_2\mathbf{C}. \quad (12.8)$$

By complete analogy with the electrical case,

$$\mathbf{C} = -Gm_1\mathbf{r}_1/r_1^3 \quad (12.9)$$

and the gravitational field produced by several masses is

$$\mathbf{C} = \mathbf{C}_1 + \mathbf{C}_2 + \mathbf{C}_3 + \cdots \quad (12.10)$$

In Chapter 7, in working out a case of planetary motion, we used this principle in essence. We simply added all the force vectors to get the resultant force on a planet. If we divide out the mass of the planet in question, we get Eq. (12.10).

Equations (12.6) and (12.10) express what is known as *the principle of superposition* of fields. This principle states that the total field due to all the sources is the sum of the fields due to each source. So far as we know today, for electricity this is an absolutely guaranteed law, which is true even when the force law is complicated because of the motions of the charges. There are apparent violations, but more careful analysis has always shown these to be due to the overlooking of certain moving charges. However, although the principle of superposition applies exactly for electrical forces, it is not exact for gravity if the field is too strong, and Newton's equation (12.10) is only approximate, according to Einstein's gravitational theory.

Closely related to electrical force is another kind, called magnetic force, and this too is analyzed in terms of a field. Some of the qualitative relations between electrical and magnetic forces can be illustrated by an experiment with an electron-ray tube (Fig. 12-3). At one end of such a tube is a source that emits a stream of electrons. Within the tube are arrangements for accelerating the electrons to a high speed and sending some of them in a narrow beam to a fluorescent screen at the other end of the tube. A spot of light glows in the center of the screen where the electrons strike, and this enables us to trace the electron path. On the way to the screen the electron beam passes through a narrow space between a pair of parallel metal plates, which are arranged, say, horizontally. A voltage can be applied across the plates, so that either plate can be made negative at will. When such a voltage is present, there is an electric field between the plates.

The first part of the experiment is to apply a negative voltage to the lower plate, which means that extra electrons have been placed on the lower plate. Since like charges repel, the light spot on the screen instantly shifts upward. (We could also say this in another way—that the electrons “felt” the field, and responded by deflecting upward.) We next reverse the voltage, making the *upper* plate negative. The light spot on the screen now jumps below the center, showing that the electrons in the beam were repelled by those in the plate above them. (Or we could say again

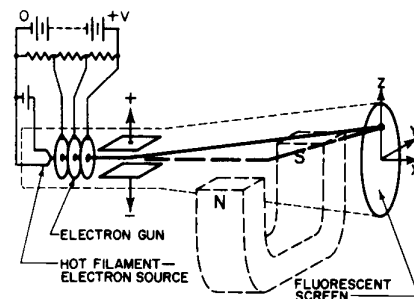


Fig. 12-3. An electron-beam tube.

that the electrons had “responded” to the field, which is now in the reverse direction.)

The second part of the experiment is to disconnect the voltage from the plates and test the effect of a magnetic field on the electron beam. This is done by means of a horseshoe magnet, whose poles are far enough apart to more or less straddle the tube. Suppose we hold the magnet below the tube in the same orientation as the letter U, with its poles up and part of the tube in between. We note that the light spot is deflected, say, upward, as the magnet approaches the tube from below. So it appears that the magnet repels the electron beam. However, it is not that simple, for if we invert the magnet without reversing the poles side-for-side, and now approach the tube from above, the spot still moves *upward*, so the electron beam is *not* repelled; instead, it appears to be attracted this time. Now we start again, restoring the magnet to its original U orientation and holding it below the tube, as before. Yes, the spot is still deflected upward; but now turn the magnet 180 degrees around a vertical axis, so that it is still in the U position but the poles are reversed side-for-side. Behold, the spot now jumps downward, and stays down, even if we invert the magnet and approach from above, as before.

To understand this peculiar behavior, we have to have a new combination of forces. We explain it thus: Across the magnet from one pole to the other there is a *magnetic field*. This field has a direction which is always away from one particular pole (which we could mark) and toward the other. Inverting the magnet did not change the direction of the field, but reversing the poles side-for-side did reverse its direction. For example, if the electron velocity were horizontal in the x -direction and the magnetic field were also horizontal but in the y -direction, the magnetic force *on the moving electrons* would be in the z -direction, i.e., up or down, depending on whether the field was in the positive or negative y -direction.

Although we shall not at the present time give the correct law of force between charges moving in an arbitrary manner, one relative to the other, because it is too complicated, we shall give one aspect of it: the complete law of the forces *if the fields are known*. The force on a charged object depends upon its motion; if, when the object is standing still at a given place, there is some force, this is taken to be proportional to the charge, the coefficient being what we call the *electric field*. When the object moves the force may be different, and the correction, the new “piece” of force, turns out to be dependent exactly *linearly on the velocity*, but at *right angles* to $\mathbf{v}$ and to another vector quantity which we call the *magnetic induction* $\mathbf{B}$. If the components of the electric field $\mathbf{E}$ and the magnetic induction $\mathbf{B}$ are, respectively, (E_x, E_y, E_z) and (B_x, B_y, B_z) , and if the velocity $\mathbf{v}$ has the components (v_x, v_y, v_z) , then the total electric and magnetic force on a moving charge q has the components

$$\begin{aligned} F_x &= q(E_x + v_y B_z - v_z B_y), \\ F_y &= q(E_y + v_z B_x - v_x B_z), \\ F_z &= q(E_z + v_x B_y - v_y B_x). \end{aligned} \quad (12.11)$$

If, for instance, the only component of the magnetic field were B_y and the only component of the velocity were v_x , then the only term left in the magnetic force would be a force in the z -direction, at right angles to both $\mathbf{B}$ and $\mathbf{v}$.

12-5 Pseudo forces

The next kind of force we shall discuss might be called a pseudo force. In Chapter 11 we discussed the relationship between two people, Joe and Moe, who use different coordinate systems. Let us suppose that the positions of a particle as measured by Joe are x and by Moe are x' ; then the laws are as follows:

$$x = x' + s, \quad y = y', \quad z = z',$$

where s is the displacement of Moe's system relative to Joe's. If we suppose that

the laws of motion are correct for Joe, how do they look for Moe? We find first, that

$$dx/dt = dx'/dt + ds/dt.$$

Previously, we considered the case where s was constant, and we found that s made no difference in the laws of motion, since $ds/dt = 0$; ultimately, therefore, the laws of physics were the same in both systems. But another case we can take is that $s = ut$, where u is a uniform velocity in a straight line. Then s is not constant, and ds/dt is not zero, but is u , a constant. However, the acceleration d^2x/dt^2 is still the same as d^2x'/dt^2 , because $du/dt = 0$. This proves the law that we used in Chapter 10, namely, that if we move in a straight line with uniform velocity the laws of physics will look the same to us as when we are standing still. That is the Galilean transformation. But we wish to discuss the interesting case where s is still more complicated, say $s = at^2/2$. Then $ds/dt = at$ and $d^2s/dt^2 = a$, a uniform acceleration; or in a still more complicated case, the acceleration might be a function of time. This means that although the laws of force from the point of view of Joe would look like

$$m \frac{d^2x}{dt^2} = F_x,$$

the laws of force as looked upon by Moe would appear as

$$m \frac{d^2x'}{dt^2} = F_x' - ma.$$

That is, since Moe's coordinate system is accelerating with respect to Joe's, the extra term ma comes in, and Moe will have to correct his forces by that amount in order to get Newton's laws to work. In other words, here is an apparent, mysterious new force of unknown origin which arises, of course, because Moe has the wrong coordinate system. This is an example of a pseudo force; other examples occur in coordinate systems that are *rotating*.

Another example of pseudo force is what is often called "centrifugal force." An observer in a rotating coordinate system, e.g., in a rotating box, will find mysterious forces, not accounted for by any known origin of force, throwing things outward toward the walls. These forces are due merely to the fact that the observer does not have Newton's coordinate system, which is the simplest coordinate system.

Pseudo force can be illustrated by an interesting experiment in which we push a jar of water along a table, with acceleration. Gravity, of course, acts downward on the water, but because of the horizontal acceleration there is also a pseudo force acting horizontally and in a direction opposite to the acceleration. The resultant of gravity and pseudo force makes an angle with the vertical, and during the acceleration the surface of the water will be perpendicular to the resultant force, i.e., inclined at an angle with the table, with the water standing higher in the rearward side of the jar. When the push on the jar stops and the jar decelerates because of friction, the pseudo force is reversed, and the water stands higher in the forward side of the jar (Fig. 12-4).

One very important feature of pseudo forces is that they are always proportional to the masses; the same is true of gravity. The possibility exists, therefore, that *gravity itself is a pseudo force*. Is it not possible that perhaps gravitation is due simply to the fact that we do not have the right coordinate system? After all, we can always get a force proportional to the mass if we imagine that a body is accelerating. For instance, a man shut up in a box that is standing still on the earth finds himself held to the floor of the box with a certain force that is proportional to his mass. But if there were no earth at all and the box were standing still, the man inside would float in space. On the other hand, if there were no earth at all and something were *pulling* the box along with an acceleration g , then the man in the box, analyzing physics, would find a pseudo force which would pull him to the floor, just as gravity does.



Fig. 12-4. Illustration of a pseudo force

Einstein put forward the famous hypothesis that accelerations give an imitation of gravitation, that the forces of acceleration (the pseudo forces) *cannot be distinguished* from those of gravity; it is not possible to tell how much of a given force is gravity and how much is pseudo force.

It might seem all right to consider gravity to be a pseudo force, to say that we are all held down because we are accelerating upward, but how about the people in Madagascar, on the other side of the earth—are they accelerating too? Einstein found that gravity could be considered a pseudo force only at one point at a time, and was led by his considerations to suggest that the *geometry of the world* is more complicated than ordinary Euclidean geometry. The present discussion is only qualitative, and does not pretend to convey anything more than the general idea. To give a rough idea of how gravitation could be the result of pseudo forces, we present an illustration which is purely geometrical and does not represent the real situation. Suppose that we all lived in two dimensions, and knew nothing of a third. We think we are on a plane, but suppose we are really on the surface of a sphere. And suppose that we shoot an object along the ground, with no forces on it. Where will it go? It will appear to go in a straight line, but it has to remain on the surface of a sphere, where the shortest distance between two points is along a great circle; so it goes along a great circle. If we shoot another object similarly, but in another direction, it goes along another great circle. Because we think we are on a plane, we expect that these two bodies will continue to diverge linearly with time, but careful observation will show that if they go far enough they move closer together again, as though they were attracting each other. But they are *not* attracting each other—there is just something “weird” about this geometry. This particular illustration does not describe correctly the way in which Euclid’s geometry is “weird,” but it illustrates that if we distort the geometry sufficiently it is possible that all gravitation is related in some way to pseudo forces; that is the general idea of the Einsteinian theory of gravitation.

12-6 Nuclear forces

We conclude this chapter with a brief discussion of the only other known forces, which are called *nuclear forces*. These forces are within the nuclei of atoms, and although they are much discussed, no one has ever calculated the force between two nuclei, and indeed at present there is no known law for nuclear forces. These forces have a very tiny range which is just about the same as the size of the nucleus, perhaps 10^{-13} centimeter. With particles so small and at such a tiny distance, only the quantum-mechanical laws are valid, not the Newtonian laws. In nuclear analysis we no longer think in terms of forces, and in fact we can replace the force concept with a concept of the energy of interaction of two particles, a subject that will be discussed later. Any formula that can be written for nuclear forces is a rather crude approximation which omits many complications; one might be somewhat as follows: forces within a nucleus do not vary inversely as the square of the distance, but die off exponentially over a certain distance r , as expressed by $F = (1/r^2) \exp(-r/r_0)$, where the distance r_0 is of the order of 10^{-13} centimeter. In other words, the forces disappear as soon as the particles are any great distance apart, although they are very strong within the 10^{-13} centimeter range. So far as they are understood today, the laws of nuclear force are very complex; we do not understand them in any simple way, and the whole problem of analyzing the fundamental machinery behind nuclear forces is unsolved. Attempts at a solution have led to the discovery of numerous strange particles, the π -mesons, for example, but the origin of these forces remains obscure.

Work and Potential Energy (A)

13-1 Energy of a falling body

In Chapter 4 we discussed the conservation of energy. In that discussion, we did not use Newton's laws, but it is, of course, of great interest to see how it comes about that energy is in fact conserved in accordance with these laws. For clarity we shall start with the simplest possible example, and then develop harder and harder examples.

The simplest example of the conservation of energy is a vertically falling object, one that moves only in a vertical direction. An object which changes its height under the influence of gravity alone has a kinetic energy T (or K.E.) due to its motion during the fall, and a potential energy mgh , abbreviated U or P.E., whose sum is constant:

$$\frac{1}{2}mv^2 + mgh = \text{const},$$

K E P E

or

$$T + U = \text{const.} \quad (13.1)$$

Now we would like to show that this statement is true. What do we mean, show it is true? From Newton's Second Law we can easily tell how the object moves, and it is easy to find out how the velocity varies with time, namely, that it increases proportionally with the time, and that the height varies as the square of the time. So if we measure the height from a zero point where the object is stationary, it is no miracle that the height turns out to be equal to the square of the velocity times a number of constants. However, let us look at it a little more closely.

Let us find out *directly* from Newton's Second Law how the kinetic energy should change, by taking the derivative of the kinetic energy with respect to time and then using Newton's laws. When we differentiate $\frac{1}{2}mv^2$ with respect to time, we obtain

$$\frac{dT}{dt} = \frac{d}{dt} \left(\frac{1}{2}mv^2 \right) = \frac{1}{2}m2v \frac{dv}{dt} = mv \frac{dv}{dt}, \quad (13.2)$$

since m is assumed constant. But from Newton's Second Law, $m(dv/dt) = F$, so that

$$dT/dt = Fv. \quad (13.3)$$

In general, it will come out to be $\mathbf{F} \cdot \mathbf{v}$, but in our one-dimensional case let us leave it as the force times the velocity.

Now in our simple example the force is constant, equal to $-mg$, a vertical force (the minus sign means that it acts downward), and the velocity, of course, is the rate of change of the vertical position, or height h , with time. Thus the rate of change of the kinetic energy is $-mg(dh/dt)$, which quantity, miracle of miracles, is the rate of change of something else! It is the time rate of change of mgh ! Therefore, as time goes on, the changes in kinetic energy and in the quantity mgh are equal and opposite, so that the sum of the two quantities remains constant. Q.E.D.

We have shown, from Newton's second law of motion, that energy is conserved for constant forces when we add the potential energy mgh to the kinetic energy $\frac{1}{2}mv^2$. Now let us look into this further and see whether it can be generalized, and thus advance our understanding. Does it work only for a freely falling body, or is it more general? We expect from our discussion of the conservation of energy

13-1 Energy of a falling body

13-2 Work done by gravity

13-3 Summation of energy

13-4 Gravitational field of large objects

that it would work for an object moving from one point to another in some kind of frictionless curve, under the influence of gravity (Fig. 13-1). If the object reaches a certain height h from the original height H , then the same formula should again be right, even though the velocity is now in some direction other than the vertical. We would like to understand *why* the law is still correct. Let us follow the same analysis, finding the time rate of change of the kinetic energy. This will again be $mv(dv/dt)$, but $m(dv/dt)$ is the rate of change of the magnitude of the momentum, i.e., the *force in the direction of motion*—the tangential force F_t . Thus

$$\frac{dT}{dt} = mv \frac{dv}{dt} = F_t v.$$

Now the speed is the rate of change of distance along the curve, ds/dt , and the tangential force F_t is not mg but is weaker by the ratio of the distance ds along the path to the vertical distance dh . In other words,

$$F_t = -mg \sin \theta = -mg \frac{dh}{ds},$$

so that

$$F_t \frac{ds}{dt} = -mg \left(\frac{dh}{ds} \right) \left(\frac{ds}{dt} \right) = -mg \frac{dh}{dt},$$

since the ds 's cancel. Thus we get $-mg(dh/dt)$, which is equal to the rate of change of mgh , as before.

In order to understand exactly how the conservation of energy works in general in mechanics, we shall now discuss a number of concepts which will help us to analyze it.

First, we discuss the rate of change of kinetic energy in general in three dimensions. The kinetic energy in three dimensions is

$$T = \frac{1}{2}m(v_x^2 + v_y^2 + v_z^2).$$

When we differentiate this with respect to time, we get three terrifying terms:

$$\frac{dT}{dt} = m \left(v_x \frac{dv_x}{dt} + v_y \frac{dv_y}{dt} + v_z \frac{dv_z}{dt} \right). \quad (13.4)$$

But $m(dv_x/dt)$ is the force F_x acting on the object in the x -direction. Thus the right side of Eq. (13.4) is $F_x v_x + F_y v_y + F_z v_z$. We recall our vector analysis and recognize this as $\mathbf{F} \cdot \mathbf{v}$; therefore

$$dT/dt = \mathbf{F} \cdot \mathbf{v}. \quad (13.5)$$

This result can be derived more quickly as follows: if $\mathbf{a}$ and $\mathbf{b}$ are two vectors, both of which may depend upon the time, the derivative of $\mathbf{a} \cdot \mathbf{b}$ is, in general,

$$d(\mathbf{a} \cdot \mathbf{b})/dt = \mathbf{a} \cdot d\mathbf{b}/dt + (d\mathbf{a}/dt) \cdot \mathbf{b}. \quad (13.6)$$

We then use this in the form $\mathbf{a} = \mathbf{b} = \mathbf{v}$:

$$\frac{d(\frac{1}{2}mv^2)}{dt} = \frac{d(\frac{1}{2}m\mathbf{v} \cdot \mathbf{v})}{dt} = m \frac{d\mathbf{v}}{dt} \cdot \mathbf{v} = \mathbf{F} \cdot \mathbf{v} = \mathbf{F} \cdot \frac{d\mathbf{s}}{dt}. \quad (13.7)$$

Because the concepts of kinetic energy, and energy in general, are so important, various names have been given to the important terms in equations such as these. $\frac{1}{2}mv^2$ is, as we know, called *kinetic energy*. $\mathbf{F} \cdot \mathbf{v}$ is called *power*: the force acting on an object times the velocity of the object (vector "dot" product) is the power being delivered to the object by that force. We thus have a marvelous theorem: *the rate of change of kinetic energy of an object is equal to the power expended by the forces acting on it.*

However, to study the conservation of energy, we want to analyze this still more closely. Let us evaluate the change in kinetic energy in a very short time dt . If we multiply both sides of Eq. (13.7) by dt , we find that the differential change in

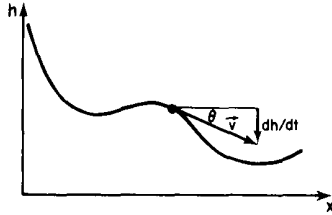


Fig. 13-1. An object moving on a frictionless curve under the influence of gravity.

the kinetic energy is the force “dot” the differential distance moved:

$$dT = \mathbf{F} \cdot d\mathbf{s}. \quad (13.8)$$

If we now integrate, we get

$$\Delta T = \int_1^2 \mathbf{F} \cdot d\mathbf{s}. \quad (13.9)$$

What does this mean? It means that if an object is moving *in any way* under the influence of a force, moving in some kind of curved path, then the change in K.E. when it goes from one point to another along the curve is equal to the integral of the component of the force along the curve times the differential displacement $d\mathbf{s}$, the integral being carried out from one point to the other. This integral also has a name; it is called the *work done by the force on the object*. We see immediately that *power equals work done per second*. We also see that it is only a component of force *in the direction of motion* that contributes to the work done. In our simple example the forces were only vertical, and had only a single component, say F_z , equal to $-mg$. No matter how the object moves in those circumstances, falling in a parabola for example, $\mathbf{F} \cdot d\mathbf{s}$, which can be written as $F_x dx + F_y dy + F_z dz$, has nothing left of it but $F_z dz = -mg dz$, because the other components of force are zero. Therefore, in our simple case,

$$\int_1^2 \mathbf{F} \cdot d\mathbf{s} = \int_{z_1}^{z_2} -mg dz = -mg(z_2 - z_1), \quad (13.10)$$

so again we find that it is only the *vertical height* from which the object falls that counts toward the potential energy.

A word about units. Since forces are measured in newtons, and we multiply by a distance in order to obtain work, work is measured in *newton · meters* (n · m), but people do not like to say newton-meters, they prefer to say *joules* (j). A newton-meter is called a joule; work is measured in joules. Power, then, is joules per second, and that is also called a *watt* (w). If we multiply watts by time, the result is the work done. The work done by the electrical company in our houses, technically, is equal to the watts times the time. That is where we get things like kilowatt hours, 1000 watts times 3600 seconds, or 3.6×10^6 joules.

Now we take another example of the law of conservation of energy. Consider an object which initially has kinetic energy and is moving very fast, and which slides against the floor with friction. It stops. At the start the kinetic energy is *not* zero, but at the end it *is* zero; there is work done by the forces, because whenever there is friction there is always a component of force in a direction opposite to that of the motion, and so energy is steadily lost. But now let us take a mass on the end of a pivot swinging in a vertical plane in a gravitational field with no friction. What happens here is different, because when the mass is going up the force is downward, and when it is coming down, the force is also downward. Thus $\mathbf{F} \cdot d\mathbf{s}$ has one sign going up and another sign coming down. At each corresponding point of the downward and upward paths the values of $\mathbf{F} \cdot d\mathbf{s}$ are exactly equal in size but of opposite sign, so the net result of the integral will be zero for this case. Thus the kinetic energy with which the mass comes back to the bottom is the same as it had when it left, that is the principle of the conservation of energy. (Note that when there are friction forces the conservation of energy seems at first sight to be invalid. We have to find another *form* of energy. It turns out, in fact, that *heat* is generated in an object when it rubs another with friction, but at the moment we supposedly do not know that.)

13-2 Work done by gravity

The next problem to be discussed is much more difficult than the above; it has to do with the case when the forces are not constant, or simply vertical, as they were in the cases we have worked out. We want to consider a planet, for example, moving around the sun, or a satellite in the space around the earth.

We shall first consider the motion of an object which starts at some point 1 and falls, say, *directly* toward the sun or toward the earth (Fig. 13-2). Will there be a law of conservation of energy in these circumstances? The only difference is that in this case, the force is *changing* as we go along, it is not just a constant. As we know, the force is GM/r^2 times the mass m , where m is the mass that moves. Now certainly when a body falls toward the earth, the kinetic energy increases as the distance fallen increases, just as it does when we do not worry about the variation of force with height. The question is whether it is possible to find another formula for potential energy different from mgh , a different function of distance away from the earth, so that conservation of energy will still be true.

This one-dimensional case is easy to treat because we know that the change in the kinetic energy is equal to the integral, from one end of the motion to the other, of $-GMm/r^2$ times the displacement dr :

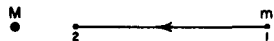


Fig. 13-2. A small mass m falls under the influence of gravity toward a large mass M .

$$T_2 - T_1 = - \int_1^2 GMm \frac{dr}{r^2}. \quad (13.11)$$

There are no cosines needed for this case because the force and the displacement are in the same direction. It is easy to integrate dr/r^2 ; the result is $-1/r$, so Eq. (13.11) becomes

$$T_2 - T_1 = +GMm \left(\frac{1}{r_2} - \frac{1}{r_1} \right). \quad (13.12)$$

Thus we have a different formula for potential energy. Equation (13.12) tells us that the quantity $(\frac{1}{2}mv^2 - GMm/r)$ calculated at point 1, at point 2, or at any other place, has a constant value.

We now have the formula for the potential energy in a gravitational field for vertical motion. Now we have an interesting problem. Can we make *perpetual motion* in a gravitational field? The gravitational field varies; in different places it is in different directions and has different strengths. Could we do something like this, using a fixed, frictionless track: start at some point and lift an object out to some other point, then move it around an arc to a third point, then lower it a certain distance, then move it in at a certain slope and pull it out some other way, so that when we bring it back to the starting point, a certain amount of work has been done by the gravitational force, and the kinetic energy of the object is increased? Can we design the curve so that it comes back moving a little bit faster than it did before, so that it goes around and around and around, and gives us perpetual motion? Since perpetual motion is impossible, we ought to find out that this is also impossible. We ought to discover the following proposition: since there is no friction the object should come back with neither higher nor lower velocity—it should be able to keep going around and around any closed path. Stated in another way, *the total work done in going around a complete cycle should be zero* for gravity forces, because if it is not zero we can get energy out by going around. (If the work turns out to be less than zero, so that we get less speed when we go around one way, then we merely go around the other way, because the forces, of course, depend only upon the position, not upon the direction; if one way is plus, the other way would be minus, so unless it is zero we will get perpetual motion by going around either way.)

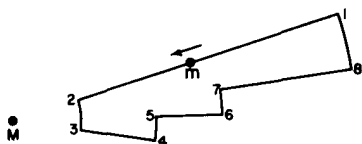


Fig. 13-3. A closed path in a gravitational field.

Is the work really zero? Let us try to demonstrate that it is. First we shall explain more or less why it is zero, and then we shall examine it a little better mathematically. Suppose that we use a simple path such as that shown in Fig. 13-3, in which a small mass is carried from point 1 to point 2, and then is made to go around a circle to 3, back to 4, then to 5, 6, 7, and 8, and finally back to 1. All of the lines are either purely radial or circular, with M as the center. How much work is done in carrying m around this path? Between points 1 and 2, it is GMm times the difference of $1/r$ between these two points:

$$W_{12} = \int_1^2 \mathbf{F} \cdot d\mathbf{s} = \int_1^2 -GMm \frac{dr}{r^2} = -GMm \left(\frac{1}{r_2} - \frac{1}{r_1} \right).$$

From 2 to 3 the force is exactly at right angles to the curve, so that $W_{23} \equiv 0$. The work from 3 to 4 is

$$W_{34} = \int_3^4 \mathbf{F} \cdot d\mathbf{s} = -GMm \left(\frac{1}{r_4} - \frac{1}{r_3} \right).$$

In the same fashion, we find that $W_{45} = 0$, $W_{56} = -GMm(1/r_6 - 1/r_5)$, $W_{67} = 0$, $W_{78} = -GMm(1/r_8 - 1/r_7)$, and $W_{81} = 0$. Thus

$$W = GMm \left(\frac{1}{r_1} - \frac{1}{r_2} + \frac{1}{r_3} - \frac{1}{r_4} + \frac{1}{r_5} - \frac{1}{r_6} + \frac{1}{r_7} - \frac{1}{r_8} \right).$$

But we note that $r_2 = r_3$, $r_4 = r_5$, $r_6 = r_7$, and $r_8 = r_1$. Therefore $W = 0$.

Of course we may wonder whether this is too trivial a curve. What if we use a *real* curve? Let us try it on a real curve. First of all, we might like to assert that a real curve could always be imitated sufficiently well by a series of sawtooth jiggles like those of Fig. 13-4, and that therefore, etc., Q.E.D., but without a little analysis, it is not obvious at first that the work done going around even a small triangle is zero. Let us magnify one of the triangles, as shown in Fig. 13-4. Is the work done in going from a to b and b to c on a triangle the same as the work done in going directly from a to c ? Suppose that the force is acting in a certain direction; let us take the triangle such that the side bc is in this direction, just as an example. We also suppose that the triangle is so small that the force is essentially constant over the entire triangle. What is the work done in going from a to c ? It is

$$W_{ac} = \int_a^c \mathbf{F} \cdot d\mathbf{s} = Fs \cos \theta,$$

since the force is constant. Now let us calculate the work done in going around the other two sides of the triangle. On the vertical side ab the force is perpendicular to ds , so that here the work is zero. On the horizontal side bc ,

$$W_{bc} = \int_b^c \mathbf{F} \cdot d\mathbf{s} = Fx.$$

Thus we see that the work done in going along the sides of a small triangle is the same as that done going on a slant, because $s \cos \theta$ is equal to x . We have proved previously that the answer is zero for any path composed of a series of notches like those of Fig. 13-3, and also that we do the same work if we cut across the corners instead of going along the notches (so long as the notches are fine enough, and we can always make them very fine); therefore, *the work done in going around any path in a gravitational field is zero*.

This is a very remarkable result. It tells us something we did not previously know about planetary motion. It tells us that when a planet moves around the sun (without any other objects around, no other forces) it moves in such a manner that the square of the speed at any point minus some constants divided by the radius at that point is always the same at every point on the orbit. For example, the closer the planet is to the sun, the faster it is going, but by how much? By the following amount: if instead of letting the planet go around the sun, we were to change the direction (but not the magnitude) of its velocity and make it move radially, and then we let it fall from some special radius to the radius of interest, the new speed would be the same as the speed it had in the actual orbit, because this is just another example of a complicated path. So long as we come back to the same distance, the kinetic energy will be the same. So, whether the motion is the real, undisturbed one, or is changed in direction by channels, by frictionless constraints, the kinetic energy with which the planet arrives at a point will be the same.

Thus, when we make a numerical analysis of the motion of the planet in its orbit, as we did earlier, we can check whether or not we are making appreciable errors by calculating this constant quantity, the energy, at every step, and it should not change. For the orbit of Table 9-2 the energy does change,* it changes by

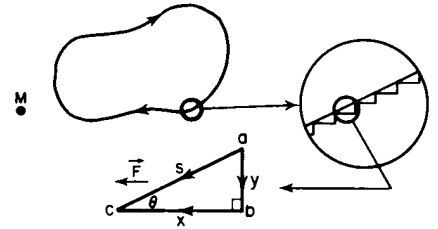


Fig. 13-4. A "smooth" closed path, showing a magnified segment of it approximated by a series of radial and circumferential steps, and an enlarged view of one step.

* The energy is $\frac{1}{2}(v_x^2 + v_y^2) - 1/r$ in the units of Table 9-2.

some 1.5 percent from the beginning to the end. Why? Either because for the numerical method we use finite intervals, or else because we made a slight mistake somewhere in arithmetic.

Let us consider the energy in another case: the problem of a mass on a spring. When we displace the mass from its balanced position, the restoring force is proportional to the displacement. In those circumstances, can we work out a law for conservation of energy? Yes, because the work done by such a force is

$$W = \int_0^x F dx = \int_0^x -kx dx = -\frac{1}{2}kx^2. \quad (13.13)$$

Therefore, for a mass on a spring we have that the kinetic energy of the oscillating mass plus $\frac{1}{2}kx^2$ is a constant. Let us see how this works. We pull the mass down; it is standing still and so its speed is zero. But x is not zero, x is at its maximum, so there is some energy, the potential energy, of course. Now we release the mass and things begin to happen (the details not to be discussed), but at any instant the kinetic plus potential energy must be a constant. For example, after the mass is on its way past the original equilibrium point, the position x equals zero, but that is when it has its biggest v^2 , and as it gets more x^2 it gets less v^2 , and so on. So the balance of x^2 and v^2 is maintained as the mass goes up and down. Thus we have another rule now, that the potential energy for a spring is $\frac{1}{2}kx^2$, if the force is $-kx$.

13-3 Summation of energy

Now we go on to the more general consideration of what happens when there are large numbers of objects. Suppose we have the complicated problem of many objects, which we label $i = 1, 2, 3, \dots$, all exerting gravitational pulls on each other. What happens then? We shall prove that if we add the kinetic energies of all the particles, and add to this the sum, over all *pairs* of particles, of their mutual gravitational potential energy, $-GMm/r_{ij}$, the total is a constant:

$$\sum_i \frac{1}{2}m_i v_i^2 + \sum_{(\text{pairs } ij)} -\frac{Gm_i m_j}{r_{ij}} = \text{const.} \quad (13.14)$$

How do we prove it? We differentiate each side with respect to time and get zero. When we differentiate $\frac{1}{2}m_i v_i^2$, we find derivatives of the velocity that are the forces, just as in Eq. (13.5). We replace these forces by the law of force that we know from Newton's law of gravity and then we notice that what is left is the same as the time derivative of

$$\sum_{\text{pairs}} -\frac{Gm_i m_j}{r_{ij}}.$$

The time derivative of the kinetic energy is

$$\begin{aligned} \frac{d}{dt} \sum_i \frac{1}{2}m_i v_i^2 &= \sum_i m_i \mathbf{v}_i \cdot \frac{d\mathbf{v}_i}{dt} \\ &= \sum_i \mathbf{F}_i \cdot \mathbf{v}_i \\ &= \sum_i \left(\sum_j -\frac{Gm_i m_j \mathbf{r}_{ij}}{r_{ij}^3} \right) \cdot \mathbf{v}_i. \end{aligned} \quad (13.15)$$

The time derivative of the potential energy is

$$\frac{d}{dt} \sum_{\text{pairs}} -\frac{Gm_i m_j}{r_{ij}} = \sum_{\text{pairs}} \left(+\frac{Gm_i m_j}{r_{ij}^2} \right) \left(\frac{dr_{ij}}{dt} \right).$$

But

$$r_{ij} = \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2 + (z_i - z_j)^2},$$

so that

$$\begin{aligned}
\frac{d\mathbf{r}_{ij}}{dt} &= \frac{1}{2r_{ij}} \left[2(x_i - x_j) \left(\frac{dx_i}{dt} - \frac{dx_j}{dt} \right) \right. \\
&\quad + 2(y_i - y_j) \left(\frac{dy_i}{dt} - \frac{dy_j}{dt} \right) \\
&\quad \left. + 2(z_i - z_j) \left(\frac{dz_i}{dt} - \frac{dz_j}{dt} \right) \right] \\
&= \mathbf{r}_{ij} \cdot \frac{\mathbf{v}_i - \mathbf{v}_j}{r_{ij}} \\
&= \mathbf{r}_{ij} \cdot \frac{\mathbf{v}_i}{r_{ij}} + \mathbf{r}_{ji} \cdot \frac{\mathbf{v}_j}{r_{ji}},
\end{aligned}$$

since $\mathbf{r}_{ij} = -\mathbf{r}_{ji}$, while $r_{ij} = r_{ji}$. Thus

$$\frac{d}{dt} \sum_{\text{pairs}} -\frac{Gm_i m_j}{r_{ij}} = \sum_{\text{pairs}} \left[\frac{Gm_i m_j \mathbf{r}_{ij}}{r_{ij}^3} \cdot \mathbf{v}_i + \frac{Gm_j m_i \mathbf{r}_{ji}}{r_{ji}^3} \cdot \mathbf{v}_j \right]. \quad (13.16)$$

Now we must note carefully what $\sum_i \{\sum_j\}$ and $\sum_{\text{pairs}}$ mean. In Eq. (13.15), $\sum_i \{\sum_j\}$ means that i takes on all values $i = 1, 2, 3, \dots$ in turn, and for each value of i , the index j takes on all values except i . Thus if $i = 3$, j takes on the values $1, 2, 4, \dots$

In Eq. (13.16), on the other hand, $\sum_{\text{pairs}}$ means that given values of i and j occur only once. Thus the particle pair 1 and 3 contributes only one term to the sum. To keep track of this, we might agree to let i range over all values $1, 2, 3, \dots$, and for each i let j range only over values *greater* than i . Thus if $i = 3$, j could only have values $4, 5, 6, \dots$. But we notice that for each i, j value there are two contributions to the sum, one involving $\mathbf{v}_i$, and the other $\mathbf{v}_j$, and that these terms have the same appearance as those of Eq. (13.14), where *all* values of i and j (except $i = j$) are included in the sum. Therefore, by matching the terms one by one, we see that Eqs. (13.16) and (13.15) are precisely the same, but of opposite sign, so that the time derivative of the kinetic plus potential energy is indeed zero. Thus we see that, for many objects, *the kinetic energy is the sum of the contributions from each individual object*, and that the potential energy is also simple, it being also just a sum of contributions, the energies between all the pairs. We can understand *why* it should be the energy of every pair this way: Suppose that we want to find the total amount of work that must be done to bring the objects to certain distances from each other. We may do this in several steps, bringing them in from infinity where there is no force, one by one. First we bring in number one, which requires no work, since no other objects are yet present to exert force on it. Next we bring in number two, which does take some work, namely $W_{12} = -Gm_1 m_2 / r_{12}$. Now, and this is an important point, suppose we bring in the next object to position three. At any moment the force on number 3 can be written as the sum of two forces—the force exerted by number 1 and that exerted by number 2. Therefore *the work done is the sum of the works done by each*, because if $\mathbf{F}_3$ can be resolved into the sum of two forces,

$$\mathbf{F}_3 = \mathbf{F}_{13} + \mathbf{F}_{23},$$

then the work is

$$\int \mathbf{F}_3 \cdot d\mathbf{s} = \int \mathbf{F}_{13} \cdot d\mathbf{s} + \int \mathbf{F}_{23} \cdot d\mathbf{s} = W_{13} + W_{23}.$$

That is, the work done is the sum of the work done against the first force and the second force, as if each acted independently. Proceeding in this way, we see that the total work required to assemble the given configuration of objects is precisely the value given in Eq. (13.14) as the potential energy. It is because gravity obeys the principle of superposition of forces that we can write the potential energy as a sum over each pair of particles.

13-4 Gravitational field of large objects

Now we shall calculate the fields which are met in a few physical circumstances involving *distributions of mass*. We have not so far considered distributions of mass, only particles, so it is interesting to calculate the forces when they are produced by more than just one particle. First we shall find the gravitational force on a mass that is produced by a plane sheet of material, infinite in extent. The force on a unit mass at a given point P , produced by this sheet of material (Fig. 13-5), will of course be directed toward the sheet. Let the distance of the point from the sheet be a , and let the amount of mass per unit area of this huge sheet be μ . We shall suppose μ to be constant; it is a uniform sheet of material. Now, what small field dC is produced by the mass dm lying between ρ and $\rho + d\rho$ from the point O of the sheet nearest point P ? Answer: $dC = G(dm/r^3)$. But this field is directed along r , and we know that only the x -component of it will remain when we add all the little vector dC 's to produce C . The x -component of dC is

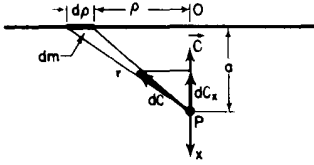


Fig. 13-5. The gravitational force F on a mass point produced by an infinite plane sheet of matter.

$$dC_x = G \frac{dmr_x}{r^3} = G \frac{dma}{r^3}.$$

Now all masses dm which are at the same distance r from P will yield the same dC_x , so we may at once write for dm the total mass in the *ring* between ρ and $\rho + d\rho$, namely $dm = \mu 2\pi\rho d\rho$ ($2\pi\rho d\rho$ is the area of a ring of radius ρ and width $d\rho$, if $d\rho \ll \rho$). Thus

$$dC_x = G\mu 2\pi\rho \frac{d\rho a}{r^3}.$$

Then, since $r^2 = \rho^2 + a^2$, $\rho d\rho = r dr$. Therefore,

$$C_x = 2\pi G\mu a \int_a^\infty \frac{dr}{r^2} = 2\pi G\mu a \left(\frac{1}{a} - \frac{1}{\infty} \right) = 2\pi G\mu. \quad (13.17)$$

Thus the force is independent of distance a ! Why? Have we made a mistake? One might think that the farther away we go, the weaker the force would be. But no! If we are close, most of the matter is pulling at an unfavorable angle; if we are far away, more of the matter is situated more favorably to exert a pull toward the plane. At any distance, the matter which is most effective lies in a certain cone. When we are farther away the force is smaller by the inverse square, but in the same cone, in the same angle, there is much *more matter*, larger by just the square of the distance! This analysis can be made rigorous by just noticing that the differential contribution in any given cone is in fact independent of the distance, because of the reciprocal variation of the strength of the force from a given mass, and the amount of mass included in the cone, with changing distance. The force is not really constant of course, because when we go on the other side of the sheet it is reversed in sign.

We have also, in effect, solved an electrical problem: if we have an electrically charged plate, with an amount σ of charge per unit area, then the electric field at a point outside the sheet is equal to $\sigma/2\epsilon_0$, and is in the outward direction if the sheet is positively charged, and inward if the sheet is negatively charged. To prove this, we merely note that G , gravity, plays the same role as $1/4\pi\epsilon_0$ for electricity.

Now suppose that we have two plates, with a positive charge $+\sigma$ on one and a negative charge $-\sigma$ on another at a distance D from the first. What is the field? Outside the two plates it is zero. Why? Because one attracts and the other repels, the force being *independent of distance*, so that the two balance out! Also, the force *between* the two plates is clearly twice as great as that from one plate, namely $E = \sigma/\epsilon_0$, and is directed from the positive plate to the negative one.

Now we come to a most interesting and important problem, whose solution we have been assuming all the time, namely, that the force produced by the earth at a point on the surface or outside it is the same as if all the mass of the earth were located at its center. The validity of this assumption is not obvious, because when we are close, some of the mass is very close to us, and some is farther away,

and so on. When we add the effects all together, it seems a miracle that the net force is exactly the same as we would get if we put all the mass in the middle!

We now demonstrate the correctness of this miracle. In order to do so, however, we shall consider a thin uniform hollow shell instead of the whole earth. Let the total mass of the shell be m , and let us calculate the *potential energy* of a particle of mass m' a distance R away from the sphere (Fig. 13-6) and show that the potential energy is the same as it would be if the mass m were a point at the center. (The potential energy is easier to work with than is the field because we do not have to worry about angles, we merely add the potential energies of all the pieces of mass.) If we call x the distance of a certain plane section from the center, then all the mass that is in a slice dx is at the same distance r from P , and the potential energy due to this ring is $-Gm' dm/r$. How much mass is in the small slice dx ? An amount

$$dm = 2\pi y \mu ds = \frac{2\pi y \mu dx}{\sin \theta} = \frac{2\pi y \mu dx a}{y} = 2\pi a \mu dx,$$

where $\mu = m/4\pi a^2$ is the surface density of mass on the spherical shell. (It is a general rule that the area of a zone of a sphere is proportional to its axial width.) Therefore the potential energy due to dm is

$$dW = -\frac{Gm' dm}{r} = -\frac{Gm' 2\pi a \mu dx}{r}.$$

But we see that

$$\begin{aligned} r^2 &= y^2 + (R - x)^2 = y^2 + x^2 + R^2 - 2Rx \\ &= a^2 + R^2 - 2Rx. \end{aligned}$$

Thus

$$2r dr = -2R dx$$

or

$$\frac{dx}{r} = \frac{dr}{R}.$$

Therefore,

$$dW = -\frac{Gm' 2\pi a \mu dr}{R},$$

and so

$$\begin{aligned} W &= -\frac{Gm' 2\pi a \mu}{R} \int_{R-a}^{R+a} dr \\ &= -\frac{Gm' 2\pi a \mu}{R} 2a = -\frac{Gm' (4\pi a^2 \mu)}{R} \\ &= -\frac{Gm' m}{R}. \end{aligned} \tag{13.18}$$

Thus, for a thin spherical shell, the potential energy of a mass m' , external to the shell, is the same as though the mass of the shell were concentrated at its center. The earth can be imagined as a series of spherical shells, each one of which contributes an energy which depends only on its mass and the distance from the center; adding them all together we get the *total mass*, and therefore the earth acts as though all the material were at the center!

But notice what happens if our point is on the *inside* of the shell. Making the same calculation, but with P on the inside, we still get the difference of the two r 's, but now in the form $a + R - (a - R) = 2R$, or twice the distance from the center. In other words, W comes out to be $W = -Gm'm/a$, which is *independent of R* and independent of position, i.e., the same energy no matter *where* we are inside. Therefore no force; no work is done when we move about inside. If the potential energy is the same no matter where an object is placed inside the sphere, there can be no force on it. So there is no force inside, there is only a force outside, and the force outside is the same as though the mass were all at the center.

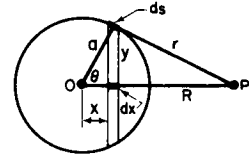


Fig. 13-6. A thin spherical shell of mass or charge.

Work and Potential Energy (conclusion)

14-1 Work

In the preceding chapter we have presented a great many new ideas and results that play a central role in physics. These ideas are so important that it seems worth while to devote a whole chapter to a closer examination of them. In the present chapter we shall not repeat the "proofs" or the specific tricks by which the results were obtained, but shall concentrate instead upon a discussion of the ideas themselves.

In learning any subject of a technical nature where mathematics plays a role, one is confronted with the task of understanding and storing away in the memory a huge body of facts and ideas, held together by certain relationships which can be "proved" or "shown" to exist between them. It is easy to confuse the proof itself with the relationship which it establishes. Clearly, the important thing to learn and to remember is the relationship, not the proof. In any particular circumstance we can either say "it can be shown that" such and such is true, or we can show it. In almost all cases, the particular proof that is used is concocted, first of all, in such form that it can be written quickly and easily on the chalkboard or on paper, and so that it will be as smooth-looking as possible. Consequently, the proof may look deceptively simple, when in fact, the author might have worked for hours trying different ways of calculating the same thing until he has found the neatest way, so as to be able to show that it can be shown in the shortest amount of time! The thing to be remembered, when seeing a proof, is not the proof itself, but rather that it *can be shown* that such and such is true. Of course, if the proof involves some mathematical procedures or "tricks" that one has not seen before, attention should be given not to the trick exactly, but to the mathematical idea involved.

It is certain that in all the demonstrations that are made in a course such as this, not one has been remembered from the time when the author studied freshman physics. Quite the contrary: he merely remembers that such and such is true, and to explain how it can be shown he invents a demonstration at the moment it is needed. Anyone who has really learned a subject should be able to follow a similar procedure, but it is no use remembering the proofs. That is why, in this chapter, we shall avoid the proofs of the various statements made previously, and merely summarize the results.

The first idea that has to be digested is *work done by a force*. The physical word "work" is not the word in the ordinary sense of "Workers of the world unite!," but is a different idea. Physical work is expressed as $\int \mathbf{F} \cdot d\mathbf{s}$, called "the line integral of $\mathbf{F}$ dot $d\mathbf{s}$," which means that if the force, for instance, is in one direction and the object on which the force is working is displaced in a certain direction, then *only the component of force in the direction of the displacement* does any work. If, for instance, the force were constant and the displacement were a finite distance Δs , then the work done in moving the constant force through that distance is only the component of force along Δs times Δs . The rule is "force times distance," but we really mean only the component of force in the direction of the displacement times Δs or, equivalently, the component of displacement in the direction of force times F . It is evident that no work whatsoever is done by a force which is at right angles to the displacement.

Now if the vector displacement $\Delta \mathbf{s}$ is resolved into components, in other words, if the actual displacement is Δs and we want to consider it effectively as a component of displacement Δx in the x -direction, Δy in the y -direction, and Δz

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in the z -direction, then the work done in carrying an object from one place to another can be calculated in three parts, by calculating the work done along x , along y , and along z . The work done in going along x involves only that component of force, namely F_x , and so on, so the work is $F_x \Delta x + F_y \Delta y + F_z \Delta z$. When the force is not constant, and we have a complicated curved motion, then we must resolve the path into a lot of little Δs 's, add the work done in carrying the object along each Δs , and take the limit as Δs goes to zero. This is the meaning of the "line integral."

Everything we have just said is contained in the formula $W = \int \mathbf{F} \cdot d\mathbf{s}$. It is all very well to say that it is a marvelous formula, but it is another thing to understand what it means, or what some of the consequences are.

The word "work" in physics has a meaning so different from that of the word as it is used in ordinary circumstances that it must be observed carefully that there are some peculiar circumstances in which it appears not to be the same. For example, according to the physical definition of work, if one holds a hundred-pound weight off the ground for a while, he is doing no work. Nevertheless, everyone knows that he begins to sweat, shake, and breathe harder, as if he were running up a flight of stairs. Yet running upstairs *is* considered as doing work (in running *downstairs*, one gets work out of the world, according to physics), but in simply holding an object in a fixed position, no work is done. Clearly, the physical definition of work differs from the physiological definition, for reasons we shall briefly explore.

It is a fact that when one holds a weight he has to do "physiological" work. Why should he sweat? Why should he need to consume food to hold the weight up? Why is the machinery inside him operating at full throttle, just to hold the weight up? Actually, the weight could be held up with no effort by just placing it on a table; then the table, quietly and calmly, without any supply of energy, is able to maintain the same weight at the same height! The physiological situation is something like the following. There are two kinds of muscles in the human body and in other animals: one kind, called *striated* or *skeletal* muscle, is the type of muscle we have in our arms, for example, which is under voluntary control; the other kind, called *smooth* muscle, is like the muscle in the intestines or, in the clam, the greater adductor muscle that closes the shell. The smooth muscles work very slowly, but they can hold a "set"; that is to say, if the clam tries to close its shell in a certain position, it will hold that position, even if there is a very great force trying to change it. It will hold a position under load for hours and hours without getting tired because it is very much like a table holding up a weight, it "sets" into a certain position, and the molecules just lock there temporarily with no work being done, no effort being generated by the clam. The fact that we have to generate effort to hold up a weight is simply due to the design of striated muscle. What happens is that when a nerve impulse reaches a muscle fiber, the fiber gives a little twitch and then relaxes, so that when we hold something up, enormous volleys of nerve impulses are coming in to the muscle, large numbers of twitches are maintaining the weight, while the other fibers relax. We can see this, of course: when we hold a heavy weight and get tired, we begin to shake. The reason is that the volleys are coming irregularly, and the muscle is tired and not reacting fast enough. Why such an inefficient scheme? We do not know exactly why, but evolution has not been able to develop *fast* smooth muscle. Smooth muscle would be much more effective for holding up weights because you could just stand there and it would lock in; there would be no work involved and no energy would be required. However, it has the disadvantage that it is very slow-operating.

Returning now to physics, we may ask *why* we want to calculate the work done. The answer is that it is interesting and useful to do so, since the work done on a particle by the resultant of all the forces acting on it is exactly equal to the change in kinetic energy of that particle. That is, if an object is being pushed, it picks up speed, and

$$\Delta(v^2) = \frac{2}{m} \mathbf{F} \cdot \Delta \mathbf{s}.$$

14-2 Constrained motion

Another interesting feature of forces and work is this: suppose that we have a sloping or a curved track, and a particle that must move along the track, but without friction. Or we may have a pendulum with a string and a weight; the string constrains the weight to move in a circle about the pivot point. The pivot point may be changed by having the string hit a peg, so that the path of the weight is along two circles of different radii. These are examples of what we call *fixed, frictionless constraints*.

In motion with a fixed frictionless constraint, no work is done by the constraint because the forces of constraint are always at right angles to the motion. By the "forces of constraint" we mean those forces which are applied to the object directly by the constraint itself—the contact force with the track, or the tension in the string.

The forces involved in the motion of a particle on a slope moving under the influence of gravity are quite complicated, since there is a constraint force, a gravitational force, and so on. However, if we base our calculation of the motion on conservation of energy and *the gravitational force alone*, we get the right result. This seems rather strange, because it is not strictly the right way to do it—we should use the *resultant* force. Nevertheless, the work done by the gravitational force alone will turn out to be the change in the kinetic energy, because the work done by the constraint part of the force is zero (Fig. 14-1).

The important feature here is that if a force can be analyzed as the sum of two or more "pieces" then the work done by the resultant force in going along a certain curve is the sum of the works done by the various "component" forces into which the force is analyzed. Thus if we analyze the force as being the vector sum of several effects, gravitational plus constraint forces, etc., or the x -component of all forces and the y -component of all forces, or any other way that we wish to split it up, then the work done by the net force is equal to the sum of the works done by all the parts into which we have divided the force in making the analysis.

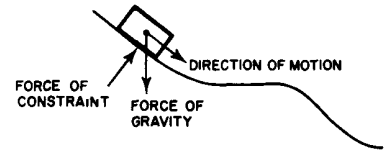


Fig. 14-1. Forces acting on a sliding body (no friction).

14-3 Conservative forces

In nature there are certain forces, that of gravity, for example, which have a very remarkable property which we call "conservative" (no political ideas involved, it is again one of those "crazy words"). If we calculate how much work is done by a force in moving an object from one point to another along some curved path, in general the work depends upon the curve, but in special cases it does not. If it does not depend upon the curve, we say that the force is a conservative force. In other words, if the integral of the force times the distance in going from position 1 to position 2 in Fig. 14-2 is calculated along curve *A* and then along *B*, we get the same number of joules, and if this is true for this pair of points on *every* curve, and if the same proposition works *no matter which pair of points* we use, then we say the force is conservative. In such circumstances, the work integral going from 1 to 2 can be evaluated in a simple manner, and we can give a formula for the result. Ordinarily it is not this easy, because we also have to specify the curve, but when we have a case where the work does not depend on the curve, then, of course, the work depends only upon the *positions* of 1 and 2.

To demonstrate this idea, consider the following. We take a "standard" point *P*, at an arbitrary location (Fig. 14-2). Then, the work line-integral from 1 to 2, which we want to calculate, can be evaluated as the work done in going from 1 to *P* plus the work done in going from *P* to 2, because the forces are conservative and the work does not depend upon the curve. Now, the work done in going from position *P* to a particular position in space is a function of that position in space. Of course it really depends on *P* also, but we hold the arbitrary point *P* fixed permanently for the analysis. If that is done, then the work done in going from point *P* to point 2 is some function of the final position of 2. It depends upon where 2 is; if we go to some other point we get a different answer.

We shall call this function of position $-U(x, y, z)$, and when we wish to refer to some particular point 2 whose coordinates are (x_2, y_2, z_2) , we shall write $U(2)$,

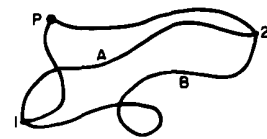


Fig. 14-2. Possible paths between two points in a field of force.

as an abbreviation for $U(x_2, y_2, z_2)$. The work done in going from point 1 to point P can be written also by going the *other way* along the integral, reversing all the ds 's. That is, the work done in going from 1 to P is *minus* the work done in going from the point P to 1:

$$\int_1^P \mathbf{F} \cdot d\mathbf{s} = \int_P^1 \mathbf{F} \cdot (-d\mathbf{s}) = - \int_P^1 \mathbf{F} \cdot d\mathbf{s}.$$

Thus the work done in going from P to 1 is $-U(1)$, and from P to 2 the work is $-U(2)$. Therefore the integral from 1 to 2 is equal to $-U(2)$ plus $[-U(1)$ backwards], or $+U(1) - U(2)$:

$$\begin{aligned} U(1) &= - \int_P^1 \mathbf{F} \cdot d\mathbf{s}, & U(2) &= - \int_P^2 \mathbf{F} \cdot d\mathbf{s}, \\ \int_1^2 \mathbf{F} \cdot d\mathbf{s} &= U(1) - U(2). \end{aligned} \quad (14.1)$$

The quantity $U(1) - U(2)$ is called the change in the potential energy, and we call U the potential energy. We shall say that when the object is located at position 2, it has potential energy $U(2)$ and at position 1 it has potential energy $U(1)$. If it is located at position P , it has zero potential energy. If we had used any other point, say Q , instead of P , it would turn out (and we shall leave it to you to demonstrate) that the *potential energy is changed only by the addition of a constant*. Since the conservation of energy depends only upon *changes*, it does not matter if we add a constant to the potential energy. Thus the point P is arbitrary.

Now, we have the following two propositions: (1) that the work done by a force is equal to the change in kinetic energy of the particle, but (2) mathematically, for a conservative force, the work done is minus the change in a function U which we call the potential energy. As a consequence of these two, we arrive at the proposition that *if only conservative forces act, the kinetic energy T plus the potential energy U remains constant*:

$$T + U = \text{constant}. \quad (14.2)$$

Let us now discuss the formulas for the potential energy for a number of cases. If we have a gravitational field that is uniform, if we are not going to heights comparable with the radius of the earth, then the force is a constant vertical force and the work done is simply the force times the vertical distance. Thus

$$U(z) = mgz, \quad (14.3)$$

and the point P which corresponds to zero potential energy happens to be any point in the plane $z = 0$. We could also have said that the potential energy is $mg(z - 6)$ if we had wanted to—all the results would, of course, be the same in our analysis except that the value of the potential energy at $z = 0$ would be $-mg6$. It makes no difference, because only *differences* in potential energy count.

The energy needed to compress a linear spring a distance x from an equilibrium point is

$$U(x) = \frac{1}{2}kx^2, \quad (14.4)$$

and the zero of potential energy is at the point $x = 0$, the equilibrium position of the spring. Again we could add any constant we wish.

The potential energy of gravitation for point masses M and m , a distance r apart, is

$$U(r) = -GMm/r. \quad (14.5)$$

The constant has been chosen here so that the potential is zero at infinity. Of course the same formula applies to electrical charges, because it is the same law:

$$U(r) = q_1q_2/4\pi\epsilon_0r. \quad (14.6)$$

Now let us actually use one of these formulas, to see whether we understand what it means. *Question*: How fast do we have to shoot a rocket away from the

earth in order for it to leave? *Solution:* The kinetic plus potential energy must be a constant; when it “leaves,” it will be millions of miles away, and if it is just barely able to leave, we may suppose that it is moving with zero speed out there, just barely going. Let a be the radius of the earth, and M its mass. The kinetic plus potential energy is then initially given by $\frac{1}{2}mv^2 - GmM/a$. At the end of the motion the two energies must be equal. The kinetic energy is taken to be zero at the end of the motion, because it is supposed to be just barely drifting away at essentially zero speed, and the potential energy is GmM divided by infinity, which is zero. So everything is zero on one side and that tells us that the square of the velocity must be $2GM/a$. But GM/a^2 is what we call the acceleration of gravity, g . Thus

$$v^2 = 2ga.$$

At what speed must a satellite travel in order to keep going around the earth? We worked this out long ago and found that $v^2 = GM/a$. Therefore to go *away* from the earth, we need $\sqrt{2}$ times the velocity we need to just go *around* the earth near its surface. We need, in other words, *twice as much energy* (because energy goes as the square of the velocity) to leave the earth as we do to go around it. Therefore the first thing that was done historically with satellites was to get one to go around the earth, which requires a speed of five miles per second. The next thing was to send a satellite away from the earth permanently; this required twice the energy, or about seven miles per second.

Now, continuing our discussion of the characteristics of potential energy, let us consider the interaction of two molecules, or two atoms, two oxygen atoms for instance. When they are very far apart, the force is one of attraction, which varies as the inverse seventh power of the distance, and when they are very close the force is a very large repulsion. If we integrate the inverse seventh power to find the work done, we find that the potential energy U , which is a function of the radial distance between the two oxygen atoms, varies as the inverse sixth power of the distance for large distances.

If we sketch the curve of the potential energy $U(r)$ as in Fig. 14-3, we thus start out at large r with an inverse sixth power, but if we come in sufficiently near we reach a point d where there is a minimum of potential energy. The minimum of potential energy at $r = d$ means this: if we start at d and move a small distance, a very small distance, the work done, which is the change in potential energy when we move this distance, is nearly zero, because there is very little change in potential energy at the bottom of the curve. Thus there is no force at this point, and so it is the equilibrium point. Another way to see that it is the equilibrium point is that it takes work to move away from d in either direction. When the two oxygen atoms have settled down, so that no more energy can be liberated from the force between them, they are in the lowest energy state, and they will be at this separation d . This is the way an oxygen molecule looks when it is cold. When we heat it up, the atoms shake and move farther apart, and we can in fact break them apart, but to do so takes a certain amount of work or energy, which is the potential energy difference between $r = d$ and $r = \infty$. When we try to push the atoms very close together the energy goes up very rapidly, because they repel each other.

The reason we bring this out is that the idea of *force* is not particularly suitable for quantum mechanics; there the idea of *energy* is most natural. We find that although forces and velocities “dissolve” and disappear when we consider the more advanced forces between nuclear matter and between molecules and so on, the energy concept remains. Therefore we find curves of potential energy in quantum mechanics books, but very rarely do we ever see a curve for the force between two molecules, because by that time people who are doing analyses are thinking in terms of energy rather than of force.

Next we note that if several conservative forces are acting on an object at the same time, then the potential energy of the object is the sum of the potential energies from each of the separate forces. This is the same proposition that we mentioned before, because if the force can be represented as a vector sum of forces, then the work done by the total force is the sum of the works done by the partial

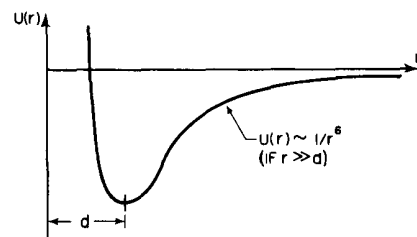


Fig. 14-3. The potential energy between two atoms as a function of the distance between them.

forces, and it can therefore be analyzed as changes in the potential energies of each of them separately. Thus the total potential energy is the sum of all the little pieces.

We could generalize this to the case of a system of many objects interacting with one another, like Jupiter, Saturn, Uranus, etc., or oxygen, nitrogen, carbon, etc., which are acting with respect to one another in pairs due to forces all of which are conservative. In these circumstances the kinetic energy in the entire system is simply the sum of the kinetic energies of all of the particular atoms or planets or whatever, and the potential energy of the system is the sum, over the pairs of particles, of the potential energy of mutual interaction of a single pair, as though the others were not there. (This is really not true for molecular forces, and the formula is somewhat more complicated; it certainly is true for Newtonian gravitation, and it is true as an approximation for molecular forces. For molecular forces there is a potential energy, but it is sometimes a more complicated function of the positions of the atoms than simply a sum of terms from pairs.) In the special case of gravity, therefore, the potential energy is the sum, over all the pairs i and j , of $-Gm_i m_j / r_{ij}$, as was indicated in Eq. (13.14). Equation (13.14) expressed mathematically the following proposition: that the total kinetic energy plus the total potential energy does not change with time. As the various planets wheel about, and turn and twist and so on, if we calculate the total kinetic energy and the total potential energy we find that the total remains constant.

14-4 Nonconservative forces

We have spent a considerable time discussing conservative forces; what about nonconservative forces? We shall take a deeper view of this than is usual, and state that there are no nonconservative forces! As a matter of fact, all the fundamental forces in nature appear to be conservative. This is not a consequence of Newton's laws. In fact, so far as Newton himself knew, the forces could be nonconservative, as friction apparently is. When we say friction *apparently* is, we are taking a modern view, in which it has been discovered that all the deep forces, the forces between the particles at the most fundamental level, are conservative.

If, for example, we analyze a system like that great globular star cluster that we saw a picture of, with the thousands of stars all interacting, then the formula for the total potential energy is simply one term plus another term, etc., summed over all pairs of stars, and the kinetic energy is the sum of the kinetic energies of all the individual stars. But the globular cluster as a whole is drifting in space too, and, if we were far enough away from it and did not see the details, could be thought of as a single object. Then if forces were applied to it, some of those forces might end up driving it forward as a whole, and we would see the center of the whole thing moving. On the other hand, some of the forces can be, so to speak, "wasted" in increasing the kinetic or potential energy of the "particles" inside. Let us suppose, for instance, that the action of these forces expands the whole cluster and makes the particles move faster. The total energy of the whole thing is really conserved, but seen from the outside with our crude eyes which cannot see the confusion of motions inside, and just thinking of the kinetic energy of the motion of the whole object as though it were a single particle, it would appear that energy is not conserved, but this is due to a lack of appreciation of what it is that we see. And that, it turns out, is the case: the total energy of the world, kinetic plus potential, is a constant when we look closely enough.

When we study matter in the finest detail at the atomic level, it is not always easy to separate the total energy of a thing into two parts, kinetic energy and potential energy, and such separation is not always necessary. It is *almost* always possible to do it, so let us say that it *is* always possible, and that the potential-plus-kinetic energy of the world is constant. Thus the total potential-plus-kinetic energy inside the whole world is constant, and if the "world" is a piece of isolated material, the energy is constant if there are no external forces. But as we have seen, some of the kinetic and potential energy of a thing may be internal, for instance the internal molecular motions, in the sense that we do not notice it. We know that in a glass of water everything is jiggling around, all the parts are moving

all the time, so there is a certain kinetic energy inside, which we ordinarily may not pay any attention to. We do not notice the motion of the atoms, which produces heat, and so we do not call it kinetic energy, but heat is primarily kinetic energy. Internal potential energy may also be in the form, for instance, of chemical energy: when we burn gasoline energy is liberated because the potential energies of the atoms in the new atomic arrangement are lower than in the old arrangement. It is not strictly possible to treat heat as being pure kinetic energy, for a little of the potential gets in, and vice versa for chemical energy, so we put the two together and say that the total kinetic and potential energy inside an object is partly heat, partly chemical energy, and so on. Anyway, all these different forms of internal energy are sometimes considered as "lost" energy in the sense described above; this will be made clearer when we study thermodynamics.

As another example, when friction is present it is not true that kinetic energy is lost, even though a sliding object stops and the kinetic energy seems to be lost. The kinetic energy is not lost because, of course, the atoms inside are jiggling with a greater amount of kinetic energy than before, and although we cannot see that, we can measure it by determining the temperature. Of course if we disregard the heat energy, then the conservation of energy theorem will appear to be false.

Another situation in which energy conservation appears to be false is when we study only part of a system. Naturally, the conservation of energy theorem will appear not to be true if something is interacting with something else on the outside and we neglect to take that interaction into account.

In classical physics potential energy involved only gravitation and electricity, but now we have nuclear energy and other energies also. Light, for example, would involve a new form of energy in the classical theory, but we can also, if we want to, imagine that the energy of light is the kinetic energy of a photon, and then our formula (14.2) would still be right.

14-5 Potentials and fields

We shall now discuss a few of the ideas associated with potential energy and with the idea of a *field*. Suppose we have two large objects *A* and *B* and a third very small one which is attracted gravitationally by the two, with some resultant force *F*. We have already noted in Chapter 12 that the gravitational force on a particle can be written as its mass, *m*, times another vector, *C*, which is dependent only upon the *position* of the particle:

$$\mathbf{F} = m\mathbf{C}.$$

We can analyze gravitation, then, by imagining that there is a certain vector *C* at every position in space which "acts" upon a mass which we may place there, but which is there itself whether we actually supply a mass for it to "act" on or not. *C* has three components, and each of those components is a function of (*x*, *y*, *z*), a function of position in space. Such a thing we call a *field*, and we say that the objects *A* and *B* generate the field, i.e., they "make" the vector *C*. When an object is put in a field, the force on it is equal to its mass times the value of the field vector at the point where the object is put.

We can also do the same with the potential energy. Since the potential energy, the integral of (**force**) · (*ds*) can be written as *m* times the integral of the (**field**) · (*ds*), a mere change of scale, we see that the potential energy *U*(*x*, *y*, *z*) of an object located at a point (*x*, *y*, *z*) in space can be written as *m* times another function which we may call the *potential* *Ψ*. The integral $\int \mathbf{C} \cdot d\mathbf{s} = -\Psi$, just as $\int \mathbf{F} \cdot d\mathbf{s} = -U$; there is only a scale factor between the two:

$$U = -\int \mathbf{F} \cdot d\mathbf{s} = -m \int \mathbf{C} \cdot d\mathbf{s} = m\Psi. \quad (14.7)$$

By having this function *Ψ*(*x*, *y*, *z*) at every point in space, we can immediately calculate the potential energy of an object at any point in space, namely, *U*(*x*, *y*, *z*) = *mΨ*(*x*, *y*, *z*)—rather a trivial business, it seems. But it is not really trivial, because it is sometimes much nicer to describe the field by giving the value

of Ψ everywhere in space instead of having to give $\mathbf{C}$. Instead of having to write three complicated components of a vector function, we can give instead the scalar function Ψ . Furthermore, it is much easier to calculate Ψ than any given component of $\mathbf{C}$ when the field is produced by a number of masses, for since the potential is a scalar we merely add, without worrying about direction. Also, the field $\mathbf{C}$ can be recovered easily from Ψ , as we shall shortly see. Suppose we have point masses $m_1, m_2, \dots$ at the points 1, 2, $\dots$ and we wish to know the potential Ψ at some arbitrary point p . This is simply the sum of the potentials at P due to the individual masses taken one by one:

$$\Psi(p) = \sum_i -\frac{Gm_i}{r_{ip}}, \quad i = 1, 2, \dots \quad (14.8)$$

In the last chapter we used this formula, that the potential is the sum of the potentials from all the different objects, to calculate the potential due to a spherical shell of matter by adding the contributions to the potential at a point from all parts of the shell. The result of this calculation is shown graphically in Fig. 14-4. It is negative, having the value zero at $r = \infty$ and varying as $1/r$ down to the radius a , and then is constant inside the shell. Outside the shell the potential is $-Gm/r$, where m is the mass of the shell, which is exactly the same as it would have been if all the mass were located at the center. But it is not *everywhere* exactly the same, for inside the shell the potential turns out to be $-Gm/a$, and is a constant! *When the potential is constant, there is no field*, or when the potential energy is constant there is no force, because if we move an object from one place to another anywhere inside the sphere the work done by the force is exactly zero. Why? Because the work done in moving the object from one place to the other is equal to minus the change in the potential energy (or, the corresponding field integral is the change of the potential). But the potential energy is the *same* at any two points inside, so there is zero change in potential energy, and therefore no work is done in going between any two points inside the shell. The only way the work can be zero for all directions of displacement is that there is no force at all.

This gives us a clue as to how we can obtain the force or the field, given the potential energy. Let us suppose that the potential energy of an object is known at the position (x, y, z) and we want to know what the force on the object is. It will not do to know the potential at only this one point, as we shall see; it requires knowledge of the potential at neighboring points as well. Why? How can we calculate the x -component of the force? (If we can do this, of course, we can also find the y - and z -components, and we will then know the whole force.) Now, if we were to move the object a small distance Δx , the work done by the force on the object would be the x -component of the force times Δx , if Δx is sufficiently small, and this should equal the change in potential energy in going from one point to the other:

$$\Delta W = -\Delta U = F_x \Delta x. \quad (14.9)$$

We have merely used the formula $\int \mathbf{F} \cdot d\mathbf{s} = -\Delta U$, but for a *very short* path. Now we divide by Δx and so find that the force is

$$F_x = -\Delta U / \Delta x. \quad (14.10)$$

Of course this is not exact. What we really want is the limit of (14.10) as Δx gets smaller and smaller, because it is only *exactly* right in the limit of infinitesimal Δx . This we recognize as the derivative of U with respect to x , and we would be inclined, therefore, to write $-dU/dx$. But U depends on x, y , and z , and the mathematicians have invented a different symbol to remind us to be very careful when we are differentiating such a function, so as to remember that we are considering that *only x varies*, and y and z do not vary. Instead of a d they simply make a "backwards 6," or ∂ . (A ∂ should have been used in the beginning of calculus because we always want to cancel that d , but we never want to cancel a ∂ !) So they write $\partial U / \partial x$, and furthermore, in moments of duress, if they want to be *very* careful, they put a line beside it with a little yz at the bottom ($\partial U / \partial x|_{yz}$),

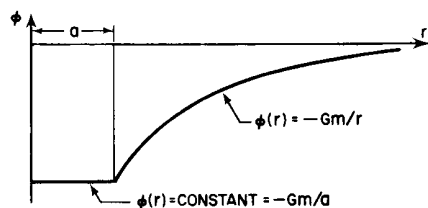


Fig. 14-4. Potential due to a spherical shell of radius a .

which means "Take the derivative of U with respect to x , keeping y and z constant." Most often we leave out the remark about what is kept constant because it is usually evident from the context, so we usually do not use the line with the y and z . However, *always* use a ∂ instead of a d as a warning that it is a derivative with some other variables kept constant. This is called a *partial derivative*; it is a derivative in which we vary only x .

Therefore, we find that the force in the x -direction is minus the partial derivative of U with respect to x :

$$F_x = -\partial U / \partial x. \quad (14.11)$$

In a similar way, the force in the y -direction can be found by differentiating U with respect to y , keeping x and z constant, and the third component, of course, is the derivative with respect to z , keeping y and x constant:

$$F_y = -\partial U / \partial y, \quad F_z = -\partial U / \partial z. \quad (14.12)$$

This is the way to get from the potential energy to the force. We get the *field* from the *potential* in exactly the same way:

$$C_x = -\partial \Psi / \partial x, \quad C_y = -\partial \Psi / \partial y, \quad C_z = -\partial \Psi / \partial z. \quad (14.13)$$

Incidentally, we shall mention here another notation, which we shall not actually use for quite a while: Since $\mathbf{C}$ is a vector and has x -, y -, and z -components, the symbolized $\partial/\partial x$, $\partial/\partial y$, and $\partial/\partial z$ which produce the x -, y -, and z -components are something like vectors. The mathematicians have invented a glorious new symbol, ∇ , called "grad" or "gradient" which is not a quantity but an *operator* which makes a vector from a scalar. It has the following "components": The x -component of this "grad" is $\partial/\partial x$, the y -component is $\partial/\partial y$, and the z -component is $\partial/\partial z$, and then we have the fun of writing our formulas this way:

$$\mathbf{F} = -\nabla U, \quad \mathbf{C} = -\nabla \Psi. \quad (14.14)$$

Using ∇ gives us a quick way of testing whether we have a real vector equation or not, but actually Eq. (14.14) means precisely the same as Eqs. (14.11) and (14.12); it is just another way of writing them, and since we do not want to write three equations every time, we just write ∇U instead.

One more example of fields and potentials has to do with the electrical case. In the case of electricity the force on a stationary object is the charge times the electric field: $\mathbf{F} = q\mathbf{E}$. (In general, of course, the x -component of force in an electrical problem has also a part which depends on the magnetic field. It is easy to show from Eq. (12.10) that the force on a particle due to magnetic fields is always at right angles to its velocity, and also at right angles to the field. Since the force due to magnetism on a moving charge is at right angles to the velocity, *no work is done* by the magnetism on the moving charge because the motion is at right angles to the force. Therefore, in calculating theorems of kinetic energy in electric and magnetic fields we can disregard the contribution from the magnetic field, since it does not change the kinetic energy.) We suppose that there is only an electric field. Then we can calculate the energy, or work done, in the same way as for gravity, and calculate a quantity ϕ which is minus the integral of $\mathbf{E} \cdot d\mathbf{s}$, from the arbitrary fixed point to the point where we make the calculation, and then the potential energy in an electric field is just charge times this quantity ϕ :

$$\phi(\mathbf{r}) = \int \mathbf{E} \cdot d\mathbf{s},$$

$$U = q\phi.$$

Let us take, as an example, the case of two parallel metal plates, each with a surface charge of $\pm\sigma$ per unit area. This is called a parallel-plate capacitor. We found previously that there is zero force outside the plates and that there is a constant electric field between them, directed from $+$ to $-$ and of magnitude σ/ϵ_0 (Fig. 14-5). We would like to know how much work would be done in

carrying a charge from one plate to the other. The work would be the (force) $\cdot$ (ds) integral, which can be written as charge times the potential value at plate 1 minus that at plate 2:

$$W = \int_1^2 \mathbf{F} \cdot d\mathbf{s} = q(\phi_1 - \phi_2).$$

We can actually work out the integral because the force is constant, and if we call the separation of the plates d , then the integral is easy:

$$\int_1^2 \mathbf{F} \cdot d\mathbf{s} = \frac{q\sigma}{\epsilon_0} \int_1^2 dx = \frac{q\sigma d}{\epsilon_0}.$$

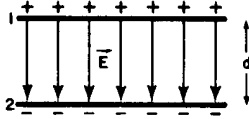


Fig. 14-5. Field between parallel plates.

The difference in potential, $\Delta\phi = \sigma d/\epsilon_0$, is called the *voltage difference*, and ϕ is measured in volts. When we say a pair of plates is charged to a certain voltage, what we mean is that the difference in electrical potential of the two plates is so-and-so many volts. For a capacitor made of two parallel plates carrying a surface charge $\pm\sigma$, the voltage, or difference in potential, of the pair of plates is $\sigma d/\epsilon_0$.

The Special Theory of Relativity

15-1 The principle of relativity

For over 200 years the equations of motion enunciated by Newton were believed to describe nature correctly, and the first time that an error in these laws was discovered, the way to correct it was also discovered. Both the error and its correction were discovered by Einstein in 1905.

Newton's Second Law, which we have expressed by the equation

$$F = d(mv)/dt,$$

was stated with the tacit assumption that m is a constant, but we now know that this is not true, and that the mass of a body increases with velocity. In Einstein's corrected formula m has the value

$$m = \frac{m_0}{\sqrt{1 - v^2/c^2}}, \quad (15.1)$$

where the "rest mass" m_0 represents the mass of a body that is not moving and c is the speed of light, which is about 3×10^5 km · sec⁻¹ or about 186,000 mi · sec⁻¹.

For those who want to learn just enough about it so they can solve problems, that is all there is to the theory of relativity—it just changes Newton's laws by introducing a correction factor to the mass. From the formula itself it is easy to see that this mass increase is very small in ordinary circumstances. If the velocity is even as great as that of a satellite, which goes around the earth at 5 mi/sec, then $v/c = 5/186,000$: putting this value into the formula shows that the correction to the mass is only one part in two to three billion, which is nearly impossible to observe. Actually, the correctness of the formula has been amply confirmed by the observation of many kinds of particles, moving at speeds ranging up to practically the speed of light. However, because the effect is ordinarily so small, it seems remarkable that it was discovered theoretically before it was discovered experimentally. Empirically, at a sufficiently high velocity, the effect is very large, but it was not discovered that way. Therefore it is interesting to see how a law that involved so delicate a modification (at the time when it was first discovered) was brought to light by a combination of experiments and physical reasoning. Contributions to the discovery were made by a number of people, the final result of whose work was Einstein's discovery.

There are really two Einstein theories of relativity. This chapter is concerned with the Special Theory of Relativity, which dates from 1905. In 1915 Einstein published an additional theory, called the General Theory of Relativity. This latter theory deals with the extension of the Special Theory to the case of the law of gravitation; we shall not discuss the General Theory here.

The principle of relativity was first stated by Newton, in one of his corollaries to the laws of motion: "The motions of bodies included in a given space are the same among themselves, whether that space is at rest or moves uniformly forward in a straight line." This means, for example, that if a space ship is drifting along at a uniform speed, all experiments performed in the space ship and all the phenomena in the space ship will appear the same as if the ship were not moving, provided, of course, that one does not look outside. That is the meaning of the principle of relativity. This is a simple enough idea, and the only question is whether it is *true* that in all experiments performed inside a moving system the laws of physics

15-1 The principle of relativity

15-2 The Lorentz transformation

15-3 The Michelson-Morley experiment

15-4 Transformation of time

15-5 The Lorentz contraction

15-6 Simultaneity

15-7 Four-vectors

15-8 Relativistic dynamics

15-9 Equivalence of mass and energy

will appear the same as they would if the system were standing still. Let us first investigate whether Newton's laws appear the same in the moving system.

Suppose that Moe is moving in the x -direction with a uniform velocity u , and he measures the position of a certain point, shown in Fig. 15-1. He designates the " x -distance" of the point in his coordinate system as x' . Joe is at rest, and measures the position of the same point, designating its x -coordinate in his system as x . The relationship of the coordinates in the two systems is clear from the diagram. After time t Moe's origin has moved a distance ut , and if the two systems originally coincided,

$$\begin{aligned}x' &= x - ut, \\y' &= y, \\z' &= z, \\t' &= t.\end{aligned}\tag{15.2}$$

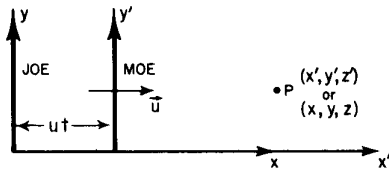


Fig. 15-1. Two coordinate systems in uniform relative motion along their x -axes.

If we substitute this transformation of coordinates into Newton's laws we find that these laws transform to the same laws in the primed system; that is, the laws of Newton are of the same form in a moving system as in a stationary system, and therefore it is impossible to tell, by making mechanical experiments, whether the system is moving or not.

The principle of relativity has been used in mechanics for a long time. It was employed by various people, in particular Huygens, to obtain the rules for the collision of billiard balls, in much the same way as we used it in Chapter 10 to discuss the conservation of momentum. In the past century interest in it was heightened as the result of investigations into the phenomena of electricity, magnetism, and light. A long series of careful studies of these phenomena by many people culminated in Maxwell's equations of the electromagnetic field, which describe electricity, magnetism, and light in one uniform system. However, the Maxwell equations did *not* seem to obey the principle of relativity. That is, if we transform Maxwell's equations by the substitution of equations 15.2, *their form does not remain the same*; therefore, in a moving space ship the electrical and optical phenomena should be different from those in a stationary ship. Thus one could use these optical phenomena to determine the speed of the ship; in particular, one could determine the absolute speed of the ship by making suitable optical or electrical measurements. One of the consequences of Maxwell's equations is that if there is a disturbance in the field such that light is generated, these electromagnetic waves go out in all directions equally and at the same speed c , or 186,000 mi/sec. Another consequence of the equations is that if the source of the disturbance is moving, the light emitted goes through space at the same speed c . This is analogous to the case of sound, the speed of sound waves being likewise independent of the motion of the source.

This independence of the motion of the source, in the case of light, brings up an interesting problem:

Suppose we are riding in a car that is going at a speed u , and light from the rear is going past the car with speed c . Differentiating the first equation in (15.2) gives

$$dx'/dt = dx/dt - u,$$

which means that according to the Galilean transformation the apparent speed of the passing light, as we measure it in the car, should not be c but should be $c - u$. For instance, if the car is going 100,000 mi/sec, and the light is going 186,000 mi/sec, then apparently the light going past the car should go 86,000 mi/sec. In any case, by measuring the speed of the light going past the car (if the Galilean transformation is correct for light), one could determine the speed of the car. A number of experiments based on this general idea were performed to determine the velocity of the earth, but they all failed—they gave *no velocity at all*. We shall discuss one of these experiments in detail, to show exactly what was done and what was the matter; something *was* the matter, of course, something was wrong with the equations of physics. What could it be?

15-2 The Lorentz transformation

When the failure of the equations of physics in the above case came to light, the first thought that occurred was that the trouble must lie in the new Maxwell equations of electrodynamics, which were only 20 years old at the time. It seemed almost obvious that these equations must be wrong, so the thing to do was to change them in such a way that under the Galilean transformation the principle of relativity would be satisfied. When this was tried, the new terms that had to be put into the equations led to predictions of new electrical phenomena that did not exist at all when tested experimentally, so this attempt had to be abandoned. Then it gradually became apparent that Maxwell's laws of electrodynamics were correct, and the trouble must be sought elsewhere.

In the meantime, H. A. Lorentz noticed a remarkable and curious thing when he made the following substitutions in the Maxwell equations:

$$\begin{aligned}x' &= \frac{x - ut}{\sqrt{1 - u^2/c^2}}, \\y' &= y, \\z' &= z, \\t' &= \frac{t - ux/c^2}{\sqrt{1 - u^2/c^2}},\end{aligned}\tag{15.3}$$

namely, Maxwell's equations remain in the same form when this transformation is applied to them! Equations (15.3) are known as a *Lorentz transformation*. Einstein, following a suggestion originally made by Poincaré, then proposed that *all the physical laws* should be of such a kind that they *remain unchanged under a Lorentz transformation*. In other words, we should change, not the laws of electrodynamics, but the laws of mechanics. How shall we change Newton's laws so that *they* will remain unchanged by the Lorentz transformation? If this goal is set, we then have to rewrite Newton's equations in such a way that the conditions we have imposed are satisfied. As it turned out, the only requirement is that the mass m in Newton's equations must be replaced by the form shown in Eq. (15.1). When this change is made, Newton's laws and the laws of electrodynamics will harmonize. Then if we use the Lorentz transformation in comparing Moe's measurements with Joe's, we shall never be able to detect whether either is moving, because the form of all the equations will be the same in both coordinate systems!

It is interesting to discuss what it means that we replace the old transformation between the coordinates and time with a new one, because the old one (Galilean) seems to be self-evident, and the new one (Lorentz) looks peculiar. We wish to know whether it is logically and experimentally possible that the new, and not the old, transformation can be correct. To find that out, it is not enough to study the laws of mechanics but, as Einstein did, we too must analyze our ideas of *space* and *time* in order to understand this transformation. We shall have to discuss these ideas and their implications for mechanics at some length, so we say in advance that the effort will be justified, since the results agree with experiment.

15-3 The Michelson-Morley experiment

As mentioned above, attempts were made to determine the absolute velocity of the earth through the hypothetical "ether" that was supposed to pervade all space. The most famous of these experiments is one performed by Michelson and Morley in 1887. It was 18 years later before the negative results of the experiment were finally explained, by Einstein.

The Michelson-Morley experiment was performed with an apparatus like that shown schematically in Fig. 15-2. This apparatus is essentially comprised of a light source A , a partially silvered glass plate B , and two mirrors C and E , all mounted on a rigid base. The mirrors are placed at equal distances L from B . The plate B splits an oncoming beam of light, and the two resulting beams con-

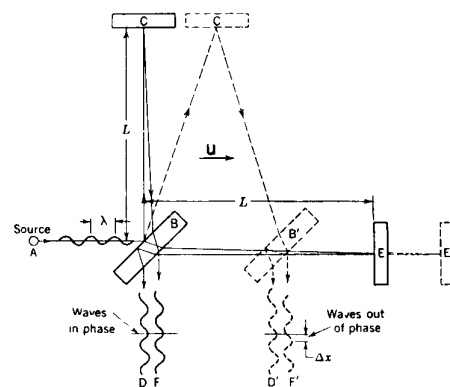


Fig. 15-2. Schematic diagram of the Michelson-Morley experiment.

tinue in mutually perpendicular directions to the mirrors, where they are reflected back to B . On arriving back at B , the two beams are recombined as two superposed beams, D and F . If the time taken for the light to go from B to E and back is the same as the time from B to C and back, the emerging beams D and F will be in phase and will reinforce each other, but if the two times differ slightly, the beams will be slightly out of phase and interference will result. If the apparatus is "at rest" in the ether, the times should be precisely equal, but if it is moving toward the right with a velocity u , there should be a difference in the times. Let us see why.

First, let us calculate the time required for the light to go from B to E and back. Let us say that the time for light to go from plate B to mirror E is t_1 , and the time for the return is t_2 . Now, while the light is on its way from B to the mirror, the apparatus moves a distance ut_1 , so the light must traverse a distance $L + ut_1$, at the speed c . We can also express this distance as ct_1 , so we have

$$ct_1 = L + ut_1, \quad \text{or} \quad t_1 = L/(c - u).$$

(This result is also obvious from the point of view that the velocity of light relative to the apparatus is $c - u$, so the time is the length L divided by $c - u$.) In a like manner, the time t_2 can be calculated. During this time the plate B advances a distance ut_2 , so the return distance of the light is $L - ut_2$. Then we have

$$ct_2 = L - ut_2, \quad \text{or} \quad t_2 = L/(c + u).$$

Then the total time is

$$t_1 + t_2 = 2Lc/(c^2 - u^2).$$

For convenience in later comparison of times we write this as

$$t_1 + t_2 = \frac{2L/c}{1 - u^2/c^2}. \quad (15.4)$$

Our second calculation will be of the time t_3 for the light to go from B to the mirror C . As before, during time t_3 the mirror C moves to the right a distance ut_3 to the position C' ; in the same time, the light travels a distance ct_3 along the hypotenuse of a triangle, which is BC' . For this right triangle we have

$$(ct_3)^2 = L^2 + (ut_3)^2$$

or

$$L^2 = c^2 t_3^2 - u^2 t_3^2 = (c^2 - u^2) t_3^2,$$

from which we get

$$t_3 = L/\sqrt{c^2 - u^2}.$$

For the return trip from C' the distance is the same, as can be seen from the symmetry of the figure; therefore the return time is also the same, and the total time is $2t_3$. With a little rearrangement of the form we can write

$$2t_3 = \frac{2L}{\sqrt{c^2 - u^2}} = \frac{2L/c}{\sqrt{1 - u^2/c^2}}. \quad (15.5)$$

We are now able to compare the times taken by the two beams of light. In expressions (15.4) and (15.5) the numerators are identical, and represent the time that would be taken if the apparatus were at rest. In the denominators, the term u^2/c^2 will be small, unless u is comparable in size to c . The denominators represent the modifications in the times caused by the motion of the apparatus. And behold, these modifications are *not the same*—the time to go to C and back is a little less than the time to E and back, even though the mirrors are equidistant from B , and all we have to do is to measure that difference with precision.

Here a minor technical point arises—suppose the two lengths L are not exactly equal? In fact, we surely cannot make them exactly equal. In that case we simply turn the apparatus 90 degrees, so that BC is in the line of motion and BE is perpendicular to the motion. Any small difference in length then becomes

unimportant, and what we look for is a *shift* in the interference fringes when we rotate the apparatus.

In carrying out the experiment, Michelson and Morley oriented the apparatus so that the line BE was nearly parallel to the earth's motion in its orbit (at certain times of the day and night). This orbital speed is about 18 miles per second, and any "ether drift" should be at least that much at some time of the day or night and at some time during the year. The apparatus was amply sensitive to observe such an effect, but no time difference was found—the velocity of the earth through the ether could not be detected. The result of the experiment was null.

The result of the Michelson-Morley experiment was very puzzling and most disturbing. The first fruitful idea for finding a way out of the impasse came from Lorentz. He suggested that material bodies contract when they are moving, and that this foreshortening is only in the direction of the motion, and also, that if the length is L_0 when a body is at rest, then when it moves with speed u parallel to its length, the new length, which we call $L_{||}$ (L -parallel), is given by

$$L_{||} = L_0 \sqrt{1 - u^2/c^2}. \quad (15.6)$$

When this modification is applied to the Michelson-Morley interferometer apparatus the distance from B to C does not change, but the distance from B to E is shortened to $L\sqrt{1 - u^2/c^2}$. Therefore Eq. (15.5) is not changed, but the L of Eq. (15.4) must be changed in accordance with Eq. (15.6). When this is done we obtain

$$t_1 + t_2 = \frac{(2L/c) \sqrt{1 - u^2/c^2}}{1 - u^2/c^2} = \frac{2L/c}{\sqrt{1 - u^2/c^2}}. \quad (15.7)$$

Comparing this result with Eq. (15.5), we see that $t_1 + t_2 = 2t_3$. So if the apparatus shrinks in the manner just described, we have a way of understanding why the Michelson-Morley experiment gives no effect at all. Although the contraction hypothesis successfully accounted for the negative result of the experiment, it was open to the objection that it was invented for the express purpose of explaining away the difficulty, and was too artificial. However, in many other experiments to discover an ether wind, similar difficulties arose, until it appeared that nature was in a "conspiracy" to thwart man by introducing some new phenomenon to undo every phenomenon that he thought would permit a measurement of u .

It was ultimately recognized, as Poincaré pointed out, that *a complete conspiracy is itself a law of nature!* Poincaré then proposed that there *is* such a law of nature, that it is not possible to discover an ether wind by *any* experiment; that is, there is no way to determine an absolute velocity.

15-4 Transformation of time

In checking out whether the contraction idea is in harmony with the facts in other experiments, it turns out that everything is correct provided that the *times* are also modified, in the manner expressed in the fourth equation of the set (15.3). That is because the time t_3 , calculated for the trip from B to C and back, is not the same when calculated by a man performing the experiment in a moving space ship as when calculated by a stationary observer who is watching the space ship. To the man in the ship the time is simply $2L/c$, but to the other observer it is $(2L/c)/\sqrt{1 - u^2/c^2}$ (Eq. 15.5). In other words, when the outsider sees the man in the space ship lighting a cigar, all the actions appear to be slower than normal, while to the man inside, everything moves at a normal rate. So not only must the lengths shorten, but also the time-measuring instruments ("clocks") must apparently slow down. That is, when the clock in the space ship records 1 second elapsed, as seen by the man in the ship, it shows $1/\sqrt{1 - u^2/c^2}$ second to the man outside.

This slowing of the clocks in a moving system is a very peculiar phenomenon, and is worth an explanation. In order to understand this, we have to watch the machinery of the clock and see what happens when it is moving. Since that is

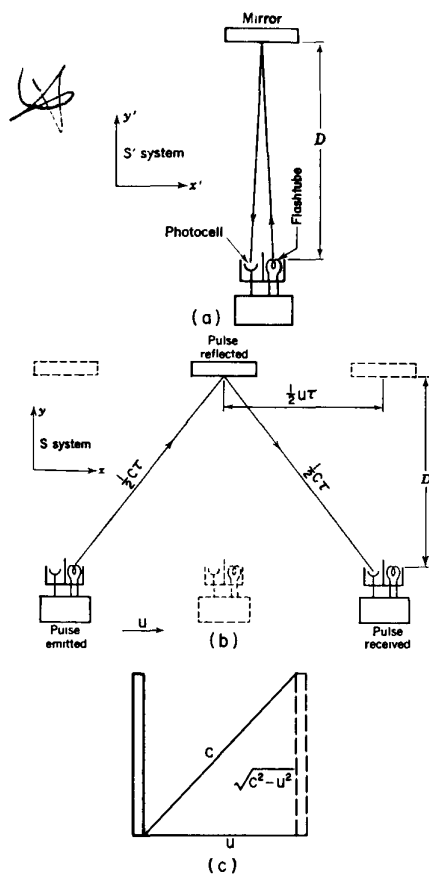


Fig. 15-3. (a) A "light clock" at rest in the S' system. (b) The same clock, moving through the S system. (c) Illustration of the diagonal path taken by the light beam in a moving "light clock."

rather difficult, we shall take a very simple kind of clock. The one we choose is rather a silly kind of clock, but it will work in principle: it is a rod (meter stick) with a mirror at each end, and when we start a light signal between the mirrors, the light keeps going up and down, making a click every time it comes down, like a standard ticking clock. We build two such clocks, with exactly the same lengths, and synchronize them by starting them together; then they agree always thereafter, because they are the same in length, and light always travels with speed c . We give one of these clocks to the man to take along in his space ship, and he mounts the rod perpendicular to the direction of motion of the ship; then the length of the rod will not change. How do we know that perpendicular lengths do not change? The men can agree to make marks on each other's y -meter stick as they pass each other. By symmetry, the two marks must come at the same y - and y' -coordinates, since otherwise, when they get together to compare results, one mark will be above or below the other, and so we could tell who was really moving.

Now let us see what happens to the moving clock. Before the man took it aboard, he agreed that it was a nice, standard clock, and when he goes along in the space ship he will not see anything peculiar. If he did, he would know he was moving—if anything at all changed because of the motion, he could tell he was moving. But the principle of relativity says this is impossible in a uniformly moving system, so nothing has changed. On the other hand, when the external observer looks at the clock going by, he sees that the light, in going from mirror to mirror, is "really" taking a zigzag path, since the rod is moving sidewise all the while. We have already analyzed such a zigzag motion in connection with the Michelson-Morley experiment. If in a given time the rod moves forward a distance proportional to u in Fig. 15-3, the distance the light travels in the same time is proportional to c , and the vertical distance is therefore proportional to $\sqrt{c^2 - u^2}$.

That is, it takes a *longer* time for light to go from end to end in the moving clock than in the stationary clock. Therefore the apparent time between clicks is longer for the moving clock, in the same proportion as shown in the hypotenuse of the triangle (that is the source of the square root expressions in our equations). From the figure it is also apparent that the greater u is, the more slowly the moving clock appears to run. Not only does this particular kind of clock run more slowly, but if the theory of relativity is correct, any other clock, operating on any principle whatsoever, would also appear to run slower, and in the same proportion—we can say this without further analysis. Why is this so?

To answer the above question, suppose we had two other clocks made exactly alike with wheels and gears, or perhaps based on radioactive decay, or something else. Then we adjust these clocks so they both run in precise synchronism with our first clocks. When light goes up and back in the first clocks and announces its arrival with a click, the new models also complete some sort of cycle, which they simultaneously announce by some doubly coincident flash, or bong, or other signal. One of these clocks is taken into the space ship, along with the first kind. Perhaps *this* clock will not run slower, but will continue to keep the same time as its stationary counterpart, and thus disagree with the other moving clock. Ah no, if that should happen, the man in the ship could use this mismatch between his two clocks to determine the speed of his ship, which we have been supposing is impossible. *We need not know anything about the machinery* of the new clock that might cause the effect—we simply know that whatever the reason, it will appear to run slow, just like the first one.

Now if *all* moving clocks run slower, if no way of measuring time gives anything but a slower rate, we shall just have to say, in a certain sense, that *time itself* appears to be slower in a space ship. All the phenomena there—the man's pulse rate, his thought processes, the time he takes to light a cigar, how long it takes to grow up and get old—all these things must be slowed down in the same proportion, because he cannot tell he is moving. The biologists and medical men sometimes say it is not quite certain that the time it takes for a cancer to develop will be longer in a space ship, but from the viewpoint of a modern physicist it is nearly certain; otherwise one could use the rate of cancer development to determine the speed of the ship!

A very interesting example of the slowing of time with motion is furnished by mu-mesons (muons), which are particles that disintegrate spontaneously after an average lifetime of 2.2×10^{-6} sec. They come to the earth in cosmic rays, and can also be produced artificially in the laboratory. Some of them disintegrate in midair, but the remainder disintegrate only after they encounter a piece of material and stop. It is clear that in its short lifetime a muon cannot travel, even at the speed of light, much more than 600 meters. But although the muons are created at the top of the atmosphere, some 10 kilometers up, yet they are actually found in a laboratory down here, in cosmic rays. How can that be? The answer is that different muons move at various speeds, some of which are very close to the speed of light. While from their own point of view they live only about $2 \mu\text{sec}$, from our point of view they live considerably longer—enough longer that they may reach the earth. The factor by which the time is increased has already been given as $1/\sqrt{1 - u^2/c^2}$. The average life has been measured quite accurately for muons of different velocities, and the values agree closely with the formula.

We do not know why the meson disintegrates or what its machinery is, but we do know its behavior satisfies the principle of relativity. That is the utility of the principle of relativity—it permits us to make predictions, even about things that otherwise we do not know much about. For example, before we have any idea at all about what makes the meson disintegrate, we can still predict that when it is moving at nine-tenths of the speed of light, the apparent length of time that it lasts is $(2.2 \times 10^{-6})/\sqrt{1 - 9^2/10^2}$ sec; and our prediction works—that is the good thing about it.

15-5 The Lorentz contraction

Now let us return to the Lorentz transformation (15.3) and try to get a better understanding of the relationship between the (x, y, z, t) and the (x', y', z', t') coordinate systems, which we shall call the S and S' systems, or Joe and Moe systems, respectively. We have already noted that the first equation is based on the Lorentz suggestion of contraction along the x -direction; how can we prove that a contraction takes place? In the Michelson-Morley experiment, we now appreciate that the *transverse* arm BC cannot change length, by the principle of relativity; yet the null result of the experiment demands that the *times* must be equal. So, in order for the experiment to give a null result, the longitudinal arm BE must appear shorter, by the square root $\sqrt{1 - u^2/c^2}$. What does this contraction mean, in terms of measurements made by Joe and Moe? Suppose that Moe, moving with the S' system in the x -direction, is measuring the x' -coordinate of some point with a meter stick. He lays the stick down x' times, so he thinks the distance is x' meters. From the viewpoint of Joe in the S system, however, Moe is using a foreshortened ruler, so the “real” distance measured is $x'\sqrt{1 - u^2/c^2}$ meters. Then if the S' system has travelled a distance ut away from the S system, the S observer would say that the same point, measured in his coordinates, is at a distance $x = x'\sqrt{1 - u^2/c^2} + ut$, or

$$x' = \frac{x - ut}{\sqrt{1 - u^2/c^2}},$$

which is the first equation of the Lorentz transformation.

15-6 Simultaneity

In an analogous way, because of the difference in time scales, the denominator expression is introduced into the fourth equation of the Lorentz transformation. The most interesting term in that equation is the ux/c^2 in the numerator, because that is quite new and unexpected. Now what does that mean? If we look at the situation carefully we see that events that occur at two separated places at the same time, as seen by Moe in S' , do *not* happen at the same time as viewed by Joe in S . If one event occurs at point x_1 at time t_0 and the other event at x_2 and t_0 (the same

time), we find that the two corresponding times t'_1 and t'_2 differ by an amount

$$t'_2 - t'_1 = \frac{u(x_1 - x_2)/c^2}{\sqrt{1 - u^2/c^2}}.$$

This circumstance is called "failure of simultaneity at a distance," and to make the idea a little clearer let us consider the following experiment.

Suppose that a man moving in a space ship (system S') has placed a clock at each end of the ship and is interested in making sure that the two clocks are in synchronism. How can the clocks be synchronized? There are many ways. One way, involving very little calculation, would be first to locate exactly the midpoint between the clocks. Then from this station we send out a light signal which will go both ways at the same speed and will arrive at both clocks, clearly, at the same time. This simultaneous arrival of the signals can be used to synchronize the clocks. Let us then suppose that the man in S' synchronizes his clocks by this particular method. Let us see whether an observer in system S would agree that the two clocks are synchronous. The man in S' has a right to believe they are, because he does not know that he is moving. But the man in S reasons that since the ship is moving forward, the clock in the front end was running away from the light signal, hence the light had to go more than halfway in order to catch up; the rear clock, however, was advancing to meet the light signal, so this distance was shorter. Therefore the signal reached the rear clock first, although the man in S' thought that the signals arrived simultaneously. We thus see that when a man in a space ship thinks the times at two locations are simultaneous, equal values of t' in his coordinate system must correspond to *different* values of t in the other coordinate system!

15-7 Four-vectors

Let us see what else we can discover in the Lorentz transformation. It is interesting to note that the transformation between the x 's and t 's is analogous in form to the transformation of the x 's and y 's that we studied in Chapter 11 for a rotation of coordinates. We then had

$$\begin{aligned} x' &= x \cos \theta + y \sin \theta, \\ y' &= y \cos \theta - x \sin \theta, \end{aligned} \quad (15.8)$$

in which the new x' mixes the old x and y , and the new y' also mixes the old x and y ; similarly, in the Lorentz transformation we find a new x' which is a mixture of x and t , and a new t' which is a mixture of t and x . So the Lorentz transformation is analogous to a rotation, only it is a "rotation" in *space and time*, which appears to be a strange concept. A check of the analogy to rotation can be made by calculating the quantity

$$x'^2 + y'^2 + z'^2 - c^2 t'^2 = x^2 + y^2 + z^2 - c^2 t^2. \quad (15.9)$$

In this equation the first three terms on each side represent, in three-dimensional geometry, the square of the distance between a point and the origin (surface of a sphere) which remains unchanged (invariant) regardless of rotation of the coordinate axes. Similarly, Eq. (15.9) shows that there is a certain combination which includes time, that is invariant to a Lorentz transformation. Thus, the analogy to a rotation is complete, and is of such a kind that vectors, i.e., quantities involving "components" which transform the same way as the coordinates and time, are also useful in connection with relativity.

Thus we contemplate an extension of the idea of vectors, which we have so far considered to have only space components, to include a time component. That is, we expect that there will be vectors with four components, three of which are like the components of an ordinary vector, and with these will be associated a fourth component, which is the analog of the time part.

This concept will be analyzed further in the next chapters, where we shall find that if the ideas of the preceding paragraph are applied to momentum, the transformation gives three space parts that are like ordinary momentum components, and a fourth component, the time part, which is the *energy*.

15-8 Relativistic dynamics

We are now ready to investigate, more generally, what form the laws of mechanics take under the Lorentz transformation. [We have thus far explained how length and time change, but not how we get the modified formula for m (Eq. 15.1). We shall do this in the next chapter.] To see the consequences of Einstein's modification of m for Newtonian mechanics, we start with the Newtonian law that force is the rate of change of momentum, or

$$\mathbf{F} = d(m\mathbf{v})/dt.$$

Momentum is still given by $m\mathbf{v}$, but when we use the new m this becomes

$$\mathbf{p} = m\mathbf{v} = \frac{m_0\mathbf{v}}{\sqrt{1 - v^2/c^2}}. \quad (15.10)$$

This is Einstein's modification of Newton's laws. Under this modification, if action and reaction are still equal (which they may not be in detail, but are in the long run), there will be conservation of momentum in the same way as before, but the quantity that is being conserved is not the old $m\mathbf{v}$ with its constant mass, but instead is the quantity shown in (15.10), which has the modified mass. When this change is made in the formula for momentum, conservation of momentum still works.

Now let us see how momentum varies with speed. In Newtonian mechanics it is proportional to the speed and, according to (15.10), over a considerable range of speed, but small compared with c , it is nearly the same in relativistic mechanics, because the square-root expression differs only slightly from 1. But when v is almost equal to c , the square-root expression approaches zero, and the momentum therefore goes toward infinity.

What happens if a constant force acts on a body for a long time? In Newtonian mechanics the body keeps picking up speed until it goes faster than light. But this is impossible in relativistic mechanics. In relativity, the body keeps picking up, not speed, but momentum, which can continually increase because the mass is increasing. After a while there is practically no acceleration in the sense of a change of velocity, but the momentum continues to increase. Of course, whenever a force produces very little change in the velocity of a body, we say that the body has a great deal of inertia, and that is exactly what our formula for relativistic mass says (see Eq. 15.10)—it says that the inertia is very great when v is nearly as great as c . As an example of this effect, to deflect the high-speed electrons in the synchrotron that is used here at Caltech, we need a magnetic field that is 2000 times stronger than would be expected on the basis of Newton's laws. In other words, the mass of the electrons in the synchrotron is 2000 times as great as their normal mass, and is as great as that of a proton! That m should be 2000 times m_0 means that $1 - v^2/c^2$ must be $1/4,000,000$, and that means that v^2/c^2 differs from 1 by one part in 4,000,000, or that v differs from c by one part in 8,000,000, so the electrons are getting pretty close to the speed of light. If the electrons and light were both to start from the synchrotron (estimated as 700 feet away) and rush out to Bridge Lab, which would arrive first? The light, of course, because light always travels faster.* How much earlier? That is too hard to tell—instead, we tell by what distance the light is ahead: it is about $1/1000$ of an inch, or $\frac{1}{4}$ the thickness of a piece of paper! When the electrons are going that fast their masses are enormous, but their speed cannot exceed the speed of light.

* The electrons would actually win the race versus *visible* light because of the index of refraction of air. A gamma ray would make out better.

Now let us look at some further consequences of relativistic change of mass. Consider the motion of the molecules in a small tank of gas. When the gas is heated, the speed of the molecules is increased, and therefore the mass is also increased and the gas is heavier. An approximate formula to express the increase of mass, for the case when the velocity is small, can be found by expanding $m_0/\sqrt{1 - v^2/c^2} = m_0(1 - v^2/c^2)^{-1/2}$ in a power series, using the binomial theorem. We get

$$m_0(1 - v^2/c^2)^{-1/2} = m_0(1 + \frac{1}{2}v^2/c^2 + \frac{3}{8}v^4/c^4 + \dots).$$

We see clearly from the formula that the series converges rapidly when v is small, and the terms after the first two or three are negligible. So we can write

$$m \cong m_0 + \frac{1}{2}m_0v^2\left(\frac{1}{c^2}\right) \quad (15.11)$$

in which the second term on the right expresses the increase of mass due to molecular velocity. When the temperature increases the v^2 increases proportionately, so we can say that the increase in mass is proportional to the increase in temperature. But since $\frac{1}{2}m_0v^2$ is the kinetic energy in the old-fashioned Newtonian sense, we can also say that the increase in mass of all this body of gas is equal to the increase in kinetic energy divided by c^2 , or $\Delta m = \Delta(\text{K.E.})/c^2$.

15-9 Equivalence of mass and energy

The above observation led Einstein to the suggestion that the mass of a body can be expressed more simply than by the formula (15.1), if we say that the mass is equal to the total energy content divided by c^2 . If Eq. (15.11) is multiplied by c^2 the result is

$$mc^2 = m_0c^2 + \frac{1}{2}m_0v^2 + \dots \quad (15.12)$$

Here, the term on the left expresses the total energy of a body, and we recognize the last term as the ordinary kinetic energy. Einstein interpreted the large constant term, m_0c^2 , to be part of the total energy of the body, an intrinsic energy known as the "rest energy."

Let us follow out the consequences of assuming, with Einstein, that *the energy of a body always equals mc^2* . As an interesting result, we shall find the formula (15.1) for the variation of mass with speed, which we have merely assumed up to now. We start with the body at rest, when its energy is m_0c^2 . Then we apply a force to the body, which starts it moving and gives it kinetic energy; therefore, since the energy has increased, the mass has increased—this is implicit in the original assumption. So long as the force continues, the energy and the mass both continue to increase. We have already seen (Chapter 13) that the rate of change of energy with time equals the force times the velocity, or

$$\frac{dE}{dt} = \mathbf{F} \cdot \mathbf{v}. \quad (15.13)$$

We also have (Chapter 9, Eq. 9.1) that $F = d(mv)/dt$. When these relations are put together with the definition of E , Eq. (15.13) becomes

$$\frac{d(mc^2)}{dt} = \mathbf{v} \cdot \frac{d(m\mathbf{v})}{dt}. \quad (15.14)$$

We wish to solve this equation for m . To do this we first use the mathematical trick of multiplying both sides by $2m$, which changes the equation to

$$c^2(2m) \frac{dm}{dt} = 2m\mathbf{v} \cdot \frac{d(m\mathbf{v})}{dt}. \quad (15.15)$$

We need to get rid of the derivatives, which can be accomplished by integrating

both sides. The quantity $(2m) dm/dt$ can be recognized as the time derivative of m^2 , and $(2mv) \cdot d(mv)/dt$ is the time derivative of $(mv)^2$. So, Eq. (15.15) is the same as

$$c^2 \frac{d(m^2)}{dt} = \frac{d(m^2 v^2)}{dt}. \quad (15.16)$$

If the derivatives of two quantities are equal, the quantities themselves differ at most by a constant, say C . This permits us to write

$$m^2 c^2 = m^2 v^2 + C. \quad (15.17)$$

We need to define the constant C more explicitly. Since Eq. (15.17) must be true for all velocities, we can choose a special case where $v = 0$, and say that in this case the mass is m_0 . Substituting these values into Eq. (15.17) gives

$$m_0^2 c^2 = 0 + C.$$

We can now use this value of C in Eq. (15.17), which becomes

$$m^2 c^2 = m^2 v^2 + m_0^2 c^2. \quad (15.18)$$

Dividing by c^2 and rearranging terms gives

$$m^2 (1 - v^2/c^2) = m_0^2,$$

from which we get

$$m = m_0 / \sqrt{1 - v^2/c^2}. \quad (15.19)$$

This is the formula (15.1), and is exactly what is necessary for the agreement between mass and energy in Eq. (15.12).

Ordinarily these energy changes represent extremely slight changes in mass, because most of the time we cannot generate much energy from a given amount of material; but in an atomic bomb of explosive energy equivalent to 20 kilotons of TNT, for example, it can be shown that the dirt after the explosion is lighter by 1 gram than the initial mass of the reacting material, because of the energy that was released, i.e., the released energy had a mass of 1 gram, according to the relationship $\Delta E = \Delta(mc^2)$. This theory of equivalence of mass and energy has been beautifully verified by experiments in which matter is annihilated—converted totally to energy: An electron and a positron come together at rest, each with a rest mass m_0 . When they come together they disintegrate and two gamma rays emerge, each with the measured energy of $m_0 c^2$. This experiment furnishes a direct determination of the energy associated with the existence of the rest mass of a particle.

Relativistic Energy and Momentum

16-1 Relativity and the philosophers

In this chapter we shall continue to discuss the principle of relativity of Einstein and Poincaré, as it affects our ideas of physics and other branches of human thought.

Poincaré made the following statement of the principle of relativity: "According to the principle of relativity, the laws of physical phenomena must be the same for a fixed observer as for an observer who has a uniform motion of translation relative to him, so that we have not, nor can we possibly have, any means of discerning whether or not we are carried along in such a motion."

When this idea descended upon the world, it caused a great stir among philosophers, particularly the "cocktail-party philosophers," who say, "Oh, it is very simple: Einstein's theory says all is relative!" In fact, a surprisingly large number of philosophers, not only those found at cocktail parties (but rather than embarrass them, we shall just call them "cocktail-party philosophers"), will say, "That all is relative is a consequence of Einstein, and it has profound influences on our ideas." In addition, they say "It has been demonstrated in physics that phenomena depend upon your frame of reference." We hear that a great deal, but it is difficult to find out what it means. Probably the frames of reference that were originally referred to were the coordinate systems which we use in the analysis of the theory of relativity. So the fact that "things depend upon your frame of reference" is supposed to have had a profound effect on modern thought. One might well wonder why, because, after all, that things depend upon one's point of view is so simple an idea that it certainly cannot have been necessary to go to all the trouble of the physical relativity theory in order to discover it. That what one sees depends upon his frame of reference is certainly known to anybody who walks around, because he sees an approaching pedestrian first from the front and then from the back; there is nothing deeper in most of the philosophy which is said to have come from the theory of relativity than the remark that "A person looks different from the front than from the back." The old story about the elephant that several blind men describe in different ways is another example, perhaps, of the theory of relativity from the philosopher's point of view.

But certainly there must be deeper things in the theory of relativity than just this simple remark that "A person looks different from the front than from the back." Of course relativity is deeper than this, because *we can make definite predictions with it*. It certainly would be rather remarkable if we could predict the behavior of nature from such a simple observation alone.

There is another school of philosophers who feel very uncomfortable about the theory of relativity, which asserts that we cannot determine our absolute velocity without looking at something outside, and who would say, "It is obvious that one cannot measure his velocity without looking outside. It is self-evident that it is *meaningless* to talk about the velocity of a thing without looking outside; the physicists are rather stupid for having thought otherwise, but it has just dawned on them that this is the case. If only we philosophers had realized what the problems were that the physicists had, we could have decided immediately by brain-work that it is impossible to tell how fast one is moving without looking outside, and we could have made an enormous contribution to physics." These philosophers are always with us, struggling in the periphery to try to tell us something, but they never really understand the subtleties and depths of the problem.

16-1 Relativity and the philosophers

16-2 The twin paradox

16-3 Transformation of velocities

16-4 Relativistic mass

16-5 Relativistic energy

Our inability to detect absolute motion is a result of *experiment* and not a result of plain thought, as we can easily illustrate. In the first place, Newton believed that it was true that one could not tell how fast he is going if he is moving with uniform velocity in a straight line. In fact, Newton first stated the principle of relativity, and one quotation made in the last chapter was a statement of Newton's. Why then did the philosophers not make all this fuss about "all is relative," or whatever, in Newton's time? Because it was not until Maxwell's theory of electrodynamics was developed that there were physical laws that suggested that one *could* measure his velocity without looking outside; soon it was found *experimentally* that one could *not*.

Now, *is* it absolutely, definitely, philosophically *necessary* that one should not be able to tell how fast he is moving without looking outside? One of the consequences of relativity was the development of a philosophy which said, "You can only define what you can measure! Since it is self-evident that one cannot measure a velocity without seeing what he is measuring it relative to, therefore it is clear that there is no *meaning* to absolute velocity. The physicists should have realized that they can talk only about what they can measure." But *that is the whole problem*: whether or not one *can define* absolute velocity is the same as the problem of whether or not one *can detect in an experiment*, without looking outside, whether he is moving. In other words, whether or not a thing is measurable is not something to be decided *a priori* by thought alone, but something that can be decided only by experiment. Given the fact that the velocity of light is 186,000 mi/sec, one will find few philosophers who will calmly state that it is self-evident that if light goes 186,000 mi/sec inside a car, and the car is going 100,000 mi/sec, that the light also goes 186,000 mi/sec past an observer on the ground. That is a shocking fact to them; the very ones who claim it is obvious find, when you give them a specific fact, that it is not obvious.

Finally, there is even a philosophy which says that one cannot detect *any* motion except by looking outside. It is simply not true in physics. True, one cannot perceive a *uniform* motion in a *straight line*, but if the whole room were *rotating* we would certainly know it, for everybody would be thrown to the wall—there would be all kinds of "centrifugal" effects. That the earth is turning on its axis can be determined without looking at the stars, by means of the so-called Foucault pendulum, for example. Therefore it is not true that "all is relative"; it is only *uniform velocity* that cannot be detected without looking outside. Uniform *rotation* about a fixed axis *can* be. When this is told to a philosopher, he is very upset that he did not really understand it, because to him it seems impossible that one should be able to determine rotation about an axis without looking outside. If the philosopher is good enough, after some time he may come back and say, "I understand. We really do not have such a thing as absolute rotation; we are really rotating *relative to the stars*, you see. And so some influence exerted by the stars on the object must cause the centrifugal force."

Now, for all we know, that is true; we have no way, at the present time, of telling whether there would have been centrifugal force if there were no stars and nebulae around. We have not been able to do the experiment of removing all the nebulae and then measuring our rotation, so we simply do not know. We must admit that the philosopher may be right. He comes back, therefore, in delight and says, "It is absolutely necessary that the world ultimately turn out to be this way: *absolute* rotation means nothing; it is only *relative* to the nebulae." Then we say to him, "Now, my friend, is it or is it not obvious that uniform velocity in a straight line, *relative to the nebulae* should produce no effects inside a car?" Now that the motion is no longer absolute, but is a motion *relative to the nebulae*, it becomes a mysterious question, and a question that can be answered only by experiment.

What, then, *are* the philosophic influences of the theory of relativity? If we limit ourselves to influences in the sense of *what kind of new ideas and suggestions* are made to the physicist by the principle of relativity, we could describe some of them as follows. The first discovery is, essentially, that even those ideas which have been held for a very long time and which have been very accurately verified might be wrong. It was a shocking discovery, of course, that Newton's laws are

wrong, after all the years in which they seemed to be accurate. Of course it is clear, not that the experiments were wrong, but that they were done over only a limited range of velocities, so small that the relativistic effects would not have been evident. But nevertheless, we now have a much more humble point of view of our physical laws—everything *can* be wrong!

Secondly, if we have a set of “strange” ideas, such as that time goes slower when one moves, and so forth, whether we *like* them or do *not* like them is an irrelevant question. The only relevant question is whether the ideas are consistent with what is found experimentally. In other words, the “strange ideas” need only agree with *experiment*, and the only reason that we have to discuss the behavior of clocks and so forth is to demonstrate that although the notion of the time dilation is strange, it is *consistent* with the way we measure time.

Finally, there is a third suggestion which is a little more technical but which has turned out to be of enormous utility in our study of other physical laws, and that is to *look at the symmetry of the laws* or, more specifically, to look for the ways in which the laws can be transformed and leave their form the same. When we discussed the theory of vectors, we noted that the fundamental laws of motion are not changed when we rotate the coordinate system, and now we learn that they are not changed when we change the space and time variables in a particular way, given by the Lorentz transformation. So this idea of studying the patterns or operations under which the fundamental laws are not changed has proved to be a very useful one.

16-2 The twin paradox

To continue our discussion of the Lorentz transformation and relativistic effects, we consider a famous so-called “paradox” of Peter and Paul, who are supposed to be twins, born at the same time. When they are old enough to drive a space ship, Paul flies away at very high speed. Because Peter, who is left on the ground, sees Paul going so fast, all of Paul’s clocks appear to go slower, his heart beats go slower, his thoughts go slower, everything goes slower, from Peter’s point of view. Of course, Paul notices nothing unusual, but if he travels around and about for a while and then comes back, he will be younger than Peter, the man on the ground! That is actually right; it is one of the consequences of the theory of relativity which has been clearly demonstrated. Just as the mu-mesons last longer when they are moving, so also will Paul last longer when he is moving. This is called a “paradox” only by the people who believe that the principle of relativity means that *all motion* is relative; they say, “Heh, heh, heh, from the point of view of Paul, can’t we say that *Peter* was moving and should therefore appear to age more slowly? By symmetry, the only possible result is that both should be the same age when they meet.” But in order for them to come back together and make the comparison, Paul must either stop at the end of the trip and make a comparison of clocks or, more simply, he has to come back, and the one who comes back must be the man who was moving, and he knows this, because he had to turn around. When he turned around, all kinds of unusual things happened in his space ship—the rockets went off, things jammed up against one wall, and so on—while Peter felt nothing.

So the way to state the rule is to say that *the man who has felt the accelerations*, who has seen things fall against the walls, and so on, is the one who would be the younger; that is the difference between them in an “absolute” sense, and it is certainly correct. When we discussed the fact that moving mu-mesons live longer, we used as an example their straight-line motion in the atmosphere. But we can also make mu-mesons in a laboratory and cause them to go in a curve with a magnet, and even under this accelerated motion, they last exactly as much longer as they do when they are moving in a straight line. Although no one has arranged an experiment explicitly so that we can get rid of the paradox, one could compare a mu-meson which is left standing with one that had gone around a complete circle, and it would surely be found that the one that went around the circle lasted longer. Although we have not actually carried out an experiment using a complete

circle, it is really not necessary, of course, because everything fits together all right. This may not satisfy those who insist that every single fact be demonstrated directly, but we confidently predict the result of the experiment in which Paul goes in a complete circle.

16-3 Transformation of velocities

The main difference between the relativity of Einstein and the relativity of Newton is that the laws of transformation connecting the coordinates and times between relatively moving systems are different. The correct transformation law, that of Lorentz, is

$$\begin{aligned}x' &= \frac{x - ut}{\sqrt{1 - u^2/c^2}}, \\y' &= y, \\z' &= z, \\t' &= \frac{t - ux/c^2}{\sqrt{1 - u^2/c^2}}.\end{aligned}\tag{16.1}$$

These equations correspond to the relatively simple case in which the relative motion of the two observers is along their common x -axes. Of course other directions of motion are possible, but the most general Lorentz transformation is rather complicated, with all four quantities mixed up together. We shall continue to use this simpler form, since it contains all the essential features of relativity.

Let us now discuss more of the consequences of this transformation. First, it is interesting to solve these equations in reverse. That is, here is a set of linear equations, four equations with four unknowns, and they can be solved in reverse, for x, y, z, t in terms of x', y', z', t' . The result is very interesting, since it tells us how a system of coordinates "at rest" looks from the point of view of one that is "moving." Of course, since the motions are relative and of uniform velocity, the man who is "moving" can say, if he wishes, that it is really the other fellow who is moving and he himself who is at rest. And since he is moving in the opposite direction, he should get the same transformation, but with the opposite sign of velocity. That is precisely what we find by manipulation, so that is consistent. If it did not come out that way, we would have real cause to worry!

$$\begin{aligned}x &= \frac{x' + ut'}{\sqrt{1 - u^2/c^2}}, \\y &= y', \\z &= z', \\t &= \frac{t' + ux'/c^2}{\sqrt{1 - u^2/c^2}}.\end{aligned}\tag{16.2}$$

Next we discuss the interesting problem of the addition of velocities in relativity. We recall that one of the original puzzles was that light travels at 186,000 mi/sec in all systems, even when they are in relative motion. This is a special case of the more general problem exemplified by the following. Suppose that an object inside a space ship is going at 100,000 mi/sec and the space ship itself is going at 100,000 mi/sec; how fast is the object inside the space ship moving from the point of view of an observer outside? We might want to say 200,000 mi/sec, which is faster than the speed of light. This is very unnerving, because it is not supposed to be going faster than the speed of light! The general problem is as follows.

Let us suppose that the object inside the ship, from the point of view of the man inside, is moving with velocity v , and that the space ship itself has a velocity u with respect to the ground. We want to know with what velocity v_x this object is moving from the point of view of the man on the ground. This is, of course, still but a special case in which the motion is in the x -direction. There will also be a

transformation for velocities in the y -direction, or for any angle; these can be worked out as needed. Inside the space ship the velocity is v_x , which means that the displacement x is equal to the velocity times the time:

$$x' = v_x t'. \quad (16.3)$$

Now we have only to calculate what the position and time are from the point of view of the outside observer for an object which has the relation (16.2) between x' and t' . So we simply substitute (16.3) into (16.2), and obtain

$$x = \frac{v_x t' + u t'}{\sqrt{1 - u^2/c^2}}. \quad (16.4)$$

But here we find x expressed in terms of t' . In order to get the velocity as seen by the man on the outside, we must divide *his distance* by *his time*, not by the *other man's time*! So we must also calculate the *time* as seen from the outside, which is

$$t = \frac{t' + u(v_x t')/c^2}{\sqrt{1 - u^2/c^2}}. \quad (16.5)$$

Now we must find the ratio of x to t , which is

$$v_x = \frac{x}{t} = \frac{u + v_x}{1 + uv_x/c^2}, \quad (16.6)$$

the square roots having cancelled. This is the law that we seek: the resultant velocity, the “summing” of two velocities, is not just the algebraic sum of two velocities (we know that it cannot be or we get in trouble), but is “corrected” by $1 + uv/c^2$.

Now let us see what happens. Suppose that you are moving inside the space ship at half the speed of light, and that the space ship itself is going at half the speed of light. Thus u is $\frac{1}{2}c$ and v is $\frac{1}{2}c$, but in the denominator uv is one-fourth, so that

$$v = \frac{\frac{1}{2}c + \frac{1}{2}c}{1 + \frac{1}{4}} = \frac{4c}{5}.$$

So, in relativity, “half” and “half” does not make “one,” it makes only “4/5.” Of course low velocities can be added quite easily in the familiar way, because so long as the velocities are small compared with the speed of light we can forget about the $(1 + uv/c^2)$ factor; but things are quite different and quite interesting at high velocity.

Let us take a limiting case. Just for fun, suppose that inside the space ship the man was observing *light itself*. In other words, $v = c$, and yet the space ship is moving. How will it look to the man on the ground? The answer will be

$$v = \frac{u + c}{1 + uc/c^2} = c \frac{u + c}{u + c} = c.$$

Therefore, if something is moving at the speed of light inside the ship, it will appear to be moving at the speed of light from the point of view of the man on the ground too! This is good, for it is, in fact, what the Einstein theory of relativity was designed to do in the first place—so it had *better* work!

Of course, there are cases in which the motion is not in the direction of the uniform translation. For example, there may be an object inside the ship which is just moving “upward” with the velocity v_y , with respect to the ship, and the ship is moving “horizontally.” Now, we simply go through the same thing, only using y 's instead of x 's, with the result

$$y = y' = v_y t',$$

so that if $v_x = 0$,

$$v_y = \frac{y}{t} = v_y \sqrt{1 - u^2/c^2}. \quad (16.7)$$

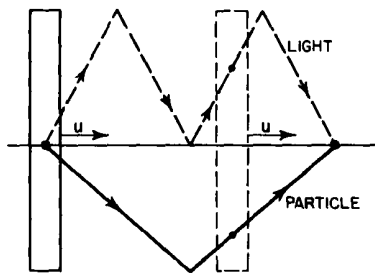


Fig 16-1. Trajectories described by a light ray and particle inside a moving clock.

Thus a sidewise velocity is no longer v_y , but $v_y\sqrt{1 - u^2/c^2}$. We found this result by substituting and combining the transformation equations, but we can also see the result directly from the principle of relativity for the following reason (it is always good to look again to see whether we can see the reason). We have already (Fig. 15-3) discussed how a possible clock might work when it is moving; the light appears to travel at an angle at the speed c in the fixed system, while it simply goes vertically with the same speed in the moving system. We found that the *vertical component* of the velocity in the fixed system is less than that of light by the factor $\sqrt{1 - u^2/c^2}$ (see Eq. 15-3). But now suppose that we let a material particle go back and forth in this same "clock," but at some integral fraction $1/n$ of the speed of light (Fig. 16-1). Then when the particle has gone back and forth once, the light will have gone exactly n times. That is, each "click" of the "particle" clock will coincide with each n th "click" of the light clock. *This fact must still be true when the whole system is moving*, because the physical phenomenon of coincidence will be a coincidence in any frame. Therefore, since the speed c_y is less than the speed of light, the speed v_y of the particle must be slower than the corresponding speed by the same square-root ratio! That is why the square root appears in any vertical velocity.

16-4 Relativistic mass

We learned in the last chapter that the mass of an object increases with velocity, but no demonstration of this was given, in the sense that we made no arguments analogous to those about the way clocks have to behave. However, we *can* show that, as a consequence of relativity plus a few other reasonable assumptions, the mass must vary in this way. (We have to say "a few other assumptions" because we cannot prove anything unless we have some laws which we assume to be true, if we expect to make meaningful deductions.) To avoid the need to study the transformation laws of force, we shall analyze a *collision*, where we need know nothing about the laws of force, except that we shall assume the conservation of momentum and energy. Also, we shall assume that the momentum of a particle which is moving is a vector and is always directed in the direction of the velocity. However, we shall not assume that the momentum is a *constant* times the velocity, as Newton did, but only that it is some *function* of velocity. We thus write the momentum vector as a certain coefficient times the vector velocity:

$$\mathbf{p} = m_v \mathbf{v}. \quad (16.8)$$

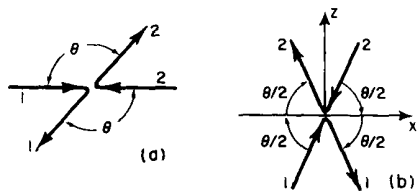


Fig. 16-2. Two views of an elastic collision between equal objects moving at the same speed in opposite directions.

We put a subscript v on the coefficient to remind us that it is a function of velocity, and we shall agree to call this coefficient m_v the "mass." Of course, when the velocity is small, it is the same mass that we would measure in the slow-moving experiments that we are used to. Now we shall try to demonstrate that the formula for m_v must be $m_0/\sqrt{1 - v^2/c^2}$, by arguing from the principle of relativity that the laws of physics must be the same in every coordinate system.

Suppose that we have two particles, like two protons, that are absolutely equal, and they are moving toward each other with exactly equal velocities. Their total momentum is zero. Now what can happen? After the collision, their directions of motion must be exactly opposite to each other, because if they are not exactly opposite, there will be a nonzero total vector momentum, and momentum would not have been conserved. Also they must have the same speeds, since they are exactly similar objects; in fact, they must have the same speed they started with, since we suppose that the energy is conserved in these collisions. So the diagram of an elastic collision, a reversible collision, will look like Fig. 16-2(a): all the arrows are the same length, all the speeds are equal. We shall suppose that such collisions can always be arranged, that any angle θ can occur, and that any speed could be used in such a collision. Next, we notice that this same collision can be viewed differently by turning the axes, and just for convenience we *shall* turn the axes, so that the horizontal splits it evenly, as in Fig. 16-2(b). It is the same collision redrawn, only with the axes turned.

Now here is the real trick: let us look at this collision from the point of view of someone riding along in a car that is moving with a speed equal to the horizontal component of the velocity of one particle. Then how does the collision look? It looks as though particle 1 is just going straight up, because it has lost its horizontal component, and it comes straight down again, also because it does not have that component. That is, the collision appears as shown in Fig. 16-3(a). Particle 2, however, was going the other way, and as we ride past it appears to fly by at some terrific speed and at a smaller angle, but we can appreciate that the angles before and after the collision are the *same*. Let us denote by u the horizontal component of the velocity of particle 2, and by w the vertical velocity of particle 1.

Now the question is, what is the vertical velocity $u \tan \alpha$? If we knew that, we could get the correct expression for the momentum, using the law of conservation of momentum in the vertical direction. Clearly, the horizontal component of the momentum is conserved: it is the same before and after the collision for both particles, and is zero for particle 1. So we need use the conservation law only for the upward velocity $u \tan \alpha$. But we *can* get the upward velocity, simply by looking at the same collision going the other way! If we look at the collision of Fig. 16-3(a) from a car to the left moving with speed u , we see the same collision, except "turned over," as shown in Fig. 16-3(b). Now particle 2 is the one that goes up and down with speed w , and particle 1 has picked up the horizontal speed u . Of course, now we *know* what the velocity $u \tan \alpha$ is: it is $w\sqrt{1 - u^2/c^2}$ (see Eq. 16.7). We know that the change in the vertical momentum of the vertically moving particle is

$$\Delta p = 2m_w w$$

(2, because it moves up and back down). The obliquely moving particle has a certain velocity v whose components we have found to be u and $w\sqrt{1 - u^2/c^2}$, and whose mass is m_v . The change in *vertical* momentum of this particle is therefore $\Delta p' = 2m_v w\sqrt{1 - u^2/c^2}$ because, in accordance with our assumed law (16.8), the momentum component is always the mass corresponding to the magnitude of the velocity times the component of the velocity in the direction of interest. Thus in order for the total momentum to be zero the vertical momenta must cancel and the ratio of the mass moving with speed v and the mass moving with speed w must therefore be

$$\frac{m_w}{m_v} = \sqrt{1 - u^2/c^2}. \quad (16.9)$$

Let us take the limiting case that w is infinitesimal. If w is very tiny indeed, it is clear that v and u are practically equal. In this case, $m_w \rightarrow m_0$ and $m_v \rightarrow m_u$. The grand result is

$$m_u = \frac{m_0}{\sqrt{1 - u^2/c^2}}. \quad (16.10)$$

It is an interesting exercise now to check whether or not Eq. (16.9) is indeed true for arbitrary values of w , assuming that Eq. (16.10) is the right formula for the mass. Note that the velocity v needed in Eq. (16.9) can be calculated from the right-angle triangle:

$$v^2 = u^2 + w^2(1 - u^2/c^2).$$

It will be found to check out automatically, although we used it only in the limit of small w .

Now, let us accept that momentum is conserved and that the mass depends upon the velocity according to (16.10) and go on to find what else we can conclude. Let us consider what is commonly called an *inelastic collision*. For simplicity, we shall suppose that two objects of the same kind, moving oppositely with equal speeds w , hit each other and stick together, to become some new, stationary object, as shown in Fig. 16-4(a). The mass m of each corresponds to w , which, as we know, is $m_0/\sqrt{1 - w^2/c^2}$. If we assume the conservation of momentum and the principle of relativity, we can demonstrate an interesting fact about the mass of the new object which has been formed. We imagine an infinitesimal velocity

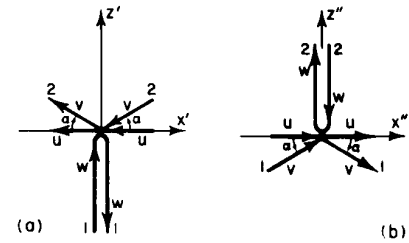


Fig. 16-3. Two more views of the collision, from moving cars.

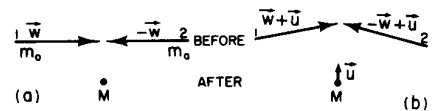


Fig. 16-4. Two views of an inelastic collision between equally massive objects.

u at right angles to w (we can do the same with finite values of u , but it is easier to understand with an infinitesimal velocity), then look at this same collision as we ride by in an elevator at the velocity $-u$. What we see is shown in Fig. 16-4(b). The composite object has an unknown mass M . Now object 1 moves with an upward component of velocity u and a horizontal component which is practically equal to w , and so also does object 2. After the collision we have the mass M moving upward with velocity u , considered very small compared with the speed of light, and also small compared with w . Momentum must be conserved, so let us estimate the momentum in the upward direction before and after the collision. Before the collision we have $p \sim 2m_w u$, and after the collision the momentum is evidently $p' = M_u u$, but M_u is essentially the same as M_0 because u is so small. These momenta must be equal because of the conservation of momentum, and therefore

$$M_0 = 2m_w. \quad (16.11)$$

The mass of the object which is formed when two equal objects collide must be twice the mass of the objects which come together. You might say, "Yes, of course, that is the conservation of mass." But not "Yes, of course," so easily, because *these masses have been enhanced* over the masses that they would be if they were standing still, yet they still contribute, to the total M , not the mass they have when standing still, but *more*. Astonishing as that may seem, in order for the conservation of momentum to work when two objects come together, the mass that they form must be greater than the rest masses of the objects, even though the objects are at rest after the collision!

16-5 Relativistic energy

In the last chapter we demonstrated that as a result of the dependence of the mass on velocity and Newton's laws, the changes in the kinetic energy of an object resulting from the total work done by the forces on it always comes out to be

$$\Delta T = (m_u - m_0)c^2 = \frac{m_0 c^2}{\sqrt{1 - u^2/c^2}} - m_0 c^2. \quad (16.12)$$

We even went further, and guessed that the total energy is the total mass times c^2 . Now we continue this discussion.

Suppose that our two equally massive objects that collide can still be "seen" inside M . For instance, a proton and a neutron are "stuck together," but are still moving about inside of M . Then, although we might at first expect the mass M to be $2m_0$, we have found that it is not $2m_0$, but $2m_w$. Since $2m_w$ is what is put in, but $2m_0$ are the rest masses of the things inside, the *excess* mass of the composite object is equal to the kinetic energy brought in. This means, of course, that *energy has inertia*. In the last chapter we discussed the heating of a gas, and showed that because the gas molecules are moving and moving things are heavier, when we put energy into the gas its molecules move faster and so the gas gets heavier. But in fact the argument is completely general, and our discussion of the inelastic collision shows that the mass is there whether or not it is *kinetic* energy. In other words, if two particles come together and produce potential or any other form of energy; if the pieces are slowed down by climbing hills, doing work against internal forces, or whatever; then it is still true that the mass is the total energy that has been put in. So we see that the conservation of mass which we have deduced above is equivalent to the conservation of energy, and therefore there is no place in the theory of relativity for strictly inelastic collisions, as there was in Newtonian mechanics. According to Newtonian mechanics it is all right for two things to collide and so form an object of mass $2m_0$ which is in no way distinct from the one that would result from putting them together slowly. Of course we know from the law of conservation of energy that there is more kinetic energy inside, but that does not affect the mass, according to Newton's laws. But now we see that this is impossible; because of the kinetic energy involved in the collision, the resulting

object will be *heavier*; therefore, it will be a *different* object. When we put the objects together gently they make something whose mass is $2m_0$; when we put them together forcefully, they make something whose mass is greater. When the mass is different, we can *tell* that it is different. So, necessarily, the conservation of energy must go along with the conservation of momentum in the theory of relativity.

This has interesting consequences. For example, suppose that we have an object whose mass M is measured, and suppose something happens so that it flies into two equal pieces moving with speed w , so that they each have a mass m_w . Now suppose that these pieces encounter enough material to slow them up until they stop; then they will have mass m_0 . How much energy will they have given to the material when they have stopped? Each will give an amount $(m_w - m_0)c^2$, by the theorem that we proved before. This much energy is left in the material in some form, as heat, potential energy, or whatever. Now $2m_w = M$, so the liberated energy is $E = (M - 2m_0)c^2$. This equation was used to estimate how much energy would be liberated under fission in the atomic bomb, for example. (Although the fragments are not exactly equal, they are nearly equal.) The mass of the uranium atom was known—it had been measured ahead of time—and the atoms into which it split, iodine, xenon, and so on, all were of known mass. By masses, we do not mean the masses while the atoms are moving, we mean the masses when the atoms are *at rest*. In other words, both M and m_0 are known. So by subtracting the two numbers one can calculate how much energy will be released if M can be made to split in “half.” For this reason poor old Einstein was called the “father” of the atomic bomb in all the newspapers. Of course, all that meant was that he could tell us ahead of time how much energy would be released if we told him what process would occur. The energy that should be liberated when an atom of uranium undergoes fission was estimated about six months before the first direct test, and as soon as the energy was in fact liberated, someone measured it directly (and if Einstein’s formula had not worked, they would have measured it anyway), and the moment they measured it they no longer needed the formula. Of course, we should not belittle Einstein, but rather should criticize the newspapers and many popular descriptions of what causes what in the history of physics and technology. The problem of how to get the thing to occur in an effective and rapid manner is a completely different matter.

The result is just as significant in chemistry. For instance, if we were to weigh the carbon dioxide molecule and compare its mass with that of the carbon and the oxygen, we could find out how much energy would be liberated when carbon and oxygen form carbon dioxide. The only trouble here is that the differences in masses are so small that it is technically very difficult to do.

Now let us turn to the question of whether we should add m_0c^2 to the kinetic energy and say from now on that the total energy of an object is mc^2 . First, if we can still *see* the component pieces of rest mass m_0 inside M , then we could say that some of the mass M of the compound object is the mechanical rest mass of the parts, part of it is kinetic energy of the parts, and part of it is potential energy of the parts. But we have discovered, in nature, particles of various kinds which undergo reactions just like the one we have treated above, in which with all the study in the world, *we cannot see the parts inside*. For instance, when a K -meson disintegrates into two pions it does so according to the law (16.11), but the idea that a K is made out of 2 π ’s is a useless idea, because it also disintegrates into 3 π ’s!

Therefore we have a *new idea*: we do not have to know what things are made of inside; we cannot and need not identify, inside a particle, which of the energy is rest energy of the parts into which it is going to disintegrate. It is not convenient and often not possible to separate the total mc^2 energy of an object into rest energy of the inside pieces, kinetic energy of the pieces, and potential energy of the pieces; instead, we simply speak of the *total energy* of the particle. We “shift the origin” of energy by adding a constant m_0c^2 to everything, and say that the total energy of a particle is the mass in motion times c^2 , and when the object is standing still, the energy is the mass at rest times c^2 .

Finally, we find that the velocity v , momentum P , and total energy E are related in a rather simple way. That the mass in motion at speed v is the mass m_0 at rest divided by $\sqrt{1 - v^2/c^2}$, surprisingly enough, is rarely used. Instead, the following relations are easily proved, and turn out to be very useful:

$$E^2 - P^2c^2 = m_0^2c^4 \quad (16.13)$$

and

$$Pc = Ev/c. \quad (16.14)$$

Space-Time

17-1 The geometry of space-time

The theory of relativity shows us that the relationships of positions and times as measured in one coordinate system and another are not what we would have expected on the basis of our intuitive ideas. It is very important that we thoroughly understand the relations of space and time implied by the Lorentz transformation, and therefore we shall consider this matter more deeply in this chapter.

The Lorentz transformation between the positions and times (x, y, z, t) as measured by an observer “standing still,” and the corresponding coordinates and time (x', y', z', t') measured inside a “moving” space ship, moving with velocity u are

$$\begin{aligned}x' &= \frac{x - ut}{\sqrt{1 - u^2/c^2}}, \\y' &= y, \\z' &= z, \\t' &= \frac{t - ux/c^2}{\sqrt{1 - u^2/c^2}}.\end{aligned}\tag{17.1}$$

Let us compare these equations with Eq. (11.5), which also relates measurements in two systems, one of which in this instance is *rotated* relative to the other:

$$\begin{aligned}x' &= x \cos \theta + y \sin \theta, \\y' &= y \cos \theta - x \sin \theta, \\z' &= z.\end{aligned}\tag{17.2}$$

In this particular case, Moe and Joe are measuring with axes having an angle θ between the x' - and x -axes. In each case, we note that the “primed” quantities are “mixtures” of the “unprimed” ones: the new x' is a mixture of x and y , and the new y' is also a mixture of x and y .

An analogy is useful: When we look at an object, there is an obvious thing we might call the “apparent width,” and another we might call the “depth.” But the two ideas, width and depth, are not *fundamental* properties of the object, because if we step aside and look at the same thing from a different angle, we get a different width and a different depth, and we may develop some formulas for computing the new ones from the old ones and the angles involved. Equations (17.2) are these formulas. One might say that a given depth is a kind of “mixture” of all depth and all width. If it were impossible ever to move, and we always saw a given object from the same position, then this whole business would be irrelevant—we would always see the “true” width and the “true” depth, and they would appear to have quite different qualities, because one appears as a subtended optical angle and the other involves some focusing of the eyes or even intuition; they would seem to be very different things and would never get mixed up. It is because we *can* walk around that we realize that depth and width are, somehow or other, just two different aspects of the same thing.

Can we not look at the Lorentz transformations in the same way? Here also we have a mixture—of positions and the time. A difference between a space measurement and a time measurement produces a new space measurement. In other words, in the space measurements of one man there is mixed in a little bit of the time, as seen by the other. Our analogy permits us to generate this idea: The “reality” of

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an object that we are looking at is somehow greater (speaking crudely and intuitively) than its “width” and its “depth” because *they* depend upon *how* we look at it; when we move to a new position, our brain immediately recalculates the width and the depth. But our brain does not immediately recalculate coordinates and time when we move at high speed, because we have had no effective experience of going nearly as fast as light to appreciate the fact that time and space are also of the same nature. It is as though we were always stuck in the position of having to look at just the width of something, not being able to move our heads appreciably one way or the other; if we could, we understand now, we would see some of the other man’s time—we would see “behind,” so to speak, a little bit.

Thus we shall try to think of objects in a new kind of world, of space and time mixed together, in the same sense that the objects in our ordinary space-world are real, and can be looked at from different directions. We shall then consider that objects occupying space and lasting for a certain length of time occupy a kind of a “blob” in a new kind of world, and that we look at this “blob” from different points of view when we are moving at different velocities. This new world, this geometrical entity in which the “blobs” exist by occupying position and taking up a certain amount of time, is called *space-time*. A given point (x, y, z, t) in space-time is called an *event*. Imagine, for example, that we plot the x -positions horizontally, y and z in two other directions, both mutually at “right angles” and at “right angles” to the paper (!), and time, vertically. Now, how does a moving particle, say, look on such a diagram? If the particle is standing still, then it has a certain x , and as time goes on, it has the same x , the same x , the same x ; so its “path” is a line that runs parallel to the t -axis (Fig. 17-1 a). On the other hand, if it drifts outward, then as the time goes on x increases (Fig. 17-1 b). So a particle, for example, which starts to drift out and then slows up should have a motion something like that shown in Fig. 17-1(c). A particle, in other words, which is permanent and does not disintegrate is represented by a line in space-time. A particle which disintegrates would be represented by a forked line, because it would turn into two other things which would start from that point.

What about light? Light travels at the speed c , and that would be represented by a line having a certain fixed slope (Fig. 17-1 d).

Now according to our new idea, if a given event occurs to a particle, say if it suddenly disintegrates at a certain space-time point into two new ones which follow some new tracks, and this interesting event occurred at a certain value of x and a certain value of t , then we would expect that, if this makes any sense, we just have to take a new pair of axes and turn them, and that will give us the new t and the new x in our new system, as shown in Fig. 17-2(a). But this is wrong, because Eq. (17.1) is not *exactly* the same mathematical transformation as Eq. (17.2). Note, for example, the difference in sign between the two, and the fact that one is written in terms of $\cos \theta$ and $\sin \theta$, while the other is written with algebraic quantities. (Of course, it is not impossible that the algebraic quantities could be written as cosine and sine, but actually they cannot.) But still, the two expressions *are* very similar. As we shall see, it is not really possible to think of space-time as a real, ordinary geometry because of that difference in sign. In fact, although we shall not emphasize this point, it turns out that a man who is moving has to use a set of axes which are inclined equally to the light ray, using a special kind of projection parallel to the x' - and t' -axes, for his x' and t' , as shown in Fig. 17-2(b). We shall not deal with the geometry, since it does not help much; it is easier to work with the equations.

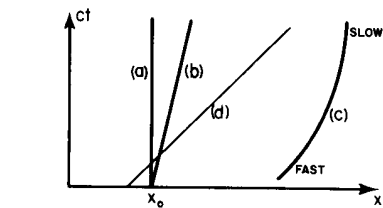


Fig. 17-1. Three particle paths in space-time: (a) a particle at rest at $x = x_0$; (b) a particle which starts at $x = x_0$ and moves with constant speed; (c) a particle which starts at high speed but slows down.

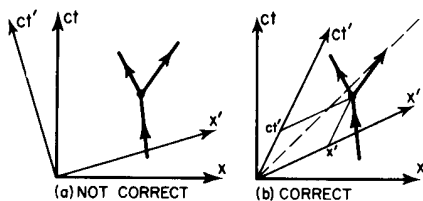


Fig. 17-2. Two views of a disintegrating particle.

17-2 Space-time intervals

Although the geometry of space-time is not Euclidean in the ordinary sense, there *is* a geometry which is very similar, but peculiar in certain respects. If this idea of geometry is right, there ought to be some functions of coordinates and time which are independent of the coordinate system. For example, under ordinary rotations, if we take two points, one at the origin, for simplicity, and the other one somewhere else, both systems would have the same origin, and the distance from

here to the other point is the same in both. That is one property that is independent of the particular way of measuring it. The square of the distance is $x^2 + y^2 + z^2$. Now what about space-time? It is not hard to demonstrate that we have here, also, something which stays the same, namely, the combination $c^2t^2 - x^2 - y^2 - z^2$ is the same before and after the transformation:

$$c^2t'^2 - x'^2 - y'^2 - z'^2 = c^2t^2 - x^2 - y^2 - z^2. \quad (17.3)$$

This quantity is therefore something which, like the distance, is “real” in some sense; it is called the *interval* between the two space-time points, one of which is, in this case, at the origin. (Actually, of course, it is the interval squared, just as $x^2 + y^2 + z^2$ is the distance squared.) We give it a different name because it is in a different geometry, but the interesting thing is only that some signs are reversed and there is a c in it.

Let us get rid of the c ; that is an absurdity if we are going to have a wonderful space with x 's and y 's that can be interchanged. One of the confusions that could be caused by someone with no experience would be to measure widths, say, by the angle subtended at the eye, and measure depth in a different way, like the strain on the muscles needed to focus them, so that the depths would be measured in feet and the widths in meters. Then one would get an enormously complicated mess of equations in making transformations such as (17.2), and would not be able to see the clarity and simplicity of the thing for a very simple technical reason, that the same thing is being measured in two different units. Now in Eqs. (17.1) and (17.3) nature is telling us that time and space are equivalent; time becomes space; *they should be measured in the same units*. What distance is a “second”? It is easy to figure out from (17.3) what it is. It is 3×10^8 meters, *the distance that light would go in one second*. In other words, if we were to measure all distances and times in the same units, seconds, then our unit of distance would be 3×10^8 meters, and the equations would be simpler. Or another way that we could make the units equal is to measure time in meters. What is a meter of time? A meter of time is the time it takes for light to go one meter, and is therefore $1/3 \times 10^{-8}$ sec, or 3.3 billionths of a second! We would like, in other words, to put all our equations in a system of units in which $c = 1$. If time and space are measured in the same units, as suggested, then the equations are obviously much simplified. They are

$$\begin{aligned} x' &= \frac{x - ut}{\sqrt{1 - u^2}}, \\ y' &= y, \\ z' &= z, \\ t' &= \frac{t - ux}{\sqrt{1 - u^2}}. \end{aligned} \quad (17.4)$$

$$t'^2 - x'^2 - y'^2 - z'^2 = t^2 - x^2 - y^2 - z^2. \quad (17.5)$$

If we are ever unsure or “frightened” that after we have this system with $c = 1$ we shall never be able to get our equations right again, the answer is quite the opposite. It is much easier to remember them without the c 's in them, and it is always easy to put the c 's back, by looking after the dimensions. For instance, in $\sqrt{1 - u^2}$, we know that we cannot subtract a velocity squared, which has units, from the pure number 1, so we know that we must divide u^2 by c^2 in order to make that unitless, and that is the way it goes.

The difference between space-time and ordinary space, and the character of an interval as related to the distance, is very interesting. According to formula (17.5), if we consider a point which in a given coordinate system had zero time, and only space, then the interval squared would be negative and we would have an imaginary interval, the square root of a negative number. Intervals can be either real or imaginary in the theory. The square of an interval may be either positive or negative, unlike distance, which has a positive square. When an interval is imaginary, we say that the two points have a *space-like interval* between them

(instead of imaginary), because the interval is more like space than like time. On the other hand, if two objects are at the same place in a given coordinate system, but differ only in time, then the square of the time is positive and the distances are zero and the interval squared is positive; this is called a *time-like interval*. In our diagram of space-time, therefore, we would have a representation something like this: at 45° there are two lines (actually, in four dimensions these will be “cones,” called light cones) and points on these lines are all at zero interval from the origin. Where light goes from a given point is always separated from it by a zero interval, as we see from Eq. (17.5). Incidentally, we have just proved that if light travels with speed c in one system, it travels with speed c in another, for if the interval is the same in both systems, i.e., zero in one and zero in the other, then to state that the propagation speed of light is invariant is the same as saying that the interval is zero.

17-3 Past, present, and future

The space-time region surrounding a given space-time point can be separated into three regions, as shown in Fig. 17-3. In one region we have space-like intervals, and in two regions, time-like intervals. Physically, these three regions into which space-time around a given point is divided have an interesting physical relationship to that point: a physical object or a signal can get from a point in region 2 to the event O by moving along at a speed less than the speed of light. Therefore events in this region can affect the point O , can have an influence on it from the past. In fact, of course, an object at P on the negative t -axis is precisely in the “past” with respect to O ; it is the same space-point as O , only earlier. What happened there then, affects O now. (Unfortunately, that is the way life is.) Another object at Q can get to O by moving with a certain speed less than c , so if this object were in a space ship and moving, it would be, again, the past of the same space-point. That is, in another coordinate system, the axis of time might go through both O and Q . So all points of region 2 are in the “past” of O , and anything that happens in this region *can* affect O . Therefore region 2 is sometimes called the *affective past*, or affecting past; it is the locus of all events which can affect point O in any way.

Region 3, on the other hand, is a region which we can affect *from* O , we can “hit” things by shooting “bullets” out at speeds less than c . So this is the world whose future can be affected by us, and we may call that the *affective future*. Now the interesting thing about all the rest of space-time, i.e., region 1, is that we can neither affect it now *from* O , nor can it affect us now *at* O , because nothing can go faster than the speed of light. Of course, what happens at R *can* affect us *later*; that is, if the sun is exploding “right now,” it takes eight minutes before we know about it, and it cannot possibly affect us before then.

What we mean by “right now” is a mysterious thing which we cannot define and we cannot affect, but it can affect us later, or we could have affected it if we had done something far enough in the past. When we look at the star Alpha Centauri, we see it as it was four years ago; we might wonder what it is like “now.” “Now” means at the same time from our special coordinate system. We can only see Alpha Centauri by the light that has come from our past, up to four years ago, but we do not know what it is doing “now”; it will take four years before what it is doing “now” can affect us. Alpha Centauri “now” is an idea or concept of our mind; it is not something that is really definable physically at the moment, because we have to wait to observe it; we cannot even define it right “now.” Furthermore, the “now” depends on the coordinate system. If, for example, Alpha Centauri were moving, an observer there would not agree with us because he would put his axes at an angle, and his “now” would be a *different* time. We have already talked about the fact that simultaneity is not a unique thing.

There are fortune tellers, or people who tell us they can know the future, and there are many wonderful stories about the man who suddenly discovers that he has knowledge about the affective future. Well, there are lots of paradoxes produced by that because if we know something is going to happen, then we can make

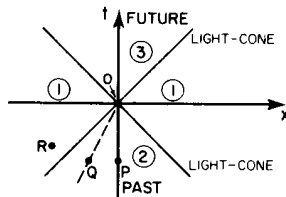


Fig. 17-3. The space-time region surrounding a point at the origin.

sure we will avoid it by doing the right thing at the right time, and so on. But actually there is no fortune teller who can even tell us the *present*! There is no one who can tell us what is really happening right now, at any reasonable distance, because that is unobservable. We might ask ourselves this question, which we leave to the student to try to answer: Would any paradox be produced if it were suddenly to become possible to know things that are in the space-like intervals of region 1?

17-4 More about four-vectors

Let us now return to our consideration of the analogy of the Lorentz transformation and rotations of the space axes. We have learned the utility of collecting together other quantities which have the same transformation properties as the coordinates, to form what we call *vectors*, directed lines. In the case of ordinary rotations, there are many quantities that transform the same way as x , y , and z under rotation: for example, the velocity has three components, an x , y , and z -component; when seen in a different coordinate system, none of the components is the same, instead they are all transformed to new values. But, somehow or other, the velocity "itself" has a greater reality than do any of its particular components, and we represent it by a directed line.

We therefore ask: Is it or is it not true that there are quantities which transform, or which are related, in a moving system and in a nonmoving system, in the same way as x , y , z , and t ? From our experience with vectors, we know that three of the quantities, like x , y , z , would constitute the three components of an ordinary space-vector, but the fourth quantity would look like an ordinary scalar under space rotation, because it does not change so long as we do not go into a moving coordinate system. Is it possible, then, to associate with some of our known "three-vectors" a fourth object, that we could call the "time component," in such a manner that the four objects together would "rotate" the same way as position and time in space-time? We shall now show that there is, indeed, at least one such thing (there are many of them, in fact): *the three components of momentum, and the energy as the time component, transform together* to make what we call a "four-vector." In demonstrating this, since it is quite inconvenient to have to write c 's everywhere, we shall use the same trick concerning units of the energy, the mass, and the momentum, that we used in Eq. (17.4). Energy and mass, for example, differ only by a factor c^2 which is merely a question of units, so we can say energy is the mass. Instead of having to write the c^2 , we put $E = m$, and then, of course, if there were any trouble we would put in the right amounts of c so that the units would straighten out in the last equation, but not in the intermediate ones.

Thus our equations for energy and momentum are

$$\begin{aligned} E &= m = m_0/\sqrt{1-v^2}, \\ \mathbf{p} &= m\mathbf{v} = m_0\mathbf{v}/\sqrt{1-v^2}. \end{aligned} \quad (17.6)$$

Also in these units, we have

$$E^2 - p^2 = m_0^2. \quad (17.7)$$

For example, if we measure energy in electron volts, what does a mass of 1 electron volt mean? It means the mass whose rest energy is 1 electron volt, that is, m_0c^2 is one electron volt. For example, the rest mass of an electron is 0.511×10^6 ev.

Now what would the momentum and energy look like in a new coordinate system? To find out, we shall have to transform Eq. (17.6), which we can do because we know how the velocity transforms. Suppose that, as we measure it, an object has a velocity v , but we look upon the same object from the point of view of a space ship which itself is moving with a velocity u , and in that system we use a prime to designate the corresponding thing. In order to simplify things at first, we shall take the case that the velocity v is in the direction of u . (Later, we can do the more general case.) What is v' , the velocity as seen from the space ship? It is the

composite velocity, the “difference” between v and u . By the law which we worked out before,

$$v' = \frac{v - u}{1 - uv}. \quad (17.8)$$

Now let us calculate the new energy E' , the energy as the fellow in the space ship would see it. He would use the same rest mass, of course, but he would use v' for the velocity. What we have to do is square v' , subtract it from one, take the square root, and take the reciprocal:

$$\begin{aligned} v'^2 &= \frac{v^2 - 2uv + u^2}{1 - 2uv + u^2v^2}, \\ 1 - v'^2 &= \frac{1 - 2uv + u^2v^2 - v^2 + 2uv - u^2}{1 - 2uv + u^2v^2} \\ &= \frac{1 - v^2 - u^2 + u^2v^2}{1 - 2uv + u^2v^2} \\ &= \frac{(1 - v^2)(1 - u^2)}{(1 - uv)^2}. \end{aligned}$$

Therefore

$$\frac{1}{\sqrt{1 - v'^2}} = \frac{1 - uv}{\sqrt{1 - v^2} \sqrt{1 - u^2}}. \quad (17.9)$$

The energy E' is then simply m_0 times the above expression. But we want to express the energy in terms of the unprimed energy and momentum, and we note that

$$E' = \frac{m_0 - m_0uv}{\sqrt{1 - v^2} \sqrt{1 - u^2}} = \frac{(m_0/\sqrt{1 - v^2}) - (m_0v/\sqrt{1 - v^2})u}{\sqrt{1 - u^2}},$$

or

$$E' = \frac{E - up_x}{\sqrt{1 - u^2}}, \quad (17.10)$$

which we recognize as being exactly of the same form as

$$t' = \frac{t - ux}{\sqrt{1 - u^2}}.$$

Next we must find the new momentum p'_x . This is just the energy E times v' , and is also simply expressed in terms of E and p :

$$p'_x = E'v' = \frac{m_0(1 - uv)}{\sqrt{1 - v^2} \sqrt{1 - u^2}} \cdot \frac{v - u}{(1 - uv)} = \frac{m_0v - m_0u}{\sqrt{1 - v^2} \sqrt{1 - u^2}}.$$

Thus

$$p'_x = \frac{p_x - uE}{\sqrt{1 - u^2}}, \quad (17.11)$$

which we recognize as being of precisely the same form as

$$x' = \frac{x - ut}{\sqrt{1 - u^2}}.$$

Thus the transformations for the new energy and momentum in terms of the old energy and momentum are exactly the same as the transformations for t' in terms of t and x , and x' in terms of x and t : all we have to do is, every time we see t in (17.4) substitute E , and every time we see x substitute p_x , and then the equations (17.4) will become the same as Eqs. (17.10) and (17.11). This would imply, if everything works right, an additional rule that $p'_y = p_y$ and that $p'_z = p_z$. To prove this would require our going back and studying the case of motion up and down. Actually, we did study the case of motion up and down in the last

chapter. We analyzed a complicated collision and we noticed that, in fact, the transverse momentum is *not* changed when viewed from a moving system; so we have already verified that $p'_y = p_y$ and $p'_z = p_z$. The complete transformation, then, is

$$\begin{aligned} p'_x &= \frac{p_x - uE}{\sqrt{1 - u^2}}, \\ p'_y &= p_y, \\ p'_z &= p_z \\ E' &= \frac{E - up_x}{\sqrt{1 - u^2}}. \end{aligned} \quad (17.12)$$

In these transformations, therefore, we have discovered four quantities which transform like x , y , z , and t , and which we call the *four-vector momentum*. Since the momentum is a four-vector, it can be represented on a space-time diagram of a moving particle as an “arrow” tangent to the path, as shown in Fig. 17-4. This arrow has a time component equal to the energy, and its space components represent its three-vector momentum; this arrow is more “real” than either the energy or the momentum, because those just depend on how we look at the diagram.

17-5 Four-vector algebra

The notation for four-vectors is different than it is for three-vectors. In the case of three-vectors, if we were to talk about the ordinary three-vector momentum we would write it $\mathbf{p}$. If we wanted to be more specific, we could say it has three components which are, for the axes in question, p_x , p_y , and p_z , or we could simply refer to a general component as p_i , and say that i could either be x , y , or z , and that these are the three components; that is, imagine that i is any one of three directions, x , y , or z . The notation that we use for four-vectors is analogous to this: we write p_μ for the four-vector, and μ stands for the *four* possible directions t , x , y , or z .

We could, of course, use any notation we want; do not laugh at notations; invent them, they are powerful. In fact, mathematics is, to a large extent, invention of better notations. The whole idea of a four-vector, in fact, is an improvement in notation so that the transformations can be remembered easily. A_μ , then, is a general four-vector, but for the special case of momentum, the p_t is identified as the energy, p_x is the momentum in the x -direction, p_y is that in the y -direction, and p_z is that in the z -direction. To add four-vectors, we add the corresponding components.

If there is an equation among four-vectors, then the equation is true for *each component*. For instance, if the law of conservation of three-vector momentum is to be true in particle collisions, i.e., if the sum of the momenta for a large number of interacting or colliding particles is to be a constant, that must mean that the sums of all momenta in the x -direction, in the y -direction, and in the z -direction, for all the particles, must each be constant. This law alone would be impossible in relativity because it is *incomplete*; it is like talking about only two of the components of a three-vector. It is incomplete because if we rotate the axes, we mix the various components, so we must include all three components in our law. Thus, in relativity, we must complete the law of conservation of momentum by extending it to include the *time* component. This is *absolutely necessary* to go with the other three, or there cannot be relativistic invariance. The *conservation of energy* is the fourth equation which goes with the conservation of momentum to make a valid four-vector relationship in the geometry of space and time. Thus the law of conservation of energy and momentum in four-dimensional notation is

$$\sum_{\text{particles in}} p_\mu = \sum_{\text{particles out}} p_\mu \quad (17.13)$$

or, in a slightly different notation

$$\sum_i p_{i\mu} = \sum_j p_{j\mu}, \quad (17.14)$$

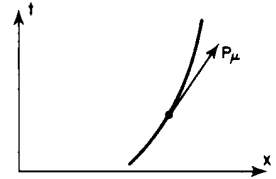


Fig. 17-4. The four-vector momentum of a particle.

where $i = 1, 2, \dots$ refers to the particles going into the collision, $j = 1, 2, \dots$ refers to the particles coming out of the collision, and $\mu = x, y, z$, or t . You say, "In which axes?" It makes no difference. The law is true for each component, using *any* axes.

In vector analysis we discussed one other thing, the dot product of two vectors. Let us now consider the corresponding thing in space-time. In ordinary rotation we discovered there was an unchanged quantity $x^2 + y^2 + z^2$. In four dimensions, we find that the corresponding quantity is $t^2 - x^2 - y^2 - z^2$ (Eq. 17.3). How can we write that? One way would be to write some kind of four-dimensional thing with a square dot between, like $A_\mu \diamond B_\mu$; one of the notations which is actually used is

$$\sum'_\mu A_\mu A_\mu = A_t^2 - A_x^2 - A_y^2 - A_z^2. \quad (17.15)$$

The prime on $\sum$ means that the first term, the "time" term, is positive, but the other three terms have minus signs. This quantity, then, will be the same in any coordinate system, and we may call it the square of the length of the four-vector. For instance, what is the square of the length of the four-vector momentum of a single particle? This will be equal to $p_t^2 - p_x^2 - p_y^2 - p_z^2$ or, in other words, $E^2 - p^2$, because we know that p_t is E . What is $E^2 - p^2$? It must be something which is the same in every coordinate system. In particular, it must be the same for a coordinate system which is moving right along with the particle, in which the particle is standing still. If the particle is standing still, it would have no momentum. So in that coordinate system, it is purely its energy, which is the same as its rest mass. Thus $E^2 - p^2 = m_0^2$. So we see that the square of the length of this vector, the four-vector momentum, is equal to m_0^2 .

From the square of a vector, we can go on to invent the "dot product," or the product which is a scalar: if a_μ is one four-vector and b_μ is another four-vector, then the scalar product is

$$\sum'_\mu a_\mu b_\mu = a_t b_t - a_x b_x - a_y b_y - a_z b_z. \quad (17.16)$$

It is the same in all coordinate systems.

Finally, we shall mention certain things whose rest mass m_0 is zero. A photon of light, for example. A photon is like a particle, in that it carries an energy and a momentum. The energy of a photon is a certain constant, called Planck's constant, times the frequency of the photon: $E = h\nu$. Such a photon also carries a momentum, and the momentum of a photon (or of any other particle, in fact) is h divided by the wavelength: $p = h/\lambda$. But, for a photon, there is a definite relationship between the frequency and the wavelength: $\nu = c/\lambda$. (The number of waves per second, times the wavelength of each, is the distance that the light goes in one second, which, of course, is c .) Thus we see immediately that the energy of a photon must be the momentum times c , or if $c = 1$, *the energy and momentum are equal*. That is to say, the rest mass is zero. Let us look at that again; that is quite curious. If it is a particle of zero rest mass, what happens when it stops? *It never stops!* It always goes at the speed c . The usual formula for energy is $m_0/\sqrt{1 - v^2}$. Now can we say that $m_0 = 0$ and $v = 1$, so the energy is 0? We *cannot* say that it is zero; the photon really can (and does) have energy even though it has no rest mass, but this it possesses by perpetually going at the speed of light!

We also know that the momentum of any particle is equal to its total energy times its velocity: if $c = 1$, $p = vE$ or, in ordinary units, $p = vE/c^2$. For any particle moving at the speed of light, $p = E$ if $c = 1$. The formulas for the energy of a photon as seen from a moving system are, of course, given by Eq. (17.12), but for the momentum we must substitute the energy times c (or times 1 in this case). The different energies after transformation means that there are different frequencies. This is called the Doppler effect, and one can calculate it easily from Eq. (17.12), using also $E = p$ and $E = h\nu$.

As Minkowski said, "Space of itself, and time of itself will sink into mere shadows, and only a kind of union between them shall survive."

Rotation in Two Dimensions

18-1 The center of mass

In the previous chapters we have been studying the mechanics of points, or small particles whose internal structure does not concern us. For the next few chapters we shall study the application of Newton's laws to more complicated things. When the world becomes more complicated, it also becomes more interesting, and we shall find that the phenomena associated with the mechanics of a more complex object than just a point are really quite striking. Of course these phenomena involve nothing but combinations of Newton's laws, but it is sometimes hard to believe that only $F = ma$ is at work.

The more complicated objects we deal with can be of several kinds: water flowing, galaxies whirling, and so on. The simplest "complicated" object to analyze, at the start, is what we call a *rigid body*, a solid object that is turning as it moves about. However, even such a simple object may have a most complex motion, and we shall therefore first consider the simplest aspects of such motion, in which an extended body rotates about a *fixed axis*. A given point on such a body then moves in a plane perpendicular to this axis. Such rotation of a body about a fixed axis is called *plane rotation* or rotation in two dimensions. We shall later generalize the results to three dimensions, but in doing so we shall find that, unlike the case of ordinary particle mechanics, rotations are subtle and hard to understand unless we first get a solid grounding in two dimensions.

The first interesting theorem concerning the motion of complicated objects can be observed at work if we throw an object made of a lot of blocks and spokes, held together by strings, into the air. Of course we know it goes in a parabola, because we studied that for a particle. But now our object is *not* a particle; it wobbles and it jiggles, and so on. It does go in a parabola though; one can see that. *What* goes in a parabola? Certainly not the point on the corner of the block, because that is jiggling about; neither is it the end of the wooden stick, or the middle of the wooden stick, or the middle of the block. But *something* goes in a parabola, there is an effective "center" which moves in a parabola. So our first theorem about complicated objects is to demonstrate that there *is* a mean position which is mathematically definable, but not necessarily a point of the material itself, which goes in a parabola. That is called the theorem of the center of the mass, and the proof of it is as follows.

We may consider any object as being made of lots of little particles, the atoms, with various forces among them. Let i represent an index which defines one of the particles. (There are millions of them, so i goes to 10^{23} , or something.) Then the force on the i th particle is, of course, the mass times the acceleration of that particle:

$$\mathbf{F}_i = m_i(d^2\mathbf{r}_i/dt^2). \quad (18.1)$$

In the next few chapters our moving objects will be ones in which all the parts are moving at speeds very much slower than the speed of light, and we shall use the nonrelativistic approximation for all quantities. In these circumstances the mass is constant, so that

$$\mathbf{F}_i = d^2(m_i\mathbf{r}_i)/dt^2 \quad (18.2)$$

If we now add the force on all the particles, that is, if we take the sum of all the $\mathbf{F}_i$'s for all the different indexes, we get the total force, $\mathbf{F}$. On the other side of the

18-1 The center of mass

18-2 Rotation of a rigid body

18-3 Angular momentum

18-4 Conservation of angular momentum

equation, we get the same thing as though we added before the differentiation:

$$\sum_i \mathbf{F}_i = \mathbf{F} = \frac{d^2(\sum_i m_i \mathbf{r}_i)}{dt^2}. \quad (18.3)$$

Therefore the total force is the second derivative of the masses times their positions, added together.

Now the total force on all the particles is the same as the *external* force. Why? Although there are all kinds of forces on the particles because of the strings, the wiggles, the pullings and pushings, and the atomic forces, and who knows what, and we have to add all these together, we are rescued by Newton's Third Law. Between any two particles the action and reaction are equal, so that when we add all the equations together, if any two particles have forces between them it cancels out in the sum; therefore the net result is only those forces which arise from other particles which are not included in whatever object we decide to sum over. So if Eq. (18.3) is the sum over a certain number of the particles, which together are called "the object," then the *external* force on the total object is equal to the sum of *all* the forces on all its constituent particles.

Now it would be nice if we could write Eq. (18.3) as the total mass times some acceleration. We can. Let us say M is the sum of all the masses, i.e., the total mass. Then if we *define* a certain vector $\mathbf{R}$ to be

$$\mathbf{R} = \sum_i m_i \mathbf{r}_i / M, \quad (18.4)$$

Eq. (18.3) will be simply

$$\mathbf{F} = d^2(M\mathbf{R})/dt^2 = M(d^2\mathbf{R}/dt^2), \quad (18.5)$$

since M is a constant. Thus we find that the external force is the total mass times the acceleration of an imaginary point whose location is $\mathbf{R}$. This point is called the *center of mass* of the body. It is a point somewhere in the "middle" of the object, a kind of average $\mathbf{r}$ in which the different $\mathbf{r}_i$'s have weights or importances proportional to the masses.

We shall discuss this important theorem in more detail in a later chapter, and we shall therefore limit our remarks to two points: First, if the external forces are zero, if the object were floating in empty space, it might whirl, and jiggle, and twist, and do all kinds of things. But the *center of mass*, this artificially invented, calculated position, somewhere in the middle, *will move with a constant velocity*. In particular, if it is initially at rest, it will stay at rest. So if we have some kind of a box, perhaps a space ship, with people in it, and we calculate the location of the center of mass and find it is standing still, then the center of mass will continue to stand still if no external forces are acting on the box. Of course, the space ship may move a little in space, but that is because the people are walking back and forth inside; when one walks toward the front, the ship goes toward the back so as to keep the average position of all the masses in exactly the same place.

Is rocket propulsion therefore absolutely impossible because one cannot move the center of mass? No; but of course we find that to propel an interesting part of the rocket, an uninteresting part must be thrown away. In other words, if we start with a rocket at zero velocity and we spit some gas out the back end, then this little blob of gas goes one way as the rocket ship goes the other, but the center of mass is still exactly where it was before. So we simply move the part that we are interested in against the part we are not interested in.

The second point concerning the center of mass, which is the reason we introduced it into our discussion at this time, is that it may be treated separately from the "internal" motions of an object, and may therefore be ignored in our discussion of rotation.

18-2 Rotation of a rigid body

Now let us discuss rotations. Of course an ordinary object does not simply rotate, it wobbles, shakes, and bends, so to simplify matters we shall discuss the motion of a nonexistent ideal object which we call a *rigid body*. This means an

object in which the forces between the atoms are so strong, and of such character, that the little forces that are needed to move it do not bend it. Its shape stays essentially the same as it moves about. If we wish to study the motion of such a body, and agree to ignore the motion of its center of mass, there is only one thing left for it to do, and that is to *turn*. We have to describe that. How? Suppose there is some line in the body which stays put (perhaps it includes the center of mass and perhaps not), and the body is rotating about this particular line as an axis. How do we define the rotation? That is easy enough, for if we mark a point somewhere on the object, anywhere except on the axis, we can always tell exactly where the object is, if we only know where this point has gone to. The only thing needed to describe the position of that point is an *angle*. So rotation consists of a study of the variations of the angle with time.

In order to study rotation, we observe the angle through which a body has turned. Of course, we are not referring to any particular angle *inside* the object itself; it is not that we draw some angle *on* the object. We are talking about the *angular change of the position* of the whole thing, from one time to another.

First, let us study the kinematics of rotations. The angle will change with time, and just as we talked about position and velocity in one dimension, we may talk about angular position and angular velocity in plane rotation. In fact, there is a very interesting relationship between rotation in two dimensions and one-dimensional displacement, in which almost every quantity has its analog. First, we have the angle θ which defines how far the body has gone *around*; this replaces the distance y , which defines how far it has gone *along*. In the same manner, we have a velocity of turning, $\omega = d\theta/dt$, which tells us how much the angle changes in a second, just as $v = ds/dt$ describes how fast a thing moves, or how far it moves in a second. If the angle is measured in radians, then the angular velocity ω will be so and so many radians per second. The greater the angular velocity, the faster the object is turning, the faster the angle changes. We can go on: we can differentiate the angular velocity with respect to time, and we can call $\alpha = d\omega/dt = d^2\theta/dt^2$ the angular acceleration. That would be the analog of the ordinary acceleration.

Now of course we shall have to relate the dynamics of rotation to the laws of dynamics of the particles of which the object is made, so we must find out how a particular particle moves when the angular velocity is such and such. To do this, let us take a certain particle which is located at a distance r from the axis and say it is in a certain location $P(x, y)$ at a given instant, in the usual manner (Fig. 18-1). If at a moment Δt later the angle of the whole object has turned through $\Delta\theta$, then this particle is carried with it. It is at the same radius away from O as it was before, but is carried to Q . The first thing we would like to know is how much the distance x changes and how much the distance y changes. If OP is called r , then the length PQ is $r\Delta\theta$, because of the way angles are defined. The change in x , then, is simply the projection of $r\Delta\theta$ in the x -direction:

$$\Delta x = -PQ \sin \theta = -r \Delta\theta \cdot (y/r) = -y \Delta\theta. \quad (18.6)$$

Similarly,

$$\Delta y = +x \Delta\theta. \quad (18.7)$$

If the object is turning with a given angular velocity ω , we find, by dividing both sides of (18.6) and (18.7) by Δt , that the velocity of the particle is

$$v_x = -\omega y \quad \text{and} \quad v_y = +\omega x. \quad (18.8)$$

Of course if we want to find the magnitude of the velocity, we just write

$$v = \sqrt{v_x^2 + v_y^2} = \sqrt{\omega^2 y^2 + \omega^2 x^2} = \omega \sqrt{x^2 + y^2} = \omega r. \quad (18.9)$$

It should not be mysterious that the value of the magnitude of this velocity is ωr ; in fact, it should be self-evident, because the distance that it moves is $r\Delta\theta$ and the distance it moves per second is $r\Delta\theta/\Delta t$, or $r\omega$.

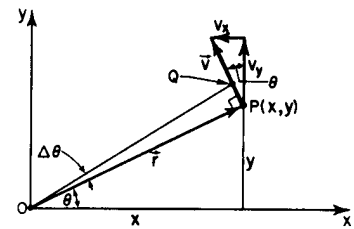


Fig. 18-1. Kinematics of two-dimensional rotation.

Let us now move on to consider the *dynamics* of rotation. Here a new concept, *force*, must be introduced. Let us inquire whether we can invent something which we shall call the *torque* (L. *torquere*, to twist) which bears the same relationship to rotation as force does to linear movement. A force is the thing that is needed to make linear motion, and the thing that makes something rotate is a “rotary force” or a “twisting force,” i.e., a torque. Qualitatively, a torque is a “twist”; what is a torque quantitatively? We shall get to the theory of torques quantitatively by studying the *work* done in turning an object, for one very nice way of defining a force is to say how much work it does when it acts through a given displacement. We are going to try to maintain the analogy between linear and angular quantities by equating the work that we do when we turn something a little bit when there are forces acting on it, to the *torque* times the *angle* it turns through. In other words, the definition of the torque is going to be so arranged that the theorem of work has an absolute analog: force times distance is work, and torque times angle is going to be work. That tells us what torque is. Consider, for instance, a rigid body of some kind with various forces acting on it, and an axis about which the body rotates. Let us at first concentrate on one force and suppose that this force is applied at a certain point (x, y) . How much work would be done if we were to turn the object through a very small angle? That is easy. The work done is

$$\Delta W = F_x \Delta x + F_y \Delta y. \quad (18.10)$$

We need only to substitute Eqs. (18.6) and (18.7) for Δx and Δy to obtain

$$\Delta W = (xF_y - yF_x) \Delta \theta. \quad (18.11)$$

That is, the amount of work that we have done is, in fact, equal to the angle through which we have turned the object, multiplied by a strange-looking combination of the force and the distance. This “strange combination” is what we call the torque. So, defining the change in work as the torque times the angle, we now have the formula for torque in terms of the forces. (Obviously, torque is not a completely new idea independent of Newtonian mechanics—torque must have a definite definition in terms of the force.)

When there are several forces acting, the work that is done is, of course, the sum of the works done by all the forces, so that ΔW will be a whole lot of terms, all added together, for all the forces, *each of which is proportional, however, to $\Delta \theta$* . We can take the $\Delta \theta$ outside and therefore can say that the change in the work is equal to the sum of all the torques due to all the different forces that are acting, times $\Delta \theta$. This sum we might call the total torque, τ . Thus torques add by the ordinary laws of algebra, but we shall later see that this is only because we are working in a plane. It is like one-dimensional kinematics, where the forces simply add algebraically, but only because they are all in the same direction. It is more complicated in three dimensions. Thus, for two-dimensional rotation,

$$\tau_i = x_i F_{yi} - y_i F_{xi} \quad (18.12)$$

and

$$\tau = \sum \tau_i. \quad (18.13)$$

It must be emphasized that the torque is about a given axis. If a different axis is chosen, so that all the x_i and y_i are changed, the value of the torque is (usually) changed too.

Now we pause briefly to note that our foregoing introduction of torque, through the idea of work, gives us a most important result for an object in equilibrium: if all the forces on an object are in balance both for translation and rotation, not only is the net *force* zero, but the total of all the *torques* is also zero, because if an object is in equilibrium, *no work is done by the forces for a small displacement*. Therefore, since $\Delta W = \tau \Delta \theta = 0$, the sum of all the torques must be zero. So there are two conditions for equilibrium: that the sum of the forces is zero, and that the sum of the torques is zero. Prove that it suffices to be sure that the sum of torques about any *one* axis (in two dimensions) is zero.

Now let us consider a single force, and try to figure out, geometrically, what this strange thing $xF_y - yF_x$ amounts to. In Fig. 18-2 we see a force $\mathbf{F}$ acting at a point $\mathbf{r}$. When the object has rotated through a small angle $\Delta\theta$, the work done, of course, is the component of force in the direction of the displacement times the displacement. In other words, it is only the tangential component of the force that counts, and this must be multiplied by the distance $r \Delta\theta$. Therefore we see that the torque is also equal to the tangential component of force (perpendicular to the radius) times the radius. That makes sense in terms of our ordinary idea of the torque, because if the force were completely radial, it would not put any "twist" on the body; it is evident that the twisting effect should involve only the part of the force which is not pulling out from the center, and that means the tangential component. Furthermore, it is clear that a given force is more effective on a long arm than near the axis. In fact, if we take the case where we push right on the axis, we are not twisting at all! So it makes sense that the amount of twist, or torque, is proportional both to the radial distance and to the tangential component of the force.

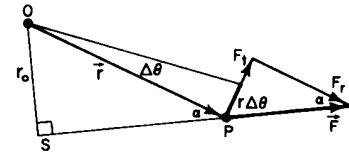


Fig. 18-2. The torque produced by a force.

There is still a third formula for the torque which is very interesting. We have just seen that the torque is the force times the radius times the sine of the angle α , in Fig. 18-2. But if we extend the line of action of the force and draw the line OS , the perpendicular distance to the line of action of the force (the *lever arm* of the force) we notice that this lever arm is shorter than r in just the same proportion as the tangential part of the force is less than the total force. Therefore the formula for the torque can also be written as the magnitude of the force times the length of the lever arm.

The torque is also often called the *moment* of the force. The origin of this term is obscure, but it may be related to the fact that "moment" is derived from the Latin *movimentum*, and that the capability of a force to move an object (using the force on a lever or crowbar) increases with the length of the lever arm. In mathematics "moment" means weighted by how far away it is from an axis.

18-3 Angular momentum

Although we have so far considered only the special case of a rigid body, the properties of torques and their mathematical relationships are interesting also even when an object is not rigid. In fact, we can prove a very remarkable theorem: just as external force is the rate of change of a quantity p , which we call the total momentum of a collection of particles, so the external torque is the rate of change of a quantity L which we call the *angular momentum* of the group of particles.

To prove this, we shall suppose that there is a system of particles on which there are some forces acting and find out what happens to the system as a result of the torques due to these forces. First, of course, we should consider just *one* particle. In Fig. 18-3 is one particle of mass m , and an axis O ; the particle is not necessarily rotating in a circle about O , it may be moving in an ellipse, like a planet going around the sun, or in some other curve. It is moving somehow, and there are forces on it, and it accelerates according to the usual formula that the x -component of force is the mass times the x -component of acceleration, etc. But let us see what the *torque* does. The torque equals $xF_y - yF_x$, and the force in the x - or y -direction is the mass times the acceleration in the x - or y -direction:

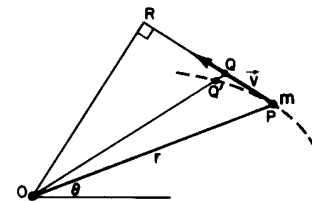


Fig. 18-3. A particle moves about an axis O .

$$\begin{aligned}\tau &= xF_y - yF_x \\ &= xm(d^2y/dt^2) - ym(d^2x/dt^2).\end{aligned}\quad (18.14)$$

Now, although this does not appear to be the derivative of any simple quantity, it is in fact the derivative of the quantity $xm(dy/dt) - ym(dx/dt)$:

$$\begin{aligned}\frac{d}{dt} \left[xm \left(\frac{dy}{dt} \right) - ym \left(\frac{dx}{dt} \right) \right] &= xm \left(\frac{d^2y}{dt^2} \right) + \left(\frac{dx}{dt} \right) m \left(\frac{dy}{dt} \right) \\ &\quad - ym \left(\frac{d^2x}{dt^2} \right) - \left(\frac{dy}{dt} \right) m \left(\frac{dx}{dt} \right) = xm \left(\frac{d^2y}{dt^2} \right) - ym \left(\frac{d^2x}{dt^2} \right).\end{aligned}\quad (18.15)$$

So it is true that the torque is the rate of change of something with time! So we pay attention to the “something,” we give it a name: we call it L , the angular momentum:

$$\begin{aligned} L &= xm(dy/dt) - ym(dx/dt) \\ &= xp_y - yp_x. \end{aligned} \quad (18.16)$$

Although our present discussion is nonrelativistic, the second form for L given above is relativistically correct. So we have found that there is also a rotational analog for the momentum, and that this analog, the angular momentum, is given by an expression in terms of the components of linear momentum that is just like the formula for torque in terms of the force components! Thus, if we want to know the angular momentum of a particle about an axis, we take only the component of the momentum that is tangential, and multiply it by the radius. In other words, what counts for angular momentum is not how fast it is going *away* from the origin, but how much it is going *around* the origin. Only the tangential part of the momentum counts for angular momentum. Furthermore, the farther out the line of the momentum extends, the greater the angular momentum. And also, because the geometrical facts are the same whether the quantity is labeled p or F , it is true that there is a lever arm (*not* the same as the lever arm of the force on the particle!) which is obtained by extending the line of the *momentum* and finding the perpendicular distance to the axis. Thus the angular momentum is the magnitude of the momentum times the momentum lever arm. So we have three formulas for angular momentum, just as we have three formulas for the torque:

$$\begin{aligned} L &= xp_y - yp_x \\ &= rp_{\text{tang}} \\ &= p \cdot \text{lever arm}. \end{aligned} \quad (18.17)$$

Like torque, angular momentum depends upon the position of the axis about which it is to be calculated.

Before proceeding to a treatment of more than one particle, let us apply the above results to a planet going around the sun. In which direction is the force? The force is toward the sun. What, then, is the torque on the object? Of course, this depends upon where we take the axis, but we get a very simple result if we take it at the sun itself, for the torque is the force times the lever arm, or the component of force perpendicular to r , times r . But there is no tangential force, so there is no torque about an axis at the sun! Therefore, the angular momentum of the planet going around the sun must remain constant. Let us see what that means. The tangential component of velocity, times the mass, times the radius, will be constant, because that is the angular momentum, and the rate of change of the angular momentum is the torque, and, in this problem, the torque is zero. Of course since the mass is also a constant, this means that the tangential velocity times the radius is a constant. But this is something we already knew for the motion of a planet. Suppose we consider a small amount of time Δt . How far will the planet move when it moves from P to Q (Fig. 18-3)? How much *area* will it sweep through? Disregarding the very tiny area $QQ'P$ compared with the much larger area OPQ , it is simply half the base PQ times the height, OR . In other words, the area that is swept through in unit time will be equal to the velocity times the lever arm of the velocity (times one-half). Thus the rate of change of area is proportional to the angular momentum, which is constant. So Kepler's law about equal areas in equal times is a word description of the statement of the law of conservation of angular momentum, when there is no torque produced by the force.

18-4 Conservation of angular momentum

Now we shall go on to consider what happens when there is a large number of particles, when an object is made of many pieces with many forces acting between them and on them from the outside. Of course, we already know that, about any given fixed axis, the torque on the i th particle (which is the force on the i th particle

times the lever arm of that force) is equal to the rate of change of the angular momentum of that particle, and that the angular momentum of the i th particle is its momentum times its momentum lever arm. Now suppose we add the torques τ_i for all the particles and call it the total torque τ . Then this will be the rate of change of the sum of the angular momenta of all the particles L_i , and that defines a new quantity which we call the total angular momentum L . Just as the total *momentum* of an object is the sum of the momenta of all the parts, so the angular momentum is the sum of the angular momenta of all the parts. Then the rate of change of the total L is the total torque:

$$\tau = \sum \tau_i = \sum \frac{dL_i}{dt} = \frac{dL}{dt}. \quad (18.18)$$

Now it might seem that the total torque is a complicated thing. There are all those internal forces and all the outside forces to be considered. But, if we take Newton's law of action and reaction to say, not simply that the action and reaction are equal, but also that they are *directed exactly oppositely along the same line* (Newton may or may not actually have said this, but he tacitly assumed it), then the two *torques* on the reacting objects, due to their mutual interaction, will be equal and opposite because the lever arms for any axis are equal. Therefore the internal torques balance out pair by pair, and so we have the remarkable theorem that the *rate of change of the total angular momentum about any axis is equal to the external torque about that axis!*

$$\tau = \sum \tau_i = \tau_{\text{ext}} = dL/dt. \quad (18.19)$$

Thus we have a very powerful theorem concerning the motion of large collections of particles, which permits us to study the over-all motion without having to look at the detailed machinery inside. This theorem is true for any collection of objects, whether they form a rigid body or not.

One extremely important case of the above theorem is the *law of conservation of angular momentum*: if no external torques act upon a system of particles, the angular momentum remains constant.

A special case of great importance is that of a rigid body, that is, an object of a definite shape that is just turning around. Consider an object that is fixed in its geometrical dimensions, and which is rotating about a fixed axis. Various parts of the object bear the same relationship to one another at all times. Now let us try to find the total angular momentum of this object. If the mass of one of its particles is m_i , and its position or location is at (x_i, y_i) , then the problem is to find the angular momentum of that particle, because the total angular momentum is the sum of the angular momenta of all such particles in the body. For an object going around in a circle, the angular momentum, of course, is the mass times the velocity times the distance from the axis, and the velocity is equal to the angular velocity times the distance from the axis:

$$L_i = m_i v_i r_i = m_i r_i^2 \omega, \quad (18.20)$$

or, summing over all the particles i , we get

$$L = I\omega, \quad (18.21)$$

where

$$I = \sum_i m_i r_i^2. \quad (18.22)$$

This is the analog of the law that the momentum is mass times velocity. Velocity is replaced by angular velocity, and we see that the mass is replaced by a new thing which we call the *moment of inertia* I , which is analogous to the mass. Equations (18.21) and (18.22) say that a body has inertia for turning which depends, not just on the masses, but on *how far away they are* from the axis. So, if we have two objects of the same mass, when we put the masses farther away from the axis, the inertia for turning will be higher. This is easily demonstrated by the apparatus

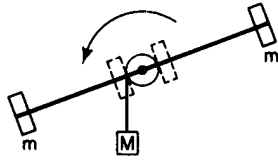


Fig. 18-4. The "inertia for turning" depends upon the lever arm of the masses.

shown in Fig. 18-4, where a weight M is kept from falling very fast because it has to turn the large weighted rod. At first, the masses m are close to the axis, and M speeds up at a certain rate. But when we change the moment of inertia by putting the two masses m much farther away from the axis, then we see that M accelerates much less rapidly than it did before, because the body has much more inertia against turning. The moment of inertia is the inertia against turning, and is the sum of the contributions of all the masses, times their distances *squared*, from the axis.

There is one important difference between mass and moment of inertia which is very dramatic. The mass of an object never changes, but its moment of inertia *can* be changed. If we stand on a frictionless rotatable stand with our arms outstretched, and hold some weights in our hands as we rotate slowly, we may change our moment of inertia by drawing our arms in, but our mass does not change. When we do this, all kinds of wonderful things happen, because of the law of the conservation of angular momentum: If the external torque is zero, then the angular momentum, the moment of inertia times omega, remains constant. Initially, we were rotating with a large moment of inertia I_1 at a low angular velocity ω_1 , and the angular momentum was $I_1\omega_1$. Then we changed our moment of inertia by pulling our arms in, say to a smaller value I_2 . Then the product $I\omega$, which has to stay the same because the total angular momentum has to stay the same, was $I_2\omega_2$. So $I_1\omega_1 = I_2\omega_2$. That is, if we *reduce* the moment of inertia, we have to *increase* the angular velocity.

Center of Mass; Moment of Inertia

19-1 Properties of the center of mass

In the previous chapter we found that if a great many forces are acting on a complicated mass of particles, whether the particles comprise a rigid or a nonrigid body, or a cloud of stars, or anything else, and we find the sum of all the forces (that is, of course, the external forces, because the internal forces balance out), then if we consider the body as a whole, and say it has a total mass M , there is a certain point "inside" the body, called the *center of mass*, such that the net resulting external force produces an acceleration of this point, just as though the whole mass were concentrated there. Let us now discuss the center of mass in a little more detail.

The location of the center of mass (abbreviated CM) is given by the equation

$$\mathbf{R}_{\text{CM}} = \frac{\sum m_i \mathbf{r}_i}{\sum m_i}. \quad (19.1)$$

This is, of course, a vector equation which is really three equations, one for each of the three directions. We shall consider only the x -direction, because if we can understand that one, we can understand the other two. What does $X_{\text{CM}} = \sum m_i x_i / \sum m_i$ mean? Suppose for a moment that the object is divided into little pieces, all of which have the same mass m ; then the total mass is simply the number N of pieces times the mass of one piece, say one gram, or any unit. Then this equation simply says that we add all the x 's, and then divide by the number of things that we have added: $X_{\text{CM}} = m \sum x_i / mN = \sum x_i / N$. In other words, X_{CM} is the average of all the x 's, if the masses are equal. But suppose one of them were twice as heavy as the others. Then in the sum, that x would come in twice. This is easy to understand, for we can think of this double mass as being split into two equal ones, just like the others; then in taking the average, of course, we have to count that x twice because there are two masses there. Thus X is the average position, in the x -direction, of all the masses, every mass being counted a number of times proportional to the mass, as though it were divided into "little grams." From this it is easy to prove that X must be somewhere between the largest and the smallest x , and, therefore lies inside the envelope including the entire body. It does not have to be in the *material* of the body, for the body could be a circle, like a hoop, and the center of mass is in the center of the hoop, not in the hoop itself.

Of course, if an object is symmetrical in some way, for instance, a rectangle, so that it has a plane of symmetry, the center of mass lies somewhere on the plane of symmetry. In the case of a rectangle there are two planes, and that locates it uniquely. But if it is just any symmetrical object, then the center of gravity lies somewhere on the axis of symmetry, because in those circumstances there are as many positive as negative x 's.

Another interesting proposition is the following very curious one. Suppose that we imagine an object to be made of two pieces, A and B (Fig. 19-1). Then the center of mass of the whole object can be calculated as follows. First, find the center of mass of piece A , and then of piece B . Also, find the total mass of each piece, M_A and M_B . Then consider a new problem, in which a *point* mass M_A is at the center of mass of object A , and another *point* mass M_B is at the center of mass of object B . The center of mass of these two point masses is then the center of mass of the whole object. In other words, if the centers of mass of various parts

19-1 Properties of the center of mass

19-2 Locating the center of mass

19-3 Finding the moment of inertia

19-4 Rotational kinetic energy

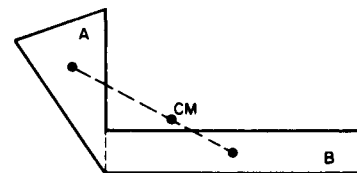


Fig. 19-1. The CM of a compound body lies on the line joining the CM's of the two composite parts.

of an object have been worked out, we do not have to start all over again to find the center of mass of the whole object; we just have to put the pieces together, treating each one as a point mass situated at the center of mass of that piece. Let us see why that is. Suppose that we wanted to calculate the center of mass of a complete object, some of whose particles are considered to be members of object A and some members of object B . The total sum $\sum m_i x_i$ can then be split into two pieces—the sum $\sum_A m_i x_i$ for the A object only, and the sum $\sum_B m_i x_i$ for object B only. Now if we were computing the center of mass of object A alone, we would have exactly the first of these sums, and we know that this by itself is $M_A X_A$, the total mass of all the particles in A times the position of the center of mass of A , because that is the theorem of the center of mass, applied to object A . In the same manner, just by looking at object B , we get $M_B X_B$, and of course, adding the two yields MX :

$$\begin{aligned} MX_{\text{CM}} &= \sum_A m_i x_i + \sum_B m_i x_i \\ &= M_A X_A + M_B X_B. \end{aligned} \quad (19.2)$$

Now since M is evidently the sum of M_A and M_B , we see that Eq. (19.2) can be interpreted as a special example of the center of mass formula for two point objects, one of mass M_A located at X_A and the other of mass M_B located at X_B .

The theorem concerning the motion of the center of mass is very interesting, and has played an important part in the development of our understanding of physics. Suppose we assume that Newton's law is right for the small component parts of a much larger object. Then this theorem shows that Newton's law is also correct for the larger object, even if we do not study the details of the object, but only the total force acting on it and its mass. In other words, Newton's law has the peculiar property that if it is right on a certain small scale, then it will be right on a larger scale. If we do not consider a baseball as a tremendously complex thing, made of myriads of interacting particles, but study only the motion of the center of mass and the external forces on the ball, we find $\mathbf{F} = m\mathbf{a}$, where $\mathbf{F}$ is the external force on the baseball, m is its mass, and $\mathbf{a}$ is the acceleration of its center of mass. So $\mathbf{F} = m\mathbf{a}$ is a law which reproduces itself on a larger scale. (There ought to be a good word, out of the Greek, perhaps, to describe a law which reproduces the same law on a larger scale.)

Of course, one might suspect that the first laws that would be discovered by human beings would be those that would reproduce themselves on a larger scale. Why? Because the actual scale of the fundamental gears and wheels of the universe are of atomic dimensions, which are so much finer than our observations that we are nowhere near that scale in our ordinary observations. So the first things that we would discover must be true for objects of no special size relative to an atomic scale. If the laws for small particles did not reproduce themselves on a larger scale, we would not discover those laws very easily. What about the reverse problem? Must the laws on a small scale be the same as those on a larger scale? Of course it is not necessarily so in nature, that at an atomic level the laws have to be the same as on a large scale. Suppose that the true laws of motion of atoms were given by some strange equation which does *not* have the property that when we go to a larger scale we reproduce the same law, but instead has the property that if we go to a larger scale, we can *approximate it by a certain expression* such that, if we extend that expression up and up, it keeps reproducing itself on a larger and larger scale. That is possible, and in fact that is the way it works. Newton's laws are the "tail end" of the atomic laws, extrapolated to a very large size. The actual laws of motion of particles on a fine scale are very peculiar, but if we take large numbers of them and compound them, they approximate, but *only* approximate, Newton's laws. Newton's laws then permit us to go on to a higher and higher scale, and it still seems to be the same law. In fact, it becomes more and more accurate as the scale gets larger and larger. This self-reproducing factor of Newton's laws is thus really not a fundamental feature of nature, but is an important historical feature. We would never discover the fundamental laws of the atomic particles at first observation because the first observations are much too crude. In fact, it turns

out that the fundamental atomic laws, which we call quantum mechanics, are quite different from Newton's laws, and are difficult to understand because all our direct experiences are with large-scale objects and the small-scale atoms behave like nothing we see on a large scale. So we cannot say, "An atom is just like a planet going around the sun," or anything like that. It is like *nothing* we are familiar with because there is *nothing like it*. As we apply quantum mechanics to larger and larger things, the laws about the behavior of many atoms together do *not* reproduce themselves, but produce *new laws*, which are Newton's laws, which then continue to reproduce themselves from, say, micro-microgram size, which still is billions and billions of atoms, on up to the size of the earth, and above.

Let us now return to the center of mass. The center of mass is sometimes called the center of gravity, for the reason that, in many cases, gravity may be considered uniform. Let us suppose that we have small enough dimensions that the gravitational force is not only proportional to the mass, but is everywhere parallel to some fixed line. Then consider an object in which there are gravitational forces on each of its constituent masses. Let m_i be the mass of one part. Then the gravitational force on that part is m_i times g . Now the question is, where can we apply a single force to balance the gravitational force on the whole thing, so that the entire object, if it is a rigid body, will not turn? The answer is that this force must go through the center of mass, and we show this in the following way. In order that the body will not turn, the torque produced by all the forces must add up to zero, because if there is a torque, there is a change of angular momentum, and thus a rotation. So we must calculate the total of all the torques on all the particles, and see how much torque there is about any given axis; it should be zero if this axis is at the center of mass. Now, measuring x horizontally and y vertically, we know that the torques are the forces in the y -direction, times the lever arm x (that is to say, the force times the lever arm around which we want to measure the torque). Now the total torque is the sum

$$\tau = \sum m_i g x_i = g \sum m_i x_i, \quad (19.3)$$

so if the total torque is to be zero, the sum $\sum m_i x_i$ must be zero. But $\sum m_i x_i = MX$, the total mass times the distance of the center of mass from the axis. Thus the x -distance of the center of mass from the axis is zero.

Of course, we have checked the result only for the x -distance, but if we use the true center of mass the object will balance in any position, because if we turned it 90 degrees, we would have y 's instead of x 's. In other words, when an object is supported at its center of mass, there is no torque on it because of a parallel gravitational field. In case the object is so large that the nonparallelism of the gravitational forces is significant, then the center where one must apply the balancing force is not simple to describe, and it departs slightly from the center of mass. That is why one must distinguish between the center of mass and the center of gravity. The fact that an object supported exactly at the center of mass will balance in all positions has another interesting consequence. If, instead of gravitation, we have a pseudoforce due to acceleration, we may use exactly the same mathematical procedure to find the position to support it so that there are no torques produced by the inertial force of acceleration. Suppose that the object is held in some manner inside a box, and that the box, and everything contained in it, is accelerating. We know that, from the point of view of someone at rest relative to this accelerating box, there will be an effective force due to inertia. That is, to make the object go along with the box, we have to push on it to accelerate it, and this force is "balanced" by the "force of inertia," which is a pseudoforce equal to the mass times the acceleration of the box. To the man in the box, this is the same situation as if the object were in a uniform gravitational field whose " g " value is equal to the acceleration a . Thus the inertial force due to accelerating an object has no torque about the center of mass.

This fact has a very interesting consequence. In an inertial frame that is not accelerating, the torque is always equal to the rate of change of the angular momentum. However, about an axis through the center of mass of an object which

is accelerating, it is *still true* that the torque is equal to the rate of change of the angular momentum. Even if the center of mass is accelerating, we may still choose one special axis, namely, one passing through the center of mass, such that it will still be true that the torque is equal to the rate of change of angular momentum around that axis. Thus the theorem that torque equals the rate of change of angular momentum is true in two general cases: (1) a fixed axis in inertial space, (2) an axis through the center of mass, even though the object may be accelerating.

19-2 Locating the center of mass

The mathematical techniques for the calculation of centers of mass are in the province of a mathematics course, and such problems provide good exercise in integral calculus. After one has learned calculus, however, and wants to know how to locate centers of mass, it is nice to know certain tricks which can be used to do so. One such trick makes use of what is called the theorem of Pappus. It works like this: if we take any closed area in a plane and generate a solid by moving it through space such that each point is always moved perpendicular to the plane of the area, the resulting solid has a total volume equal to the area of the cross section times the distance that the center of mass moved! Certainly this is true if we move the area in a straight line perpendicular to itself, but if we move it in a circle or in some other curve, then it generates a rather peculiar volume. For a curved path, the outside goes around farther, and the inside goes around less, and these effects balance out. So if we want to locate the center of mass of a plane sheet of uniform density, we can remember that the volume generated by spinning it about an axis is the distance that the center of mass goes around, times the area of the sheet.

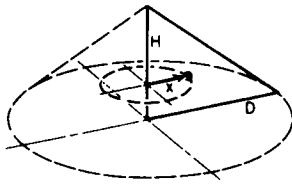


Fig. 19-2. A right triangle and a right circular cone generated by rotating the triangle.

For example, if we wish to find the center of mass of a right triangle of base D and height H (Fig. 19-2), we might solve the problem in the following way. Imagine an axis along H , and rotate the triangle about that axis through a full 360 degrees. This generates a cone. The distance that the x -coordinate of the center of mass has moved is $2\pi x$. The area which is being moved is the area of the triangle, $\frac{1}{2}HD$. So the x -distance of the center of mass times the area of the triangle is the volume swept out, which is of course $\pi D^2 H/3$. Thus $(2\pi x)(\frac{1}{2}HD) = 1/3\pi D^2 H$, or $x = D/3$. In a similar manner, by rotating about the other axis, or by symmetry, we find $y = H/3$. In fact, the center of mass of any uniform triangular area is where the three medians, the lines from the vertices through the centers of the opposite sides, all meet. That point is $1/3$ of the way along each median. *Clue:* Slice the triangle up into a lot of little pieces, each parallel to a base. Note that the median line bisects every piece, and therefore the center of mass must lie on this line.

Now let us try a more complicated figure. Suppose that it is desired to find the position of the center of mass of a uniform semicircular disc—a disc sliced in half. Where is the center of mass? For a full disc, it is at the center, of course, but a half-disc is more difficult. Let r be the radius and x be the distance of the center of mass from the straight edge of the disc. Spin it around this edge as axis to generate a sphere. Then the center of mass has gone around $2\pi x$, the area is $\pi r^2/2$ (because it is only half a circle). The volume generated is, of course, $4\pi r^3/3$, from which we find that

$$(2\pi x)(\frac{1}{2}\pi r^2) = 4\pi r^3/3,$$

or

$$x = 4r/3\pi.$$

There is another theorem of Pappus which is a special case of the above one, and therefore equally true. Suppose that, instead of the solid semicircular disc, we have a semicircular piece of wire with uniform mass density along the wire, and we want to find its center of mass. In this case there is no mass in the interior, only on the wire. Then it turns out that the *area* which is swept by a plane curved line, when it moves as before, is the distance that the center of mass moves times the *length* of the line. (The line can be thought of as a very narrow area, and the previous theorem can be applied to it.)

19-3 Finding the moment of inertia

Now let us discuss the problem of finding the *moments of inertia* of various objects. The formula for the moment of inertia about the z -axis of an object is

$$I = \sum m_i(x_i^2 + y_i^2)$$

or

$$I = \int (x^2 + y^2) dm = \int (x^2 + y^2) \rho dv. \quad (19.4)$$

That is, we must sum the masses, each one multiplied by the square of its distance ($x_i^2 + y_i^2$) from the axis. Note that it is not the three-dimensional distance, only the two-dimensional distance squared, even for a three-dimensional object. For the most part, we shall restrict ourselves to two-dimensional objects, but the formula for rotation about the z -axis is just the same in three dimensions.

As a simple example, consider a rod rotating about a perpendicular axis through one end (Fig. 19-3). Now we must sum all the masses times the x -distances squared (the y 's being all zero in this case). What we mean by "the sum," of course, is the integral of x^2 times the little elements of mass. If we divide the rod into small elements of length dx , the corresponding elements of mass are proportional to dx , and if dx were the length of the whole rod the mass would be M . Therefore

$$dm = M dx/L$$

and so

$$I = \int_0^L x^2 \frac{M dx}{L} = \frac{M}{L} \int_0^L x^2 dx = \frac{ML^2}{3}. \quad (19.5)$$

The dimensions of moment of inertia are always mass times length squared, so all we really had to work out was the factor $1/3$.

Now what is I if the rotation axis is at the center of the rod? We could just do the integral over again, letting x range from $-\frac{1}{2}L$ to $+\frac{1}{2}L$. But let us notice a few things about the moment of inertia. We can imagine the rod as two rods, each of mass $M/2$ and length $L/2$; the moments of inertia of the two small rods are equal, and are both given by the formula (19.5). Therefore the moment of inertia is

$$I = \frac{2(M/2)(L/2)^2}{3} = \frac{ML^2}{12}. \quad (19.6)$$

Thus it is much easier to turn a rod about its center, than to swing it around an end.

Of course, we could go on to compute the moments of inertia of various other bodies of interest. However, while such computations provide a certain amount of important exercise in the calculus, they are not basically of interest to us as such. There is, however, an interesting theorem which is very useful. Suppose we have an object, and we want to find its moment of inertia around some axis. That means we want the inertia needed to carry it by rotation about that axis. Now if we support the object on pivots at the center of mass, so that the object does not turn as it rotates about the axis (because there is no torque on it from inertial effects, and therefore it will not turn when we start moving it), then the forces needed to swing it around are the same as though all the mass were concentrated at the center of mass, and the moment of inertia would be simply $I_1 = MR_{CM}^2$, where R_{CM} is the distance from the axis to the center of mass. But of course that is not the right formula for the moment of inertia of an object which is really being rotated as it revolves, because not only is the center of it moving in a circle, which would contribute an amount I_1 to the moment of inertia, but also we must turn it about its center of mass. So it is not unreasonable that we must add to I_1 the moment of inertia I_c about the center of mass. So it is a good guess that the total moment of inertia about any axis will be

$$I = I_c + MR_{CM}^2. \quad (19.7)$$

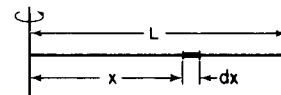


Fig. 19-3. A straight rod of length L rotating about an axis through one end.

This theorem is called the *parallel-axis theorem*, and may be easily proved. The moment of inertia about any axis is the mass times the sum of the x_i 's and the y_i 's, each squared: $I = \sum (x_i^2 + y_i^2)m_i$. We shall concentrate on the x 's, but of course the y 's work the same way. Now x is the distance of a particular point mass from the origin, but let us consider how it would look if we measured x' from the CM, instead of x from the origin. To get ready for this analysis, we write

$$x_i = x'_i + X_{\text{CM}}.$$

Then we just square this to find

$$x_i^2 = x'^2_i + 2X_{\text{CM}}x'_i + X_{\text{CM}}^2.$$

So, when this is multiplied by m_i and summed over all i , what happens? Taking the constants outside the summation sign, we get

$$I_x = \sum m_i x'^2_i + 2X_{\text{CM}} \sum m_i x'_i + X_{\text{CM}}^2 \sum m_i.$$

The third sum is easy; it is just MX_{CM}^2 . In the second sum there are two pieces, one of them is $\sum m_i x'_i$, which is the total mass times the x' -coordinate of the center of mass. But this contributes nothing, because x' is *measured* from the center of mass, and in these axes the average position of all the particles, weighted by the masses, is zero. The first sum, of course, is the x part of I_c . Thus we arrive at Eq. (19.7), just as we guessed.

Let us check (19.7) for one example. Let us just see whether it works for the rod. For an axis through one end, the moment of inertia should be $ML^2/3$, for we calculated that. The center of mass of a rod, of course, is in the center of the rod, at a distance $L/2$. Therefore we should find that $ML^2/3 = ML^2/12 + M(L/2)^2$. Since one-quarter plus one-twelfth is one-third, we have made no fundamental error.

Incidentally, we did not really need to use an integral to find the moment of inertia (19.5). If we simply assume that it is ML^2 times γ , an unknown coefficient, and then use the argument about the two halves to get $\frac{1}{4}\gamma$ for (19.6), then from our argument about transferring the axes we could prove that $\gamma = \frac{1}{4}\gamma + \frac{1}{4}$, so γ must be $1/3$. There is always another way to do it!

In applying the parallel-axis theorem, it is of course important to remember that the axis for I_c *must be parallel* to the axis about which the moment of inertia is wanted.

One further property of the moment of inertia is worth mentioning because it is often helpful in finding the moment of inertia of certain kinds of objects. This property is that if one has a *plane figure* and a set of coordinate axes with origin in the plane and z -axis perpendicular to the plane, then the moment of inertia of this figure about the z -axis is equal to the sum of the moments of inertia about the x - and y -axes. This is easily proved by noting that

$$I_z = \sum m_i (y_i^2 + z_i^2) = \sum m_i y_i^2$$

(since $z_i = 0$). Similarly,

$$I_y = \sum m_i (x_i^2 + z_i^2) = \sum m_i x_i^2,$$

but

$$\begin{aligned} I_z &= \sum m_i (x_i^2 + y_i^2) = \sum m_i x_i^2 + \sum m_i y_i^2 \\ &= I_x + I_y. \end{aligned}$$

As an example, the moment of inertia of a uniform rectangular plate of mass M , width w , and length L , about an axis perpendicular to the plate and through its center is simply

$$I = M(w^2 + L^2)/12,$$

because its moment of inertia about an axis in its plane and parallel to its length is $Mw^2/12$, i.e., just as for a rod of length w , and the moment of inertia about the other axis in its plane is $ML^2/12$, just as for a rod of length L .

To summarize, the moment of inertia of an object about a given axis, which we shall call the z -axis, has the following properties:

- (1) The moment of inertia is

$$I_z = \sum_i m_i(x_i^2 + y_i^2) = \int (x^2 + y^2) dm.$$

- (2) If the object is made of a number of parts, each of whose moment of inertia is known, the total moment of inertia is the sum of the moments of inertia of the pieces.
- (3) The moment of inertia about any given axis is equal to the moment of inertia about a parallel axis through the CM plus the total mass times the square of the distance from the axis to the CM.
- (4) If the object is a plane figure, the moment of inertia about an axis perpendicular to the plane is equal to the sum of the moments of inertia about any two mutually perpendicular axes lying in the plane and intersecting at the perpendicular axis.

The moments of inertia of a number of elementary shapes having uniform mass densities are given in Table 19-1, and the moments of inertia of some other objects, which may be deduced from Table 19-1, using the above properties, are given in Table 19-2.

Table 19-1

Object	z -axis	I_z
Thin rod, length L	$\perp$ rod at center	$ML^2/12$
Thin concentric circular ring, radii r_1 and r_2	$\perp$ ring at center	$M(r_1^2 + r_2^2)/2$
Sphere, radius r	through center	$2Mr^2/5$

Table 19-2

Object	z -axis	I_z
Rect. sheet, sides a, b	$\parallel b$ at center	$Ma^2/12$
Rect. sheet, sides a, b	$\perp$ sheet at center	$M(a^2 + b^2)/12$
Thin annular ring, radii r_1, r_2	any diameter	$M(r_1^2 + r_2^2)/4$
Rect. parallelepiped, sides a, b, c	$\parallel c$, through center	$M(a^2 + b^2)/12$
Rt. circ. cyl., radius r , length L	$\parallel L$, through center	$Mr^2/2$
Rt. circ. cyl., radius r , length L	$\perp L$, through center	$M(r^2/4 + L^2/12)$

19-4 Rotational kinetic energy

Now let us go on to discuss dynamics further. In the analogy between linear motion and angular motion that we discussed in Chapter 18, we used the work theorem, but we did not talk about kinetic energy. What is the kinetic energy of a rigid body, rotating about a certain axis with an angular velocity ω ? We can immediately guess the correct answer by using our analogies. The moment of inertia corresponds to the mass, angular velocity corresponds to velocity, and so the kinetic energy ought to be $\frac{1}{2}I\omega^2$, and indeed it is, as will now be demonstrated. Suppose the object is rotating about some axis so that each point has a velocity whose magnitude is ωr_i , where r_i is the radius from the particular point to the axis.

Then if m_i is the mass of that point, the total kinetic energy of the whole thing is just the sum of the kinetic energies of all of the little pieces:

$$T = \frac{1}{2} \sum m_i v_i^2 = \frac{1}{2} \sum m_i (r_i \omega)^2.$$

Now ω^2 is a constant, the same for all points. Thus

$$T = \frac{1}{2} \omega^2 \sum m_i r_i^2 = \frac{1}{2} I \omega^2. \quad (19.8)$$

At the end of Chapter 18 we pointed out that there are some interesting phenomena associated with an object which is not rigid, but which changes from one rigid condition with a definite moment of inertia, to another rigid condition. Namely, in our example of the turntable, we had a certain moment of inertia I_1 with our arms stretched out, and a certain angular velocity ω_1 . When we pulled our arms in, we had a different moment of inertia, I_2 , and a different angular velocity, ω_2 , but again we were "rigid." The angular momentum remained constant, since there was no torque about the vertical axis of the turntable. This means that $I_1 \omega_1 = I_2 \omega_2$. Now what about the energy? That is an interesting question. With our arms pulled in, we turn faster, but our moment of inertia is less, and it looks as though the energies might be equal. But they are not, because what *does* balance is $I\omega$, not $I\omega^2$. So if we compare the kinetic energy before and after, the kinetic energy before is $\frac{1}{2} I_1 \omega_1^2 = \frac{1}{2} L \omega_1$, where $L = I_1 \omega_1 = I_2 \omega_2$ is the angular momentum. Afterward, by the same argument, we have $T = \frac{1}{2} L \omega_2$, and since $\omega_2 > \omega_1$ the kinetic energy of rotation is greater than it was before. So we had a certain energy when our arms were out, and when we pulled them in, we were turning faster and had more kinetic energy. What happened to the theorem of the conservation of energy? Somebody must have done some work. We did work! When did we do any work? When we move a weight horizontally, we do not do any work. If we hold a thing out and pull it in, we do not do any work. But that is when we are not rotating! When we *are* rotating, there is centrifugal force on the weights. They are trying to fly out, so when we are going around we have to pull the weights in against the centrifugal force. So, the work we do against the centrifugal force ought to agree with the difference in rotational energy, and of course it does. That is where the extra kinetic energy comes from.

There is still another interesting feature which we can treat only descriptively, as a matter of general interest. This feature is a little more advanced, but is worth pointing out because it is quite curious and produces many interesting effects.

Consider that turntable experiment again. Consider the body and the arms separately, from the point of view of the man who is rotating. After the weights are pulled in, the whole object is spinning faster, but observe, *the central part of the body is not changed*, yet it is turning faster after the event than before. So, if we were to draw a circle around the inner body, and consider only objects inside the circle, *their* angular momentum would *change*; they are going faster. Therefore there must be a torque exerted on the body while we pull in our arms. No torque can be exerted by the centrifugal force, because that is radial. So that means that among the forces that are developed in a rotating system, centrifugal force is not the entire story, *there is another force*. This other force is called *Coriolis force*, and it has the very strange property that when we move something in a rotating system, it seems to be pushed sidewise. Like the centrifugal force, it is an apparent force. But if we live in a system that is rotating, and move something radially, we find that we must also push it sidewise to move it radially. This sidewise push which we have to exert is what turned our body around.

Now let us develop a formula to show how this Coriolis force really works. Suppose Moe is sitting on a carousel that appears to him to be stationary. But from the point of view of Joe, who is standing on the ground and who knows the right laws of mechanics, the carousel is going around. Suppose that we have drawn a radial line on the carousel, and that Moe is moving some mass radially along this line. We would like to demonstrate that a sidewise force is required to do that. We can do this by paying attention to the angular momentum of the mass. It is

always going around with the same angular velocity ω , so that the angular momentum is

$$L = mv_{\text{tang}}r = m\omega r \cdot r = m\omega r^2.$$

So when the mass is close to the center, it has relatively little angular momentum, but if we move it to a new position farther out, if we increase r , m has more angular momentum, so *a torque must be exerted* in order to move it along the radius. (To walk along the radius in a carousel, one has to lean over and push sidewise. Try it sometime.) The torque that is required is the rate of change of L with time as m moves along the radius. If m moves only along the radius, ω stays constant, so that the torque is

$$\tau = F_c r = \frac{dL}{dt} = \frac{d(m\omega r^2)}{dt} = 2m\omega r \frac{dr}{dt},$$

where F_c is the Coriolis force. What we really want to know is what sidewise *force* has to be exerted by Moe in order to move m out at speed $v_r = dr/dt$. This is $F_c = \tau/r = 2m\omega v_r$.

Now that we have a formula for the Coriolis force, let us look at the situation a little more carefully, to see whether we can understand the origin of this force from a more elementary point of view. We note that the Coriolis force is the same at every radius, and is evidently present even at the origin! But it is especially easy to understand it at the origin, just by looking at what happens from the inertial system of Joe, who is standing on the ground. Figure 19-4 shows three successive views of m just as it passes the origin at $t = 0$. Because of the rotation of the carousel, we see that m does not move in a straight line, but in a *curved path* tangent to a diameter of the carousel where $r = 0$. In order for m to go in a curve, there must be a force to accelerate it in absolute space. This is the Coriolis force.

This is not the only case in which the Coriolis force occurs. We can also show that if an object is moving with constant speed around the circumference of a circle, there is also a Coriolis force. Why? Moe sees a velocity v_M around the circle. On the other hand, Joe sees m going around the circle with the velocity $v_J = v_M + \omega r$, because m is also carried by the carousel. Therefore we know what the force really is, namely, the total centripetal force due to the velocity v_J , or mv_J^2/r ; that is the actual force. Now from Moe's point of view, this centripetal force has three pieces. We may write it all out as follows:

$$F_r = -\frac{mv_J^2}{r} = -\frac{mv_M^2}{r} - 2mv_M\omega - m\omega^2 r.$$

Now, F_r is the force that Moe would see. Let us try to understand it. Would Moe appreciate the first term? "Yes," he would say, "even if I were not turning, there would be a centripetal force if I were to run around a circle with velocity v_M ." This is simply the centripetal force that Moe would expect, having nothing to do with rotation. In addition, Moe is quite aware that there is another centripetal force that would act even on objects which are standing still on his carousel. This is the third term. But there is another term in addition to these, namely the second term, which is again $2m\omega v$. The Coriolis force F_c was tangential when the velocity was radial, and now it is radial when the velocity is tangential. In fact, one expression has a minus sign relative to the other. The force is always in the same direction, relative to the velocity, no matter in which direction the velocity is. The force is at right angles to the velocity, and of magnitude $2m\omega v$.

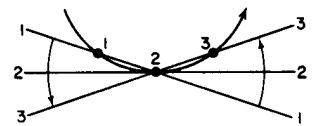


Fig. 19-4. Three successive views of a point moving radially on a rotating turntable.

Rotation in space

20-1 Torques in three dimensions

In this chapter we shall discuss one of the most remarkable and amusing consequences of mechanics, the behavior of a rotating wheel. In order to do this we must first extend the mathematical formulation of rotational motion, the principles of angular momentum, torque, and so on, to three-dimensional space. We shall not *use* these equations in all their generality and study all their consequences, because this would take many years, and we must soon turn to other subjects. In an introductory course we can present only the fundamental laws and apply them to a very few situations of special interest.

First, we notice that if we have a rotation in three dimensions, whether of a rigid body or any other system, what we deduced for two dimensions is still right. That is, it is still true that $xF_y - yF_x$ is the torque “in the xy -plane,” or the torque “around the z -axis.” It also turns out that this torque is still equal to the rate of change of $xp_y - yp_x$, for if we go back over the derivation of Eq. (18.15) from Newton’s laws we see that we did not have to assume that the motion was in a plane; when we differentiate $xp_y - yp_x$, we get $xF_y - yF_x$, so this theorem is still right. The quantity $xp_y - yp_x$, then, we call the angular momentum belonging to the xy -plane, or the angular momentum about the z -axis. This being true, we can use any other pair of axes and get another equation. For instance, we can use the yz -plane, and it is clear from symmetry that if we just substitute y for x and z for y , we would find $yF_z - zF_y$ for the torque and $yp_z - zp_y$ would be the angular momentum associated with the yz -plane. Of course we could have another plane, the zx -plane, and for this we would find $zF_x - xF_z = d/dt (zp_x - xp_z)$.

That these three equations can be deduced for the motion of a single particle is quite clear. Furthermore, if we added such things as $xp_y - yp_x$ together for many particles and called it the total angular momentum, we would have three kinds for the three planes xy , yz , and zx , and if we did the same with the forces, we would talk about the torque in the planes xy , yz , and zx also. Thus we would have laws that the external torque associated with any plane is equal to the rate of change of the angular momentum associated with that plane. This is just a generalization of what we wrote in two dimensions.

But now one may say, “Ah, but there are more planes; after all, can we not take some other plane at some angle, and calculate the torque on that plane from the forces? Since we would have to write another set of equations for every such plane, we would have a lot of equations!” Interestingly enough, it turns out that if we were to work out the combination $x'F_{y'} - y'F_{x'}$ for another plane, measuring the x' , $F_{y'}$, etc., in that plane, the result can be written as some *combination* of the three expressions for the xy -, yz - and zx -planes. There is nothing new. In other words, if we know what the three torques in the xy -, yz -, and zx -planes are, then the torque in any other plane, and correspondingly the angular momentum also, can be written as some combination of these: six percent of one and ninety-two percent of another, and so on. This property we shall now analyze.

Suppose that in the xyz -axes, Joe has worked out all his torques and his angular momenta in his planes. But Moe has axes x' , y' , z' in some other direction. To make it a little easier, we shall suppose that only the x - and y -axes have been turned. Moe’s x' and y' are new, but his z' happens to be the same. That is, he has new planes, let us say, for yz and zx . He therefore has new torques and angular momenta which he would work out. For example, his torque in the $x'y'$ -plane would be equal to $x'F_{y'} - y'F_{x'}$ and so forth. What we must now do is to find the relationship between the new torques and the old torques, so we will be able to make a

20-1 Torques in three dimensions

20-2 The rotation equations using cross products

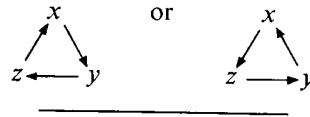
20-3 The gyroscope

20-4 Angular momentum of a solid body

connection from one set of axes to the other. Someone may say, "That looks just like what we did with vectors." And indeed, that is exactly what we are intending to do. Then he may say, "Well, isn't torque just a vector?" It *does* turn out to be a vector, but we do not know that right away without making an analysis. So in the following steps we shall make the analysis. We shall not discuss every step in detail, since we only want to illustrate how it works. The torques calculated by Joe are

$$\begin{aligned}\tau_{xy} &= xF_y - yF_x, \\ \tau_{yz} &= yF_z - zF_y, \\ \tau_{zx} &= zF_x - xF_z.\end{aligned}\tag{20.1}$$

We digress at this point to note that in such cases as this one may get the wrong sign for some quantity if the coordinates are not handled in the right way. Why not write $\tau_{yz} = zF_y - yF_z$? The problem arises from the fact that a coordinate system may be either "right-handed" or "left-handed." Having chosen (arbitrarily) a sign for, say τ_{xy} , then the correct expressions for the other two quantities may always be found by interchanging the letters xyz in either order



Moe now calculates the torques in his system:

$$\begin{aligned}\tau_{x'y'} &= x'F_{y'} - y'F_{x'}, \\ \tau_{y'z'} &= y'F_{z'} - z'F_{y'}, \\ \tau_{z'x'} &= z'F_{x'} - x'F_{z'}.\end{aligned}\tag{20.2}$$

Now we suppose that one coordinate system is rotated by a fixed angle θ , such that the z - and z' -axes are the same. (This angle θ has nothing to do with rotating objects or what is going on inside the coordinate system. It is merely the relationship between the axes used by one man and the axes used by the other, and is supposedly constant.) Thus the coordinates of the two systems are related by

$$\begin{aligned}x' &= x \cos \theta + y \sin \theta, \\ y' &= y \cos \theta - x \sin \theta, \\ z' &= z.\end{aligned}\tag{20.3}$$

Likewise, because force is a vector it transforms into the new system in the same way as do x , y , and z , since a thing is a vector if and only if the various components transform in the same way as x , y , and z :

$$\begin{aligned}F_{x'} &= F_x \cos \theta + F_y \sin \theta, \\ F_{y'} &= F_y \cos \theta - F_x \sin \theta, \\ F_{z'} &= F_z.\end{aligned}\tag{20.4}$$

Now we can find out how the torque transforms by merely substituting for x' , y' , and z' the expressions (20.3), and for $F_{x'}$, $F_{y'}$, $F_{z'}$ those given by (20.4), all into (20.2). So, we have a rather long string of terms for $\tau_{x'y'}$ and (rather surprisingly at first) it turns out that it comes right down to $xF_y - yF_x$, which we recognize to be the torque in the xy -plane:

$$\begin{aligned}\tau_{x'y'} &= (x \cos \theta + y \sin \theta)(F_y \cos \theta - F_x \sin \theta) \\ &\quad - (y \cos \theta - x \sin \theta)(F_x \cos \theta + F_y \sin \theta) \\ &= xF_y(\cos^2 \theta + \sin^2 \theta) - yF_x(\sin^2 \theta + \cos^2 \theta) \\ &\quad + xF_x(-\sin \theta \cos \theta + \sin \theta \cos \theta) \\ &\quad + yF_y(\sin \theta \cos \theta - \sin \theta \cos \theta) \\ &= xF_y - yF_x = \tau_{xy}.\end{aligned}\tag{20.5}$$

That result is clear, for if we only turn our axes *in the plane*, the twist around z in that plane is no different than it was before, because it is the same plane! What will be more interesting is the expression for $\tau_{y'z'}$, because that is a new plane. We now do exactly the same thing with the $y'z'$ -plane, and it comes out as follows:

$$\begin{aligned}\tau_{y'z'} &= (y \cos \theta - x \sin \theta)F_z \\ &\quad - z(F_y \cos \theta - F_x \sin \theta) \\ &= (yF_z - zF_y) \cos \theta + (zF_x - xF_z) \sin \theta \\ &= \tau_{yz} \cos \theta + \tau_{zx} \sin \theta.\end{aligned}\tag{20.6}$$

Finally, we do it for $z'x'$:

$$\begin{aligned}\tau_{z'x'} &= z(F_x \cos \theta + F_y \sin \theta) \\ &\quad - (x \cos \theta + y \sin \theta)F_z \\ &= (zF_x - xF_z) \cos \theta - (yF_z - zF_y) \sin \theta \\ &= \tau_{zx} \cos \theta - \tau_{yz} \sin \theta.\end{aligned}\tag{20.7}$$

We wanted to get a rule for finding torques in new axes in terms of torques in old axes, and now we have the rule. How can we ever remember that rule? If we look carefully at (20.5), (20.6), and (20.7), we see that there is a close relationship between these equations and the equations for x , y , and z . If, somehow, we could call τ_{xy} the z -component of something, let us call it the z -component of τ , then it would be all right; we would understand (20.5) as a vector transformation, since the z -component would be unchanged, as it should be. Likewise, if we associate with the yz -plane the x -component of our newly invented vector, and with the zx -plane, the y -component, then these transformation expressions would read

$$\begin{aligned}\tau_{z'} &= \tau_z, \\ \tau_{x'} &= \tau_x \cos \theta + \tau_y \sin \theta, \\ \tau_{y'} &= \tau_y \cos \theta - \tau_x \sin \theta,\end{aligned}\tag{20.8}$$

which is just the rule for vectors!

Therefore we have proved that we may identify the combination of $xF_y - yF_x$ with what we ordinarily call the z -component of a certain artificially invented vector. Although a torque is a twist on a plane, and it has no *a priori* vector character, mathematically it does behave like a vector. This vector is at right angles to the plane of the twist, and its length is proportional to the strength of the twist. The three components of such a quantity will transform like a real vector.

So we represent torques by vectors; with each plane on which the torque is supposed to be acting, we associate a line at right angles, by a rule. But "at right angles" leaves the sign unspecified. To get the sign right, we must adopt a rule which will tell us that if the torque were in a certain sense on the xy -plane, then the axis that we want to associate with it is in the "up" z -direction. That is, somebody has to define "right" and "left" for us. Supposing that the coordinate system is x , y , z in a right-hand system, then the rule will be the following: if we think of the twist as if we were turning a screw having a right-hand thread, then the direction of the vector that we will associate with that twist is in the direction that the screw would advance.

Why is torque a vector? It is a miracle of good luck that we can associate a single axis with a plane, and therefore that we can associate a vector with the torque; it is a special property of three-dimensional space. In two dimensions, the torque is an ordinary scalar, and there need be no direction associated with it. In three dimensions, it is a vector. If we had four dimensions, we would be in great difficulty, because (if we had time, for example, as the fourth dimension) we would not only have planes like xy , yz , and zx , we would also have tx -, ty -, and tz -planes. There would be *six* of them, and one cannot represent six quantities as one vector in four dimensions.

We will be living in three dimensions for a long time, so it is well to notice that the foregoing mathematical treatment did not depend upon the fact that x

was position and F was force; it only depended on the transformation laws for vectors. Therefore if, instead of x , we used the x -component of some other vector, it is not going to make any difference. In other words, if we were to calculate $a_x b_y - a_y b_x$, where $\mathbf{a}$ and $\mathbf{b}$ are vectors, and call it the z -component of some new quantity c , then these new quantities form a vector $\mathbf{c}$. We need a mathematical notation for the relationship of the new vector, with its three components, to the vectors $\mathbf{a}$ and $\mathbf{b}$. The notation that has been devised for this is $\mathbf{c} = \mathbf{a} \times \mathbf{b}$. We have then, in addition to the ordinary scalar product in the theory of vector analysis, a new kind of product, called the *vector product*. Thus, if $\mathbf{c} = \mathbf{a} \times \mathbf{b}$, this is the same as writing

$$\begin{aligned} c_x &= a_y b_z - a_z b_y, \\ c_y &= a_z b_x - a_x b_z, \\ c_z &= a_x b_y - a_y b_x. \end{aligned} \quad (20.9)$$

If we reverse the order of $\mathbf{a}$ and $\mathbf{b}$, calling $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{b}$, $\mathbf{a}$, we would have the sign of $\mathbf{c}$ reversed, because c_z would be $b_x a_y - b_y a_x$. Therefore the cross product is unlike ordinary multiplication, where $ab = ba$; for the cross product, $\mathbf{b} \times \mathbf{a} = -\mathbf{a} \times \mathbf{b}$. From this, we can prove at once that if $\mathbf{a} = \mathbf{b}$, the cross product is zero. Thus, $\mathbf{a} \times \mathbf{a} = 0$.

The cross product is very important for representing the features of rotation, and it is important that we understand the geometrical relationship of the three vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$. Of course the relationship in components is given in Eq. (20.9) and from that one can determine what the relationship is in geometry. The answer is, first, that the vector $\mathbf{c}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$. (Try to calculate $\mathbf{c} \cdot \mathbf{a}$, and see if it does not reduce to zero.) Second, the magnitude of $\mathbf{c}$ turns out to be the magnitude of $\mathbf{a}$ times the magnitude of $\mathbf{b}$ times the sine of the angle between the two. In which direction does $\mathbf{c}$ point? Imagine that we turn $\mathbf{a}$ into $\mathbf{b}$ through an angle less than 180° ; a screw with a right-hand thread turning in this way will advance in the direction of $\mathbf{c}$. The fact that we say a *right-hand* screw instead of a *left-hand* screw is a convention, and is a perpetual reminder that if $\mathbf{a}$ and $\mathbf{b}$ are “honest” vectors in the ordinary sense, the new kind of “vector” which we have created by $\mathbf{a} \times \mathbf{b}$ is artificial, or slightly different in its character from $\mathbf{a}$ and $\mathbf{b}$, because it was made up with a special rule. If $\mathbf{a}$ and $\mathbf{b}$ are called ordinary vectors, we have a special name for them, we call them *polar vectors*. Examples of such vectors are the coordinate $\mathbf{r}$, force $\mathbf{F}$, momentum $\mathbf{p}$, velocity $\mathbf{v}$, electric field $\mathbf{E}$, etc.; these are ordinary polar vectors. Vectors which involve just one cross product in their definition are called *axial vectors* or *pseudovectors*. Examples of pseudovectors are, of course, torque $\boldsymbol{\tau}$ and the angular momentum $\mathbf{L}$. It also turns out that the angular velocity $\boldsymbol{\omega}$ is a pseudovector, as is the magnetic field $\mathbf{B}$.

In order to complete the mathematical properties of vectors, we should know all the rules for their multiplication, using dot and cross products. In our applications at the moment, we will need very little of this, but for the sake of completeness we shall write down all of the rules for vector multiplication so that we can use the results later. These are

$$\begin{aligned} (a) \quad & \mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}, \\ (b) \quad & (\alpha \mathbf{a}) \times \mathbf{b} = \alpha(\mathbf{a} \times \mathbf{b}), \\ (c) \quad & \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}, \\ (d) \quad & \mathbf{a} \times (\mathbf{b} \times \mathbf{c}) = \mathbf{b}(\mathbf{a} \cdot \mathbf{c}) - \mathbf{c}(\mathbf{a} \cdot \mathbf{b}), \\ (e) \quad & \mathbf{a} \times \mathbf{a} = 0, \\ (f) \quad & \mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = 0. \end{aligned} \quad (20.10)$$

20-2 The rotation equations using cross products

Now let us ask whether any equations in physics can be written using the cross product. The answer, of course, is that a great many equations can be so written. For instance, we see immediately that the torque is equal to the position

vector cross the force:

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}. \quad (20.11)$$

This is a vector summary of the three equations $\tau_x = yF_z - zF_y$, etc. By the same token, the angular momentum vector, if there is only one particle present, is the distance from the origin multiplied by the vector momentum:

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}. \quad (20.12)$$

For three-dimensional space rotation, the dynamical law analogous to the law $\mathbf{F} = d\mathbf{p}/dt$ of Newton, is that the torque vector is the rate of change with time of the angular momentum vector:

$$\boldsymbol{\tau} = d\mathbf{L}/dt. \quad (20.13)$$

If we sum (20.13) over many particles, the external torque on a system is the rate of change of the total angular momentum:

$$\boldsymbol{\tau}_{\text{ext}} = d\mathbf{L}_{\text{tot}}/dt. \quad (20.14)$$

Another theorem: If the total external torque is zero, then the total vector angular momentum of the system is a constant. This is called the law of *conservation of angular momentum*. If there is no torque on a given system, its angular momentum cannot change.

What about angular velocity? Is it a vector? We have already discussed turning a solid object about a fixed axis, but for a moment suppose that we are turning it simultaneously about *two* axes. It might be turning about an axis inside a box, while the box is turning about some other axis. The net result of such combined motions is that the object simply turns about some new axis! The wonderful thing about this new axis is that it can be figured out this way. If the rate of turning in the xy -plane is written as a vector in the z -direction whose length is equal to the rate of rotation in the plane, and if another vector is drawn in the y -direction, say, which is the rate of rotation in the zx -plane, then if we add these together as a vector, the magnitude of the result tells us how fast the object is turning, and the direction tells us in what plane, by the rule of the parallelogram. That is to say, simply, angular velocity *is* a vector, where we draw the magnitudes of the rotations in the three planes as projections at right angles to those planes.*

As a simple application of the use of the angular velocity vector, we may evaluate the power being expended by the torque acting on a rigid body. The power, of course, is the rate of change of work with time; in three dimensions, the power turns out to be $P = \boldsymbol{\tau} \cdot \boldsymbol{\omega}$.

All the formulas that we wrote for plane rotation can be generalized to three dimensions. For example, if a rigid body is turning about a certain axis with angular velocity $\boldsymbol{\omega}$, we might ask, "What is the velocity of a point at a certain radial position $\mathbf{r}$?" We shall leave it as a problem for the student to show that the velocity of a particle in a rigid body is given by $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$, where $\boldsymbol{\omega}$ is the angular velocity and $\mathbf{r}$ is the position. Also, as another example of cross products, we had a formula for Coriolis force, which can also be written using cross products: $\mathbf{F}_c = 2m\mathbf{v} \times \boldsymbol{\omega}$. That is, if a particle is moving with velocity $\mathbf{v}$ in a coordinate system which is, in fact, rotating with angular velocity $\boldsymbol{\omega}$, and we want to think in terms of the rotating coordinate system, then we have to add the pseudoforce $\mathbf{F}_c$.

20-3 The gyroscope

Let us now return to the law of conservation of angular momentum. This law may be demonstrated with a rapidly spinning wheel, or gyroscope, as follows (see Fig. 20-1). If we sit on a swivel chair and hold the spinning wheel with its axis horizontal, the wheel has an angular momentum about the horizontal axis.

* That this is true can be derived by compounding the displacements of the particles of the body during an infinitesimal time Δt . It is not self-evident, and is left to those who are interested to try to figure it out.

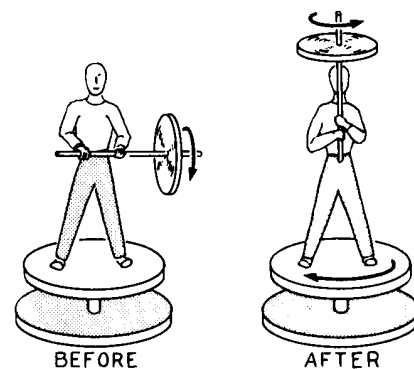


Fig. 20-1. *Before*: axis is horizontal; moment about vertical axis = 0. *After*: axis is vertical; momentum about vertical axis is still zero; man and chair spin in direction opposite to spin of the wheel.

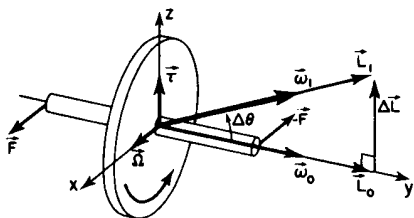


Fig. 20-2. A gyroscope.

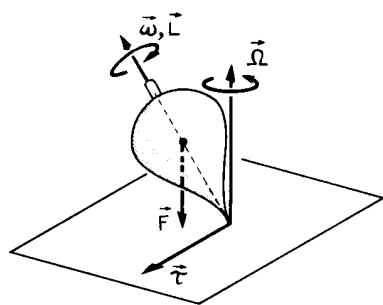


Fig. 20-3. A rapidly spinning top. Note that the direction of the torque vector is the direction of the precession.

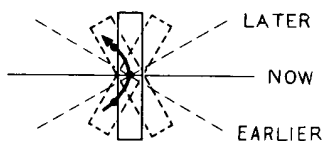


Fig. 20-4. The motion of particles in the spinning wheel of Fig. 20-2, whose axis is turning, is in curved lines.

Angular momentum around a *vertical* axis cannot change because of the (frictionless) pivot of the chair, so if we turn the axis of the wheel into the vertical, then the wheel would have angular momentum about the vertical axis, because it is now spinning about this axis. But the *system* (wheel, yourself, and chair) *cannot* have a vertical component, so we and the chair have to turn in the direction opposite to the spin of the wheel, to balance it.

First let us analyze in more detail the thing we have just described. What is surprising, and what we must understand, is the origin of the forces which turn us and the chair around as we turn the axis of the gyroscope toward the vertical. Figure 20-2 shows the wheel spinning rapidly about the *y*-axis. Therefore its angular velocity is about that axis and, it turns out, its angular momentum is likewise in that direction. Now suppose that we wish to rotate the wheel about the *x*-axis at a small angular velocity Ω ; what forces are required? After a short time Δt , the axis has turned to a new position, tilted at an angle $\Delta\theta$ with the horizontal. Since the major part of the angular momentum is due to the spin on the axis (very little is contributed by the slow turning), we see that the angular momentum vector has changed. What is the change in angular momentum? The angular momentum does not change in *magnitude*, but it does change in *direction* by an amount $\Delta\theta$. The magnitude of the vector ΔL is thus $\Delta L = L_0 \Delta\theta$, so that the torque, which is the time rate of change of the angular momentum, is $\tau = \Delta L / \Delta t = L_0 \Delta\theta / \Delta t = L_0 \Omega$. Taking the directions of the various quantities into account, we see that

$$\tau = \Omega \times L_0. \quad (20.15)$$

Thus, if Ω and L_0 are both horizontal, as shown in the figure, τ is *vertical*. To produce such a torque, horizontal forces F and $-F$ must be applied at the ends of the axle. How are these forces applied? By our hands, as we try to rotate the axis of the wheel into the vertical direction. But Newton's Third Law demands that equal and opposite forces (and equal and opposite *torques*) act on *us*. This causes us to rotate in the opposite sense about the vertical axis *z*.

This result can be generalized for a rapidly spinning top. In the familiar case of a spinning top, gravity acting on its center of mass furnishes a torque about the point of contact with the floor (see Fig. 20-3). This torque is in the horizontal direction, and causes the top to precess with its axis moving in a circular cone about the vertical. If Ω is the (vertical) angular velocity of precession, we again find that

$$\tau = d\mathbf{L}/dt = \Omega \times L_0.$$

Thus, when we apply a torque to a rapidly spinning top, the direction of the precessional motion is in the direction of the torque, or at right angles to the forces producing the torque.

We may now claim to understand the precession of gyroscopes, and indeed we do, mathematically. However, this is a mathematical thing which, in a sense, appears as a "miracle." It will turn out, as we go to more and more advanced physics, that many simple things can be deduced mathematically more rapidly than they can be really understood in a fundamental or simple sense. This is a strange characteristic, and as we get into more and more advanced work there are circumstances in which mathematics will produce results which *no one* has really been able to understand in any direct fashion. An example is the Dirac equation, which appears in a very simple and beautiful form, but whose consequences are hard to understand. In our particular case, the precession of a top looks like some kind of a miracle involving right angles and circles, and twists and right-hand screws. What we should try to do is to understand it in a more physical way.

How can we explain the torque in terms of the real forces and the accelerations? We note that when the wheel is precessing, the particles that are going around the wheel are not really moving in a plane because the wheel is precessing (see Fig. 20-4). As we explained previously (Fig. 19-4), the particles which are crossing through the precession axis are moving in *curved paths*, and this requires application of a lateral force. This is supplied by our pushing on the axle, which then com-

municates the force to the rim through the spokes. “Wait,” someone says, “what about the particles that are going back on the other side?” It does not take long to decide that there must be a force in the *opposite direction* on that side. The net force that we have to apply is therefore zero. The *forces* balance out, but one of them must be applied at one side of the wheel, and the other must be applied at the other side of the wheel. We could apply these forces directly, but because the wheel is solid we are allowed to do it by pushing on the axle, since forces can be carried up through the spokes.

What we have so far proved is that if the wheel is precessing, it can balance the torque due to gravity or some other applied torque. But all we have shown is that this is *a* solution of an equation. That is, if the torque is given, and *if we get the spinning started right*, then the wheel will precess smoothly and uniformly. But we have not proved (and it is not true) that a uniform precession is the *most general* motion a spinning body can undergo as the result of a given torque. The general motion involves also a “wobbling” about the mean precession. This “wobbling” is called *nutation*.

Some people like to say that when one exerts a torque on a gyroscope, it turns and it precesses, and that the torque *produces* the precession. It is very strange that when one suddenly lets go of a gyroscope, it does not *fall* under the action of gravity, but moves sidewise instead! Why is it that the *downward* force of the gravity, which we *know* and *feel*, makes it go *sidewise*? All the formulas in the world like (20.15) are not going to tell us, because (20.15) is a special equation, valid only after the gyroscope is precessing nicely. What really happens, in detail, is the following. If we were to hold the axis absolutely fixed, so that it cannot precess in any manner (but the top is spinning) then there is no torque acting, not even a torque from gravity, because it is balanced by our fingers. But if we suddenly let go, then there will instantaneously be a torque from gravity. Anyone in his right mind would think that the top would fall, and that is what it starts to do, as can be seen if the top is not spinning too fast.

The gyro actually does fall, as we would expect. But as soon as it falls, it is then turning, and if this turning were to continue, a torque would be required. In the absence of a torque in this direction, the gyro begins to “fall” in the direction opposite that of the missing force. This gives the gyro a component of motion around the vertical axis, as it would have in steady precession. But the actual motion “overshoots” the steady precessional velocity, and the axis actually rises again to the level from which it started. The path followed by the end of the axle is a cycloid (the path followed by a pebble that is stuck in the tread of an automobile tire). Ordinarily, this motion is too quick for the eye to follow, and it damps out quickly because of the friction in the gimbal bearings, leaving only the steady precessional drift (Fig. 20-5). The slower the wheel spins, the more obvious the nutation is.

When the motion settles down, the axis of the gyro is a little bit lower than it was at the start. Why? (These are the more complicated details, but we bring them in because we do not want the reader to get the idea that the gyroscope is an absolute miracle. It *is* a wonderful thing, but it is not a miracle.) If we were holding the axis absolutely horizontally, and suddenly let go, then the simple precession equation would tell us that it precesses, that it goes around in a horizontal plane. But that is impossible! Although we neglected it before, it is true that the wheel has *some* moment of inertia about the precession axis, and if it is moving about that axis, even slowly, it has a weak angular momentum about the axis. Where did it come from? If the pivots are perfect, there is no torque about the vertical axis. How then *does* it get to precess if there is no change in the angular momentum? The answer is that the cycloidal motion of the end of the axis damps down to the average, steady motion of the center of the equivalent rolling circle. That is, it settles down a little bit low. Because it is low, the spin angular momentum now has a small vertical component, which is exactly what is needed for the precession. So you see it has to go down a little, in order to go around. It has to yield a little bit to the gravity; by turning its axis down a little bit, it maintains the rotation about the vertical axis. That, then, is the way a gyroscope works.

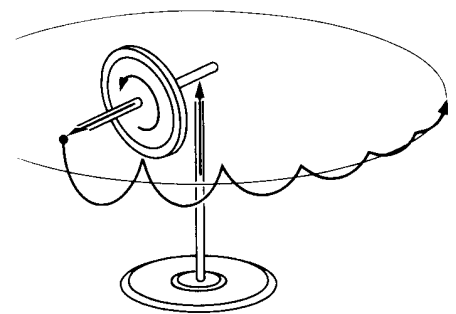


Fig. 20-5. Actual motion of tip of axis of gyroscope under gravity just after releasing axis previously held fixed.

20-4 Angular momentum of a solid body

Before we leave the subject of rotations in three dimensions, we shall discuss, at least qualitatively, a few effects that occur in three-dimensional rotations that are not self-evident. The main effect is that, in general, the angular momentum of a rigid body is *not necessarily* in the same direction as the angular velocity. Consider a wheel that is fastened onto a shaft in a lopsided fashion, but with the axis through the center of gravity, to be sure (Fig. 20-6). When we spin the wheel around the axis, anybody knows that there will be shaking at the bearings because of the lopsided way we have it mounted. Qualitatively, we know that in the rotating system there is centrifugal force acting on the wheel, trying to throw its mass as far as possible from the axis. This tends to line up the plane of the wheel so that it is perpendicular to the axis. To resist this tendency, a torque is exerted by the bearings. If there is a torque exerted by the bearings, there must be a rate of change of angular momentum. How can there be a rate of change of angular momentum when we are simply turning the wheel about the axis? Suppose we break the angular velocity ω into components ω_1 and ω_2 perpendicular and parallel to the plane of the wheel. What is the angular momentum? The moments of inertia about these two axes are *different*, so the angular momentum components, which (in these particular, special axes only) are equal to the moments of inertia times the corresponding angular velocity components, are in a *different ratio* than are the angular velocity components. Therefore the angular momentum vector is in a direction in space *not* along the axis. When we turn the object, we have to turn the angular momentum vector in space, so we must exert torques on the shaft.

Although it is much too complicated to prove here, there is a very important and interesting property of the moment of inertia which is easy to describe and to use, and which is the basis of our above analysis. This property is the following: Any rigid body, even an irregular one like a potato, possesses three mutually perpendicular axes through the CM, such that the moment of inertia about one of these axes has the greatest possible value for any axis through the CM, the moment of inertia about another of the axes has the *minimum* possible value, and the moment of inertia about the third axis is intermediate between these two (or equal to one of them). These axes are called the *principal axes* of the body, and they have the important property that if the body is rotating about one of them, its angular momentum is in the same *direction* as the angular velocity. For a body having axes of symmetry, the principal axes are along the symmetry axes.

If we take the x -, y -, and z -axes along the principal axes, and call the corresponding principal moments of inertia A , B , and C , we may easily evaluate the angular momentum and the kinetic energy of rotation of the body for any angular velocity ω . If we resolve ω into components ω_x , ω_y , and ω_z along the x -, y -, z -axes, and use unit vectors $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$, also along x , y , z , we may write the angular momentum as

$$\mathbf{L} = A\omega_x\mathbf{i} + B\omega_y\mathbf{j} + C\omega_z\mathbf{k}. \quad (20.16)$$

The kinetic energy of rotation is

$$\begin{aligned} \text{KE} &= \frac{1}{2}(A\omega_x^2 + B\omega_y^2 + C\omega_z^2) \\ &= \frac{1}{2}\mathbf{L} \cdot \boldsymbol{\omega}. \end{aligned} \quad (20.17)$$

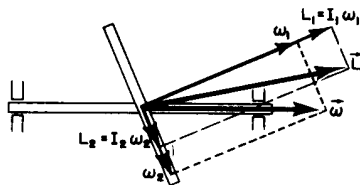


Fig. 20-6. The angular momentum of a rotating body is not necessarily parallel to the angular velocity.

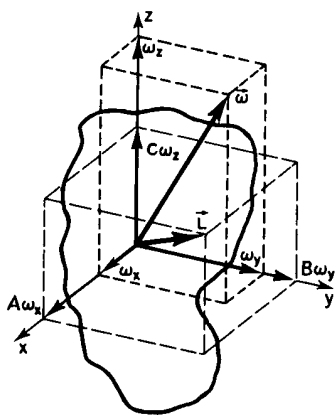


Fig. 20-7. The angular velocity and angular momentum of a rigid body ($A > B > C$).

The Harmonic Oscillator

21-1 Linear differential equations

In the study of physics, usually the course is divided into a series of subjects, such as mechanics, electricity, optics, etc., and one studies one subject after the other. For example, this course has so far dealt mostly with mechanics. But a strange thing occurs again and again: the equations which appear in different fields of physics, and even in other sciences, are often almost exactly the same, so that many phenomena have analogs in these different fields. To take the simplest example, the propagation of sound waves is in many ways analogous to the propagation of light waves. If we study acoustics in great detail we discover that much of the work is the same as it would be if we were studying optics in great detail. So the study of a phenomenon in one field may permit an extension of our knowledge in another field. It is best to realize from the first that such extensions are possible, for otherwise one might not understand the reason for spending a great deal of time and energy on what appears to be only a small part of mechanics.

The harmonic oscillator, which we are about to study, has close analogs in many other fields; although we start with a mechanical example of a weight on a spring, or a pendulum with a small swing, or certain other mechanical devices, we are really studying a certain *differential equation*. This equation appears again and again in physics and in other sciences, and in fact it is a part of so many phenomena that its close study is well worth our while. Some of the phenomena involving this equation are the oscillations of a mass on a spring; the oscillations of charge flowing back and forth in an electrical circuit; the vibrations of a tuning fork which is generating sound waves; the analogous vibrations of the electrons in an atom, which generate light waves; the equations for the operation of a servosystem, such as a thermostat trying to adjust a temperature; complicated interactions in chemical reactions; the growth of a colony of bacteria in interaction with the food supply and the poisons the bacteria produce; foxes eating rabbits eating grass, and so on; all these phenomena follow equations which are very similar to one another, and this is the reason why we study the mechanical oscillator in such detail. The equations are called *linear differential equations with constant coefficients*. A linear differential equation with constant coefficients is a differential equation consisting of a sum of several terms, each term being a derivative of the dependent variable with respect to the independent variable, and multiplied by some constant. Thus

$$a_n d^n x/dt^n + a_{n-1} d^{n-1} x/dt^{n-1} + \cdots + a_1 dx/dt + a_0 x = f(t) \quad (21.1)$$

is called a linear differential equation of order n with constant coefficients (each a_i is constant).

21-2 The harmonic oscillator

Perhaps the simplest mechanical system whose motion follows a linear differential equation with constant coefficients is a mass on a spring: first the spring stretches to balance the gravity; once it is balanced, we then discuss the vertical displacement of the mass from its equilibrium position (Fig. 21-1). We shall call this upward displacement x , and we shall also suppose that the spring is perfectly linear, in which case the force pulling back when the spring is stretched is precisely proportional to the amount of stretch. That is, the force is $-kx$ (with a

21-1 Linear differential equations

21-2 The harmonic oscillator

21-3 Harmonic motion and circular motion

21-4 Initial conditions

21-5 Forced oscillations

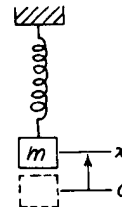


Fig. 21-1. A mass on a spring: a simple example of a harmonic oscillator.

minus sign to remind us that it pulls back). Thus the mass times the acceleration must equal $-kx$:

$$m d^2x/dt^2 = -kx. \quad (21.2)$$

For simplicity, suppose it happens (or we change our unit of time measurement) that the ratio $k/m = 1$. We shall first study the equation

$$d^2x/dt^2 = -x. \quad (21.3)$$

Later we shall come back to Eq. (21.2) with the k and m explicitly present.

We have already analyzed Eq. (21.3) in detail numerically; when we first introduced the subject of mechanics we solved this equation (see Eq. 9.12) to find the motion. By numerical integration we found a curve (Fig. 9-4) which showed that if m was initially displaced, but at rest, it would come down and go through zero; we did not then follow it any farther, but of course we know that it just keeps going up and down—it *oscillates*. When we calculated the motion numerically, we found that it went through the equilibrium point at $t = 1.570$. The length of the whole cycle is four times this long, or $t_0 = 6.28$ “sec.” This was found numerically, before we knew much calculus. We assume that in the meantime the Mathematics Department has brought forth a function which, when differentiated twice, is equal to itself with a minus sign. (There are, of course, ways of getting at this function in a direct fashion, but they are more complicated than already knowing what the answer is.) The function is $x = \cos t$. If we differentiate this we find $dx/dt = -\sin t$ and $d^2x/dt^2 = -\cos t = -x$. The function $x = \cos t$ starts, at $t = 0$, with $x = 1$, and no initial velocity; that was the situation with which we started when we did our numerical work. Now that we know that $x = \cos t$, we can calculate a *precise* value for the time at which it should pass $x = 0$. The answer is $t = \pi/2$, or 1.57108. We were wrong in the last figure because of the errors of numerical analysis, but it was very close!

Now to go further with the original problem, we restore the time units to real seconds. What is the solution then? First of all, we might think that we can get the constants k and m in by multiplying $\cos t$ by something. So let us try the equation $x = A \cos t$; then we find $dx/dt = -A \sin t$, and $d^2x/dt^2 = -A \cos t = -x$. Thus we discover to our horror that we did not succeed in solving Eq. (21.2), but we got Eq. (21.3) again! That fact illustrates one of the most important properties of linear differential equations: *if we multiply a solution of the equation by any constant, it is again a solution*. The mathematical reason for this is clear. If x is a solution, and we multiply both sides of the equation, say by A , we see that all derivatives are also multiplied by A , and therefore Ax is just as good a solution of the original equation as x was. The physics of it is the following. If we have a weight on a spring, and pull it down twice as far, the force is twice as much, the resulting acceleration is twice as great, the velocity it acquires in a given time is twice as great, the distance covered in a given time is twice as great; but it *has* to cover twice as great a distance in order to get back to the origin because it is pulled down twice as far. So it takes the *same time* to get back to the origin, irrespective of the initial displacement. In other words, with a linear equation, the motion has the same *time pattern*, no matter how “strong” it is.

That was the wrong thing to do—it only taught us that we can multiply the solution by anything, and it satisfies the same equation, but not a different equation. After a little cut and try to get to an equation with a different constant multiplying x , we find that we must alter the scale of *time*. In other words, Eq. (21.2) has a solution of the form

$$x = \cos \omega_0 t. \quad (21.4)$$

(It is important to realize that in the present case, ω_0 is not an angular velocity of a spinning body, but we run out of letters if we are not allowed to use the same letter for more than one thing.) The reason we put a subscript “0” on ω is that we are going to have more omegas before long; let us remember that ω_0 refers to the natural motion of this oscillator. Now we try Eq. (21.4) and this time we are more successful, because $dx/dt = -\omega_0 \sin \omega_0 t$ and $d^2x/dt^2 = -\omega_0^2 \cos \omega_0 t = -\omega_0^2 x$.

So at last we have solved the equation that we really wanted to solve. The equation $d^2x/dt^2 = -\omega_0^2x$ is the same as Eq. (21.2) if $\omega_0^2 = k/m$.

The next thing we must investigate is the physical significance of ω_0 . We know that the cosine function repeats itself when the angle it refers to is 2π . So $x = \cos \omega_0 t$ will repeat its motion, it will go through a complete cycle, when the "angle" changes by 2π . The quantity $\omega_0 t$ is often called the *phase* of the motion. In order to change $\omega_0 t$ by 2π , the time must change by an amount t_0 , called the *period* of one complete oscillation; of course t_0 must be such that $\omega_0 t_0 = 2\pi$. That is, $\omega_0 t_0$ must account for one cycle of the angle, and then everything will repeat itself—if we increase t by t_0 , we add 2π to the phase. Thus

$$t_0 = 2\pi/\omega_0 = 2\pi\sqrt{m/k}. \quad (21.5)$$

Thus if we had a heavier mass, it would take longer to oscillate back and forth on a spring. That is because it has more inertia, and so, while the forces are the same, it takes longer to get the mass moving. Or, if the spring is stronger, it will move more quickly, and that is right: the period is less if the spring is stronger.

Note that the period of oscillation of a mass on a spring does not depend in any way on *how* it has been started, how far down we pull it. The *period* is determined, but the amplitude of the oscillation is *not* determined by the equation of motion (21.2). The amplitude *is* determined, in fact, by how we let go of it, by what we call the *initial conditions* or starting conditions.

Actually, we have not quite found the most general possible solution of Eq. (21.2). There are other solutions. It should be clear why: because all of the cases covered by $x = a \cos \omega_0 t$ start with an initial displacement and no initial velocity. But it is possible, for instance, for the mass to start at $x = 0$, and we may then give it an impulsive kick, so that it has some speed at $t = 0$. Such a motion is not represented by a cosine—it is represented by a sine. To put it another way, if $x = \cos \omega_0 t$ is a solution, then is it not obvious that if we were to happen to walk into the room at some time (which we would call " $t = 0$ ") and saw the mass as it was passing $x = 0$, it would keep on going just the same? Therefore, $x = \cos \omega_0 t$ cannot be the most general solution; it must be possible to shift the beginning of time, so to speak. As an example, we could write the solution this way: $x = a \cos \omega_0(t - t_1)$, where t_1 is some constant. This also corresponds to shifting the origin of time to some new instant. Furthermore, we may expand

$$\cos(\omega_0 t + \Delta) = \cos \omega_0 t \cos \Delta - \sin \omega_0 t \sin \Delta,$$

and write

$$x = A \cos \omega_0 t + B \sin \omega_0 t,$$

where $A = a \cos \Delta$ and $B = -a \sin \Delta$. Any one of these forms is a possible way to write the complete, general solution of (21.2): that is, every solution of the differential equation $d^2x/dt^2 = -\omega_0^2x$ that exists in the world can be written as

$$(a) \quad x = a \cos \omega_0(t - t_1),$$

or

$$(b) \quad x = a \cos(\omega_0 t + \Delta), \quad (21.6)$$

or

$$(c) \quad x = A \cos \omega_0 t + B \sin \omega_0 t.$$

Some of the quantities in (21.6) have names: ω_0 is called the *angular frequency*; it is the number of radians by which the phase changes in a second. That is determined by the differential equation. The other constants are not determined by the equation, but by how the motion is started. Of these constants, a measures the maximum displacement attained by the mass, and is called the *amplitude* of oscillation. The constant Δ is sometimes called the *phase* of the oscillation, but that is a confusion, because other people call $\omega_0 t + \Delta$ the phase, and say the phase changes with time. We might say that Δ is a *phase shift* from some defined zero. Let us put it differently. Different Δ 's correspond to motions in different phases. That is true, but whether we want to call Δ *the* phase, or not, is another question.

21-3 Harmonic motion and circular motion

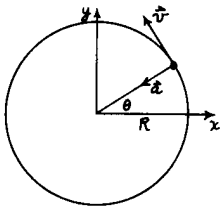


Fig. 21-2. A particle moving in a circular path at constant speed.

The fact that cosines are involved in the solution of Eq. (21.2) suggests that there might be some relationship to circles. This is artificial, of course, because there is no circle actually involved in the linear motion—it just goes up and down. We may point out that we have, in fact, already solved that differential equation when we were studying the mechanics of circular motion. If a particle moves in a circle with a constant speed v , the radius vector from the center of the circle to the particle turns through an angle whose size is proportional to the time. If we call this angle $\theta = vt/R$ (Fig. 21-2) then $d\theta/dt = \omega_0 = v/R$. We know that there is an acceleration $a = v^2/R = \omega_0^2 R$ toward the center. Now we also know that the position x , at a given moment, is the radius of the circle times $\cos \theta$, and that y is the radius times $\sin \theta$:

$$x = R \cos \theta, \quad y = R \sin \theta.$$

Now what about the acceleration? What is the x -component of acceleration, d^2x/dt^2 ? We have already worked that out geometrically; it is the magnitude of the acceleration times the cosine of the projection angle, with a minus sign because it is toward the center.

$$a_x = -a \cos \theta = -\omega^2 R \cos \theta = -\omega^2 x. \quad (21.7)$$

In other words, when a particle is moving in a circle, the horizontal component of its motion has an acceleration which is proportional to the horizontal displacement from the center. Of course we also have the solution for motion in a circle: $x = R \cos \omega_0 t$. Equation (21.7) does not depend upon the radius of the circle, so for a circle of any radius, one finds the same equation for a given ω_0 . Thus, for several reasons, we expect that the displacement of a mass on a spring will turn out to be proportional to $\cos \omega_0 t$, and will, in fact, be exactly the same motion as we would see if we looked at the x -component of the position of an object rotating in a circle with angular velocity ω_0 . As a check on this, one can devise an experiment to show that the up-and-down motion of a mass on a spring is the same as that of a point going around in a circle. In Fig. 21-3 an arc light projected on a screen casts shadows of a crank pin on a shaft and of a vertically oscillating mass, side by side. If we let go of the mass at the right time from the right place, and if the shaft speed is carefully adjusted so that the frequencies match, each should follow the other exactly. One can also check the numerical solution we obtained earlier with the cosine function, and see whether that agrees very well.

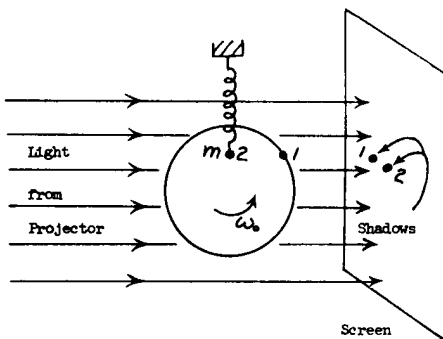


Fig. 21-3. Demonstration of the equivalence between simple harmonic motion and uniform circular motion.

Here we may point out that because uniform motion in a circle is so closely related mathematically to oscillatory up-and-down motion, we can analyze oscillatory motion in a simpler way if we imagine it to be a projection of something going in a circle. In other words, although the distance y means nothing in the oscillator problem, we may still artificially supplement Eq. (21.2) with another equation using y , and put the two together. If we do this, we will be able to analyze our one-dimensional oscillator with *circular* motions, which is a lot easier than having to solve a differential equation. The trick in doing this is to use complex numbers, a procedure we shall introduce in the next chapter.

21-4 Initial conditions

Now let us consider what determines the constants A and B , or a and Δ . Of course these are determined by how we start the motion. If we start the motion with just a small displacement, that is one type of oscillation; if we start with an initial displacement and then push up when we let go, we get still a different motion. The constants A and B , or a and Δ , or any other way of putting it, are determined, of course, by the way the motion started, not by any other features of the situation. These are called the *initial conditions*. We would like to connect the initial conditions with the constants. Although this can be done using any one of the forms (21.6), it turns out to be easiest if we use Eq. (21.6c). Suppose that at $t = 0$ we have started with an initial displacement x_0 and a certain velocity v_0 .

This is the most general way we can start the motion. (We cannot specify the *acceleration* with which it started, true, because that is determined by the spring, once we specify x_0 .) Now let us calculate A and B . We start with the equation for x ,

$$x = A \cos \omega_0 t + B \sin \omega_0 t.$$

Since we shall later need the velocity also, we differentiate x and obtain

$$v = -\omega_0 A \sin \omega_0 t + \omega_0 B \cos \omega_0 t.$$

These expressions are valid for all t , but we have special knowledge about x and v at $t = 0$. So if we put $t = 0$ into these equations, on the left we get x_0 and v_0 , because that is what x and v are at $t = 0$; also, we know that the cosine of zero is unity, and the sine of zero is zero. Therefore we get

$$x_0 = A \cdot 1 + B \cdot 0 = A$$

and

$$v_0 = -\omega_0 A \cdot 0 + \omega_0 B \cdot 1 = \omega_0 B.$$

So for this particular case we find that

$$A = x_0, \quad B = v_0/\omega_0.$$

From these values of A and B , we can get a and Δ if we wish.

That is the end of our solution, but there is one physically interesting thing to check, and that is the conservation of energy. Since there are no frictional losses, energy ought to be conserved. Let us use the formula

$$x = a \cos (\omega_0 t + \Delta);$$

then

$$v = -\omega_0 a \sin (\omega_0 t + \Delta).$$

Now let us find out what the kinetic energy T is, and what the potential energy U is. The potential energy at any moment is $\frac{1}{2}kx^2$, where x is the displacement and k is the constant of the spring. If we substitute for x , using our expression above, we get

$$U = \frac{1}{2}kx^2 = \frac{1}{2}ka^2 \cos^2 (\omega_0 t + \Delta).$$

Of course the potential energy is not constant; the potential never becomes negative, naturally—there is always some energy in the spring, but the amount of energy fluctuates with x . The kinetic energy, on the other hand, is $\frac{1}{2}mv^2$, and by substituting for v we get

$$T = \frac{1}{2}mv^2 = \frac{1}{2}m\omega_0^2 a^2 \sin^2 (\omega_0 t + \Delta).$$

Now the kinetic energy is zero when x is at the maximum, because then there is no velocity; on the other hand, it is maximal when x is passing through zero, because then it is moving fastest. This variation of the kinetic energy is just the opposite of that of the potential energy. But the total energy ought to be a constant. If we note that $k = m\omega_0^2$, we see that

$$T + U = \frac{1}{2}m\omega_0^2 a^2 [\cos^2 (\omega_0 t + \Delta) + \sin^2 (\omega_0 t + \Delta)] = \frac{1}{2}m\omega_0^2 a^2.$$

The energy is dependent on the square of the amplitude; if we have twice the amplitude, we get an oscillation which has four times the energy. The *average* potential energy is half the maximum and, therefore, half the total, and the average kinetic energy is likewise half the total energy.

21-5 Forced oscillations

Next we shall discuss the *forced harmonic oscillator*, i.e., one in which there is an external driving force acting. The equation then is the following:

$$m d^2x/dt^2 = -kx + F(t). \quad (21.8)$$

We would like to find out what happens in these circumstances. The external driving force can have various kinds of functional dependence on the time; the first one that we shall analyze is very simple—we shall suppose that the force is oscillating:

$$F(t) = F_0 \cos \omega t. \quad (21.9)$$

Notice, however, that this ω is not necessarily ω_0 : we have ω under our control; the forcing may be done at different frequencies. So we try to solve Eq. (21.8) with the special force (21.9). What is the solution of (21.8)? One special solution, (we shall discuss the more general cases later) is

$$x = C \cos \omega t, \quad (21.10)$$

where the constant is to be determined. In other words, we might suppose that if we kept pushing back and forth, the mass would follow back and forth in step with the force. We can try it anyway. So we put (21.10) into (21.9), and get

$$-m\omega^2 C \cos \omega t = -m\omega_0^2 C \cos \omega t + F_0 \cos \omega t. \quad (21.11)$$

We have also put in $k = m\omega_0^2$, so that we will understand the equation better at the end. Now because the cosine appears everywhere, we can divide it out, and that shows that (21.10) is, in fact, a solution, provided we pick C just right. The answer is that C must be

$$C = F_0 / m(\omega_0^2 - \omega^2). \quad (21.12)$$

That is, m oscillates at the same frequency as the force, but with an amplitude which depends on the frequency of the force, and also upon the frequency of the natural motion of the oscillator. It means, first, that if ω is very small compared with ω_0 , then the displacement and the force are in the same direction. On the other hand, if we shake it back and forth very fast, then (21.12) tells us that C is negative if ω is above the natural frequency ω_0 of the harmonic oscillator. (We will call ω_0 the natural frequency of the harmonic oscillator, and ω the applied frequency.) At very high frequency the denominator may become very large, and there is then not much amplitude.

Of course the solution we have found is the solution only if things are started just right, for otherwise there is a part which usually dies out after a while. This other part is called the *transient* response to $F(t)$, while (21.10) and (21.12) are called the *steady-state* response.

According to our formula (21.12), a very remarkable thing should also occur: if ω is almost exactly the same as ω_0 , then C should approach infinity. So if we adjust the frequency of the force to be “in time” with the natural frequency, then we should get an enormous displacement. This is well known to anybody who has pushed a child on a swing. It does not work very well to close our eyes and push at a certain speed at random. If we happen to get the right timing, then the swing goes very high, but if we have the wrong timing, then sometimes we may be pushing when we should be pulling, and so on, and it does not work.

If we make ω exactly equal to ω_0 , we find that it should oscillate at an *infinite* amplitude, which is, of course, impossible. The reason it does not is that something goes wrong with the equation, there are some other frictional terms, and other forces, which are not in (21.8) but which occur in the real world. So the amplitude does not reach infinity for some reason; it may be that the spring breaks!

Algebra

22-1 Addition and multiplication

In our study of oscillating systems we shall have occasion to use one of the most remarkable, almost astounding, formulas in all of mathematics. From the physicist's point of view we could bring forth this formula in two minutes or so, and be done with it. But science is as much for intellectual enjoyment as for practical utility, so instead of just spending a few minutes on this amazing jewel, we shall surround the jewel by its proper setting in the grand design of that branch of mathematics which is called elementary algebra.

Now you may ask, "What is mathematics doing in a physics lecture?" We have several possible excuses: first, of course, mathematics is an important tool, but that would only excuse us for giving the formula in two minutes. On the other hand, in theoretical physics we discover that all our laws can be written in mathematical form; and that this has a certain simplicity and beauty about it. So, ultimately, in order to understand nature it may be necessary to have a deeper understanding of mathematical relationships. But the real reason is that the subject is enjoyable, and although we humans cut nature up in different ways, and we have different courses in different departments, such compartmentalization is really artificial, and we should take our intellectual pleasures where we find them.

Another reason for looking more carefully at algebra now, even though most of us studied algebra in high school, is that that was the first time we studied it; all the equations were unfamiliar, and it was hard work, just as physics is now. Every so often it is a great pleasure to look back to see what territory has been covered, and what the great map or plan of the whole thing is. Perhaps some day somebody in the Mathematics Department will present a lecture on mechanics in such a way as to show what it was we were trying to learn in the physics course!

The subject of algebra will not be developed from the point of view of a mathematician, exactly, because the mathematicians are mainly interested in how various mathematical facts are demonstrated, and how many assumptions are absolutely required, and what is not required. They are not so interested in the result of what they prove. For example, we may find the Pythagorean theorem quite interesting, that the sum of the squares of the sides of a right triangle is equal to the square of the hypotenuse; that is an interesting fact, a curiously simple thing, which may be appreciated without discussing the question of how to prove it, or what axioms are required. So, in the same spirit, we shall describe qualitatively, if we may put it that way, the system of elementary algebra. We say *elementary* algebra because there is a branch of mathematics called *modern* algebra in which some of the rules such as $ab = ba$, are abandoned, and it is still called algebra, but we shall not discuss that.

To discuss this subject we start in the middle. We suppose that we already know what integers are, what zero is, and what it means to increase a number by one unit. You may say, "That is not in the middle!" But it is the middle from a mathematical standpoint, because we could go even further back and describe the theory of sets in order to *derive* some of these properties of integers. But we are not going in that direction, the direction of mathematical philosophy and mathematical logic, but rather in the other direction, from the assumption that we know what integers are and we know how to count.

If we start with a certain number a , an integer, and we count successively one unit b times, the number we arrive at we call $a + b$, and that defines *addition* of integers.

22-1 Addition and multiplication

22-2 The inverse operations

22-3 Abstraction and generalization

22-4 Approximating irrational numbers

22-5 Complex numbers

22-6 Imaginary exponents

Once we have defined addition, then we can consider this: if we start with nothing and add a to it, b times in succession, we call the result *multiplication of integers*; we call it b times a .

Now we can also have a *succession of multiplications*: if we start with 1 and multiply by a , b times in succession, we call that *raising to a power*: a^b .

Now as a consequence of these definitions it can be easily shown that all of the following relationships are true:

$$\begin{array}{ll}
 \text{(a)} & a + b = b + a \\
 \text{(b)} & a + (b + c) = (a + b) + c \\
 \text{(c)} & ab = ba \\
 \text{(d)} & a(b + c) = ab + ac \\
 \text{(e)} & (ab)c = a(bc) \\
 \text{(f)} & (ab)^c = a^c b^c \\
 \text{(g)} & a^b a^c = a^{(b+c)} \\
 \text{(h)} & (a^b)^c = a^{(bc)} \\
 \text{(i)} & a + 0 = a \\
 \text{(j)} & a \cdot 1 = a \\
 \text{(k)} & a^1 = a
 \end{array} \tag{22.1}$$

These results are well known and we shall not belabor the point, we merely list them. Of course, 1 and 0 have special properties; for example, $a + 0$ is a , a times 1 = a , and a to the first power is a .

In this discussion we must also assume a few other properties like continuity and ordering, which are very hard to define; we will let the rigorous theory do it. Furthermore, it is definitely true that we have written down too many “rules”; some of them may be deducible from the others, but we shall not worry about such matters.

22-2 The inverse operations

In addition to the direct operations of addition, multiplication, and raising to a power, we have also the *inverse* operations, which are defined as follows. Let us assume that a and c are given, and that we wish to find what values of b satisfy such equations as $a + b = c$, $ab = c$, $b^a = c$. If $a + b = c$, b is defined as $c - a$, which is called *subtraction*. The operation called division is also clear: if $ab = c$, then $b = c/a$ defines division—a solution of the equation $ab = c$ “backwards.” Now if we have a power $b^a = c$ and we ask ourselves, “What is b ?” it is called the a th *root* of c : $b = \sqrt[a]{c}$. For instance, if we ask ourselves the following question, “What integer, raised to the third power, equals 8?” then the answer is called the *cube root* of 8; it is 2. Because b^a and a^b are not equal, there are *two* inverse problems associated with powers, and the other inverse problem would be, “To what power must we raise 2 to get 8?” This is called taking the *logarithm*. If $a^b = c$, we write $b = \log_a c$. The fact that it has a cumbersome notation relative to the others does not mean that it is any less elementary, at least applied to integers, than the other processes. Although logarithms come late in an algebra class, in practice they are, of course, just as simple as roots; they are just a different kind of solution of an algebraic equation. The direct and inverse operations are summarized as follows:

$$\begin{array}{ll}
 \text{(a)} & \text{addition} \\
 & a + b = c \\
 \text{(b)} & \text{multiplication} \\
 & ab = c \\
 \text{(c)} & \text{power} \\
 & b^a = c \\
 \text{(d)} & \text{power} \\
 & a^b = c \\
 \text{(a')} & \text{subtraction} \\
 & b = c - a \\
 \text{(b')} & \text{division} \\
 & b = c/a \\
 \text{(c')} & \text{root} \\
 & b = \sqrt[a]{c} \\
 \text{(d')} & \text{logarithm} \\
 & b = \log_a c
 \end{array} \tag{22.2}$$

Now here is the idea. These relationships, or rules, are correct for integers, since they follow from the definitions of addition, multiplication, and raising to a power. *We are going to discuss whether or not we can broaden the class of objects*

which a , b , and c represent so that they will obey these same rules, although the processes for $a + b$, and so on, will not be definable in terms of the direct action of adding 1, for instance, or successive multiplications by integers.

22-3 Abstraction and generalization

When we try to solve simple algebraic equations using all these definitions, we soon discover some insoluble problems, such as the following. Suppose that we try to solve the equation $b = 3 - 5$. That means, according to our definition of subtraction, that we must find a number which, when added to 5, gives 3. And of course there *is* no such number, because we consider only positive integers; this is an insoluble problem. However, the plan, the great idea, is this: *abstraction and generalization*. From the whole structure of algebra, rules plus integers, we abstract the original definitions of addition and multiplication, but we leave the rules (22.1) and (22.2), and assume these to be true *in general* on a wider class of numbers, even though they are originally derived on a smaller class. Thus, rather than using integers symbolically to define the rules, we use the rules as the definition of the symbols, which then represent a more general kind of number. As an example, by working with the rules alone we can show that $3 - 5 = 0 - 2$. In fact we can show that one can make *all* subtractions, provided we define a whole set of new numbers: $0 - 1$, $0 - 2$, $0 - 3$, $0 - 4$, and so on, called the *negative integers*. Then we may use all the other rules, like $a(b + c) = ab + ac$ and so forth, to find what the rules are for multiplying negative numbers, and we will discover, in fact, that all of the rules can be maintained with negative as well as positive integers.

So we have increased the range of objects over which the rules work, but the meaning of the symbols is different.

One cannot say, for instance, that -2 times 5 really means to add 5 together successively -2 times. That means nothing. But nevertheless everything will work out all right according to the rules.

An interesting problem comes up in taking powers. Suppose that we wish to discover what $a^{(3-5)}$ means. We know only that $3 - 5$ is a solution of the problem, $(3 - 5) + 5 = 3$. Knowing that, we know that $a^{(3-5)}a^5 = a^3$. Therefore $a^{(3-5)} = a^3/a^5$, by the definition of division. With a little more work, this can be reduced to $1/a^2$. So we find that the negative powers are the reciprocals of the positive powers, but $1/a^2$ is a meaningless symbol, because if a is a positive or negative integer, the square of it is greater than 1, and we do not yet know what we mean by 1 divided by a number greater than 1!

Onward! The great plan is to continue the process of generalization; whenever we find another problem that we cannot solve we extend our realm of numbers. Consider division: we cannot find a number which is an integer, even a negative integer, which is equal to the result of dividing 3 by 5. But if we suppose that all fractional numbers also satisfy the rules, then we can talk about multiplying and adding fractions, and everything works as well as it did before.

Take another example of powers: what is $a^{3/5}$? We know only that $(3/5)5 = 3$, since that was the definition of $3/5$. So we know also that $(a^{(3/5)})^5 = a^{(3/5)(5)} = a^3$, because this is one of the rules. Then by the definition of roots we find that $a^{(3/5)} = \sqrt[5]{a^3}$.

In this way, then, we can define what we mean by putting fractions in the various symbols, by using the rules themselves to help us determine the definition—it is not arbitrary. It is a remarkable fact that all the rules still work for positive and negative integers, as well as for fractions!

We go on in the process of generalization. Are there any other equations we cannot solve? Yes, there are. For example, it is impossible to solve this equation: $b = 2^{1/2} = \sqrt{2}$. It is impossible to find a number which is rational (a fraction) whose square is equal to 2. It is very easy for us in modern days to answer this question. We know the decimal system, and so we have no difficulty in appreciating the meaning of an unending decimal as a type of approximation to the square root of 2. Historically, this idea presented great difficulty to the Greeks. To really

define *precisely* what is meant here requires that we add some substance of continuity and ordering, and it is, in fact, quite the most difficult step in the processes of generalization just at this point. It was made, formally and rigorously, by Dedekind. However, without worrying about the mathematical rigor of the thing, it is quite easy to understand that what we mean is that we are going to find a whole sequence of approximate fractions, perfect fractions (because any decimal, when stopped somewhere, is of course rational), which just keeps on going, getting closer and closer to the desired result. That is good enough for what we wish to discuss, and it permits us to involve ourselves in irrational numbers, and to calculate things like the square root of 2 to any accuracy that we desire, with enough work.

22-4 Approximating irrational numbers

The next problem comes with what happens with the irrational powers. Suppose that we want to define, for instance, $10^{\sqrt{2}}$. In principle, the answer is simple enough. If we approximate the square root of 2 to a certain number of decimal places, then the power is rational, and we can take the approximate root, using the above method, and get an *approximation* to $10^{\sqrt{2}}$. Then we may run it up a few more decimal places (it is again rational), take the appropriate root, this time a much higher root because there is a much bigger denominator in the fraction, and get a better approximation. Of course we are going to get some enormously high roots involved here, and the work is quite difficult. How can we cope with this problem?

In the computations of square roots, cube roots, and other small roots, there is an arithmetical process available by which we can get one decimal place after another. But the amount of labor needed to calculate irrational powers and the logarithms that go with them (the inverse problem) is so great that there is no simple arithmetical process we can use. Therefore tables have been built up which permit us to calculate these powers, and these are called the tables of logarithms, or the tables of powers, depending on which way the table is set up. It is merely a question of saving time; if we must raise some number to an irrational power, we can look it up rather than having to compute it. Of course, such a computation is just a technical problem, but it is an interesting one, and of great historical value. In the first place, not only do we have the problem of solving $x = 10^{\sqrt{2}}$, but we also have the problem of solving $10^x = 2$, or $x = \log_{10} 2$. This is not a problem where we have to define a new kind of number for the result, it is merely a computational problem. The answer is simply an irrational number, an unending decimal, not a new kind of a number.

Let us now discuss the problem of calculating solutions of such equations. The general idea is really very simple. If we could calculate 10^1 , and $10^{4/10}$, and $10^{1/100}$, and $10^{4/1000}$ and so on, and multiply them all together, we would get $10^{1.414\dots}$ or $10^{\sqrt{2}}$, and that is the general idea on which things work. But instead of calculating $10^{1/10}$ and so on, we shall calculate $10^{1/2}$, $10^{1/4}$, and so on. Before we start, we should explain why we make so much work with 10, instead of some other number. Of course, we realize that logarithm tables are of great practical utility, quite aside from the mathematical problem of taking roots, since with any base at all,

$$\log_b (ac) = \log_b a + \log_b c. \quad (22.3)$$

We are all familiar with the fact that one can use this fact in a practical way to multiply numbers if we have a table of logarithms. The only question is, with what base b shall we compute? It makes no difference what base is used; we can use the same principle all the time, and if we are using logarithms to any particular base, we can find logarithms to any other base merely by a change in scale, a multiplying factor. If we multiply Eq. (22.3) by 61, it is just as true, and if we had a table of logs with a base b , and somebody else multiplied all of our table by 61, there would be no essential difference. Suppose that we know the logarithms of all the numbers to the base b . In other words, we can solve the equation $b^a = c$ for any c because

we have a table. The problem is to find the logarithm of the same number c to some other base, let us say the base x . We would like to solve $x^{a'} = c$. It is easy to do, because we can always write $x = b^t$, which defines t , knowing x and b . As a matter of fact, $t = \log_b x$. Then if we put that in and solve for a' , we see that $(b^t)^{a'} = b^{ta'} = c$. In other words, ta' is the logarithm of c in base b . Thus $a' = a/t$. Thus logs to base x are just $1/t$, which is a constant, times the logs to the base, b . Therefore any log table is equivalent to any other log table if we multiply by a constant, and the constant is $1/\log_b x$. This permits us to choose a particular base, and for convenience we take the base 10. (The question may arise as to whether there is any natural base, any base in which things are somehow simpler, and we shall try to find an answer to that later. At the moment we shall just use the base 10.)

Table 22-1
Successive Square Roots of Ten

Power s	$1024\ s$	10^s	$(10^s - 1)/s$
1	1024	10.00000	9.00
1/2	512	3.16228	4.32
1/4	256	1.77828	3.113
1/8	128	1.33352	2.668
1/16	64	1.15478	2.476
1/32	32	1.074607	2.3874
1/64	16	1.036633	2.3445
1/128	8	1.018152	2.3234 ²¹¹
1/256	4	1.0090350	2.3130 ¹⁰⁴
1/512	2	1.0045073	2.3077 ⁵³
1/1024	1	1.0022511	2.3051 ²⁶
			↓ 26
$\Delta/1024$ ($\Delta \rightarrow 0$)	Δ	$1 + .0022486\Delta$	2.3025

Now let us see how to calculate logarithms. We begin by computing successive square roots of 10, by cut and try. The results are shown in Table 22-1. The powers of 10 are given in the first column, and the result, 10^s , is given in the third column. Thus $10^1 = 10$. The one-half power of 10 we can easily work out, because that is the square root of 10, and there is a known, simple process for taking square roots of any number.* Using this process, we find the first square root to be 3.16228. What good is that? It already tells us something, it tells us how to take $10^{0.5}$, so we now know at least *one* logarithm, if we happen to need the logarithm of 3.16228, we know the answer is close to 0.50000. But we must do a little bit better than that; we clearly need more information. So we take the square root again, and find $10^{1/4}$, which is 1.77828. Now we have the logarithm of more numbers than we had before, 1.250 is the logarithm of 17.78 and, incidentally, if it happens that somebody asks for $10^{0.75}$, we can get it, because that is $10^{(0.5+0.25)}$; it is therefore the product of the second and third numbers. If we can get enough numbers in column s to be able to make up almost any number, then by multiplying the proper things in column 3, we can get 10 to any power; that is the plan. So we evaluate ten successive square roots of 10, and that is the main work which is involved in the calculations.

Why don't we keep on going for more and more accuracy? Because we begin to notice something. When we raise 10 to a very small power, we get 1 plus a small amount. The reason for this is clear, because we are going to have to take the

* There is a definite arithmetic procedure, but the easiest way to find the square root of any number N is to choose some a fairly close, find N/a , average $a' = \frac{1}{2}[a + (N/a)]$, and use this average a' for the next choice for a . The convergence is very rapid—the number of significant figures doubles each time.

1000th power of $10^{1/1000}$ to get back to 10, so we had better not start with too big a number; it has to be close to 1. What we notice is that the small numbers that are added to 1 begin to look as though we are merely dividing by 2 each time; we see 1815 becomes 903, then 450, 225; so it is clear that, to an excellent approximation, if we take another root, we shall get 1.00112 something, and rather than actually *take* all the square roots, we *guess* at the ultimate limit. When we take a small fraction Δ of 1024 as Δ approaches zero, what will the answer be? Of course it will be some number close to 0.0022511Δ . Not exactly 0.0022511Δ , however—we can get a better value by the following trick: we subtract the 1, and then divide by the power s . This ought to correct all the excesses to the same value. We see that they are very closely equal. At the top of the table they are not equal, but as they come down, they get closer and closer to a constant value. What is the value? Again we look to see how the series is going, how it has changed with s . It changed by 211, by 104, by 53, by 26. These changes are obviously half of each other, very closely, as we go down. Therefore, if we kept going, the changes would be 13, 7, 3, 2 and 1, more or less, or a total of 26. Thus we have only 26 more to go, and so we find that the true number is 2.3025. (Actually, we shall later see that the *exact* number should be 2.3026, but to keep it realistic, we shall not alter anything in the arithmetic.) From this table we can now calculate any power of 10, by compounding the power out of 1024ths.

Let us now actually calculate a logarithm, because the process we shall use is where logarithm tables actually come from. The procedure is shown in Table 22-2, and the numerical values are shown in Table 22-1 (columns 2 and 3).

Table 22-2

Calculation of a logarithm: $\log_{10} 2$

$$\begin{aligned}
 2 \div 1.77828 &= 1.124682 \\
 1.124682 \div 1.074607 &= 1.046598, \text{ etc.} \\
 \therefore 2 &= (1.77828)(1.074607)(1.036633)(1.090350)(1.000573) \\
 &= 10 \left[\frac{1}{1024} (256 + 32 + 16 + 4 + 0.254) \right] = 10 \left[\frac{308.254}{1024} \right] \\
 &= 10^{0.30103} \qquad \left(\frac{573}{2249} = 0.254 \right) \\
 \therefore \log_{10} 2 &= 0.30103
 \end{aligned}$$

Suppose we want the logarithm of 2. That is, we want to know to what power we must raise 10 to get 2. Can we raise 10 to the $1/2$ power? No; that is too big. In other words, we can see that the answer is going to be bigger than $1/4$, and less than $1/2$. Let us take the factor $10^{1/4}$ out; we divide 2 by 1.778 . . . , and get 1.124 . . . , and so on, and now we know that we have taken away 0.250000 from the logarithm. The number 1.124 . . . is now the number whose logarithm we need. When we are finished we shall add back the $1/4$, or $256/1024$. Now we look in the table for the next number just below 1.124 . . . , and that is 1.074607. We therefore divide by 1.074607 and get 1.046598. From that we discover that 2 can be made up of a product of numbers that are in Table 22-1, as follows:

$$2 = (1.77828)(1.074607)(1.036633)(1.0090350)(1.000573).$$

There was one factor (1.000573) left over, naturally, which is beyond the range of our table. To get the logarithm of this factor, we use our result that $10^{\Delta/1024} \approx 1 + 2.3025 \Delta/1024$. We find $\Delta = 0.254$. Therefore our answer is 10 to the following power: $(256 + 32 + 16 + 4 + 0.254)/1024$. Adding those together, we get $308.254/1024$. Dividing, we get 0.30103, so we know that the $\log_{10} 2 = 0.30103$, which happens to be right to 5 figures!

This is how logarithms were originally computed by Mr. Briggs of Halifax, in 1620. He said, "I computed successively 54 square roots of 10." We know he

really computed only the first 27, because the rest of them can be obtained by this trick with Δ . His work involved calculating the square root of 10 twenty-seven times, which is not much more than the ten times we did; however, it was more work because he calculated to sixteen decimal places, and then reduced his answer to fourteen when he published it, so that there were no rounding errors. He made tables of logarithms to fourteen decimal places by this method, which is quite tedious. But all logarithm tables for three hundred years were borrowed from Mr. Briggs' tables by reducing the number of decimal places. Only in modern times, with the WPA and computing machines, have new tables been independently computed. There are much more efficient methods of computing logarithms today, using certain series expansions.

In the above process, we discovered something rather interesting, and that is that for very small powers ϵ we can calculate 10^ϵ easily; we have discovered that $10^\epsilon = 1 + 2.3025\epsilon$, by sheer numerical analysis. Of course this also means that $10^{n/2.3025} = 1 + n$ if n is very small. Now logarithms to any other base are merely multiples of logarithms to the base 10. The base 10 was used only because we have 10 fingers, and the arithmetic of it is easy, but if we ask for a mathematically natural base, one that has nothing to do with the number of fingers on human beings, we might try to change our *scale* of logarithms in some convenient and natural manner, and the method which people have chosen is to redefine the logarithms by multiplying all the logarithms to the base 10 by 2.3025 . . . This then corresponds to using some other base, and this is called the *natural* base, or base e . Note that $\log_e(1 + n) \approx n$, or $e^n \approx 1 + n$ as $n \rightarrow 0$.

It is easy enough to find out what e is: $e = 10^{1/2.3025}$ or $10^{0.434294\dots}$, an irrational power. Our table of the successive square roots of 10 can be used to compute, not just logarithms, but also 10 to any power, so let us use it to calculate this natural base e . For convenience we transform $0.434294\dots$ into $444.73/1024$. Now, 444.73 is $256 + 128 + 32 + 16 + 2 + 0.73$. Therefore e , since it is an exponent of a sum, will be a product of the numbers

$$(1.77828)(1.33352)(1.074607)(1.036633)(1.018152)(1.009035)(1.001643) = 2.7184.$$

(The only problem is the last one, which is 0.73, and which is not in the table, but we know that if Δ is small enough, the answer is $1 + 2.3025 \Delta$.) When we multiply all these together, we get 2.7184 (it should be 2.7183, but it is good enough). The use of such tables, then, is the way in which irrational powers and the logarithms of irrational numbers are all calculated. That takes care of the irrationals.

22-5 Complex numbers

Now it turns out that after all that work we *still* cannot solve every equation! For instance, what is the square root of -1 ? Suppose we have to find $x^2 = -1$. The square of no rational, of no irrational, of *nothing* that we have discovered so far, is equal to -1 . So we again have to generalize our numbers to a still wider class. Let us suppose that a specific solution of $x^2 = -1$ is called something, we shall call it i ; i has the property, by definition, that its square is -1 . That is about all we are going to say about it; of course, there is more than one root of the equation $x^2 = -1$. Someone could write i , but another could say, "No, I prefer $-i$. My i is minus your i ." It is just as good a solution, and since the only definition that i has is that $i^2 = -1$, it must be true that any equation we can write is equally true if the sign of i is changed everywhere. This is called taking the *complex conjugate*. Now we are going to make up numbers by adding successive i 's, and multiplying i 's by numbers, and adding other numbers, and so on, according to all of our rules. In this way we find that numbers can all be written in the form $p + iq$, where p and q are what we call *real* numbers, i.e., the numbers we have been defining up until now. The number i is called the *unit imaginary* number. Any real multiple of i is called *pure imaginary*. The most general number, a , is of the form $p + iq$ and is called a *complex number*. Things do not get any worse if, for instance, we multiply two such numbers, let us say $(r + is)(p + iq)$. Then,

using the rules, we get

$$\begin{aligned}
 (r + is)(p + iq) &= rp + r(iq) + (is)p + (is)(iq) \\
 &= rp + i(rq) + i(sp) + (ii)(sq) \\
 &= (rp - sq) + i(rq + sp),
 \end{aligned} \tag{22.4}$$

since $ii = i^2 = -1$. Therefore all the numbers that now belong in the rules (22.1) have this mathematical form.

Now you say, “This can go on forever! We have defined powers of imaginaries and all the rest, and when we are all finished, somebody else will come along with another equation which cannot be solved, like $x^6 + 3x^2 = -2$. Then we have to generalize all over again!” But it turns out that *with this one more invention*, just the square root of -1 , *every algebraic equation can be solved!* This is a fantastic fact, which we must leave to the Mathematics Department to prove. The proofs are very beautiful and very interesting, but certainly not self-evident. In fact, the most obvious supposition is that we are going to have to invent again and again and again. But the greatest miracle of all is that we do not. This is the last invention. After this invention of complex numbers, we find that the rules still work with complex numbers, and we are finished inventing new things. We can find the complex power of any complex number, we can solve any equation that is written algebraically, in terms of a finite number of those symbols. We do not find any new numbers. The square root of i , for instance, has a definite result, it is not something new; and i^i is something. We will discuss that now.

We have already discussed multiplication, and addition is also easy; if we add two complex numbers, $(p + iq) + (r + is)$, the answer is $(p + r) + i(q + s)$. Now we can add and multiply complex numbers. But the real problem, of course, is to compute *complex powers of complex numbers*. It turns out that the problem is actually no more difficult than computing complex powers of real numbers. So let us concentrate now on the problem of calculating 10 to a complex power, not just an irrational power, but $10^{(r+is)}$. Of course, we must at all times use our rules (22.1) and (22.2). Thus

$$10^{(r+is)} = 10^r 10^{is}. \tag{22.5}$$

But 10^r we already know how to compute, and we can always multiply anything by anything else; therefore the problem is to compute only 10^{is} . Let us call it some complex number, $x + iy$. Problem: given s , find x , find y . Now if

$$10^{is} = x + iy,$$

then the complex conjugate of this equation must also be true, so that

$$10^{-is} = x - iy.$$

(Thus we see that we can deduce a number of things without actually computing anything, by using our rules.) We deduce another interesting thing by multiplying these together:

$$10^{is} 10^{-is} = 10^0 = 1 = (x + iy)(x - iy) = x^2 + y^2. \tag{22.6}$$

Thus if we find x , we have y also.

Now the problem is *how* to compute 10 to an imaginary power. What guide is there? We may work over our rules until we can go no further, but here is a reasonable guide: if we can compute it for any particular s , we can get it for all the rest. If we know 10^{is} for any one s and then we want it for twice that s , we can square the number, and so on. But how can we find 10^{is} for even one special value of s ? To do so we shall make one additional assumption, which is not quite in the category of all the other rules, but which leads to reasonable results and permits us to make progress: when the power is small, we shall suppose that the “law” $10^\epsilon = 1 + 2.3025\epsilon$ is right, as ϵ gets very small, not only for real ϵ , *but for complex ϵ as well*. Therefore, we begin with the supposition that this law is true in general, and that tells us that $10^{is} = 1 + 2.3025 \cdot is$, for $s \rightarrow 0$. So we assume that if s is very small, say one part in 1024, we have a rather good approximation to 10^{is} .

Now we make a table by which we can compute *all* the imaginary powers of 10, that is, compute x and y . It is done as follows. The first power we start with is the $1/1024$ power, which we presume is very nearly $1 + 2.3025i/1024$. Thus we start with

$$10^{i/1024} = 1.00000 + 0.0022486i, \quad (22.7)$$

and if we keep multiplying the number by itself, we can get to a higher imaginary power. In fact, we may just reverse the procedure we used in making our logarithm table, and calculate the square, 4th power, 8th power, etc., of (22.7), and thus build up the values shown in Table 22-3. We notice an interesting thing, that the x numbers are positive at first, but then swing negative. We shall look into that a little bit more in a moment. But first we may be curious to find for what number s the real part of 10^{is} is zero. The y -value would be i , and so we would have $10^{is} = i$, or $is = \log_{10} i$. As an example of how to use this table, just as we calculated $\log_{10} 2$ before, let us now use Table 22-3 to find $\log_{10} i$.

Which of the numbers in Table 22-3 do we have to multiply together to get a pure imaginary result? After a little trial and error, we discover that to reduce x the most, it is best to multiply "512" by "128." This gives $0.13056 + 0.99144i$. Then we discover that we should multiply this by a number whose imaginary part is about equal to the size of the real part we are trying to remove. Thus we choose "64" whose y -value is 0.14349, since that is closest to 0.13056. This then gives $-0.01350 + 0.99993i$. Now we have overshoot, and must *divide* by $0.99996 + 0.00900i$. How do we do that? By changing the sign of i and multiplying by $0.99996 - 0.00900i$ (which works if $x^2 + y^2 = 1$). Continuing in this way, we find that the entire power to which 10 must be raised to give i is $i(512 + 128 + 64 - 4 - 2 + 0.20)/1024$, or $698.20i/1024$. If we raise 10 to that power, we can get i . Therefore $\log_{10} i = 0.68226i$.

22-6 Imaginary exponents

To further investigate the subject of taking complex imaginary powers, let us look at the powers of 10 taking *successive powers*, not doubling the power each time, in order to follow Table 22-3 further and to see what happens to those minus signs. This is shown in Table 22-4, in which we take $10^{i/8}$, and just keep multiplying it. We see that x decreases, passes through zero, swings almost to -1 (if we could get in between $p = 10$ and $p = 11$ it would obviously swing to -1), and swings back. The y -value is going back and forth too.

In Fig. 22-1 the dots represent the numbers that appear in Table 22-4, and the lines are just drawn to help you visually. So we see that the numbers x and y oscillate; 10^{is} *repeats itself*, it is a *periodic* thing, and as such, it is easy enough to explain, because if a certain power is i , then the fourth power of that would be i^2 *squared*. It would be $+1$ again, and therefore, since $10^{0.68i}$ is equal to i , by taking the fourth power we discover that $10^{2.72i}$ is equal to $+1$. Therefore, if we wanted $10^{3.00i}$, for instance, we could write it as $10^{2.72i}$ times $10^{0.28i}$. In other words, it has a period, it repeats. Of course, we recognize what the curves look like! They look like the sine and cosine, and we shall call them, for a while, the algebraic sine and algebraic cosine. However, instead of using the base 10, we shall put them into our natural base, which only changes the horizontal scale; so we denote $2.3025s$ by t , and write $10^{is} = e^{it}$, where t is a real number. Now $e^{it} = x + iy$, and we shall write this as the algebraic cosine of t plus i times the algebraic sine of t . Thus

$$e^{it} = \cos t + i \sin t. \quad (22.8)$$

What are the properties of $\cos t$ and $\sin t$? First, we know, for instance, that $x^2 + y^2$ must be 1; we have proved that before, and it is just as true for base e as for base 10. Therefore $\cos^2 t + \sin^2 t = 1$. We also know that, for small t , $e^{it} = 1 + it$, and therefore $\cos t$ is nearly 1, and $\sin t$ is nearly t , and so it goes, that *all of the various properties of these remarkable functions*, which come from taking imaginary powers, *are the same as the sine and cosine of trigonometry*.

Table 22-3

Successive Squares of

$$10^{i/1024} = 1 + 0.0022486i$$

Power is	1024s	10^{is}
$i/1024$	1	$1.00000 + 0.00225i^*$
$i/512$	2	$1.00000 + 0.00450i$
$i/256$	4	$0.99996 + 0.00900i$
$i/128$	8	$0.99984 + 0.01800i$
$i/64$	16	$0.99936 + 0.03599i$
$i/32$	32	$0.99742 + 0.07193i$
$i/16$	64	$0.98967 + 0.14349i$
$i/8$	128	$0.95885 + 0.28402i$
$i/4$	256	$0.83872 + 0.54467i$
$i/2$	512	$0.40679 + 0.91365i$
$i/1$	1024	$-0.66928 + 0.74332i$

* Should be 0.0022486i

Table 22-4

Successive Powers of $10^{i/8}$

$p = \text{power} \cdot 8i$	$10^{ip/8}$
0	$1.00000 + 0.00000i$
1	$0.95882 + 0.28402i$
2	$0.83867 + 0.54465i$
3	$0.64944 + 0.76042i$
4	$0.40672 + 0.91356i$
5	$0.13050 + 0.99146i$
6	$-0.15647 + 0.98770i$
7	$-0.43055 + 0.90260i$
8	$-0.66917 + 0.74315i$
9	$-0.85268 + 0.52249i$
10	$-0.96596 + 0.25880i$
11	$-0.99969 - 0.02620i$
12	$-0.95104 - 0.30905i$
14	$-0.62928 - 0.77717i$
16	$-0.10447 - 0.99453i$
18	$+0.45454 - 0.89098i$
20	$+0.86648 - 0.49967i$
22	$+0.99884 + 0.05287i$
24	$+0.80890 + 0.58836i$

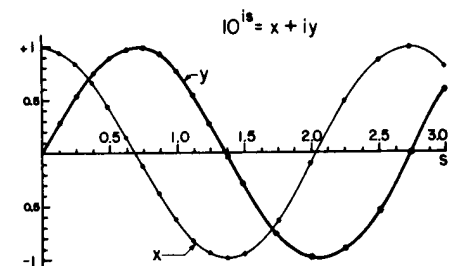


Figure 22-1

Is the period the same? Let us find out. e to what power is equal to i ? What is the logarithm of i to the base e ? We worked it out before, in the base 10 it was $0.68226i$, but when we change our logarithmic scale to e , we have to multiply by 2.3025, and if we do that it comes out 1.5709. So this will be called “algebraic $\pi/2$.” But, we see, it differs from the regular $\pi/2$ by only one place in the last point, and that, of course, is the result of errors in our arithmetic! So we have created two new functions in a purely algebraic manner, the cosine and the sine, which belong to algebra, and only to algebra. We wake up at the end to discover the very functions that are natural to geometry. So there is a connection, ultimately, between algebra and geometry.

We summarize with this, the most remarkable formula in mathematics:

$$e^{i\theta} = \cos \theta + i \sin \theta. \quad (22.9)$$

This is our jewel.

We may relate the geometry to the algebra by representing complex numbers in a plane; the horizontal position of a point is x , the vertical position of a point is y (Fig. 22-2). We represent every complex number, $x + iy$. Then if the radial distance to this point is called r and the angle is called θ , the algebraic law is that $x + iy$ is written in the form $re^{i\theta}$, where the geometrical relationships between x , y , r , and θ are as shown. This, then, is the unification of algebra and geometry.

When we began this chapter, armed only with the basic notions of integers and counting, we had little idea of the power of the processes of abstraction and generalization. Using the set of algebraic “laws,” or properties of numbers, Eq. (22.1), and the definitions of inverse operations (22.2), we have been able here, ourselves, to manufacture not only numbers but useful things like tables of logarithms, powers, and trigonometric functions (for these are what the imaginary powers of real numbers are), all merely by extracting ten successive square roots of ten!

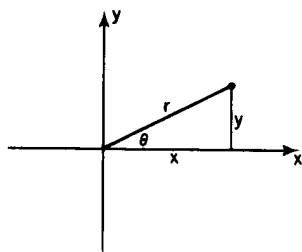


Fig. 22-2. $x + iy = re^{i\theta}$.

Resonance

23-1 Complex numbers and harmonic motion

In the present chapter we shall continue our discussion of the harmonic oscillator and, in particular, the forced harmonic oscillator, using a new technique in the analysis. In the preceding chapter we introduced the idea of complex numbers, which have real and imaginary parts and which can be represented on a diagram in which the ordinate represents the imaginary part and the abscissa represents the real part. If a is a complex number, we may write it as $a = a_r + ia_i$, where the subscript r means the real part of a , and the subscript i means the imaginary part of a . Referring to Fig. 23-1, we see that we may also write a complex number $a = x + iy$ in the form $x + iy = re^{i\theta}$, where $r^2 = x^2 + y^2 = (x + iy)(x - iy) = aa^*$. (The complex conjugate of a , written a^* , is obtained by reversing the sign of i in a .) So we shall represent a complex number in either of two forms, a real plus an imaginary part, or a magnitude r and a phase angle θ , so-called. Given r and θ , x and y are clearly $r \cos \theta$ and $r \sin \theta$ and, in reverse, given a complex number $x + iy$, $r = \sqrt{x^2 + y^2}$ and $\tan \theta = y/x$, the ratio of the imaginary to the real part.

We are going to apply complex numbers to our analysis of physical phenomena by the following trick. We have examples of things that oscillate; the oscillation may have a driving force which is a certain constant times $\cos \omega t$. Now such a force, $F = F_0 \cos \omega t$, can be written as the real part of a complex number $F = F_0 e^{i\omega t}$ because $e^{i\omega t} = \cos \omega t + i \sin \omega t$. The reason we do this is that it is easier to work with an exponential function than with a cosine. So the whole trick is to represent our oscillatory functions as the real parts of certain complex functions. The complex number F that we have so defined is not a real physical force, because no force in physics is really complex; actual forces have no imaginary part, only a real part. We shall, however, speak of the "force" $F_0 e^{i\omega t}$, but of course the actual force is the *real part* of that expression.

Let us take another example. Suppose we want to represent a force which is a cosine wave that is out of phase with a delayed phase Δ . This, of course, would be the real part of $F_0 e^{i(\omega t - \Delta)}$, but exponentials being what they are, we may write $e^{i(\omega t - \Delta)} = e^{i\omega t} e^{-i\Delta}$. Thus we see that the algebra of exponentials is much easier than that of sines and cosines; this is the reason we choose to use complex numbers. We shall often write

$$F = F_0 e^{-i\Delta} e^{i\omega t} = \hat{F} e^{i\omega t}. \quad (23.1)$$

We write a little caret (^) over the F to remind ourselves that this quantity is a complex number: here the number is

$$\hat{F} = F_0 e^{-i\Delta}.$$

Now let us solve an equation, using complex numbers, to see whether we can work out a problem for some real case. For example, let us try to solve

$$\frac{d^2 x}{dt^2} + \frac{kx}{m} = \frac{F}{m} = \frac{F_0}{m} \cos \omega t, \quad (23.2)$$

where F is the force which drives the oscillator and x is the displacement. Now, absurd though it may seem, let us suppose that x and F are actually complex numbers, for a mathematical purpose only. That is to say, x has a real part and an imaginary part times i , and F has a real part and an imaginary part times i .

23-1 Complex numbers and harmonic motion

23-2 The forced oscillator with damping

23-3 Electrical resonance

23-4 Resonance in nature

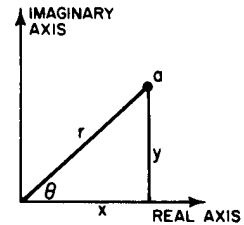


Fig. 23-1. A complex number may be represented by a point in the "complex plane."

Now if we had a solution of (23.2) with complex numbers, and substituted the complex numbers in the equation, we would get

$$\frac{d^2(x_r + ix_i)}{dt^2} + \frac{k(x_r + ix_i)}{m} = \frac{F_r + iF_i}{m}$$

or

$$\frac{d^2x_r}{dt^2} + \frac{kx_r}{m} + i\left(\frac{d^2x_i}{dt^2} + \frac{kx_i}{m}\right) = \frac{F_r}{m} + \frac{iF_i}{m}.$$

Now, since if two complex numbers are equal, their real parts must be equal *and* their complex parts must be equal, we deduce that *the real part of x satisfies the equation with the real part of the force*. We must emphasize, however, that this separation into a real part and an imaginary part is *not* valid in general, but is valid only for equations which are *linear*, that is, for equations in which x appears in every term only in the first power or the zeroth power. For instance, if there were in the equation a term λx^2 , then when we substitute $x_r + ix_i$, we would get $\lambda(x_r + ix_i)^2$, but when separated into real and imaginary parts this would yield $\lambda(x_r^2 - x_i^2)$ as the real part and $2i\lambda x_r x_i$ as the imaginary part. So we see that the real part of the equation would not involve just λx_r^2 , but also $-\lambda x_i^2$. In this case we get a different equation than the one we wanted to solve, with x_i , the completely artificial thing we introduced in our analysis, mixed in.

Let us now try our new method for the problem of the forced oscillator, that we already know how to solve. We want to solve Eq. (23.2) as before, but we say that we are going to solve

$$\frac{d^2x}{dt^2} + \frac{kx}{m} = \frac{\hat{F}e^{i\omega t}}{m}, \quad (23.3)$$

where $\hat{F}e^{i\omega t}$ is a complex number. Of course x will also be complex, but remember the rule: take the real part to find out what is really going on. So we try to solve (23.3) for the forced solution; we shall discuss other solutions later. The forced solution has the same frequency as the applied force, and has some amplitude of oscillation and some phase, and so it can be represented also by some complex number $\hat{x}$ whose magnitude represents the swing of x and whose phase represents the time delay in the same way as for the force. Now a wonderful feature of an exponential function is that $d(\hat{x}e^{i\omega t})/dt = i\omega\hat{x}e^{i\omega t}$. When we differentiate an exponential function, we bring down the exponent as a simple multiplier. The second derivative does the same thing, it brings down another $i\omega$, and so it is very simple to write immediately, by inspection, what the equation is for $\hat{x}$: every time we see a differentiation, we simply multiply by $i\omega$. (Differentiation is now as easy as multiplication! This idea of using exponentials in linear differential equations is almost as great as the invention of logarithms, in which multiplication is replaced by addition. Here differentiation is replaced by multiplication.) Thus our equation becomes

$$(i\omega)^2\hat{x} + (k\hat{x}/m) = \hat{F}/m. \quad (23.4)$$

(We have cancelled the common factor $e^{i\omega t}$.) See how simple it is! Differential equations are immediately converted, by sight, into mere algebraic equations; we virtually have the solution by sight, that

$$\hat{x} = \frac{\hat{F}/m}{(k/m) - \omega^2},$$

since $(i\omega)^2 = -\omega^2$. This may be slightly simplified by substituting $k/m = \omega_0^2$, which gives

$$\hat{x} = \hat{F}/m(\omega_0^2 - \omega^2). \quad (23.5)$$

This, of course, is the solution we had before; for since $m(\omega_0^2 - \omega^2)$ is a real number, the phase angles of $\hat{F}$ and of $\hat{x}$ are the same (or perhaps 180° apart, if $\omega^2 > \omega_0^2$), as advertised previously. The magnitude of $\hat{x}$, which measures how far it oscillates, is related to the size of the $\hat{F}$ by the factor $1/m(\omega_0^2 - \omega^2)$, and this factor becomes

enormous when ω is nearly equal to ω_0 . So we get a very strong response when we apply the right frequency ω (if we hold a pendulum on the end of a string and shake it at just the right frequency, we can make it swing very high).

23-2 The forced oscillator with damping

That, then, is how we analyze oscillatory motion with the more elegant mathematical technique. But the elegance of the technique is not at all exhibited in such a problem that can be solved easily by other methods. It is only exhibited when one applies it to more difficult problems. Let us therefore solve another, more difficult problem, which furthermore adds a relatively realistic feature to the previous one. Equation (23.5) tells us that if the frequency ω were exactly equal to ω_0 , we would have an infinite response. Actually, of course, no such infinite response occurs because some other things, like friction, which we have so far ignored, limits the response. Let us therefore add to Eq. (23.2) a friction term.

Ordinarily such a problem is very difficult because of the character and complexity of the frictional term. There are, however, many circumstances in which the frictional force is *proportional to the speed* with which the object moves. An example of such friction is the friction for slow motion of an object in oil or a thick liquid. There is no force when it is just standing still, but the faster it moves the faster the oil has to go past the object, and the greater is the resistance. So we shall assume that there is, in addition to the terms in (23.2), another term, a resistance force proportional to the velocity: $F_f = -c dx/dt$. It will be convenient, in our mathematical analysis, to write the constant c as m times γ to simplify the equation a little. This is just the same trick we use with k when we replace it by $m\omega_0^2$, just to simplify the algebra. Thus our equation will be

$$m(d^2x/dt^2) + c(dx/dt) + kx = F \quad (23.6)$$

or, writing $c = m\gamma$ and $k = m\omega_0^2$ and dividing out the mass m ,

$$(d^2x/dt^2) + \gamma(dx/dt) + \omega_0^2x = F/m. \quad (23.6a)$$

Now we have the equation in the most convenient form to solve. If γ is very small, that represents very little friction; if γ is very large, there is a tremendous amount of friction. How do we solve this new linear differential equation? Suppose that the driving force is equal to $F_0 \cos(\omega t + \Delta)$; we could put this into (23.6a) and try to solve it, but we shall instead solve it by our new method. Thus we write F as the real part of $\hat{F}e^{i\omega t}$ and x as the real part of $\hat{x}e^{i\omega t}$, and substitute these into Eq. (23.6a). It is not even necessary to do the actual substituting, for we can see by inspection that the equation would become

$$[(i\omega)^2\hat{x} + \gamma(i\omega)\hat{x} + \omega_0^2\hat{x}]e^{i\omega t} = (\hat{F}/m)e^{i\omega t}. \quad (23.7)$$

[As a matter of fact, if we tried to solve Eq. (23.6a) by our old straightforward way, we would really appreciate the magic of the “complex” method.] If we divide by $e^{i\omega t}$ on both sides, then we can obtain the response $\hat{x}$ to the given force $\hat{F}$; it is

$$\hat{x} = \hat{F}/m(\omega_0^2 - \omega^2 + i\gamma\omega). \quad (23.8)$$

Thus again $\hat{x}$ is given by $\hat{F}$ times a certain factor. There is no technical name for this factor, no particular letter for it, but we may call it R for discussion purposes:

$$R = \frac{1}{m(\omega_0^2 - \omega^2 + i\gamma\omega)}$$

and

$$\hat{x} = \hat{F}R. \quad (23.9)$$

(Although the letters γ and ω_0 are in very common use, this R has no particular name.) This factor R can either be written as $p + iq$, or as a certain magnitude ρ times $e^{i\theta}$. If it is written as a certain magnitude times $e^{i\theta}$, let us see what it means.

Now $\hat{F} = F_0 e^{i\Delta}$, and the actual force F is the real part of $F_0 e^{i\Delta} e^{i\omega t}$, that is, $F_0 \cos(\omega t + \Delta)$. Next, Eq. (23.9) tells us that $\hat{x}$ is equal to $\hat{F}R$. So, writing $R = \rho e^{i\theta}$ as another name for R , we get

$$\hat{x} = R\hat{F} = \rho e^{i\theta} F_0 e^{i\Delta} = \rho F_0 e^{i(\theta + \Delta)}.$$

Finally, going even further back, we see that the physical x , which is the real part of the complex $\hat{x}$, is equal to the real part of $\rho F_0 e^{i(\theta + \Delta)} e^{i\omega t}$. But ρ and F_0 are real, and the real part of $e^{i(\theta + \Delta + \omega t)}$ is simply $\cos(\omega t + \Delta + \theta)$. Thus

$$x = \rho F_0 \cos(\omega t + \Delta + \theta). \quad (23.10)$$

This tells us that the amplitude of the response is the magnitude of the force F multiplied by a certain magnification factor, ρ ; this gives us the “amount” of oscillation. It also tells us, however, that x is not oscillating in phase with the force, which has the phase Δ , but is shifted by an extra amount θ . Therefore ρ and θ represent the size of the response and the phase shift of the response.

Now let us work out what ρ is. If we have a complex number, the square of the magnitude is equal to the number times its complex conjugate; thus

$$\begin{aligned} \rho^2 &= \frac{1}{m^2(\omega_0^2 - \omega^2 + i\gamma\omega)(\omega_0^2 - \omega^2 - i\gamma\omega)} \\ &= \frac{1}{m^2[(\omega^2 - \omega_0^2)^2 + \gamma^2\omega^2]}. \end{aligned} \quad (23.11)$$

In addition, the phase angle θ is easy to find, for if we write

$$1/R = 1/\rho e^{i\theta} = (1/\rho)e^{-i\theta} = m(\omega_0^2 - \omega^2 + i\gamma\omega),$$

we see that

$$\tan \theta = -\gamma\omega/(\omega_0^2 - \omega^2). \quad (23.12)$$

It is minus because $\tan(-\theta) = -\tan \theta$. A negative value for θ results for all ω , and this corresponds to the displacement x lagging the force F .

Figure 23-2 shows how ρ^2 varies as a function of frequency (ρ^2 is physically more interesting than ρ , because ρ^2 is proportional to the square of the amplitude, or more or less to the *energy* that is developed in the oscillator by the force). We see that if γ is very small, then $1/(\omega_0^2 - \omega^2)^2$ is the most important term, and the response tries to go up toward infinity when ω equals ω_0 . Now the “infinity” is not actually infinite because if $\omega = \omega_0$, then $1/\gamma^2\omega^2$ is still there. The phase shift varies as shown in Fig. 23-3.

In certain circumstances we get a slightly different formula than (23.8), also called a “resonance” formula, and one might think that it represents a different phenomenon, but it does not. The reason is that if γ is very small the most interesting part of the curve is near $\omega = \omega_0$, and we may replace (23.8) by an approximate formula which is very accurate if γ is small and ω is near ω_0 . Since $\omega_0^2 - \omega^2 = (\omega_0 - \omega)(\omega_0 + \omega)$, if ω is near ω_0 this is nearly the same as $2\omega_0(\omega_0 - \omega)$ and $\gamma\omega$ is nearly the same as $\gamma\omega_0$. Using these in (23.8), we see that $\omega_0^2 - \omega^2 + i\gamma\omega \approx 2\omega_0(\omega_0 - \omega + i\gamma/2)$, so that

$$\hat{x} \approx \hat{F}/2m\omega_0(\omega_0 - \omega + i\gamma/2) \quad \text{if } \gamma \ll \omega_0 \text{ and } \omega \approx \omega_0. \quad (23.13)$$

It is easy to find the corresponding formula for ρ^2 . It is

$$\rho^2 \approx 1/4m^2\omega_0^2[(\omega_0 - \omega)^2 + \gamma^2/4].$$

We shall leave it to the student to show the following: if we call the maximum height of the curve of ρ^2 vs. ω one unit, and we ask for the width $\Delta\omega$ of the curve, at one half the maximum height, the full width at half the maximum height of the curve is $\Delta\omega = \gamma$, supposing that γ is small. The resonance is sharper and sharper as the frictional effects are made smaller and smaller.

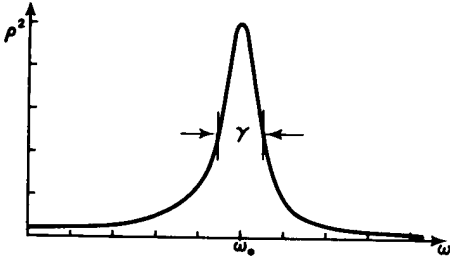


Fig. 23-2. Plot of ρ^2 versus ω .

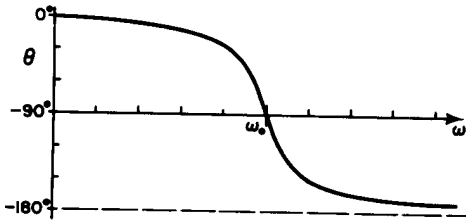


Fig. 23-3. Plot of θ versus ω .

As another measure of the width, some people use a quantity Q which is defined as $Q = \omega_0/\gamma$. The narrower the resonance, the higher the Q : $Q = 1000$ means a resonance whose width is only 1000th of the frequency scale. The Q of the resonance curve shown in Fig. 23-2 is 5.

The importance of the resonance phenomenon is that it occurs in many other circumstances, and so the rest of this chapter will describe some of these other circumstances.

23-3 Electrical resonance

The simplest and broadest technical applications of resonance are in electricity. In the electrical world there are a number of objects which can be connected to make electric circuits. These *passive circuit elements*, as they are often called, are of three main types, although each one has a little bit of the other two mixed in. Before describing them in greater detail, let us note that the whole idea of our mechanical oscillator being a mass on the end of a spring is only an approximation. All the mass is not actually at the “mass”; some of the mass is in the inertia of the spring. Similarly, all of the spring is not at the “spring”; the mass itself has a little elasticity, and although it may appear so, it is not *absolutely* rigid, and as it goes up and down, it flexes ever so slightly under the action of the spring pulling it. The same thing is true in electricity. There is an approximation in which we can lump things into “circuit elements” which are assumed to have pure, ideal characteristics. It is not the proper time to discuss that approximation here, we shall simply assume that it is true in the circumstances.

The three main kinds of circuit elements are the following. The first is called a *capacitor* (Fig. 23-4); an example is two plane metallic plates spaced a very small distance apart by an insulating material. When the plates are charged there is a certain voltage difference, that is, a certain difference in potential, between them. The same difference of potential appears between the terminals A and B , because if there were any difference along the connecting wire, electricity would flow right away. So there is a certain voltage difference V between the plates if there is a certain electric charge $+q$ and $-q$ on them, respectively. Between the plates there will be a certain electric field; we have even found a formula for it (Chapters 13 and 14):

$$V = \sigma d / \epsilon_0 = qd / \epsilon_0 A, \quad (23.14)$$

where d is the spacing and A is the area of the plates. Note that the potential difference is a linear function of the charge. If we do not have parallel plates, but insulated electrodes which are of any other shape, the difference in potential is still precisely proportional to the charge, but the constant of proportionality may not be so easy to compute. However, all we need to know is that the potential difference across a capacitor is *proportional to the charge*: $V = q/C$; the proportionality constant is $1/C$, where C is the *capacitance* of the object.

The second kind of circuit element is called a *resistor*; it offers resistance to the flow of electrical current. It turns out that metallic wires and many other substances resist the flow of electricity in this manner: if there is a voltage difference across a piece of some substance, there exists an electric current $I = dq/dt$ that is proportional to the electric voltage difference:

$$V = RI = R dq/dt. \quad (23.15)$$

The proportionality coefficient is called the *resistance* R . This relationship may already be familiar to you; it is Ohm's law.

If we think of the charge q on a capacitor as being analogous to the displacement x of a mechanical system, we see that the current, $I = dq/dt$, is analogous to velocity, $1/C$ is analogous to a spring constant k , and R is analogous to the resistive coefficient γ . Now it is very interesting that there exists another circuit element which is the analog of *mass*! This is a coil which builds up a magnetic field within itself when there is a current in it. A *changing* magnetic field develops in the coil a voltage that is proportional to dI/dt (this is how a transformer works, in

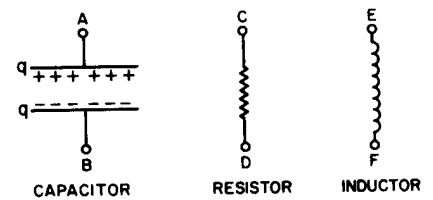


Fig. 23-4. The three passive circuit elements.

fact). The magnetic field is proportional to a current, and the induced voltage (so-called) in such a coil is proportional to the rate of change of the current:

$$V = L dI/dt = L d^2q/dt^2. \quad (23.16)$$

The coefficient L is the *self-inductance*, and is analogous to the mass in a mechanical oscillating circuit.

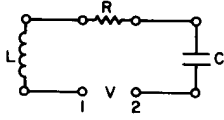


Fig. 23-5. An oscillatory electrical circuit with resistance, inductance, and capacitance.

Suppose we make a circuit in which we have connected the three circuit elements in series (Fig. 23-5); then the voltage across the whole thing from 1 to 2 is the work done in carrying a charge through, and it consists of the sum of several pieces: across the inductor, $V_L = L d^2q/dt^2$; across the resistance, $V_R = R dq/dt$; across the capacitor, $V_c = q/C$. The sum of these is equal to the applied voltage, V :

$$L d^2q/dt^2 + R dq/dt + q/C = V(t). \quad (23.17)$$

Now we see that this equation is exactly the same as the mechanical equation (23.6), and of course it can be solved in exactly the same manner. We suppose that $V(t)$ is oscillatory: we are driving the circuit with a generator with a pure sine wave oscillation. Then we can write our $V(t)$ as a complex $\hat{V}$ with the understanding that it must be ultimately multiplied by $e^{i\omega t}$, and the real part taken in order to find the true V . Likewise, the charge q can thus be analyzed, and then in exactly the same manner as in Eq. (23.8) we write the corresponding equation: the second derivative of $\hat{q}$ is $(i\omega)^2\hat{q}$; the first derivative is $(i\omega)\hat{q}$. Thus Eq. (23.17) translates to

$$\left[L(i\omega)^2 + R(i\omega) + \frac{1}{C} \right] \hat{q} = \hat{V}$$

or

$$\hat{q} = \frac{\hat{V}}{L(i\omega)^2 + R(i\omega) + \frac{1}{C}}$$

which we can write in the form

$$\hat{q} = \hat{V}/L(\omega_0^2 - \omega^2 + i\gamma\omega), \quad (23.18)$$

where $\omega_0^2 = 1/LC$ and $\gamma = R/L$. It is exactly the same denominator as we had in the mechanical case, with exactly the same resonance properties! The correspondence between the electrical and mechanical cases is outlined in Table 23-1.

Table 23-1

General characteristic	Mechanical property	Electrical property
indep. variable	time (t)	time (t)
dep. variable	position (x)	charge (q)
inertia	mass (m)	inductance (L)
resistance	drag coeff. ($c = \gamma m$)	resistance ($R = \gamma L$)
stiffness	stiffness (k)	(capacitance) $^{-1}$ ($1/C$)
resonant frequency	$\omega_0^2 = k/m$	$\omega_0^2 = 1/LC$
period	$t_0 = 2\pi\sqrt{m/k}$	$t_0 = 2\pi\sqrt{LC}$
figure of merit	$Q = \omega_0/\gamma$	$Q = \omega_0 L/R$

We must mention a small technical point. In the electrical literature, a different notation is used. (From one field to another, the subject is not really any different, but the way of writing the notations is often different.) First, j is commonly used instead of i in electrical engineering, to denote $\sqrt{-1}$. (After all, i must be the current!) Also, the engineers would rather have a relationship between $\hat{V}$ and $\hat{I}$ than between $\hat{V}$ and $\hat{q}$, just because they are more used to it that way. Thus, since $\hat{I} = d\hat{q}/dt = i\omega\hat{q}$, we can just substitute $\hat{I}/i\omega$ for $\hat{q}$ and get

$$\hat{V} = (i\omega L + R + 1/i\omega C)\hat{I} = \hat{Z}\hat{I}. \quad (23.19)$$

Another way is to rewrite Eq. (23.17), so that it looks more familiar; one often sees it written this way:

$$L \frac{dI}{dt} + RI + (1/C) \int^t I dt = V(t). \quad (23.20)$$

At any rate, we find the relation (23.19) between voltage $\hat{V}$ and current $\hat{I}$ which is just the same as (23.18) except divided by $i\omega$, and that produces Eq. (23.19). The quantity $R + i\omega L + 1/i\omega C$ is a complex number, and is used so much in electrical engineering that it has a name: it is called the *complex impedance*, $\hat{Z}$. Thus we can write $\hat{V} = \hat{Z}\hat{I}$. The reason that the engineers like to do this is that they learned something when they were young: $V = RI$ for resistances, when they only knew about resistances and DC. Now they have become more educated and have AC circuits, so they want the equation to look the same. Thus they write $\hat{V} = \hat{Z}\hat{I}$, the only difference being that the resistance is replaced by a more complicated thing, a complex quantity. So they insist that they cannot use what everyone else in the world uses for imaginary numbers, they have to use a j for that; it is a miracle that they did not insist also that the letter Z be an R ! (Then they get into trouble when they talk about current densities, for which they also use j . The difficulties of science are to a large extent the difficulties of notations, the units, and all the other artificialities which are invented by man, not by nature.)

23-4 Resonance in nature

Although we have discussed the electrical case in detail, we could also bring up case after case in many fields, and show exactly how the resonance equation is the same. There are many circumstances in nature in which something is "oscillating" and in which the resonance phenomenon occurs. We said that in an earlier chapter; let us now demonstrate it. If we walk around our study, pulling books off the shelves and simply looking through them to find an example of a curve that corresponds to Fig. 23-2 and comes from the same equation, what do we find? Just to demonstrate the wide range obtained by taking the smallest possible sample, it takes only five or six books to produce quite a series of phenomena which show resonances.

The first two are from mechanics, the first on a large scale: the atmosphere of the whole earth. If the atmosphere, which we suppose surrounds the earth evenly on all sides, is pulled to one side by the moon or, rather, squashed prolate into a double tide, and if we could then let it go, it would go sloshing up and down; it is an oscillator. This oscillator is *driven* by the moon, which is effectively revolving about the earth; any one component of the force, say in the x -direction, has a cosine component, and so the response of the earth's atmosphere to the tidal pull of the moon is that of an oscillator. The expected response of the atmosphere is shown in Fig. 23-6, curve b (curve a is another theoretical curve under discussion in the book from which this is taken out of context). Now one might think that we only have one point on this resonance curve, since we only have the one frequency, corresponding to the rotation of the earth under the moon, which occurs at a period of 12.42 hours—12 hours for the earth (the tide is a double bump), plus a little more because the moon is going around. But from the *size* of the atmospheric tides, and from the *phase*, the amount of delay, we can get both ρ and θ . From those we can get ω_0 and γ , and thus draw the entire curve! This is an example of very poor science. From two numbers we obtain two numbers, and from those two numbers we draw a beautiful curve, which of course goes through the very point that determined the curve! It is of no use *unless we can measure something else*, and in the case of geophysics that is often very difficult. But in this particular case there is another thing which we can show theoretically must have the same timing as the natural frequency ω_0 : that is, if someone disturbed the atmosphere, it would oscillate with the frequency ω_0 . Now there *was* such a sharp disturbance in 1883; the Krakatoa volcano exploded and half the island blew off, and it made such a terrific explosion in the atmosphere that the period of oscillation of the atmosphere could be measured. It came out to $10\frac{1}{2}$ hours. The ω_0 obtained from

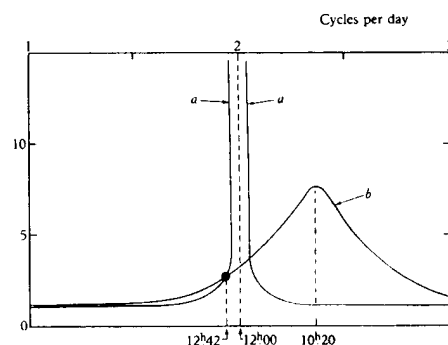


Fig. 23-6. Response of the atmosphere to external excitation. a is the required response if the atmospheric S_2 -tide is of gravitational origin; peak amplification is 100:1. b is derived from observed magnification and phase of M_2 -tide. [Munk and MacDonald, "Rotation of the Earth," Cambridge University Press (1960)]

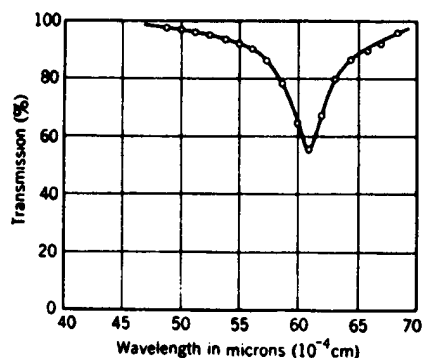


Fig. 23-7. Transmission of infrared radiation through a thin (0.17μ) sodium chloride film. [After R. B. Barnes, *Z. Physik* **75**, 723 (1932). Kittel, *Introduction to Solid State Physics*, Wiley, 1956.]

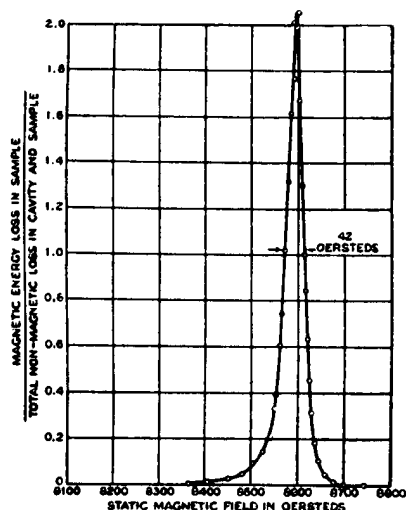


Fig. 23-8. Magnetic energy loss in paramagnetic organic compound as function of applied magnetic field intensity. [Holden et al., *Phys. Rev.* **75**, 1614 (1949)]

Fig. 23-6 comes out 10 hours and 20 minutes, so there we have at least one check on the reality of our understanding of the atmospheric tides.

Next we go to the small scale of mechanical oscillation. This time we take a sodium chloride crystal, which has sodium ions and chlorine ions next to each other, as we described in an early chapter. These ions are electrically charged, alternately plus and minus. Now there is an interesting oscillation possible. Suppose that we could drive all the plus charges to the right and all the negative charges to the left, and let go; they would then oscillate back and forth, the sodium lattice against the chlorine lattice. How can we ever drive such a thing? That is easy, for if we apply an electric field on the crystal, it will push the plus charge one way and the minus charge the other way! So, by having an external electric field we can perhaps get the crystal to oscillate. The frequency of the electric field needed is so high, however, that it corresponds to *infrared radiation*! So we try to find a resonance curve by measuring the absorption of infrared light by sodium chloride. Such a curve is shown in Fig. 23-7. The abscissa is not frequency, but is given in terms of wavelength, but that is just a technical matter, of course, since for a wave there is a definite relation between frequency and wavelength; so it is really a frequency scale, and a certain frequency corresponds to the resonant frequency.

But what about the width? What determines the width? There are many cases in which the width that is seen on the curve is not really the natural width γ that one would have theoretically. There are two reasons why there can be a wider curve than the theoretical curve. If the objects do not all have the same frequency, as might happen if the crystal were strained in certain regions, so that in those regions the oscillation frequency were slightly different than in other regions, then what we have is many resonance curves on top of each other; so we apparently get a wider curve. The other kind of width is simply this: perhaps we cannot measure the frequency precisely enough—if we open the slit of the spectrometer fairly wide, so although we thought we had only one frequency, we actually had a certain range $\Delta\omega$, then we may not have the resolving power needed to see a narrow curve. Offhand, we cannot say whether the width in Fig. 23-7 is natural, or whether it is due to inhomogeneities in the crystal or the finite width of the slit of the spectrometer.

Now we turn to a more esoteric example, and that is the swinging of a magnet. If we have a magnet, with north and south poles, in a constant magnetic field, the N end of the magnet will be pulled one way and the S end the other way, and there will in general be a torque on it, so it will vibrate about its equilibrium position, like a compass needle. However, the magnets we are talking about are *atoms*. These atoms have an angular momentum, the torque does not produce a simple motion in the direction of the field, but instead, of course, a *precession*. Now, looked at from the side, any one component is “swinging,” and we can disturb or drive that swinging and measure an absorption. The curve in Fig. 23-8 represents a typical such resonance curve. What has been done here is slightly different technically. The frequency of the lateral field that is used to drive this swinging is always kept the same, while we would have expected that the investigators would vary that and plot the curve. They could have done it that way, but technically it was easier for them to leave the frequency ω fixed, and change the strength of the constant magnetic field, which corresponds to changing ω_0 in our formula. They have plotted the resonance curve against ω_0 . Anyway, this is a typical resonance with a certain ω_0 and γ .

Now we go still further. Our next example has to do with atomic nuclei. The motions of protons and neutrons in nuclei are oscillatory in certain ways, and we can demonstrate this by the following experiment. We bombard a lithium atom with protons, and we discover that a certain reaction, producing γ -rays, actually has a very sharp maximum typical of resonance. We note in Fig. 23-9, however, one difference from other cases: the horizontal scale is not a frequency, it is an *energy*! The reason is that in quantum mechanics what we think of classically as the energy will turn out to be really related to a frequency of a wave amplitude. When we analyze something which in simple large-scale physics has to do with a frequency, we find that when we do quantum-mechanical experiments with atomic

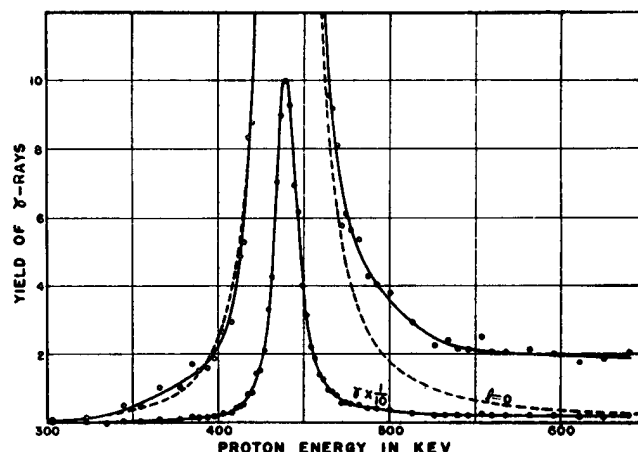


Fig. 23-9. The intensity of gamma-radiation from lithium as a function of the energy of the bombarding protons. The dashed curve is a theoretical one calculated for protons with an angular momentum $\ell = 0$. [Bonner and Evans, *Phys. Rev.* **73**, 666 (1948)]

matter, we get the corresponding curve as a function of energy. In fact, this curve is a demonstration of this relationship, in a sense. It shows that frequency and energy have some deep interrelationship, which of course they do.

Now we turn to another example which also involves a nuclear energy level, but now a much, much narrower one. The ω_0 in Fig. 23-10 corresponds to an energy of 100,000 electron volts, while the width γ is approximately 10^{-5} electron volt; in other words, this has a Q of 10^{10} ! When this curve was measured it was the largest Q of any oscillator that had ever been measured. It was measured by Dr. Mössbauer, and it was the basis of his Nobel prize. The horizontal scale here is velocity, because the technique for obtaining the slightly different frequencies was to use the Doppler effect, by moving the source relative to the absorber. One can see how delicate the experiment is when we realize that the speed involved is a few centimeters per second! On the actual scale of the figure, zero frequency would correspond to a point about 10^{10} cm to the left—slightly off the paper!

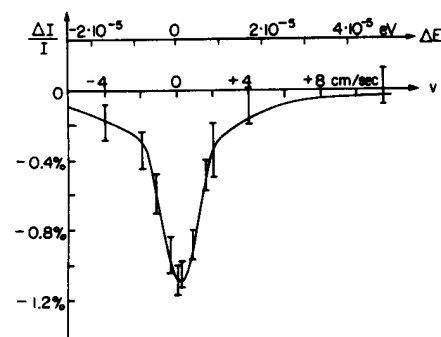


Fig. 23-10. [Courtesy of Dr. R. Mössbauer]

Finally, if we look in an issue of the *Physical Review*, say that of January 1, 1962, will we find a resonance curve? Every issue has a resonance curve, and Fig. 23-11 is the resonance curve for this one. This resonance curve turns out to be very interesting. It is the resonance found in a certain reaction among strange particles, a reaction in which a K^- and a proton interact. The resonance is detected by seeing how many of some kinds of particles come out, and depending on what and how many come out, one gets different curves, but of the same shape and with the peak at the same energy. We thus determine that there is a resonance at a certain energy for the K^- meson. That presumably means that there is some kind of a state, or condition, corresponding to this resonance, which can be attained by putting together a K^- and a proton. This is a new particle, or resonance. Today we do not know whether to call a bump like this a “particle” or simply a resonance. When there is a very *sharp* resonance, it corresponds to a very *definite* energy, just as though there were a particle of that energy present in nature. When the resonance gets wider, then we do not know whether to say there is a particle which does not last very long, or simply a resonance in the reaction probability. In the second chapter, this point is made about the particles, but when the second chapter was written this resonance was not known, so our chart should now have still another particle in it!

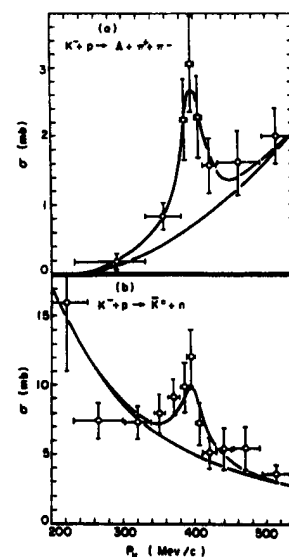


Fig. 23-11. Momentum dependence of the cross section for the reactions (a) $K^- + p \rightarrow \Lambda + \pi^+ + \pi^-$ and (b) $K^- + p \rightarrow K^0 + n$. The lower curves in (a) and (b) represent the presumed nonresonant backgrounds, while the upper curves contain in addition the superposed resonance. [Ferro-Luzzi *et al.*, *Phys. Rev., Lett.* **8**, 28 (1962)]

Transients

24-1 The energy of an oscillator

Although this chapter is entitled “transients,” certain parts of it are, in a way, part of the last chapter on forced oscillation. One of the features of a forced oscillation which we have not yet discussed is the *energy* in the oscillation. Let us now consider that energy.

In a mechanical oscillator, how much kinetic energy is there? It is proportional to the square of the velocity. Now we come to an important point. Consider an arbitrary quantity A , which may be the velocity or something else that we want to discuss. When we write $A = \hat{A}e^{i\omega t}$, a complex number, the true and honest A , in the physical world, is only the *real part*; therefore if, for some reason, we want to use the *square* of A , it is not right to square the complex number and then take the real part, because the real part of the square of a complex number is not just the square of the real part, but also involves the *imaginary* part. So when we wish to find the energy we have to get away from the complex notation for a while to see what the inner workings are.

Now the true physical A is the real part of $A_0 e^{i(\omega t + \Delta)}$, that is, $A = A_0 \cos(\omega t + \Delta)$, where $\hat{A}$, the complex number, is written as $A_0 e^{i\Delta}$. Now the square of this real physical quantity is $A^2 = A_0^2 \cos^2(\omega t + \Delta)$. The square of the quantity, then, goes up and down from a maximum to zero, like the square of the cosine. The square of the cosine has a maximum of 1 and a minimum of 0, and its average value is 1/2.

In many circumstances we are not interested in the energy at any specific moment during the oscillation; for a large number of applications we merely want the average of A^2 , the *mean* of the square of A over a period of time large compared with the period of oscillation. In those circumstances, the average of the cosine squared may be used, so we have the following theorem: if A is represented by a complex number, then the mean of A^2 is equal to $\frac{1}{2}A_0^2$. Now A_0^2 is the square of the magnitude of the complex $\hat{A}$. (This can be written in many ways—some people like to write $|\hat{A}|^2$; others write, $\hat{A}\hat{A}^*$, $\hat{A}$ times its complex conjugate.) We shall use this theorem several times.

Now let us consider the energy in a forced oscillator. The equation for the forced oscillator is

$$m d^2x/dt^2 + \gamma m dx/dt + m\omega_0^2 x = F(t). \quad (24.1)$$

In our problem, of course, $F(t)$ is a cosine function of t . Now let us analyze the situation: how much work is done by the outside force F ? The work done by the force per second, i.e., the power, is the force times the velocity. (We know that the differential work in a time dt is $F dx$, and the power is $F dx/dt$.) Thus

$$P = F \frac{dx}{dt} = m \left[\left(\frac{dx}{dt} \right) \left(\frac{d^2x}{dt^2} \right) + \omega_0^2 x \left(\frac{dx}{dt} \right) \right] + \gamma m \left(\frac{dx}{dt} \right)^2. \quad (24.2)$$

But the first two terms on the right can also be written as $d/dt[\frac{1}{2}m(dx/dt)^2 + \frac{1}{2}m\omega_0^2 x^2]$, as is immediately verified by differentiating. That is to say, the term in brackets is a pure derivative of two terms that are easy to understand—one is the kinetic energy of motion, and the other is the potential energy of the spring. Let us call this quantity the *stored energy*, that is, the energy stored in the oscillation. Suppose that we want the average power over many cycles when the oscillator is being forced and has been running for a long time. In the long run, the stored

24-1 The energy of an oscillator

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energy does not change—its derivative gives zero average effect. In other words, if we average the power in the long run, *all the energy ultimately ends up in the resistive term* $\gamma m(dx/dt)^2$. There is *some* energy stored in the oscillation, but that does not change with time, if we average over many cycles. Therefore the mean power $\langle P \rangle$ is

$$\langle P \rangle = \langle \gamma m(dx/dt)^2 \rangle. \quad (24.3)$$

Using our method of writing complex numbers, and our theorem that $\langle A^2 \rangle = \frac{1}{2}A_0^2$, we may find this mean power. Thus if $x = \hat{x}e^{i\omega t}$, then $dx/dt = i\omega\hat{x}e^{i\omega t}$. Therefore, in these circumstances, the average power could be written as

$$\langle P \rangle = \frac{1}{2}\gamma m\omega^2 x_0^2. \quad (24.4)$$

In the notation for electrical circuits, dx/dt is replaced by the current I (I is dq/dt , where q corresponds to x), and $m\gamma$ corresponds to the resistance R . Thus the rate of the energy loss—the power used up by the forcing function—is the resistance in the circuit times the average square of the current:

$$\langle P \rangle = R\langle I^2 \rangle = R \cdot \frac{1}{2}I_0^2. \quad (24.5)$$

This energy, of course, goes into heating the resistor; it is sometimes called the heating loss or the Joule heating.

Another interesting feature to discuss is how much energy is *stored*. That is not the same as the power, because although power was at first used to store up some energy, after that the system keeps on absorbing power, insofar as there are any heating (resistive) losses. At any moment there is a certain amount of stored energy, so we would like to calculate the mean stored energy $\langle E \rangle$ also. We have already calculated what the average of $(dx/dt)^2$ is, so we find

$$\begin{aligned} \langle E \rangle &= \frac{1}{2}m\langle (dx/dt)^2 \rangle + \frac{1}{2}m\omega_0^2\langle x^2 \rangle \\ &= \frac{1}{2}m(\omega^2 + \omega_0^2)\frac{1}{2}x_0^2. \end{aligned} \quad (24.6)$$

Now, when an oscillator is very efficient, and if ω is near ω_0 , so that $|\hat{x}|$ is large, the stored energy is very high—we can get a large stored energy from a relatively small force. The force does a great deal of work in getting the oscillation going, but then to keep it steady, all it has to do is to fight the friction. The oscillator can have a great deal of energy if the friction is very low, and even though it is oscillating strongly, not much energy is being lost. The efficiency of an oscillator can be measured by how much energy is stored, compared with how much work the force does per oscillation.

How does the stored energy compare with the amount of work that is done in one cycle? This is called the Q of the system, and Q is defined as 2π times the mean stored energy, divided by the work done per cycle. (If we were to say the work done per *radian* instead of per cycle, then the 2π disappears.)

$$Q = 2\pi \frac{\frac{1}{2}m(\omega^2 + \omega_0^2) \cdot \langle x^2 \rangle}{\gamma m\omega^2 \langle x^2 \rangle \cdot 2\pi/\omega} = \frac{\omega^2 + \omega_0^2}{2\gamma\omega}. \quad (24.7)$$

Q is not a very useful number unless it is very large. When it is relatively large, it gives a measure of how good the oscillator is. People have tried to define Q in the simplest and most useful way; various definitions differ a bit from one another, but if Q is very large, all definitions are in agreement. The most generally accepted definition is Eq. (24.7), which depends on ω . For a good oscillator, close to resonance, we can simplify (24.7) a little by setting $\omega = \omega_0$, and we then have $Q = \omega_0/\gamma$, which is the definition of Q that we used before.

What is Q for an electrical circuit? To find out, we merely have to translate L for m , R for $m\gamma$, and $1/C$ for $m\omega_0^2$ (see Table 23-1). The Q at resonance is $L\omega/R$, where ω is the resonance frequency. If we consider a circuit with a high Q , that means that the amount of energy stored in the oscillation is very large compared with the amount of work done per cycle by the machinery that drives the oscillations.

24-2 Damped oscillations

We now turn to our main topic of discussion: transients. By a *transient* is meant a solution of the differential equation when there is no force present, but when the system is not simply at rest. (Of course, if it is standing still at the origin with no force acting, that is a nice problem—it stays there!) Suppose the oscillation starts another way: say it was driven by a force for a while, and then we turn off the force. What happens then? Let us first get a rough idea of what will happen for a very high Q system. So long as a force is acting, the stored energy stays the same, and there is a certain amount of work done to maintain it. Now suppose we turn off the force, and no more work is being done; then the losses which are eating up the energy of the supply are no longer eating up its energy—there *is* no more driver. The losses will have to consume, so to speak, the energy that is stored. Let us suppose that $Q/2\pi = 1000$. Then the work done per cycle is $1/1000$ of the stored energy. Is it not reasonable, since it is oscillating with no driving force, that in one cycle the system will still lose a thousandth of its energy E , which ordinarily would have been supplied from the outside, and that it will continue oscillating, always losing $1/1000$ of its energy per cycle? So, as a guess, for a relatively high Q system, we would suppose that the following equation might be roughly right (we will later do it exactly, and it will turn out that it *was* right!):

$$dE/dt = -\omega E/Q. \quad (24.8)$$

This is rough because it is true only for large Q . In each radian the system loses a fraction $1/Q$ of the stored energy E . Thus in a given amount of time dt the energy will change by an amount $\omega dt/Q$, since the number of radians associated with the time dt is ωdt . What is the frequency? Let us suppose that the system moves so nicely, with hardly any force, that if we let go it will oscillate at essentially the same frequency all by itself. So we will guess that ω is the resonant frequency ω_0 . Then we deduce from Eq. (24.8) that the stored energy will vary as

$$E = E_0 e^{-\omega_0 t/Q} = E_0 e^{-\gamma t}. \quad (24.9)$$

This would be the measure of the *energy* at any moment. What would the formula be, roughly, for the amplitude of the oscillation as a function of the time? The same? No! The amount of energy in a spring, say, goes as the *square* of the displacement; the kinetic energy goes as the *square* of the velocity; so the total energy goes as the *square* of the displacement. Thus the displacement, the amplitude of oscillation, will decrease half as fast because of the square. In other words, we guess that the solution for the damped transient motion will be an oscillation of frequency close to the resonance frequency ω_0 , in which the amplitude of the sine-wave motion will diminish as $e^{-\gamma t/2}$:

$$x = A_0 e^{-\gamma t/2} \cos \omega_0 t. \quad (24.10)$$

This equation and Fig. 24-1 give us an idea of what we should expect; now let us try to analyze the motion *precisely* by solving the differential equation of the motion itself.

So, starting with Eq. (24.1), with no outside force, how do we solve it? Being physicists, we do not have to worry about the *method* as much as we do about what the solution *is*. Armed with our previous experience, let us try as a solution an exponential curve, $x = Ae^{i\alpha t}$. (Why do we try this? It is the easiest thing to differentiate!) We put this into (24.1) (with $F(t) = 0$), using the rule that each time we differentiate x with respect to time, we multiply by $i\alpha$. So it is really quite simple to substitute. Thus our equation looks like this:

$$(-\alpha^2 + i\gamma\alpha + \omega_0^2)Ae^{i\alpha t} = 0. \quad (24.11)$$

The net result must be zero for *all times*, which is impossible unless (a) $A = 0$, which is no solution at all—it stands still, or (b)

$$-\alpha^2 + i\alpha\gamma + \omega_0^2 = 0. \quad (24.12)$$

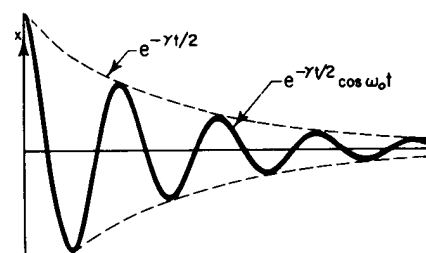


Fig. 24-1. A damped cosine oscillation.

If we can solve this and find an α , then we will have a solution in which A need not be zero!

$$\alpha = i\gamma/2 \pm \sqrt{\omega_0^2 - \gamma^2/4}. \quad (24.13)$$

For a while we shall assume that γ is fairly small compared with ω_0 , so that $\omega_0^2 - \gamma^2/4$ is definitely positive, and there is nothing the matter with taking the square root. The only bothersome thing is that we get *two* solutions! Thus

$$\alpha_1 = i\gamma/2 + \sqrt{\omega_0^2 - \gamma^2/4} = i\gamma/2 + \omega_\gamma \quad (24.14)$$

and

$$\alpha_2 = i\gamma/2 - \sqrt{\omega_0^2 - \gamma^2/4} = i\gamma/2 - \omega_\gamma. \quad (24.15)$$

Let us consider the first one, supposing that we had not noticed that the square root has two possible values. Then we know that a solution for x is $x_1 = Ae^{i\alpha_1 t}$, where A is any constant whatever. Now, in substituting α_1 , because it is going to come so many times and it takes so long to write, we shall call $\sqrt{\omega_0^2 - \gamma^2/4} = \omega_\gamma$. Thus $i\alpha_1 = -\gamma/2 + i\omega_\gamma$, and we get $x = Ae^{(-\gamma/2 + i\omega_\gamma)t}$, or what is the same, because of the wonderful properties of an exponential,

$$x_1 = Ae^{-\gamma t/2} e^{i\omega_\gamma t}. \quad (24.16)$$

First, we recognize this as an oscillation, an oscillation at a frequency ω_γ which is not *exactly* the frequency ω_0 , but is rather close to ω_0 if it is a good system. Second, the amplitude of the oscillation is decreasing exponentially! If we take, for instance, the real part of (24.16), we get

$$x_1 = Ae^{-\gamma t/2} \cos \omega_\gamma t. \quad (24.17)$$

This is very much like our guessed-at solution (24.10), except that the frequency really is ω_γ . This is the only error, so it is the same thing—we have the right idea. But everything is *not* all right! What is not all right is that *there is another solution*.

The other solution is α_2 , and we see that the difference is only that the sign of ω_γ is reversed:

$$x_2 = Be^{-\gamma t/2} e^{-i\omega_\gamma t}. \quad (24.18)$$

What does this mean? We shall soon prove that if x_1 and x_2 are each a possible solution of Eq. (24.1) with $F = 0$, then $x_1 + x_2$ is also a solution of the same equation! So the general solution x is of the mathematical form

$$x = e^{-\gamma t/2} (Ae^{i\omega_\gamma t} + Be^{-i\omega_\gamma t}). \quad (24.19)$$

Now we may wonder why we bother to give this other solution, since we were so happy with the first one all by itself. What is the extra one for, because of course we know we should only take the real part? *We* know that we must take the real part, but how did the *mathematics* know that we only wanted the real part? When we had a nonzero driving force $F(t)$, we put in an *artificial* force to go with it, and the *imaginary* part of the equation, so to speak, was driven in a definite way. But when we put $F(t) \equiv 0$, our convention that x should be only the real part of whatever we write down is purely our own, and the mathematical equations do not know it yet. The physical world *has* a real solution, but the answer that we were so happy with before is not real, it is *complex*. The *equation* does not know that we are arbitrarily going to take the real part, so it has to present us, so to speak, with a complex conjugate type of solution, so that by putting them together we can *make a truly real* solution; that is what α_2 is doing for us. In order for x to be real, $Be^{-i\omega_\gamma t}$ will have to be the complex conjugate of $Ae^{i\omega_\gamma t}$, so that the imaginary parts disappear. So it turns out that B is the complex conjugate of A , and our real solution is

$$x = e^{-\gamma t/2} (Ae^{i\omega_\gamma t} + A^* e^{-i\omega_\gamma t}). \quad (24.20)$$

So our real solution is an oscillation with a *phase shift* and a damping—just as advertised.

24-3 Electrical transients

Now let us see if the above really works. We construct the electrical circuit shown in Fig. 24-2, in which we apply to an oscilloscope the voltage across the inductance L after we suddenly turn on a voltage by closing the switch S . It is an oscillatory circuit, and it generates a transient of some kind. It corresponds to a circumstance in which we suddenly apply a force and the system starts to oscillate. It is the electrical analog of a damped mechanical oscillator, and we watch the oscillation on an oscilloscope, where we should see the curves that we were trying to analyze. (The horizontal motion of the oscilloscope is driven at a uniform speed, while the vertical motion is the voltage across the inductor. The rest of the circuit is only a technical detail. We would like to repeat the experiment many, many times, since the persistence of vision is not good enough to see only one trace on the screen. So we do the experiment again and again by closing the switch 60 times a second; each time we close the switch, we also start the oscilloscope horizontal sweep, and it draws the curve over and over.) In Figs. 24-3 to 24-6 we see examples of damped oscillations, actually photographed on an oscilloscope screen. Figure 24-3 shows a damped oscillation in a circuit which has a high Q , a small γ . It does not die out very fast; it oscillates many times on the way down.

But let us see what happens as we decrease Q , so that the oscillation dies out more rapidly. We can decrease Q by increasing the resistance R in the circuit. When we increase the resistance in the circuit, it dies out faster (Fig. 24-4). Then if we increase the resistance in the circuit still more, it dies out faster still (Fig. 24-5). But when we put in more than a certain amount, we cannot see any oscillation at all! The question is, is this because our eyes are not good enough? If we increase the resistance still more, we get a curve like that of Fig. 24-6, which does not appear to have any oscillations, except perhaps one. Now, how can we explain that by mathematics?

The resistance is, of course, proportional to the γ term in the mechanical device. Specifically, γ is R/L . Now if we increase the γ in the solutions (24.14) and (24.15) that we were so happy with before, chaos sets in when $\gamma/2$ exceeds ω_0 ; we must write it a different way, as

$$i\gamma/2 + i\sqrt{\gamma^2/4 - \omega_0^2} \quad \text{and} \quad i\gamma/2 - i\sqrt{\gamma^2/4 - \omega_0^2}.$$

Those are now the two solutions and, following the same line of mathematical reasoning as previously, we again find two solutions: $e^{i\alpha_1 t}$ and $e^{i\alpha_2 t}$. If we now substitute for α_1 , we get

$$x = Ae^{-(\gamma/2 + \sqrt{\gamma^2/4 - \omega_0^2})t},$$

a nice exponential decay with no oscillations. Likewise, the other solution is

$$x = Be^{-(\gamma/2 - \sqrt{\gamma^2/4 - \omega_0^2})t}.$$

Note that the square root cannot exceed $\gamma/2$, because even if $\omega_0 = 0$, one term just equals the other. But ω_0^2 is taken away from $\gamma^2/4$, so the square root is less than $\gamma/2$, and the term in parentheses is, therefore, always a positive number. Thank goodness! Why? Because if it were negative, we would find e raised to a *positive* factor times t , which would mean it was exploding! In putting more and more resistance into the circuit, we know it is not going to explode—quite the contrary. So now we have two solutions, each one by itself a dying exponential, but one having a much faster “dying rate” than the other. The general solution is of course a combination of the two; the coefficients in the combination depending upon how the motion starts—what the initial conditions of the problem are. In the particular way this circuit happens to be starting, the A is negative and the B is positive, so we get the difference of two exponential curves.

Now let us discuss how we can find the two coefficients A and B (or A and A^*), if we know how the motion was started.

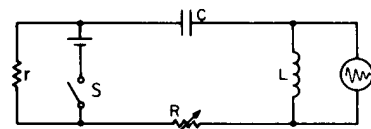


Fig. 24-2. An electrical circuit for demonstrating transients.

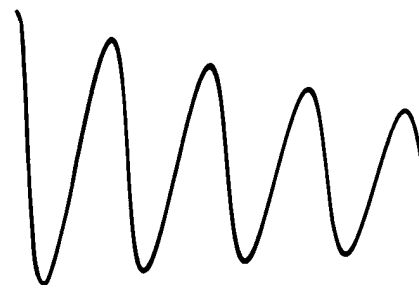


Figure 24-3

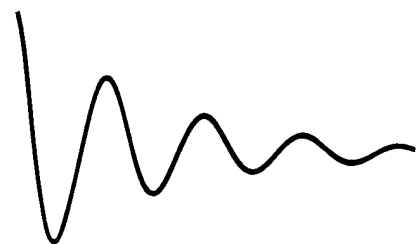


Figure 24-4

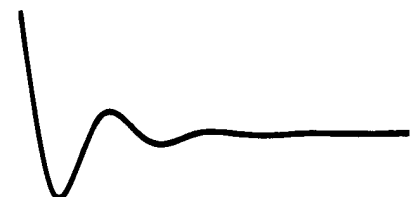


Figure 24-5



Figure 24-6

Suppose that at $t = 0$ we know that $x = x_0$, and $dx/dt = v_0$. If we put $t = 0$, $x = x_0$, and $dx/dt = v_0$ into the expressions

$$x = e^{-\gamma t/2}(Ae^{i\omega_\gamma t} + A^*e^{-i\omega_\gamma t}),$$

$$dx/dt = e^{-\gamma t/2}[(-\gamma/2 + i\omega_\gamma)Ae^{i\omega_\gamma t} + (-\gamma/2 - i\omega_\gamma)A^*e^{-i\omega_\gamma t}],$$

we find, since $e^0 = e^{i0} = 1$,

$$x_0 = A + A^* = 2A_R,$$

$$v_0 = (-\gamma/2)(A + A^*) + i\omega_\gamma(A - A^*)$$

$$= -\gamma x_0/2 + i\omega_\gamma(2iA_I),$$

where $A = A_R + iA_I$, and $A^* = A_R - iA_I$. Thus we find

$$A_R = x_0/2$$

and

$$A_I = (v_0 + \gamma x_0/2)/2\omega_\gamma. \quad (24.21)$$

This completely determines A and A^* , and therefore the complete curve of the transient solution, in terms of how it begins. Incidentally, we can write the solution another way if we note that

$$e^{i\theta} + e^{-i\theta} = 2 \cos \theta \quad \text{and} \quad e^{i\theta} - e^{-i\theta} = 2i \sin \theta.$$

We may then write the complete solution as

$$x = e^{-\gamma t/2} \left[x_0 \cos \omega_\gamma t + \frac{v_0 + \gamma x_0/2}{\omega_\gamma} \sin \omega_\gamma t \right], \quad (24.22)$$

where $\omega_\gamma = +\sqrt{\omega_0^2 - \gamma^2/4}$. This is the mathematical expression for the way an oscillation dies out. We shall not make direct use of it, but there are a number of points we should like to emphasize that are true in more general cases.

First of all the behavior of such a system with no external force is expressed by a sum, or superposition, of pure exponentials in time (which we wrote as $e^{i\alpha t}$). This is a good solution to try in such circumstances. The values of α may be complex in general, the imaginary parts representing damping. Finally the intimate mathematical relation of the sinusoidal and exponential function discussed in Chapter 22 often appears physically as a change from oscillatory to exponential behavior when some physical parameter (in this case resistance, γ) exceeds some critical value.

Linear Systems and Review

25-1 Linear differential equations

In this chapter we shall discuss certain aspects of oscillating systems that are found somewhat more generally than just in the particular systems we have been discussing. For our particular system, the differential equation that we have been solving is

$$m \frac{d^2x}{dt^2} + \gamma m \frac{dx}{dt} + m\omega_0^2 x = F(t). \quad (25.1)$$

Now this particular combination of “operations” on the variable x has the interesting property that if we substitute $(x + y)$ for x , then we get the sum of the same operations on x and y ; or, if we multiply x by a , then we get just a times the same combination. This is easy to prove. Just as a “shorthand” notation, because we get tired of writing down all those letters in (25.1), we shall use the symbol $\underline{L}(x)$ instead. When we see this, it means the left-hand side of (25.1), with x substituted in. With this system of writing, $\underline{L}(x + y)$ would mean the following:

$$\underline{L}(x + y) = m \frac{d^2(x + y)}{dt^2} + \gamma m \frac{d(x + y)}{dt} + m\omega_0^2(x + y). \quad (25.2)$$

(We underline the $\underline{L}$ so as to remind ourselves that it is not an ordinary function.) We sometimes call this an *operator notation*, but it makes no difference what we call it, it is just “shorthand.”

Our first statement was that

$$\underline{L}(x + y) = \underline{L}(x) + \underline{L}(y), \quad (25.3)$$

which of course follows from the fact that $a(x + y) = ax + ay$, $d(x + y)/dt = dx/dt + dy/dt$, etc.

Our second statement was, for constant a ,

$$\underline{L}(ax) = a\underline{L}(x). \quad (25.4)$$

[Actually, (25.3) and (25.4) are very closely related, because if we put $x + x$ into (25.3), this is the same as setting $a = 2$ in (25.4), and so on.]

In more complicated problems, there may be more derivatives, and more terms in $\underline{L}$; the question of interest is whether the two equations (25.3) and (25.4) are maintained or not. If they are, we call such a problem a *linear* problem. In this chapter we shall discuss some of the properties that exist because the system is linear, to appreciate the generality of some of the results that we have obtained in our special analysis of a special equation.

Now let us study some of the properties of linear differential equations, having illustrated them already with the specific equation (25.1) that we have studied so closely. The first property of interest is this: suppose that we have to solve the differential equation for a transient, the free oscillation with no driving force. That is, we want to solve

$$\underline{L}(x) = 0. \quad (25.5)$$

Suppose that, by some hook or crook, we have found a particular solution, which we shall call x_1 . That is, we have an x_1 for which $\underline{L}(x_1) = 0$. Now we notice that ax_1 is also a solution to the same equation; we can multiply this special solution by any constant whatever, and get a new solution. In other words, if we had a

25-1 Linear differential equations

25-2 Superposition of solutions

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25-4 Analogs in physics

25-5 Series and parallel impedances

motion of a certain “size,” then a motion twice as “big” is again a solution. *Proof:* $\underline{L}(ax_1) = a\underline{L}(x_1) = a \cdot 0 = 0$.

Next, suppose that, by hook or by crook, we have not only found *one* solution x_1 , but also another solution, x_2 . (Remember that when we substituted $x = e^{iat}$ for finding the transients, we found *two* values for α , that is, two solutions, x_1 and x_2 .) Now let us show that the combination $(x_1 + x_2)$ is also a solution. In other words, if we put $x = x_1 + x_2$, x is again a solution of the equation. Why? Because, if $\underline{L}(x_1) = 0$ and $\underline{L}(x_2) = 0$, then $\underline{L}(x_1 + x_2) = \underline{L}(x_1) + \underline{L}(x_2) = 0 + 0 = 0$. So if we have found a number of solutions for the motion of a linear system we can add them together.

Combining these two ideas, we see, of course, that we can also add six of one and two of the other: if x_1 is a solution, so is αx_1 . Therefore any sum of these two solutions, such as $(\alpha x_1 + \beta x_2)$, is also a solution. If we happen to be able to find three solutions, then we find that any combination of the three solutions is again a solution, and so on. It turns out that the number of what we call *independent solutions** that we have obtained for our oscillator problem is only *two*. The number of independent solutions that one finds in the general case depends upon what is called the number of *degrees of freedom*. We shall not discuss this in detail now, but if we have a second-order differential equation, there are only two independent solutions, and we have found both of them; so we have the most general solution.

Now let us go on to another proposition, which applies to the situation in which the system is subjected to an outside force. Suppose the equation is

$$\underline{L}(x) = F(t), \quad (25.6)$$

and suppose that we have found a special solution of it. Let us say that Joe’s solution is x_J , and that $\underline{L}(x_J) = F(t)$. Suppose we want to find yet another solution; suppose we add to Joe’s solution one of those that was a solution of the free equation (25.5), say x_1 . Then we see by (25.3) that

$$\underline{L}(x_J + x_1) = \underline{L}(x_J) + \underline{L}(x_1) = F(t) + 0 = F(t). \quad (25.7)$$

Therefore, to the “forced” solution we can add any “free” solution, and we still have a solution. The free solution is called a *transient* solution.

When we have no force acting, and suddenly turn one on, we do not immediately get the steady solution that we solved for with the sine wave solution, but for a while there is a transient which sooner or later dies out, if we wait long enough. The “forced” solution does not die out, since it keeps on being driven by the force. Ultimately, for long periods of time, the solution is unique, but initially the motions are different for different circumstances, depending on how the system was started.

25-2 Superposition of solutions

Now we come to another interesting proposition. Suppose that we have a certain particular driving force F_a (let us say an oscillatory one with a certain $\omega = \omega_a$, but our conclusions will be true for any functional form of F_a) and we have solved for the forced motion (with or without the transients; it makes no difference). Now suppose some other force is acting, let us say F_b , and we solve the same problem, but for this different force. Then suppose someone comes along and says, “I have a new problem for you to solve; I have the force $F_a + F_b$.” Can we do it? Of course we can do it, because the solution is the sum of the two solutions x_a and x_b for the forces taken separately—a most remarkable circumstance indeed. If we use (25.3), we see that

$$\underline{L}(x_a + x_b) = \underline{L}(x_a) + \underline{L}(x_b) = F_a(t) + F_b(t). \quad (25.8)$$

* Solutions which cannot be expressed as linear combinations of each other are called independent.

This is an example of what is called the *principle of superposition* for linear systems, and it is very important. It means the following: if we have a complicated force which can be broken up in any convenient manner into a sum of separate pieces, each of which is in some way simple, in the sense that for each special piece into which we have divided the force we can solve the equation, then the answer is available for the *whole* force, because we may simply add the pieces of the *solution* back together, in the same manner as the total *force* is compounded out of pieces (Fig. 25-1).

Let us give another example of the principle of superposition. In Chapter 12 we said that it was one of the great facts of the laws of electricity that if we have a certain distribution of charges q_a and calculate the electric field E_a arising from these charges at a certain place P , and if, on the other hand, we have another set of charges q_b and we calculate the field E_b due to these at the corresponding place, then if both charge distributions are present at the same time, the field E at P is the *sum* of E_a due to one set plus E_b due to the other. In other words, if we know the field due to a certain charge, then the field due to many charges is merely the vector sum of the fields of these charges taken individually. This is exactly analogous to the above proposition that if we know the result of two given forces taken at one time, then if the force is considered as a sum of them, the response is a sum of the corresponding individual responses.

The reason why this is true in electricity is that the great laws of electricity, Maxwell's equations, which determine the electric field, turn out to be differential equations which are *linear*, i.e., which have the property (25.3). What corresponds to the force is the *charge* generating the electric field, and the equation which determines the electric field in terms of the charge is linear.

As another interesting example of this proposition, let us ask how it is possible to "tune in" to a particular radio station at the same time as all the radio stations are broadcasting. The radio station transmits, fundamentally, an oscillating electric field of very high frequency which acts on our radio antenna. It is true that the amplitude of the oscillation of the field is changed, modulated, to carry the signal of the voice, but that is very slow, and we are not going to worry about it. When one hears "This station is broadcasting at a frequency of 780 kilocycles," this indicates that 780,000 oscillations per second is the frequency of the electric field of the station antenna, and this drives the electrons up and down at that frequency in our antenna. Now at the same time we may have another radio station in the same town radiating at a different frequency, say 550 kilocycles per second; then the electrons in our antenna are also being driven by that frequency. Now the question is, how is it that we can separate the signals coming into the one radio at 780 kilocycles from those coming in at 550 kilocycles? We certainly do not hear both stations at the same time.

By the principle of superposition, the response of the electric circuit in the radio, the first part of which is a linear circuit, to the forces that are acting due to the electric field $F_a + F_b$, is $x_a + x_b$. It therefore looks as though we will never disentangle them. In fact, the very proposition of superposition seems to insist that we cannot *avoid* having both of them in our system. But remember, for a *resonant* circuit, the response curve, the amount of x per unit F , as a function of the frequency, looks like Fig. 25-3. If it were a very high Q circuit, the response would show a very sharp maximum. Now suppose that the two stations are comparable in strength, that is, the two *forces* are of the same magnitude. The *response* that we get is the sum of x_a and x_b . But, in Fig. 25-3, x_a is tremendous, while x_b is small. So, in spite of the fact that the two signals are equal in strength, when they go through the sharp resonant circuit of the radio tuned for ω_a , the frequency of the transmission of one station, then the response to this station is much greater than to the other. Therefore the complete response, with both signals acting, is almost all made up of ω_a , and we have selected the station we want.

Now what about the tuning? How do we tune it? We change ω_0 by changing the L or the C of the circuit, because the frequency of the circuit has to do with the combination of L and C . In particular, most radios are built so that one can change the capacitance. When we retune the radio, we can make a new setting of

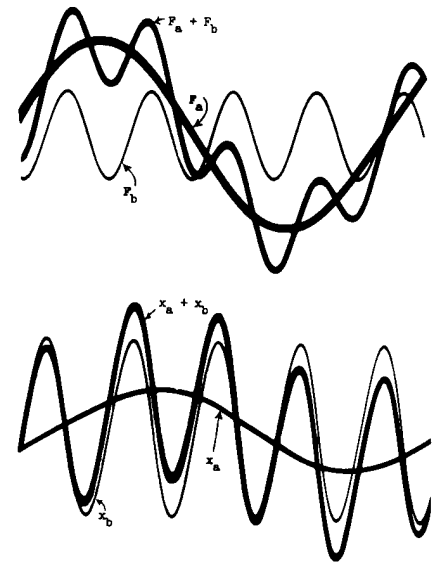


Fig. 25-1. An example of the principle of superposition for linear systems.

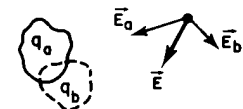


Fig. 25-2. The principle of superposition in electrostatics.

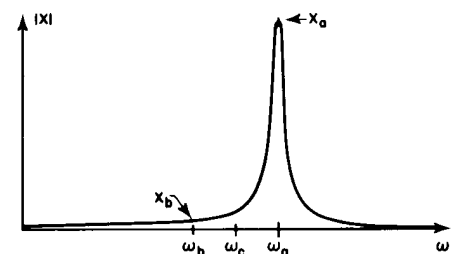


Fig. 25-3. A sharply tuned resonance curve.

the dial, so that the natural frequency of the circuit is shifted, say, to ω_c . In those circumstances we hear neither one station nor the other; we get silence, provided there is no other station at frequency ω_c . If we keep on changing the capacitance until the resonance curve is at ω_b , then of course we hear the other station. That is how radio tuning works; it is again the principle of superposition, combined with a resonant response.*

To conclude this discussion, let us describe qualitatively what happens if we proceed further in analyzing a linear problem with a given force, when the force is quite complicated. Out of the many possible procedures, there are two especially useful general ways that we can solve the problem. One is this: suppose that we can solve it for special known forces, such as sine waves of different frequencies. We know it is child's play to solve it for sine waves. So we have the so-called "child's play" cases. Now the question is whether our very complicated force can be represented as the sum of two or more "child's play" forces. In Fig. 25-1 we already had a fairly complicated curve, and of course we can make it more complicated still if we add in more sine waves. So it is certainly possible to obtain very complicated curves. And, in fact, the reverse is also true: practically every curve can be obtained by adding together *infinite numbers* of sine waves of different wavelengths (or frequencies) for each one of which we know the answer. We just have to know how much of each sine wave to put in to make the given F , and then our answer, x , is the corresponding sum of the F sine waves, each multiplied by its effective ratio of x to F . This method of solution is called the method of *Fourier transforms* or *Fourier analysis*. We are not going to actually carry out such an analysis just now; we only wish to describe the idea involved.

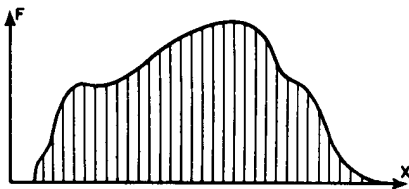


Fig. 25-4. A complicated force may be treated as a succession of sharp impulses.

Another way in which our complicated problem can be solved is the following very interesting one. Suppose that, by some tremendous mental effort, it were possible to solve our problem for a special force, namely an *impulse*. The force is quickly turned on and then off; it is all over. Actually we need only solve for an impulse of some unit strength, any other strength can be gotten by multiplication by an appropriate factor. We know that the response x for an impulse is a damped oscillation. Now what can we say about some other force, for instance a force like that of Fig. 25-4?

Such a force can be likened to a succession of blows with a hammer. First there is no force, and all of a sudden there is a steady force—impulse, impulse, impulse, impulse, . . . and then it stops. In other words, we imagine the continuous force to be a series of impulses, very close together. Now, we know the result for an impulse, so the result for a whole series of impulses will be a whole series of damped oscillations: it will be the curve for the first impulse, and then (slightly later) we add to that the curve for the second impulse, and the curve for the third impulse, and so on. Thus we can represent, mathematically, the complete solution for arbitrary functions if we know the answer for an impulse. We get the answer for any other force simply by integrating. This method is called the *Green's function method*. A Green's function is a response to an impulse, and the method of analyzing any force by putting together the response of impulses is called the Green's function method.

The physical principles involved in both of these schemes are so simple, involving just the linear equation, that they can be readily understood, but the *mathematical* problems that are involved, the complicated integrations and so on, are a little too advanced for us to attack right now. You will most likely return to this some day when you have had more practice in mathematics. But the *idea* is very simple indeed.

Finally, we make some remarks on why *linear* systems are so important. The answer is simple: because we can solve them! So most of the time we solve linear

* In modern superheterodyne receivers the actual operation is more complex. The amplifiers are all tuned to a fixed frequency (called IF frequency) and an oscillator of variable tunable frequency is combined with the input signal in a *nonlinear* circuit to produce a new frequency (the difference of signal and oscillator frequency) equal to the IF frequency, which is then amplified. This will be discussed in Chapter 50.

problems. Second (and most important), it turns out that *the fundamental laws of physics are often linear*. The Maxwell equations for the laws of electricity are linear, for example. The great laws of quantum mechanics turn out, so far as we know, to be linear equations. *That* is why we spend so much time on linear equations: because if we understand linear equations, we are ready, in principle, to understand a lot of things.

We mention another situation where linear equations are found. When displacements are small, many functions can be *approximated* linearly. For example, if we have a simple pendulum, the correct equation for its motion is

$$d^2\theta/dt^2 = -(g/L) \sin \theta. \quad (25.9)$$

This equation can be solved by elliptic functions, but the easiest way to solve it is numerically, as was shown in Chapter 9 on Newton's Laws of Motion. A non-linear equation cannot be solved, ordinarily, any other way *but* numerically. Now for small θ , $\sin \theta$ is practically equal to θ , and we have a linear equation. It turns out that there are many circumstances where small effects are linear: for the example here the swing of a pendulum through small arcs. As another example, if we pull a little bit on a spring, the force is proportional to the extension. If we pull hard, we break the spring, and the force is a completely different function of the distance! Linear equations are important. In fact they are *so* important that perhaps fifty percent of the time we are solving linear equations in physics and in engineering.

25-3 Oscillations in linear systems

Let us now review the things we have been talking about in the past few chapters. It is very easy for the physics of oscillators to become obscured by the mathematics. The physics is actually very simple, and if we may forget the mathematics for a moment we shall see that we can understand almost everything that happens in an oscillating system. First, if we have only the spring and the weight, it is easy to understand why the system oscillates—it is a consequence of inertia. We pull the mass down and the force pulls it back up; as it passes zero, which is the place it likes to be, it cannot just suddenly stop; because of its momentum it keeps on going and swings to the other side, and back and forth. So, if there were no friction, we would surely *expect* an oscillatory motion, and indeed we get one. But if there is even a little bit of friction, then on the return cycle, the swing will not be quite as high as it was the first time.

Now what happens, cycle by cycle? That depends on the kind and amount of friction. Suppose that we could concoct a kind of friction force that always remains in the same proportion to the other forces, of inertia and in the spring, as the amplitude of oscillation varies. In other words, for smaller oscillations the friction should be weaker than for big oscillations. Ordinary friction does not have this property, so a special kind of friction must be carefully invented for the very purpose of creating a friction that is directly proportional to the velocity—so that for big oscillations it is stronger and for small oscillations it is weaker. If we happen to have that kind of friction, then at the end of each successive cycle the system is in the same condition as it was at the start, except a little bit smaller. All the forces are smaller in the same proportion: the spring force is reduced, the inertial effects are lower because the accelerations are now weaker, and the friction is less too, by our careful design. When we actually have that kind of friction, we find that each oscillation is exactly the same as the first one, except reduced in amplitude. If the first cycle dropped the amplitude, say, to 90 percent of what it was at the start, the next will drop it to 90 percent of 90 percent, and so on: *the sizes of the oscillations are reduced by the same fraction of themselves in every cycle*. An exponential function is a curve which does just that. It changes by the same factor in each equal interval of time. That is to say, if the amplitude of one cycle, relative to the preceding one, is called a , then the amplitude of the next is a^2 , and of the next, a^3 . So the amplitude is some constant raised to a power equal to the number of cycles traversed:

$$A = A_0 a^n. \quad (25.10)$$

But of course $n \sim t$, so it is perfectly clear that the general solution will be some kind of an oscillation, sine or cosine ωt , times an amplitude which goes as b^t more or less. But b can be written as e^{-c} , if b is positive and less than 1. So this is why the solution looks like $e^{-ct} \cos \omega t$. It is very simple.

What happens if the friction is not so artificial; for example, ordinary rubbing on a table, so that the friction force is a certain constant amount, and is independent of the size of the oscillation that reverses its direction each half-cycle? Then the equation is no longer linear, it becomes hard to solve, and must be solved by the numerical method given in Chapter 2, or by considering each half-cycle separately. The numerical method is the most powerful method of all, and can solve any equation. It is only when we have a simple problem that we can use mathematical analysis.

Mathematical analysis is not the grand thing it is said to be; it solves only the simplest possible equations. As soon as the equations get a little more complicated, just a shade—they cannot be solved analytically. But the numerical method, which was advertised at the beginning of the course, can take care of any equation of physical interest.

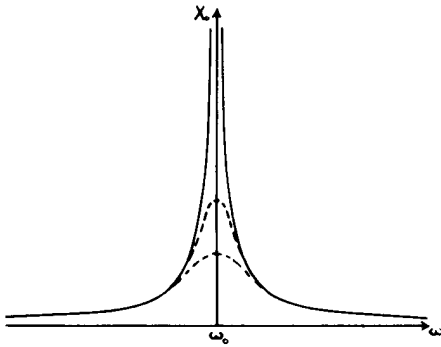


Fig. 25-5. Resonance curves with various amounts of friction present.

Next, what about the resonance curve? Why is there a resonance? First, imagine for a moment that there is no friction, and we have something which could oscillate by itself. If we tapped the pendulum just right each time it went by, of course we could make it go like mad. But if we close our eyes and do not watch it, and tap at arbitrary equal intervals, what is going to happen? Sometimes we will find ourselves tapping when it is going the wrong way. When we happen to have the timing just right, of course, each tap is given at just the right time, and so it goes higher and higher and higher. So without friction we get a curve which looks like the solid curve in Fig. 25-5 for different frequencies. Qualitatively, we understand the resonance curve; in order to get the exact shape of the curve it is probably just as well to do the mathematics. The curve goes toward infinity as $\omega \rightarrow \omega_0$, where ω_0 is the natural frequency of the oscillator.

Now suppose there is a little bit of friction; then when the displacement of the oscillator is small, the friction does not affect it much; the resonance curve is the same, except when we are near resonance. Instead of becoming infinite near resonance, the curve is only going to get so high that the work done by our tapping each time is enough to compensate for the loss of energy by friction during the cycle. So the top of the curve is rounded off—it does not go to infinity. If there is more friction, the top of the curve is rounded off still more. Now someone might say, “I thought the widths of the curves depended on the friction.” That is because the curve is usually plotted so that the top of the curve is called one unit. However, the mathematical expression is even simpler to understand if we just plot all the curves on the same scale; then all that happens is that the friction cuts down the top! If there is less friction, we can go farther up into that little pinnacle before the friction cuts it off, so it looks relatively narrow. That is, the higher the peak of the curve, the narrower the width at half the maximum height.

Finally, we take the case where there is an enormous amount of friction. It turns out that if there is too much friction, the system does not oscillate at all. The energy in the spring is barely able to move it against the frictional force, and so it slowly oozes down to the equilibrium point.

25-4 Analogs in physics

The next aspect of this review is to note that masses and springs are not the only linear systems; there are others. In particular, there are electrical systems called linear circuits, in which we find a complete analog to mechanical systems. We did not learn exactly *why* each of the objects in an electrical circuit works in the way it does—that is not to be understood at the present moment; we may assert it as an experimentally verifiable fact that they behave as stated.

For example, let us take the simplest possible circumstance. We have a piece of wire, which is just a resistance, and we have applied to it a difference in potential, V . Now the V means this: if we carry a charge q through the wire from one terminal

to another terminal, the work done is qV . The higher the voltage difference, the more work was done when the charge, as we say, “falls” from the high potential end of the terminal to the low potential end. So charges release energy in going from one end to the other. Now the charges do not simply fly from one end straight to the other end; the atoms in the wire offer some resistance to the current, and this resistance obeys the following law for almost all ordinary substances: if there is a current I , that is, so and so many charges per second tumbling down, the number per second that comes tumbling through the wire is proportional to how hard we push them—in other words, proportional to how much voltage there is:

$$V = IR = R(dq/dt). \quad (25.11)$$

The coefficient R is called the *resistance*, and the equation is called Ohm’s Law. The unit of resistance is the ohm; it is equal to one volt per ampere. In mechanical situations, to get such a frictional force in proportion to the velocity is difficult; in an electrical system it is very easy, and this law is extremely accurate for most metals.

We are often interested in how much work is done per second, the power loss, or the energy liberated by the charges as they tumble down the wire. When we carry a charge q through a voltage V , the work is qV , so the work done per second would be $V(dq/dt)$, which is the same as VI , or also $IR \cdot I = I^2R$. This is called the *heating loss*—this is how much heat is generated in the resistance per second, by the conservation of energy. It is this heat that makes an ordinary incandescent light bulb work.

Of course, there are other interesting properties of mechanical systems, such as the mass (inertia), and it turns out that there is an electrical analog to inertia also. It is possible to make something called an *inductor*, having a property called *inductance*, such that a current, once started through the inductance, *does not want to stop*. It requires a voltage in order to change the current! If the current is constant, there is no voltage across an inductance. DC circuits do not know anything about inductance; it is only when we *change* the current that the effects of inductance show up. The equation is

$$V = L(dI/dt) = L(d^2q/dt^2), \quad (25.12)$$

and the unit of inductance, called the *henry*, is such that one volt applied to an inductance of one henry produces a change of one ampere per second in the current. Equation (25.12) is the analog of Newton’s law for electricity, if we wish: V corresponds to F , L corresponds to m , and I corresponds to velocity! All of the consequent equations for the two kinds of systems will have the same derivations because, in all the equations, we can change any letter to its corresponding analog letter and we get the same equation; everything we deduce will have a correspondence in the two systems.

Now what electrical thing corresponds to the mechanical spring, in which there was a force proportional to the stretch? If we start with $F = kx$ and replace $F \rightarrow V$ and $x \rightarrow q$, we get $V = \alpha q$. It turns out that there *is* such a thing, in fact it is the only one of the three circuit elements we can really understand, because we did study a pair of parallel plates, and we found that if there were a charge of certain equal, opposite amounts on each plate, the electric field between them would be proportional to the size of the charge. So the work done in moving a unit charge across the gap from one plate to the other is precisely proportional to the charge. This work is the *definition* of the voltage difference, and it is the line integral of the electric field from one plate to another. It turns out, for historical reasons, that the constant of proportionality is not called C , but $1/C$. It could have been called C , but it was not. So we have

$$V = q/C. \quad (25.13)$$

The unit of capacitance, C , is the farad; a charge of one coulomb on each plate of a one-farad capacitor yields a voltage difference of one volt.

There are our analogies, and the equation corresponding to the oscillating circuit becomes the following, by direct substitution of L for m , q for x , etc:

$$m(d^2x/dt^2) + \gamma m(dx/dt) + kx = F, \quad (25.14)$$

$$L(d^2q/dt^2) + R(dq/dt) + q/C = V. \quad (25.15)$$

Now everything we learned about (25.14) can be transformed to apply to (25.15). Every *consequence* is the same; so much the same that there is a brilliant thing we can do.

Suppose we have a mechanical system which is quite complicated, not just one mass on a spring, but several masses on several springs, all hooked together. What do we do? Solve it? Perhaps; but look, we can make an *electrical* circuit which will have the same equations as the thing we are trying to analyze! For instance, if we wanted to analyze a mass on a spring, why can we not build an electrical circuit in which we use an inductance proportional to the mass, a resistance proportional to the corresponding $m\gamma$, $1/C$ proportional to k , all in the same ratio? Then, of course, this electrical circuit will be the exact analog of our mechanical one, in the sense that whatever q does, in response to V (V also is made to correspond to the forces that are acting), so the x would do in response to the force! So if we have a complicated thing with a whole lot of interconnecting elements, we can interconnect a whole lot of resistances, inductances, and capacitances, to *imitate* the mechanically complicated system. What is the advantage to that? One problem is just as hard (or as easy) as the other, because they are exactly equivalent. The advantage is not that it is any easier to solve the *mathematical equations* after we discover that we have an electrical circuit (although that *is* the method used by electrical engineers!), but instead, the real reason for looking at the analog is that it is easier to *make* the electrical circuit, and to *change* something in the system.

Suppose we have designed an automobile, and want to know how much it is going to shake when it goes over a certain kind of bumpy road. We build an electrical circuit with inductances to represent the inertia of the wheels, spring constants as capacitances to represent the springs of the wheels, and resistors to represent the shock absorbers, and so on for the other parts of the automobile. Then we need a bumpy road. All right, we apply a *voltage* from a generator, which represents such and such a kind of bump, and then look at how the left wheel jiggles by measuring the charge on some capacitor. Having measured it (it is easy to do), we find that it is bumping too much. Do we need more shock absorber, or less shock absorber? With a complicated thing like an automobile, do we actually change the shock absorber, and solve it all over again? No!, we simply turn a dial; dial number ten is shock absorber number three, so we put in more shock absorber. The bumps are worse—all right, we try less. The bumps are still worse; we change the stiffness of the spring (dial 17), and we adjust all these things *electrically*, with merely the turn of a knob.

This is called an *analog computer*. It is a device which imitates the problem that we want to solve by making another problem, which has the same equation, but in another circumstance of nature, and which is easier to build, to measure, to adjust, and to destroy!

25-5 Series and parallel impedances

Finally, there is an important item which is not quite in the nature of review. This has to do with an electrical circuit in which there is more than one circuit element. For example, when we have an inductor, a resistor, and a capacitor connected as in Fig. 24-2, we note that all the charge went through every one of the three, so that the current in such a singly connected thing is the same at all points along the wire. Since the current is the same in each one, the voltage across R is IR , the voltage across L is $L(dI/dt)$, and so on. So, the total voltage drop is the sum of these, and this leads to Eq. (25.15). Using complex numbers, we found that we could solve the equation for the steady-state motion in response to a

sinusoidal force. We thus found that $\hat{V} = \hat{Z}\hat{I}$. Now $\hat{Z}$ is called the *impedance* of this particular circuit. It tells us that if we apply a sinusoidal voltage, $\hat{V}$, we get a current $\hat{I}$.

Now suppose we have a more complicated circuit which has two pieces, which by themselves have certain impedances, $\hat{Z}_1$ and $\hat{Z}_2$, and we put them in *series* (Fig. 25-6a) and apply a voltage. What happens? It is now a little more complicated, but if $\hat{I}$ is the current through $\hat{Z}_1$, the voltage difference across $\hat{Z}_1$ is $\hat{V}_1 = \hat{I}\hat{Z}_1$; similarly, the voltage across $\hat{Z}_2$ is $\hat{V}_2 = \hat{I}\hat{Z}_2$. *The same current goes through both.* Therefore the total voltage is the sum of the voltages across the two sections and is equal to $\hat{V} = \hat{V}_1 + \hat{V}_2 = (\hat{Z}_1 + \hat{Z}_2)\hat{I}$. This means that the voltage on the complete circuit can be written $\hat{V} = \hat{I}\hat{Z}_s$ where the $\hat{Z}_s$ of the combined system in series is the sum of the two $\hat{Z}$'s of the separate pieces:

$$\hat{Z}_s = \hat{Z}_1 + \hat{Z}_2. \quad (25.16)$$

This is not the only way things may be connected. We may also connect them in another way, called a *parallel* connection (Fig. 25-6b). Now we see that a given voltage across the terminals, if the connecting wires are perfect conductors, is effectively applied to both of the impedances, and will cause currents in each independently. Therefore the current through $\hat{Z}_1$ is equal to $\hat{I}_1 = \hat{V}/\hat{Z}_1$. The current in $\hat{Z}_2$ is $\hat{I}_2 = \hat{V}/\hat{Z}_2$. It is the *same voltage*. Now the total current which is supplied to the terminals is the *sum* of the currents in the two sections: $\hat{I} = \hat{V}/\hat{Z}_1 + \hat{V}/\hat{Z}_2$. This can be written as

$$\hat{V} = \frac{\hat{I}}{(\hat{I}/\hat{Z}_1) + (\hat{I}/\hat{Z}_2)} = \hat{I}\hat{Z}_p.$$

Thus

$$1/\hat{Z}_p = 1/\hat{Z}_1 + 1/\hat{Z}_2. \quad (25.17)$$

More complicated circuits can sometimes be simplified by taking pieces of them, working out the succession of impedances of the pieces, and combining the circuit together step by step, using the above rules. If we have any kind of circuit with many impedances connected in all kinds of ways, and if we include the voltages in the form of little generators having no impedance (when we pass charge through it, the generator adds a voltage V), then the following principles apply: (1) At any junction, the sum of the currents into a junction is zero. That is, all the current which comes in must come back out. (2) If we carry a charge around any loop, and back to where it started, the net work done is zero. These rules are called *Kirchhoff's laws* for electrical circuits. Their systematic application to complicated circuits often simplifies the analysis of such circuits. We mention them here in conjunction with Eqs. (25.16) and (25.17), in case you have already come across such circuits that you need to analyze in laboratory work. They will be discussed again in more detail next year.

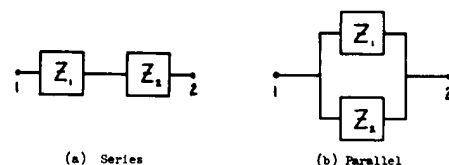


Fig. 25-6. Two impedances, connected in series and in parallel.

Optics: The Principle of Least Time

26-1 Light

This is the first of a number of chapters on the subject of *electromagnetic radiation*. Light, with which we see, is only one small part of a vast spectrum of the same kind of thing, the various parts of this spectrum being distinguished by different values of a certain quantity which varies. This variable quantity could be called the “wavelength.” As it varies in the visible spectrum, the light apparently changes color from red to violet. If we explore the spectrum systematically, from long wavelengths toward shorter ones, we would begin with what are usually called *radiowaves*. Radiowaves are technically available in a wide range of wavelengths, some even longer than those used in regular broadcasts; regular broadcasts have wavelengths corresponding to about 500 meters. Then there are the so-called “short waves,” i.e., radar waves, millimeter waves, and so on. There are no actual boundaries between one range of wavelengths and another, because nature did not present us with sharp edges. The number associated with a given name for the waves are only approximate and, of course, so are the names we give to the different ranges.

Then, a long way down through the millimeter waves, we come to what we call the *infrared*, and thence to the visible spectrum. Then going in the other direction, we get into a region which is called the *ultraviolet*. Where the ultraviolet stops, the x-rays begin, but we cannot define precisely where this is; it is roughly at 10^{-8} m, or 10^{-2} μ . These are “soft” x-rays; then there are ordinary x-rays and very hard x-rays; then γ -rays, and so on, for smaller and smaller values of this dimension called the wavelength.

Within this vast range of wavelengths, there are three or more regions of approximation which are especially interesting. In one of these, a condition exists in which the wavelengths involved are very small compared with the dimensions of the equipment available for their study; furthermore, the photon energies, using the quantum theory, are small compared with the energy sensitivity of the equipment. Under these conditions we can make a rough first approximation by a method called *geometrical optics*. If, on the other hand, the wavelengths are comparable to the dimensions of the equipment, which is difficult to arrange with visible light but easier with radiowaves, and if the photon energies are still negligibly small, then a very useful approximation can be made by studying the behavior of the waves, still disregarding the quantum mechanics. This method is based on the *classical theory of electromagnetic radiation*, which will be discussed in a later chapter. Next, if we go to very short wavelengths, where we can disregard the wave character but the photons have a very *large* energy compared with the sensitivity of our equipment, things get simple again. This is the simple *photon* picture, which we will describe only very roughly. The complete picture, which unifies the whole thing into one model, will not be available to us for a long time.

In this chapter our discussion is limited to the geometrical optics region, in which we forget about the wavelength and the photon character of the light, which will all be explained in due time. We do not even bother to say what the light *is*, but just find out *how it behaves* on a large scale compared with the dimensions of interest. All this must be said in order to emphasize the fact that what we are going to talk about is only a very crude approximation; this is one of the chapters that we shall have to “unlearn” again. But we shall very quickly unlearn it, because we shall almost immediately go on to a more accurate method.

26-1 Light

26-2 Reflection and refraction

26-3 Fermat's principle of least time

26-4 Applications of Fermat's principle

26-5 A more precise statement of Fermat's principle

26-6 How it works

Although geometrical optics is just an approximation, it is of very great importance technically and of great interest historically. We shall present this subject more historically than some of the others in order to give some idea of the development of a physical theory or physical idea.

First, light is, of course, familiar to everybody, and has been familiar since time immemorial. Now one problem is, by what process do we *see* light? There have been many theories, but it finally settled down to one, which is that there is something which enters the eye—which bounces off objects into the eye. We have heard that idea so long that we accept it, and it is almost impossible for us to realize that very intelligent men have proposed contrary theories—that something comes out of the eye and feels for the object, for example. Some other important observations are that, as light goes from one place to another, it goes in *straight lines*, if there is nothing in the way, and that the rays do not seem to interfere with one another. That is, light is crisscrossing in all directions in the room, but the light that is passing across our line of vision does not affect the light that comes to us from some object. This was once a most powerful argument against the corpuscular theory; it was used by Huygens. If light were like a lot of arrows shooting along, how could other arrows go through them so easily? Such philosophical arguments are not of much weight. One could always say that light is made up of arrows which go through each other!

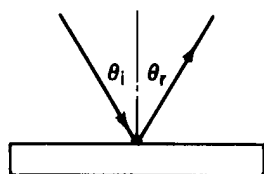


Fig. 26-1. The angle of incidence is equal to the angle of reflection.

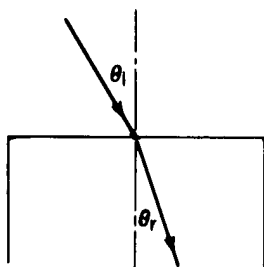


Fig. 26-2. A light ray is refracted when it passes from one medium into another.

26-2 Reflection and refraction

The discussion above gives enough of the basic *idea* of geometrical optics—now we have to go a little further into the quantitative features. Thus far we have light going only in straight lines between two points; now let us study the behavior of light when it hits various materials. The simplest object is a mirror, and the law for a mirror is that when the light hits the mirror, it does not continue in a straight line, but bounces off the mirror into a new straight line, which changes when we change the inclination of the mirror. The question for the ancients was, what is the relation between the two angles involved? This is a very simple relation, discovered long ago. The light striking a mirror travels in such a way that the two angles, between each beam and the mirror, are equal. For some reason it is customary to measure the angles from the normal to the mirror surface. Thus the so-called law of reflection is

$$\theta_i = \theta_r. \quad (26.1)$$

That is a simple enough proposition, but a more difficult problem is encountered when light goes from one medium into another, for example from air into water; here also, we see that it does not go in a straight line. In the water the ray is at an inclination to its path in the air; if we change the angle θ_i so that it comes down more nearly vertically, then the angle of “breakage” is not as great. But if we tilt the beam of light at quite an angle, then the deviation angle is very large. The question is, what is the relation of one angle to the other? This also puzzled the ancients for a long time, and here they never found the answer! It is, however, one of the few places in all of Greek physics that one may find any experimental results listed. Claudius Ptolemy made a list of the angle in water for each of a number of different angles in air. Table 26-1 shows the angles in the air, in degrees, and the corresponding angle as measured in the water. (Ordinarily it is said that Greek scientists never did any experiments. But it would be impossible to obtain this table of values without knowing the right law, except by experiment. It should be noted, however, that these do not represent independent careful measurements for each angle but only some numbers interpolated from a few measurements, for they all fit perfectly on a parabola.)

This, then, is one of the important steps in the development of physical law: first we observe an effect, then we measure it and list it in a table; then we try to find the *rule* by which one thing can be connected with another. The above numerical table was made in 140 A.D., but it was not until 1621 that someone finally found the rule connecting the two angles! The rule, found by Willebrord

Table 26-1

Angle in air	Angle in water
10°	8°
20°	15-1/2°
30°	22-1/2°
40°	29°
50°	35°
60°	40-1/2°
70°	45-1/2°
80°	50°

Snell, a Dutch mathematician, is as follows: if θ_i is the angle in air and θ_r is the angle in the water, then it turns out that the sine of θ_i is equal to some constant multiple of the sine of θ_r :

$$\sin \theta_i = n \sin \theta_r. \quad (26.2)$$

For water the number n is approximately 1.33. Equation (26.2) is called *Snell's law*; it permits us to *predict* how the light is going to bend when it goes from air into water. Table 26-2 shows the angles in air and in water according to Snell's law. Note the remarkable agreement with Ptolemy's list.

Table 26-2

Angle in air	Angle in water
10°	7-1/2°
20°	15°
30°	22°
40°	29°
50°	35°
60°	40-1/2°
70°	45°
80°	48°

26-3 Fermat's principle of least time

Now in the further development of science, we want more than just a formula. First we have an observation, then we have numbers that we measure, then we have a law which summarizes all the numbers. But the real *glory* of science is that we can find a way of thinking such that the law is *evident*.

The first way of thinking that made the law about the behavior of light evident was discovered by Fermat in about 1650, and it is called *the principle of least time*, or *Fermat's principle*. His idea is this: that out of all possible paths that it might take to get from one point to another, light takes the path which requires the *shortest time*.

Let us first show that this is true for the case of the mirror, that this simple principle contains both the law of straight-line propagation and the law for the mirror. So, we are growing in our understanding! Let us try to find the solution to the following problem. In Fig. 26-3 are shown two points, A and B , and a plane mirror, MM' . What is the way to get from A to B in the shortest time? The answer is to go straight from A to B ! But if we add the extra rule that the light has to *strike the mirror* and come back in the shortest time, the answer is not so easy. One way would be to go as quickly as possible to the mirror and then go to B , on the path ADB . Of course, we then have a long path DB . If we move over a little to the right, to E , we slightly increase the first distance, but we greatly *decrease* the second one, and so the total path length, and therefore the travel time, is less. How can we find the point C for which the time is the shortest? We can find it very nicely by a geometrical trick.

We construct on the other side of MM' an artificial point B' , which is the same distance below the plane MM' as the point B is above the plane. Then we draw the line EB' . Now because BFM is a right angle and $BF = FB'$, EB is equal to EB' . Therefore the sum of the two distances, $AE + EB$, which is proportional to the time it will take if the light travels with constant velocity, is also the sum of the two lengths $AE + EB'$. Therefore the problem becomes, when is the sum of these two lengths the least? The answer is easy: when the line goes through point C as a *straight line* from A to B' ! In other words, we have to find the point where we go toward the artificial point, and that will be the correct one. Now if ACB' is a straight line, then angle BCF is equal to angle $B'CF$ and thence to angle ACM . Thus the statement that the angle of incidence equals the angle of reflection is equivalent to the statement that the light goes to the mirror in such a way that it comes back to the point B' in the *least possible time*. Originally, the statement was made by Hero of Alexandria that the light travels in such a way that it goes to the mirror and to the other point in the shortest possible *distance*, so it is not a modern theory. It was this that inspired Fermat to suggest to himself that perhaps refraction operated on a similar basis. But for refraction, light obviously does not use the path of shortest *distance*, so Fermat tried the idea that it takes the shortest *time*.

Before we go on to analyze refraction, we should make one more remark about the mirror. If we have a source of light at the point B and it sends light toward the mirror, then we see that the light which goes to A from the point B comes to A in exactly the same manner as it would have come to A if there were an object at B' , and *no* mirror. Now of course the eye detects only the light which enters it physically, so if we have an object at B and a mirror which makes the light come

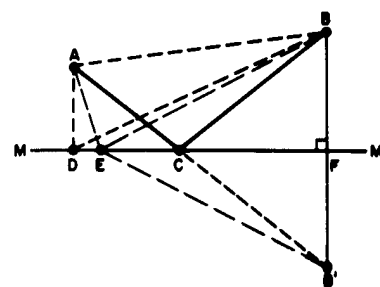


Fig. 26-3. Illustration of the principle of least time.

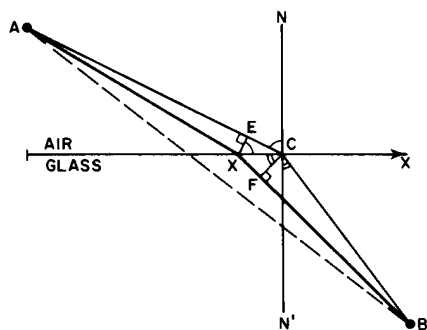


Fig. 26-4. Illustration of Fermat's principle for refraction.

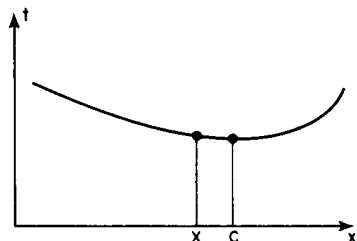


Fig. 26-5. The minimum time corresponds to point C, but nearby points correspond to nearly the same time.

into the eye in exactly the same manner as it would have come into the eye if the object were at B' , then the eye-brain system interprets that, assuming it does not know too much, as *being* an object at B' . So the illusion that there is an object behind the mirror is merely due to the fact that the light which is entering the eye is entering in exactly the same manner, physically, as it would have entered had there *been* an object back there (except for the dirt on the mirror, and our knowledge of the existence of the mirror, and so on, which is corrected in the brain).

Now let us demonstrate that the principle of least time will give Snell's law of refraction. We must, however, make an assumption about the speed of light in water. We shall assume that the speed of light in water is lower than the speed of light in air by a certain factor, n .

In Fig. 26-4, our problem is again to go from A to B in the *shortest time*. To illustrate that the best thing to do is not just to go in a straight line, let us imagine that a beautiful girl has fallen out of a boat, and she is screaming for help in the water at point B . The line marked X is the shoreline. We are at point A on land, and we see the accident, and we can run and can also swim. But we can run faster than we can swim. What do we do? Do we go in a straight line? (Yes, no doubt!) However, by using a little more intelligence we would realize that it would be advantageous to travel a little greater distance on land in order to decrease the distance in the water, because we go so much slower in the water. (Following this line of reasoning out, we would say the right thing to do is to *compute* very carefully what should be done!) At any rate, let us try to show that the final solution to the problem is the path ACB , and that this path takes the shortest time of all possible ones. If it is the shortest path, that means that if we take any other, it will be longer. So, if we were to plot the time it takes against the position of point X , we would get a curve something like that shown in Fig. 26-5, where point C corresponds to the shortest of all possible times. This means that if we move the point X to points *near* C , in the first approximation there is essentially *no change* in time because the slope is zero at the bottom of the curve. So our way of finding the law will be to consider that we move the place by a very small amount, and to demand that there be essentially no change in time. (Of course there is an infinitesimal change of a *second* order; we ought to have a positive increase for displacements in either direction from C .) So we consider a nearby point X and we calculate how long it would take to go from A to B by the two paths, and compare the new path with the old path. It is very easy to do. We want the difference, of course, to be nearly zero if the distance XC is short. First, look at the path on land. If we draw a perpendicular XE , we see that this path is shortened by the amount EC . Let us say we gain by not having to go that extra distance. On the other hand, in the water, by drawing a corresponding perpendicular, CF , we find that we have to go the extra distance XF , and that is what we lose. Or, in *time*, we gain the time it would have taken to go the distance EC , but we lose the time it would have taken to go the distance XF . Those times must be equal since, in the first approximation, there is to be no change in time. But supposing that in the water the speed is $1/n$ times as fast as in air, then we must have

$$EC = n \cdot XF. \quad (26.3)$$

Therefore we see that when we have the right point, $XC \sin EXC = n \cdot XC \sin XCF$ or, cancelling the common hypotenuse length XC and noting that

$$EXC = ECN = \theta_i \quad \text{and} \quad XCF = BCN' = \theta_r,$$

we have

$$\sin \theta_i = n \sin \theta_r. \quad (26.4)$$

So we see that to get from one point to another in the least time when the ratio of speeds is n , the light should enter at such an angle that the ratio of the sines of the angles θ_i and θ_r is the ratio of the speeds in the two media.

26-4 Applications of Fermat's principle

Now let us consider some of the interesting consequences of the principle of least time. First is the principle of reciprocity. If to go from A to B we have found the path of the least time, then to go in the opposite direction (assuming that light goes at the same speed in any direction), the shortest time will be the same path, and therefore, if light can be sent one way, it can be sent the other way.

An example of interest is a glass block with plane parallel faces, set at an angle to a light beam. Light, in going through the block from a point A to a point B (Fig. 26-6) does not go through in a straight line, but instead it decreases the time in the block by making the angle in the block less inclined, although it loses a little bit in the air. The beam is simply displaced parallel to itself because the angles in and out are the same.

A third interesting phenomenon is the fact that when we see the sun setting, it is already below the horizon! It does not *look* as though it is below the horizon, but it is (Fig. 26-7). The earth's atmosphere is thin at the top and dense at the bottom. Light travels more slowly in air than it does in a vacuum, and so the light of the sun can get to point S beyond the horizon more quickly if, instead of just going in a straight line, it avoids the dense regions where it goes slowly by getting through them at a steeper tilt. When it appears to go below the horizon, it is actually already well below the horizon. Another example of this phenomenon is the mirage that one often sees while driving on hot roads. One sees "water" on the road, but when he gets there, it is as dry as the desert! The phenomenon is the following. What we are really seeing is the sky light "reflected" on the road: light from the sky, heading for the road, can end up in the eye, as shown in Fig. 26-8. Why? The air is very hot just above the road but it is cooler up higher. Hotter air is more expanded than cooler air and is thinner, and this decreases the speed of light less. That is to say, light goes faster in the hot region than in the cool region. Therefore, instead of the light deciding to come in the straightforward way, it also has a least-time path by which it goes into the region where it goes faster for awhile, in order to save time. So, it can go in a curve.

As another important example of the principle of least time, suppose that we would like to arrange a situation where we have all the light that comes out of one point, P , collected back together at another point, P' (Fig. 26-9). That means, of course, that the light can go in a straight line from P to P' . That is all right. But how can we arrange that not only does it go straight, but also so that the light starting out from P toward Q also ends up at P' ? We want to bring all the light back to what we call a *focus*. How? If the light always takes the path of least time, then certainly it should not want to go over all these other paths. The only way that the light can be perfectly satisfied to take several adjacent paths is to make those times *exactly equal*! Otherwise, it would select the one of least time. Therefore the problem of making a focusing system is merely to arrange a device so that it takes the same time for the light to go on *all* the different paths!

This is easy to do. Suppose that we had a piece of glass in which light goes slower than it does in the air (Fig. 26-10). Now consider a ray which goes in air in the path PQP' . That is a longer path than from P directly to P' and no doubt takes a longer time. But if we were to insert a piece of glass of just the right thickness (we shall later figure out how thick) it might exactly compensate the excess time that it would take the light to go at an angle! In those circumstances we can arrange that the time the light takes to go straight through is the same as the time it takes to go in the path PQP' . Likewise, if we take a ray $PRR'P'$ which is partly inclined, it is not quite as long as PQP' , and we do not have to compensate as much as for the straight one, but we do have to compensate somewhat. We end up with a piece of glass that looks like Fig. 26-10. With this shape, all the light which comes from P will go to P' . This, of course, is well known to us, and we call such a device a converging *lens*. In the next chapter we shall actually calculate what shape the lens has to have to make a perfect focus.

Take another example: suppose we wish to arrange some mirrors so that the light from P always goes to P' (Fig. 26-11). On any path, it goes to some mirror

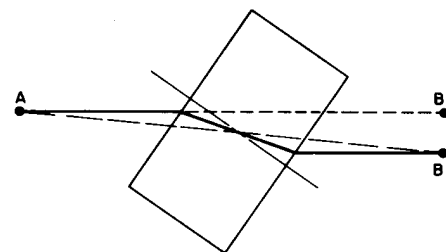


Fig. 26-6. A beam of light is offset as it passes through a transparent block.

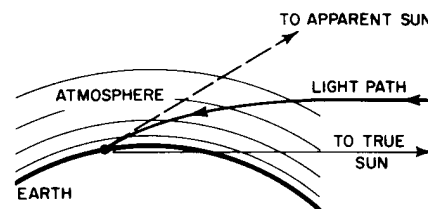


Fig. 26-7. Near the horizon, the apparent sun is higher than the true sun by about $1/2$ degree.

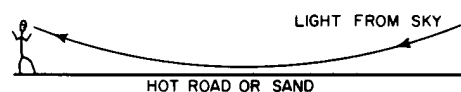


Fig. 26-8. A mirage.

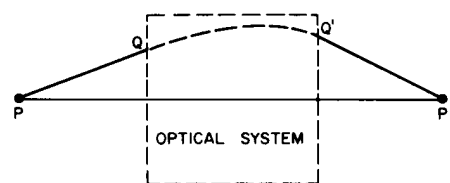


Fig. 26-9. An optical "black box."

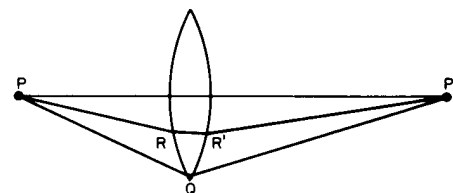


Fig. 26-10. A focusing optical system.

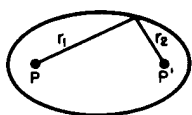


Fig. 26-11. An ellipsoidal mirror.

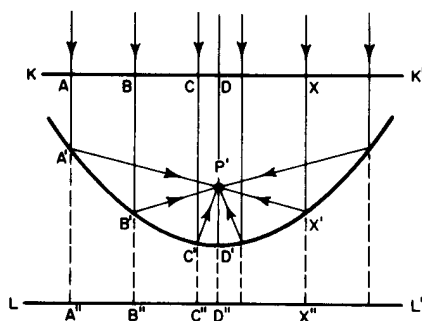


Fig. 26-12. A paraboloidal mirror.

and comes back, and all times must be equal. Here the light always travels in air, so the time and the distance are proportional. Therefore the statement that all the times are the same is the same as the statement that the total distance is the same. Thus the sum of the two distances r_1 and r_2 must be a constant. An *ellipse* is that curve which has the property that the sum of the distances from two points is a constant for every point on the ellipse; thus we can be sure that the light from one focus will come to the other.

The same principle works for gathering the light of a star. The great 200-inch Palomar telescope is built on the following principle. Imagine a star billions of miles away; we would like to cause all the light that comes in to come to a focus. Of course we cannot draw the rays that go all the way up to the star, but we still want to check whether the times are equal. Of course we know that when the various rays have arrived at some plane KK' , perpendicular to the rays, all the times in this plane are equal (Fig. 26-12). The rays must then come down to the mirror and proceed toward P' in equal times. That is, we must find a curve which has the property that the sum of the distances $XX' + X'P'$ is a constant, no matter where X is chosen. An easy way to find it is to extend the length of the line XX' down to a plane LL' . Now if we arrange our curve so that $A'A'' = A'P'$, $B'B'' = B'P'$, $C'C'' = C'P'$, and so on, we will have our curve, because then of course, $AA' + A'P' = AA' + A'A''$ will be constant. Thus our curve is the locus of all points equidistant from a line and a point. Such a curve is called a *parabola*; the mirror is made in the shape of a parabola.

The above examples illustrate the principle upon which such optical devices can be designed. The exact curves can be calculated using the principle that, to focus perfectly, the travel times must be exactly equal for all light rays, as well as being less than for any other nearby path.

We shall discuss these focusing optical devices further in the next chapter; let us now discuss the further development of the theory. When a new theoretical principle is developed, such as the principle of least time, our first inclination might be to say, "Well, that is very pretty; it is delightful; but the question is, does it help at all in understanding the physics?" Someone may say, "Yes, look at how many things we can now understand!" Another says, "Very well, but I can understand mirrors, too. I need a curve such that every tangent plane makes equal angles with the two rays. I can figure out a lens, too, because every ray that comes to it is bent through an angle given by Snell's law." Evidently the statement of least time and the statement that angles are equal on reflection, and that the sines of the angles are proportional on refraction, are the same. So is it merely a philosophical question, or one of beauty? There can be arguments on both sides.

However, the importance of a powerful principle is that *it predicts new things*.

It is easy to show that there are a number of new things predicted by Fermat's principle. First, suppose that there are *three* media, glass, water, and air, and we perform a refraction experiment and measure the index n for one medium against another. Let us call n_{12} the index of air (1) against water (2); n_{13} the index of air (1) against glass (3). If we measured water against glass, we should find another index, which we shall call n_{23} . But there is no *a priori* reason why there should be any connection between n_{12} , n_{13} , and n_{23} . On the other hand, according to the idea of least time, there *is* a definite relationship. The index n_{12} is the ratio of two things, the speed in air to the speed in water; n_{13} is the ratio of the speed in air to the speed in glass; n_{23} is the ratio of the speed in water to the speed in glass. Therefore we cancel out the air, and get

$$n_{23} = \frac{v_2}{v_3} = \frac{v_1/v_3}{v_1/v_2} = \frac{n_{13}}{n_{12}}. \quad (26.5)$$

In other words, we *predict* that the index for a new pair of materials can be obtained from the indexes of the individual materials, both against air or against vacuum. So if we measure the speed of light in all materials, and from this get a single number for each material, namely its index relative to vacuum, called n_i

(n_1 is the speed in air relative to the speed in vacuum, etc.), then our formula is easy. The index for any two materials i and j is

$$n_{ij} = \frac{v_i}{v_j} = \frac{n_j}{n_i}. \quad (26.6)$$

Using only Snell's law, there is no basis for a prediction of this kind.* But of course this prediction works. The relation (26.5) was known very early, and was a very strong argument for the principle of least time.

Another argument for the principle of least time, another prediction, is that if we *measure* the speed of light in water, it will be lower than in air. This is a prediction of a completely different type. It is a brilliant prediction, because all we have so far measured are *angles*; here we have a theoretical prediction which is quite different from the observations from which Fermat deduced the idea of least time. It turns out, in fact, that the speed in water *is* slower than the speed in air, by just the proportion that is needed to get the right index!

26-5 A more precise statement of Fermat's principle

Actually, we must make the statement of the principle of least time a little more accurately. It was not stated correctly above. It is *incorrectly* called the principle of least time and we have gone along with the incorrect description for convenience, but we must now see what the correct statement is. Suppose we had a mirror as in Fig. 26-3. What makes the light think it has to go to the mirror? The path of *least* time is clearly AB . So some people might say, "Sometimes it is a maximum time." It is *not* a maximum time, because certainly a curved path would take a still longer time! The correct statement is the following: a ray going in a certain particular path has the property that if we make a small change (say a one percent shift) in the ray in any manner whatever, say in the location at which it comes to the mirror, or the shape of the curve, or anything, there will be *no* first-order change in the time; there will be only a *second*-order change in the time. In other words, the principle is that light takes a path such that there are many other paths nearby which take almost exactly the *same* time.

The following is another difficulty with the principle of least time, and one which people who do not like this kind of a theory could never stomach. With Snell's theory we can "understand" light. Light goes along, it sees a surface, it bends because it does something at the surface. The idea of causality, that it goes from one point to another, and another, and so on, is easy to understand. But the principle of least time is a completely different philosophical principle about the way nature works. Instead of saying it is a causal thing, that when we do one thing, something else happens, and so on, it says this: we set up the situation, and *light* decides which is the shortest time, or the extreme one, and chooses that path. But *what* does it do, *how* does it find out? Does it *smell* the nearby paths, and check them against each other? The answer is, yes, it does, in a way. That is the feature which is, of course, not known in geometrical optics, and which is involved in the idea of *wavelength*; the wavelength tells us approximately how far away the light must "smell" the path in order to check it. It is hard to demonstrate this fact on a large scale with light, because the wavelengths are so terribly short. But with radiowaves, say 3-cm waves, the distances over which the radiowaves are checking are larger. If we have a source of radiowaves, a detector, and a slit, as in Fig. 26-13, the rays of course go from S to D because it is a straight line, and if we close down the slit it is all right—they still go. But now if we move the detector aside to D' , the waves will not go through the wide slit from S to D' , because they check several paths nearby, and say, "No, my friend, those all correspond to different times." On the other hand, if we *prevent* the radiation from checking the paths by closing the slit down to a very narrow crack, then there is but one path available, and the

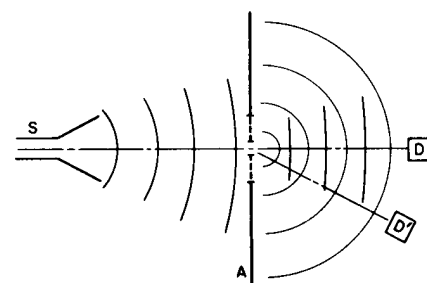


Fig. 26-13. The passage of radio-waves through a narrow slit.

* Although it can be deduced if the additional assumption is made that adding a layer of one substance to the surface of another does not change the eventual angle of refraction in the latter material.

radiation takes it! With a narrow slit, more radiation reaches D' than reaches it with a wide slit!

One can do the same thing with light, but it is hard to demonstrate on a large scale. The effect can be seen under the following simple conditions. Find a small, bright light, say an unfrosted bulb in a street light far away or the reflection of the sun in a curved automobile bumper. Then put two fingers in front of one eye, so as to look through the crack, and squeeze the light to zero very gently. You will see that the image of the light, which was a little dot before, becomes quite elongated, and even stretches into a long line. The reason is that the fingers are very close together, and the light which is supposed to come in a straight line is spread out at an angle, so that when it comes into the eye it comes in from several directions. Also you will notice, if you are very careful, side maxima, a lot of fringes along the edges too. Furthermore, the whole thing is colored. All of this will be explained in due time, but for the present it is a demonstration that light does not always go in straight lines, and it is one that is very easily performed.

26-6 How it works

Finally, we give a very crude view of what actually happens, how the whole thing really works, from what we now believe is the correct, quantum-dynamically accurate viewpoint, but of course only qualitatively described. In following the light from A to B in Fig. 26-3, we find that the light does not seem to be in the form of waves at all. Instead the rays seem to be made up of photons, and they actually produce clicks in a photon counter, if we are using one. The brightness of the light is proportional to the average number of photons that come in per second, and what we calculate is the *chance* that a photon gets from A to B , say by hitting the mirror. The *law* for that chance is the following very strange one. Take any path and find the time for that path; then make a complex number, or draw a little complex vector, $pe^{i\theta}$, whose angle θ is proportional to the time. The number of turns per second is the frequency of the light. Now take another path; it has, for instance, a different time, so the vector for it is turned through a different angle—the angle being always proportional to the time. Take *all* the available paths and add on a little vector for each one; then the answer is that the chance of arrival of the photon is proportional to the square of the length of the final vector, from the beginning to the end!

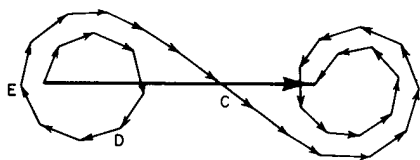


Fig. 26-14. The summation of probability amplitudes for many neighboring paths.

Now let us show how this implies the principle of least time for a mirror. We consider all rays, all possible paths ADB , AEB , ACB , etc., in Fig. 26-3. The path ADB makes a certain small contribution, but the next path, AEB , takes a quite different time, so its angle θ is quite different. Let us say that point C corresponds to minimum time, where if we change the paths the times do not change. So for awhile the times do change, and then they begin to change less and less as we get near point C (Fig. 26-14). So the arrows which we have to add are coming almost exactly at the same angle for awhile near C , and then gradually the time begins to increase again, and the phases go around the other way, and so on. Eventually, we have quite a tight knot. The total probability is the distance from one end to the other, squared. *Almost all of that accumulated probability occurs in the region where all the arrows are in the same direction* (or in the same phase). All the contributions from the paths which have very *different* times as we change the path, cancel themselves out by pointing in different directions. That is why, if we hide the extreme parts of the mirror, it still reflects almost exactly the same, because all we did was to take out a piece of the diagram inside the spiral ends, and that makes only a very small change in the light. So this is the relationship between the ultimate picture of photons with a probability of arrival depending on an accumulation of arrows, and the principle of least time.

Geometrical Optics

27-1 Introduction

In this chapter we shall discuss some elementary applications of the ideas of the previous chapter to a number of practical devices, using the approximation called *geometrical optics*. This is a most useful approximation in the practical design of many optical systems and instruments. Geometrical optics is either very simple or else it is very complicated. By that we mean that we can either study it only superficially, so that we can design instruments roughly, using rules that are so simple that we hardly need deal with them here at all, since they are practically of high school level, or else, if we want to know about the small errors in lenses and similar details, the subject gets so complicated that it is too advanced to discuss here! If one has an actual, detailed problem in lens design, including analysis of aberrations, then he is advised to read about the subject or else simply to trace the rays through the various surfaces (which is what the book tells how to do), using the law of refraction from one side to the other, and to find out where they come out and see if they form a satisfactory image. People have said that this is too tedious, but today, with computing machines, it is the right way to do it. One can set up the problem and make the calculation for one ray after another very easily. So the subject is really ultimately quite simple, and involves no new principles. Furthermore, it turns out that the rules of either elementary or advanced optics are seldom characteristic of other fields, so that there is no special reason to follow the subject very far, with one important exception.

The most advanced and abstract theory of geometrical optics was worked out by Hamilton, and it turns out that this has very important applications in mechanics. It is actually even more important in mechanics than it is in optics, and so we leave Hamilton's theory for the subject of advanced analytical mechanics, which is studied in the senior year or in graduate school. So, appreciating that geometrical optics contributes very little, except for its own sake, we now go on to discuss the elementary properties of simple optical systems on the basis of the principles outlined in the last chapter.

In order to go on, we must have one geometrical formula, which is the following: if we have a triangle with a small altitude h and a long base d , then the diagonal s (we are going to need it to find the difference in time between two different routes) is longer than the base (Fig. 27-1). How much longer? The difference $\Delta = s - d$ can be found in a number of ways. One way is this. We see that $s^2 - d^2 = h^2$, or $(s - d)(s + d) = h^2$. But $s - d = \Delta$, and $s + d \sim 2s$. Thus

$$\Delta \sim h^2/2s. \quad (27.1)$$

This is all the geometry we need to discuss the formation of images by curved surfaces!

27-2 The focal length of a spherical surface

The first and simplest situation to discuss is a single refracting surface, separating two media with different indices of refraction (Fig. 27-2). We leave the case of arbitrary indices of refraction to the student, because *ideas* are always the most important thing, not the specific situation, and the problem is easy enough to do in any case. So we shall suppose that, on the left, the speed is 1 and on the right it is $1/n$, where n is the index of refraction. The light travels more slowly in the glass by a factor n .

27-1 Introduction

27-2 The focal length of a spherical surface

27-3 The focal length of a lens

27-4 Magnification

27-5 Compound lenses

27-6 Aberrations

27-7 Resolving power

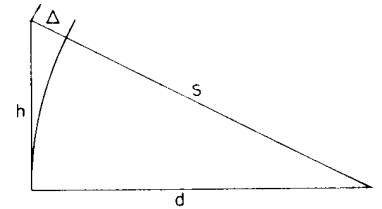


Figure 27-1

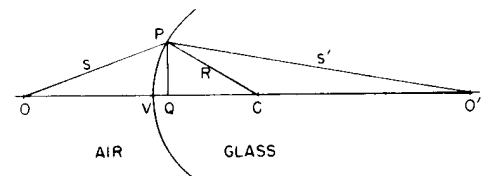


Fig. 27-2. Focusing by a single refracting surface.

Now suppose that we have a point at O , at a distance s from the front surface of the glass, and another point O' at a distance s' inside the glass, and we desire to arrange the curved surface in such a manner that every ray from O which hits the surface, at any point P , will be bent so as to proceed toward the point O' . For that to be true, we have to shape the surface in such a way that the time it takes for the light to go from O to P , that is, the distance OP divided by the speed of light (the speed here is unity), plus $n \cdot O'P$, which is the time it takes to go from P to O' , is equal to a constant independent of the point P . This condition supplies us with an equation for determining the surface. The answer is that the surface is a very complicated fourth-degree curve, and the student may entertain himself by trying to calculate it by analytic geometry. It is simpler to try a special case that corresponds to $s \rightarrow \infty$, because then the curve is a second-degree curve and is more recognizable. It is interesting to compare this curve with the parabolic curve we found for a focusing mirror when the light is coming from infinity.

So the proper surface cannot easily be made—to focus the light from one point to another requires a rather complicated surface. It turns out in practice that we do not try to make such complicated surfaces ordinarily, but instead we make a compromise. Instead of trying to get *all* the rays to come to a focus, we arrange it so that only the rays fairly close to the axis OO' come to a focus. The farther ones may deviate if they want to, unfortunately, because the ideal surface is complicated, and we use instead a spherical surface with the right curvature at the axis. It is so much easier to fabricate a sphere than other surfaces that it is profitable for us to find out what happens to rays striking a spherical surface, supposing that only the rays near the axis are going to be focused perfectly. Those rays which are near the axis are sometimes called *paraxial rays*, and what we are analyzing are the conditions for the focusing of paraxial rays. We shall discuss later the errors that are introduced by the fact that all rays are not always close to the axis.

Thus, supposing P is close to the axis, we drop a perpendicular PQ such that the height PQ is h . For a moment, we imagine that the surface is a plane passing through P . In that case, the time needed to go from O to P would exceed the time from O to Q , and also, the time from P to O' would exceed the time from Q to O' . But that is why the glass must be curved, because the total excess time must be compensated by the delay in passing from V to Q ! Now the excess time along route OP is $h^2/2s$, and the excess time on the other route is $nh^2/2s'$. This excess time, which must be matched by the delay in going along VQ , differs from what it would have been in a vacuum, because there is a medium present. In other words, the time to go from V to Q is not as if it were straight in the air, but it is slower by the factor n , so that the excess delay in this distance is then $(n - 1)VQ$. And now, how large is VQ ? If the point C is the center of the sphere and if its radius is R , we see by the same formula that the distance VQ is equal to $h^2/2R$. Therefore we discover that the law that connects the distances s and s' , and that gives us the radius of curvature R of the surface that we need, is

$$(h^2/2s) + (nh^2/2s') = (n - 1)h^2/2R \quad (27.2)$$

or

$$(1/s) + (n/s') = (n - 1)/R. \quad (27.3)$$

If we have a position O and another position O' , and want to focus light from O to O' , then we can calculate the required radius of curvature R of the surface by this formula.

Now it turns out, interestingly, that the same lens, with the same curvature R , will focus for other distances, namely, for any pair of distances such that the sum of the two reciprocals, one multiplied by n , is a constant. Thus a given lens will (so long as we limit ourselves to paraxial rays) focus not only from O to O' , but between an infinite number of other pairs of points, so long as those pairs of points bear the relationship that $1/s + n/s'$ is a constant, characteristic of the lens.

In particular, an interesting case is that in which $s \rightarrow \infty$. We can see from the formula that as one s increases, the other decreases. In other words, if point O

goes out, point O' comes in, and vice versa. As point O goes toward infinity, point O' keeps moving in until it reaches a certain distance, called the *focal length* f' , inside the material. If parallel rays come in, they will meet the axis at a distance f' . Likewise, we could imagine it the other way. (Remember the reciprocity rule: if light will go from O to O' , of course it will also go from O' to O .) Therefore, if we had a light source inside the glass, we might want to know where the focus is. In particular, if the light in the glass were at infinity (same problem) where would it come to a focus outside? This distance is called f . Of course, we can also put it the other way. If we had a light source at f and the light went through the surface, then it would go out as a parallel beam. We can easily find out what f and f' are:

$$n/f' = (n - 1)/R \quad \text{or} \quad f' = Rn/(n - 1), \quad (27.4)$$

$$1/f = (n - 1)/R \quad \text{or} \quad f = R/(n - 1). \quad (27.5)$$

We see an interesting thing: if we divide each focal length by the corresponding index of refraction we get the same result! This theorem, in fact, is general. It is true of any system of lenses, no matter how complicated, so it is interesting to remember. We did not prove here that it is general—we merely noted it for a single surface, but it happens to be true in general that the two focal lengths of a system are related in this way. Sometimes Eq. (27.3) is written in the form

$$1/s + n/s' = 1/f. \quad (27.6)$$

This is more useful than (27.3) because we can measure f more easily than we can measure the curvature and index of refraction of the lens: if we are not interested in designing a lens or in knowing how it got that way, but simply lift it off a shelf, the interesting quantity is f , not the n and the 1 and the R !

Now an interesting situation occurs if s becomes less than f . What happens then? If $s < f$, then $(1/s) > (1/f)$, and therefore s' is negative; our equation says that the light will focus only with a negative value of s' , whatever that means! It does mean something very interesting and very definite. It is still a useful formula, in other words, even when the numbers are negative. What it means is shown in Fig. 27-3. If we draw the rays which are diverging from O , they will be bent, it is true, at the surface, and they will not come to a focus, because O is so close in that they are “beyond parallel.” However, they diverge as if they had come from a point O' outside the glass. This is an apparent image, sometimes called a *virtual image*. The image O' in Fig. 27-2 is called a *real image*. If the light really comes to a point, it is a real image. But if the light *appears* to be coming from a point, a fictitious point different from the original point, it is a virtual image. So when s' comes out negative, it means that O' is on the other side of the surface, and everything is all right.

Now consider the interesting case where R is equal to infinity; then we have $(1/s) + (n/s') = 0$. In other words, $s' = -ns$, which means that if we look from a dense medium into a rare medium and see a point in the rare medium, it appears to be deeper by a factor n . Likewise, we can use the same equation backwards, so that if we look into a plane surface at an object that is at a certain distance inside the dense medium, it will appear as though the light is coming from not as far back (Fig. 27-4). When we look at the bottom of a swimming pool from above, it does not look as deep as it really is, by a factor $3/4$, which is the reciprocal of the index of refraction of water.

We could go on, of course, to discuss the spherical mirror. But if one appreciates the ideas involved, he should be able to work it out for himself. Therefore we leave it to the student to work out the formula for the spherical mirror, but we mention that it is well to adopt certain conventions concerning the distances involved:

- (1) The object distance s is positive if the point O is to the left of the surface.
- (2) The image distance s' is positive if the point O' is to the right of the surface.
- (3) The radius of curvature of the surface is positive if the center is to the right of the surface.

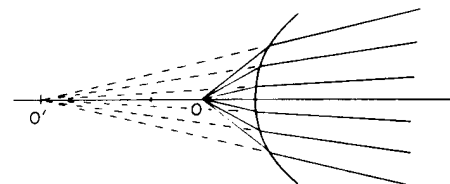


Fig. 27-3. A virtual image.

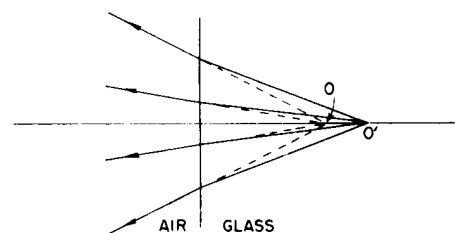


Fig. 27-4. A plane surface re-images the light from O' to O .

In Fig. 27-2, for example, s , s' , and R are all positive; in Fig. 27-3, s and R are positive, but s' is negative. If we had used a concave surface, our formula (27.3) would still give the correct result if we merely make R a negative quantity.

In working out the corresponding formula for a mirror, using the above conventions, you will find that if you put $n = -1$ throughout the formula (27.3) (as though the material behind the mirror had an index -1), the right formula for a mirror results!

Although the derivation of formula (27.3) is simple and elegant, using least time, one can of course work out the same formula using Snell's law, remembering that the angles are so small that the sines of angles can be replaced by the angles themselves.

27-3 The focal length of a lens

Now we go on to consider another situation, a very practical one. Most of the lenses that we use have two surfaces, not just one. How does this affect matters? Suppose that we have two surfaces of different curvature, with glass filling the space between them (Fig. 27-5). We want to study the problem of focusing from a point O to an alternate point O' . How can we do that? The answer is this: First, use formula (27.3) for the first surface, forgetting about the second surface. This will tell us that the light which was diverging from O will appear to be converging or diverging, depending on the sign, from some other point, say O'' . Now we consider a new problem. We have a different surface, between glass and air, in which rays are converging toward a certain point O' . Where will they actually converge? We use the same formula again! We find that they converge at O''' . Thus, if necessary, we can go through 75 surfaces by just using the same formula in succession, from one to the next!

There are some rather high-class formulas that would save us considerable energy in the few times in our lives that we might have to chase the light through five surfaces, but it is easier just to chase it through five surfaces when the problem arises than it is to memorize a lot of formulas, because it may be we will never have to chase it through any surfaces at all!

In any case, the principle is that when we go through one surface we find a new position, a new focal point, and then take that point as the starting point for the next surface, and so on. In order to actually do this, since on the second surface we are going from n to 1 rather than from 1 to n , and since in many systems there is more than one kind of glass, so that there are indices $n_1, n_2, \dots$, we really need a generalization of formula (27.3) for a case where there are two different indices, n_1 and n_2 , rather than only n . Then it is not difficult to prove that the general form of (27.3) is

$$(n_1/s) + (n_2/s') = (n_2 - n_1)/R. \quad (27.7)$$

Particularly simple is the special case in which the two surfaces are very close together—so close that we may ignore small errors due to the thickness. If we draw the lens as shown in Fig. 27-6, we may ask this question: How must the lens be built so as to focus light from O to O' ? Suppose the light comes exactly to the edge of the lens, at point P . Then the excess time in going from O to O' is $(n_1 h^2/2s) + (n_1 h^2/2s')$, ignoring for a moment the presence of the thickness T of glass of index n_2 . Now, to make the time for the direct path equal to that for the path OPO' , we have to use a piece of glass whose thickness T at the center is such that the delay introduced in going through this thickness is enough to compensate for the excess time above. Therefore the thickness of the lens at the center must be given by the relationship

$$(n_1 h^2/2s) + (n_1 h^2/2s') = (n_2 - n_1)T. \quad (27.8)$$

We can also express T in terms of the radii R_1 and R_2 of the two surfaces. Paying attention to our convention (3), we thus find, for $R_1 < R_2$ (a convex lens),

$$T = (h^2/2R_1) - (h^2/2R_2). \quad (27.9)$$

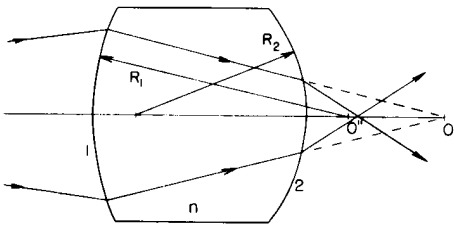


Fig. 27-5. Image formation by a two-surface lens.

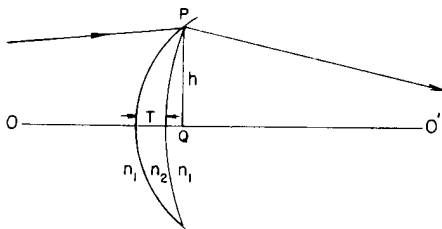


Fig. 27-6. A thin lens with two positive radii.

Therefore, we finally get

$$(n_1/s) + (n_1/s') = (n_2 - n_1)(1/R_1 - 1/R_2). \quad (27.10)$$

Now we note again that if one of the points is at infinity, the other will be at a point which we will call the focal length f . The focal length f is given by

$$1/f = (n - 1)(1/R_1 - 1/R_2), \quad (27.11)$$

where $n = n_2/n_1$.

Now, if we take the opposite case, where s goes to infinity, we see that s' is at the focal length f' . This time the focal lengths are equal. (This is another special case of the general rule that the ratio of the two focal lengths is the ratio of the indices of refraction in the two media in which the rays focus. In this particular optical system, the initial and final indices are the same, so the two focal lengths are equal.)

Forgetting for a moment about the actual formula for the focal length, if we bought a lens that somebody designed with certain radii of curvature and a certain index, we could measure the focal length, say, by seeing where a point at infinity focuses. Once we had the focal length, it would be better to write our equation in terms of the focal length directly, and the formula then is

$$(1/s) + (1/s') = 1/f. \quad (27.12)$$

Now let us see how the formula works and what it implies in different circumstances. First, it implies that if s or s' is infinite the other one is f . That means that parallel light focuses at a distance f , and this in effect *defines* f . Another interesting thing it says is that both points move in the same direction. If one moves to the right, the other does also. Another thing it says is that s and s' are equal if they are both equal to $2f$. In other words, if we want a symmetrical situation, we find that they will both focus at a distance $2f$.

27-4 Magnification

So far we have discussed the focusing action only for points on the axis. Now let us discuss also the imaging of objects not exactly on the axis, but a little bit off, so that we can understand the properties of *magnification*. When we set up a lens so as to focus light from a small filament onto a "point" on a screen, we notice that on the screen we get a "picture" of the same filament, except of a larger or smaller size than the true filament. This must mean that the light comes to a focus from *each point* of the filament. In order to understand this a little better, let us analyze the thin lens system shown schematically in Fig. 27-7. We know the following facts:

- (1) Any ray that comes in parallel on one side proceeds toward a certain particular point called the focus on the other side, at a distance f from the lens.
- (2) Any ray that arrives at the lens from the focus on one side comes out parallel to the axis on the other side.

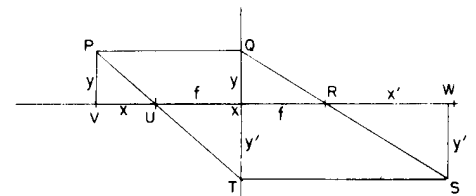


Fig. 27-7. The geometry of imaging by a thin lens.

This is all we need to establish formula (27.12) by geometry, as follows: Suppose we have an object at some distance x from the focus; let the height of the object be y . Then we know that one of the rays, namely PQ , will be bent so as to pass through the focus R on the other side. Now if the lens will focus point P at all, we can find out where if we find out where just one other ray goes, because the new focus will be where the two intersect again. We need only use our ingenuity to find the exact direction of *one* other ray. But we remember that a parallel ray goes through the focus and *vice versa*: a ray which goes through the focus will come out parallel! So we draw ray PT through U . (It is true that the actual rays which are doing the focusing may be much more limited than the two we have drawn, but they are harder to figure, so we make believe that we can make this ray.) Since it would come out parallel, we draw TS parallel to XW . The intersection S is the point we need. This will determine the correct place and the correct height. Let

us call the height y' and the distance from the focus, x' . Now we may derive a lens formula. Using the similar triangles PVU and TXU , we find

$$\frac{y'}{f} = \frac{y}{x}. \quad (27.13)$$

Similarly, from triangles SWR and QXR , we get

$$\frac{y'}{x'} = \frac{y}{f}. \quad (27.14)$$

Solving each for y'/y , we find that

$$\frac{y'}{y} = \frac{x'}{f} = \frac{f}{x}. \quad (27.15)$$

Equation (27.15) is the famous lens formula; in it is everything we need to know about lenses: It tells us the magnification, y'/y , in terms of the distances and the focal lengths. It also connects the two distances x and x' with f :

$$xx' = f^2, \quad (27.16)$$

which is a much neater form to work with than Eq. (27.12). We leave it to the student to demonstrate that if we call $s = x + f$ and $s' = x' + f$, Eq. (27.12) is the same as Eq. (27.16).

27-5 Compound lenses

Without actually deriving it, we shall briefly describe the general result when we have a number of lenses. If we have a system of several lenses, how can we possibly analyze it? That is easy. We start with some object and calculate where its image is for the first lens, using formula (27.16) or (27.12) or any other equivalent formula, or by drawing diagrams. So we find an image. Then we treat this image as the source for the next lens, and use the second lens with whatever its focal length is to again find an image. We simply chase the thing through the succession of lenses. That is all there is to it. It involves nothing new in principle, so we shall not go into it. However, there is a very interesting net result of the effects of any sequence of lenses on light that starts and ends up in the same medium, say air. Any optical instrument—a telescope or a microscope with any number of lenses and mirrors—has the following property: There exist two planes, called the *principal planes* of the system (these planes are often fairly close to the first surface of the first lens and the last surface of the last lens), which have the following properties: (1) If light comes into the system parallel from the first side, it comes out at a certain focus, at a distance from the *second* principal plane equal to the focal length, just as though the system were a thin lens situated at this plane. (2) If parallel light comes in the other way, it comes to a focus at the same distance f from the *first* principal plane, again as if a thin lens were situated there. (See Fig. 27-8.)

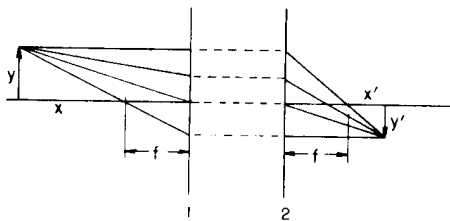


Fig. 27-8. Illustration of the principal planes of an optical system.

Of course, if we measure the distances x and x' , and y and y' as before, the formula (27.16) that we have written for the thin lens is absolutely general, provided that we measure the focal length from the principal planes and not from the center of the lens. It so happens that for a thin lens the principal planes are coincident. It is just as though we could take a thin lens, slice it down the middle, and separate it, and not notice that it was separated. Every ray that comes in pops out immediately on the other side of the second plane from the same point as it went into the first plane! The principal planes and the focal length may be found either by experiment or by calculation, and then the whole set of properties of the optical system are described. It is very interesting that the result is not complicated when we are all finished with such a big, complicated optical system.

27-6 Aberrations

Before we get too excited about how marvelous lenses are, we must hasten to add that there are also serious limitations, because of the fact that we have limited ourselves, strictly speaking, to paraxial rays, the rays near the axis. A real lens having a finite size will, in general, exhibit *aberrations*. For example, a ray that is on the axis, of course, goes through the focus; a ray that is very close to the axis will still come to the focus very well. But as we go farther out, the ray begins to deviate from the focus, perhaps by falling short, and a ray striking near the top edge comes down and misses the focus by quite a wide margin. So, instead of getting a point image, we get a smear. This effect is called *spherical aberration*, because it is a property of the spherical surfaces we use in place of the right shape. This could be remedied, for any specific object distance, by re-forming the shape of the lens surface, or perhaps by using several lenses arranged so that the aberrations of the individual lenses tend to cancel each other.

Lenses have another fault: light of different colors has different speeds, or different indices of refraction, in the glass, and therefore the focal length of a given lens is different for different colors. So if we image a white spot, the image will have colors, because when we focus for the red, the blue is out of focus, or vice versa. This property is called *chromatic aberration*.

There are still other faults. If the object is off the axis, then the focus really isn't perfect any more, when it gets far enough off the axis. The easiest way to verify this is to focus a lens and then tilt it so that the rays are coming in at a large angle from the axis. Then the image that is formed will usually be quite crude, and there may be no place where it focuses well. There are thus several kinds of errors in lenses that the optical designer tries to remedy by using many lenses to compensate each other's errors.

How careful do we have to be to eliminate aberrations? Is it possible to make an absolutely perfect optical system? Suppose we had built an optical system that is supposed to bring light exactly to a point. Now, arguing from the point of view of least time, can we find a condition on how perfect the system has to be? The system will have some kind of an entrance opening for the light. If we take the farthest ray from the axis that can come to the focus (if the system is perfect, of course), the times for all rays are exactly equal. But nothing is perfect, so the question is, how wrong can the time be for this ray and not be worth correcting any further? That depends on how perfect we want to make the image. But suppose we want to make the image as perfect as it possibly can be made. Then, of course, our impression is that we have to arrange that every ray takes as nearly the same time as possible. But it turns out that this is not true, that beyond a certain point we are trying to do something that is too fine, because the theory of geometrical optics does not work!

Remember that the principle of least time is not an accurate formulation, unlike the principle of conservation of energy or the principle of conservation of momentum. The principle of least time is only an *approximation*, and it is interesting to know how much error can be allowed and still not make any apparent difference. The answer is that if we have arranged that between the maximal ray—the worst ray, the ray that is farthest out—and the central ray, the difference in time is less than about the period that corresponds to one oscillation of the light, then there is no use improving it any further. Light is an oscillatory thing with a definite frequency that is related to the wavelength, and if we have arranged that the time difference for different rays is less than about a period, there is no use going any further.

27-7 Resolving power

Another interesting question—a very important technical question with all optical instruments—is how much *resolving power* they have. If we build a microscope, we want to see the objects that we are looking at. That means, for instance, that if we are looking at a bacterium with a spot on each end, we want to *see* that

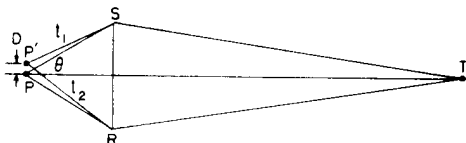


Fig. 27-9. The resolving power of an optical system.

there are two dots when we magnify them. One might think that all we have to do is to get enough magnification—we can always add another lens, and we can always magnify again and again, and with the cleverness of designers, all the spherical aberrations and chromatic aberrations can be cancelled out, and there is no reason why we cannot keep on magnifying the image. So the limitations of a microscope are not that it is impossible to build a lens that magnifies more than 2000 diameters. We can build a system of lenses that magnifies 10,000 diameters, but we *still* could not see two points that are too close together because of the limitations of geometrical optics, because of the fact that least time is not precise.

To discover the rule that determines how far apart two points have to be so that at the image they appear as separate points can be stated in a very beautiful way associated with the time it takes for different rays. Suppose that we disregard the aberrations now, and imagine that for a particular point P (Fig. 27-9) all the rays from object to image T take exactly the same time. (It is not true, because it is not a perfect system, but that is another problem.) Now take another nearby point, P' , and ask whether its image will be distinct from T . In other words, whether we can make out the difference between them. Of course, according to geometrical optics, there should be two point images, but what we see may be rather smeared and we may not be able to make out that there are two points. The condition that the second point is focused in a distinctly different place from the first one is that the two times for the extreme rays $P'ST$ and $P'RT$ on each side of the big opening of the lenses to go from one end to the other, must *not* be equal from the two possible object points to a given image point. Why? Because, if the times were equal, of course both would *focus* at the same point. So the times are not going to be equal. But by how much do they have to differ so that we can say that both do *not* come to a common focus, so that we can distinguish the two image points? The general rule for the resolution of any optical instrument is this: two different point sources can be resolved only if one source is focused at such a point that the times for the maximal rays from the other source to reach that point, as compared with its own true image point, differ by more than one period. It is necessary that the difference in time between the top ray and the bottom ray to the *wrong* focus shall exceed a certain amount, namely, approximately the period of oscillation of the light:

$$t_2 - t_1 > 1/\nu, \quad (27.17)$$

where ν is the frequency of the light (number of oscillations per second; also speed divided by wavelength). If the distance of separation of the two points is called D , and if the opening angle of the lens is called θ , then one can demonstrate that (27.17) is exactly equivalent to the statement that D must exceed $\lambda/n \sin \theta$, where n is the index of refraction at P and λ is the wavelength. The smallest things that we can see are therefore approximately the wavelength of light. A corresponding formula exists for telescopes, which tells us the smallest difference in angle between two stars that can just be distinguished.*

* The angle is about λ/D , where D is the lens diameter. Can you see why?

Electromagnetic Radiation

28-1 Electromagnetism

The most dramatic moments in the development of physics are those in which great syntheses take place, where phenomena which previously had appeared to be different are suddenly discovered to be but different aspects of the same thing. The history of physics is the history of such syntheses, and the basis of the success of physical science is mainly that we are *able* to synthesize.

Perhaps the most dramatic moment in the development of physics during the 19th century occurred to J. C. Maxwell one day in the 1860's, when he combined the laws of electricity and magnetism with the laws of the behavior of light. As a result, the properties of light were partly unravelled—that old and subtle stuff that is so important and mysterious that it was felt necessary to arrange a special creation for it when writing Genesis. Maxwell could say, when he was finished with his discovery, “Let there be electricity and magnetism, and there is light!”

For this culminating moment there was a long preparation in the gradual discovery and unfolding of the laws of electricity and magnetism. This story we shall reserve for detailed study next year. However, the story is, briefly, as follows. The gradually discovered properties of electricity and magnetism, of electric forces of attraction and repulsion, and of magnetic forces, showed that although these forces were rather complex, they all fell off inversely as the square of the distance. We know, for example, that the simple coulomb law for stationary charges is that the electric force field varies inversely as the square of the distance. As a consequence, for sufficiently great distances there is very little influence of one system of charges on another. Maxwell noted that the equations or the laws that had been discovered up to this time were mutually inconsistent when he tried to put them all together, and in order for the whole system to be consistent, he had to add another term to his equations. With this new term there came an amazing prediction, which was that a part of the electric and magnetic fields would fall off much more slowly with the distance than the inverse square, namely, inversely as the first power of the distance! And so he realized that electric currents in one place can affect other charges far away, and he predicted the basic effects with which we are familiar today—radio transmission, radar, and so on.

It seems a miracle that someone talking in Europe can, with mere electrical influences, be heard thousands of miles away in Los Angeles. How is it possible? It is because the fields do not vary as the inverse square, but only inversely as the first power of the distance. Finally, then, even light itself was recognized to be electric and magnetic influences extending over vast distances, generated by an almost incredibly rapid oscillation of the electrons in the atoms. All these phenomena we summarize by the word *radiation* or, more specifically, *electromagnetic radiation*, there being one or two other kinds of radiation also. Almost always, radiation means electromagnetic radiation.

And thus is the universe knit together. The atomic motions of a distant star still have sufficient influence at this great distance to set the electrons in our eye in motion, and so we know about the stars. If this law did not exist, we would all be literally in the dark about the exterior world! And the electric surgings in a galaxy five billion light years away—which is the farthest object we have found so far—can still influence in a significant and detectable way the currents in the great “dish” in front of a radio telescope. And so it is that we see the stars and the galaxies.

28-1 Electromagnetism

28-2 Radiation

28-3 The dipole radiator

28-4 Interference

This remarkable phenomenon is what we shall discuss in the present chapter. At the beginning of this course in physics we outlined a broad picture of the world, but we are now better prepared to understand some aspects of it, and so we shall now go over some parts of it again in greater detail. We begin by describing the position of physics at the end of the 19th century. All that was then known about the fundamental laws can be summarized as follows.

First, there were laws of forces: one force was the law of gravitation, which we have written down several times; the force on an object of mass m , due to another of mass M , is given by

$$\mathbf{F} = GmMe_r/r^2, \quad (28.1)$$

where $\mathbf{e}_r$ is a unit vector directed from m to M , and r is the distance between them.

Next, the laws of electricity and magnetism, as known at the end of the 19th century, are these: the electrical forces acting on a charge q can be described by two fields, called $\mathbf{E}$ and $\mathbf{B}$, and the velocity $\mathbf{v}$ of the charge q , by the equation

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (28.2)$$

To complete this law, we have to say what the formulas for $\mathbf{E}$ and $\mathbf{B}$ are in a given circumstance: if a number of charges are present, $\mathbf{E}$ and the $\mathbf{B}$ are each the sum of contributions, one from each individual charge. So if we can find the $\mathbf{E}$ and $\mathbf{B}$ produced by a single charge, we need only to add all the effects from all the charges in the universe to get the total $\mathbf{E}$ and $\mathbf{B}$! This is the principle of superposition.

What is the formula for the electric and magnetic field produced by one individual charge? It turns out that this is very complicated, and it takes a great deal of study and sophistication to appreciate it. But that is not the point. We write down the law now only to impress the reader with the beauty of nature, so to speak, i.e., that it is *possible* to summarize all the fundamental knowledge on one page, with notations that he is now familiar with. This law for the fields of an individual charge is complete and accurate, so far as we know (except for quantum mechanics) but it looks rather complicated. We shall not study all the pieces now; we only write it down to give an impression, to show that it can be written, and so that we can see ahead of time roughly what it looks like. As a matter of fact, the most *useful* way to write the correct laws of electricity and magnetism is not the way we shall now write them, but involves what are called *field equations*, which we shall learn about next year. But the mathematical notations for these are different and new, and so we write the law in an inconvenient form for calculation, but in notations that we now know.

The electric field, $\mathbf{E}$, is given by

$$\mathbf{E} = \frac{-q}{4\pi\epsilon_0} \left[\frac{\mathbf{e}_{r'}}{r'^2} + \frac{r'}{c} \frac{d}{dt} \left(\frac{\mathbf{e}_{r'}}{r'^2} \right) + \frac{1}{c^2} \frac{d^2}{dt^2} \mathbf{e}_{r'} \right]. \quad (28.3)$$

What do the various terms tell us? Take the first term, $\mathbf{E} = -q\mathbf{e}_{r'}/4\pi\epsilon_0 r'^2$. That, of course, is Coulomb's law, which we already know: q is the charge that is producing the field; $\mathbf{e}_{r'}$ is the unit vector in the direction from the point P where $\mathbf{E}$ is measured, r is the distance from P to q . But, Coulomb's law is wrong. The discoveries of the 19th century showed that influences cannot travel faster than a certain fundamental speed c , which we now call the speed of light. It is not correct that the first term is Coulomb's law, not only because it is not possible to know where the charge is *now* and at what distance it is *now*, but also because the only thing that can affect the field at a given place and time is the behavior of the charges in the *past*. How *far* in the past? The time delay, or *retarded time*, so-called, is the time it takes, at speed c , to get from the charge to the field point P . The delay is r'/c .

So to allow for this time delay, we put a little prime on r , meaning how far away it *was* when the information now arriving at P left q . Just for a moment suppose that the charge carried a light, and that the light could only come to P at the speed c . Then when we look at q , we would not see where it is now, of course, but where it *was* at some earlier time. What appears in our formula is the *apparent*

direction $\theta_{r'}$ —the direction it used to be—the so-called *retarded* direction—and at the *retarded* distance r' . That would be easy enough to understand, too, but it is also wrong. The whole thing is much more complicated.

There are several more terms. The next term is as though nature were trying to allow for the fact that the effect is retarded, if we might put it very crudely. It suggests that we should calculate the delayed coulomb field and add a correction to it, which is its rate of change times the time delay that we use. Nature seems to be attempting to guess what the field at the present time is going to be, by taking the rate of change and multiplying by the time that is delayed. But we are not yet through. There is a third term—the second derivative, with respect to t , of the unit vector in the direction of the charge. Now the formula *is* finished, and that is all there is to the electric field from an arbitrarily moving charge.

The magnetic field is given by

$$\mathbf{B} = -\mathbf{e}_{r'} \times \mathbf{E}/c. \quad (28.4)$$

We have written these down only for the purpose of showing the beauty of nature or, in a way, the power of mathematics. We do not pretend to understand *why* it is possible to write so much in such a small space, but (28.3) and (28.4) contain the machinery by which electric generators work, how light operates, all the phenomena of electricity and magnetism. Of course, to complete the story we also need to know something about the behavior of the materials involved—the properties of matter—which are not described properly by (28.3).

To finish with our description of the world of the 19th century we must mention one other great synthesis which occurred in that century, one with which Maxwell had a great deal to do also, and that was the synthesis of the phenomena of heat and mechanics. We shall study that subject soon.

What had to be added in the 20th century was that the dynamical laws of Newton were found to be all wrong, and quantum mechanics had to be introduced to correct them. Newton's laws are approximately valid when the scale of things is sufficiently large. These quantum-mechanical laws, combined with the laws of electricity, have only recently been combined to form a set of laws called *quantum electrodynamics*. In addition, there were discovered a number of new phenomena, of which the first was radioactivity, discovered by Becquerel in 1898—he just sneaked it in under the 19th century. This phenomenon of radioactivity was followed up to produce our knowledge of nuclei and new kinds of forces that are not gravitational and not electrical, but new particles with different interactions, a subject which has still not been unravelled.

For those purists who know more (the professors who happen to be reading this), we should add that when we say that (28.3) is a complete expression of the knowledge of electrodynamics, we are not being entirely accurate. There was a problem that was not quite solved at the end of the 19th century. When we try to calculate the field from all the charges *including the charge itself that we want the field to act on*, we get into trouble trying to find the distance, for example, of a charge from itself, and dividing something by that distance, which is zero. The problem of how to handle the part of this field which is generated by the very charge on which we want the field to act is not yet solved today. So we leave it there; we do not have a complete solution to that puzzle yet, and so we shall avoid the puzzle for as long as we can.

28-2 Radiation

That, then, is a summary of the world picture. Now let us use it to discuss the phenomena called radiation. To discuss these phenomena, we must select from Eq. (28.3) only that piece which varies inversely as the distance and not as the square of the distance. It turns out that when we finally do find that piece, it is so simple in its form that it is legitimate to study optics and electrodynamics in an elementary way by taking it as “the law” of the electric field produced by a moving charge far away. We shall take it temporarily as a given law which we will learn about in detail next year.

Of the terms appearing in (28.3), the first one evidently goes inversely as the square of the distance, and the second is only a correction for delay, so it is easy to show that both of them vary inversely as the square of the distance. All of the effects we are interested in come from the third term, which is not very complicated, after all. What this term says is: look at the charge and note the direction of the unit vector (we can project the end of it onto the surface of a unit sphere). As the charge moves around, the unit vector wiggles, and *the acceleration of that unit vector is what we are looking for*. That is all. Thus

$$\mathbf{E} = \frac{-q}{4\pi\epsilon_0 c^2} \frac{d^2 \mathbf{e}_{r'}}{dt^2} \quad (28.5)$$

is a statement of the laws of radiation, because that is the only important term when we get far enough away that the fields are varying inversely as the distance. (The parts that go as the square have fallen off so much that we are not interested in them.)

Now we can go a little bit further in studying (28.5) to see what it means. Suppose a charge is moving in any manner whatsoever, and we are observing it from a distance. We imagine for a moment that in a sense it is “lit up” (although it is light that we are trying to explain); we imagine it as a little white dot. Then we would see this white dot running around. But we don’t see *exactly* how it is running around right *now*, because of the delay that we have been talking about. What counts is how it was moving *earlier*. The unit vector $\mathbf{e}_{r'}$ is pointed toward the apparent position of the charge. Of course, the end of $\mathbf{e}_{r'}$ goes on a slight curve, so that its acceleration has two components. One is the transverse piece, because the end of it goes up and down, and the other is a radial piece because it stays on a sphere. It is easy to demonstrate that the latter is much smaller and varies as the inverse square of r when r is very great. This is easy to see, for when we imagine that we move a given source farther and farther away, then the wiggles of $\mathbf{e}_{r'}$ look smaller and smaller, inversely as the distance, but the radial component of acceleration is varying much more rapidly than inversely as the distance. So for practical purposes all we have to do is project the motion on a plane at unit distance. Therefore we find the following rule: Imagine that we look at the moving charge and that everything we see is delayed—like a painter trying to paint a scene on a screen at a unit distance. A real painter, of course, does not take into account the fact that light is going at a certain speed, but paints the world as he sees it. We want to see what his picture would look like. So we see a dot, representing the charge, moving about in the picture. The acceleration of that dot is proportional to the electric field. That is all—all we need.

Thus Eq. (28.5) is the complete and correct formula for radiation; even relativity effects are all contained in it. However, we often want to apply it to a still simpler circumstance in which the charges are moving only a small distance at a relatively slow rate. Since they are moving slowly, they do not move an appreciable distance from where they start, so that the delay time is practically constant. Then the law is still simpler, because the delay time is fixed. Thus we imagine that the charge is executing a very tiny motion at an effectively constant distance. The delay at the distance r is r/c . Then our rule becomes the following: If the charged object is moving in a very small motion and it is laterally displaced by the distance $x(t)$, then the angle that the unit vector $\mathbf{e}_{r'}$ is displaced is x/r , and since r is practically constant, the x -component of $d^2 \mathbf{e}_{r'}/dt^2$ is simply the acceleration of x itself at an earlier time, and so finally we get the law we want, which is

$$E_x(t) = \frac{-q}{4\pi\epsilon_0 c^2 r} a_x \left(t - \frac{r}{c} \right). \quad (28.6)$$

Only the component of a_x perpendicular to the line of sight is important. Let us see why that is. Evidently, if the charge is moving in and out straight at us, the unit vector in that direction does not wiggle at all, and it has no acceleration. So it is only the sidewise motion which is important, only the acceleration that we see projected on the screen.

28-3 The dipole radiator

As our fundamental “law” of electromagnetic radiation, we are going to assume that (28.6) is true, i.e., that the electric field produced by an accelerating charge which is moving nonrelativistically at a very large distance r approaches that form. The electric field varies inversely as r and is proportional to the acceleration of the charge, projected onto the “plane of sight,” and this acceleration is not today’s acceleration, but the acceleration that it had at an earlier time, the amount of delay being a time, r/c . In the remainder of this chapter we shall discuss this law so that we can understand it better physically, because we are going to use it to understand all of the phenomena of light and radio propagation, such as reflection, refraction, interference, diffraction, and scattering. It is the central law, and is all we need. All the rest of Eq. (28.3) was written down only to set the stage, so that we could appreciate where (28.6) fits and how it comes about.

We shall discuss (28.3) further next year. In the meantime, we shall accept it as true, but not just on a theoretical basis. We may devise a number of experiments which illustrate the character of the law. In order to do so, we need an accelerating charge. It should be a single charge, but if we can make a great many charges move together, all the same way, we know that the field will be the sum of the effects of each of the individual charges; we just add them together. As an example, consider two pieces of wire connected to a generator, as shown in Fig. 28-1. The idea is that the generator makes a potential difference, or a field, which pulls electrons away from piece A and pushes them into B at one moment, and then, an infinitesimal time later, it reverses the effect and pulls the electrons out of B and pumps them back into A ! So in these two wires charges, let us say, are accelerating upward in wire A and upward in wire B for one moment, and a moment later they are accelerating downward in wire A and downward in wire B . The fact that we need two wires and a generator is merely that this is a way of doing it. The net result is that we merely have a charge accelerating up and down as though A and B were one single wire. A wire that is very short compared with the distance light travels in one oscillation period is called an *electric dipole oscillator*. Thus we have the circumstance that we need to apply our law, which tells us that this charge makes an electric field, and so we need an instrument to detect an electric field, and the instrument we use is the same thing—a pair of wires like A and B ! If an electric field is applied to such a device, it will produce a force which will pull the electrons up on both wires or down on both wires. This signal is detected by means of a rectifier mounted between A and B , and a tiny, fine wire carries the information into an amplifier, where it is amplified so we can hear the audio-frequency tone with which the radiofrequency is modulated. When this probe feels an electric field, there will be a loud noise coming out of the loudspeaker, and when there is no electric field driving it, there will be no noise.

Because the room in which the waves we are measuring has other objects in it, our electric field will shake electrons in these other objects; the electric field makes these other charges go up and down, and in going up and down, these also produce an effect on our probe. Thus for a successful experiment we must hold things fairly close together, so that the influences from the walls and from ourselves—the reflected waves—are relatively small. So the phenomena will not turn out to appear to be precisely and perfectly in accord with Eq. (28.6), but will be close enough that we shall be able to appreciate the law.

Now we turn the generator on and hear the audio signal. We find a strong field when the detector D is parallel to the generator G at point 1 (Fig. 28-2). We find the same amount of field also at any other azimuth angle about the axis of G , because it has no directional effects. On the other hand, when the detector is at 3 the field is zero. That is all right, because our formula said that the field should be the acceleration of the charge *projected perpendicular* to the line of sight. Therefore when we look down on G , the charge is moving toward and away from D , and there is no effect. So that checks the first rule, that there is no effect when the charge is moving directly toward us. Secondly, the formula says that the electric field should be perpendicular to r and in the plane of G and r ; so if we put D at 1 but rotate it 90°, we should get no signal. And this is just what we find, the

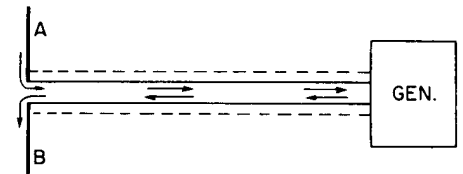


Fig. 28-1. A high-frequency signal generator drives charges up and down on two wires.

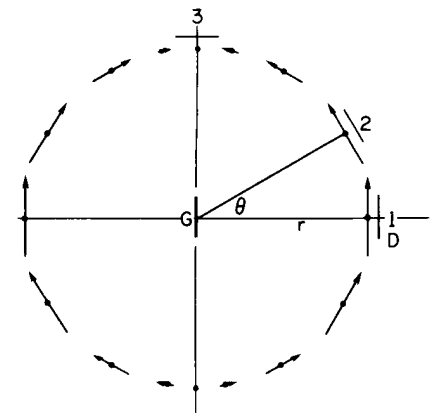


Fig. 28-2. The instantaneous electric field on a sphere centered at a localized, linearly oscillating charge.

electric field is indeed vertical, and not horizontal. When we move D to some intermediate angle, we see that the strongest signal occurs when it is oriented as shown, because although G is vertical, it does not produce a field that is simply parallel to itself—it is the *projection of the acceleration perpendicular to the line of sight* that counts. The signal is weaker at 2 than it is at 1, because of the projection effect.

28-4 Interference

Next, we may test what happens when we have two sources side by side a few centimeters apart (Fig. 28-3). The law is that the two sources should add their effects at point 1 when both of the sources are connected to the same generator and are both moving up and down the same way, so that the total electric field is the sum of the two and is twice as strong as it was before.

Now comes an interesting possibility. Suppose we make the charges in S_1 and S_2 both accelerate up and down, but delay the timing of S_2 so that they are 180° out of phase. Then the field produced by S_1 will be in one direction and the field produced by S_2 will be in the opposite direction at any instant, and therefore we should get *no* effect at point 1. The phase of oscillation is neatly adjustable by means of a pipe which is carrying the signal to S_2 . By changing the length of this pipe we change the time it takes the signal to arrive at S_2 and thus we change the phase of that oscillation. By adjusting this length, we can indeed find a place where there is no more signal left, in spite of the fact that both S_1 and S_2 are moving! The fact that they are both moving can be checked, because if we cut one out, we can see the motion of the other. So the two of them together can produce *zero* if everything is adjusted correctly.

Now, it is very interesting to show that the addition of the two fields is in fact a *vector* addition. We have just checked it for up and down motion, but let us check two nonparallel directions. First, we restore S_1 and S_2 to the same phase; that is, they are again moving together. But now we turn S_1 through 90° , as shown in Fig. 28-4. Now we should have at point 1 the sum of two effects, one of which is vertical and the other horizontal. The electric field is the vector sum of these two in-phase signals—they are both strong at the same time and go through zero together; the total field should be a signal R at 45° . If we turn D to get the maximum noise, it should be at about 45° , and not vertical. And if we turn it at right angles to that direction, we should get zero, which is easy to measure. Indeed, we observe just such behavior!

Now, how about the retardation? How can we demonstrate that the signal is retarded? We could, with a great deal of equipment, measure the time at which it arrives, but there is another, very simple way. Referring again to Fig. 28-3, suppose that S_1 and S_2 are in phase. They are both shaking together, and they produce equal electric fields at point 1. But suppose we go to a certain place 2 which is closer to S_2 and farther from S_1 . Then, in accordance with the principle that the acceleration should be retarded by an amount equal to r/c , if the retardations are not equal, the signals are no longer in phase. Thus it should be possible to find a position at which the distances of D from S_1 and S_2 differ by some amount Δ , in such a manner that there is no net signal. That is, the distance Δ is to be the distance light goes in one-half an oscillation of the generator. We may go still further, and find a point where the difference is greater by a whole cycle; that is to say, the signal from the first antenna reaches point 3 with a delay in time that is greater than that of the second antenna by just the length of time it takes for the electric current to oscillate once, and therefore the two electric fields produced at 3 are in phase again. At point 3 the signal is strong again.

This completes our discussion of the experimental verification of some of the important features of Eq. (28.6). Of course we have not really checked the $1/r$ variation of the electric field strength, or the fact that there is also a magnetic field that goes along with the electric field. To do so would require rather sophisticated techniques and would hardly add to our understanding at this point. In any case, we have checked those features that are of the greatest importance for our later applications, and we shall come back to study some of the other properties of electromagnetic waves next year.

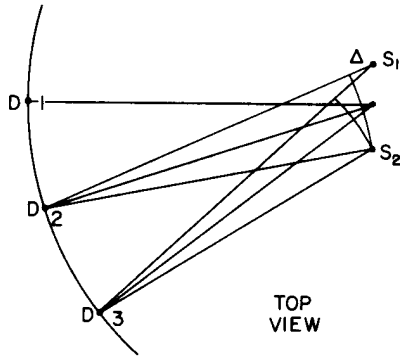


Fig. 28-3. Illustration of interference of sources.

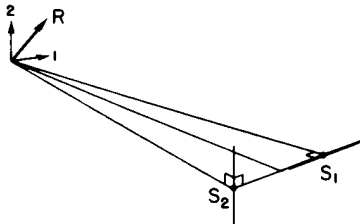


Fig. 28-4. Illustration of the vector character of the combination of sources.

Interference

29-1 Electromagnetic waves

In this chapter we shall discuss the subject of the preceding chapter more mathematically. We have qualitatively demonstrated that there are maxima and minima in the radiation field from two sources, and our problem now is to describe the field in mathematical detail, not just qualitatively.

We have already physically analyzed the meaning of formula (28.6) quite satisfactorily, but there are a few points to be made about it mathematically. In the first place, if a charge is accelerating up and down along a line, in a motion of very small amplitude, the field at some angle θ from the axis of the motion is in a direction at right angles to the line of sight and in the plane containing both the acceleration and the line of sight (Fig. 29-1). If the distance is called r , then at time t the electric field has the magnitude

$$E(t) = \frac{-qa(t - r/c) \sin \theta}{4\pi\epsilon_0 c^2 r}, \quad (29.1)$$

where $a(t - r/c)$ is the acceleration at the time $(t - r/c)$, called the *retarded* acceleration.

Now it would be interesting to draw a picture of the field under different conditions. The thing that is interesting, of course, is the factor $a(t - r/c)$, and to understand it we can take the simplest case, $\theta = 90^\circ$, and plot the field graphically. What we had been thinking of before is that we stand in one position and ask how the field there changes with time. But instead of that, we are now going to see what the field looks like at different positions in space at a given instant. So what we want is a "snapshot" picture which tells us what the field is in different places. Of course it depends upon the acceleration of the charge. Suppose that the charge at first had some particular motion: it was initially standing still, and it suddenly accelerated in some manner, as shown in Fig. 29-2, and then stopped. Then, a little bit later, we measure the field at a different place. Then we may assert that the field will appear as shown in Fig. 29-3. At each point the field is determined by the acceleration of the charge at an earlier time, the amount earlier being the delay r/c . The field at farther and farther points is determined by the acceleration at earlier and earlier times. So the curve in Fig. 29-3 is really, in a sense, a "reversed" plot of the acceleration as a function of time; the distance is related to time by a constant scale factor c , which we often take as unity. This is easily seen by considering the mathematical behavior of $a(t - r/c)$. Evidently, if we add a little time Δt , we get the same value for $a(t - r/c)$ as we would have if we had subtracted a little distance: $\Delta r = -c \Delta t$.

Stated another way: if we add a little time Δt , we can restore $a(t - r/c)$ to its former value by *adding* a little distance $\Delta r = c \Delta t$. That is, as time goes on *the field moves as a wave outward from the source*. That is the reason why we sometimes say light is propagated as waves. It is equivalent to saying that the field is delayed, or to saying that the electric field is moving outward as time goes on.

An interesting special case is that where the charge q is moving up and down in an oscillatory manner. The case which we studied experimentally in the last chapter was one in which the displacement x at any time t was equal to a certain constant x_0 , the magnitude of the oscillation, times $\cos \omega t$. Then the acceleration is

$$a = -\omega^2 x_0 \cos \omega t = a_0 \cos \omega t, \quad (29.2)$$

29-1 Electromagnetic waves

29-2 Energy of radiation

29-3 Sinusoidal waves

29-4 Two dipole radiators

29-5 The mathematics of interference

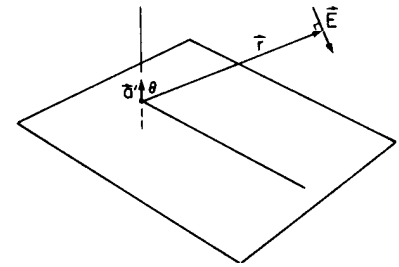


Fig. 29-1. The electric field $\mathbf{E}$ due to a positive charge whose retarded acceleration is a' .

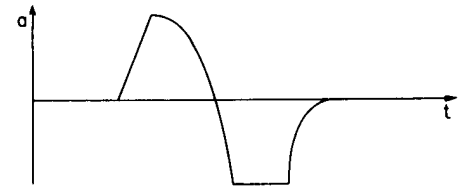


Fig. 29-2. The acceleration of a certain charge as a function of time.

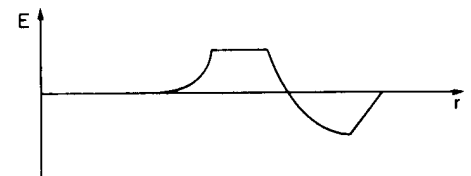


Fig. 29-3. The electric field as a function of position at a later time. (The $1/r$ variation is ignored.)

where a_0 is the maximum acceleration, $-\omega^2 x_0$. Putting this formula into (29.1), we find

$$E = -q \sin \theta \frac{a_0 \cos \omega(t - r/c)}{4\pi\epsilon_0 r c^2}. \quad (29.3)$$

Now, ignoring the angle θ and the constant factors, let us see what that looks like as a function of position or as a function of time.

29-2 Energy of radiation

First of all, at any particular moment or in any particular place, the strength of the field varies inversely as the distance r , as we mentioned previously. Now we must point out that the *energy* content of a wave, or the energy effects that such an electric field can have, are proportional to the *square* of the field, because if, for instance, we have some kind of a charge or an oscillator in the electric field, then if we let the field act on the oscillator, it makes it move. If this is a linear oscillator, the acceleration, velocity, and displacement produced by the electric field acting on the charge are all proportional to the field. So the kinetic energy which is developed in the charge is proportional to the *square* of the field. So we shall take it that the energy that a field can deliver to a system is proportional somehow to the square of the field.

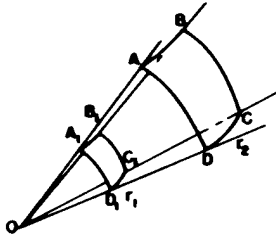


Fig. 29-4. The energy flowing within the cone OABCD is independent of the distance r at which it is measured.

This means that the energy that the source can deliver decreases as we get farther away; in fact, it varies *inversely as the square of the distance*. But that has a very simple interpretation: if we wanted to pick up all the energy we could from the wave in a certain cone at a distance r_1 (Fig. 29-4), and we do the same at another distance r_2 , we find that the amount of energy per unit area at any one place goes inversely as the square of r , but the area of the surface intercepted by the cone goes *directly* as the square of r . So the energy that we can take out of the wave within a given conical angle is the same, no matter how far away we are! In particular, the total energy that we could take out of the whole wave by putting absorbing oscillators all around is a certain fixed amount. So the fact that the amplitude of E varies as $1/r$ is the same as saying that there is an energy flux which is never lost, an energy which goes on and on, spreading over a greater and greater effective area. Thus we see that after a charge has oscillated, it has lost some energy which it can never recover; the energy keeps going farther and farther away without diminution. So if we are far enough away that our basic approximation is good enough, the charge cannot recover the energy which has been, as we say, radiated away. Of course the energy still exists somewhere, and is available to be picked up by other systems. We shall study this energy "loss" further in Chapter 32.

Let us now consider more carefully how the wave (29.3) varies as a function of time at a given place, and as a function of position at a given time. Again we ignore the $1/r$ variation and the constants.

29-3 Sinusoidal waves

First let us fix the position r , and watch the field as a function of time. It is oscillatory at the angular frequency ω . The angular frequency ω can be defined as the *rate of change of phase with time* (radians per second). We have already studied such a thing, so it should be quite familiar to us by now. The *period* is the time needed for one oscillation, one complete cycle, and we have worked that out too; it is $2\pi/\omega$, because ω times the period is one cycle of the cosine.

Now we introduce a new quantity which is used a great deal in physics. This has to do with the opposite situation, in which we fix t and look at the wave as a function of distance r . Of course we notice that, as a function of r , the wave (29.3) is also oscillatory. That is, aside from $1/r$, which we are ignoring, we see that E oscillates as we change the position. So, in analogy with ω , we can define a quantity called the *wave number*, symbolized as k . This is defined as the *rate of change of phase with distance* (radians per meter). That is, as we move in space at a fixed time, the phase changes.

There is another quantity that corresponds to the period, and we might call it the period in space, but it is usually called the wavelength, symbolized λ . The wavelength is the distance occupied by one complete cycle. It is easy to see, then, that the wavelength is $2\pi/k$, because k times the wavelength would be the number of radians that the whole thing changes, being the product of the rate of change of the radians per meter, times the number of meters, and we must make a 2π change for one cycle. So $k\lambda = 2\pi$ is exactly analogous to $\omega t_0 = 2\pi$.

Now in our particular wave there is a definite relationship between the frequency and the wavelength, but the above definitions of k and ω are actually quite general. That is, the wavelength and the frequency may not be related in the same way in other physical circumstances. However, in our circumstance the rate of change of phase with distance is easily determined, because if we call $\phi = \omega(t - r/c)$ the phase, and differentiate (partially) with respect to distance r , the rate of change, $\partial\phi/\partial r$, is

$$\left| \frac{\partial\phi}{\partial r} \right| = k = \frac{\omega}{c}. \quad (29.4)$$

There are many ways to represent the same thing, such as

$$\lambda = ct_0 \quad (29.5) \quad \lambda\nu = c \quad (29.7)$$

$$\omega = ck \quad (29.6) \quad \omega\lambda = 2\pi c \quad (29.8)$$

Why is the wavelength equal to c times the period? That's very easy, of course, because if we sit still and wait for one period to elapse, the waves, travelling at the speed c , will move a distance ct_0 , and will of course have moved over just one wavelength.

In a physical situation other than that of light, k is not necessarily related to ω in this simple way. If we call the distance along an axis x , then the formula for a cosine wave moving in a direction x with a wave number k and an angular frequency ω will be written in general as $\cos(\omega t - kx)$.

Now that we have introduced the idea of wavelength, we may say something more about the circumstances in which (29.1) is a legitimate formula. We recall that the field is made up of several pieces, one of which varies inversely as r , another part which varies inversely as r^2 , and others which vary even faster. It would be worth while to know in what circumstances the $1/r$ part of the field is the most important part, and the other parts are relatively small. Naturally, the answer is "if we go 'far enough' away," because terms which vary inversely as the square ultimately become negligible compared with the $1/r$ term. How far is "far enough"? The answer is, qualitatively, that the other terms are of order λ/r smaller than the $1/r$ term. Thus, so long as we are beyond a few wavelengths, (29.1) is an excellent approximation to the field. Sometimes the region beyond a few wavelengths is called the "wave zone."

29-4 Two dipole radiators

Next let us discuss the mathematics involved in combining the effects of two oscillators to find the net field at a given point. This is very easy in the few cases that we considered in the previous chapter. We shall first describe the effects qualitatively, and then more quantitatively. Let us take the simple case, where the oscillators are situated with their centers in the same horizontal plane as the detector, and the line of vibration is vertical.

Figure 29-5(a) represents the top view of two such oscillators, and in this particular example they are half a wavelength apart in a N-S direction, and are oscillating together in the same phase, which we call zero phase. Now we would like to know the intensity of the radiation in various directions. By the intensity we mean the amount of energy that the field carries past us per second, which is proportional to the square of the field, averaged in time. So the thing to look at, when we want to know how bright the light is, is the square of the electric field, not the electric field itself. (The electric field tells the strength of the force felt by a

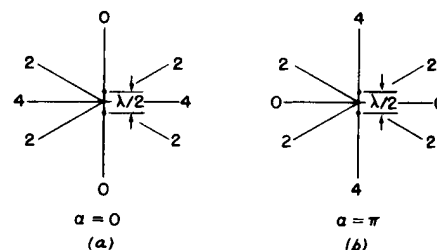


Fig. 29-5. The intensities in various directions from two dipole oscillators one-half wavelength apart. Left: in phase ($\alpha = 0$). Right: one-half period out of phase ($\alpha = \pi$).

stationary charge, but the amount of energy that is going past, in watts per square meter, is proportional to the square of the electric field. We shall derive the constant of proportionality in the next chapter.) If we look at the array from the W side, both oscillators contribute equally and in phase, so the electric field is twice as strong as it would be from a single oscillator. Therefore *the intensity is four times as strong as it would be if there were only one oscillator.* (The numbers in Fig. 29-5 represent how strong the intensity would be in this case, compared with what it would be if there were only a single oscillator of unit strength.) Now, in either the N or S direction along the line of the oscillators, since they are half a wavelength apart, the effect of one oscillator turns out to be out of phase by exactly half an oscillation from the other, and therefore the fields add to zero. At a certain particular intermediate angle (in fact, at 30°) the intensity is 2, and it falls off, 4, 2, 0, and so forth. We have to learn how to find these numbers at other angles. It is a question of adding two oscillations with different phases.

Let us quickly look at some other cases of interest. Suppose the oscillators are again one-half a wavelength apart, but the phase α of one is set half a period behind the other in its oscillation (Fig. 29-5b). In the W direction the intensity is now zero, because one oscillator is “pushing” when the other one is “pulling.” But in the N direction the signal from the near one comes at a certain time, and that of the other comes half a period later. But the latter was *originally* half a period behind in timing, and therefore it is now exactly *in time* with the first one, and so the intensity in this direction is 4 units. The intensity in the direction at 30° is still 2, as we can prove later.

Now we come to an interesting case which shows up a possibly useful feature. Let us remark that one of the reasons that phase relations of oscillators are interesting is for beaming radio transmitters. For instance, if we build an antenna system and want to send a radio signal, say, to Hawaii, we set the antennas up as in Fig. 29-5(a) and we broadcast with our two antennas in phase, because Hawaii is to the west of us. Then we decide that tomorrow we are going to broadcast toward Alberta, Canada. Since that is north, not west, all we have to do is to reverse the phase of one of our antennas, and we can broadcast to the north. So we can build antenna systems with various arrangements. Ours is one of the simplest possible ones; we can make them much more complicated, and by changing the phases in the various antennas we can send the beams in various directions and send most of the power in the direction in which we wish to transmit, without ever moving the antenna! In both of the preceding cases, however, while we are broadcasting toward Alberta we are wasting a lot of power on Easter Island, and it would be interesting to ask whether it is possible to send it in only *one* direction. At first sight we might think that with a pair of antennas of this nature the result is always going to be symmetrical. So let us consider a case that comes out unsymmetrical, to show the possible variety.

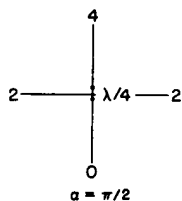


Fig. 29-6. A pair of dipole antennas giving maximum power in one direction.

If the antennas are separated by one-quarter wavelength, and if the N one is one-fourth period behind the S one in time, then what happens (Fig. 29-6)? In the W direction we get 2, as we will see later. In the S direction we get *zero*, because the signal from S comes at a certain time; that from N comes 90° later in *time*, but it is already 90° behind in its built-in phase, therefore it arrives, altogether, 180° out of phase, and there is no effect. On the other hand, in the N direction, the N signal arrives earlier than the S signal by 90° in time, because it is a quarter wavelength closer. But its phase is set so that it is oscillating 90° *behind* in time, which just compensates the delay difference, and therefore the two signals appear *together* in phase, making the field strength twice as large, and the energy four times as great.

Thus, by using some cleverness in spacing and phasing our antennas, we can send the power all in one direction. But still it is distributed over a great range of angles. Can we arrange it so that it is focused still more sharply in a particular direction? Let us consider the case of Hawaii again, where we are sending the beam east and west but it is spread over quite an angle, because even at 30° we are still getting half the intensity—we are wasting the power. Can we do better than that? Let us take a situation in which the separation is ten wavelengths (Fig.

29–7), which is more nearly comparable to the situation in which we experimented in the previous chapter, with separations of several wavelengths rather than a small fraction of a wavelength. Here the picture is quite different.

If the oscillators are ten wavelengths apart (we take the in-phase case to make it easy), we see that in the E–W direction, they are in phase, and we get a strong intensity, four times what we would get if one of them were there alone. On the other hand, at a very small angle away, the arrival times differ by 180° and the intensity is zero. To be precise, if we draw a line from each oscillator to a distant point and the difference Δ in the two distances is $\lambda/2$, half an oscillation, then they will be out of phase. So this first null occurs when that happens. (The figure is not drawn to scale; it is only a rough sketch.) This means that we do indeed have a very sharp beam in the direction we want, because if we just move over a little bit we lose all our intensity. Unfortunately for practical purposes, if we were thinking of making a radio broadcasting array and we doubled the distance Δ , then we would be a whole cycle out of phase, which is the same as being exactly *in* phase again! Thus we get many successive maxima and minima, just as we found with the $2\frac{1}{2}\lambda$ spacing in Chapter 28.

Now how can we arrange to get rid of all these extra maxima, or “lobes,” as they are called? We could get rid of the unwanted lobes in a rather interesting way. Suppose that we were to place another set of antennas between the two that we already have. That is, the outside ones are still 10λ apart, but between them, say every 2λ , we have put another antenna, and we drive them all in phase. There are now six antennas, and if we looked at the intensity in the E–W direction, it would, of course, be much higher with six antennas than with one. The field would be six times and the intensity thirty-six times as great (the square of the field). We get 36 units of intensity in that direction. Now if we look at neighboring points, we find a zero as before, roughly, but if we go farther, to where we used to get a big “bump,” we get a much smaller “bump” now. Let us try to see why.

The reason is that although we might expect to get a big bump when the distance Δ is exactly equal to the wavelength, it is true that dipoles 1 and 6 are then in phase and are cooperating in trying to get some strength in that direction. But numbers 3 and 4 are roughly $\frac{1}{2}$ a wavelength out of phase with 1 and 6, and although 1 and 6 push together, 3 and 4 push together too, but in opposite phase. Therefore there is very little intensity in this direction—but there is something; it does not balance exactly. This kind of thing keeps on happening; we get very little bumps, and we have the strong beam in the direction where we want it. But in this particular example, something else will happen: namely, since the distance between successive dipoles is 2λ , it is possible to find an angle where the distance δ between successive dipoles is exactly one wavelength, so that the effects from all of them are in phase again. Each one is delayed relative to the next one by 360° , so they all come back in phase, and we have another strong beam in that direction! It is easy to avoid this in practice because it is possible to put the dipoles closer than one wavelength apart. If we put in more antennas, closer than one wavelength apart, then this cannot happen. But the fact that this *can* happen at certain angles, if the spacing is bigger than one wavelength, is a very interesting and useful phenomenon in other applications—not radio broadcasting, but in *diffraction gratings*.

29–5 The mathematics of interference

Now we have finished our analysis of the phenomena of dipole radiators qualitatively, and we must learn how to analyze them quantitatively. To find the effect of two sources at some particular angle in the most general case, where the two oscillators have some intrinsic relative phase α from one another and the strengths A_1 and A_2 are not equal, we find that we have to add two cosines having the same frequency, but with different phases. It is very easy to find this phase difference; it is made up of a delay due to the difference in distance, and the intrinsic, built-in phase of the oscillation. Mathematically, we have to find the sum R of two waves: $R = A_1 \cos(\omega t + \phi_1) + A_2 \cos(\omega t + \phi_2)$. How do we do it?

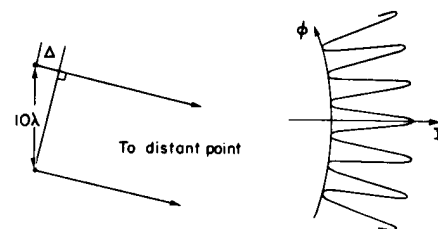


Fig. 29–7. The intensity pattern for two dipoles separated by 10λ .

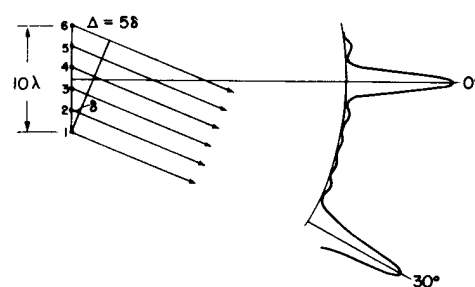


Fig. 29–8. A six-dipole antenna array and part of its intensity pattern.

It is really very easy, and we presume that we already know how to do it. However, we shall outline the procedure in some detail. First, we can, if we are clever with mathematics and know enough about cosines and sines, simply work it out. The easiest such case is the one where A_1 and A_2 are equal, let us say they are both equal to A . In those circumstances, for example (we could call this the trigonometric method of solving the problem), we have

$$R = A[\cos(\omega t + \phi_1) + \cos(\omega t + \phi_2)]. \quad (29.9)$$

Once, in our trigonometry class, we may have learned the rule that

$$\cos A + \cos B = 2 \cos \frac{1}{2}(A + B) \cos \frac{1}{2}(A - B). \quad (29.10)$$

If we know that, then we can immediately write R as

$$R = 2A \cos \frac{1}{2}(\phi_1 - \phi_2) \cos(\omega t + \frac{1}{2}\phi_1 + \frac{1}{2}\phi_2). \quad (29.11)$$

So we find that we have an oscillatory wave with a new phase and a new amplitude. In general, the result *will* be an oscillatory wave with a new amplitude A_R , which we may call the resultant amplitude, oscillating at the same frequency but with a phase difference ϕ_R , called the resultant phase. In view of this, our particular case has the following result: that the resultant amplitude is

$$A_R = 2A \cos \frac{1}{2}(\phi_1 - \phi_2), \quad (29.12)$$

and the resultant phase is the average of the two phases, and we have completely solved our problem.

Now suppose that we cannot remember that the sum of two cosines is twice the cosine of half the sum times the cosine of half the difference. Then we may use another method of analysis which is more geometrical. Any cosine function of ωt can be considered as the horizontal projection of a *rotating vector*. Suppose there were a vector $\mathbf{A}_1$ of length A_1 rotating with time, so that its angle with the horizontal axis is $\omega t + \phi_1$. (We shall leave out the ωt in a minute, and see that it makes no difference.) Suppose that we take a snapshot at the time $t = 0$, although, in fact, the picture is rotating with angular velocity ω (Fig. 29-9). The projection of $\mathbf{A}_1$ along the horizontal axis is precisely $A_1 \cos(\omega t + \phi_1)$. Now at $t = 0$ the second wave could be represented by another vector, $\mathbf{A}_2$, of length A_2 and at an angle ϕ_2 , and also rotating. They are both rotating with the same angular velocity ω , and therefore the *relative* positions of the two are fixed. The system goes around like a rigid body. The horizontal projection of $\mathbf{A}_2$ is $A_2 \cos(\omega t + \phi_2)$. But we know from the theory of vectors that if we add the two vectors in the ordinary way, by the parallelogram rule, and draw the resultant vector $\mathbf{A}_R$, the x-component of the resultant is the sum of the x-components of the other two vectors. That solves our problem. It is easy to check that this gives the correct result for the special case we treated above, where $A_1 = A_2 = A$. In this case, we see from Fig. 29-9 that $\mathbf{A}_R$ lies midway between $\mathbf{A}_1$ and $\mathbf{A}_2$ and makes an angle $\frac{1}{2}(\phi_2 - \phi_1)$ with each. Therefore we see that $A_R = 2A \cos \frac{1}{2}(\phi_2 - \phi_1)$, as before. Also, as we see from the triangle, the phase of $\mathbf{A}_R$, as it goes around, is the average angle of $\mathbf{A}_1$ and $\mathbf{A}_2$ when the two amplitudes are equal. Clearly, we can also solve for the case where the amplitudes are not equal, just as easily. We can call that the *geometrical* way of solving the problem.

There is still another way of solving the problem, and that is the *analytical* way. That is, instead of having actually to draw a picture like Fig. 29-9, we can write something down which says the same thing as the picture: instead of drawing the vectors, we write a *complex number* to represent each of the vectors. The real parts of the complex numbers are the actual physical quantities. So in our particular case the waves could be written in this way: $A_1 e^{i(\omega t + \phi_1)}$ [the real part of this is $A_1 \cos(\omega t + \phi_1)$] and $A_2 e^{i(\omega t + \phi_2)}$. Now we can add the two:

$$R = A_1 e^{i(\omega t + \phi_1)} + A_2 e^{i(\omega t + \phi_2)} = (A_1 e^{i\phi_1} + A_2 e^{i\phi_2}) e^{i\omega t} \quad (29.13)$$

or

$$\hat{R} = A_1 e^{i\phi_1} + A_2 e^{i\phi_2} = A_R e^{i\phi_R}. \quad (29.14)$$

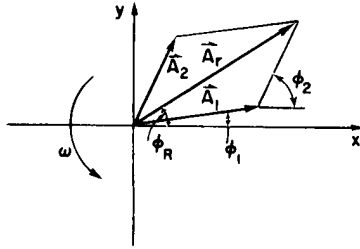


Fig. 29-9. A geometrical method for combining two cosine waves. The entire diagram is thought of as rotating counter-clockwise with angular frequency ω .

This solves the problem that we wanted to solve, because it represents the result as a complex number of magnitude A_R and phase ϕ_R .

To see how this method works, let us find the amplitude A_R which is the “length” of $\hat{R}$. To get the “length” of a complex quantity, we always multiply the quantity by its complex conjugate, which gives the length squared. The complex conjugate is the same expression, but with the sign of the i ’s reversed. Thus we have

$$A_R^2 = (A_1 e^{i\phi_1} + A_2 e^{i\phi_2})(A_1 e^{-i\phi_1} + A_2 e^{-i\phi_2}). \quad (29.15)$$

In multiplying this out, we get $A_1^2 + A_2^2$ (here the e ’s cancel), and for the cross terms we have

$$A_1 A_2 (e^{i(\phi_1 - \phi_2)} + e^{i(\phi_2 - \phi_1)}).$$

Now

$$e^{i\theta} + e^{-i\theta} = \cos \theta + i \sin \theta + \cos \theta - i \sin \theta.$$

That is to say, $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$. Our final result is therefore

$$A_R^2 = A_1^2 + A_2^2 + 2A_1 A_2 \cos(\phi_2 - \phi_1). \quad (29.16)$$

As we see, this agrees with the length of A_R in Fig. 29-9, using the rules of trigonometry.

Thus the sum of the two effects has the intensity A_1^2 we would get with one of them alone, plus the intensity A_2^2 we would get with the other one alone, plus a correction. This correction we call the *interference effect*. It is really only the difference between what we get simply by adding the intensities, and what actually happens. We call it interference whether it is positive or negative. (Interference in ordinary language usually suggests opposition or hindrance, but in physics we often do not use language the way it was originally designed!) If the interference term is positive, we call that case *constructive* interference, horrible though it may sound to anybody other than a physicist! The opposite case is called *destructive* interference.

Now let us see how to apply our general formula (29.16) for the case of two oscillators to the special situations which we have discussed qualitatively. To apply this general formula, it is only necessary to find what phase difference, $\phi_1 - \phi_2$, exists between the signals arriving at a given point. (It depends only on the phase difference, of course, and not on the phase itself.) So let us consider the case where the two oscillators, of equal amplitude, are separated by some distance d and have an intrinsic relative phase α . (When one is at phase zero, the phase of the other is α .) Then we ask what the intensity will be in some azimuth direction θ from the E-W line. [Note that this is *not* the same θ as appears in (29.1). We are torn between using an unconventional symbol like $\mathcal{U}$, or the conventional symbol θ (Fig. 29-10).] The phase relationship is found by noting that the difference in distance from P to the two oscillators is $d \sin \theta$, so that the phase difference contribution from this is the number of wavelengths in $d \sin \theta$, multiplied by 2π . (Those who are more sophisticated might want to multiply the wave number k , which is the rate of change of phase with distance, by $d \sin \theta$; it is exactly the same.) The phase difference due to the distance difference is thus $2\pi d \sin \theta / \lambda$, but, due to the timing of the oscillators, there is an additional phase α . So the phase difference at arrival would be

$$\phi_2 - \phi_1 = \alpha + 2\pi d \sin \theta / \lambda. \quad (29.17)$$

This takes care of all the cases. Thus all we have to do is substitute this expression into (29.16) for the case $A_1 = A_2$, and we can calculate all the various results for two antennas of equal intensity.

Now let us see what happens in our various cases. The reason we know, for example, that the intensity is 2 at 30° in Fig. 29-5 is the following: the two oscillators are $\frac{1}{2}\lambda$ apart, so at 30° , $d \sin \theta = \lambda/4$. Thus $\phi_2 - \phi_1 = 2\pi\lambda/4\lambda = \pi/2$, and so the interference term is zero. (We are adding two vectors at 90° .) The result is the hypotenuse of a 45° right-angle triangle, which is $\sqrt{2}$ times the unit amplitude; squaring it, we get twice the intensity of one oscillator alone. All the other cases can be worked out in this same way.

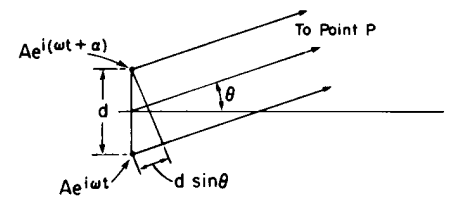


Fig. 29-10. Two oscillators of equal amplitude, with a phase difference α between them.

Diffraction

30-1 The resultant amplitude due to n equal oscillators

This chapter is a direct continuation of the previous one, although the name has been changed from *Interference* to *Diffraction*. No one has ever been able to define the difference between interference and diffraction satisfactorily. It is just a question of usage, and there is no specific, important physical difference between them. The best we can do, roughly speaking, is to say that when there are only a few sources, say two, interfering, then the result is usually called interference, but if there is a large number of them, it seems that the word diffraction is more often used. So, we shall not worry about whether it is interference or diffraction, but continue directly from where we left off in the middle of the subject in the last chapter.

Thus we shall now discuss the situation where there are n equally spaced oscillators, all of equal amplitude but different from one another in phase, either because they are driven differently in phase, or because we are looking at them at an angle such that there is a difference in time delay. For one reason or another, we have to add something like this:

$$R = A[\cos \omega t + \cos(\omega t + \phi) + \cos(\omega t + 2\phi) + \cdots + \cos(\omega t + (n-1)\phi)], \quad (30.1)$$

where ϕ is the phase difference between one oscillator and the next one, as seen in a particular direction. Specifically, $\phi = \alpha + 2\pi d \sin \theta / \lambda$. Now we must add all the terms together. We shall do this geometrically. The first one is of length A , and it has zero phase. The next is also of length A and it has a phase equal to ϕ . The next one is again of length A and it has a phase equal to 2ϕ , and so on. So we are evidently going around an equiangular polygon with n sides (Fig. 30-1).

Now the vertices, of course, all lie on a circle, and we can find the net amplitude most easily if we find the radius of that circle. Suppose that Q is the center of the circle. Then we know that the angle OQS is just a phase angle ϕ . (This is because the radius QS bears the same geometrical relation to A_2 as QO bears to A_1 , so they form an angle ϕ between them.) Therefore the radius r must be such that $A = 2r \sin \phi/2$, which fixes r . But the large angle OQT is equal to $n\phi$, and we thus find that $A_R = 2r \sin n\phi/2$. Combining these two results to eliminate r , we get

$$A_R = A \frac{\sin n\phi/2}{\sin \phi/2}. \quad (30.2)$$

The resultant intensity is thus

$$I = I_0 \frac{\sin^2 n\phi/2}{\sin^2 \phi/2}. \quad (30.3)$$

Now let us analyze this expression and study some of its consequences. In the first place, we can check it for $n = 1$. It checks: $I = I_0$. Next, we check it for $n = 2$: writing $\sin \phi = 2 \sin \phi/2 \cos \phi/2$, we find that $A_R = 2A \cos \phi/2$, which agrees with (29.12).

Now the idea that led us to consider the addition of several sources was that we might get a much stronger intensity in one direction than in another; that the nearby maxima which would have been present if there were only two sources will have gone down in strength. In order to see this effect, we plot the curve that comes from (30.3), taking n to be enormously large and plotting the region near

30-1 The resultant amplitude due to n equal oscillators

30-2 The diffraction grating

30-3 Resolving power of a grating

30-4 The parabolic antenna

30-5 Colored films; crystals

30-6 Diffraction by opaque screens

30-7 The field of a plane of oscillating charges

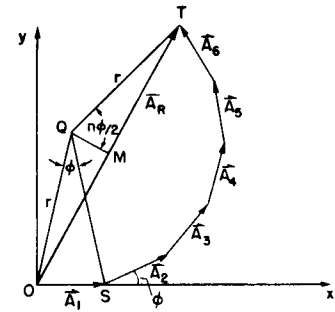


Fig. 30-1. The resultant amplitude of $n = 6$ equally spaced sources with net successive phase differences ϕ .

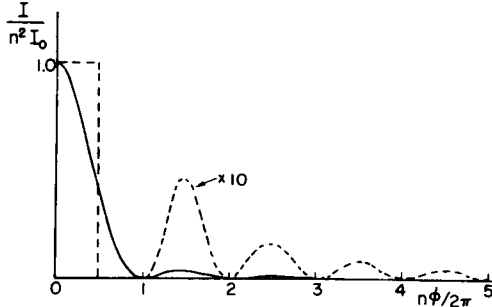


Fig. 30-2. The intensity as a function of phase angle for a large number of oscillators of equal strength.

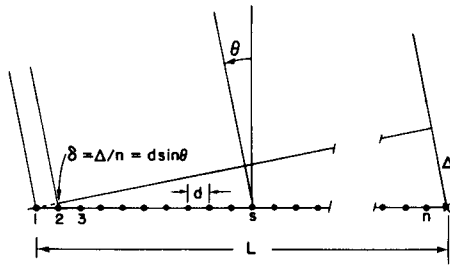


Fig. 30-3. A linear array of n equal oscillators, driven with phases $\alpha_s = s\alpha$.

$\phi = 0$. In the first place, if ϕ is exactly 0, we have $0/0$, but if ϕ is infinitesimal, the ratio of the two sines squared is simply n^2 , since the sine and the angle are approximately equal. Thus the intensity of the maximum of the curve is equal to n^2 times the intensity of one oscillator. That is easy to see, because if they are all in phase, then the little vectors have no relative angle and all n of them add up so the amplitude is n times, and the intensity n^2 times, stronger.

As the phase ϕ increases, the ratio of the two sines begins to fall off, and the first time it reaches zero is when $n\phi/2 = \pi$, because $\sin \pi = 0$. In other words, $\phi = 2\pi/n$ corresponds to the first minimum in the curve (Fig. 30-2). In terms of what is happening with the arrows in Fig. 30-1, the first minimum occurs when all the arrows come back to the starting point; that means that the total accumulated angle in all the arrows, the total phase difference between the first and last oscillator, must be 2π to complete the circle.

Now we go to the next maximum, and we want to see that it is really much smaller than the first one, as we had hoped. We shall not go precisely to the maximum position, because both the numerator and the denominator of (30.3) are variant, but $\sin \phi/2$ varies quite slowly compared with $\sin n\phi/2$ when n is large, so when $\sin n\phi/2 = 1$ we are very close to the maximum. The next maximum of $\sin^2 n\phi/2$ comes at $n\phi/2 = 3\pi/2$, or $\phi = 3\pi/n$. This corresponds to the arrows having traversed the circle one and a half times. On putting $\phi = 3\pi/n$ into the formula to find the size of the maximum, we find that $\sin^2 3\pi/2 = 1$ in the numerator (because that is why we picked this angle), and in the denominator we have $\sin^2 3\pi/2n$. Now if n is sufficiently large, then this angle is very small and the sine is equal to the angle; so for all practical purposes, we can put $\sin 3\pi/2n = 3\pi/2n$. Thus we find that the intensity at this maximum is $I = I_0(4n^2/9\pi^2)$. But $n^2 I_0$ was the maximum intensity, and so we have $4/9\pi^2$ times the maximum intensity, which is about 0.047, less than 5 percent, of the maximum intensity! Of course there are decreasing intensities farther out. So we have a very sharp central maximum with very weak subsidiary maxima on the sides.

It is possible to prove that the area of the whole curve, including all the little bumps, is equal to $2\pi n I_0$, or twice the area of the dotted rectangle in Fig. 30-2.

Now let us consider further how we may apply Eq. (30.3) in different circumstances, and try to understand what is happening. Let us consider our sources to be all on a line, as drawn in Fig. 30-3. There are n of them, all spaced by a distance d , and we shall suppose that the intrinsic relative phase, one to the next, is α . Then if we are observing in a given direction θ from the normal, there is an additional phase $2\pi d \sin \theta / \lambda$ because of the time delay between each successive two, which we talked about before. Thus

$$\begin{aligned}\phi &= \alpha + 2\pi d \sin \theta / \lambda \\ &= \alpha + kd \sin \theta.\end{aligned}\quad (30.4)$$

First, we shall take the case $\alpha = 0$. That is, all oscillators are in phase, and we want to know what the intensity is as a function of the angle θ . In order to find out, we merely have to put $\phi = kd \sin \theta$ into formula (30.3) and see what happens. In the first place, there is a maximum when $\phi = 0$. That means that when all the oscillators are in phase there is a strong intensity in the direction $\theta = 0$. On the other hand, an interesting question is, where is the first minimum? That occurs when $\phi = 2\pi/n$. In other words, when $2\pi d \sin \theta / \lambda = 2\pi/n$, we get the first minimum of the curve. If we get rid of the 2π 's so we can look at it a little better, it says that

$$nd \sin \theta = \lambda. \quad (30.5)$$

Now let us understand physically why we get a minimum at that position. nd is the total length L of the array. Referring to Fig. 30-3, we see that $nd \sin \theta = L \sin \theta = \Delta$. What (30.5) says is that when Δ is equal to one wavelength, we get a minimum. Now why do we get a minimum when $\Delta = \lambda$? Because the contributions of the various oscillators are then uniformly distributed in phase from 0° to

360°. The arrows (Fig. 30-1) are going around a whole circle—we are adding equal vectors in all directions, and such a sum is zero. So when we have an angle such that $\Delta = \lambda$, we get a minimum. That is the first minimum.

There is another important feature about formula (30.3), which is that if the angle ϕ is increased by any multiple of 2π , it makes no difference to the formula. So we will get other strong maxima at $\phi = 2\pi, 4\pi, 6\pi$, and so forth. Near each of these great maxima the pattern of Fig. 30-2 is repeated. We may ask ourselves, what is the geometrical circumstance that leads to these other great maxima? The condition is that $\phi = 2\pi m$, where m is any integer. That is, $2\pi d \sin \theta / \lambda = 2\pi m$. Dividing by 2π , we see that

$$d \sin \theta = m\lambda. \quad (30.6)$$

This looks like the other formula, (30.5). No, that formula was $nd \sin \theta = \lambda$. The difference is that here we have to look at the *individual sources*, and when we say $d \sin \theta = m\lambda$, that means that we have an angle θ such that $\delta = m\lambda$. In other words, each source is now contributing a certain amount, and successive ones are out of phase by a whole multiple of 360°, and therefore are contributing *in phase*, because out of phase by 360° is the same as being in phase. So they all contribute in phase and produce just as good a maximum as the one for $m = 0$ that we discussed before. The subsidiary bumps, the whole shape of the pattern, is just like the one near $\phi = 0$, with exactly the same minima on each side, etc. Thus such an array will send beams in various directions—each beam having a strong central maximum and a certain number of weak “side lobes.” The various strong beams are referred to as the zero-order beam, the first-order beam, etc., according to the value of m . m is called the *order* of the beam.

We call attention to the fact that if d is less than λ , Eq. (30.6) can have no solution except $m = 0$, so that if the spacing is too small there is only one possible beam, the zero-order one centered at $\theta = 0$. (Of course, there is also a beam in the opposite direction.) In order to get subsidiary great maxima, we must have the spacing d of the array greater than one wavelength.

30-2 The diffraction grating

In technical work with antennas and wires it is possible to arrange that all the phases of the little oscillators, or antennas, are equal. The question is whether and how we can do a similar thing with light. We cannot at the present time literally make little optical-frequency radio stations and hook them up with infinitesimal wires and drive them all with a given phase. But there is a very easy way to do what amounts to the same thing.

Suppose that we had a lot of parallel wires, equally spaced at a spacing d , and a radiofrequency source very far away, practically at infinity, which is generating an electric field which arrives at each one of the wires at the same phase (it is so far away that the time delay is the same for all of the wires). (One can work out cases with curved arrays, but let us take a plane one.) Then the external electric field will drive the electrons up and down in each wire. That is, the field which is coming from the original source will shake the electrons up and down, and in moving, these represent *new generators*. This phenomenon is called scattering: a light wave from some source can induce a motion of the electrons in a piece of material, and these motions generate their own waves. Therefore all that is necessary is to set up a lot of wires, equally spaced, drive them with a radiofrequency source far away, and we have the situation that we want, without a whole lot of special wiring. If the incidence is normal, the phases will be equal, and we will get exactly the circumstance we have been discussing. Therefore, if the wire spacing is greater than the wavelength, we will get a strong intensity of scattering in the normal direction, and in certain other directions given by (30.6).

This can also be done with light! Instead of wires, we use a flat piece of glass and make notches in it such that each of the notches scatters a little differently than the rest of the glass. If we then shine light on the glass, each one of the notches

will represent a source, and if we space the lines very finely, but not closer than a wavelength (which is technically almost impossible anyway), then we would expect a miraculous phenomenon: the light not only will pass straight through, but there will also be a strong beam at a finite angle, depending on the spacing of the notches! Such objects have actually been made and are in common use—they are called *diffraction gratings*.

In one of its forms, a diffraction grating consists of nothing but a plane glass sheet, transparent and colorless, with scratches on it. There are often several hundred scratches to the millimeter, *very* carefully arranged so as to be equally spaced. The effect of such a grating can be seen by arranging a projector so as to throw a narrow, vertical line of light (the image of a slit) onto a screen. When we put the grating into the beam, with its scratches vertical, we see that the line is still there but, in addition, on each side we have *another* strong patch of light which is *colored*. This, of course, is the slit image spread out over a wide angular range, because the angle θ in (30.6) depends upon λ , and lights of different colors, as we know, correspond to different frequencies, and therefore different wavelengths. The longest visible wavelength is red, and since $d \sin \theta = \lambda$, that requires a larger θ . And we do, in fact, find that red is at a greater angle out from the central image! There should also be a beam on the other side, and indeed we see one on the screen. Then, there might be another solution of (30.6) when $m = 2$. We do see that there is something vaguely there—very weak—and there are even other beams beyond.

We have just argued that all these beams ought to be of the same strength, but we see that they actually are not and, in fact, not even the first ones on the right and left are equal! The reason is that the grating has been carefully built to do just this. How? If the grating consists of very fine notches, infinitesimally wide, spaced evenly, then all the intensities would indeed be equal. But, as a matter of fact, although we have taken the simplest case, we could also have considered an array of *pairs* of antennas, in which each member of the pair has a certain strength and some relative phase. In this case, it is possible to get intensities which are different in the different orders. A grating is often made with little “sawtooth” cuts instead of little symmetrical notches. By carefully arranging the “sawteeth,” more light may be sent into one particular order of spectrum than into the others. In a practical grating, we would like to have as much light as possible in one of the orders. This may seem a complicated point to bring in, but it is a very clever thing to do, because it makes the grating more useful.

So far, we have taken the case where all the phases of the sources are equal. But we also have a formula for ϕ when the phases differ from one to the next by an angle α . That requires wiring up our antennas with a slight phase shift between each one. Can we do that with light? Yes, we can do it very easily, for suppose that there were a source of light at infinity, *at an angle* such that the light is coming in at an angle θ_{in} , and let us say that we wish to discuss the scattered beam, which is leaving at an angle θ_{out} . The θ_{out} is the same θ as we have had before, but the θ_{in} is merely a means for arranging that the phase of each source is different: the light coming from the distant driving source first hits one scratch, then the next, then the next, and so on, with a phase shift from one to the other, which, as we see, is $\alpha = -d \sin \theta_{in}/\lambda$. Therefore we have the formula for a grating in which light both comes in and goes out at an angle:

$$\phi = 2\pi d \sin \theta_{out}/\lambda - 2\pi d \sin \theta_{in}/\lambda. \quad (30.7)$$

Let us try to find out where we get strong intensity in these circumstances. The condition for strong intensities is, of course, that ϕ should be a multiple of 2π . There are several interesting points to be noted.

One case of rather great interest is that which corresponds to $m = 0$, where d is less than λ ; in fact, this is the only solution. In this case we see that $\sin \theta_{out} = \sin \theta_{in}$, which means that the light comes out in the *same direction* as the light which was exciting the grating. We might think that the light “goes right through.” No, it is *different light* that we are talking about. The light that goes right through is from the original source; what we are talking about is the new light *which is*

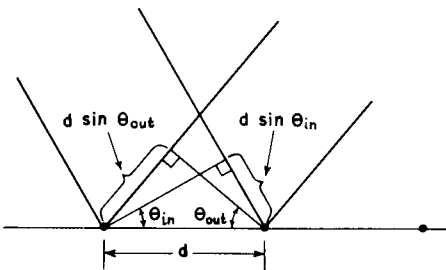


Fig. 30-4. The path difference for rays scattered from adjacent rulings of a grating is $d \sin \theta_{out} - d \sin \theta_{in}$.

generated by scattering. It turns out that the scattered light is going in the same direction as the original light, in fact it can interfere with it—a feature which we will study later.

There is another solution for this same case. For a given θ_{in} , θ_{out} may be the supplement of θ_{in} . So not only do we get a beam in the same direction as the incoming beam but also one in another direction, which, if we consider it carefully, is such that the *angle of incidence is equal to the angle of scattering*. This we call the *reflected beam*.

So we begin to understand the basic machinery of reflection: the light that comes in generates motions of the atoms in the reflector, and the reflector then regenerates a *new wave*, and one of the solutions for the direction of scattering, the *only* solution if the spacing of the scatterers is small compared with one wavelength, is that the angle at which the light comes out is equal to the angle at which it comes in!

Next, we discuss the special case when $d \rightarrow 0$. That is, we have just a solid piece of material, so to speak, but of finite length. In addition, we want the phase shift from one scatterer to the next to go to zero. In other words, we put more and more antennas between the other ones, so that each of the phase differences is getting smaller, but the number of antennas is increasing in such a way that the total phase difference, between one end of the line and the other, is constant. Let us see what happens to (30.3) if we keep the difference in phase $n\phi$ from one end to the other constant (say $n\phi = \Phi$), letting the number go to infinity and the phase shift ϕ of each one go to zero. But now ϕ is so small that $\sin \phi = \phi$, and if we also recognize $n^2 I_0$ as I_m , the maximum intensity at the center of the beam, we find

$$I = 4I_m \sin^2 \frac{1}{2}\Phi / \Phi^2. \quad (30.8)$$

This limiting case is what is shown in Fig. 30-2.

In such circumstances we find the same general kind of a picture as for finite spacing with $d > \lambda$; all the side lobes are practically the same as before, but there are no higher-order maxima. If the scatterers are all in phase, we get a maximum in the direction $\theta_{\text{out}} = 0$, and a minimum when the distance Δ is equal to λ , just as for finite d and n . So we can even analyze a *continuous* distribution of scatterers or oscillators, by using integrals instead of summing.

As an example, suppose there were a long line of oscillators, with the charge oscillating along the direction of the line (Fig. 30-5). From such an array the greatest intensity is perpendicular to the line. There is a little bit of intensity up and down from the equatorial plane, but it is very slight. With this result, we can handle a more complicated situation. Suppose we have a set of such lines, each producing a beam only in a plane perpendicular to the line. To find the intensity in various directions from a series of long wires, instead of infinitesimal wires, is the same problem as it was for infinitesimal wires, so long as we are in the central plane perpendicular to the wires; we just add the contribution from each of the long wires. That is why, although we actually analyzed only tiny antennas, we might as well have used a grating with long, narrow slots. Each of the long slots produces an effect only in its own direction, not up and down, but they are all set next to each other horizontally, so they produce interference that way.

Thus we can build up more complicated situations by having various distributions of scatterers in lines, planes, or in space. The first thing we did was to consider scatterers in a line, and we have just extended the analysis to strips; we can work it out by just doing the necessary summations, adding the contributions from the individual scatterers. The principle is always the same.

30-3 Resolving power of a grating

We are now in a position to understand a number of interesting phenomena. For example, consider the use of a grating for separating wavelengths. We noticed that the whole spectrum was spread out on the screen, so a grating can be used as an instrument for separating light into its different wavelengths. One of the

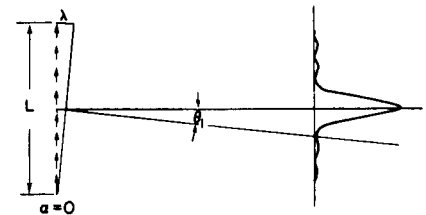


Fig. 30-5. The intensity pattern of a continuous line of oscillators has a single strong maximum and many weak "side lobes."

interesting questions is: supposing that there were two sources of slightly different frequency, or slightly different wavelength, how close together in wavelength could they be such that the grating would be unable to tell that there were really two different wavelengths there? The red and the blue were clearly separated. But when one wave is red and the other is slightly redder, very close, how close can they be? This is called the *resolving power* of the grating, and one way of analyzing the problem is as follows. Suppose that for light of a certain color we happen to have the maximum of the diffracted beam occurring at a certain angle. If we vary the wavelength the phase $2\pi d \sin \theta / \lambda$ is different, so of course the maximum occurs at a different angle. That is why the red and blue are spread out. How different in angle must it be in order for us to be able to see it? If the two maxima are exactly on top of each other, of course we cannot see them. If the maximum of one is far enough away from the other, then we can see that there is a double bump in the distribution of light. In order to be able to just make out the double bump, the following simple criterion, called *Rayleigh's criterion*, is usually used (Fig. 30-6). It is that the first minimum from one bump should sit at the maximum of the other. Now it is very easy to calculate, when one minimum sits on the other maximum, how much the difference in wavelength is. The best way to do it is geometrically.

In order to have a maximum for wavelength λ' , the distance Δ (Fig. 30-3) must be $n\lambda'$, and if we are looking at the m th-order beam, it is $mn\lambda'$. In other words, $2\pi d \sin \theta / \lambda' = 2\pi m$, so $nd \sin \theta$, which is Δ , is λ' times n , or $mn\lambda'$. For the other beam, of wavelength λ , we want to have a *minimum* at this angle. That is, we want Δ to be exactly one wavelength λ more than $mn\lambda$. That is, $\Delta = mn\lambda + \lambda = mn\lambda'$. Thus if $\lambda' = \lambda + \Delta\lambda$, we find

$$\Delta\lambda / \lambda = 1 / mn. \quad (30.9)$$

The ratio $\lambda / \Delta\lambda$ is called the *resolving power* of a grating; we see that it is equal to the total number of lines in the grating, times the order. It is not hard to prove that this formula is equivalent to the formula that the error in frequency is equal to the reciprocal time difference between extreme paths that are allowed to interfere.*

$$\Delta\nu = 1/T.$$

In fact, that is the best way to remember it, because the general formula works not only for gratings, but for any other instrument whatsoever, while the special formula (30.9) depends on the fact that we are using a grating.

30-4 The parabolic antenna

Now let us consider another problem in resolving power. This has to do with the antenna of a radio telescope, used for determining the position of radio sources in the sky, i.e., how large they are in angle. Of course if we use any old antenna and find signals, we would not know from what direction they came. We are very interested to know whether the source is in one place or another. One way we can find out is to lay out a whole series of equally spaced dipole wires on the Australian landscape. Then we take all the wires from these antennas and feed them into the same receiver, in such a way that all the delays in the feed lines are equal. Thus the receiver receives signals from all of the dipoles in phase. That is, it adds all the waves from every one of the dipoles in the same phase. Now what happens? If the source is directly above the array, at infinity or nearly so, then its radiowaves will excite all the antennas in the same phase, so they all feed the receiver together.

Now suppose that the radio source is at a slight angle θ from the vertical. Then the various antennas are receiving signals a little out of phase. The receiver adds all these out-of-phase signals together, and so we get nothing, if the angle

* In our case $T = \Delta / c = mn\lambda / c$, where c is the speed of light. The frequency $\nu = c / \lambda$, so $\Delta\nu = c \Delta\lambda / \lambda^2$.

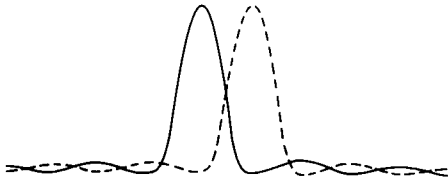


Fig. 30-6. Illustration of the Rayleigh criterion. The maximum of one pattern falls on the first minimum of the other.

θ is too big. How big may the angle be? *Answer:* we get zero if the angle $\Delta/L = \theta$ (Fig. 30-3) corresponds to a 360° phase shift, that is, if Δ is the wavelength λ . This is because the vector contributions form together a complete polygon with zero resultant. The smallest angle that can be resolved by an antenna array of length L is $\theta = \lambda/L$. Notice that the receiving pattern of an antenna such as this is exactly the same as the intensity distribution we would get if we turned the receiver around and made it into a transmitter. This is an example of what is called a *reciprocity principle*. It turns out, in fact, to be generally true for any arrangement of antennas, angles, and so on, that if we first work out what the relative intensities would be in various directions if the receiver were a transmitter instead, then the relative directional sensitivity of a receiver with the same external wiring, the same array of antennas, is the same as the relative intensity of emission would be if it were a transmitter.

Some radio antennas are made in a different way. Instead of having a whole lot of dipoles in a long line, with a lot of feed wires, we may arrange them not in a line but in a curve, and put the receiver at a certain point where it can detect the scattered waves. This curve is cleverly designed so that if the radiowaves are coming down from above, and the wires scatter, making a new wave, the wires are so arranged that the scattered waves reach the receiver all at the same time (Fig. 26-12). In other words, the curve is a *parabola*, and when the source is exactly on its axis, we get a very strong intensity at the focus. In this case we understand very clearly what the resolving power of such an instrument is. The arranging of the antennas on a parabolic curve is not an essential point. It is only a convenient way to get all the signals to the same point with no relative delay and without feed wires. The angle such an instrument can resolve is still $\theta = \lambda/L$, where L is the separation of the first and last antennas. It does not depend on the spacing of the antennas and they may be very close together or in fact be all one piece of metal. Now we are describing a telescope mirror, of course. We have found the resolving power of a telescope! (Sometimes the resolving power is written $\theta = 1.22\lambda/L$, where L is the diameter of the telescope. The reason that it is not exactly λ/L is this: when we worked out that $\theta = \lambda/L$, we assumed that all the lines of dipoles were equal in strength, but when we have a circular telescope, which is the way we usually arrange a telescope, not as much signal comes from the outside edges, because it is not like a square, where we get the same intensity all along a side. We get somewhat less because we are using only part of the telescope there; thus we can appreciate that the effective diameter is a little shorter than the true diameter, and that is what the 1.22 factor tells us. In any case, it seems a little pedantic to put such precision into the resolving power formula.*)

30-5 Colored films; crystals

The above, then, are some of the effects of interference obtained by adding the various waves. But there are a number of other examples, and even though we do not understand the fundamental mechanism yet, we will some day, and we can understand even now how the interference occurs. For example, when a light wave hits a surface of a material with an index n , let us say at normal incidence, some of the light is reflected. The *reason* for the reflection we are not in a position to understand right now; we shall discuss it later. But suppose we know that some of the light is reflected both on entering and leaving a refracting medium. Then, if we look at the reflection of a light source in a thin film, we see the sum of two waves; if the thicknesses are small enough, these two waves will produce an interference, either constructive or destructive, depending on the signs of the phases. It might be, for instance, that for red light, we get an enhanced reflection, but for

* This is because Rayleigh's criterion is a rough idea in the first place. It tells you where it begins to get very hard to tell whether the image was made by one or by two stars. Actually, if sufficiently careful measurements of the exact intensity distribution over the diffracted image spot can be made, the fact that two sources make the spot can be proved even if θ is less than λ/L .

blue light, which has a different wavelength, perhaps we get a destructively interfering reflection, so that we see a bright red reflection. If we change the thickness, i.e., if we look at another place where the film is thicker, it may be reversed, the red interfering and the blue not, so it is bright blue, or green, or yellow, or whatnot. So we see *colors* when we look at thin films and the colors change if we look at different angles, because we can appreciate that the timings are different at different angles. Thus we suddenly appreciate another hundred thousand situations involving the colors that we see on oil films, soap bubbles, etc. at different angles. But the principle is all the same: we are only adding waves at different phases.

As another important application of diffraction, we may mention the following. We used a grating and we saw the diffracted image on the screen. If we had used monochromatic light, it would have been at a certain specific place. Then there were various higher-order images also. From the positions of the images, we could tell how far apart the lines on the grating were, if we knew the wavelength of the light. From the difference in intensity of the various images, we could find out the shape of the grating scratches, whether the grating was made of wires, sawtooth notches, or whatever, *without being able to see them*. This principle is used to discover the positions of the *atoms in a crystal*. The only complication is that a crystal is three-dimensional; it is a repeating three-dimensional array of atoms. We cannot use ordinary light, because we must use something whose wavelength is less than the space between the atoms or we get no effect; so we must use radiation of very short wavelength, i.e., x-rays. So, by shining x-rays into a crystal and by noticing how intense is the reflection in the various orders, we can determine the arrangement of the atoms inside without ever being able to see them with the eye! It is in this way that we know the arrangement of the atoms in various substances, which permitted us to draw those pictures in the first chapter, showing the arrangement of atoms in salt, and so on. We shall later come back to this subject and discuss it in more detail, and therefore we say no more about this most remarkable idea at present.

30-6 Diffraction by opaque screens

Now we come to a very interesting situation. Suppose that we have an opaque sheet with holes in it, and a light on one side of it. We want to know what the intensity is on the other side. What most people say is that the light shines through the holes, and produces an effect on the other side. It will turn out that one gets the right answer, to an excellent approximation, if he assumes that there are sources distributed with uniform density across the open holes, and that the phases of these sources are the same as they would have been if the opaque material were absent. Of course, actually there are *no* sources at the holes, in fact that is the only place that there are *certainly* no sources. Nevertheless, we get the correct diffraction patterns by considering the holes to be the only places that there *are* sources; that is a rather peculiar fact. We shall explain later why this is true, but for now let us just suppose that it is.

In the theory of diffraction there is another kind of diffraction that we should briefly discuss. It is usually not discussed in an elementary course as early as this, only because the mathematical formulas involved in adding these little vectors are a little elaborate. Otherwise it is exactly the same as we have been doing all along. All the interference phenomena are the same; there is nothing very much more advanced involved, only the circumstances are more complicated and it is harder to add the vectors together, that is all.

Suppose that we have light coming in from infinity, casting a shadow of an object. Figure 30-7 shows a screen on which the shadow of an object AB is made by a light source very far away compared with one wavelength. Now we would expect that outside the shadow, the intensity is all bright, and inside it, it is all dark. As a matter of fact, if we plot the intensity as a function of position near the shadow edge, the intensity rises and then overshoots, and wobbles, and oscillates about in a very peculiar manner near this edge (Fig. 30-8). We now shall discuss the reason for this. If we use the theorem that we have not yet proved, then we can

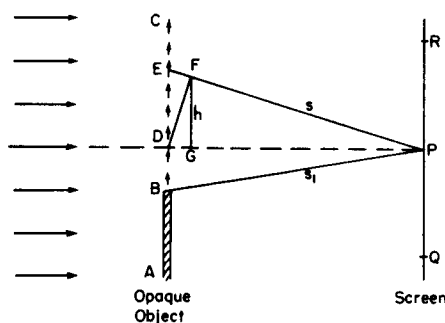


Fig. 30-7. A distant light source casts a shadow of an opaque object on a screen.